Probability

— *EYNTKA* Series —

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Introduction

Probability theory, as this book develops it, is measure theory read through a single normalization: the ambient measure has total mass one. That one constraint changes what the theory is about. A set of measure at most one reads as an event of a definite likelihood; complementation becomes quantitative through $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; and a vocabulary opens up that has no useful analogue for an infinite measure, the language of events that hold *almost surely* or recur *infinitely often*, of variables that are *independent*, of an average value that a random quantity takes in the long run. The measure theory itself is taken as built. What this book installs on top of it is the probabilistic structure and, above all, the theorems that structure makes available: the laws of large numbers, the central limit theorem, and the convergence and stopping theorems for martingales.

The book is written for a reader who has met measure theory at the level of *Measure Theory* [2]: the Lebesgue integral, the convergence theorems (monotone, dominated, Fatou), product measures and Fubini, the L^p spaces, and the Radon-Nikodym theorem. No prior probability is assumed. Where a construction is purely measure-theoretic, the Caratheodory extension, the existence of Lebesgue-Stieltjes measures, the Radon-Nikodym derivative, it is cited to that volume rather than rebuilt; the effort here goes into what is probabilistic, the independence and the limit theorems that separate the subject from general integration theory.

The journey has three parts. Part I fixes the objects: the probability space and the tools that pin a measure down from partial information, the random variable and the independence that is probability's own contribution to measure theory, and expectation as the integral against $\mathbb{P}$ with the inequalities every later estimate leans on. Part II is the classical core, the asymptotics of a sum of many independent contributions: the weak and strong laws of large numbers, weak convergence and the characteristic function that linearizes it, the central limit theorem, and, at the fine end of the scale, the law of the iterated logarithm. Part III turns to information that arrives over time: conditional expectation, built as a Radon-Nikodym derivative and read as an orthogonal projection, and then the martingales it supports, whose optional-stopping and convergence theorems close the book at the threshold of continuous time.

A thread from number theory runs through the middle part and is worth flagging, since it is not where a reader expects probability to lead. The number of distinct prime factors of a random integer obeys a law of large numbers (Hardy-Ramanujan) and then a central limit theorem (Erdos-Kac); the size of the typical fluctuation is governed, like a random walk's, by the iterated logarithm. These appear not as digressions but as the theory's own examples, worked with the same machinery as the coin-tossing that motivates it. The fair coin, and the simple random walk it generates, is the running example throughout: it is where each abstraction is first tested and where the final chapter's martingales do their sharpest work.

Each section ends with exercises, worked in full in an accompanying answer block; they are part of the exposition, not an appendix to it. Cross-references are live, and a result is stated where it is first needed and cited thereafter rather than restated. The reader in a hurry can follow the main line through the boxed definitions and theorems and the paragraph of motivation before each; the reader with time will find the proofs complete and the examples substantive.

0.0.1 Prerequisites and Place in the Series This volume builds on *Measure Theory* [2]. No later volume currently lists it as a prerequisite. The reader has access to every completed volume of the series but is not assumed to have read all of them; prerequisites are cited where invoked.

Notation and Conventions

The measure-theoretic notation follows *Measure Theory* [2]. The conventions particular to this book are collected here.

- A *probability space* is written $(\Omega, \mathcal{F}, \mathbb{P})$, with $\mathbb{P}(\Omega) = 1$. Events are elements of $\mathcal{F}$; "$\mathbb{P}$-almost surely", abbreviated *a.s.*, means outside a $\mathbb{P}$-null set. Random variables are measurable maps on Ω, named by capitals X, Y, Z.
- Expectation is $\mathbb{E}X = \int_\Omega X \, d\mathbb{P}$, and $\mathbb{E}[X; A] = \mathbb{E}[X \mathbb{1}_A]$ over an event A. Here $\mathbb{1}_A$ (or $\mathbb{1}$) is the indicator of A. Variance and covariance are Var and Cov. The law (distribution) of X is the pushforward measure $\mathbb{P} \circ X^{-1}$ on the target.
- Independence is written $X \perp\!\!\!\perp Y$ (and likewise for σ-algebras or events). The conditioning bar is $\big|$, as in $\mathbb{E}[X \big| \mathcal{G}]$ and $\mathbb{P}(A \big| \mathcal{G})$.
- Standard laws use upright operator names: $\mathrm{Ber}(p)$, $\mathrm{Bin}(n, p)$, $\mathrm{Poi}(\lambda)$, and Unif for the uniform law; the Gaussian law is $\mathcal{N}(\mu, \sigma^2)$, with $\mathcal{N}(0, 1)$ standard.
- Modes of convergence are named in words or by the standard arrows, "almost surely", "in probability", "in L^p", and "in distribution" (equivalently weak convergence). Filtrations are increasing families $(\mathcal{F}_n)_{n \geq 0}$ of sub-σ-algebras, and a stopping time is written τ.
- Function declarations use $f \colon X \to Y$; set-builder and conditioning both use the bar $\big|$. Structural divisions are referenced by name (Part I, and so on), never by a hardcoded numeral. The single hyphen is the only dash in the text.

Part I

Foundations of Probability

Probability theory, in the form developed in this book, is measure theory under a particular reading: a measure of total mass one, whose measurable sets are named *events* and whose integral is named *expectation.* The measure-theoretic foundation is taken as given, drawn from *Measure Theory* [2]; the task of this Part is not to rebuild it but to install the probabilistic vocabulary, to isolate the structure that the normalization $\mathbb{P}(\Omega) = 1$ makes available, and to show why that structure is worth having.

The development proceeds in three movements. Chapter 1 fixes the probability space and two instruments that let a probability measure be determined by partial information: the π-λ theorem, which controls when agreement on a generating class forces agreement everywhere, and the Borel-Cantelli lemmas, which convert a sum of probabilities into a statement about how often events recur. The chapters that follow introduce random variables together with the independence that separates probability from general measure theory, and then expectation as the Lebesgue integral against $\mathbb{P}$, with the inequalities on which nearly every later estimate rests.

1

Probability Spaces

A probability space is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ in which the total measure $\mathbb{P}(\Omega)$ equals one. The definition adds nothing to measure theory beyond that normalization, and yet the normalization is what the whole subject turns on. It makes the measure of a set a number in $[0, 1]$ that reads as a degree of certainty; it makes complementation quantitative, through $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; and it makes available a family of statements, such as the assertion that an event holds *almost surely* or that a sequence of events occurs *infinitely often*, that have no useful analogue when the ambient measure is infinite.

This chapter installs the apparatus in four steps. The first fixes the probability space itself and the elementary identities (monotonicity, inclusion-exclusion, the union bound, continuity along monotone sequences of events) that follow from countable additivity once the mass is one. The second isolates the π-λ theorem, which lets a probability measure be pinned down by its values on a class of sets closed under intersection. The third constructs the measures the rest of the book depends on: the correspondence between probability measures on $\mathbb{R}$ and their distribution functions, and the Kolmogorov extension that assembles an infinite sequence of independent trials, the fair-coin-tossing space chief among them. The fourth proves the Borel-Cantelli lemmas and reads them against that coin-tossing space. Throughout, the measure-theoretic constructions themselves (Carathéodory extension, the existence of Lebesgue-Stieltjes measures) are cited to *Measure Theory* [2] rather than rederived.

1.1 From Measure Spaces to Probability Spaces

Probability begins with a single act of normalization. A measure space $(\Omega, \mathcal{F}, \mu)$ carries a σ-algebra of measurable sets and a countably additive measure on them; nothing in that structure singles out a scale, and the total mass $\mu(\Omega)$ may be any value in $[0, \infty]$. Fixing that total mass at one turns the measure into a probability. The axioms are otherwise unchanged, and every theorem of measure theory continues to hold word for word. What changes is the reading: a set is now an *event*, its measure is the *chance* that the event occurs, and the number lives in $[0, 1]$ where it can be interpreted as a degree of certainty. This section fixes that vocabulary, records the elementary identities that the normalization makes available, and proves that probability is continuous along monotone sequences of events. None of the measure-theoretic content is rebuilt here; the σ-algebra axioms and countable additivity are recalled from *Measure Theory* [2] and put to probabilistic use.

Definition 1.1.1: Probability Space

A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ in which $\mathbb{P}(\Omega) = 1$. Explicitly, Ω is a nonempty set, $\mathcal{F}$ is a σ-algebra of subsets of Ω, and $\mathbb{P}\colon \mathcal{F} \to [0, 1]$ is a function satisfying

1. $\mathbb{P}(\Omega) = 1$;
2. (σ-additivity) for every sequence $(A_n)_{n\geq 1}$ of pairwise disjoint sets in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mathbb{P}(A_n)$$

The set Ω is the *sample space* and its elements ω are *sample points* (or *outcomes*). A set $A \in \mathcal{F}$ is an *event*, and $\mathbb{P}(A)$ is the *probability* of A. The event Ω is the *sure event* and $\emptyset$ is the *impossible event*; an event N with $\mathbb{P}(N) = 0$ is a *null event*. A statement about ω that holds for all ω outside some null event is said to hold *almost surely* (abbreviated *a.s.*).

The definition asks the reader to hold two pictures at once. Measure-theoretically, $\mathbb{P}$ is a finite measure whose total mass happens to be one, and every result proved for finite measures (monotone convergence, dominated convergence, the construction of product measures) applies to it verbatim. Probabilistically, $\mathbb{P}$ is a bookkeeping device for uncertainty: $\mathbb{P}(A) = 1$ says A is certain, $\mathbb{P}(A) = 0$ says A is negligible, and intermediate values interpolate between the two. The two pictures never conflict because the axioms are identical; the second is a way of reading the first. The requirement that $\mathcal{F}$ be a σ-algebra is exactly the requirement that the events form a class closed under the logical operations one performs on statements: “not A” is the complement A^c, “A or B” is the union, “A and B” is the intersection, and closure under countable unions permits “at least one of infinitely many events” to itself be an event. Countable additivity is what ties the numbers to those operations: the chance of a disjoint alternative is the sum of the chances.

The phrase *almost surely* is the probabilistic reading of “almost everywhere” from measure theory, and it is used throughout the subject. Because $\mathbb{P}(\Omega) = 1$, an event of probability one is the complement of a null event, so “A holds almost surely” and “A^c is null” say the same thing. This symmetry between full probability and null probability, invisible when the total mass is infinite, is precisely what the normalization gives; the interpretive prose closing this section returns to why it matters.

Before developing any theory, it is worth seeing the definition instantiated on the two spaces that ground the entire subject: a discrete space, where $\mathbb{P}$ is built from a list of weights, and the unit interval, where $\mathbb{P}$ is Lebesgue measure. Every finite computation in probability descends from the first, and the second is the prototype of a continuous distribution.

Example 1.1.2: Discrete Probability Spaces

Let Ω be a finite or countable set and let $\mathcal{F} = 2^\Omega$ be the power set, the largest σ-algebra on Ω. A *probability mass function* (pmf) on Ω is a function $p\colon \Omega \to [0,1]$ with

$$\sum_{\omega\in\Omega} p(\omega) = 1$$

Define $\mathbb{P}\colon \mathcal{F} \to [0,1]$ by

$$\mathbb{P}(A) = \sum_{\omega\in A} p(\omega)$$

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. Indeed $\mathbb{P}(\Omega) = \sum_{\omega\in\Omega} p(\omega) = 1$, and if (A_n) are pairwise disjoint then rearranging the (absolutely convergent, nonnegative) double sum gives

$$\mathbb{P}\left(\bigcup_n A_n\right) = \sum_{\omega\in\bigcup_n A_n} p(\omega) = \sum_n \sum_{\omega\in A_n} p(\omega) = \sum_n \mathbb{P}(A_n)$$

which is σ-additivity.

Take a single fair die: $\Omega = \{1,2,3,4,5,6\}$ with the uniform pmf $p(\omega) = 1/6$. The event "the roll is even" is $A = \{2,4,6\}$, and

$$\mathbb{P}(A) = p(2) + p(4) + p(6) = \frac{3}{6} = \frac{1}{2}$$

The event "the roll is at least 5" is $B = \{5,6\}$, so $\mathbb{P}(B) = 2/6 = 1/3$. The events A and B overlap in $A \cap B = \{6\}$ with $\mathbb{P}(A\cap B) = 1/6$, so the event "even or at least 5" has probability

$$\mathbb{P}(A\cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A\cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$$

anticipating the inclusion-exclusion identity proved below.

For a countable example, take $\Omega = \mathbb{N} = \{1,2,3,\dots\}$ with the geometric pmf $p(n) = 2^{-n}$. This is a valid pmf because $\sum_{n\geq 1} 2^{-n} = 1$. The event "the outcome is even" is $A = \{2,4,6,\dots\}$, and

$$\mathbb{P}(A) = \sum_{k\geq 1} 2^{-2k} = \frac{1/4}{1-1/4} = \frac{1}{3}$$

Here $\mathbb{P}(\{n\}) = 2^{-n} > 0$ for every n, so no single outcome is null; since every weight is positive, the only null event is the empty set.

The discrete case is deceptively complete: on a countable Ω every probability measure on the power set arises from a pmf, since $p(\omega) = \mathbb{P}(\{\omega\})$ recovers the weights and countable additivity recovers $\mathbb{P}$ from them. The situation changes decisively once Ω is uncountable, where no pmf on individual points can capture a measure that assigns zero to every singleton. The unit interval is the first and most important such space.

Example 1.1.3: The Uniform Distribution on the Unit Interval
Let $\Omega = [0,1]$, let $\mathcal{F} = \mathcal{B}([0,1])$ be the Borel σ-algebra (the restriction of $\mathcal{B}(\mathbb{R})$ to $[0,1]$), and let $\mathbb{P} = \lambda|_{[0,1]}$ be Lebesgue measure restricted to $[0,1]$. The existence of Lebesgue measure, a translation-invariant Borel measure with $\lambda([a,b]) = b - a$, is proved via the Carathéodory extension in *Measure Theory* [2] and is not rebuilt here. Because $\lambda([0,1]) = 1 - 0 = 1$, the triple $([0,1], \mathcal{B}([0,1]), \lambda)$ is a probability space, the *uniform distribution* $\mathrm{Unif}([0,1])$.

Probabilities are lengths. The event "the point lands in the left third" is $[0,1/3]$, with

$$\mathbb{P}([0,1/3]) = \lambda([0,1/3]) = \frac{1}{3}$$

Every singleton is null: $\mathbb{P}(\{x\}) = \lambda(\{x\}) = 0$ for each $x \in [0,1]$, since a point has length zero. By countable additivity the same holds for any countable set, so

$$\mathbb{P}(\mathbb{Q} \cap [0,1]) = \sum_{q \in \mathbb{Q} \cap [0,1]} \mathbb{P}(\{q\}) = 0$$

A point drawn uniformly at random is therefore irrational almost surely. This is the phenomenon a pmf cannot model: the mass is spread so thinly that no individual outcome carries any of it, yet intervals carry mass in proportion to their length. The reader should keep both examples in view; the discrete space is where finite combinatorial probability lives, and the uniform space is the seed from which every absolutely continuous distribution on $\mathbb{R}$ will be grown in section 1.3.

With the definition and its two anchoring examples in place, the elementary calculus of probabilities follows from countable additivity together with the one new fact $\mathbb{P}(\Omega) = 1$. Each identity below is a statement of measure theory specialized to total mass one, but the specialization is exactly what makes the identities quantitative rather than qualitative. The most consequential is complementation: in a general measure space one can say only that $\mu(A^c) = \mu(\Omega) - \mu(A)$ when $\mu(A) < \infty$, an equation with an infinite right-hand side when $\mu(\Omega) = \infty$; here it reads $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, turning "the event fails" into a number.

Proposition 1.1.4: Elementary Properties of Probability

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A, B, A_1, A_2, \dots \in \mathcal{F}$. Then:

1. $\mathbb{P}(\emptyset) = 0$.
2. (Finite additivity) if $A_1, \dots, A_n$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i)$.
3. (Complementation) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
4. (Monotonicity) if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$, and moreover $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.
5. (Inclusion-exclusion, two sets) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
6. (Inclusion-exclusion, general) for events $A_1, \dots, A_n$,
$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$
7. (Boole's inequality / the union bound) for any sequence (A_n),
$$\mathbb{P}\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n \geq 1} \mathbb{P}(A_n)$$

***Proof*:**
Part (1): the impossible event. The empty set is disjoint from itself, so applying σ-additivity to the sequence $\emptyset, \emptyset, \dots$ gives $\mathbb{P}(\emptyset) = \sum_{n \geq 1} \mathbb{P}(\emptyset)$. A nonnegative real number equal to the sum of infinitely many copies of itself must be zero, so $\mathbb{P}(\emptyset) = 0$.

Part (2): finite additivity. Given pairwise disjoint $A_1, \dots, A_n$, pad the list to an infinite sequence by setting $A_m = \emptyset$ for $m > n$. The padded sequence is pairwise disjoint and has the same union, so σ-additivity together with part (1) gives

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i) + \sum_{m > n} \mathbb{P}(\emptyset) = \sum_{i=1}^n \mathbb{P}(A_i)$$

Part (3): complementation. The events A and A^c are disjoint with union Ω, so finite additivity and axiom (1) of definition 1.1.1 give $\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(\Omega) = 1$. Rearranging yields $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Part (4): monotonicity. If $A \subseteq B$, then B is the disjoint union of A and $B \setminus A$, so finite additivity gives $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. Hence $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$, and since $\mathbb{P}(B \setminus A) \geq 0$ it follows that $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Part (5): inclusion-exclusion for two sets. Decompose $A \cup B$ into the three pairwise disjoint pieces $A \setminus B$, $A \cap B$, and $B \setminus A$. Finite additivity gives $\mathbb{P}(A \cup B) = \mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A)$.

By part (4) applied to $A \cap B \subseteq A$ and to $A \cap B \subseteq B$,

$$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B), \qquad \mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Substituting and simplifying gives $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Part (6): the general inclusion-exclusion identity. Argue by induction on n. The case $n = 1$ is trivial and $n = 2$ is part (5). Assume the identity for $n - 1$ events, and write $C = \bigcup_{i=1}^{n-1} A_i$ so that $\bigcup_{i=1}^{n} A_i = C \cup A_n$. By part (5),

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \mathbb{P}(C) + \mathbb{P}(A_n) - \mathbb{P}(C \cap A_n)$$

Now $C \cap A_n = \bigcup_{i=1}^{n-1}(A_i \cap A_n)$ is a union of $n-1$ events, so the inductive hypothesis applies both to $\mathbb{P}(C)$ and to $\mathbb{P}(C \cap A_n)$. Expanding, $\mathbb{P}(C)$ contributes every term of the claimed identity whose index set omits n, while $\mathbb{P}(A_n) - \mathbb{P}(C \cap A_n)$ contributes every term whose index set contains n: the summand $\mathbb{P}(A_n)$ is the singleton $\{n\}$, and each term $-(-1)^{k-2}\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_{k-1}} \cap A_n) = (-1)^{k-1}\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_{k-1}} \cap A_n)$ from $-\mathbb{P}(C \cap A_n)$ is the term indexed by $\{i_1, \ldots, i_{k-1}, n\}$ with the correct sign. Every subset of $\{1, \ldots, n\}$ is accounted for exactly once, giving the identity for n.

Part (7): Boole's inequality. Disjointify the sequence: set $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. The B_n are pairwise disjoint, each $B_n \subseteq A_n$, and $\bigcup_n B_n = \bigcup_n A_n$. By σ-additivity and monotonicity (part (4)),

$$\mathbb{P}\left(\bigcup_{n \geq 1} A_n\right) = \mathbb{P}\left(\bigcup_{n \geq 1} B_n\right) = \sum_{n \geq 1} \mathbb{P}(B_n) \leq \sum_{n \geq 1} \mathbb{P}(A_n)$$

which is the union bound, completing the proof.

Two of these deserve a second look. Complementation, part (3), is the identity that makes probability worth its own name: it converts a lower bound on $\mathbb{P}(A)$ into an upper bound on $\mathbb{P}(A^c)$ and back, so that showing an event is nearly certain is the same as showing its negation is nearly impossible. This is the move behind every argument that concludes "the bad event has small probability, hence the good event holds with high probability." The union bound, part (7), is the crudest and most frequently used inequality in the subject: it controls the chance that *any* of a list of events occurs by the sum of their individual chances, with no independence or disjointness needed. Its power is that it costs nothing; its weakness is that it is lossy when the events overlap heavily. The first Borel-Cantelli lemma in section 1.4 is little more than the union bound applied to the tail of a convergent series.

Countable additivity was stated as an axiom, but its most useful consequences concern not disjoint families but *monotone* ones, sequences of events that increase or decrease to a limit. That a probability responds continuously to such sequences is the analytic fact the subject rests on: it is what lets a probability about an infinite process be computed as the limit of probabilities about finite truncations.

Proposition 1.1.5: Continuity of Probability

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(A_n)_{n\geq 1}$ be a sequence of events.

1. (Continuity from below) if $A_n \uparrow A$, meaning $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$, then $\mathbb{P}(A_n) \to \mathbb{P}(A)$.
2. (Continuity from above) if $A_n \downarrow A$, meaning $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_n A_n$, then $\mathbb{P}(A_n) \to \mathbb{P}(A)$.
3. (Fatou for events) in general,
$$\mathbb{P}\left(\liminf_n A_n\right) \leq \liminf_n \mathbb{P}(A_n) \leq \limsup_n \mathbb{P}(A_n) \leq \mathbb{P}\left(\limsup_n A_n\right)$$
where $\liminf_n A_n = \bigcup_n \bigcap_{k\geq n} A_k$ and $\limsup_n A_n = \bigcap_n \bigcup_{k\geq n} A_k$.

***Proof*:**
Part (1): continuity from below. Split the increasing union into disjoint annuli. Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Because the A_n increase, the B_n are pairwise disjoint, $\bigcup_{k=1}^n B_k = A_n$, and $\bigcup_{k\geq 1} B_k = A$. By σ-additivity,

$$\mathbb{P}(A) = \sum_{k\geq 1} \mathbb{P}(B_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}(A_n)$$

where the middle equality is the definition of the sum of a series and the last uses finite additivity to recognize $\sum_{k=1}^n \mathbb{P}(B_k) = \mathbb{P}(A_n)$.

Part (2): continuity from above. Pass to complements to reduce to part (1); this is the step where finiteness of the total mass is used. If $A_n \downarrow A$, then $A_n^c \uparrow A^c$, so part (1) gives $\mathbb{P}(A_n^c) \to \mathbb{P}(A^c)$. By complementation (proposition 1.1.4(3)), $\mathbb{P}(A_n^c) = 1 - \mathbb{P}(A_n)$ and $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, so $1 - \mathbb{P}(A_n) \to 1 - \mathbb{P}(A)$, whence $\mathbb{P}(A_n) \to \mathbb{P}(A)$. The finiteness $\mathbb{P}(\Omega) = 1$ is essential: subtracting from the total mass is meaningless when it is infinite, and continuity from above fails for infinite measures, as Lebesgue measure on the shrinking sets $[n, \infty)$ shows, each of infinite measure while their intersection is empty.

Part (3): Fatou for events. The two inner inequalities $\liminf_n \mathbb{P}(A_n) \leq \limsup_n \mathbb{P}(A_n)$ hold for any sequence of reals, so only the two outer ones require proof, and they are symmetric under complementation; it suffices to prove the right-hand inequality $\limsup_n \mathbb{P}(A_n) \leq \mathbb{P}(\limsup_n A_n)$.

Set $C_n = \bigcup_{k\geq n} A_k$. Then $C_1 \supseteq C_2 \supseteq \cdots$ with $\bigcap_n C_n = \limsup_n A_n$, so continuity from above (part (2)) gives $\mathbb{P}(C_n) \to \mathbb{P}(\limsup_n A_n)$. For each fixed n and each $k \geq n$ the inclusion $A_k \subseteq C_n$ gives $\mathbb{P}(A_k) \leq \mathbb{P}(C_n)$ by monotonicity, hence $\sup_{k\geq n} \mathbb{P}(A_k) \leq \mathbb{P}(C_n)$. Letting $n \to \infty$,

$$\limsup_n \mathbb{P}(A_n) = \lim_{n\to\infty} \sup_{k\geq n} \mathbb{P}(A_k) \leq \lim_{n\to\infty} \mathbb{P}(C_n) = \mathbb{P}\left(\limsup_n A_n\right)$$

For the left-hand inequality, apply what was just proved to the complements A_n^c: since

$\limsup_n A_n^c = (\liminf_n A_n)^c$, complementation converts $\limsup_n \mathbb{P}(A_n^c) \le \mathbb{P}(\limsup_n A_n^c)$ into $\mathbb{P}(\liminf_n A_n) \le \liminf_n \mathbb{P}(A_n)$, completing the proof.

Parts (1) and (2) are the two halves of a single principle: probability commutes with monotone limits of events. The asymmetry between them (part (2) needs the finiteness of the total mass while part (1) does not) is the sharpest way the normalization $\mathbb{P}(\Omega) = 1$ enters at the level of theory. Continuity from above is the tool that computes the probability of an intersection over an infinite family as a limit of finite intersections, and it is used constantly: the probability that a random walk never returns to the origin, that a sequence of trials produces no success ever, that a supremum stays below a level for all time, each is an intersection handled by part (2). Part (3), the event-level analogue of Fatou's lemma, is what turns the lim sup and lim inf of a sequence of events, the events "A_n occurs infinitely often" and "A_n occurs eventually", into objects whose probabilities can be bounded; section 1.4 is built on exactly this.

The whole of this section rests on one distinction. Measure theory supplies the machinery (the σ-algebra, countable additivity, monotone continuity); probability supplies a single constraint, $\mathbb{P}(\Omega) = 1$, and reaps a disproportionate return. The return is that complementation becomes quantitative. In an infinite measure space the complement of a finite-measure set is uninformative, its measure is typically infinite, and there is no dual relationship between a set being large and its complement being small. Normalizing the total mass to one installs exactly that duality: $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$ makes "A is nearly certain" and "A^c is nearly impossible" interchangeable, and it is this interchangeability that the entire asymptotic theory of the subject exploits. The laws of large numbers, the central limit theorem, and the martingale convergence theorems are all, at the level of logic, statements that some event has probability one, proved by showing its complement has probability zero.

The second dividend is linguistic and just as consequential. Because the total mass is one, "probability zero" and "probability one" are dual notions rather than a set and its uninformative complement, and this duality makes two families of statements meaningful. The first is the almost-sure language already fixed in definition 1.1.1: a property holds *almost surely* when its failure set is null, so that from the point of view of $\mathbb{P}$ the property holds everywhere that matters. The second is the infinitely-often language: given a sequence of events (A_n), the event $\limsup_n A_n$ that "A_n occurs for infinitely many n" is a genuine event whose probability the Fatou inequality of proposition 1.1.5 already lets one bound, and whose exact value the Borel-Cantelli lemmas of section 1.4 will determine. Neither language survives the loss of normalization: in an infinite measure space there is no scale against which "the failure set is negligible" or "this happens with overwhelming certainty" can be calibrated. The one axiom that separates probability from measure theory is the one that makes its characteristic statements sayable.

Exercise 1.1.1

1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A, B \in \mathcal{F}$. Prove the *Bonferroni inequality* $\mathbb{P}(A \cap B) \ge \mathbb{P}(A) + \mathbb{P}(B) - 1$, and deduce that if $\mathbb{P}(A) = \mathbb{P}(B) = 0.9$ then $\mathbb{P}(A \cap B) \ge 0.8$. Show also that the general lower bound $\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) \ge \sum_{i=1}^n \mathbb{P}(A_i) - (n-1)$ holds.
2. On the uniform space $\mathrm{Unif}([0,1])$ of example 1.1.3, let $A_n = [0, 1/2 + 1/(n+2)]$ and $B_n = (1/n, 1]$. Identify $\lim_n A_n$ and $\lim_n B_n$ as monotone limits, then compute $\mathbb{P}(A_n)$, $\mathbb{P}(B_n)$, and the limiting probabilities directly, verifying proposition 1.1.5 in each case.
3. A number is drawn from $\{1, 2, \ldots, 100\}$ uniformly (each with probability $1/100$). Using inclusion-exclusion, compute the probability that the number is divisible by 2, 3, or 5.

4. Let (A_n) be events with $\sum_n \mathbb{P}(A_n^c) < 1$. Prove that $\mathbb{P}\left(\bigcap_n A_n\right) \geq 1 - \sum_n \mathbb{P}(A_n^c)$, and deduce that $\mathbb{P}\left(\bigcap_n A_n\right) > 0$. (This is the complementary form of the union bound and is the mechanism behind many "almost surely all of them hold" arguments.)

5. Give an example of a measure space $(\Omega, \mathcal{F}, \mu)$ and a decreasing sequence $A_n \downarrow \emptyset$ of measurable sets for which $\mu(A_n) \not\to \mu(\emptyset)$. Explain precisely which hypothesis in the proof of continuity from above (proposition 1.1.5(2)) fails, and why the failure cannot occur on a probability space.

6. Call an event A *almost sure* if $\mathbb{P}(A) = 1$. Prove that a countable intersection of almost-sure events is almost sure, but that an uncountable intersection need not be, by exhibiting an uncountable family of almost-sure events on $\mathrm{Unif}([0,1])$ whose intersection is empty.

1.2 Generating Classes and Uniqueness

A probability measure is a function on an entire σ-algebra, and a σ-algebra is an unwieldy object: the Borel sets of $\mathbb{R}$ cannot be listed, described by a formula, or built in countably many steps from the intervals. In practice one never specifies a measure by giving its value on every measurable set. One specifies it on a small, concrete class: on intervals when the space is $\mathbb{R}$, on cylinders when the space is a sequence space, on rectangles when the space is a product. The measure on all the other sets is then supposed to be forced. This section makes "forced" precise. The question it answers is: when does agreement of two probability measures on a class of simple sets compel agreement on the whole σ-algebra those sets generate?

The naive hope is that agreement on any generating class suffices. It does not. Two distinct probability measures can agree on a class that generates the full σ-algebra, provided that class is poorly behaved. What rescues the hope is a single closure property: if the generating class is closed under finite intersections, agreement propagates. The reason lies in Dynkin's π-λ theorem, the structural engine of this section and one of the two or three most frequently invoked lemmas in all of probability. It reappears in section 1.3 to pin down a measure on $\mathbb{R}$ from its distribution function, and again in chapter 1 and beyond whenever independence is checked on generators rather than on all events.

Definition 1.2.1: π-System

Let Ω be a set. A collection $\mathcal{P}$ of subsets of Ω is a *π-system* if it is closed under finite intersection: whenever $A, B \in \mathcal{P}$, also $A \cap B \in \mathcal{P}$.

The condition is deliberately weak. A π-system need not contain Ω, need not contain $\emptyset$, and need not be closed under complement or union; it asks only that the intersection of two of its members stay inside it (and hence, by induction, that any finite intersection do so). This is exactly the closure a generating class of "basic" sets tends to have for free. Intervals of the form $(-\infty, x]$ intersect to give another such interval; measurable rectangles $A \times B$ intersect to give the rectangle $(A \cap C) \times (B \cap D)$; cylinder sets fixing finitely many coordinates intersect to give a cylinder fixing the union of those coordinates. Each of these is the natural class on which a measure is prescribed, and each is a π-system.

Example 1.2.2: Half-Open Intervals and a Non-Example

On $\Omega = \mathbb{R}$, the collection

$$\mathcal{P} = \{(a,b] \big| a \le b,\ a,b \in \mathbb{R}\} \cup \{\emptyset\}$$

of half-open intervals is a π-system: the intersection of $(a,b]$ and $(c,d]$ is $(\max\{a,c\}, \min\{b,d\}]$ when that interval is nonempty and is $\emptyset$ otherwise, so it lies in $\mathcal{P}$ in either case.

By contrast, the collection $\mathcal{U}$ of all open subsets of $\mathbb{R}$ that are *unions of two disjoint open intervals* is not a π-system. Take $A = (0,1) \cup (2,3)$ and $B = (0,1) \cup (4,5)$, both in $\mathcal{U}$; their intersection is $A \cap B = (0,1)$, a single interval, which is not a union of two disjoint intervals and so lies outside $\mathcal{U}$. Failing the intersection condition, $\mathcal{U}$ is exactly the kind of generating class on which agreement of two measures would carry no force.

The complementary notion supplies the closure a π-system lacks, but in a form tailored to what a measure respects rather than to what a σ-algebra respects. A measure is additive on disjoint sets and continuous along increasing sequences; it does not directly "see" arbitrary unions. The λ-system axioms encode precisely these two operations, plus the presence of the whole space.

Definition 1.2.3: λ-System (Dynkin System)

Let Ω be a set. A collection $\mathcal{L}$ of subsets of Ω is a *λ-system* (also called a *Dynkin system*) if

1. $\Omega \in \mathcal{L}$;
2. $\mathcal{L}$ is closed under proper differences: if $A, B \in \mathcal{L}$ and $A \subseteq B$, then $B \setminus A \in \mathcal{L}$;
3. $\mathcal{L}$ is closed under increasing countable unions: if $A_1 \subseteq A_2 \subseteq \cdots$ with each $A_n \in \mathcal{L}$, then $\bigcup_n A_n \in \mathcal{L}$.

Two immediate consequences fix the shape of the axioms. Taking $A = A$ and $B = \Omega$ in (2) shows $\mathcal{L}$ is closed under complement, $A^c = \Omega \setminus A \in \mathcal{L}$; combined with (1) this gives $\emptyset = \Omega^c \in \mathcal{L}$. And a λ-system is closed under *disjoint* finite unions: if $A, B \in \mathcal{L}$ are disjoint, then $A \subseteq B^c$, so $B^c \setminus A = (A \cup B)^c \in \mathcal{L}$ by (2), whence $A \cup B \in \mathcal{L}$ by complementation. What a λ-system need not do is take the union of two *overlapping* sets, and that single gap is the whole point: it is the one closure a σ-algebra has that a λ-system may lack, and filling it is exactly what the π-system contributes below.

The comparison with a σ-algebra is worth stating outright. A collection is a σ-algebra if and only if it is both a π-system and a λ-system. One direction is clear: a σ-algebra contains Ω, is closed under complement (hence under proper differences), and is closed under all countable unions (hence increasing ones), so it is a λ-system; and it is closed under intersection, so it is a π-system. Conversely, a λ-system that is also a π-system is closed under arbitrary finite union, since $A \cup B = (A^c \cap B^c)^c$ combines complement (from the λ axioms) with intersection (from the π axiom); given closure under finite union and complement, closure under countable union follows by writing $\bigcup_n A_n = \bigcup_n (A_1 \cup \cdots \cup A_n)$ as an increasing union and invoking (3). This equivalence is the target the whole argument aims at.

Example 1.2.4: A λ-System That Is Not a σ-Algebra

Let $\Omega = \{1,2,3,4\}$ and let $\mathcal{L}$ consist of Ω, $\emptyset$, and every subset of even cardinality:

$$\mathcal{L} = \{\emptyset, \Omega\} \cup \{S \subseteq \Omega \big| |S| = 2\}$$

This is a λ-system. It contains Ω; it is closed under complement, since the complement of a two-element set is a two-element set and the complements of $\emptyset$ and Ω are again in the list; and closure under proper differences and increasing unions follows (on a finite space the increasing-union axiom is vacuous beyond finite unions of an increasing chain, and each such union is one of the listed sets). But $\mathcal{L}$ is not a π-system, and hence not a σ-algebra: the two-element sets $\{1, 2\}$ and $\{2, 3\}$ lie in $\mathcal{L}$, yet their intersection $\{2\}$ has odd cardinality and is absent. This is the smallest picture of the phenomenon the π-λ theorem controls: a family respecting complements and disjoint unions but not intersections.

As with σ-algebras, an arbitrary intersection of λ-systems is again a λ-system: each axiom is a condition preserved under intersection of the collections. Consequently, for any class $\mathcal{C}$ of subsets of Ω there is a smallest λ-system containing $\mathcal{C}$, namely the intersection of all λ-systems that contain $\mathcal{C}$ (the power set is one, so the intersection is over a nonempty family). Write $\lambda(\mathcal{C})$ for this *λ-system generated by $\mathcal{C}$*. Since every σ-algebra containing $\mathcal{C}$ is in particular a λ-system containing $\mathcal{C}$, and $\sigma(\mathcal{C})$ is the smallest σ-algebra containing $\mathcal{C}$, one always has $\lambda(\mathcal{C}) \subseteq \sigma(\mathcal{C})$. The content of Dynkin's theorem is that when $\mathcal{C}$ is a π-system, the inclusion is an equality.

Theorem 1.2.5: Dynkin's π-λ Theorem

Let Ω be a set and let $\mathcal{P}$ be a π-system on Ω. Then every λ-system containing $\mathcal{P}$ contains $\sigma(\mathcal{P})$. Equivalently, the λ-system generated by $\mathcal{P}$ equals the σ-algebra it generates:

$$\lambda(\mathcal{P}) = \sigma(\mathcal{P})$$

***Proof*:**
The two formulations are equivalent. If every λ-system containing $\mathcal{P}$ contains $\sigma(\mathcal{P})$, then in particular the smallest one, $\lambda(\mathcal{P})$, contains $\sigma(\mathcal{P})$; together with the reverse inclusion $\lambda(\mathcal{P}) \subseteq \sigma(\mathcal{P})$ noted above, this gives equality. Conversely, if $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$, then any λ-system $\mathcal{L} \supseteq \mathcal{P}$ satisfies $\mathcal{L} \supseteq \lambda(\mathcal{P}) = \sigma(\mathcal{P})$ by minimality of $\lambda(\mathcal{P})$. It therefore suffices to prove the equation.

The inclusion $\lambda(\mathcal{P}) \subseteq \sigma(\mathcal{P})$ holds for any class, as observed before the statement. The work is the reverse inclusion, $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$. By the equivalence recorded above, a λ-system that is also a π-system is a σ-algebra; so if $\lambda(\mathcal{P})$ were closed under intersection, it would be a σ-algebra containing $\mathcal{P}$, hence would contain the *smallest* such σ-algebra $\sigma(\mathcal{P})$, and the reverse inclusion would follow. The entire proof reduces to one claim: $\lambda(\mathcal{P})$ is closed under finite intersection. Write $\mathcal{L} = \lambda(\mathcal{P})$ for brevity.

The obstacle is that $\mathcal{L}$ is defined by a minimality property, not by an explicit description, so intersection-closure cannot be checked element by element. The device that circumvents this is the "good sets" argument, run twice. For a set $A \subseteq \Omega$, define its *class of good partners*

$$\mathcal{G}_A = \{B \subseteq \Omega \big| A \cap B \in \mathcal{L}\}$$

the sets whose intersection with A stays inside $\mathcal{L}$. The goal, that $A \cap B \in \mathcal{L}$ for all $A, B \in \mathcal{L}$, is precisely the statement that $\mathcal{L} \subseteq \mathcal{G}_A$ for every $A \in \mathcal{L}$. The strategy is to prove this by the minimality of $\mathcal{L}$: show $\mathcal{G}_A$ is a λ-system and contains $\mathcal{P}$, so that $\mathcal{L} = \lambda(\mathcal{P}) \subseteq \mathcal{G}_A$.

Step 1: $\mathcal{G}_A$ is a λ-system whenever $A \in \mathcal{L}$. Fix $A \in \mathcal{L}$ and verify the three axioms for $\mathcal{G}_A$.

1. $\Omega \in \mathcal{G}_A$, since $A \cap \Omega = A \in \mathcal{L}$.
2. Suppose $B, C \in \mathcal{G}_A$ with $B \subseteq C$. Then $A \cap B$ and $A \cap C$ lie in $\mathcal{L}$, and $A \cap B \subseteq A \cap C$, so by the proper-difference axiom in $\mathcal{L}$,
$$A \cap (C \setminus B) = (A \cap C) \setminus (A \cap B) \in \mathcal{L}$$
hence $C \setminus B \in \mathcal{G}_A$.
3. Suppose $B_1 \subseteq B_2 \subseteq \cdots$ with each $B_n \in \mathcal{G}_A$. Then $A \cap B_1 \subseteq A \cap B_2 \subseteq \cdots$ is an increasing sequence in $\mathcal{L}$, and by the increasing-union axiom in $\mathcal{L}$,
$$A \cap \Big(\bigcup_n B_n \Big) = \bigcup_n (A \cap B_n) \in \mathcal{L}$$
hence $\bigcup_n B_n \in \mathcal{G}_A$.

Each axiom for $\mathcal{G}_A$ was transported directly from the corresponding axiom for $\mathcal{L}$ through the operation $B \mapsto A \cap B$, which is why the λ-system axioms, and not the σ-algebra axioms, are the ones that make the argument run: intersection distributes over the proper difference and over increasing unions, but not over arbitrary unions.

Step 2: $\mathcal{P} \subseteq \mathcal{G}_A$ for every $A \in \mathcal{P}$. This is the only place the hypothesis on $\mathcal{P}$ is used. Fix $A \in \mathcal{P}$. For any $B \in \mathcal{P}$, the π-system property gives $A \cap B \in \mathcal{P} \subseteq \mathcal{L}$, so $B \in \mathcal{G}_A$. Thus $\mathcal{P} \subseteq \mathcal{G}_A$. Combined with Step 1, $\mathcal{G}_A$ is a λ-system containing $\mathcal{P}$, so by minimality of $\mathcal{L} = \lambda(\mathcal{P})$,
$$\mathcal{L} \subseteq \mathcal{G}_A \qquad \text{for every } A \in \mathcal{P}$$
Unpacking the definition of $\mathcal{G}_A$, this says: for every $A \in \mathcal{P}$ and every $B \in \mathcal{L}$, one has $A \cap B \in \mathcal{L}$.

Step 3: bootstrap to all of $\mathcal{L}$. The conclusion of Step 2 is symmetric in a way that lets the roles of $\mathcal{P}$ and $\mathcal{L}$ be swapped. It states that $A \cap B \in \mathcal{L}$ whenever $A \in \mathcal{P}$ and $B \in \mathcal{L}$; reading this with B fixed, it says that every $A \in \mathcal{P}$ is a good partner of B, that is, $\mathcal{P} \subseteq \mathcal{G}_B$ for every $B \in \mathcal{L}$. Now Step 1 applies to $B \in \mathcal{L}$ just as it did to $A \in \mathcal{L}$: $\mathcal{G}_B$ is a λ-system. So $\mathcal{G}_B$ is a λ-system containing $\mathcal{P}$, and minimality of $\mathcal{L}$ gives
$$\mathcal{L} \subseteq \mathcal{G}_B \qquad \text{for every } B \in \mathcal{L}$$
Unpacking once more: for every $B \in \mathcal{L}$ and every $A \in \mathcal{L}$, one has $A \cap B \in \mathcal{L}$. This is exactly the intersection-closure of $\mathcal{L}$.

With $\mathcal{L} = \lambda(\mathcal{P})$ now closed under finite intersection, it is a π-system as well as a λ-system, hence a σ-algebra by the equivalence recorded before the theorem. It contains $\mathcal{P}$, so it contains the smallest σ-algebra over $\mathcal{P}$, giving $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$. Together with the reverse inclusion this yields $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$, as we sought to show.

The mechanism deserves a second look, because the two-step "good sets" pattern recurs throughout measure theory and probability. The proof never constructs $\lambda(\mathcal{P})$ or lists its members; it works entirely through the universal property that $\lambda(\mathcal{P})$ is contained in every λ-system over $\mathcal{P}$. To show every member of $\lambda(\mathcal{P})$ has a property, one exhibits a λ-system, defined by that property, which contains $\mathcal{P}$; minimality then forces $\lambda(\mathcal{P})$ inside it. The π-system hypothesis enters at exactly one point, Step 2, where it seeds the argument with a single verified intersection; the two applications of minimality then propagate that seed first across $\mathcal{L}$ in the second slot and then across $\mathcal{L}$ in the first. Remove the π-system hypothesis and Step 2 collapses, which is why the half-open intervals succeed where the unions of two intervals from example 1.2.2 would fail.

The payoff is the uniqueness statement that motivated the section. A probability measure is determined by its values on a π-system that generates the σ-algebra, and the proof is a single application of Dynkin's theorem: the set on which two measures agree is a λ-system.

Corollary 1.2.6: Uniqueness from a Generating π-System

Let $(\Omega, \mathcal{F}, \mathbb{P})$ and $(\Omega, \mathcal{F}, \mathbb{Q})$ be probability spaces on the same measurable space (definition 1.1.1), and let $\mathcal{P} \subseteq \mathcal{F}$ be a π-system with $\sigma(\mathcal{P}) = \mathcal{F}$. If $\mathbb{P}(A) = \mathbb{Q}(A)$ for every $A \in \mathcal{P}$, then $\mathbb{P} = \mathbb{Q}$; that is, $\mathbb{P}(A) = \mathbb{Q}(A)$ for every $A \in \mathcal{F}$.

***Proof*:**
Let $\mathcal{L} = \{A \in \mathcal{F} \big| \mathbb{P}(A) = \mathbb{Q}(A)\}$ be the set of events on which the two measures agree. By hypothesis $\mathcal{P} \subseteq \mathcal{L}$. The claim is that $\mathcal{L}$ is a λ-system; granting this, Dynkin's theorem (theorem 1.2.5) gives $\mathcal{F} = \sigma(\mathcal{P}) \subseteq \mathcal{L}$, and since $\mathcal{L} \subseteq \mathcal{F}$ by construction, $\mathcal{L} = \mathcal{F}$, which is the assertion $\mathbb{P} = \mathbb{Q}$.

It remains to verify the three λ-system axioms for $\mathcal{L}$. This is where the total mass being finite and equal to 1 is used; the argument for proper differences below subtracts one finite number from another and would not survive an infinite total mass.

1. $\Omega \in \mathcal{L}$, since $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$ by the normalization of a probability measure.

2. Suppose $A, B \in \mathcal{L}$ with $A \subseteq B$. Then $B \setminus A \in \mathcal{F}$, and using finite additivity together with the finiteness of the total mass,

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A) = \mathbb{Q}(B) - \mathbb{Q}(A) = \mathbb{Q}(B \setminus A)$$

   the middle equality holding because $A, B \in \mathcal{L}$. Hence $B \setminus A \in \mathcal{L}$.

3. Suppose $A_1 \subseteq A_2 \subseteq \cdots$ with each $A_n \in \mathcal{L}$, and set $A = \bigcup_n A_n$. By continuity from below (proposition 1.1.5), applied to each measure,

$$\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n) = \lim_{n\to\infty} \mathbb{Q}(A_n) = \mathbb{Q}(A)$$

   where the middle equality holds termwise because $A_n \in \mathcal{L}$. Hence $A \in \mathcal{L}$.

Thus $\mathcal{L}$ is a λ-system containing the π-system $\mathcal{P}$, and the theorem completes the proof.

The corollary is the working form of the whole section, and the caveat inside its proof is the reason the π-system hypothesis cannot be dropped. Two measures agreeing on a mere generating class need not agree everywhere, because the agreement set $\mathcal{L}$ is a λ-system but not automatically a σ-algebra: it is closed under proper differences and increasing limits, the operations the additivity and continuity of a measure supply, but not under overlapping unions. The π-system property is exactly the extra ingredient that, through Dynkin's theorem, upgrades this λ-system to the full σ-algebra. The other silent hypothesis is finiteness: the proof of axiom (2) subtracts $\mathbb{P}(A)$ from $\mathbb{P}(B)$, which is legitimate only because both are finite. For σ-finite measures the corollary still holds, provided the π-system contains an increasing sequence exhausting Ω with finite measure, but the bare statement above is a fact about finite (here, probability) measures.

The section closes with the example that this book will lean on hardest. It identifies a concrete π-system generating the Borel σ-algebra of the line, and reads off the uniqueness consequence that a probability measure on $\mathbb{R}$ is fixed by its values on half-lines. That consequence is what makes the distribution function of section 1.3 a faithful record of the measure.

Example 1.2.7: Half-Lines Determine a Measure on $\mathbb{R}$

Consider the collection of left-infinite closed half-lines,

$$\mathcal{P} = \{(-\infty, x] \big| x \in \mathbb{R}\}$$

Two claims: $\mathcal{P}$ is a π-system, and $\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R})$.

$\mathcal{P}$ is a π-system. For $x, y \in \mathbb{R}$,

$$(-\infty, x] \cap (-\infty, y] = (-\infty, \min\{x, y\}]$$

which is again a member of $\mathcal{P}$. So $\mathcal{P}$ is closed under pairwise, hence finite, intersection. (It is very far from a λ-system: it is not closed under complement, and it does not contain $\mathbb{R}$ itself, reaching $\mathbb{R}$ only through the increasing union $\bigcup_n (-\infty, n]$, which shows why the increasing-union closure of a λ-system is doing real work.)

$\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R})$. Each half-line $(-\infty, x]$ is a closed set, hence Borel, so $\mathcal{P} \subseteq \mathcal{B}(\mathbb{R})$ and therefore $\sigma(\mathcal{P}) \subseteq \mathcal{B}(\mathbb{R})$. For the reverse inclusion it suffices to produce every open interval, since the open intervals generate $\mathcal{B}(\mathbb{R})$ (they form a base for the topology of $\mathbb{R}$, and every open set is a countable union of open intervals). First, complements give the right-open half-lines $(x, \infty) = \mathbb{R} \setminus (-\infty, x] \in \sigma(\mathcal{P})$. Next, a countable increasing union yields the open half-lines from below:

$$(-\infty, x) = \bigcup_{n=1}^{\infty} \Big(-\infty, x - \tfrac{1}{n} \Big] \in \sigma(\mathcal{P})$$

Intersecting a right-open with an open half-line gives every bounded open interval,

$$(a, b) = (a, \infty) \cap (-\infty, b) \in \sigma(\mathcal{P})$$

for $a < b$. Since $\sigma(\mathcal{P})$ contains every open interval, it contains the σ-algebra they generate, which is $\mathcal{B}(\mathbb{R})$. Hence $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\mathcal{P})$, and equality holds.

Consequence. By corollary 1.2.6, a Borel probability measure $\mathbb{P}$ on $\mathbb{R}$ is determined by the numbers $\mathbb{P}\big((-\infty, x]\big)$ for $x \in \mathbb{R}$: if two Borel probability measures assign the same value to every half-line, they agree on all of $\mathcal{B}(\mathbb{R})$. This single function of x is enough to encode the entire measure. It is the distribution function of $\mathbb{P}$, and the correspondence it sets up between measures and functions is the subject of section 1.3.

Exercise 1.2.1

1. Show directly that a collection $\mathcal{A}$ of subsets of Ω is a σ-algebra if and only if it is simultaneously a π-system and a λ-system. (Reproduce the argument from the text, filling in every step: in particular, deduce closure under complement and under arbitrary finite and countable unions.)

2. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ be the outcomes of a die. Exhibit a π-system $\mathcal{P}$ on Ω with $\sigma(\mathcal{P}) = 2^\Omega$ (the full power set) using as few sets as you can, and prove your $\mathcal{P}$ generates the power set. Then exhibit a generating class of Ω (still with $\sigma(\mathcal{P}) = 2^\Omega$) that is *not* a π-system.

3. Give two distinct probability measures $\mathbb{P} \neq \mathbb{Q}$ on a finite space Ω together with a class $\mathcal{C} \subseteq 2^\Omega$ such that $\sigma(\mathcal{C}) = 2^\Omega$ and $\mathbb{P}(C) = \mathbb{Q}(C)$ for every $C \in \mathcal{C}$. Explain, by pointing to the exact step of corollary 1.2.6 that fails, why this does not contradict the uniqueness corollary.

4. The half-open intervals $\mathcal{P} = \{(a, b] \big| a \le b\} \cup \{\emptyset\}$ of example 1.2.2 form a π-system. Prove that $\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R})$, and conclude that a Borel probability measure on $\mathbb{R}$ is determined by the values $\mathbb{P}\big((a, b]\big)$ for all $a \le b$.

5. Let $\mathbb{P}$ and $\mathbb{Q}$ be Borel probability measures on $\mathbb{R}$ with the same distribution function, meaning $\mathbb{P}\big((-\infty, x]\big) = \mathbb{Q}\big((-\infty, x]\big)$ for all $x \in \mathbb{R}$. Using example 1.2.7, prove $\mathbb{P}\big((a, b]\big) = \mathbb{Q}\big((a, b]\big)$ for all $a < b$ and $\mathbb{P}(\{x\}) = \mathbb{Q}(\{x\})$ for every x, expressing each in terms of the distribution function.

6. Let $\mathcal{P}$ be a π-system on Ω and let $\mathcal{L}$ be a λ-system with $\mathcal{P} \subseteq \mathcal{L}$. Suppose, in addition, that Ω is a *countable* union of sets from $\mathcal{P}$. Where in the proof of theorem 1.2.5 is the hypothesis $\Omega \in \mathcal{P}$ (as opposed to $\Omega \in \mathcal{L}$) used, and why is it not needed? Confirm that the theorem's conclusion $\sigma(\mathcal{P}) \subseteq \mathcal{L}$ holds even when $\Omega \notin \mathcal{P}$.

1.3 Constructing Probability Measures

The previous section established a uniqueness principle: a probability measure is determined by its values on a π-system that generates the relevant σ-algebra (corollary 1.2.6). Uniqueness is only half of what a working probabilist needs. The other half is existence, a way to produce a measure with prescribed behaviour, and existence cannot be settled inside the same generating class: knowing what values a measure ought to take on the half-lines does not by itself guarantee that a countably additive measure with those values exists. This section supplies existence for the two constructions on which the rest of the book stands. The first is the passage from a probability measure on $\mathbb{R}$ to its distribution function and back, which turns the abstract object $\mathbb{P}$ into a single monotone function of a real variable. The second is the Kolmogorov extension, which assembles an infinite sequence of independent trials out of the individual trials, and with it the fair-coin-tossing space that serves as the running example of Part I. In both cases the pattern is the same: build the measure by hand on a small, tractable class of sets, hand the countably additive extension to the Carathéodory machinery of *Measure Theory* [2], and use the π-system uniqueness of corollary 1.2.6 to pin the extension down.

A probability measure on $\mathbb{R}$ carries a great deal of information, one number for every Borel set. Most of that information is redundant. Because the half-lines $(-\infty, x]$ form a π-system generating $\mathcal{B}(\mathbb{R})$ (example 1.2.7), the values $\mathbb{P}\big((-\infty, x]\big)$ for $x \in \mathbb{R}$ already determine $\mathbb{P}$ completely. Packaging those values as a single function of x produces a concrete, finite-dimensional-looking handle on an otherwise infinite-dimensional object. The next definition isolates the three properties such a function must have.

Definition 1.3.1: Distribution Function

A *distribution function* (or *cumulative distribution function*) is a function $F\colon \mathbb{R} \to [0,1]$ satisfying:

1. *Monotonicity.* F is nondecreasing: $x \le y$ implies $F(x) \le F(y)$.
2. *Right-continuity.* For every $x \in \mathbb{R}$, $\lim\limits_{y \downarrow x} F(y) = F(x)$.
3. *Normalization at the ends.* $\lim\limits_{x\to-\infty} F(x) = 0$ and $\lim\limits_{x\to+\infty} F(x) = 1$.

Given a Borel probability measure $\mathbb{P}$ on $\mathbb{R}$, its *distribution function* is

$$F(x) = \mathbb{P}\big((-\infty, x]\big), \qquad x \in \mathbb{R}$$

The three conditions are exactly the traces left on the half-line values by the axioms of a probability measure, and it is worth seeing why before proving they characterize distribution functions completely. Monotonicity is the monotonicity of $\mathbb{P}$: if $x \le y$ then $(-\infty, x] \subseteq (-\infty, y]$, so $F(x) \le F(y)$. Right-continuity is continuity from above (proposition 1.1.5) applied to the decreasing sets $(-\infty, x + 1/n]$, whose intersection is $(-\infty, x]$. The two limits at the ends record that the increasing sets $(-\infty, n]$ exhaust $\mathbb{R}$ while the decreasing sets $(-\infty, -n]$ shrink to $\emptyset$, so continuity of $\mathbb{P}$ forces the values 1 and 0 respectively. The asymmetry between right-continuity (present) and left-continuity (absent) is the single most consequential feature of the definition: F can jump, and it jumps precisely where $\mathbb{P}$ places an atom. The left limit $F(x^-) = \lim_{y\uparrow x} F(y)$ equals $\mathbb{P}\big((-\infty, x)\big)$, so the jump is

$$F(x) - F(x^-) = \mathbb{P}(\{x\})$$

which is the mass the measure assigns to the single point x. A distribution function is continuous at x if and only if $\{x\}$ is a null event.

Example 1.3.2: Distribution Functions of Standard Laws

Three distribution functions illustrate the range of behaviour the axioms permit.

1. *A point mass.* Let $\mathbb{P} = \delta_0$ be the measure putting all its mass at 0, so $\mathbb{P}(A) = \mathbb{1}_A(0)$. Its distribution function is the unit step

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 & x \ge 0 \end{cases}$$

which jumps by 1 at the origin. The jump equals $\mathbb{P}(\{0\}) = 1$, and F is right-continuous there because $(-\infty, 0]$ contains the atom while $(-\infty, x)$ for $x \le 0$ does not. This is the extreme case of a purely atomic law.

2. *The uniform law on* $[0,1]$. Let $\mathbb{P} = \mathrm{Unif}[0,1]$ be Lebesgue measure restricted to $[0,1]$ (example 1.1.3). Then

$$F(x) = \begin{cases} 0 & x < 0 \\ x & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

a continuous, piecewise-linear function. Continuity everywhere records that this law has no atoms: every single point is null.

3. *A mixed law.* Fix $p \in (0,1)$ and let $\mathbb{P} = p\,\delta_0 + (1-p)\,\mathrm{Unif}[0,1]$. Its distribution function is $F(x) = 0$ for $x < 0$, then $F(x) = p + (1-p)x$ for $0 \le x \le 1$, then $F(x) = 1$ for $x > 1$. It has a jump of size p at the origin and rises continuously thereafter, the archetype of a law that is neither purely atomic nor atomless.

Each of these functions is nondecreasing, right-continuous, and runs from 0 to 1, so each is a distribution function in the sense of definition 1.3.1. The content of the next theorem is the converse: every function with those three properties, no matter how it was written down, is the distribution function of one and only one Borel probability measure on $\mathbb{R}$. This is the statement that makes the definition useful, because it lets a measure be specified by simply writing down a monotone right-continuous function with the correct end values, with no separate verification that a measure realizing it exists.

Theorem 1.3.3: Measure-Distribution Function Correspondence

The assignment $\mathbb{P} \mapsto F$, where $F(x) = \mathbb{P}\big((-\infty, x]\big)$, is a bijection between the Borel probability measures on $\mathbb{R}$ and the distribution functions in the sense of definition 1.3.1. Explicitly:

1. For every Borel probability measure $\mathbb{P}$ on $\mathbb{R}$, the function $F(x) = \mathbb{P}\big((-\infty, x]\big)$ is a distribution function.

2. Distinct probability measures have distinct distribution functions.

3. Every distribution function F is the distribution function of some Borel probability measure $\mathbb{P}$ on $\mathbb{R}$.

***Proof*:**
The three claims are proved in turn; parts (1) and (2) are proved in full here, and part (3) invokes the Lebesgue-Stieltjes construction from *Measure Theory* [2].

Part (1): the distribution function is well-defined. Let $\mathbb{P}$ be a Borel probability measure and set $F(x) = \mathbb{P}\big((-\infty, x]\big)$. Each value lies in $[0,1]$ because $\mathbb{P}$ is a probability measure. For monotonicity, if $x \le y$ then $(-\infty, x] \subseteq (-\infty, y]$, and monotonicity of $\mathbb{P}$ (proposition 1.1.4) gives $F(x) \le F(y)$. For right-continuity, fix x and take any sequence $y_n \downarrow x$. The sets $(-\infty, y_n]$ decrease to $\bigcap_n (-\infty, y_n] = (-\infty, x]$, and since $\mathbb{P}$ is finite, continuity from above (proposition 1.1.5) yields $F(y_n) = \mathbb{P}\big((-\infty, y_n]\big) \to \mathbb{P}\big((-\infty, x]\big) = F(x)$; as the sequence was arbitrary, $\lim_{y \downarrow x} F(y) = F(x)$. For the end behaviour, the sets $(-\infty, n]$ increase to $\mathbb{R}$, so continuity from below gives $F(n) \to \mathbb{P}(\mathbb{R}) = 1$, and by monotonicity $F(x) \to 1$ as $x \to +\infty$; the sets $(-\infty, -n]$ decrease to $\emptyset$, so continuity from above gives $F(-n) \to \mathbb{P}(\emptyset) = 0$, and by monotonicity $F(x) \to 0$ as $x \to -\infty$. Thus F is a distribution function.

Part (2): injectivity. Suppose two Borel probability measures $\mathbb{P}$ and $\mathbb{Q}$ have the same distribution function, so $\mathbb{P}\big((-\infty, x]\big) = \mathbb{Q}\big((-\infty, x]\big)$ for every $x \in \mathbb{R}$. The half-lines

$$\mathcal{P} = \big\{(-\infty, x] \big| x \in \mathbb{R}\big\}$$

form a π-system, since the intersection of two half-lines is again a half-line, $(-\infty, x] \cap (-\infty, y] = (-\infty, x \wedge y]$, and this π-system generates $\mathcal{B}(\mathbb{R})$ (example 1.2.7). Two probability measures agreeing on a generating π-system agree on the whole generated σ-algebra (corollary 1.2.6), so $\mathbb{P} = \mathbb{Q}$ on $\mathcal{B}(\mathbb{R})$. Distinct measures therefore have distinct distribution functions.

Part (3): surjectivity, via Lebesgue-Stieltjes. Let F be an arbitrary distribution function. The existence of a Borel measure $\mathbb{P}$ with $\mathbb{P}\big((a,b]\big) = F(b) - F(a)$ for all $a \le b$ is the Lebesgue-Stieltjes construction: one defines a premeasure on the algebra of finite disjoint unions of half-open intervals by $(a,b] \mapsto F(b) - F(a)$, checks that monotonicity and right-continuity of F make it a countably additive premeasure there, and applies the Carathéodory extension theorem to obtain a measure on $\mathcal{B}(\mathbb{R})$. This construction is carried out in full in *Measure Theory* [2]; it is the upstream machinery this book cites rather than rederives. The end normalizations $F(-\infty) = 0$ and $F(+\infty) = 1$ give

$$\mathbb{P}(\mathbb{R}) = \lim_{b\to\infty} \big(F(b) - F(-b)\big) = 1 - 0 = 1$$

so the resulting $\mathbb{P}$ is a probability measure. Finally its distribution function is

$$\mathbb{P}\big((-\infty, x]\big) = \lim_{a\to-\infty} \big(F(x) - F(a)\big) = F(x)$$

so $\mathbb{P}$ realizes F. Combined with part (2), which shows this $\mathbb{P}$ is the only such measure, the assignment $\mathbb{P} \mapsto F$ is a bijection, completing the proof.

The theorem is the reason distribution functions, rather than measures, are the everyday language of probability on $\mathbb{R}$. A law can be prescribed by a formula for F; questions about the measure translate into questions about a single monotone function; and the analytic operations of Part I and later Parts, differentiating F to obtain a density, reading off jumps as atoms, taking limits of F_n to express convergence in distribution, all become operations on ordinary real functions. The proof splits the labour cleanly: uniqueness is a probabilistic fact, flowing entirely from the π-system principle of corollary 1.2.6, whereas existence is a measure-theoretic fact, delegated to the Carathéodory construction. Every construction in this section will exhibit the same split.

The correspondence just proved manufactures probability measures on the line one at a time. The subject, however, is about what happens when a chance experiment is repeated: a coin tossed again and again, a random walk stepping forward at each tick of a clock, a sequence of measurements accumulating without end. To model such an experiment one needs a *single* probability space carrying an infinite sequence of trials at once, on which statements like "the average of the first n outcomes converges" or "some pattern recurs infinitely often" are events with well-defined probabilities. Building the first n trials is easy: it is a finite product measure, supplied by the product construction of *Measure Theory* [2]. The difficulty is passing to the infinite product, coordinatizing infinitely many trials on one space in a way that is consistent with every finite truncation. The Kolmogorov extension theorem does exactly this. The events one can describe using only finitely many coordinates, the *cylinder sets*, form an algebra on which a candidate measure is transparent; the theorem extends that candidate to the full product σ-algebra.

Fix a sequence of probability spaces $(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ for $n \ge 1$, one for each trial. The infinite product sample space is

$$\Omega = \prod_{n\ge1} \Omega_n = \big\{\omega = (\omega_1, \omega_2, \dots) \big| \omega_n \in \Omega_n \text{ for all } n\big\}$$

whose points are entire infinite sequences of outcomes. For each n the coordinate projection $\pi_n \colon \Omega \to \Omega_n$ sends ω to ω_n. The events that depend on only finitely many trials are the ones that can be pinned down directly.

Definition 1.3.4: Cylinder Set and the Product σ-Algebra

A *cylinder set* (or *finite-dimensional cylinder*) in $\Omega = \prod_{n\geq 1} \Omega_n$ is a set of the form

$$C = \{\omega \in \Omega \big| \omega_1 \in A_1, \dots, \omega_m \in A_m\} = A_1 \times \cdots \times A_m \times \prod_{n>m} \Omega_n$$

for some $m \geq 1$ and measurable sets $A_i \in \mathcal{F}_i$. The collection $\mathcal{A}$ of all finite disjoint unions of cylinder sets is an algebra on Ω. The *product σ-algebra* is $\mathcal{F} = \bigotimes_{n\geq 1} \mathcal{F}_n = \sigma(\mathcal{A})$, the smallest σ-algebra making every coordinate projection π_n measurable.

A cylinder constrains the first m coordinates and leaves all later ones free, so it is the event "the first m trials land in $A_1, \dots, A_m$". Its natural probability is the finite product $\mathbb{P}_1(A_1) \cdots \mathbb{P}_m(A_m)$, which is precisely the probability the finite product measure on $(\Omega_1, \mathcal{F}_1) \times \cdots \times (\Omega_m, \mathcal{F}_m)$ assigns to $A_1 \times \cdots \times A_m$. The cylinder sets themselves are not closed under union, but they are closed under intersection (an intersection of two cylinders is a cylinder, constraining as many coordinates as either does), so they form a π-system, and the finite disjoint unions of cylinders form the algebra $\mathcal{A}$ they generate. The theorem below builds a probability measure on $\mathcal{F}$ whose value on each cylinder is the finite product.

Theorem 1.3.5: Kolmogorov Extension Theorem (Product Form)

Let $(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)_{n\geq 1}$ be a sequence of probability spaces. There exists a unique probability measure $\mathbb{P}$ on the product space $(\Omega, \mathcal{F}) = \left(\prod_n \Omega_n, \bigotimes_n \mathcal{F}_n\right)$ such that for every cylinder set

$$C = \{\omega \big| \omega_1 \in A_1, \dots, \omega_m \in A_m\}, \qquad A_i \in \mathcal{F}_i$$

one has

$$\mathbb{P}(C) = \prod_{i=1}^{m} \mathbb{P}_i(A_i)$$

This $\mathbb{P}$ is the *infinite product measure*, written $\mathbb{P} = \bigotimes_{n\geq 1} \mathbb{P}_n$.

***Proof*:**
The argument has three movements: define the candidate on the algebra of cylinders, verify countable additivity there (the one step this book proves in full), extend by Carathéodory, and pin the extension down by π-system uniqueness.

Step 1: the premeasure on the algebra of cylinders. For a cylinder $C = \{\omega_1 \in A_1, \dots, \omega_m \in A_m\}$ define

$$\mathbb{P}_0(C) = \prod_{i=1}^{m} \mathbb{P}_i(A_i)$$

This is unambiguous: if the same cylinder is written with $m' > m$ coordinates by adjoining $A_{m+1} = \Omega_{m+1}, \dots, A_{m'} = \Omega_{m'}$, the extra factors are $\mathbb{P}_i(\Omega_i) = 1$ and the product is unchanged. On a finite disjoint union $\bigsqcup_j C_j$ of cylinders set $\mathbb{P}_0\left(\bigsqcup_j C_j\right) = \sum_j \mathbb{P}_0(C_j)$; a standard refinement argument, writing any two representations of an element of $\mathcal{A}$ over a common finite set of coordinates and comparing them against the finite product measure on those coordinates, shows this is well-defined and finitely additive, with $\mathbb{P}_0(\Omega) = 1$.

Step 2: countable additivity on the algebra. By a standard equivalence for finitely additive set functions on an algebra, finite additivity together with continuity at $\emptyset$ gives countable additivity. It therefore suffices to show: if $(D_k)_{k\geq 1} \subseteq \mathcal{A}$ is a decreasing sequence of cylinder-algebra sets with $\bigcap_k D_k = \emptyset$, then $\mathbb{P}_0(D_k) \to 0$. Argue by contraposition: suppose $\mathbb{P}_0(D_k) \geq \varepsilon > 0$ for all k; it will be shown that $\bigcap_k D_k \neq \emptyset$.

Each D_k depends on finitely many coordinates, and passing to a subsequence (which only helps, since the D_k decrease) one may assume D_k depends on the first m_k coordinates with $m_1 \leq m_2 \leq \cdots$. Write $\mu_{(j)}$ for the finite product measure $\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_j$ on $\Omega_1 \times \cdots \times \Omega_j$, supplied by *Measure Theory* [2]. For a point $x \in \Omega_1$ let D_k^x denote the section of D_k obtained by fixing the first coordinate at x; it is a set in the cylinder algebra of $\prod_{n\geq 2} \Omega_n$, and Fubini's theorem for the finite products gives

$$\mathbb{P}_0(D_k) = \int_{\Omega_1} \mathbb{P}_0'\big(D_k^x\big)\, \mathbb{P}_1(dx)$$

where $\mathbb{P}_0'$ is the analogous premeasure on $\prod_{n\geq 2} \Omega_n$. Since the integrand is bounded by 1 and the integral is at least ε, the set $G_k = \{x \in \Omega_1 \big| \mathbb{P}_0'(D_k^x) \geq \varepsilon/2\}$ has $\mathbb{P}_1(G_k) \geq \varepsilon/2$. The G_k decrease with k, and each has measure at least $\varepsilon/2$, so continuity from above of the probability measure $\mathbb{P}_1$ (proposition 1.1.5) forces $\bigcap_k G_k \neq \emptyset$. Choose $x_1 \in \bigcap_k G_k$. Then for every k the section $D_k^{x_1}$ has $\mathbb{P}_0'(D_k^{x_1}) \geq \varepsilon/2 > 0$, and the $D_k^{x_1}$ are a decreasing sequence in the cylinder algebra of the shifted product. Repeating the argument on the second coordinate produces x_2 with all twice-sectioned sets of premeasure at least $\varepsilon/4$, and inductively a sequence $x_1, x_2, x_3, \ldots$ with $x_n \in \Omega_n$ such that for every k the set D_k contains the cylinder obtained by fixing the first as-many-as-needed coordinates at $x_1, x_2, \ldots$. Concretely, the point $\omega^* = (x_1, x_2, x_3, \ldots) \in \Omega$ lies in D_k for every k: each D_k constrains only the first m_k coordinates, and those coordinates of ω^* were chosen to keep the corresponding section nonempty, hence to satisfy D_k's constraint. Therefore $\omega^* \in \bigcap_k D_k$, so the intersection is nonempty. This contradicts $\bigcap_k D_k = \emptyset$ and establishes continuity at $\emptyset$; combined with Step 1, $\mathbb{P}_0$ is a countably additive probability premeasure on the algebra $\mathcal{A}$.

Step 3: extension and uniqueness. By the Carathéodory extension theorem of *Measure Theory* [2], a countably additive premeasure on an algebra $\mathcal{A}$ extends to a measure on $\sigma(\mathcal{A}) = \mathcal{F}$; since $\mathbb{P}_0(\Omega) = 1$, the extension $\mathbb{P}$ is a probability measure, and it agrees with $\mathbb{P}_0$ on every cylinder, so $\mathbb{P}(C) = \prod_i \mathbb{P}_i(A_i)$ as claimed. For uniqueness, the cylinder sets form a π-system generating $\mathcal{F}$, and any two probability measures with the stated values agree on that π-system; by corollary 1.2.6 they agree on all of $\mathcal{F}$. Hence $\mathbb{P}$ is the unique such measure, completing the proof.

The theorem delivers exactly the object the subject requires: a single probability space on which infinitely many independent trials live simultaneously. The independence is built into the product formula, since the value of $\mathbb{P}$ on a cylinder factors across coordinates, and this is the source of every independence statement in the book, promoted in the next chapter from events to σ-algebras and random variables. The proof again splits along the line seen in theorem 1.3.3: the substantive probabilistic work is Step 2, the verification that the cylinder premeasure is countably additive, carried out here in full by a sectioning argument that reduces the infinite-dimensional statement to repeated applications of continuity from above on one coordinate at a time; the Carathéodory extension in Step 3 is cited, and uniqueness is again the π-system principle. The product form proved here is the case of Kolmogorov's theorem needed for independent sequences; the general theorem allows the $\mathbb{P}_n$ to be replaced by a consistent family of finite-dimensional distributions that need not factor as a product, and the same sectioning argument, run against the consistent family rather than the product, proves that version as well.

The construction is applied to the example that anchors the rest of Part I. Repeated tosses of a fair coin are the simplest nontrivial infinite experiment, and the Kolmogorov extension turns them into a concrete probability space that every later chapter revisits.

Example 1.3.6: The Fair-Coin-Tossing Space

Take each factor to be the two-point space $\Omega_n = \{0,1\}$ with its power set $\mathcal{F}_n = 2^{\{0,1\}}$ and the fair Bernoulli law $\mathbb{P}_n = \mathrm{Ber}(1/2)$, so $\mathbb{P}_n(\{1\}) = \mathbb{P}_n(\{0\}) = 1/2$. Here 1 records "heads" and 0 "tails". The infinite product space is

$$\Omega = \{0,1\}^{\mathbb{N}} = \big\{\omega = (\omega_1, \omega_2, \dots) \big| \omega_n \in \{0,1\}\big\}$$

the set of all infinite 0-1 sequences, equipped with the product σ-algebra $\mathcal{F} = \bigotimes_n 2^{\{0,1\}}$ and the product measure $\mathbb{P} = \bigotimes_n \mathrm{Ber}(1/2)$ furnished by theorem 1.3.5. This is the *fair-coin-tossing space.*

For each n let

$$H_n = \{\omega \in \Omega \big| \omega_n = 1\}$$

be the event "the n-th toss is heads". Each H_n is a cylinder constraining a single coordinate, so theorem 1.3.5 gives

$$\mathbb{P}(H_n) = \mathbb{P}_n(\{1\}) = \tfrac{1}{2}$$

The events $H_1, H_2, \dots$ are independent in the sense to be developed fully in section 1.4: for any finite set of indices $\{n_1, \dots, n_k\}$, the intersection $H_{n_1} \cap \cdots \cap H_{n_k}$ is a cylinder constraining those k coordinates, and the product formula gives

$$\mathbb{P}\big(H_{n_1} \cap \cdots \cap H_{n_k}\big) = \prod_{i=1}^{k} \mathbb{P}_{n_i}(\{1\}) = \left(\tfrac{1}{2}\right)^k = \prod_{i=1}^{k} \mathbb{P}(H_{n_i})$$

A finite pattern is computed the same way. The probability that the first three tosses read heads, tails, heads, that is the event $H_1 \cap H_2^c \cap H_3$, is the value of $\mathbb{P}$ on the cylinder $\{\omega_1 = 1, \omega_2 = 0, \omega_3 = 1\}$, namely

$$\mathbb{P}\big(H_1 \cap H_2^c \cap H_3\big) = \mathbb{P}_1(\{1\})\,\mathbb{P}_2(\{0\})\,\mathbb{P}_3(\{1\}) = \tfrac{1}{2} \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{8}$$

and by the same reasoning any prescribed outcome of the first m tosses has probability 2^{-m}. Every finite coin-tossing calculation is thus a value of the product measure on a single cylinder.

The coin-tossing space is the paradigm of an independent sequence, and its uses recur throughout the book. In section 1.4 it is the space on which the Borel-Cantelli lemmas are read, deciding for instance whether arbitrarily long runs of heads occur with probability one; in the limit theory to come it becomes the source of the strong law of large numbers (the fraction of heads in the first n tosses converges to $1/2$) and, after rescaling, of the central limit theorem. That so much rests on one construction is a direct consequence of theorem 1.3.5: once the infinite product exists as a genuine probability space, every asymptotic question about the sequence of tosses is a question about the single measure $\mathbb{P}$ on $\{0,1\}^{\mathbb{N}}$.

The two constructions of this section, the distribution-function correspondence and the Kolmogorov extension, look different, one about a single real random quantity, the other about an infinite sequence of them, but they are the same argument twice. In each, a probability measure is prescribed on a π-system (the half-lines in one case, the cylinders in the other), the existence of a countably additive measure realizing those values is a Carathéodory extension cited to *Measure Theory* [2], and the uniqueness of the result is the π-system principle of corollary 1.2.6. This is the recurring shape of every existence-and-uniqueness result in probability: build on a generating class, extend by measure theory, and pin down by uniqueness on the generating class. The next section takes the coin-tossing space just constructed and asks its first asymptotic questions.

Exercise 1.3.1

1. Let F be the distribution function of a Borel probability measure $\mathbb{P}$ on $\mathbb{R}$. Express each of the following in terms of F and its left limits $F(x^-) = \lim_{y\uparrow x} F(y)$: (a) $\mathbb{P}\big((a,b]\big)$; (b) $\mathbb{P}\big([a,b]\big)$; (c) $\mathbb{P}\big(\{a\}\big)$; (d) $\mathbb{P}\big((a,b)\big)$.
2. Show directly from definition 1.3.1 that a distribution function F has at most countably many discontinuities. Deduce that a Borel probability measure on $\mathbb{R}$ has at most countably many atoms (points x with $\mathbb{P}(\{x\}) > 0$).
3. Let $F(x) = 0$ for $x < 0$, $F(x) = \frac{1}{2} + \frac{1}{2}x$ for $0 \le x \le 1$, and $F(x) = 1$ for $x > 1$. Verify that F is a distribution function, and describe the measure $\mathbb{P}$ it corresponds to under theorem 1.3.3. Compute $\mathbb{P}(\{0\})$, $\mathbb{P}\big([0, \frac{1}{2}]\big)$, and $\mathbb{P}\big((\frac{1}{2}, 1]\big)$.
4. On the fair-coin-tossing space $\{0,1\}^{\mathbb{N}}$ of example 1.3.6, let E be the event "the first head occurs on an even-numbered toss". Express E as a countable disjoint union of cylinder events and compute $\mathbb{P}(E)$.
5. Fix a single point $\omega = (\omega_1, \omega_2, \dots) \in \{0,1\}^{\mathbb{N}}$ in the coin-tossing space. Using theorem 1.3.5 and continuity from above (proposition 1.1.5), show that $\mathbb{P}(\{\omega\}) = 0$. Contrast this with the fact that $\mathbb{P}(\Omega) = 1$ and comment on what it says about the size of a single infinite outcome sequence.
6. Let $(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ be probability spaces, and suppose the cylinder premeasure $\mathbb{P}_0$ from theorem 1.3.5 were defined only to be *finitely* additive on the algebra of cylinders, with no continuity-at-$\emptyset$ hypothesis. Explain, referring to Step 2 of the proof of theorem 1.3.5, why finite additivity alone does not yield a countably additive extension, and identify precisely which property of the coordinate measures $\mathbb{P}_n$ Step 2 uses to rule this out.

1.4 The Borel-Cantelli Lemmas

The three preceding sections built the static apparatus of a probability space: the axioms, the uniqueness of a measure from a generating class, and the construction of concrete measures culminating in the fair-coin-tossing space of example 1.3.6. This section takes the first step toward the asymptotic theory that occupies the rest of the book. Given an infinite sequence of events $A_1, A_2, \dots$, one asks not whether any single A_n occurs but how many of them occur, and in particular whether infinitely many occur. The event "A_n happens for infinitely many n" is the natural asymptotic object attached to a sequence of events, and the two Borel-Cantelli lemmas are a nearly complete dichotomy for its probability: a convergent sum $\sum_n \mathbb{P}(A_n)$ forces that probability to 0, and, under independence, a divergent sum forces it to 1. This is the first place in the book where the convergence or divergence of a series of probabilities controls an almost-sure statement, a mechanism that recurs in the strong law of large numbers and throughout the martingale theory.

The dichotomy also introduces independence into the narrative. The second lemma is the first result that fails without it, and to state it precisely requires an event-level notion of independent events. That minimal notion is defined here and reused, unchanged, by the general independence of σ-algebras and random variables developed in the next chapter, which builds on it rather than replacing it.

The starting point is the two extremal events attached to a sequence. The pattern is familiar from the limit superior and limit inferior of a sequence of real numbers, transported to the lattice of events under union and intersection.

Definition 1.4.1: Limit Superior and Limit Inferior of Events

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(A_n)_{n\geq 1}$ be a sequence of events. The *limit superior* of the sequence is the event

$$\limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

and the *limit inferior* is the event

$$\liminf_n A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$$

Both are events (each is a countable combination of members of $\mathcal{F}$). The limit superior is read "A_n occurs infinitely often" and written $\{A_n \text{ i.o.}\}$; the limit inferior is read "A_n occurs eventually" (that is, for all sufficiently large n).

To see that these readings are correct, fix a sample point $\omega \in \Omega$. By definition $\omega \in \limsup_n A_n$ means that for every n there exists some $k \geq n$ with $\omega \in A_k$; equivalently, the set $\{k \big| \omega \in A_k\}$ is infinite, so ω lies in infinitely many of the A_k. Likewise $\omega \in \liminf_n A_n$ means that for some n, $\omega \in A_k$ for every $k \geq n$; equivalently, ω fails to lie in only finitely many of the A_k, so it belongs to all of them from some point onward. The inclusion $\liminf_n A_n \subseteq \limsup_n A_n$ is immediate: an event that occurs eventually certainly occurs infinitely often. Passing to complements, De Morgan's laws give the useful identities

$$\left(\limsup_n A_n\right)^c = \liminf_n A_n^c \qquad \text{and} \qquad \left(\liminf_n A_n\right)^c = \limsup_n A_n^c$$

which say in words that "not infinitely often" is the same as "eventually in the complement", and vice versa. These identities are the bridge between the two lemmas below: proving that $\mathbb{P}(A_n \text{ i.o.}) = 1$ is the same as proving that $\mathbb{P}(A_n^c \text{ eventually}) = 0$.

A concrete instance grounds the definition before the theory begins.

Example 1.4.2: Runs in a Coin-Tossing Sequence

Work in the fair-coin-tossing space $\Omega = \{0,1\}^{\mathbb{N}}$ of example 1.3.6, whose sample point $\omega = (\omega_1, \omega_2, \dots)$ records an infinite sequence of tosses, with $\omega_i = 1$ read as "heads on toss i". Let $A_n = \{\omega \big| \omega_n = 1\}$ be the event that toss n is heads, so $\mathbb{P}(A_n) = 1/2$ for every n. Then

$$\limsup_n A_n = \{\omega \big| \omega_n = 1 \text{ for infinitely many } n\}$$

is the event that heads appears infinitely often, and its complement $\liminf_n A_n^c = \{\omega \big| \omega_n = 0 \text{ eventually}\}$ is the event that all tosses are tails from some point on. Intuitively the latter is negligible and the former is certain; the second Borel-Cantelli lemma will make "$\mathbb{P}(A_n \text{ i.o.}) = 1$" a theorem. By contrast, $\liminf_n A_n = \{\omega \big| \omega_n = 1 \text{ eventually}\}$ is the event that only finitely many tails occur, which by the same reasoning has probability 0.

The second Borel-Cantelli lemma requires that the events be independent, and independence at the level of events is the minimal notion needed. It is defined here in the form that makes the product rule hold across every finite subfamily.

Definition 1.4.3: Independence of Events

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A finite or countable family of events $(A_i)_{i \in I}$ is *independent* if for every finite subset $S \subseteq I$,

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i) \tag{1.1}$$

In particular, two events A and B are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, written $A \perp\!\!\!\perp B$.

Two points deserve emphasis. First, the condition (1.1) is imposed on *every* finite subfamily, not merely on the whole family or on pairs. Pairwise independence, in which (1.1) holds only for two-element subsets S, is strictly weaker: three events can be pairwise independent yet fail the three-fold product rule, and mutual independence is what the arguments below use. Second, this is deliberately the *event-level* notion and nothing more. The general theory of independence, of σ-algebras and of random variables, with the π-system criterion that reduces checking independence to a generating class, is developed in the next chapter and builds on this definition rather than replacing it; the present section needs only definition 1.4.3 to state and prove the second lemma. A basic closure property, that independence survives passing to complements, is what powers that proof, and it is worth isolating.

Lemma 1.4.4: Independence Is Preserved Under Complementation

Let $(A_i)_{i \in I}$ be an independent family of events. Then the family obtained by replacing any subset of the A_i with their complements is again independent. In particular, if $(A_i)_{i \in I}$ is independent then so is $(A_i^c)_{i \in I}$.

***Proof*:**
It suffices to show that replacing a single event by its complement preserves the defining condition (1.1), since the general case follows by replacing complements one at a time. Fix an index $j \in I$ and a finite subset $S \subseteq I$; we must check the product rule for the family with A_j replaced by A_j^c. If $j \notin S$ there is nothing to prove, so assume $j \in S$ and write $S = \{j\} \cup T$ with $j \notin T$. Set $B = \bigcap_{i \in T} A_i$, and split it along A_j by

$$B = (B \cap A_j) \cup (B \cap A_j^c)$$

a partition into disjoint events. Finite additivity (proposition 1.1.4) gives $\mathbb{P}(B \cap A_j^c) = \mathbb{P}(B) -$

$\mathbb{P}(B \cap A_j)$. Applying independence of the original family to B and to $B \cap A_j$,

$$\mathbb{P}(B) = \prod_{i \in T} \mathbb{P}(A_i) \qquad \text{and} \qquad \mathbb{P}(B \cap A_j) = \mathbb{P}(A_j) \prod_{i \in T} \mathbb{P}(A_i)$$

Subtracting,

$$\mathbb{P}(B \cap A_j^c) = (1 - \mathbb{P}(A_j)) \prod_{i \in T} \mathbb{P}(A_i) = \mathbb{P}(A_j^c) \prod_{i \in T} \mathbb{P}(A_i)$$

where the last equality uses $\mathbb{P}(A_j^c) = 1 - \mathbb{P}(A_j)$ from proposition 1.1.4. This is exactly the product rule for the family with A_j replaced by A_j^c over the subset S, completing the proof.

The lemma is the reason independence and the "infinitely often" event interact so cleanly. The event $\{A_n \text{ i.o.}\}$ is built from the complements A_n^c through its own complement $\liminf_n A_n^c$, and lemma 1.4.4 guarantees that if the A_n are independent the A_n^c inherit that structure, so the product rule remains available after the De Morgan passage.

With the vocabulary in place, the first lemma is the "easy" half of the dichotomy, in that it needs no independence hypothesis at all. It says that if the probabilities $\mathbb{P}(A_n)$ are summable then, with probability one, only finitely many of the A_n occur. The mechanism is that the tail $\sum_{k \geq n} \mathbb{P}(A_k)$ of a convergent series can be made arbitrarily small, and the union bound converts this tail into a bound on the probability that any event past index n occurs.

Theorem 1.4.5: First Borel-Cantelli Lemma

Let $(A_n)_{n \geq 1}$ be a sequence of events in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$$

then $\mathbb{P}(\limsup_n A_n) = 0$; that is, with probability one only finitely many of the A_n occur.

***Proof*:**
Write $B_n = \bigcup_{k \geq n} A_k$ for the event that some event of index at least n occurs. By definition $\limsup_n A_n = \bigcap_n B_n$, and the sets B_n decrease: $B_1 \supseteq B_2 \supseteq \cdots$, since dropping the earliest terms can only shrink the union. Because $\limsup_n A_n \subseteq B_n$ for every n, monotonicity of $\mathbb{P}$ (proposition 1.1.4) gives

$$\mathbb{P}\left(\limsup_n A_n\right) \leq \mathbb{P}(B_n) = \mathbb{P}\left(\bigcup_{k \geq n} A_k\right)$$

for each fixed n. Countable subadditivity, the union bound of proposition 1.1.4, bounds the right-hand side by the tail of the series,

$$\mathbb{P}\left(\bigcup_{k \geq n} A_k\right) \leq \sum_{k=n}^{\infty} \mathbb{P}(A_k)$$

Combining the two displays, $\mathbb{P}(\limsup_n A_n) \leq \sum_{k \geq n} \mathbb{P}(A_k)$ for every n. The hypothesis $\sum_n \mathbb{P}(A_n) < \infty$ means precisely that the tail $\sum_{k \geq n} \mathbb{P}(A_k)$ tends to 0 as $n \to \infty$. The left-hand side does not depend on n, so letting $n \to \infty$ forces

$$\mathbb{P}\left(\limsup_n A_n\right) \leq \lim_{n\to\infty} \sum_{k=n}^{\infty} \mathbb{P}(A_k) = 0$$

Since a probability is nonnegative, equality holds, as we sought to show.

The proof used only monotonicity and subadditivity, which is why no independence was needed. The conclusion is a genuine almost-sure statement extracted from a purely quantitative hypothesis on the sizes of the events: summability of the probabilities is a condition one can often verify by a direct estimate, and it delivers the qualitative fact that the events stop occurring. This one-directional implication is already enough for many applications, and the alert reader will notice that it says nothing when the series diverges. A divergent sum leaves the probability of $\{A_n \text{ i.o.}\}$ entirely undetermined in general, and it is exactly here that independence enters to pin it down.

The converse direction is false without further hypotheses. It is easy to arrange a divergent sum $\sum_n \mathbb{P}(A_n) = \infty$ with $\mathbb{P}(A_n \text{ i.o.}) < 1$: take a single event A with $0 < \mathbb{P}(A) < 1$ and set $A_n = A$ for all n, so that $\sum_n \mathbb{P}(A_n) = \infty$ while $\limsup_n A_n = A$ has probability $\mathbb{P}(A) < 1$. The events here are as far from independent as possible; they are identical. The second Borel-Cantelli lemma says that independence is exactly the hypothesis that rescues the converse: for independent events, a divergent sum forces the "infinitely often" event to have probability one.

Theorem 1.4.6: Second Borel-Cantelli Lemma

Let $(A_n)_{n\geq 1}$ be a sequence of *independent* events (definition 1.4.3) in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$$

then $\mathbb{P}(\limsup_n A_n) = 1$; that is, with probability one infinitely many of the A_n occur.

***Proof*:**
It is equivalent, and more convenient, to prove that the complement has probability 0. By the De Morgan identities established after definition 1.4.1,

$$\left(\limsup_n A_n\right)^c = \liminf_n A_n^c = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c$$

Write $C_n = \bigcap_{k\geq n} A_k^c$ for the event that no event of index at least n occurs, so that $(\limsup_n A_n)^c = \bigcup_n C_n$. Countable subadditivity (proposition 1.1.4) reduces the claim to showing $\mathbb{P}(C_n) = 0$ for every fixed n, since then $\mathbb{P}(\bigcup_n C_n) \leq \sum_n \mathbb{P}(C_n) = 0$.

Step 1: A finite truncation. Fix n and, for $m \geq n$, set $C_{n,m} = \bigcap_{k=n}^{m} A_k^c$, the finite intersection. These events decrease to C_n as $m \to \infty$: $C_{n,n} \supseteq C_{n,n+1} \supseteq \cdots$ and $\bigcap_{m\geq n} C_{n,m} = C_n$. Continuity from above along this decreasing sequence (proposition 1.1.5), applicable because $\mathbb{P}(C_{n,n}) \leq 1 < \infty$, gives

$$\mathbb{P}(C_n) = \lim_{m\to\infty} \mathbb{P}(C_{n,m}) = \lim_{m\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{m} A_k^c\right)$$

The advantage of the truncation is that the finite intersection factors, which the infinite one does not directly.

Step 2: Factoring by independence. Since the A_k are independent, so are the complements A_k^c by lemma 1.4.4. Applying the product rule (1.1) to the finite subfamily indexed by $\{n, n+1, \ldots, m\}$,

$$\mathbb{P}\left(\bigcap_{k=n}^{m} A_k^c\right) = \prod_{k=n}^{m} \mathbb{P}(A_k^c) = \prod_{k=n}^{m} (1 - \mathbb{P}(A_k))$$

using $\mathbb{P}(A_k^c) = 1 - \mathbb{P}(A_k)$ from proposition 1.1.4.

Step 3: The elementary inequality and divergence. For every real x the inequality $1 - x \le e^{-x}$ holds, being the tangent-line bound to the convex function e^{-x} at $x = 0$. Applying it with $x = \mathbb{P}(A_k) \in [0,1]$ to each factor,

$$\prod_{k=n}^{m} (1 - \mathbb{P}(A_k)) \le \prod_{k=n}^{m} e^{-\mathbb{P}(A_k)} = \exp\left(-\sum_{k=n}^{m} \mathbb{P}(A_k)\right)$$

Now let $m \to \infty$. The hypothesis $\sum_k \mathbb{P}(A_k) = \infty$ means the partial sums $\sum_{k=n}^{m} \mathbb{P}(A_k)$ diverge to $+\infty$ as $m \to \infty$ (the finitely many omitted terms $\mathbb{P}(A_1), \ldots, \mathbb{P}(A_{n-1})$ do not affect divergence of the tail), so the exponential on the right tends to 0. Combining the three steps,

$$\mathbb{P}(C_n) = \lim_{m\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{m} A_k^c\right) \le \lim_{m\to\infty} \exp\left(-\sum_{k=n}^{m} \mathbb{P}(A_k)\right) = 0$$

so $\mathbb{P}(C_n) = 0$. As this holds for every n, subadditivity gives $\mathbb{P}((\limsup_n A_n)^c) \le \sum_n \mathbb{P}(C_n) = 0$, and therefore $\mathbb{P}(\limsup_n A_n) = 1$, completing the proof.

Placing the two lemmas side by side gives a clean picture. For an independent sequence, the probability of $\{A_n \text{ i.o.}\}$ is a strict dichotomy governed by the series $\sum_n \mathbb{P}(A_n)$: it is 0 when the series converges and 1 when it diverges, with no intermediate value possible. This is a first instance of a zero-one law, a phenomenon in which a natural asymptotic event has probability either 0 or 1 and never anything strictly between; the tail σ-algebra and Kolmogorov's 0-1 law of the next chapter explain why such events are forced to be trivial. The role of independence is sharp and asymmetric. The first lemma is independence-free, so the convergent side of the dichotomy holds for any sequence; the second lemma needs independence, as the constant-sequence counterexample above shows the divergent side can fail badly without it. Independence is what converts a statement about the sizes $\mathbb{P}(A_n)$ of individual events into a statement about their joint occurrence.

The dichotomy has a vivid reading on the coin-tossing space, the "infinite monkey" phenomenon that a fixed finite pattern, and indeed arbitrarily long runs of heads, must recur infinitely often in an infinite sequence of fair tosses. The intuition is folklore; the second Borel-Cantelli lemma makes it a theorem, provided the events are arranged to be independent by cutting the sequence into disjoint blocks.

Example 1.4.7: Arbitrarily Long Runs of Heads Occur Almost Surely

Work in the fair-coin-tossing space $\Omega = \{0,1\}^{\mathbb{N}}$ of example 1.3.6, where the coordinates $\omega_1, \omega_2, \ldots$ are independent events each of probability 1/2, "heads" meaning $\omega_i = 1$. Independence of the individual coordinate events is exactly the product structure of the measure built in example 1.3.6. Fix a target run length $\ell \ge 1$; the claim is that with probability one the sequence contains a run of ℓ consecutive heads infinitely often, and hence, ℓ being arbitrary, that it contains runs of *every* length infinitely often.

The disjoint-blocks device. Partition the tosses into consecutive blocks of length ℓ: the n-th block occupies coordinates $(n-1)\ell+1, (n-1)\ell+2, \ldots, n\ell$. Let

$$A_n = \{\omega \big| \omega_{(n-1)\ell+1} = \omega_{(n-1)\ell+2} = \cdots = \omega_{n\ell} = 1\}$$

be the event that the n-th block is all heads, a run of ℓ heads. Because A_n depends only on the ℓ coordinates in block n, and distinct blocks use disjoint coordinates, the events $(A_n)_{n\geq 1}$ are independent (a finite intersection of them constrains disjoint sets of coordinates, and the product measure factors over disjoint coordinates). Each has probability

$$\mathbb{P}(A_n) = \left(\tfrac{1}{2}\right)^{\ell} = 2^{-\ell}$$

the same positive constant for every n. Consequently

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} 2^{-\ell} = \infty$$

since it is a sum of infinitely many copies of the fixed positive number $2^{-\ell}$. The events are independent and their probabilities sum to infinity, so the second Borel-Cantelli lemma (theorem 1.4.6) applies:

$$\mathbb{P}\left(\limsup_{n} A_n\right) = 1$$

With probability one, infinitely many blocks are entirely heads. Each such block is in particular a run of at least ℓ consecutive heads occurring in the sequence, so almost every ω contains a run of ℓ heads infinitely often.

From one length to all lengths. The event just shown to have probability one, "a run of length ℓ occurs infinitely often", will be denoted R_ℓ. A countable intersection of probability-one events again has probability one: if $\mathbb{P}(R_\ell) = 1$ for every ℓ then $\mathbb{P}\left(\bigcap_{\ell\geq 1} R_\ell\right) = 1$, because the complement is the countable union $\bigcup_\ell R_\ell^c$ of null events, which is null by subadditivity (proposition 1.1.4). Hence with probability one the sequence contains, for every ℓ, a run of ℓ consecutive heads occurring infinitely often. The same argument applies verbatim to any fixed finite pattern in place of the all-heads block: if $w \in \{0,1\}^\ell$ is any word of length ℓ, the block events "block n equals w" are independent with common probability $2^{-\ell} > 0$, so every fixed finite pattern appears infinitely often almost surely. This is the rigorous content of the infinite-monkey heuristic: over an infinite sequence of independent fair trials, every finite configuration that is not impossible is certain to recur without end.

The passage from "length ℓ" to "all lengths" and to "all patterns" illustrates a technique used constantly in the sequel: an almost-sure statement that holds for each member of a countable family holds simultaneously for the whole family, because the exceptional null sets union to a null set. The disjoint-blocks device is equally characteristic. The individual coordinate-run events, overlapping runs starting at each position, are not independent, so the second lemma does not apply to them directly; cutting the sequence into disjoint blocks manufactures an independent subsequence to which it does apply, at the cost of possibly missing some runs that straddle a block boundary. Missing those runs only weakens the conclusion, which is why the device is legitimate, and it is the standard way to extract an independent subsequence from a dependent one.

1.4.8 Foreshadowing: The Law of the Iterated Logarithm The two lemmas also give the sharpest almost-sure theorem in the classical theory. Coding each fair toss as a step ± 1 and writing S_n for the position after n steps, the law of the iterated logarithm pins the fluctuations of S_n to the exact scale $\sqrt{2n \log\log n}$:

$$\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n\log\log n}} = 1 \quad \text{almost surely}$$

Its upper bound is a first Borel-Cantelli argument and its lower bound a second. The theorem is proved as a capstone of the limit theory later in the book.

Exercise 1.4.1

1. Let $(A_n)_{n\geq 1}$ be arbitrary events. Prove directly from definition 1.4.1, without invoking either Borel-Cantelli lemma, the inclusion $\liminf_n A_n \subseteq \limsup_n A_n$, and give a sequence of events for which the inclusion is strict.
2. A sequence of events satisfies $\mathbb{P}(A_n) = 1/n^2$ for every n. Show that with probability one only finitely many of the A_n occur. Does the conclusion require the A_n to be independent?
3. In the fair-coin-tossing space of example 1.3.6, let $A_n = \{\omega_n = 1\}$ be the event of heads on toss n. Using the second Borel-Cantelli lemma, show that $\mathbb{P}(A_n \text{ i.o.}) = 1$ and, by applying the same reasoning to the complements, that with probability one both heads and tails occur infinitely often.
4. Give three events A, B, C that are pairwise independent (each pair satisfies the two-event product rule) but not independent as a family, that is, $\mathbb{P}(A\cap B\cap C) \neq \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Explain in one sentence why the proof of theorem 1.4.6 would break if only pairwise independence were assumed.
5. Let (A_n) be independent with $\mathbb{P}(A_n) = p_n$. Prove the sharpened dichotomy: $\mathbb{P}(A_n \text{ i.o.})$ equals 0 if $\sum_n p_n < \infty$ and equals 1 if $\sum_n p_n = \infty$, so the probability is never strictly between 0 and 1. Which lemma supplies each case?
6. (Second lemma without complementation.) Let (A_n) be independent with $\sum_n \mathbb{P}(A_n) = \infty$. Using the inequality $1 - x \leq e^{-x}$ directly on the events $C_{n,m} = \bigcap_{k=n}^m A_k^c$, show that $\mathbb{P}(\bigcup_{k=n}^\infty A_k) = 1$ for every n, and deduce $\mathbb{P}(A_n \text{ i.o.}) = 1$. Explain where independence and where continuity from above (proposition 1.1.5) are used.

2

Random Variables and Distributions

Chapter 1 built the stage: a probability space, the uniqueness of a measure from a generating class, and the construction of concrete measures. Probability, though, is rarely done on the sample space directly; it is done through *random variables*, the measurable functions on Ω, and the measures they induce on the line. A random variable transports the abstract space to a concrete range, usually $\mathbb{R}$, where its distribution function (section 1.3) is a faithful and computable record of its behaviour. This chapter installs random variables and their laws, and then develops the notion that most sharply separates probability from general measure theory: independence.

The four sections move from the single variable to the joint structure of many. section 2.1 defines a random variable, its law as a pushforward measure, and the σ-algebra it generates. Section 2.2 lifts the event-level independence of section 1.4 to σ-algebras and random variables, and proves the criterion that makes independence checkable on generators, resting on the π-λ theorem of section 1.2. Section 2.3 identifies independence with a product law, records the form of Fubini's theorem the subject needs, and uses the Kolmogorov extension of section 1.3 to build an infinite independent sequence with prescribed marginals, the setting for the limit theorems of the next Part. Section 2.4 closes with the tail σ-algebra and Kolmogorov's zero-one law, which explains why the Borel-Cantelli events of section 1.4 were forced to have probability zero or one in the first place.

2.1 Random Variables and Their Laws

A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ (definition 1.1.1) is rarely the object of direct interest. The outcome $\omega \in \Omega$ is an idealized record of "everything that happened", and one seldom cares about all of it at once. What one measures is a *feature* of the outcome: the number of heads in a hundred tosses, the position of a walker after n steps, the energy of a configuration. Each such feature is a function of ω, and each carries the outcome from the abstract sample space to a concrete range, almost always the real line. This section installs the two objects that make this transport rigorous and useful. The first is the random variable itself, a measurable map that pulls events on the range back to events on Ω. The second is its *law*, the measure it induces on the range, which is the only aspect of a random variable that probability can access. The distinction between the two, the map versus the distribution it induces, is the organizing theme of the whole subject, and it is worth getting exactly right at the outset.

The one property a feature must have to be usable is that the questions one asks of it correspond to events. If X records a numerical measurement, then "X lies in the range B" must be an event, that is, a member of $\mathcal{F}$, so that it has a probability. Demanding this for every Borel set B is exactly the requirement that X be a measurable map into $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. This is the measure theory of [2] read with a probabilist's emphasis: the measurability of a function is not a technical nuisance but the precise condition under which its values can be assigned probabilities.

Definition 2.1.1: Random Variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *random variable* is a measurable map

$$X\colon (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

that is, a function $X\colon \Omega \to \mathbb{R}$ such that $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$. We write $\{X \in B\} = X^{-1}(B) = \{\omega \in \Omega \big| X(\omega) \in B\}$ for the corresponding event, and abbreviate $\{X \leq x\} = X^{-1}((-\infty, x])$.
More generally, a *random vector* is a measurable map $X\colon (\Omega, \mathcal{F}) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, and a *random element* of a measurable space $(E, \mathcal{E})$ is a measurable map $X\colon (\Omega, \mathcal{F}) \to (E, \mathcal{E})$. An *extended-real random variable* takes values in $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$, where $\overline{\mathbb{R}} = [-\infty, +\infty]$ carries the Borel σ-algebra generated by its order topology.

The measurability condition asks for $X^{-1}(B) \in \mathcal{F}$ for *every* Borel set B, an uncountable family of conditions. In practice one never checks all of them, because it suffices to check the half-lines. The sets $(-\infty, x]$ for $x \in \mathbb{R}$ form a π-system that generates $\mathcal{B}(\mathbb{R})$ (example 1.2.7), and the collection of sets whose preimage lies in $\mathcal{F}$ is itself a σ-algebra; a σ-algebra containing a generating family contains everything it generates. This is the working criterion.

Proposition 2.1.2: Measurability from Half-Lines

A function $X\colon \Omega \to \mathbb{R}$ is a random variable if and only if $\{X \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

***Proof*:**
Necessity is immediate: $(-\infty, x] \in \mathcal{B}(\mathbb{R})$, so if X is a random variable then $\{X \leq x\} = X^{-1}((-\infty, x]) \in \mathcal{F}$. For sufficiency, let

$$\mathcal{G} = \{B \subseteq \mathbb{R} \big| X^{-1}(B) \in \mathcal{F}\}$$

Preimage commutes with complements and countable unions, $X^{-1}(B^c) = X^{-1}(B)^c$ and $X^{-1}\big(\bigcup_n B_n\big) = \bigcup_n X^{-1}(B_n)$, and $X^{-1}(\mathbb{R}) = \Omega \in \mathcal{F}$, so $\mathcal{G}$ is a σ-algebra. By hypothesis $\mathcal{G}$ contains every half-line $(-\infty, x]$. The half-lines form a π-system generating $\mathcal{B}(\mathbb{R})$ (example 1.2.7), and a σ-algebra containing a generating family contains the σ-algebra it generates, so $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{G}$. Thus $X^{-1}(B) \in \mathcal{F}$ for every Borel B, completing the proof.

The same argument works for any generating π-system, so one is free to test on whichever generators are convenient: open intervals, closed intervals, or the rays (x, ∞) all serve equally. The reduction to half-lines is also the link between random variables and the distribution functions of section 1.3: the half-line events $\{X \leq x\}$ are exactly the events whose probabilities assemble into the distribution function, which is why that one family suffices to pin down everything. Interpreted probabilistically, a random variable is an observable on the outcome, a measurement whose every sensible query is an event with a probability attached.

Once a random variable is in hand, the probability measure on Ω can be pushed forward to the range. The result is the single most important object attached to a random variable, because it is all that probability ever sees. Two random variables built on entirely different sample spaces, or the same variable recorded before and after an irrelevant reshuffling of Ω, are indistinguishable to probability if they induce the same measure on $\mathbb{R}$. That induced measure is the law.

Definition 2.1.3: Law (Distribution) of a Random Variable

The *law*, or *distribution*, of a random variable $X\colon \Omega \to \mathbb{R}$ is the pushforward of $\mathbb{P}$ along X, the set function

$$\mu_X(B) = \mathbb{P}\big(X^{-1}(B)\big) = \mathbb{P}(X \in B), \qquad B \in \mathcal{B}(\mathbb{R})$$

on $\mathcal{B}(\mathbb{R})$. For a random vector $X\colon \Omega \to \mathbb{R}^d$ the law μ_X is defined the same way on $\mathcal{B}(\mathbb{R}^d)$ and is called the *joint law* of the components.

The defining feature of the law is that it repackages X as a measure on the range, discarding the sample space entirely. The next proposition confirms that μ_X is a genuine probability measure and connects it to the distribution function of section 1.3, closing the loop: every random variable produces a distribution function, and by the correspondence of that section a distribution function determines a unique Borel probability measure. The law and the distribution function are therefore two encodings of the same information.

Proposition 2.1.4: The Law is a Probability Measure Determined by its Distribution Function

Let $X\colon \Omega \to \mathbb{R}$ be a random variable with law μ_X.

1. μ_X is a Borel probability measure on $\mathbb{R}$.
2. The function $F_X(x) = \mu_X\big((-\infty, x]\big) = \mathbb{P}(X \leq x)$ is a distribution function in the sense of definition 1.3.1: it is nondecreasing, right-continuous, and satisfies $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$.
3. The law μ_X is the unique Borel probability measure on $\mathbb{R}$ with distribution function F_X; that is, μ_X is determined by F_X.

***Proof*:**
We prove the three parts in turn.

1. That μ_X is a measure is the general fact that a pushforward of a measure along a measurable map is a measure, which we record here in the present notation. First $\mu_X(B) = \mathbb{P}(X^{-1}(B)) \geq 0$ for every Borel B, and $\mu_X(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$, so μ_X is nonnegative and normalized. For countable additivity, let $(B_n)_{n\geq 1}$ be disjoint Borel sets. Their preimages $X^{-1}(B_n)$ are disjoint, since $X^{-1}(B_m) \cap X^{-1}(B_n) = X^{-1}(B_m \cap B_n) = X^{-1}(\emptyset) = \emptyset$ for $m \neq n$, and $X^{-1}\big(\bigsqcup_n B_n\big) = \bigsqcup_n X^{-1}(B_n)$. Countable additivity of $\mathbb{P}$ gives
$$\mu_X\Big(\bigsqcup_n B_n\Big) = \mathbb{P}\Big(\bigsqcup_n X^{-1}(B_n)\Big) = \sum_n \mathbb{P}\big(X^{-1}(B_n)\big) = \sum_n \mu_X(B_n)$$
so μ_X is a Borel probability measure.

2. Each property of F_X transfers from a property of the measure μ_X established in part (1). For monotonicity, if $x \leq y$ then $(-\infty, x] \subseteq (-\infty, y]$, so $F_X(x) = \mu_X((-\infty, x]) \leq \mu_X((-\infty, y]) = F_X(y)$. For right-continuity, fix x and take any sequence $x_n \downarrow x$; then $(-\infty, x_n] \downarrow (-\infty, x]$, and continuity of μ_X along decreasing sequences (proposition 1.1.5, applicable since μ_X is finite) gives $F_X(x_n) = \mu_X((-\infty, x_n]) \to \mu_X((-\infty, x]) = F_X(x)$. For the limits at infinity, $(-\infty, n] \uparrow \mathbb{R}$ as $n \to +\infty$ and $(-\infty, -n] \downarrow \emptyset$ as $n \to +\infty$, so continuity from below and above yields $F_X(n) \to \mu_X(\mathbb{R}) = 1$ and $F_X(-n) \to \mu_X(\emptyset) = 0$; monotonicity upgrades these to the full limits. Thus F_X satisfies definition 1.3.1.

3. By part (2), F_X is a distribution function, so theorem 1.3.3 applies: there is a unique Borel probability measure on $\mathbb{R}$ whose distribution function is F_X. The measure μ_X has distribution function F_X by the definition of F_X, so it is that unique measure. Hence μ_X is determined by F_X, as we sought to show.

Part (3) has a consequence that governs how the subject is practised: to specify the distribution of a random variable, one need only write down its distribution function (or, when it exists, a density or a probability mass function that generates F_X). Every probability of the form $\mathbb{P}(X \in B)$ is then determined, even though B ranges over all Borel sets and F_X records only the half-line values. This is the payoff of the uniqueness theorem of section 1.2 in its most-used form. It also justifies a piece of standard language: two random variables X and Y, possibly on different probability spaces, are said to be *equal in distribution*, written $X \overset{d}{=} Y$, when $\mu_X = \mu_Y$, and by part (3) this holds if and only if $F_X = F_Y$. Equality in distribution is far weaker than equality of functions: it says the two variables are *distributionally indistinguishable*, that no probabilistic question phrased through the law can tell them apart, while as maps on their respective sample spaces they may have nothing to do with one another.

Definition 2.1.5: Generated σ-Algebra

The *σ-algebra generated by* a random variable $X\colon \Omega \to \mathbb{R}$ is the preimage

$$\sigma(X) = X^{-1}\big(\mathcal{B}(\mathbb{R})\big) = \{X^{-1}(B) \big| B \in \mathcal{B}(\mathbb{R})\}$$

For a family $(X_i)_{i\in I}$ of random variables, $\sigma(X_i : i \in I)$ is the smallest σ-algebra on Ω making every X_i measurable, equivalently the σ-algebra generated by $\bigcup_{i\in I} \sigma(X_i)$.

That $\sigma(X)$ is a σ-algebra is the same computation as in the proof of proposition 2.1.2: preimage commutes with complement and countable union, so the image of a σ-algebra under X^{-1} is a σ-algebra. It is contained in $\mathcal{F}$ precisely because X is a random variable, and it is the smallest sub-σ-algebra of $\mathcal{F}$ with respect to which X is measurable. The reading of $\sigma(X)$ is informational: it is exactly the collection of events whose occurrence can be decided from the value of X alone. Knowing $X(\omega)$ tells one whether each event $X^{-1}(B)$ occurred and nothing more, so $\sigma(X)$ is "the information carried by X". This interpretation is the seed of the theory of conditioning and filtrations, where a growing family of σ-algebras models information accumulating over time.

The informational reading raises a precise question. If a second random variable Y is measurable with respect to $\sigma(X)$, that is, if the value of Y is decidable from the value of X, then intuitively Y ought to be a function of X. The Doob-Dynkin lemma confirms this exactly: $\sigma(X)$-measurability is the same as being a Borel function of X. The lemma is the rigorous form of the slogan "$\sigma(X)$ is the information in X", and it is proved by the standard machine of measure theory, building the function g up from indicators through simple functions to measurable limits.

Lemma 2.1.6: Doob-Dynkin

Let $X\colon \Omega \to \mathbb{R}$ be a random variable and let $Y\colon \Omega \to \mathbb{R}$ be $\sigma(X)$-measurable. Then there exists a Borel measurable function $g\colon \mathbb{R} \to \mathbb{R}$ such that $Y = g(X)$, that is, $Y(\omega) = g(X(\omega))$ for every $\omega \in \Omega$.

***Proof*:**
The strategy is the standard machine: prove the claim first for indicators of $\sigma(X)$-sets, extend by linearity to simple functions, and pass to the limit for a general measurable Y. The only structural input is the shape of $\sigma(X)$: by definition 2.1.5, a set $A \in \sigma(X)$ is exactly $A = X^{-1}(B)$ for some $B \in \mathcal{B}(\mathbb{R})$.

Step 1: Indicators. Let $A \in \sigma(X)$ and write $A = X^{-1}(B)$ with $B \in \mathcal{B}(\mathbb{R})$. Then for every ω,

$$\mathbb{1}_A(\omega) = \mathbb{1}\{X(\omega) \in B\} = (\mathbb{1}_B \circ X)(\omega)$$

so with $g = \mathbb{1}_B$, a Borel function on $\mathbb{R}$, we have $\mathbb{1}_A = g(X)$. Thus every indicator of a $\sigma(X)$-set is a Borel function of X.

Step 2: Simple functions. Let $Y = \sum_{k=1}^m c_k \mathbb{1}_{A_k}$ be a $\sigma(X)$-measurable simple function, with $c_k \in \mathbb{R}$ and $A_k \in \sigma(X)$. By Step 1 each $A_k = X^{-1}(B_k)$ for some $B_k \in \mathcal{B}(\mathbb{R})$, and $\mathbb{1}_{A_k} = \mathbb{1}_{B_k}(X)$. Setting $g = \sum_{k=1}^m c_k \mathbb{1}_{B_k}$, a finite linear combination of Borel functions and hence Borel, gives

$$Y = \sum_{k=1}^m c_k \mathbb{1}_{B_k}(X) = g(X)$$

so every $\sigma(X)$-measurable simple function is a Borel function of X.

Step 3: Nonnegative measurable functions. Suppose first $Y \geq 0$ is $\sigma(X)$-measurable. There is an increasing sequence of $\sigma(X)$-measurable simple functions $Y_n \uparrow Y$ pointwise; this is the standard approximation, for instance

$$Y_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\{k/2^n \leq Y < (k+1)/2^n\}} + n\,\mathbb{1}_{\{Y \geq n\}}$$

each term of which is $\sigma(X)$-measurable because Y is. By Step 2, for each n there is a Borel function $g_n : \mathbb{R} \to \mathbb{R}$ with $Y_n = g_n(X)$. Define

$$g(t) = \limsup_{n\to\infty} g_n(t), \qquad t \in \mathbb{R}$$

which is Borel as a pointwise limsup of Borel functions. For every $\omega \in \Omega$ we have $g_n(X(\omega)) = Y_n(\omega) \to Y(\omega)$, so the ordinary limit exists along the point $t = X(\omega)$ and equals its limsup; hence

$$g(X(\omega)) = \limsup_{n\to\infty} g_n(X(\omega)) = \lim_{n\to\infty} Y_n(\omega) = Y(\omega)$$

Thus $Y = g(X)$ with g Borel. (The function g is only constrained on the range of X; its values elsewhere are immaterial, which is why the ambient limsup causes no harm.)

Step 4: General measurable functions. For arbitrary $\sigma(X)$-measurable Y, split into positive and negative parts $Y = Y^+ - Y^-$, both nonnegative and $\sigma(X)$-measurable. By Step 3 there are Borel functions g^+, g^- with $Y^+ = g^+(X)$ and $Y^- = g^-(X)$. The events $\{Y^+ > 0\}$ and $\{Y^- > 0\}$ are disjoint, so at each ω at most one of $g^+(X(\omega)), g^-(X(\omega))$ is nonzero, and $g = g^+ - g^-$, a Borel function, satisfies $g(X) = g^+(X) - g^-(X) = Y^+ - Y^- = Y$. This exhibits the required Borel g, completing the proof.

The lemma is used constantly and in both directions. Read forward, it says any quantity computable from X, however elaborately, is literally $g(X)$ for some measurable g, so working with $\sigma(X)$-measurable random variables and working with Borel functions of X are the same activity. Read backward, it is the tool that lets one *define* a conditional expectation, a projection, or a best predictor as a function of the observed data and know it is a legitimate random variable. The same statement holds verbatim with X a random vector and g a Borel function on $\mathbb{R}^d$, and, more generally, with X a random element of an arbitrary measurable space $(E, \mathcal{E})$ in place of $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, so that g is a Borel function on E; the proof is unchanged, since the standard machine used only the description $A = X^{-1}(B)$ of the generating sets. The extension is recorded once here so that later chapters may invoke it in the vector case without comment.

The definitions above are abstract; the following examples ground them in the distributions that occupy the rest of the book. Each computes the law μ_X and the distribution function F_X explicitly, so that the passage from the map X to its distribution can be watched in concrete cases.

Example 2.1.7: Indicators, Discrete, and Absolutely Continuous Laws
Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a fixed probability space.

1. *Indicators and simple random variables.* For an event $A \in \mathcal{F}$, the indicator $\mathbb{1}_A : \Omega \to \mathbb{R}$, equal to 1 on A and 0 on A^c, is a random variable: $\{\mathbb{1}_A \leq x\}$ is $\emptyset$, A^c, or Ω according as $x < 0$, $0 \leq x < 1$, or $x \geq 1$, all in $\mathcal{F}$. Its law is the two-point measure $\mu_{\mathbb{1}_A} = \mathbb{P}(A^c)\,\delta_0 + \mathbb{P}(A)\,\delta_1$,

where δ_c is the unit mass at c, and its distribution function is the step function

$$F_{\mathbb{1}_A}(x) = \begin{cases} 0, & x < 0 \\ \mathbb{P}(A^c), & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

More generally, a *simple random variable* $X = \sum_{k=1}^{m} c_k \mathbb{1}_{A_k}$ with distinct values $c_1 < \cdots < c_m$ and $A_k = \{X = c_k\}$ a measurable partition of Ω has law $\mu_X = \sum_{k=1}^{m} \mathbb{P}(A_k)\, \delta_{c_k}$ and distribution function $F_X(x) = \sum_{k: c_k \le x} \mathbb{P}(A_k)$, a step function that jumps by $\mathbb{P}(A_k)$ at each c_k.

2. *Discrete random variables and probability mass functions.* A random variable X is *discrete* if it takes values in a countable set $S = \{s_1, s_2, \dots\} \subseteq \mathbb{R}$. Its *probability mass function* is $p_X(s) = \mathbb{P}(X = s)$; these are nonnegative and sum to $\sum_{s \in S} p_X(s) = \mathbb{P}(X \in S) = 1$. The law is the countable sum of point masses

$$\mu_X = \sum_{s \in S} p_X(s)\, \delta_s$$

so that $\mu_X(B) = \sum_{s \in S \cap B} p_X(s)$, and the distribution function is the step function $F_X(x) = \sum_{s \le x} p_X(s)$, which jumps by $p_X(s)$ at each atom s and is constant between atoms. For a concrete instance, a Poisson variable with $S = \{0, 1, 2, \dots\}$ and $p_X(k) = e^{-\lambda}\lambda^k/k!$ has $\mu_X = e^{-\lambda} \sum_{k \ge 0} (\lambda^k/k!)\, \delta_k$.

3. *Absolutely continuous random variables and densities.* A random variable X is *absolutely continuous* if there is a nonnegative Borel function $f\colon \mathbb{R} \to [0, \infty)$, the *density*, with $\int_{\mathbb{R}} f\, d\lambda = 1$ (Lebesgue measure λ) such that

$$\mu_X(B) = \int_B f\, d\lambda, \qquad B \in \mathcal{B}(\mathbb{R})$$

Then the distribution function is the integral of the density, $F_X(x) = \int_{(-\infty, x]} f\, d\lambda = \int_{-\infty}^{x} f(t)\, dt$, which is continuous (indeed absolutely continuous) in x; where f is continuous, the fundamental theorem of calculus gives $F_X' = f$. That μ_X so defined is a probability measure with distribution function F_X is immediate from the properties of the integral, and by proposition 2.1.4 it is the unique law with that distribution function. For a concrete instance, the standard exponential density $f(t) = e^{-t}\mathbb{1}_{[0,\infty)}(t)$ gives $F_X(x) = (1 - e^{-x})\mathbb{1}_{[0,\infty)}(x)$.

The three cases exhaust the two extremes and their point-mass analog: a purely atomic law concentrated on a countable set, a law with a density and no atoms, and the elementary building block, a single atom's worth of an indicator. A general law is a mixture of an atomic part and a continuous part (the Lebesgue decomposition of [2] applied to μ_X), so these examples are the ingredients from which every one-dimensional distribution is assembled. The next example instantiates the absolutely continuous case on the concrete space of section 1.1, the unit interval under Lebesgue measure, producing the uniform variable that will seed the constructions of section 2.3.

Example 2.1.8: The Uniform Random Variable on $[0,1]$

Take the probability space of example 1.1.3: $\Omega = [0,1]$ with $\mathcal{F} = \mathcal{B}([0,1])$ and $\mathbb{P} = \lambda$ Lebesgue measure restricted to $[0,1]$. Let $U\colon \Omega \to \mathbb{R}$ be the identity map, $U(\omega) = \omega$. It is a random variable, since for $x \in \mathbb{R}$

$$\{U \le x\} = [0,1] \cap (-\infty, x] = \begin{cases} \emptyset, & x < 0 \\ [0,x], & 0 \le x < 1 \\ [0,1], & x \ge 1 \end{cases}$$

each a Borel subset of $[0,1]$, so the half-line criterion (proposition 2.1.2) is met. Its law is

$$\mu_U(B) = \mathbb{P}(U \in B) = \lambda\big([0,1] \cap B\big)$$

that is, Lebesgue measure restricted to $[0,1]$, viewed now as a Borel measure on $\mathbb{R}$. Reading off the distribution function from the display above,

$$F_U(x) = \mathbb{P}(U \le x) = \begin{cases} 0, & x < 0 \\ x, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

which is continuous and piecewise linear, with density $f(x) = \mathbb{1}_{[0,1]}(x)$. This is the standard uniform variable, written $U \sim \mathrm{Unif}[0,1]$. It is the generic source of randomness: the map $\omega \mapsto \omega$ turns the "pick a point uniformly" space of example 1.1.3 into a random variable whose law is the uniform distribution, and section 2.3 will build every other distribution by transforming copies of it.

The last two examples make the section's central distinction visible. In example 2.1.8 the random variable U is a specific function on a specific sample space, the identity on $[0,1]$; its law μ_U is a measure on the line, the uniform distribution, that has forgotten the sample space entirely. A different construction, say the map $\omega \mapsto 1 - \omega$ on the same space, or a coordinate on the coin-tossing space of section 1.3 suitably combined, could reproduce the very same law while being a completely different function. The random variable is the map; the law is the distribution it induces. Probability speaks the language of laws, because every probability $\mathbb{P}(X \in B) = \mu_X(B)$ is computed from μ_X alone, and this is exactly why $\overset{d}{=}$, equality of laws, is the equivalence under which the subject identifies its objects. When later chapters say "let X be standard Gaussian" or "let (X_n) be i.i.d. uniform", they are specifying laws, and the sample space on which the variables are realized is chosen for convenience, never observed.

Exercise 2.1.1

1. Let $X\colon \Omega \to \mathbb{R}$ be a function such that $\{X < x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. Show that X is a random variable. (Thus the strict-inequality rays $(-\infty, x)$ generate $\mathcal{B}(\mathbb{R})$ just as well as the closed rays, and either family may be used in proposition 2.1.2.)

2. Let X, Y be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Show that $X+Y$, XY, and $\max(X,Y)$ are random variables, and that if (X_n) is a sequence of random variables then $\sup_n X_n$ and $\limsup_n X_n$ are extended-real random variables.

3. Compute the law μ_X and distribution function F_X of $X = \mathbb{1}_A - \mathbb{1}_B$, where $A, B \in \mathcal{F}$ are disjoint with $\mathbb{P}(A) = p$ and $\mathbb{P}(B) = q$. Identify the atoms of μ_X and their masses.

4. Let $U \sim \mathrm{Unif}[0,1]$ as in example 2.1.8, and let $X = -\log U$. Compute F_X and show that X is absolutely continuous with the standard exponential density $f(t) = e^{-t}\mathbb{1}_{[0,\infty)}(t)$. (This is the probability integral transform in reverse: it manufactures an exponential variable from a uniform one.)

5. Give two random variables X, Y on the *same* probability space with $X \overset{d}{=} Y$ (equal laws) but $X \neq Y$ as functions, indeed with $\mathbb{P}(X \neq Y) = 1$. Use the uniform space of example 2.1.8.

6. Let X be a random variable and $g\colon \mathbb{R} \to \mathbb{R}$ a Borel function. Show that $\sigma(g(X)) \subseteq \sigma(X)$, and give an example where the inclusion is strict. Then show that if g is a Borel bijection with Borel inverse, equality holds. (Contrast with the Doob-Dynkin lemma, lemma 2.1.6, which characterizes when a variable lies in $\sigma(X)$.)

2.2 Independence

Every structural fact established so far belongs to measure theory: a probability space is a measure space of total mass one, a random variable is a measurable function, and a law is a pushforward measure. A measure theorist handed $(\Omega, \mathcal{F}, \mathbb{P})$ would recognize all of it. Independence is the first notion that has no counterpart in general measure theory and gives probability its own character. It is the assertion that knowing the value of one random variable tells you nothing about another, and it is the hypothesis under which sums of many variables acquire the sharp asymptotics that occupy the rest of this book.

The event-level version is already in hand: two events A and B are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ (definition 1.4.3), the form used to force the Borel-Cantelli dichotomy in section 1.4. That definition is enough for events but awkward for random variables, which carry not one event but a whole σ-algebra of them, the sets $\{X \in B\}$ as B ranges over the Borel sets. The right level at which to state independence is therefore the σ-algebra: independence of random variables will mean independence of the σ-algebras they generate, and independence of events will be recovered as the special case of the tiny σ-algebras those events generate. This section defines independence once, at the level of σ-algebras, proves the criterion that makes it checkable on a generating class rather than on the full σ-algebra, and reads that criterion off for random variables as the factorization of the joint distribution function.

The definition applies the event-level product rule $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ simultaneously to every choice of one set from each σ-algebra. For a family indexed by an arbitrary set, the product rule is imposed on every *finite* subfamily, which is the only way an infinite product of probabilities could be given meaning, and which is exactly what the downstream limit theorems require.

Definition 2.2.1: Independent σ-Algebras

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(\mathcal{G}_i)_{i\in I}$ be a family of sub-σ-algebras of $\mathcal{F}$. The family is *independent* if for every finite subset $J \subseteq I$ and every choice of sets $A_i \in \mathcal{G}_i$ for $i \in J$,

$$\mathbb{P}\left(\bigcap_{i\in J} A_i\right) = \prod_{i\in J} \mathbb{P}(A_i)$$

Two sub-σ-algebras $\mathcal{G}_1, \mathcal{G}_2$ are independent, written $\mathcal{G}_1 \perp\!\!\!\perp \mathcal{G}_2$, when $\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2)$ for all $A_1 \in \mathcal{G}_1$ and $A_2 \in \mathcal{G}_2$.

The reason the finite-subset rule is the correct formulation, rather than imposing the two-at-a-time condition $\mathcal{G}_i \perp\!\!\!\perp \mathcal{G}_j$ for every pair, is that pairwise independence is strictly weaker than the joint condition: three σ-algebras can be independent in pairs while the triple fails the product rule. The definition requires the full product for every finite selection, and example 2.2.6 exhibits a concrete triple that is pairwise but not mutually independent, so the distinction is not a formality.

One immediate consequence of the definition deserves to be recorded, because it is used constantly and silently: Ω and $\emptyset$ lie in every sub-σ-algebra, and $\mathbb{P}(\Omega) = 1$, $\mathbb{P}(\emptyset) = 0$, so a factor of Ω can be inserted into or deleted from any product without changing it. Concretely, if $J' \subseteq J$ and one takes $A_i = \Omega$ for $i \in J \setminus J'$, the finite product rule over J collapses to the product rule over J'. Thus verifying the rule for every finite subset is the same as verifying it for every finite subset together with the freedom to pad with Ω, and one may always assume a selection uses a genuine set from each of finitely many coordinates.

Independence of random variables is now a definition about the σ-algebras $\sigma(X_i)$ they generate (definition 2.1.5). This is the definition around which the entire theory of sums of independent variables is built.

Definition 2.2.2: Independent Random Variables

Random variables $(X_i)_{i\in I}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, each taking values in a measurable space, are *independent* if the generated σ-algebras $\big(\sigma(X_i)\big)_{i\in I}$ are independent in the sense of definition 2.2.1. Explicitly, for every finite $J \subseteq I$ and Borel sets B_i in the respective ranges,

$$\mathbb{P}\left(\bigcap_{i\in J} \{X_i \in B_i\}\right) = \prod_{i\in J} \mathbb{P}(X_i \in B_i)$$

Events $(A_i)_{i\in I}$ are independent, in the sense of definition 1.4.3 generalized to a family, precisely when the σ-algebras $(\sigma(\mathbb{1}_{A_i}))_{i\in I}$ they generate are independent.

The two clauses of the definition agree because $\{X_i \in B_i\} = X_i^{-1}(B_i)$ runs over all of $\sigma(X_i)$ as B_i runs over the Borel sets: that is what $\sigma(X_i) = X_i^{-1}(\mathcal{B})$ means (definition 2.1.5). So the σ-algebra condition of definition 2.2.1 applied to the $\sigma(X_i)$ unwinds to the displayed factorization over Borel sets, and no information is lost by phrasing the definition either way.

It remains to see that this genuinely *generalizes* the event-level definition of section 1.4, rather than replacing it. The σ-algebra generated by a single event A is the four-element algebra

$$\sigma(A) = \{\emptyset,\ A,\ A^c,\ \Omega\}$$

and the indicator $\mathbb{1}_A$ generates the same σ-algebra, since $\mathbb{1}_A^{-1}(\mathcal{B})$ consists exactly of $\emptyset, A, A^c, \Omega$ according to whether the Borel set contains neither, only 1, only 0, or both. The following lemma records that independence of events, independence of their indicators, and independence of these four-element σ-algebras are the same condition, so definition 2.2.2 contains definition 1.4.3 as a special case.

Proposition 2.2.3: Events, Indicators, and Their σ-Algebras

Let $(A_i)_{i\in I}$ be events. The following are equivalent.

1. The events (A_i) are independent in the sense of definition 1.4.3: for every finite $J \subseteq I$, $\mathbb{P}(\bigcap_{i\in J} A_i) = \prod_{i\in J} \mathbb{P}(A_i)$.
2. The indicators $(\mathbb{1}_{A_i})$ are independent random variables.
3. The σ-algebras $(\{\emptyset, A_i, A_i^c, \Omega\})_{i\in I}$ are independent.

***Proof*:**
Statements (2) and (3) are literally the same, since $\sigma(\mathbb{1}_{A_i}) = \{\emptyset, A_i, A_i^c, \Omega\}$ by the computation of the preimages above. It remains to prove (1) $\Leftrightarrow$ (3).

That (3) implies (1) is immediate: taking the set A_i from each four-element σ-algebra in the selection recovers exactly the event product rule. For the converse, fix a finite $J \subseteq I$ and a selection of one set $C_i \in \{\emptyset, A_i, A_i^c, \Omega\}$ for each $i \in J$; the goal is $\mathbb{P}(\bigcap_{i\in J} C_i) = \prod_{i\in J} \mathbb{P}(C_i)$. If any $C_i = \emptyset$ both sides vanish, so assume every $C_i \in \{A_i, A_i^c, \Omega\}$. A factor $C_i = \Omega$ contributes 1 to the product and drops out of the intersection, so it may be discarded from J; assume then that each C_i is A_i or A_i^c.

The remaining case, that complements may be mixed in freely, is exactly lemma 1.4.4: if events satisfy the product rule, so does any family obtained by replacing some of them with their complements. Applying that lemma to the events $(A_i)_{i\in J}$ with the chosen sets C_i turns the hypothesis (1) into $\mathbb{P}(\bigcap_{i\in J} C_i) = \prod_{i\in J} \mathbb{P}(C_i)$, which is the required identity, completing the proof.

The upshot is that the reader loses nothing and gains uniformity: every statement about independent events is a statement about independent random variables applied to indicators, and the machinery developed below for σ-algebras specializes to section 1.4 without new work.

definition 2.2.1 and definition 2.2.2 are not directly usable: verifying independence of $\sigma(X)$ and $\sigma(Y)$ from the definition would require checking $\mathbb{P}(X \in B, Y \in C) = \mathbb{P}(X \in B)\mathbb{P}(Y \in C)$ for *all* Borel B and C, an uncountable family of identities. In practice one wants to check the product rule only on a small, concrete generating class, such as the half-lines $\{X \leq x\}$, and conclude that it then holds on the full σ-algebras. That this reduction is legitimate is the central technical result of the section, and it is precisely the payoff of the π-λ theorem of section 1.2: a product identity that holds on a generating π-system extends to the generated σ-algebra.

Theorem 2.2.4: The π-System Criterion for Independence

Let $(\mathcal{P}_i)_{i\in I}$ be a family of π-systems on Ω with each $\mathcal{P}_i \subseteq \mathcal{F}$, and let $\mathcal{G}_i = \sigma(\mathcal{P}_i)$ be the σ-algebra it generates. Suppose the product rule holds for every finite selection from the π-systems: for every finite $J \subseteq I$ and every choice of sets $P_i \in \mathcal{P}_i$ for $i \in J$,

$$\mathbb{P}\left(\bigcap_{i\in J} P_i\right) = \prod_{i\in J} \mathbb{P}(P_i) \tag{2.1}$$

Then the σ-algebras $(\mathcal{G}_i)_{i\in I}$ are independent.

***Proof*:**
Because independence is a condition on finite subfamilies, it suffices to fix a finite index set, which we take to be $J = \{1, \ldots, n\}$, and prove that $\mathcal{G}_1, \ldots, \mathcal{G}_n$ satisfy the product rule. So assume (2.1) for all choices $P_i \in \mathcal{P}_i$; the goal is

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \mathbb{P}(A_i) \qquad \text{for all } A_i \in \mathcal{G}_i \tag{2.2}$$

The strategy is to promote the identity from the π-systems to the σ-algebras one coordinate at a time, each promotion carried by a single application of the π-λ theorem (theorem 1.2.5). A subtle point makes this work: to run the Dynkin argument in the first coordinate, the sets in all the *other* coordinates must range over π-systems that also contain Ω, so we first enlarge each $\mathcal{P}_i$ to

$$\widetilde{\mathcal{P}}_i = \mathcal{P}_i \cup \{\Omega\}$$

This is still a π-system generating $\mathcal{G}_i$ (adjoining Ω preserves closure under finite intersection, since $\Omega \cap P = P$), and the product rule (2.1) still holds over the $\widetilde{\mathcal{P}}_i$, because inserting a factor Ω multiplies both sides by $\mathbb{P}(\Omega) = 1$ and leaves the intersection unchanged. Replacing each $\mathcal{P}_i$ by $\widetilde{\mathcal{P}}_i$, assume from now on that every $\mathcal{P}_i$ contains Ω.

Step 1: Promote the first coordinate. Fix sets $P_2 \in \mathcal{P}_2, \ldots, P_n \in \mathcal{P}_n$ once and for all, and let

$$\mathcal{L}_1 = \mathbf{Set} A \in \mathcal{F} \middle| \mathbb{P}\left(A \cap \bigcap_{i=2}^{n} P_i\right) = \mathbb{P}(A) \prod_{i=2}^{n} \mathbb{P}(P_i)$$

the collection of first-coordinate sets on which the product identity holds while coordinates $2, \ldots, n$ are frozen at the chosen π-system sets. Write $Q = \bigcap_{i=2}^{n} P_i$ and $q = \prod_{i=2}^{n} \mathbb{P}(P_i)$, so $A \in \mathcal{L}_1$ means $\mathbb{P}(A \cap Q) = \mathbb{P}(A)\, q$. Direct verification shows $\mathcal{L}_1$ is a λ-system:

1. $\Omega \in \mathcal{L}_1$, since $\mathbb{P}(\Omega \cap Q) = \mathbb{P}(Q)$ and by (2.1) applied to $\Omega, P_2, \ldots, P_n$ this equals $\mathbb{P}(\Omega)\, q = q$.
2. If $A, B \in \mathcal{L}_1$ with $A \subseteq B$, then $(B \setminus A) \cap Q = (B \cap Q) \setminus (A \cap Q)$ with $A \cap Q \subseteq B \cap Q$, so $\mathbb{P}((B \setminus A) \cap Q) = \mathbb{P}(B \cap Q) - \mathbb{P}(A \cap Q) = \mathbb{P}(B)q - \mathbb{P}(A)q = \mathbb{P}(B \setminus A)\, q$, giving $B \setminus A \in \mathcal{L}_1$.
3. If $A_k \in \mathcal{L}_1$ increase to $A = \bigcup_k A_k$, then $A_k \cap Q \uparrow A \cap Q$, and continuity of $\mathbb{P}$ along increasing sequences (proposition 1.1.5) gives $\mathbb{P}(A \cap Q) = \lim_k \mathbb{P}(A_k \cap Q) = \lim_k \mathbb{P}(A_k) q = \mathbb{P}(A)q$, so $A \in \mathcal{L}_1$.

By hypothesis (2.1), every $P_1 \in \mathcal{P}_1$ lies in $\mathcal{L}_1$; that is, $\mathcal{L}_1$ contains the π-system $\mathcal{P}_1$. The π-λ theorem (theorem 1.2.5) then forces $\mathcal{L}_1 \supseteq \sigma(\mathcal{P}_1) = \mathcal{G}_1$. Since the frozen sets $P_2, \dots, P_n$ were arbitrary, the conclusion of this step is

$$\mathbb{P}\left(A_1 \cap \bigcap_{i=2}^{n} P_i\right) = \mathbb{P}(A_1) \prod_{i=2}^{n} \mathbb{P}(P_i) \qquad \text{for all } A_1 \in \mathcal{G}_1,\ P_i \in \mathcal{P}_i \tag{2.3}$$

The first coordinate now ranges over the full σ-algebra while the rest remain in their π-systems.

Step 2: Iterate over the remaining coordinates. The identity (2.3) has exactly the same shape as the hypothesis (2.1), with $\mathcal{P}_1$ replaced by $\mathcal{G}_1$. Repeat the argument of Step 1 in the second coordinate: fix $A_1 \in \mathcal{G}_1$ and $P_3 \in \mathcal{P}_3, \dots, P_n \in \mathcal{P}_n$, and let

$$\mathcal{L}_2 = \mathbf{Set} A \in \mathcal{F} \big| \mathbb{P}\left(A_1 \cap A \cap \bigcap_{i=3}^{n} P_i\right) = \mathbb{P}(A_1)\,\mathbb{P}(A) \prod_{i=3}^{n} \mathbb{P}(P_i)$$

The same three verifications, now with $Q = A_1 \cap \bigcap_{i=3}^{n} P_i$ and $q = \mathbb{P}(A_1) \prod_{i=3}^{n} \mathbb{P}(P_i)$, show $\mathcal{L}_2$ is a λ-system, and (2.3) shows it contains $\mathcal{P}_2$. Hence $\mathcal{L}_2 \supseteq \mathcal{G}_2$, promoting the second coordinate to its full σ-algebra. Continuing coordinate by coordinate, after n applications of theorem 1.2.5 every coordinate has been promoted to its σ-algebra, which is exactly (2.2). This holds for every finite $J \subseteq I$, so $(\mathcal{G}_i)_{i \in I}$ are independent, as we sought to show.

The criterion is used throughout the subject. Every subsequent verification of independence, from the factorization of joint laws in section 2.3 to the independence of the tail σ-algebra in section 2.4, is an instance of it: one exhibits a convenient π-system generating each σ-algebra, checks the product rule on those generators, and invokes theorem 2.2.4. The proof is also a model of the π-λ method itself, freezing all but one coordinate so that the collection of good sets in the free coordinate becomes a λ-system containing a generating π-system, hence everything. A close relative of the argument, uniqueness of a measure from a generating π-system (corollary 1.2.6), gives an alternative route to the same conclusion: both sides of (2.2), as functions of A_1 with the other coordinates frozen, are finite measures of equal total mass agreeing on a generating π-system, hence equal.

For random variables the generating π-system is chosen once and for all: the half-lines. Recall from example 1.2.7 that the sets $(-\infty, x]$, $x \in \mathbb{R}$, form a π-system generating $\mathcal{B}(\mathbb{R})$, so the events $\{X \leq x\}$ form a π-system generating $\sigma(X)$. Feeding this π-system into theorem 2.2.4 converts independence of a finite family of random variables into a single scalar identity, the factorization of the joint distribution function.

Proposition 2.2.5: Independence via the Joint Distribution Function

Real random variables $X_1, \dots, X_n$ on $(\Omega, \mathcal{F}, \mathbb{P})$ are independent if and only if their joint distribution function factors:

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^{n} F_{X_i}(x_i) \qquad \text{for all } (x_1, \dots, x_n) \in \mathbb{R}^n \tag{2.4}$$

where $F_{X_i}(x) = \mathbb{P}(X_i \leq x)$ is the distribution function of X_i (proposition 2.1.4).

***Proof*:**
If $X_1, \ldots, X_n$ are independent, then taking $B_i = (-\infty, x_i]$ in definition 2.2.2 gives (2.4) directly, since $\{X_i \le x_i\} = \{X_i \in B_i\}$ and $\mathbb{P}(X_i \le x_i) = F_{X_i}(x_i)$.

Conversely, suppose (2.4) holds. For each i let

$$\mathcal{P}_i = \big\{\{X_i \le x\} \big| x \in \mathbb{R}\big\} = \big\{X_i^{-1}((-\infty, x]) \big| x \in \mathbb{R}\big\}$$

Since $(-\infty, x]$ ranges over the half-line π-system of example 1.2.7, and preimage commutes with finite intersection, $\mathcal{P}_i$ is a π-system, and it generates $\sigma(X_i) = X_i^{-1}(\mathcal{B}(\mathbb{R}))$ because the half-lines generate $\mathcal{B}(\mathbb{R})$. Assumption (2.4) is precisely the product rule (2.1) for these π-systems on the full index set $\{1, \ldots, n\}$. The hypothesis of theorem 2.2.4 requires (2.1) for every finite selection, including those over a proper subset $J \subsetneq \{1, \ldots, n\}$; these follow from the full-index case by a limiting argument, since the half-line π-systems $\mathcal{P}_i$ do not contain Ω and so cannot supply an omitted coordinate directly. Given sets $\{X_i \le x_i\}$ for $i \in J$, let each omitted coordinate $x_j \to +\infty$ (for $j \notin J$): then $\{X_j \le x_j\} \uparrow \Omega$, so the intersection over $\{1, \ldots, n\}$ increases to $\bigcap_{i \in J}\{X_i \le x_i\}$ and $F_{X_j}(x_j) \to 1$. Continuity of $\mathbb{P}$ along increasing sequences (proposition 1.1.5) passes (2.4) to the limit, giving $\mathbb{P}\big(\bigcap_{i \in J}\{X_i \le x_i\}\big) = \prod_{i \in J} F_{X_i}(x_i)$, the product rule over J. Thus (2.1) holds for every finite selection, and by theorem 2.2.4 the σ-algebras $\sigma(X_1), \ldots, \sigma(X_n)$ are independent, which is the definition of independence of $X_1, \ldots, X_n$, completing the proof.

The proposition turns an abstract condition on σ-algebras into a concrete one that can be verified by inspection or computation. When the X_i have densities the factorization descends to the densities, and when they are discrete it descends to the joint probability mass function, both recorded in the examples below. The same reduction, applied to the π-system of measurable rectangles rather than half-lines, identifies independence with a product structure on the joint law $\mu_{(X_1, \ldots, X_n)}$; that identification, and the Fubini theorem it unlocks, is the business of section 2.3.

The definitions are abstract, so the following collects the concrete instances that anchor them: the canonical independent sequence of section 1.3, the density criterion, and the cautionary triple that separates pairwise from mutual independence.

Example 2.2.6: Independent Coordinates, Product Densities, and a Pairwise Trap

1. *Coordinates of the coin-tossing space.* Let $\Omega = \{0,1\}^{\mathbb{N}}$ with the fair-coin measure $\mathbb{P}$ of example 1.3.6, and let $X_n : \Omega \to \{0, 1\}$ be the n-th coordinate, $X_n(\omega) = \omega_n$. The defining property of that measure is that for any finite set of coordinates $n_1 < \cdots < n_k$ and any bits $b_1, \ldots, b_k \in \{0, 1\}$,

$$\mathbb{P}(X_{n_1} = b_1, \ldots, X_{n_k} = b_k) = 2^{-k} = \prod_{j=1}^{k} \tfrac{1}{2} = \prod_{j=1}^{k} \mathbb{P}(X_{n_j} = b_j)$$

   Each X_n generates the four-element σ-algebra $\sigma(X_n) = \{\emptyset, \{X_n = 0\}, \{X_n = 1\}, \Omega\}$, and the events $\{X_n = b\}$ form a π-system generating it, so the displayed product rule and theorem 2.2.4 make $(X_n)_{n \in \mathbb{N}}$ an independent sequence. This is the archetypal independent sequence and the running example of the limit theory to come: the coordinates are i.i.d. fair bits.

2. *Two random variables with a product density.* Suppose the pair (X, Y) has a joint density $f_{X,Y}$ on $\mathbb{R}^2$, meaning $\mathbb{P}((X,Y) \in C) = \iint_C f_{X,Y}(x,y)\,dx\,dy$ for Borel $C \subseteq \mathbb{R}^2$, and suppose

this density factors,
$$f_{X,Y}(x,y) = g(x)\,h(y)$$
for nonnegative Borel g, h. Then X and Y are independent. To see it, integrate: the marginal density of X is $f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy = g(x)\int_{\mathbb{R}} h$, and similarly $f_Y(y) = h(y)\int_{\mathbb{R}} g$, and since $\iint f_{X,Y} = 1$ the two normalizing integrals multiply to 1. Therefore
$$F_{X,Y}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} g(s)h(t)\,dt\,ds = \left(\int_{-\infty}^{x} g\right)\left(\int_{-\infty}^{y} h\right) = F_X(x)\,F_Y(y)$$
where the last equality uses $F_X(x) = \int_{-\infty}^{x} g \cdot \int_{\mathbb{R}} h$ and $F_Y(y) = \int_{-\infty}^{y} h \cdot \int_{\mathbb{R}} g$ together with the fact just noted that the product $(\int_{\mathbb{R}} g)(\int_{\mathbb{R}} h)$ of the two full integrals equals 1; neither factor need be 1 on its own. Hence proposition 2.2.5 gives independence. Concretely, a pair of independent standard Gaussians has joint density $\frac{1}{2\pi}e^{-(x^2+y^2)/2} = \big(\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\big)\big(\frac{1}{\sqrt{2\pi}}e^{-y^2/2}\big)$, a product, matching the factorization.

3. *A pairwise but not mutually independent triple.* On $\Omega = \{0,1\}^2$ with the uniform measure (four equally likely outcomes), let X and Y be the two coordinates and let $Z = X \oplus Y$ be their sum modulo 2. Each of X, Y, Z is a fair bit: $\mathbb{P}(Z = 1) = \mathbb{P}(\{01, 10\}) = \frac{1}{2}$. Any two of the three are independent, for instance
$$\mathbb{P}(X = 1, Z = 1) = \mathbb{P}(\{10\}) = \tfrac{1}{4} = \tfrac{1}{2} \cdot \tfrac{1}{2} = \mathbb{P}(X = 1)\mathbb{P}(Z = 1)$$
and the same computation holds for every pair and every pair of values, so (X, Y, Z) is pairwise independent. But the triple is not independent, because Z is determined by X and Y: the event $\{X = 1, Y = 1, Z = 1\}$ is empty, since $1 \oplus 1 = 0 \neq 1$, so
$$\mathbb{P}(X = 1, Y = 1, Z = 1) = 0 \neq \tfrac{1}{8} = \mathbb{P}(X = 1)\mathbb{P}(Y = 1)\mathbb{P}(Z = 1)$$
The product rule fails for the full triple though it holds for every pair. This is the promised witness that pairwise independence does not imply mutual independence, and it is why definition 2.2.1 requires the product rule for every finite subfamily rather than only for pairs.

The three examples span the ways independence is met in practice: as a built-in feature of a construction (the coin-tossing coordinates, where the product rule is how the measure was defined), as a factorization to be checked (the density criterion), and as a property that can silently fail even when every pairwise instance of it holds (the parity triple). The first is the object the limit theorems act on; the second is the computational form used throughout the limit theory to come; the third is the standing warning that independence is a joint condition, not a pairwise one.

Exercise 2.2.1

1. Let $\mathcal{G}_1, \mathcal{G}_2$ be independent sub-σ-algebras, and let $A \in \mathcal{G}_1$ have $\mathbb{P}(A) \in \{0, 1\}$. Show that A is independent of every event in $\mathcal{G}_2$ directly, and conclude that any event of probability 0 or 1 is independent of every event whatsoever. What does this say about the σ-algebra $\{\emptyset, \Omega\}$?
2. Let X and Y be independent random variables and let $u, v\colon \mathbb{R} \to \mathbb{R}$ be Borel measurable. Show that $u(X)$ and $v(Y)$ are independent. (Work at the level of σ-algebras: relate $\sigma(u(X))$ to $\sigma(X)$.)
3. Suppose X and Y are independent and $X = Y$ almost surely. Show that X is almost surely

constant. (Hint: apply independence to the events $\{X \le x\}$ and $\{Y \le x\} = \{X \le x\}$ a.s. to get $F_X(x) = F_X(x)^2$.)

4. Give random variables X, Y, Z that are pairwise independent but not mutually independent and for which *all three* are non-degenerate and take more than two values, or explain why the parity example of example 2.2.6 cannot be adapted without such a modification.

5. Let $(X_i)_{i\in I}$ be independent and let $\{I_\alpha\}$ be a partition of I into disjoint blocks. Assuming the block σ-algebras $\mathcal{H}_\alpha = \sigma(X_i : i \in I_\alpha)$ are independent (a fact that follows from theorem 2.2.4; you may take it as given here), show that a function of the variables in one block is independent of a function of the variables in a disjoint block. Illustrate with the coin-tossing space: $X_1 + X_2$ is independent of X_3X_4.

6. Two events A, B with $\mathbb{P}(A), \mathbb{P}(B) \in (0, 1)$ satisfy $A \perp\!\!\!\perp B$. Can A and B be disjoint? Can $A \subseteq B$? Determine exactly when a pair related by disjointness or containment can be independent.

2.3 Product Measure, Independent Sequences, and Fubini

The previous section characterized independence through generating π-systems and, for random variables, through the factorization of the joint distribution function (proposition 2.2.5). This section translates that factorization into the language of measures on the range, where it becomes a single clean statement: independence is the assertion that the joint law is a product measure. That identification is the bridge between probability and the product-measure theory of *Measure Theory* [2], and it unlocks three things at once. First, the whole apparatus of Fubini's theorem becomes available for computing probabilities of joint events by iterated integration. Second, the law of a sum $X + Y$ of independent variables is expressed as a convolution of the marginal laws, the operation that governs how independent noise accumulates. Third, and most consequential for the rest of the book, the Kolmogorov extension of section 1.3 produces, from any single law μ on $\mathbb{R}$, an infinite sequence of independent variables each distributed as μ living on one probability space. That sequence, and the simple random walk built from it, is the object the entire limit theory of Part I onward is about.

The starting point is the passage from the abstract product-rule definition of independence to a statement purely about the measures a random vector induces on Euclidean space. For a finite family $X_1, \dots, X_n$ of real random variables, the map $X = (X_1, \dots, X_n)\colon \Omega \to \mathbb{R}^n$ is itself a random vector (definition 2.1.1), and its law $\mu_X = \mathbb{P} \circ X^{-1}$ (definition 2.1.3) is a Borel probability measure on $\mathbb{R}^n$, the *joint law* of the family. The individual laws $\mu_{X_1}, \dots, \mu_{X_n}$ on $\mathbb{R}$ are the *marginals*, recovered from μ_X by pushing forward along the coordinate projections. The product measure $\mu_{X_1} \otimes \cdots \otimes \mu_{X_n}$ on $\mathbb{R}^n$, whose existence and uniqueness on σ-finite (here, probability) factors is supplied by *Measure Theory* [2], is the unique Borel probability measure assigning to each measurable rectangle $B_1 \times \cdots \times B_n$ the product $\prod_i \mu_{X_i}(B_i)$. The theorem says that independence of the family is exactly the statement that these two measures on $\mathbb{R}^n$ coincide.

Theorem 2.3.1: Independence as a Product Law

Let $X_1, \dots, X_n$ be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ with laws $\mu_{X_1}, \dots, \mu_{X_n}$ on $\mathbb{R}$, and let $X = (X_1, \dots, X_n)$ with joint law μ_X on $\mathbb{R}^n$. Then $X_1, \dots, X_n$ are independent (definition 2.2.2) if and only if the joint law is the product of the marginals,

$$\mu_X = \mu_{X_1} \otimes \cdots \otimes \mu_{X_n}$$

the unique Borel probability measure on $\mathbb{R}^n$ with $\mu_X(B_1 \times \cdots \times B_n) = \prod_{i=1}^n \mu_{X_i}(B_i)$ for all Borel sets $B_1, \dots, B_n \subseteq \mathbb{R}$.

***Proof*:**
Write $\nu = \mu_{X_1} \otimes \cdots \otimes \mu_{X_n}$ for the product measure on $\big(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)\big)$, which *Measure Theory* [2] produces as the unique Borel probability measure on $\mathbb{R}^n$ with the stated value on measurable rectangles. Both μ_X and ν are probability measures on $\mathbb{R}^n$, and the argument turns on a single generating class.

Step 1: the rectangles are a generating π-system. Let

$$\mathcal{R} = \big\{B_1 \times \cdots \times B_n \big| B_i \in \mathcal{B}(\mathbb{R})\big\}$$

be the collection of measurable rectangles in $\mathbb{R}^n$. It is a π-system: the intersection of two rectangles is $(B_1 \cap B_1') \times \cdots \times (B_n \cap B_n')$, again a rectangle. It generates the Borel σ-algebra, $\sigma(\mathcal{R}) = \mathcal{B}(\mathbb{R}^n)$, since every open box is a rectangle and the open boxes generate $\mathcal{B}(\mathbb{R}^n)$.

Step 2: the values on rectangles. For a rectangle $R = B_1 \times \cdots \times B_n$ the preimage under X splits along coordinates,

$$X^{-1}(R) = \{X \in R\} = \bigcap_{i=1}^n \{X_i \in B_i\}$$

so by the definition of the joint law and of independence,

$$\mu_X(R) = \mathbb{P}\Big(\bigcap_{i=1}^n \{X_i \in B_i\}\Big), \qquad \nu(R) = \prod_{i=1}^n \mu_{X_i}(B_i) = \prod_{i=1}^n \mathbb{P}(X_i \in B_i)$$

The two right-hand sides are equal for all B_i precisely when the events $\{X_i \in B_i\}$ satisfy the product rule for every choice of Borel sets.

Step 3: the equivalence. Suppose first that $X_1, \dots, X_n$ are independent. Then $\{X_i \in B_i\} \in \sigma(X_i)$ for each i, and by the definition of independence of the σ-algebras $\sigma(X_i)$ (definition 2.2.2) the product rule $\mathbb{P}\big(\bigcap_i \{X_i \in B_i\}\big) = \prod_i \mathbb{P}(X_i \in B_i)$ holds. By Step 2, μ_X and ν agree on the π-system $\mathcal{R}$. Two probability measures agreeing on a generating π-system are equal (corollary 1.2.6), so $\mu_X = \nu$.

Conversely, suppose $\mu_X = \nu$. Then for every choice of Borel sets B_i, Step 2 gives $\mathbb{P}\big(\bigcap_i \{X_i \in B_i\}\big) = \prod_i \mathbb{P}(X_i \in B_i)$. This is exactly the product rule for the events $\{X_i \in B_i\} \in \sigma(X_i)$ over all finite selections of Borel sets, so by the definition of independence of the $\sigma(X_i)$ (definition 2.2.2) the family $X_1, \dots, X_n$ is independent, as we sought to show.

The theorem is the structural heart of the section. Independence, defined as an infinite family of product-rule equations, one for each choice of events, is compressed into the single equation $\mu_X = \bigotimes_i \mu_{X_i}$ between two measures on $\mathbb{R}^n$. Everything downstream is a reading of that equation. The marginals carry all the information about the individual variables; the assertion of independence is precisely that they carry *all* the information about the joint behaviour, since the joint law is then determined by them through the product. When the joint law is not a product, the difference $\mu_X - \bigotimes_i \mu_{X_i}$ measures exactly the dependence, and this deficit is the quantity later chapters learn to control. The equivalence also explains why so many independence computations reduce to checking a factorization on rectangles or half-lines: the π-system machinery of section 1.2 guarantees that agreement on that small class propagates to the full Borel σ-algebra.

One reading of the product-law identity is so useful it deserves separate billing: it lets probabilities of joint events be computed by iterated integration. The value of the product measure on a set $C \subseteq \mathbb{R}^n$ is, by the Fubini-Tonelli theorem of *Measure Theory* [2], an iterated integral of the indicator of C against the marginals one variable at a time. For independent variables, where the joint law *is* the product, this turns the abstract probability $\mathbb{P}((X_1, \dots, X_n) \in C)$ into a concrete nested integral against the individual laws. The statement below records the two-variable form, which is the version used constantly in the sequel; the n-variable form is identical with n nested integrals. Expectation is not yet available (it is the subject of chapter 3), so everything is phrased through laws and integrals against laws, never through an expectation operator.

Proposition 2.3.2: Fubini for Independent Variables

Let X, Y be independent random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ with laws μ_X, μ_Y on $\mathbb{R}$, and let (X, Y) have joint law $\mu_{(X,Y)} = \mu_X \otimes \mu_Y$ on $\mathbb{R}^2$ (theorem 2.3.1).

1. For every Borel set $C \subseteq \mathbb{R}^2$,

$$\mathbb{P}\big((X,Y) \in C\big) = \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_C(x,y)\, d\mu_Y(y)\, d\mu_X(x) = \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_C(x,y)\, d\mu_X(x)\, d\mu_Y(y)$$

   and in particular, for a product set $C = A \times B$, $\mathbb{P}(X \in A,\, Y \in B) = \mu_X(A)\, \mu_Y(B)$.

2. For every Borel function $h \colon \mathbb{R}^2 \to \mathbb{R}$ that is either nonnegative or bounded, the integral of h against the joint law factors as an iterated integral,

$$\int_{\mathbb{R}^2} h\, d\mu_{(X,Y)} = \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} h(x,y)\, d\mu_Y(y) \Big) d\mu_X(x) = \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} h(x,y)\, d\mu_X(x) \Big) d\mu_Y(y)$$

   each inner integrand being a Borel function of the free variable.

***Proof*:**
The laws μ_X and μ_Y are probability measures on $\mathbb{R}$, hence σ-finite, so the Fubini-Tonelli theorem of *Measure Theory* [2] applies to the product measure $\mu_X \otimes \mu_Y$ on $\mathbb{R}^2$. By theorem 2.3.1 the joint law $\mu_{(X,Y)}$ equals this product, so integration against $\mu_{(X,Y)}$ is integration against $\mu_X \otimes \mu_Y$.

For (2), the map $h \geq 0$ is a nonnegative Borel function on the product space, so Tonelli's half of the theorem gives that its inner integrals $x \mapsto \int_{\mathbb{R}} h(x,y)\, d\mu_Y(y)$ and $y \mapsto \int_{\mathbb{R}} h(x,y)\, d\mu_X(x)$ are Borel functions and that both iterated integrals equal $\int_{\mathbb{R}^2} h\, d(\mu_X \otimes \mu_Y)$. For bounded Borel h, write $h = h^+ - h^-$ with $h^{\pm} \geq 0$ bounded; each is integrable against the probability measure $\mu_X \otimes \mu_Y$ (its integral is at most $\sup |h| < \infty$), so Fubini's half applies to each and the iterated integrals of h agree with $\int_{\mathbb{R}^2} h \, d(\mu_X \otimes \mu_Y)$ by linearity.

Part (1) is the special case $h = \mathbb{1}_C$, a bounded (indeed $\{0,1\}$-valued) nonnegative Borel function, for which $\int_{\mathbb{R}^2} \mathbb{1}_C \, d\mu_{(X,Y)} = \mu_{(X,Y)}(C) = \mathbb{P}((X,Y) \in C)$ by the definition of the joint law. For $C = A \times B$ the inner integral collapses, $\int_{\mathbb{R}} \mathbb{1}_{A\times B}(x,y) \, d\mu_Y(y) = \mathbb{1}_A(x) \, \mu_Y(B)$, and integrating in x leaves $\mu_X(A) \, \mu_Y(B)$, completing the proof.

The proposition is the working form of Fubini in this book: a probability of a joint event, or an integral of a function of two independent variables, is computed by integrating out one variable and then the other, in either order. The recurring pattern in later chapters is to fix one variable, evaluate an inner integral against that variable's law (which often has a closed form), and then integrate the result against the second law. Two points are worth isolating. The freedom to integrate in either order is not decoration: many arguments choose the order that makes the inner integral tractable. And the hypothesis, nonnegative or bounded, is exactly what the Tonelli and Fubini halves respectively require; the general integrable case follows once expectation and L^1 are in place in chapter 3, but the nonnegative and bounded cases already cover every use in the immediate sequel, since indicators are bounded and densities are nonnegative.

The first substantial payoff is a formula for the law of a sum. When independent sources of randomness are added, their laws combine by an operation that mixes one law's mass across the other's, shifted by the running value. This operation is the convolution, and it is the reason the sum of independent variables is so much more tractable than a general function of them.

Definition 2.3.3: Convolution of Laws

Let μ, ν be Borel probability measures on $\mathbb{R}$. Their *convolution* $\mu * \nu$ is the pushforward of the product measure $\mu \otimes \nu$ under the addition map $s\colon \mathbb{R}^2 \to \mathbb{R}$, $s(x,y) = x + y$; explicitly, for every Borel set $A \subseteq \mathbb{R}$,

$$(\mu * \nu)(A) = (\mu \otimes \nu)\big(\{(x,y) \big| x + y \in A\}\big) = \int_{\mathbb{R}} \mu(A - y) \, d\nu(y) = \int_{\mathbb{R}} \nu(A - x) \, d\mu(x)$$

where $A - y = \{a - y \big| a \in A\}$. The convolution $\mu * \nu$ is again a Borel probability measure on $\mathbb{R}$.

That $\mu * \nu$ is a Borel probability measure is immediate: it is the pushforward of a probability measure under a Borel map, and pushforward preserves total mass. The two integral formulas are the two orders of iterated integration in proposition 2.3.2 applied to the indicator $\mathbb{1}_{\{x+y\in A\}}$: fixing y and integrating in x gives $\mu(A - y)$, and the outer integral in y then produces the first formula; the symmetric choice gives the second. Convolution is commutative and associative, inheriting both from the commutativity and associativity of addition together with the symmetry of the product measure, so the convolution $\mu_1 * \cdots * \mu_n$ of finitely many laws is unambiguous. The next result says this is precisely the law of a sum of independent variables.

Proposition 2.3.4: Law of a Sum is a Convolution

Let X, Y be independent random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ with laws μ_X, μ_Y. Then the law of the sum $X + Y$ is the convolution

$$\mu_{X+Y} = \mu_X * \mu_Y$$

and its distribution function (definition 1.3.1) is

$$F_{X+Y}(z) = \int_{\mathbb{R}} F_Y(z - x)\, d\mu_X(x) = \int_{\mathbb{R}} F_X(z - y)\, d\mu_Y(y)$$

***Proof*:**
The sum is the composition $X + Y = s \circ (X, Y)$ of the random vector (X, Y) with the addition map $s(x, y) = x + y$. Pushforward is functorial under composition, so the law of $X + Y$ is the pushforward of the joint law under s,

$$\mu_{X+Y} = s_* \, \mu_{(X,Y)} = s_*\big(\mu_X \otimes \mu_Y\big)$$

where the second equality is theorem 2.3.1, using independence. By definition 2.3.3 the pushforward $s_*(\mu_X \otimes \mu_Y)$ is exactly $\mu_X * \mu_Y$, giving $\mu_{X+Y} = \mu_X * \mu_Y$.

For the distribution function, take $A = (-\infty, z]$ in definition 2.3.3. Then $A - x = (-\infty, z - x]$, so $\mu_Y(A - x) = \mu_Y((-\infty, z - x]) = F_Y(z - x)$ by the definition of the distribution function, and the second integral formula of definition 2.3.3 (with the roles arranged so μ_Y is sliced) gives

$$F_{X+Y}(z) = \mu_{X+Y}\big((-\infty, z]\big) = \int_{\mathbb{R}} F_Y(z - x)\, d\mu_X(x)$$

The symmetric formula follows by exchanging the roles of X and Y, which is legitimate because convolution is commutative, completing the proof.

Convolution is the arithmetic of independent randomness: to add independent variables, convolve their laws. The distribution-function formula is the version one computes with when the laws have densities or are discrete, and it is worked out concretely below. The formula also exposes a structural fact used repeatedly: F_{X+Y} is an average of shifted copies of F_Y, weighted by the law of X, so adding an independent summand can only spread mass, never concentrate it further, a principle that returns in the central limit theorem where repeated convolution drives every law toward the Gaussian shape. When both laws have densities, the convolution of measures becomes the familiar convolution of functions, as the examples confirm.

Everything so far concerns finitely many variables. The limit theorems of the next Part are statements about *infinitely* many independent variables with a common law, and the final result of this section produces exactly that object. The input is a single Borel probability measure μ on $\mathbb{R}$, the common law each variable should have; the output is a sequence (X_n) of independent variables, each with law μ, all living on one probability space. The construction is the Kolmogorov extension of section 1.3, applied to countably many copies of the same factor.

Theorem 2.3.5: Existence of an i.i.d. Sequence

Let μ be any Borel probability measure on $\mathbb{R}$. There exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a sequence $(X_n)_{n\geq 1}$ of random variables that are independent (definition 2.2.2) and identically distributed, each with law $\mu_{X_n} = \mu$.

***Proof*:**
Take each factor to be a copy of $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$: set $\Omega_n = \mathbb{R}$, $\mathcal{F}_n = \mathcal{B}(\mathbb{R})$, and $\mathbb{P}_n = \mu$ for every $n \geq 1$. By the Kolmogorov extension theorem (theorem 1.3.5) there is a unique probability measure $\mathbb{P} = \bigotimes_n \mu$ on the product space

$$(\Omega, \mathcal{F}) = \Big(\mathbb{R}^{\mathbb{N}}, \bigotimes_{n\geq 1} \mathcal{B}(\mathbb{R})\Big)$$

whose value on every cylinder set (definition 1.3.4) is the finite product,

$$\mathbb{P}\big(\omega_1 \in B_1, \ldots, \omega_m \in B_m\big) = \prod_{i=1}^{m} \mu(B_i)$$

Let $X_n : \Omega \to \mathbb{R}$ be the n-th coordinate projection, $X_n(\omega) = \omega_n$. Each X_n is measurable, since $\{X_n \in B\} = \{\omega \big| \omega_n \in B\}$ is a cylinder for every Borel B, hence a random variable.

Identically distributed. For a Borel set $B \subseteq \mathbb{R}$ the event $\{X_n \in B\}$ is the one-coordinate cylinder constraining coordinate n to B, so $\mathbb{P}(X_n \in B) = \mu(B)$ by the product formula (with the single factor B and all others equal to $\mathbb{R}$). Thus $\mu_{X_n} = \mu$ for every n.

Independent. Fix a finite set of indices $n_1 < \cdots < n_k$ and Borel sets $B_1, \ldots, B_k$. The intersection $\bigcap_j \{X_{n_j} \in B_j\}$ is a single cylinder constraining exactly the coordinates $n_1, \ldots, n_k$ (all other coordinates free), so the product formula gives

$$\mathbb{P}\Big(\bigcap_{j=1}^{k} \{X_{n_j} \in B_j\}\Big) = \prod_{j=1}^{k} \mu(B_j) = \prod_{j=1}^{k} \mathbb{P}(X_{n_j} \in B_j)$$

The half-line sets $\{\{X_{n_j} \leq x\} \big| x \in \mathbb{R}\}$ are π-systems generating the $\sigma(X_{n_j})$ (example 1.2.7), and the display shows the product rule for all finite selections from them, so by the π-system criterion (theorem 2.2.4) the $\sigma(X_n)$ are independent. Hence (X_n) is an independent, identically distributed sequence with common law μ, as we sought to show.

The theorem removes the last existential worry from the limit theory to come. Every statement of the form "let (X_n) be i.i.d. with law μ", which opens nearly every theorem in Part I and beyond, is now backed by an actual probability space on which such a sequence lives; the coordinate realization even makes X_n a concrete function, the n-th coordinate of a random infinite sequence. This is the *canonical model* of an i.i.d. sequence: whatever the abstract space a modeller starts with, the joint behaviour of (X_n) is captured by its law on $\mathbb{R}^{\mathbb{N}}$, which is $\bigotimes_n \mu$, so nothing is lost by taking the coordinate model. The fair-coin-tossing space (example 1.3.6) is precisely this construction with $\mu = \mathrm{Ber}(1/2)$, the special case built by hand in section 1.3; the theorem generalizes it to an arbitrary common law.

With an infinite i.i.d. sequence in hand, the central object of the next Part can be named. Given i.i.d. variables $X_1, X_2, \dots$, the *partial sums*

$$S_0 = 0, \qquad S_n = X_1 + X_2 + \cdots + X_n \quad (n \geq 1)$$

form the *simple random walk* driven by the X_n. When the X_n take the values ± 1 with equal probability, obtained from the coin sequence by $X_n = 2\omega_n - 1$, the process (S_n) is the *simple symmetric random walk on* $\mathbb{Z}$: at each step it moves one unit right or left with equal probability, and S_n is its position after n steps. By proposition 2.3.4 the law of S_n is the n-fold convolution $\mu^{*n} = \mu * \cdots * \mu$ of the step law with itself, so the entire distributional evolution of the walk is governed by iterated convolution. This process, and the sums S_n for general step laws, is the running object of the limit theorems: the laws of large numbers describe the average S_n/n, and the central limit theorem describes the fluctuation $S_n/\sqrt{n}$. Its increments $X_n = S_n - S_{n-1}$ are independent by construction, which is what makes the asymptotic analysis possible.

The convolution formula is best understood through explicit computation. The following examples carry out the convolution in the three basic regimes, continuous, discrete, and the coin sequence, and identify the resulting laws.

Example 2.3.6: Convolution of Standard Laws

1. *Sum of two uniforms: the triangular law.* Let X, Y be independent, each with the uniform law $\mathrm{Unif}[0,1]$ (example 1.1.3), so $\mu_X = \mu_Y$ has density $f(x) = \mathbb{1}_{[0,1]}(x)$ with respect to Lebesgue measure. Their sum $Z = X + Y$ has law $\mu_X * \mu_Y$, and since both laws have densities the convolution has density

$$f_Z(z) = \int_{\mathbb{R}} f(x)\, f(z-x)\, dx = \int_{\mathbb{R}} \mathbb{1}_{[0,1]}(x)\, \mathbb{1}_{[0,1]}(z-x)\, dx$$

   The integrand is 1 exactly when $0 \leq x \leq 1$ and $0 \leq z - x \leq 1$, that is $\max(0, z-1) \leq x \leq \min(1, z)$, and 0 otherwise. The length of this interval is

$$f_Z(z) = \begin{cases} z, & 0 \leq z \leq 1 \\ 2 - z, & 1 \leq z \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

   the *triangular density* on $[0,2]$, rising linearly to its peak 1 at $z = 1$ and falling back. The single flat uniform, convolved with itself, already produces a peaked, tent-shaped law, the first visible step of the smoothing that convolution performs.

2. *Sum of independent Poissons is Poisson.* Let $X \sim \mathrm{Poi}(a)$ and $Y \sim \mathrm{Poi}(b)$ be independent, with laws supported on $\{0, 1, 2, \dots\}$ and mass functions $\mathbb{P}(X = j) = e^{-a}a^j/j!$ and $\mathbb{P}(Y = k) = e^{-b}b^k/k!$. For an integer $m \geq 0$ the convolution mass function is

$$\mathbb{P}(X + Y = m) = \sum_{j=0}^{m} \mathbb{P}(X = j)\, \mathbb{P}(Y = m - j) = \sum_{j=0}^{m} \frac{e^{-a}a^j}{j!} \cdot \frac{e^{-b}b^{m-j}}{(m-j)!}$$

   Pull out $e^{-(a+b)}$ and multiply and divide by $m!$,

$$\mathbb{P}(X + Y = m) = \frac{e^{-(a+b)}}{m!} \sum_{j=0}^{m} \binom{m}{j} a^j b^{m-j} = \frac{e^{-(a+b)}(a+b)^m}{m!}$$

using the binomial theorem. Thus $X + Y \sim \mathrm{Poi}(a + b)$: the Poisson family is closed under convolution, with the parameters adding.

3. *Sum of independent Bernoullis is binomial.* Let $X_1, \dots, X_n$ be independent, each $\mathrm{Ber}(p)$, so $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = 0) = 1 - p$. The sum $S_n = X_1 + \cdots + X_n$ counts the number of ones. For $0 \le k \le n$, the event $\{S_n = k\}$ is the disjoint union over the $\binom{n}{k}$ index sets of size k of the events fixing those coordinates to 1 and the rest to 0; by independence each such event has probability $p^k(1-p)^{n-k}$, so
$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$$
which is the mass function of the binomial law $\mathrm{Bin}(n, p)$. Equivalently, $\mathrm{Bin}(n, p)$ is the n-fold convolution $\mathrm{Ber}(p)^{*n}$, matching proposition 2.3.4: the law of a sum of i.i.d. steps is the iterated convolution of the step law.

4. *The coin sequence as the canonical i.i.d. sequence.* On the fair-coin-tossing space (example 1.3.6) the coordinates ω_n are, by theorem 2.3.5 with $\mu = \mathrm{Ber}(1/2)$, an i.i.d. sequence with common law $\mathrm{Ber}(1/2)$. The centered steps $X_n = 2\omega_n - 1 \in \{-1, +1\}$ are then i.i.d. with $\mathbb{P}(X_n = \pm 1) = 1/2$, and their partial sums $S_n = X_1 + \cdots + X_n$ are the simple symmetric random walk on $\mathbb{Z}$. By part (3), the number of heads $H_n = \omega_1 + \cdots + \omega_n$ in the first n tosses has law $\mathrm{Bin}(n, 1/2)$, and $S_n = 2H_n - n$ recovers the walk's position from the head count.

The four computations trace convolution across every regime the book uses. The uniform example shows the smoothing effect on densities; the Poisson and binomial examples show that the two standard discrete families are closed under convolution, the parameters combining additively; and the coin example ties the abstract i.i.d. construction back to the concrete running space of Part I, exhibiting the simple random walk as its centered partial sums. That $\mathrm{Bin}(n, p)$ is $\mathrm{Ber}(p)^{*n}$ and $\mathrm{Poi}(a) * \mathrm{Poi}(b) = \mathrm{Poi}(a + b)$ are the prototype facts the characteristic-function machinery of the next Part will later reprove in one line, but the direct convolution here is what makes the mechanism concrete.

Exercise 2.3.1

1. Let X and Y be independent random variables, each uniform on $\{1, 2, \dots, 6\}$ (a fair die). Using proposition 2.3.4, compute the mass function of $X + Y$ by convolving the two mass functions, and confirm that $\mathbb{P}(X + Y = 7) = 1/6$ is the largest value.

2. Let $X \sim \mathrm{Unif}[0, 1]$ and $Y \sim \mathrm{Unif}[0, 1]$ be independent. Use proposition 2.3.2(1) to compute $\mathbb{P}(X + Y \le 1)$ and $\mathbb{P}(X^2 + Y^2 \le 1)$ directly as double integrals against the two laws, and interpret each answer as an area.

3. Let $X \sim \mathrm{Poi}(a)$ and $Y \sim \mathrm{Poi}(b)$ be independent. Show that the conditional distribution of X given $X + Y = m$ is the binomial law $\mathrm{Bin}\left(m, \frac{a}{a+b}\right)$; that is, compute $\mathbb{P}(X = j \big| X + Y = m) = \mathbb{P}(X = j,\, X + Y = m)/\mathbb{P}(X + Y = m)$ using example 2.3.6(2). (Only elementary conditional probability $\mathbb{P}(A \big| B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ for events with $\mathbb{P}(B) > 0$ is meant here, not the conditional-expectation machinery developed later in the book.)

4. Let μ be a Borel probability measure on $\mathbb{R}$ and let (X_n) be the i.i.d. sequence with law μ furnished by theorem 2.3.5. Show that for any fixed permutation of finitely many indices, the reordered sequence has the same joint law as (X_n) (i.i.d. sequences are exchangeable).

Deduce that the law of the partial sum S_n is unchanged by permuting the summands.

5. Convolution has a unit and no nontrivial inverses among probability laws. Show that $\delta_0 * \mu = \mu$ for every law μ, where δ_0 is the point mass at 0, and that if $\mu * \nu = \delta_0$ then both μ and ν are point masses. (Hint for the second part: a sum of two independent variables can equal a constant only if each is almost surely constant; argue via the ranges of the two laws.)

6. Let X, Y be independent with $X \sim \mathrm{Unif}[0,1]$ and Y having density $g(y) = e^{-y}\mathbb{1}_{[0,\infty)}(y)$ (an exponential law). Using the density form of convolution, compute the density of $X + Y$ on the two ranges $0 \le z \le 1$ and $z \ge 1$, and check that the two pieces agree in value at $z = 1$.

2.4 The Tail Sigma-Algebra and Kolmogorov's Zero-One Law

The Borel-Cantelli lemmas of section 1.4 produced a dichotomy that now requires explanation. For an independent sequence of events, the probability that infinitely many occur is either 0 or 1, with nothing in between, and the sum of the probabilities decides which. The first lemma gave one half of the dichotomy unconditionally; the second gave the other half under independence. The suspicious feature is the dichotomy itself: why should $\{A_n \text{ i.o.}\}$ be forbidden the value 1/2? The answer is that this event belongs to a σ-algebra whose every member has probability 0 or 1, the *tail* σ-algebra of the sequence, and Kolmogorov's zero-one law is the theorem that pins this down. It is the first structural payoff of independence developed at the level of σ-algebras in section 2.2, and its reach extends well beyond Borel-Cantelli: any asymptotic feature of an independent sequence that survives altering finitely many terms is deterministic, even when the sequence itself is not.

The organizing idea is to isolate, inside the joint σ-algebra of a sequence, the events that do not depend on any finite block of coordinates. Changing $X_1, \dots, X_n$ for any n leaves such an event unaltered; only the "tail" of the sequence, its behaviour out at infinity, can affect it. This is exactly the kind of event whose probability the two Borel-Cantelli lemmas were computing.

Definition 2.4.1: The Tail Sigma-Algebra

Let $(X_n)_{n\ge1}$ be a sequence of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For each $n \ge 1$ write

$$\mathcal{T}_n = \sigma(X_n, X_{n+1}, \dots)$$

for the σ-algebra generated by the coordinates from index n onward (definition 2.1.5). The $\mathcal{T}_n$ decrease, $\mathcal{T}_1 \supseteq \mathcal{T}_2 \supseteq \cdots$, and their intersection

$$\mathcal{T} = \bigcap_{n\ge1} \mathcal{T}_n = \bigcap_{n\ge1} \sigma(X_n, X_{n+1}, \dots)$$

is the *tail* σ-algebra of the sequence. A set $T \in \mathcal{T}$ is a *tail event*, and a $\mathcal{T}$-measurable function is a *tail random variable.*

An intersection of σ-algebras is again a σ-algebra, so $\mathcal{T}$ is a σ-algebra; this is the one structural fact used without comment, and it is elementary (the defining closure properties are preserved under intersection). The content of the definition is the interpretation. A set T lies in $\mathcal{T}$ precisely when, for every n, membership of ω in T is determined by the values $X_n(\omega), X_{n+1}(\omega), \dots$ alone. Equivalently, T is unaffected by any change to a finite initial segment: if two outcomes agree from some index onward, either both lie in T or neither does. Deleting or overwriting $X_1, \dots, X_{n-1}$ costs nothing, because T already belongs to $\mathcal{T}_n$, which those coordinates do not generate. The tail σ-algebra is the mathematical home of the phrase "in the long run": it collects exactly the events and quantities that ignore any finite prefix of the sequence.

A word on why the decreasing family is intersected rather than unioned. Each $\mathcal{T}_n$ contains events that may still depend on the coordinate X_n; only by insisting that an event lie in *every* $\mathcal{T}_n$ is all dependence on any single coordinate stripped out at once. The intersection is the largest σ-algebra contained in all of them, hence the events immune to every finite truncation.

Example 2.4.2: Tail Events and a Non-Example
Fix a Borel set $B \in \mathcal{B}(\mathbb{R})$ and a sequence (X_n) of random variables.

1. *Infinitely many hits.* The event

$$\{X_n \in B \text{ i.o.}\} = \bigcap_{n\geq 1} \bigcup_{k\geq n} \{X_k \in B\}$$

in the sense of definition 1.4.1 is a tail event. For any fixed m, the inner union $\bigcup_{k\geq n}\{X_k \in B\}$ with $n \geq m$ involves only coordinates X_k with $k \geq m$, so the whole event lies in $\mathcal{T}_m$; as this holds for every m, it lies in $\mathcal{T}$. Concretely, whether $X_n \in B$ happens infinitely often cannot be changed by altering $X_1, \dots, X_{99}$: infinitude is decided by the tail, not by any finite block.

2. *Convergence of the series.* The event $\{\sum_n X_n \text{ converges}\}$ is a tail event. Convergence of a series is unaffected by discarding finitely many terms, since $\sum_{n\geq 1} X_n$ converges iff $\sum_{n\geq m} X_n$ converges for any (equivalently every) m. The latter event is measurable with respect to $X_m, X_{m+1}, \dots$, hence lies in $\mathcal{T}_m$ for every m, so it belongs to $\mathcal{T}$.

3. *Existence of the Cesàro limit.* The event

$$E = \left\{ \lim_{n\to\infty} \tfrac{1}{n} \textstyle\sum_{i=1}^n X_i \text{ exists} \right\}$$

is a tail event, and on E the limiting value is a tail random variable. For any m, altering $X_1, \dots, X_{m-1}$ changes the partial sum $\sum_{i=1}^n X_i$ by a fixed constant c_m independent of n, hence changes $\frac{1}{n}\sum_{i=1}^n X_i$ by $c_m/n \to 0$; the existence of the limit is untouched, and when the limit exists its value is unchanged. So $E \in \mathcal{T}_m$ for every m, and on E the map $\omega \mapsto \lim_n \frac{1}{n}\sum_{i=1}^n X_i(\omega)$ is measurable with respect to $X_m, X_{m+1}, \dots$ for every m, hence $\mathcal{T}$-measurable.

4. *A non-example.* The event $\{X_1 \in B\}$ is *not* a tail event (when it is nondegenerate). It lies in $\mathcal{T}_1 = \sigma(X_1, X_2, \dots)$, but not in $\mathcal{T}_2 = \sigma(X_2, X_3, \dots)$: knowing only $X_2, X_3, \dots$ says nothing about whether $X_1 \in B$. Its membership depends on precisely one coordinate, the first, and changing that coordinate flips it. Membership in a tail event must survive every such finite change, which this event does not.

The four items sort the world into two kinds of statement about a sequence: asymptotic statements, which are tail events, and finite-horizon statements, which are not. Every question that a limit theorem answers is of the first kind. Whether the averages $\frac{1}{n}S_n$ converge, whether a series of independent increments sums to a finite total, whether a rare event recurs forever: each is decided by the tail. That $\{X_1 \in B\}$ fails to be tail is not a defect but the point: it depends on finite information, and Kolmogorov's law will have nothing to say about it. The law speaks only to the asymptotic questions, and about those it speaks decisively.

There is one subtlety worth flagging before the main theorem. A tail event lies in the intersection $\mathcal{T}$, but the value of a tail random variable, such as the Cesàro limit in item (3), is defined only where the limit exists. On the complement the function may be set to any fixed convention (say 0); what matters is that on the set where it is defined it is measurable with respect to every $\mathcal{T}_n$. The zero-one law will force the set of existence to have probability 0 or 1, and where the limit exists it will be almost surely a single constant.

Everything so far used no independence. The definition of $\mathcal{T}$ and the classification of tail events hold for any sequence. Independence is what collapses the tail to the trivial dichotomy, and the mechanism is a self-referential independence argument of unusual economy. The plan is to show that $\mathcal{T}$ is independent of the σ-algebra generated by the whole sequence. Since $\mathcal{T}$ is contained in that σ-algebra, $\mathcal{T}$ ends up independent of itself, and a σ-algebra independent of itself can only contain trivial events. The engine is the π-system criterion from section 2.2, which reduces independence of σ-algebras to a product rule checked on generators.

Theorem 2.4.3: Kolmogorov's Zero-One Law

Let $(X_n)_{n\geq 1}$ be a sequence of *independent* random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ (definition 2.2.2), and let $\mathcal{T}$ be its tail σ-algebra (definition 2.4.1). Then:

1. every tail event $T \in \mathcal{T}$ has $\mathbb{P}(T) \in \{0, 1\}$; and
2. every tail random variable is almost surely constant: if $Y : \Omega \to \mathbb{R}$ (or to the extended reals) is $\mathcal{T}$-measurable, there is a constant $c \in [-\infty, \infty]$ with $\mathbb{P}(Y = c) = 1$.

***Proof*:**
Write $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ for the σ-algebra of the first n coordinates and $\mathcal{T}_n = \sigma(X_n, X_{n+1}, \dots)$ as in definition 2.4.1, so $\mathcal{T} = \bigcap_n \mathcal{T}_n$.

Step 1: The tail is independent of every finite prefix. Fix $n \geq 1$. The claim is that $\mathcal{F}_n$ and $\mathcal{T}$ are independent. Since $\mathcal{T} \subseteq \mathcal{T}_{n+1} = \sigma(X_{n+1}, X_{n+2}, \dots)$, it suffices to prove the stronger statement that $\mathcal{F}_n$ and $\mathcal{T}_{n+1}$ are independent: independence is inherited by sub-σ-algebras, being a restriction of the product rule to fewer sets, and $\mathcal{T} \subseteq \mathcal{T}_{n+1}$.

Apply the π-system criterion (theorem 2.2.4). A generating π-system for $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ is the collection

$$\mathcal{P} = \big\{\{X_1 \in B_1, \dots, X_n \in B_n\} \big| B_1, \dots, B_n \in \mathcal{B}(\mathbb{R})\big\}$$

of finite-dimensional cylinder events on the first n coordinates; it is closed under intersection (the intersection of two such events replaces each B_i by an intersection of Borel sets) and generates $\mathcal{F}_n$. Likewise, a generating π-system for $\mathcal{T}_{n+1}$ is the collection $\mathcal{Q}$ of events $\{X_{n+1} \in C_1, \dots, X_{n+m} \in C_m\}$ depending on finitely many coordinates $X_{n+1}, \dots, X_{n+m}$ (any $m \geq 1$, any Borel C_j); this too is closed under intersection and generates $\mathcal{T}_{n+1}$.

It remains to verify the product rule for one event from each system. Take $P = \{X_1 \in B_1, \dots, X_n \in B_n\} \in \mathcal{P}$ and $Q = \{X_{n+1} \in C_1, \dots, X_{n+m} \in C_m\} \in \mathcal{Q}$. The event $P \cap Q$ is $\{X_1 \in B_1, \dots, X_n \in B_n, X_{n+1} \in C_1, \dots, X_{n+m} \in C_m\}$, a cylinder event on the $n+m$ distinct coordinates $X_1, \dots, X_{n+m}$. By the independence of the sequence (X_j) (definition 2.2.2), the probability of any finite intersection of events, one from each distinct coordinate, factors over the coordinates, so

$$\mathbb{P}(P \cap Q) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i) \cdot \prod_{j=1}^{m} \mathbb{P}(X_{n+j} \in C_j) = \mathbb{P}(P)\,\mathbb{P}(Q)$$

the last equality regrouping the same factors as $\mathbb{P}(P) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i)$ and $\mathbb{P}(Q) = \prod_{j=1}^{m} \mathbb{P}(X_{n+j} \in C_j)$, again by independence of the coordinates. The product rule thus holds on $\mathcal{P} \times \mathcal{Q}$, and theorem 2.2.4 promotes it to independence of $\mathcal{F}_n$ and $\mathcal{T}_{n+1}$, hence of $\mathcal{F}_n$ and $\mathcal{T}$.

Step 2: The tail is independent of the whole sequence. Let $\mathcal{F}_\infty = \sigma(X_1, X_2, \dots) = \sigma\big(\bigcup_n \mathcal{F}_n\big)$. The union $\bigcup_n \mathcal{F}_n$ is a π-system: it is closed under intersection because the $\mathcal{F}_n$ increase, so any two of its members both lie in a common $\mathcal{F}_N$ (with N the larger index), whose closure under intersection contains the intersection. By Step 1, every $\mathcal{F}_n$ is independent of $\mathcal{T}$; hence the product rule $\mathbb{P}(F \cap T) = \mathbb{P}(F)\mathbb{P}(T)$ holds for every $F \in \bigcup_n \mathcal{F}_n$ and every $T \in \mathcal{T}$. Fixing $T \in \mathcal{T}$, the two set functions $F \mapsto \mathbb{P}(F \cap T)$ and $F \mapsto \mathbb{P}(F)\mathbb{P}(T)$ are measures on $\mathcal{F}_\infty$ (the first is $\mathbb{P}$ restricted to the trace on T, the second a scalar multiple of $\mathbb{P}$) that agree on the generating π-system $\bigcup_n \mathcal{F}_n$, which contains Ω. By the uniqueness of a measure determined on a generating π-system (corollary 1.2.6) they agree on all of $\mathcal{F}_\infty$. As T was arbitrary, $\mathcal{F}_\infty$ and $\mathcal{T}$ are independent.

Step 3: The tail is independent of itself. By construction $\mathcal{T} \subseteq \mathcal{F}_\infty$, so Step 2 says in particular that $\mathcal{T}$ is independent of $\mathcal{T}$. Applying the product rule to a tail event $T \in \mathcal{T}$ paired with itself gives

$$\mathbb{P}(T) = \mathbb{P}(T \cap T) = \mathbb{P}(T)\,\mathbb{P}(T) = \mathbb{P}(T)^2$$

The equation $p = p^2$ over the reals forces $p \in \{0, 1\}$, so $\mathbb{P}(T) \in \{0, 1\}$, which is part (1).

Step 4: Tail random variables are almost surely constant. Let Y be $\mathcal{T}$-measurable, valued in the extended reals. Its distribution function is $G(t) = \mathbb{P}(Y \le t)$. For each real t the event $\{Y \le t\}$ is a tail event, so by part (1) $G(t) \in \{0, 1\}$. The function G is nondecreasing and $\{0, 1\}$-valued on the reals; a nondecreasing $\{0, 1\}$-valued function on $\mathbb{R}$ has at most one jump, and the location of that jump (or its absence at either end) is recorded by

$$c = \inf\{t \in \mathbb{R} \big| G(t) = 1\} \in [-\infty, +\infty]$$

with the convention $\inf \emptyset = +\infty$. Three cases arise, and each fixes the almost-sure value of Y using only the values of G at finite t, never a limit at infinity.

If c is finite, then $\mathbb{P}(Y \le t) = 0$ for $t < c$ and $\mathbb{P}(Y \le t) = 1$ for $t \ge c$, the latter by right-continuity of G. Hence $\mathbb{P}(Y < c) = 0$ and $\mathbb{P}(Y \le c) = 1$, so $\mathbb{P}(Y = c) = 1$. If $c = +\infty$, then $G(t) = 0$ for every finite t, that is, $\mathbb{P}(Y \le t) = 0$ for all real t; the events $\{Y \le t\}$ increase to $\{Y < +\infty\}$ as $t \to +\infty$, so by continuity of $\mathbb{P}$ (proposition 1.1.5) $\mathbb{P}(Y < +\infty) = 0$, whence $\mathbb{P}(Y = +\infty) = 1$. Dually, if $c = -\infty$, then $G(t) = 1$ for every finite t; the events $\{Y \le t\}$ decrease to $\{Y = -\infty\}$ as $t \to -\infty$, so $\mathbb{P}(Y = -\infty) = 1$. In every case Y equals the single value $c \in [-\infty, +\infty]$ almost surely, completing the proof.

The proof turns on a single maneuver: a σ-algebra is shown independent of itself, which is only possible if it is trivial. Independence of $\mathcal{G}$ from $\mathcal{G}$ means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for all $A, B \in \mathcal{G}$, and taking $A = B$ forces $\mathbb{P}(A) = \mathbb{P}(A)^2$, hence $\mathbb{P}(A) \in \{0, 1\}$. The whole theorem is the observation that the tail σ-algebra of an independent sequence sits in this self-independent position, sandwiched between the finite prefixes it is independent of and the full σ-algebra those prefixes generate. The

π-system criterion of section 2.2 is what makes the sandwich rigorous: it lets independence be checked on the cylinder events, where the sequence's independence is literally the statement needed, and then extends automatically.

Two cautions frame the result. First, independence is essential, not decorative: for a dependent sequence the tail can be as large as one likes. If $X_n = X_1$ for all n, then every $\mathcal{T}_n$ contains $\sigma(X_1)$, so $\mathcal{T} = \sigma(X_1)$, and a tail event like $\{X_1 \leq 0\}$ can have any probability. Second, the law is a statement about probabilities, not about the events themselves: it does not say a tail event is empty or everything, only that its probability is 0 or 1. Which of the two holds is a separate computation, and supplying that computation is where Borel-Cantelli returns.

Example 2.4.4: Applications of the Zero-One Law

Let (X_n) be independent throughout.

1. *Recurrence of events and Borel-Cantelli.* Suppose $A_n \in \sigma(X_n)$ for each n, so the events A_n are independent and A_n depends only on the n-th coordinate. Then $\{A_n \text{ i.o.}\}$ is a tail event by the same reasoning as example 2.4.2(1), so $\mathbb{P}(A_n \text{ i.o.}) \in \{0, 1\}$ by theorem 2.4.3. The zero-one law guarantees the dichotomy; the two Borel-Cantelli lemmas *decide* it. If $\sum_n \mathbb{P}(A_n) < \infty$ then $\mathbb{P}(A_n \text{ i.o.}) = 0$ by the first lemma (theorem 1.4.5), and if $\sum_n \mathbb{P}(A_n) = \infty$ then $\mathbb{P}(A_n \text{ i.o.}) = 1$ by the second (theorem 1.4.6). The zero-one law thus explains the shape of the Borel-Cantelli dichotomy of section 1.4, and the lemmas fill in which side of it holds.

2. *Radius of convergence of a random power series.* Consider the random power series $\sum_n X_n z^n$ with independent coefficients (X_n), and its radius of convergence

$$R = \frac{1}{\limsup_{n\to\infty} |X_n|^{1/n}}$$

given by the Cauchy-Hadamard formula. The quantity $L = \limsup_n |X_n|^{1/n}$ is a tail random variable: altering $X_1, \ldots, X_{m-1}$ changes only finitely many terms of the sequence $|X_n|^{1/n}$ and so leaves its lim sup unchanged, whence L is $\mathcal{T}_m$-measurable for every m. By part (2) of theorem 2.4.3, L is almost surely a constant, and therefore the radius of convergence $R = 1/L$ is almost surely a constant. The circle of convergence of a random power series is deterministic, even though the series itself is random.

3. *The average is asymptotically deterministic.* Let $S_n = X_1 + \cdots + X_n$ be the partial sums (the simple random walk when the X_i are ± 1 coin flips, obtained from the coordinates ξ_n of example 1.3.6 by the rescaling $X_n = 2\xi_n - 1$). The quantity $\limsup_n \frac{1}{n} S_n$ is a tail random variable, by the constant-shift argument of example 2.4.2(3) applied to lim sup in place of lim: changing $X_1, \ldots, X_{m-1}$ adds a fixed c_m to each S_n, hence adds $c_m/n \to 0$ to $\frac{1}{n} S_n$ and leaves the lim sup untouched. By part (2) of theorem 2.4.3 it is almost surely a constant, as is the matching $\liminf_n \frac{1}{n} S_n$. So the averages $\frac{1}{n} S_n$ are trapped, almost surely, between two deterministic bounds. This is a foreshadow of the strong law of large numbers, which will identify both constants with the common mean of the X_i and thereby force $\frac{1}{n} S_n$ to converge to that mean almost surely.

The three applications trace an arc from the concrete to the structural. The first closes the loop with chapter 1: the mysterious rigidity of the Borel-Cantelli dichotomy is a consequence of the zero-one law, and the two lemmas are the computation that resolves the dichotomy the law imposes. The second shows the law acting on a continuous parameter, forcing an entire radius of convergence to be nonrandom. The third reduces the strong law of large numbers, still several chapters away, to the single quantity $\limsup_n \frac{1}{n} S_n$. The zero-one law does not evaluate this almost-sure constant, but it guarantees the value is a number rather than a distribution, and that reduction is the reason the strong law can be proved at all.

A closing perspective. Kolmogorov's law is the assertion that independence manufactures determinism out at infinity. Each individual trial is random, and any finite collection of them remains random, but the asymptotic verdicts, whether a rare event recurs forever, whether an average settles down, whether a random series converges, are settled in advance with probability one. The randomness lives entirely in the finite; the infinite is decided. This is the philosophical core of the limit theorems that occupy the next Part, and the tail σ-algebra is the precise object that separates the two regimes.

Exercise 2.4.1

1. Let $(X_n)_{n\geq 1}$ be any sequence of random variables (not assumed independent). Show that each of the following is a tail event, by exhibiting for every m a description of the event using only $X_m, X_{m+1}, \dots$:

    1. $\{\sup_n X_n = +\infty\}$;
    2. $\{\lim_n X_n \text{ exists in } \mathbb{R}\}$;
    3. $\{\sum_n X_n^2 \text{ converges}\}$.

    Then show that $\{\sup_n X_n > 5\}$ is in general *not* a tail event.

2. Let (X_n) be independent. Using theorem 2.4.3, show that the random variables $\limsup_n X_n$ and $\liminf_n X_n$ (valued in the extended reals) are each almost surely constant. Deduce that $\lim_n X_n$ exists almost surely if and only if these two constants coincide, in which case the limit is almost surely that common constant.

3. Consider the fair-coin-tossing space (example 1.3.6), with coordinates X_n taking values 0 and 1 each with probability 1/2, independently. Let A_n be the event that a run of n consecutive 1's begins at position n, that is $A_n = \{X_n = X_{n+1} = \cdots = X_{2n-1} = 1\}$. Show that $\{A_n \text{ i.o.}\}$ is a tail event, compute $\mathbb{P}(A_n)$, and use the Borel-Cantelli lemmas together with theorem 2.4.3 to decide whether $\mathbb{P}(A_n \text{ i.o.})$ is 0 or 1.

4. Give an explicit dependent sequence (X_n) for which the tail σ-algebra $\mathcal{T}$ contains an event of probability 1/2, thereby showing that the independence hypothesis in theorem 2.4.3 cannot be dropped.

5. Let (X_n) be independent, and let $\mathcal{T}$ be its tail σ-algebra. Show that for every tail event T and every event $F \in \sigma(X_1, X_2, \dots)$,

    $$\mathbb{P}(F \cap T) = \mathbb{P}(F)\,\mathbb{P}(T)$$

    and use this to give a one-line proof that $\mathbb{P}(T) \in \{0, 1\}$ by taking $F = T$. (This is the content of Steps 2 and 3 of the proof of theorem 2.4.3; the point of the exercise is to reproduce the self-independence maneuver in isolation.)

6. Let (X_n) be independent with each $X_n \geq 0$. Show that the event $\{\sum_n X_n < \infty\}$ has probability 0 or 1, and give an example of an independent sequence for which the probability is 0 and another for which it is 1. (You may use the Borel-Cantelli lemmas.)

3

Expectation

Chapters 1 and 2 built the objects of probability: the space, the random variable, its law, and independence. What has been missing is a way to attach to a random variable a single number summarizing its typical size. That number is the *expectation*, and it is nothing other than the Lebesgue integral of *Measure Theory* [2] against the probability measure $\mathbb{P}$, read with a probabilist's vocabulary: the average value of X over the outcomes, weighted by their probabilities. This chapter develops expectation and the inequalities it supports, the analytic results on which every limit theorem in the book depends.

Expectation is the integral, and its foundational properties, linearity, monotonicity, and the three convergence theorems, are those of the integral, cited to *Measure Theory* [2] and specialized to total mass one. section 3.1 fixes the definition and the change-of-variables formula that computes $\mathbb{E}[g(X)]$ from the law μ_X alone. Section 3.2 proves the inequalities that convert information about moments into control of probabilities: Markov, Chebyshev, Jensen, Hölder, and Minkowski. section 3.3 organizes the integrable variables into the spaces L^p, whose nesting on a probability space and whose L^2 Hilbert geometry recur throughout the subject. Section 3.4 closes with moments and the moment generating function, whose behaviour under sums of independent variables makes it the computational engine of the limit theorems and yields the first exponential tail bounds.

3.1 Expectation as an Integral

A random variable X records the outcome of a chance experiment as a number, and the previous chapter attached to it a law μ_X describing how that number is distributed. What is still missing is a single number summarizing the typical size of X: the value one would expect to see on average across many independent repetitions. That number is the *expectation* $\mathbb{E}[X]$, and it is not a new construction but an old one wearing probabilistic clothing. It is the Lebesgue integral of X against the probability measure $\mathbb{P}$, and every property it has, linearity, monotonicity, and its behaviour under limits, is a property of that integral specialized to a space of total mass one. This section fixes the definition, proves the change-of-variables formula that lets $\mathbb{E}[X]$ be computed from the law μ_X without any reference to the underlying space, and records the properties of the integral in the probabilistic idiom in which the rest of the book will use them.

The integral of *Measure Theory* [2] is built in three stages: first for simple functions (finite linear combinations of indicators), where the integral is the obvious weighted sum of measures; then for nonnegative measurable functions, by approximation from below; and finally for functions of either sign, by splitting into positive and negative parts and subtracting, provided the two pieces are not both infinite. Expectation inherits exactly this three-stage structure. For a nonnegative random variable the expectation always makes sense as a value in $[0, \infty]$; for a general random variable it is defined precisely when X is integrable, meaning $\mathbb{E}|X| < \infty$.

Definition 3.1.1: Expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let X be a random variable on it.

1. If $X \geq 0$, the *expectation* of X is the Lebesgue integral

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P} \in [0, \infty]$$

which is always defined, possibly equal to $+\infty$.

2. For a general (signed) random variable X, write $X = X^+ - X^-$ with $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$. If $\mathbb{E}|X| = \mathbb{E}[X^+] + \mathbb{E}[X^-] < \infty$, one says X is *integrable* and sets

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$$

If $\mathbb{E}|X| = \infty$ the expectation is left undefined (unless exactly one of $\mathbb{E}[X^+], \mathbb{E}[X^-]$ is infinite, in which case $\mathbb{E}[X]$ takes the value $+\infty$ or $-\infty$ accordingly).

The set of integrable random variables is denoted $L^1 = L^1(\Omega, \mathcal{F}, \mathbb{P})$.

The definition says nothing that the integral does not already say; its content is the reading. Think of $\mathbb{E}[X]$ as the probability-weighted average of the values $X(\omega)$ over the sample space: each outcome ω contributes its value $X(\omega)$, weighted by the infinitesimal probability $d\mathbb{P}(\omega)$, and the integral totals these contributions. For a simple random variable $X = \sum_i a_i \mathbb{1}_{A_i}$ taking finitely many values a_i on disjoint events A_i, this weighted average is visible directly: the definition unwinds to

$$\mathbb{E}[X] = \sum_i a_i \, \mathbb{P}(A_i)$$

the sum of each value times the probability of attaining it. The single most useful instance is the indicator itself. Taking $X = \mathbb{1}_A$ for an event $A \in \mathcal{F}$, so that X equals 1 on A and 0 off it, gives

$$\mathbb{E}[\mathbb{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A) \tag{3.1}$$

Expectation of an indicator is probability. This identity is the hinge between the two languages: every probability is an expectation, and every event-level statement can be rewritten as a statement about expectations of indicators. It is used so often that it is worth fixing in the reader's mind now.

A concrete instance grounds the definition immediately. On the fair coin-tossing space of example 1.3.6, let X_1 be the first coordinate coded as ± 1, so $X_1 = +1$ on the event "first toss is heads" and $X_1 = -1$

on its complement, each of probability 1/2. This is a simple random variable, and its expectation is

$$\mathbb{E}[X_1] = (+1) \cdot \tfrac{1}{2} + (-1) \cdot \tfrac{1}{2} = 0$$

The symmetric single step of the random walk has mean zero, as its symmetry requires. Fuller computations follow in example 3.1.4, once the change-of-variables formula makes them routine.

The definition computes $\mathbb{E}[X]$ as an integral over Ω, but in practice one never wants to integrate over the abstract sample space. All the probabilistic content of X lives in its law μ_X, the pushforward of $\mathbb{P}$ to $\mathbb{R}$ (definition 2.1.3), and one expects that $\mathbb{E}[X]$, and more generally the expectation of any function of X, should be computable from μ_X alone. The mechanism that makes this precise is the abstract change-of-variables formula for integration against a pushforward measure: integrating a function over the target of a measurable map, against the pushforward measure, equals integrating the composition over the source. Specialized to $X\colon \Omega \to \mathbb{R}$ and $\mathbb{P}$, it turns an integral over Ω into an integral over $\mathbb{R}$ against μ_X. In the probability literature the resulting formula is called the *law of the unconscious statistician*: to average $g(X)$ one may proceed "unconsciously," integrating g against the distribution of X without ever revisiting the sample space.

Theorem 3.1.2: Change of Variables (Law of the Unconscious Statistician)

Let X be a random variable with law μ_X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and let $g\colon \mathbb{R} \to \mathbb{R}$ be Borel measurable. Then $g(X)$ is a random variable, and:

1. If $g \geq 0$,
$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g \, d\mu_X \in [0, \infty]$$

2. For general g, the variable $g(X)$ is integrable if and only if $\int_{\mathbb{R}} |g| \, d\mu_X < \infty$, and in that case
$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g \, d\mu_X \tag{3.2}$$

In particular, taking $g(x) = x$ (the identity), X is integrable if and only if $\int_{\mathbb{R}} |x| \, d\mu_X(x) < \infty$, and then

$$\mathbb{E}[X] = \int_{\mathbb{R}} x \, d\mu_X(x) \tag{3.3}$$

***Proof*:**
First, $g(X) = g \circ X$ is a composition of the measurable maps $X\colon \Omega \to \mathbb{R}$ and $g\colon \mathbb{R} \to \mathbb{R}$, hence measurable, so it is a random variable and $\mathbb{E}[g(X)]$ is meaningful.

The identity $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g \, d\mu_X$ is the abstract change of variables for a pushforward measure, applied to the map X and the base measure $\mathbb{P}$. The law $\mu_X = X_*\mathbb{P}$ is by definition the pushforward, $\mu_X(B) = \mathbb{P}(X^{-1}(B))$ for $B \in \mathcal{B}(\mathbb{R})$ (definition 2.1.3), and integration against a pushforward satisfies

$$\int_{\mathbb{R}} g \, d(X_*\mathbb{P}) = \int_{\Omega} (g \circ X) \, d\mathbb{P}$$

for every nonnegative Borel g, and for every Borel g with $g \circ X$ integrable; this is the change-of-variables theorem for the image measure, proved in *Measure Theory* [2]. Its own proof is the standard measure-theoretic ascent, verified in turn for indicators (where it is the definition of the pushforward), extended by linearity to simple functions, raised to nonnegative g by the monotone convergence theorem, and finally to signed g by splitting into positive and negative parts.

The right side of that identity is $\int_{\mathbb{R}} g\,d\mu_X$, and the left side is $\int_\Omega g(X)\,d\mathbb{P} = \mathbb{E}[g(X)]$ by definition 3.1.1. This proves part (1) for $g \geq 0$. For general g, apply part (1) to $|g|$ to get $\mathbb{E}|g(X)| = \int_{\mathbb{R}} |g|\,d\mu_X$, so the two integrability conditions coincide; when they hold, split $g = g^+ - g^-$, apply part (1) to each nonnegative piece, and subtract to obtain (3.2). Taking $g(x) = x$ gives (3.3), completing the proof.

The formula relocates every expectation computation from the invisible space Ω to the concrete line $\mathbb{R}$, where the law μ_X is a genuine object one can write down and integrate against. Two specializations cover essentially every computation in practice, corresponding to the two standard shapes a law can take.

When X is discrete, taking values in a countable set $\{s_1, s_2, \dots\}$ with probability mass function $p_X(s) = \mathbb{P}(X = s)$, the law is the atomic measure $\mu_X = \sum_s p_X(s)\,\delta_s$, and integrating g against a sum of point masses is summation:

$$\mathbb{E}[g(X)] = \sum_s g(s)\,p_X(s) \tag{3.4}$$

valid whenever $g \geq 0$ or $\sum_s |g(s)|p_X(s) < \infty$. When instead X has a density f_X with respect to Lebesgue measure, meaning $\mu_X(B) = \int_B f_X\,dx$, integrating against μ_X becomes integrating against $f_X\,dx$:

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x)\,dx \tag{3.5}$$

under the same nonnegativity-or-integrability proviso. These two formulas, $\sum g(s)p_X(s)$ and $\int g(x)f_X(x)\,dx$, are the ones the reader will reach for; (3.2) is the theorem that certifies them and unifies the discrete and continuous cases under one statement.

Having pinned the definition and the computational formula, the properties that make expectation usable can be recorded. Every one of them is a property of the Lebesgue integral, cited to *Measure Theory* [2] and restated for random variables against a measure of total mass one. They are collected in a single proposition because they are used together and constantly: linearity and monotonicity are the algebra of expectation, and the three convergence theorems are the analysis, the results that license passing a limit through an expectation. The convergence theorems in particular are the technical core of every limit theorem to come, so their exact hypotheses deserve care.

Proposition 3.1.3: Foundational Properties of Expectation

Let X, Y and $X_1, X_2, \ldots$ be random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

1. *Linearity.* If $X, Y \in L^1$ and $a, b \in \mathbb{R}$, then $aX + bY \in L^1$ and

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$$

2. *Monotonicity.* If $X \le Y$ almost surely and both expectations exist (in particular if $X, Y \in L^1$), then $\mathbb{E}[X] \le \mathbb{E}[Y]$. In particular $X = 0$ a.s. implies $\mathbb{E}[X] = 0$, and if $X \ge 0$ a.s. with $\mathbb{E}[X] = 0$ then $X = 0$ a.s.
3. *Triangle inequality.* If $X \in L^1$ then $|\mathbb{E}[X]| \le \mathbb{E}|X|$.
4. *Monotone convergence.* If $0 \le X_1 \le X_2 \le \cdots$ almost surely and $X_n \uparrow X$ a.s., then

$$\mathbb{E}[X_n] \uparrow \mathbb{E}[X] \qquad \text{in } [0, \infty]$$

5. *Fatou's lemma.* If $X_n \ge 0$ almost surely for all n, then

$$\mathbb{E}\Big[\liminf_{n\to\infty} X_n\Big] \le \liminf_{n\to\infty} \mathbb{E}[X_n]$$

6. *Dominated convergence.* If $X_n \to X$ almost surely and there is an integrable $Y \in L^1$ with $|X_n| \le Y$ almost surely for all n, then $X \in L^1$, $\mathbb{E}|X_n - X| \to 0$, and in particular

$$\mathbb{E}[X_n] \to \mathbb{E}[X]$$

***Proof*:**
Each part is the corresponding property of the Lebesgue integral against the finite measure $\mathbb{P}$, established in *Measure Theory* [2]; what remains is to see that the probabilistic hypotheses match the measure-theoretic ones.

(1)-(3): Algebra. Linearity of the integral gives (1), noting that $|aX + bY| \le |a|\,|X| + |b|\,|Y|$ keeps $aX + bY$ integrable. Monotonicity (2) is monotonicity of the integral, valid "almost surely" because a property holding off a $\mathbb{P}$-null set does not affect the integral; the rigidity clause ($X \ge 0$ a.s. and $\mathbb{E}[X] = 0 \Rightarrow X = 0$ a.s.) is the standard fact that a nonnegative function of zero integral vanishes almost everywhere. The triangle inequality (3) is $-|X| \le X \le |X|$ fed through (2), giving $-\mathbb{E}|X| \le \mathbb{E}[X] \le \mathbb{E}|X|$.

(4)-(6): Analysis. These are the three convergence theorems. Part (4) is the monotone convergence theorem, part (5) is Fatou's lemma, and part (6) is the dominated convergence theorem, each cited to *Measure Theory* [2]. Two points of translation make the statements exact in the probabilistic setting. First, all hypotheses are stated "almost surely" rather than everywhere; since altering the X_n on a null set changes no expectation, one may pass to genuine everywhere statements by modifying each variable on a single null set, the countable union of which is again null. Second, the dominating hypothesis in (6) uses a fixed integrable envelope Y; on a probability space the simplest useful envelopes are the constants, since $\mathbb{P}(\Omega) = 1$ makes every bounded variable integrable, so a uniform bound $|X_n| \le M$ already supplies $Y \equiv M \in L^1$. The conclusion $\mathbb{E}|X_n - X| \to 0$ is convergence in L^1; it implies $\mathbb{E}[X_n] \to \mathbb{E}[X]$ by (3) applied to

$X_n - X$, as we sought to show.

The division of labour in this proposition is worth naming: parts (1) through (3) are the everyday algebra: linearity is what makes expectation tractable at all, letting one break a complicated variable into a sum and average the pieces separately, and it holds with no independence assumption whatsoever, a point easy to take for granted and important not to. Monotonicity and the triangle inequality are behind every inequality in the next section. Parts (4) through (6) are the reason expectation interacts well with limits, and they are not interchangeable: monotone convergence needs monotonicity but no domination and allows the value $+\infty$; Fatou needs only nonnegativity but yields an inequality, not an equality; dominated convergence gives the full equality but requires an integrable envelope. Knowing which of the three applies in a given situation is a large part of the working analyst's craft, and the coming limit theorems will use all three.

A caution accompanies part (6). The integrable envelope is not a technicality that can be dropped. Without it, an expectation can fail to pass to the limit even when $X_n \to X$ everywhere: on $\Omega = (0, 1)$ with Lebesgue measure (the uniform law of example 1.1.3), the variables $X_n = n\,\mathbb{1}_{(0,1/n)}$ satisfy $X_n \to 0$ at every point, yet $\mathbb{E}[X_n] = 1$ for all n, so $\mathbb{E}[X_n] \to 1 \neq 0 = \mathbb{E}[\lim X_n]$. The mass escapes to infinity through an ever-thinner spike; no integrable Y dominates the whole sequence, and the conclusion of dominated convergence indeed fails. This is exactly the phenomenon the envelope is there to forbid.

With the machinery assembled, the standard distributions can have their means computed once and for all. The following example works through the discrete and continuous computational formulas on the distributions the reader met in example 2.1.7, and closes with a formula, expectation as the integral of a tail probability, that is used in many moment estimates.

Example 3.1.4: Expectations of the Standard Distributions

Throughout, the change-of-variables formula theorem 3.1.2 in its discrete form (3.4) and density form (3.5) does all the work.

1. *Indicator.* For an event A, the variable $\mathbb{1}_A$ is Bernoulli with $p = \mathbb{P}(A)$, and (3.1) gives $\mathbb{E}[\mathbb{1}_A] = \mathbb{P}(A)$ directly. This is the anchor case; everything else is built from it by linearity.
2. *Bernoulli.* Let $X \sim \mathrm{Ber}(p)$, so $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. The discrete formula (3.4) with g the identity gives
$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p$$
The mean of an indicator-type variable is the success probability, as it must be, since $X = \mathbb{1}_{\{X=1\}}$.
3. *Uniform on* $[0, 1]$. Let $U \sim \mathrm{Unif}[0, 1]$, the variable of example 2.1.8, with density $f_U = \mathbb{1}_{[0,1]}$. The density formula (3.5) gives
$$\mathbb{E}[U] = \int_{\mathbb{R}} x\, f_U(x)\, dx = \int_0^1 x\, dx = \frac{1}{2}$$
The average value of a point dropped uniformly on the unit interval is its midpoint, matching intuition exactly.
4. *Poisson.* Let $X \sim \mathrm{Poi}(\lambda)$, so $p_X(k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$. The discrete formula

gives, after cancelling the $k = 0$ term and reindexing $j = k - 1$,

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k\, e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} = \lambda\, e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda\, e^{-\lambda} e^{\lambda} = \lambda$$

The parameter λ is the mean of the Poisson law, which is why it is often introduced as the "rate" or "expected count."

The Poisson computation exhibits a small trick, factoring one power of λ out to reindex the series into the exponential's Taylor expansion, that recurs whenever one differentiates or sums against a Poisson weight. For a nonnegative variable there is a second route to the mean that bypasses summation and integration against the law entirely, expressing $\mathbb{E}[X]$ as the total mass under the tail-probability curve $t \mapsto \mathbb{P}(X > t)$. This tail formula converts questions about the mean into questions about the probability of exceeding a threshold, which is often the quantity one controls.

Proposition 3.1.5: Tail-Integral Formula for the Mean

If $X \geq 0$ is a nonnegative random variable, then

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X > t)\, dt \tag{3.6}$$

both sides lying in $[0, \infty]$ and being simultaneously finite or infinite. The same holds with $\mathbb{P}(X > t)$ replaced by $\mathbb{P}(X \geq t)$, the two integrals agreeing because they differ only on the at most countable set of atoms of X.

***Proof*:**
The identity is a single application of Tonelli's theorem, which permits interchanging a nonnegative integral over Ω with one over $(0, \infty)$ against Lebesgue measure without any integrability hypothesis; this is the nonnegative half of the Fubini-Tonelli theorem of *Measure Theory* [2], and it is available precisely because the integrand below is nonnegative.

The mechanism is the "layer-cake" identity: a nonnegative number is the length of the interval below it, so for each fixed ω,

$$X(\omega) = \int_0^{X(\omega)} dt = \int_0^{\infty} \mathbb{1}_{\{t < X(\omega)\}}\, dt$$

Integrate both sides over Ω against $\mathbb{P}$ and exchange the order of the two integrals. The integrand $(\omega, t) \mapsto \mathbb{1}_{\{t < X(\omega)\}}$ is nonnegative and jointly measurable, so Tonelli applies:

$$\mathbb{E}[X] = \int_{\Omega} \left(\int_0^{\infty} \mathbb{1}_{\{t < X(\omega)\}}\, dt \right) d\mathbb{P}(\omega) = \int_0^{\infty} \left(\int_{\Omega} \mathbb{1}_{\{t < X(\omega)\}}\, d\mathbb{P}(\omega) \right) dt$$

The inner integral over Ω is the expectation of an indicator, and by (3.1),

$$\int_{\Omega} \mathbb{1}_{\{t < X(\omega)\}}\, d\mathbb{P}(\omega) = \mathbb{P}(X > t)$$

Substituting gives (3.6). Finally, $\{X \geq t\}$ and $\{X > t\}$ differ only when t is an atom of the law of X, and a nonnegative variable has at most countably many atoms, a Lebesgue-null set of thresholds, so replacing $>$ by $\geq$ leaves the integral unchanged, as we sought to show.

The tail formula is the first place where the layer-cake decomposition appears, and it is a template worth remembering: any nonnegative quantity built by integrating over its own sublevel sets can be turned inside out by Tonelli. The same idea, applied to $|X|^p$ in place of X, will express higher moments through tail probabilities in the next section, and the exchange of order it relies on is the reason the nonnegativity hypothesis cannot be dropped.

A single interpretive point ties the section together and clarifies what expectation "sees." The change-of-variables formula theorem 3.1.2 computes $\mathbb{E}[g(X)]$ from the law μ_X alone, with no further reference to the probability space on which X is defined. Two random variables that share a law, even ones living on entirely different sample spaces, therefore share every expectation of every function of themselves.

Proposition 3.1.6: Expectation Depends Only on the Law

If X and Y are random variables (possibly on different probability spaces) with the same law, $\mu_X = \mu_Y$, then for every Borel $g\colon \mathbb{R} \to \mathbb{R}$ with $g \geq 0$ or $g(X) \in L^1$,

$$\mathbb{E}[g(X)] = \mathbb{E}[g(Y)]$$

In particular equal-in-distribution variables have equal mean whenever the mean exists.

***Proof*:**
By theorem 3.1.2, $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g\, d\mu_X$ and $\mathbb{E}[g(Y)] = \int_{\mathbb{R}} g\, d\mu_Y$. Since $\mu_X = \mu_Y$ by hypothesis, the two integrals are literally the same integral, so the two expectations are equal, and the integrability condition $\int_{\mathbb{R}} |g|\, d\mu_X < \infty$ that governs one governs the other identically, completing the proof.

This completes what the previous chapter began. There it was seen that the law captures everything probabilistically observable about a random variable and that the sample space is an auxiliary choice one is free to discard; here the same fact is seen at the level of averages. Expectation is a functional of the distribution, not of the outcome-level description, and the probabilist may, and routinely does, replace X by any convenient variable with the same law without changing a single expectation. This freedom, to build a variable on whatever space is convenient and read its expectations off its law, underlies the constructions of independent sequences in the previous chapter and every distributional argument in the chapters ahead.

Exercise 3.1.1

1. Let X take the values $-1, 0, 2$ with probabilities $1/2, 1/4, 1/4$ respectively. Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathbb{E}[|X|]$ directly from the discrete change-of-variables formula (3.4). Verify that $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$.
2. Let $X \sim \mathrm{Poi}(\lambda)$. Using the reindexing trick from example 3.1.4, compute $\mathbb{E}[X(X-1)]$, and deduce $\mathbb{E}[X^2]$. (This is the second factorial moment; it computes the Poisson variance in the next section without a fresh series manipulation.)
3. Let $U \sim \mathrm{Unif}[0,1]$ (example 2.1.8). Compute $\mathbb{E}[U^n]$ for each integer $n \geq 0$ using (3.5), and separately recompute $\mathbb{E}[U]$ from the tail-integral formula (3.6), checking the two agree.
4. Let X be geometric with $\mathbb{P}(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, 3, \ldots$, where $0 < p < 1$. Use

the tail-integral formula in the form $\mathbb{E}[X] = \sum_{k \geq 1} \mathbb{P}(X \geq k)$ (the integer analogue of (3.6) for a $\mathbb{Z}_{\geq 0}$-valued variable) to show $\mathbb{E}[X] = 1/p$, avoiding any differentiation of a power series.

5. Give a nonnegative random variable X on $([0,1], \mathcal{B}, \mathrm{Leb})$ with $\mathbb{E}[X] = \infty$, and confirm via the tail-integral formula that $\int_0^\infty \mathbb{P}(X > t)\, dt$ diverges. (Take a variable with a heavy tail, e.g. a suitable power of $1/U$ for $U \sim \mathrm{Unif}[0,1]$.)

6. Fix $c > 0$ and let $X \geq 0$. Prove the truncated tail identity $\mathbb{E}[\min(X, c)] = \int_0^c \mathbb{P}(X > t)\, dt$ by the same layer-cake argument as in proposition 3.1.5. Explain why this shows $\mathbb{E}[\min(X, c)]$ is always finite and increases to $\mathbb{E}[X]$ as $c \to \infty$, and name the convergence theorem that justifies the last step.

3.2 Inequalities

Expectation attaches to a random variable a single number, its average. The value of that number is that it can be leveraged: knowing $\mathbb{E}[X]$, or more generally a few moments $\mathbb{E}[X^k]$, one can bound the probability that X strays from its average. This is the mechanism that makes the limit theorems possible. A law of large numbers asserts that a sample average is close to the true mean with high probability; the only way to prove such a statement from moment data is an inequality that turns "the second moment of the fluctuation is small" into "the fluctuation is rarely large". This section assembles the standard inequalities of the subject. Markov's inequality is the seed from which the others grow: Chebyshev's inequality is Markov applied to the squared fluctuation, and the Chernoff bounds of a later section will be Markov applied to an exponential. Jensen's inequality relates the expectation of a convex function to the convex function of the expectation, and the pair of Hölder and Minkowski inequalities govern the products and sums of random variables in L^p. Together they are the analytic results on which every asymptotic argument in the book rests.

We begin with the quantity these inequalities most often control, the variance. The expectation locates the center of a distribution; it says nothing about how widely the distribution is spread around that center. The natural measure of spread is the average squared deviation from the mean, and squaring rather than taking absolute values is what makes the quantity algebraically tractable and ties it to the L^2 geometry developed in the next section.

Definition 3.2.1: Variance, Covariance, and Correlation

Let X be a random variable with $\mathbb{E}[X^2] < \infty$, so that X is integrable and $\mathbb{E}[X]$ is finite. The *variance* of X is

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}X)^2\big]$$

a nonnegative real number, and the *standard deviation* of X is $\sigma(X) = \sqrt{\mathrm{Var}(X)}$. For two random variables X, Y with finite second moments, the *covariance* is

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}X)(Y - \mathbb{E}Y)\big]$$

and, when both variances are strictly positive, the *correlation coefficient* is

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\,\sigma(Y)}$$

Two random variables are *uncorrelated* when $\mathrm{Cov}(X, Y) = 0$.

The requirement $\mathbb{E}[X^2] < \infty$ is exactly what makes every quantity in the definition finite. On a probability space the second moment dominates the first: by the change-of-variables formula (theorem 3.1.2) applied to $g(x) = |x|$ and $g(x) = x^2$, together with $|x| \leq 1 + x^2$, the finiteness of $\mathbb{E}[X^2]$ forces $\mathbb{E}|X| < \infty$, so $\mathbb{E}[X]$ is a genuine real number and the centered variable $X - \mathbb{E}X$ is well defined. The covariance measures the tendency of the fluctuations of X and Y to move together: it is positive when X above its mean tends to accompany Y above its mean, negative when they move oppositely, and zero when there is no linear association. The correlation coefficient normalizes this to the interval $[-1, 1]$, a bound that is the Cauchy-Schwarz inequality in disguise (corollary 3.2.11); it strips out the units and scales of X and Y, reporting only the strength of the linear relationship.

Expanding the square turns the variance into a difference of moments, which is almost always the form used in computation.

Proposition 3.2.2: Computational Identities for Variance

Let X, Y have finite second moments and let $a, b \in \mathbb{R}$. Then

1. $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$;
2. $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$;
3. $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$, and $\mathrm{Var}(X) = \mathrm{Cov}(X, X)$;
4. $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X, Y) + \mathrm{Var}(Y)$.

***Proof*:**
Write $\mu = \mathbb{E}[X]$ and $\nu = \mathbb{E}[Y]$; these are finite as noted above. All four identities are the linearity of expectation (proposition 3.1.3) applied after multiplying out a product.

(1) Variance as a difference of moments. Expanding the square,

$$(X - \mu)^2 = X^2 - 2\mu X + \mu^2$$

Taking expectations and using linearity, together with $\mathbb{E}[\mu X] = \mu\,\mathbb{E}[X] = \mu^2$ and $\mathbb{E}[\mu^2] = \mu^2$

since μ is a constant,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - 2\mu\,\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2$$

which is (1). In particular $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2 \geq 0$ recovers the inequality $(\mathbb{E}X)^2 \leq \mathbb{E}[X^2]$.

(2) Behaviour under affine maps. Since $\mathbb{E}[aX + b] = a\mu + b$ by linearity, the centered variable is $aX + b - (a\mu + b) = a(X - \mu)$, and

$$\mathrm{Var}(aX + b) = \mathbb{E}\big[a^2(X - \mu)^2\big] = a^2\mathrm{Var}(X)$$

The additive constant b shifts the whole distribution and cannot change its spread; the multiplicative constant a rescales deviations and so rescales the variance by a^2.

(3) The covariance identity. Expanding the product,

$$(X - \mu)(Y - \nu) = XY - \nu X - \mu Y + \mu\nu$$

and taking expectations gives $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \nu\mu - \mu\nu + \mu\nu = \mathbb{E}[XY] - \mu\nu$. Setting $Y = X$ recovers (1) and shows $\mathrm{Var}(X) = \mathrm{Cov}(X, X)$.

(4) Variance of a sum. By bilinearity of the covariance, which is immediate from (3) and linearity of expectation,

$$\mathrm{Var}(X + Y) = \mathrm{Cov}(X + Y, X + Y) = \mathrm{Cov}(X, X) + 2\mathrm{Cov}(X, Y) + \mathrm{Cov}(Y, Y)$$

which is the stated identity, completing the proof.

Identity (1) is the one used most: variance is the second moment minus the square of the first. Identity (2) records the two ways constants act, a shift leaving the spread unchanged and a scaling multiplying it by the square of the factor. Identity (4) exposes the covariance as exactly the obstruction to variance being additive: the variance of a sum is the sum of the variances plus twice the covariance, so variances add precisely when the cross term vanishes, that is, when the summands are uncorrelated. The next section shows that independent variables are uncorrelated (proposition 3.3.7), which is what makes the variance of a sum of independent variables so easy to compute and drives the weak law of large numbers.

Example 3.2.3: Variance of a Bernoulli and of a Uniform

Two computations ground the definition.

A Bernoulli variable. Let $X \sim \mathrm{Ber}(p)$, so $X = 1$ with probability p and $X = 0$ with probability $1 - p$. Since $X^2 = X$ (the values $0, 1$ are fixed by squaring), $\mathbb{E}[X^2] = \mathbb{E}[X] = p$, and by proposition 3.2.2(1),

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2 = p - p^2 = p(1 - p)$$

The variance is largest at $p = 1/2$, where the outcome is most uncertain, and vanishes at $p \in \{0, 1\}$, where X is constant. This is the variance of a single fair or biased coin flip.

A uniform variable. Let $U \sim \mathrm{Unif}[0, 1]$, the coordinate on the unit-interval space (example 2.1.8). Then $\mathbb{E}[U] = 1/2$ and, by theorem 3.1.2 with $\mu_U = \lambda$ on $[0, 1]$,

$$\mathbb{E}[U^2] = \int_0^1 u^2\,du = \frac{1}{3}$$

so $\mathrm{Var}(U) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$, and the standard deviation is $1/\sqrt{12} \approx 0.289$. The spread is a fixed fraction of the length of the interval, as it must be by the affine-scaling identity proposition 3.2.2(2).

With the variance in hand, the first inequality controls how far a random variable can stray from its mean. The most basic question one can ask about a nonnegative random variable is how often it can be large. If the average $\mathbb{E}[X]$ is small, then X cannot often be large, for a single large value pulls the average up. Markov's inequality makes this quantitative, and it does so with no hypothesis beyond nonnegativity and a finite mean. Its proof is a single line, yet it is the source of nearly every concentration bound in probability.

Theorem 3.2.4: Markov's Inequality

Let X be a random variable and $a > 0$. Then

$$\mathbb{P}(|X| \geq a) \leq \frac{\mathbb{E}|X|}{a}$$

In particular, if $X \geq 0$ then $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$.

***Proof*:**
Fix $a > 0$. The pointwise inequality

$$a\,\mathbb{1}_{\{|X| \geq a\}} \leq |X|$$

holds for every outcome ω: on the event $\{|X| \geq a\}$ the left side is a and the right side is $|X| \geq a$, while off the event the left side is 0 and the right side is nonnegative. Taking expectations and using monotonicity and linearity (proposition 3.1.3), together with $\mathbb{E}[\mathbb{1}_{\{|X|\geq a\}}] = \mathbb{P}(|X| \geq a)$ from definition 3.1.1,

$$a\,\mathbb{P}(|X| \geq a) = \mathbb{E}\big[a\,\mathbb{1}_{\{|X|\geq a\}}\big] \leq \mathbb{E}|X|$$

Dividing by $a > 0$ gives the inequality. The final claim is the case $X \geq 0$, where $|X| = X$, completing the proof.

The proof is worth pausing over, because the same three-step pattern recurs throughout the subject: bound the indicator of a bad event by a nonnegative function that is large on the event, take expectations, and read off the tail bound. The inequality is sharp in the sense that equality can hold, for instance when X takes only the values 0 and a; no better bound is available from the mean alone. Its weakness is precisely that it uses only the mean: it decays like $1/a$, which is slow. To obtain faster decay one feeds Markov a random variable built to grow quickly, and the first such choice is the squared fluctuation, which yields Chebyshev's inequality.

Corollary 3.2.5: Chebyshev's Inequality

Let X have finite second moment and let $a > 0$. Then

$$\mathbb{P}\big(|X - \mathbb{E}X| \geq a\big) \leq \frac{\mathrm{Var}(X)}{a^2}$$

***Proof*:**
The event $\{|X - \mathbb{E}X| \geq a\}$ equals the event $\{(X - \mathbb{E}X)^2 \geq a^2\}$, since squaring is monotone on nonnegative numbers and $a > 0$. Apply Markov's inequality (theorem 3.2.4) to the nonnegative variable $Y = (X - \mathbb{E}X)^2$ with threshold a^2:

$$\mathbb{P}\big(|X - \mathbb{E}X| \geq a\big) = \mathbb{P}(Y \geq a^2) \leq \frac{\mathbb{E}[Y]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

where the last equality is the definition of the variance (definition 3.2.1), as we sought to show.

Chebyshev's inequality is the first genuine concentration bound: it says a random variable is within a of its mean except on an event of probability at most $\mathrm{Var}(X)/a^2$. Written with $a = k\sigma$ a multiple of the standard deviation, it reads $\mathbb{P}(|X - \mathbb{E}X| \geq k\sigma) \leq 1/k^2$, so at least 75% of the mass lies within two standard deviations and at least 89% within three, irrespective of the distribution. The decay is now $1/a^2$, quadratically faster than Markov, gave with the second moment rather than the first. This is exactly the trade that powers the weak law of large numbers: a sample average has a variance that shrinks like $1/n$, and Chebyshev converts that shrinkage into convergence in probability, as the application at the end of the section shows.

Example 3.2.6: Chebyshev for a Fair Coin Count
Let S_n count the heads in n tosses of a fair coin, so $S_n \sim \mathrm{Bin}(n, 1/2)$, with $\mathbb{E}[S_n] = n/2$ and, anticipating the additivity of variance for independent summands (proposition 3.3.7), $\mathrm{Var}(S_n) = n \cdot \frac{1}{4} = n/4$. Chebyshev's inequality with $a = \varepsilon n$ bounds the probability that the head fraction S_n/n deviates from $1/2$ by more than ε:

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| \geq \varepsilon\right) = \mathbb{P}\big(|S_n - \mathbb{E}S_n| \geq \varepsilon n\big) \leq \frac{\mathrm{Var}(S_n)}{(\varepsilon n)^2} = \frac{n/4}{\varepsilon^2 n^2} = \frac{1}{4\varepsilon^2 n}$$

For $\varepsilon = 0.1$ and $n = 10\,000$ this is at most $1/400 = 0.0025$: the fraction of heads is within 0.1 of $1/2$ with probability at least 0.9975. The bound tends to 0 as $n \to \infty$, which is the weak law of large numbers for the fair coin.

Markov and Chebyshev bound how far a random variable can wander. Jensen's inequality controls a different interaction, that between expectation and a nonlinear transformation. Expectation is linear, so it commutes with affine functions: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$. It does not commute with a general nonlinear function; but when the function is convex, the two operations relate by a definite inequality. The geometric content is that a convex function lies above each of its tangent lines, and averaging respects that one-sided bound. This is the inequality behind the ordering of the L^p norms, the arithmetic-geometric mean inequality, and countless a-priori estimates.

Recall that a function $\varphi \colon I \to \mathbb{R}$ on an interval $I \subseteq \mathbb{R}$ is *convex* if for all $x, y \in I$ and $t \in [0, 1]$,

$$\varphi\big(tx + (1-t)y\big) \leq t\,\varphi(x) + (1-t)\,\varphi(y)$$

that is, the graph of φ over any chord lies below the chord. The one fact about convex functions used below is the existence of supporting lines: at every interior point m of I there is a slope c with $\varphi(x) \geq \varphi(m) + c(x - m)$ for all $x \in I$. Such a c exists because the left and right difference quotients of a convex function are monotone, so the one-sided derivatives at m exist and any c between them supports the graph; the verification is a fact about convex functions on the line, cited to *Measure Theory* [2].

Theorem 3.2.7: Jensen's Inequality

Let $\varphi\colon I \to \mathbb{R}$ be convex on an interval $I \subseteq \mathbb{R}$, and let X be an integrable random variable with $\mathbb{P}(X \in I) = 1$ and $\mathbb{E}[X] \in I^\circ$, the interior of I. If $\varphi(X)$ is integrable, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$$

***Proof*:**
Write $m = \mathbb{E}[X]$, which lies in the interior I° by hypothesis. Let c be the slope of a supporting line to φ at m, so that

$$\varphi(x) \geq \varphi(m) + c\,(x - m) \qquad \text{for all } x \in I \tag{3.7}$$

Since $X \in I$ almost surely, substituting $x = X(\omega)$ into (3.7) gives the pointwise inequality

$$\varphi(X) \geq \varphi(m) + c\,(X - m)$$

valid almost surely. Both sides are integrable: $\varphi(X)$ by hypothesis, and the right side because X is integrable and $\varphi(m), c, m$ are constants. Taking expectations and using monotonicity and linearity (proposition 3.1.3),

$$\mathbb{E}[\varphi(X)] \geq \varphi(m) + c\,\big(\mathbb{E}[X] - m\big) = \varphi(m)$$

since $\mathbb{E}[X] - m = 0$ by the choice $m = \mathbb{E}[X]$. As $\varphi(m) = \varphi(\mathbb{E}X)$, this is the claim, as we sought to show.

The proof is the whole idea in one picture: replace the curve φ by its tangent line at the mean, a line that lies below φ everywhere and touches it at $m = \mathbb{E}[X]$. Expectation of a linear function is trivial to compute, and because the line sits below the curve pointwise, its expectation, which equals its value at the mean, sits below $\mathbb{E}[\varphi(X)]$. For a concave function the inequality reverses, since $-\varphi$ is then convex; the two directions are used constantly. A useful mnemonic: the expectation of the transform is at least the transform of the expectation, so convexity "inflates" averages.

Example 3.2.8: Jensen for Squares, Exponentials, and Logarithms
Three standard convex or concave functions illustrate the reach of the inequality.

The square. With $\varphi(x) = x^2$, convex on all of $\mathbb{R}$, Jensen gives $(\mathbb{E}X)^2 \leq \mathbb{E}[X^2]$, which is the nonnegativity of the variance (proposition 3.2.2(1)) seen from the convexity side. The gap between the two sides is exactly $\operatorname{Var}(X)$.

The exponential. With $\varphi(x) = e^x$, convex on $\mathbb{R}$, Jensen gives $e^{\mathbb{E}[X]} \leq \mathbb{E}[e^X]$. Applied to $X = \log Y$ for a positive variable Y taking finitely many values $y_1, \dots, y_n$ each with probability $1/n$, this is the arithmetic-geometric mean inequality

$$\left(\prod_{i=1}^n y_i\right)^{1/n} = e^{\frac{1}{n}\sum_i \log y_i} \leq \frac{1}{n}\sum_{i=1}^n y_i$$

the geometric mean never exceeds the arithmetic mean.

The reciprocal. With $\varphi(x) = 1/x$, convex on $(0, \infty)$, Jensen gives $1/\mathbb{E}[X] \leq \mathbb{E}[1/X]$ for a positive random variable X with $\mathbb{E}[X] < \infty$: the reciprocal of the average is at most the average of the reciprocal. This inequality reappears in the study of harmonic means and in the analysis of estimators.

The final pair of inequalities governs the algebra of L^p. Hölder's inequality bounds the expectation of a product XY by a product of norms, and it is the tool that makes the L^p norms interact; Minkowski's inequality is the triangle inequality for the L^p norm, the fact that makes $\|\cdot\|_p$ a norm at all. Both are proved from a single elementary convexity fact, Young's inequality, and both specialize at $p = q = 2$ to the Cauchy-Schwarz inequality that bounds a covariance by a product of standard deviations.

Throughout, p and q are *conjugate exponents*: real numbers with $1 < p, q < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$, with the boundary convention that $p = 1$ pairs with $q = \infty$. For $1 \leq p < \infty$ recall the L^p norm $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$. The starting point is a numerical inequality between a product and a weighted sum of powers.

Lemma 3.2.9: Young's Inequality

Let p, q be conjugate exponents with $1 < p, q < \infty$. Then for all real numbers $u, v \geq 0$,

$$uv \leq \frac{u^p}{p} + \frac{v^q}{q}$$

with equality if and only if $u^p = v^q$.

***Proof*:**
If $u = 0$ or $v = 0$ the inequality reads $0 \leq$ (nonnegative), so assume $u, v > 0$. The logarithm is concave on $(0, \infty)$, so for the weights $\frac{1}{p} + \frac{1}{q} = 1$,

$$\log\left(\frac{u^p}{p} + \frac{v^q}{q}\right) \geq \frac{1}{p}\log(u^p) + \frac{1}{q}\log(v^q) = \log u + \log v = \log(uv)$$

where the inequality is concavity of log applied to the points u^p, v^q with weights $1/p, 1/q$. Exponentiating, which preserves the inequality since exp is increasing, gives $uv \leq u^p/p + v^q/q$. Equality in the concavity step holds exactly when the two points coincide, $u^p = v^q$, completing the proof.

Young's inequality is the seed. Summing its two-variable form against a probability measure, after normalizing X and Y to unit norm, produces Hölder's inequality with no further work.

Theorem 3.2.10: Hölder's and Minkowski's Inequalities

Let p, q be conjugate exponents with $1 \leq p, q \leq \infty$.

1. *(Hölder)* For random variables X, Y,

$$\mathbb{E}|XY| \leq \|X\|_p \|Y\|_q$$

2. *(Minkowski)* For $1 \leq p < \infty$ and random variables X, Y in L^p,

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p$$

***Proof*:**
The general integral forms hold on any measure space and are proved in *Measure Theory* [2]; the proofs below specialize them to the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and are given in full.

Part (1): Hölder. Consider first $1 < p, q < \infty$. If $\|X\|_p = 0$ then $|X|^p$ has expectation 0, so $X = 0$ almost surely, hence $XY = 0$ almost surely and both sides vanish; likewise if $\|Y\|_q = 0$. If $\|X\|_p = \infty$ or $\|Y\|_q = \infty$ the right side is infinite and there is nothing to prove. So assume $0 < \|X\|_p, \|Y\|_q < \infty$, and normalize by setting

$$u = \frac{|X|}{\|X\|_p}, \qquad v = \frac{|Y|}{\|Y\|_q}$$

so that $\mathbb{E}[u^p] = 1$ and $\mathbb{E}[v^q] = 1$. Young's inequality (lemma 3.2.9) applied pointwise gives

$$uv \leq \frac{u^p}{p} + \frac{v^q}{q}$$

and taking expectations, using linearity and monotonicity (proposition 3.1.3),

$$\mathbb{E}[uv] \leq \frac{\mathbb{E}[u^p]}{p} + \frac{\mathbb{E}[v^q]}{q} = \frac{1}{p} + \frac{1}{q} = 1$$

Multiplying through by $\|X\|_p \|Y\|_q$ and recognizing $\mathbb{E}[uv] = \mathbb{E}|XY|/(\|X\|_p\|Y\|_q)$ yields $\mathbb{E}|XY| \leq \|X\|_p\|Y\|_q$. For the boundary case $p = 1, q = \infty$, write $\|Y\|_\infty = \operatorname{ess\,sup}|Y|$ for the essential supremum (developed in section 3.3); then $|XY| \leq \|Y\|_\infty |X|$ almost surely, and taking expectations gives $\mathbb{E}|XY| \leq \|Y\|_\infty \mathbb{E}|X| = \|X\|_1 \|Y\|_\infty$.

Part (2): Minkowski. The cases $p = 1$ and $\|X + Y\|_p = 0$ are immediate: for $p = 1$ it is $\mathbb{E}|X + Y| \leq \mathbb{E}|X| + \mathbb{E}|Y|$ from the pointwise triangle inequality $|X + Y| \leq |X| + |Y|$, and if $\|X + Y\|_p = 0$ the left side vanishes. So take $1 < p < \infty$ and $0 < \|X + Y\|_p < \infty$; the finiteness follows from $|X + Y|^p \leq 2^p(|X|^p + |Y|^p)$ and $X, Y \in L^p$. Split the p-th power using the triangle inequality:

$$|X + Y|^p = |X + Y|\,|X + Y|^{p-1} \leq |X|\,|X + Y|^{p-1} + |Y|\,|X + Y|^{p-1}$$

Take expectations and apply Hölder to each term with exponents p and $q = p/(p - 1)$. Since $(p - 1)q = p$, the factor $|X + Y|^{p-1}$ has

$$\left\|\, |X + Y|^{p-1} \right\|_q = \left(\mathbb{E}|X + Y|^{(p-1)q}\right)^{1/q} = \left(\mathbb{E}|X + Y|^p\right)^{1/q} = \|X + Y\|_p^{p/q}$$

so Hölder gives

$$\mathbb{E}|X+Y|^p \le \big(\|X\|_p + \|Y\|_p\big)\,\|X+Y\|_p^{p/q}$$

The left side is $\|X+Y\|_p^p$. Dividing both sides by $\|X+Y\|_p^{p/q}$, which is finite and nonzero, and using $p - p/q = p(1-1/q) = 1$, we obtain $\|X+Y\|_p \le \|X\|_p + \|Y\|_p$, completing the proof.

Hölder's inequality is the statement that the pairing $(X,Y) \mapsto \mathbb{E}[XY]$ is bounded when X is measured in L^p and Y in the conjugate L^q; it is the reason the dual of L^p is L^q, developed in the next section. Minkowski's inequality is the triangle inequality for $\|\cdot\|_p$, the last axiom needed to call $\|\cdot\|_p$ a norm and hence to speak of L^p as a normed space at all. The single most-used case is $p = q = 2$, where the two exponents coincide and Hölder becomes Cauchy-Schwarz.

Corollary 3.2.11: The Cauchy-Schwarz Inequality

For random variables X, Y with finite second moments,

$$\big|\mathbb{E}[XY]\big| \le \mathbb{E}|XY| \le \|X\|_2\,\|Y\|_2 = \sqrt{\mathbb{E}[X^2]}\,\sqrt{\mathbb{E}[Y^2]}$$

Consequently $|\mathrm{Cov}(X,Y)| \le \sigma(X)\,\sigma(Y)$, so the correlation coefficient satisfies $|\rho(X,Y)| \le 1$.

***Proof*:**
The middle inequality is Hölder (theorem 3.2.10) with $p = q = 2$, and the outer inequality $|\mathbb{E}[XY]| \le \mathbb{E}|XY|$ is proposition 3.1.3. For the covariance bound, apply the inequality to the centered variables $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$, whose second moments are $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$: this gives $|\mathrm{Cov}(X,Y)| = |\mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)]| \le \sigma(X)\sigma(Y)$. Dividing by $\sigma(X)\sigma(Y)$ when both are positive yields $|\rho| \le 1$, as we sought to show.

Cauchy-Schwarz is the inequality that makes L^2 a Hilbert space, with inner product $\langle X, Y\rangle = \mathbb{E}[XY]$; it says the inner product is bounded by the product of the lengths, so the angle between two random variables is well defined. The covariance corollary explains the range of the correlation coefficient promised after definition 3.2.1: correlation lies in $[-1,1]$, with the extremes ± 1 attained exactly when $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$ are almost surely proportional, the case of a perfect linear relationship. This geometry of L^2 is the foundation of the projection view of conditional expectation to come later in the book.

The inequalities of this section share one purpose: they convert information about moments, quantities computed by integration, into control of probabilities, statements about how often a random variable behaves in a given way. The archetype is the passage from a variance estimate to a law of large numbers, and it is worth isolating the argument, because the same three lines recur, with different inputs, throughout the limit theory of the book.

Example 3.2.12: Chebyshev, Jensen, and the Sample Average
Three applications show the inequalities at work.

A Chebyshev bound on the sample average. Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with finite variance, common mean $\mu = \mathbb{E}[X_1]$ and common variance $\sigma^2 = \mathrm{Var}(X_1)$, and form the partial sums $S_n = X_1 + \cdots + X_n$ and the sample average S_n/n. By linearity $\mathbb{E}[S_n/n] = \mu$, and because the summands are independent hence uncorrelated, the variance of the sum is the sum of the variances (proposition 3.3.7):

$$\mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n^2}\mathrm{Var}(S_n) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

using the affine-scaling identity proposition 3.2.2(2) for the first equality. Chebyshev's inequality (corollary 3.2.5) applied to S_n/n now gives, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(S_n/n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \tag{3.8}$$

The right side of (3.8) tends to 0 as $n \to \infty$ for each fixed ε. This is the *weak law of large numbers* for variables with finite variance: the sample average converges to the mean in probability. The entire proof is the variance computation followed by one application of Chebyshev; the limit theorems to come sharpen both the hypotheses and the conclusion, but this is the mechanism in miniature.

Jensen and the second moment. Applying Jensen's inequality (theorem 3.2.7) with $\varphi(x) = x^2$ gives $(\mathbb{E}X)^2 \leq \mathbb{E}[X^2]$ for any $X \in L^2$, so the mean square dominates the square of the mean, with gap exactly $\mathrm{Var}(X)$. This is the inequality that forces the standard deviation to be a meaningful scale: no random variable has its typical size smaller than the size of its mean.

Jensen and the reciprocal. For a strictly positive random variable X with $\mathbb{E}[X] < \infty$, applying Jensen with the convex $\varphi(x) = 1/x$ on $(0, \infty)$ gives

$$\mathbb{E}\left[\frac{1}{X}\right] \geq \frac{1}{\mathbb{E}[X]}$$

The average of a reciprocal is at least the reciprocal of the average. A gambler paid $1/X$ per round, where X is a positive random cost, earns on average at least the reciprocal of the average cost, and strictly more unless X is almost surely constant.

The pattern in the first application is the template for the whole of the limit theory. A quantity of interest, here the deviation of the sample average from its mean, is controlled by a moment, here the variance, and an inequality, here Chebyshev, turns the moment bound into a probability bound that improves with n. Later chapters replace the second moment by an exponential moment, obtaining the sharper Chernoff and subgaussian bounds, and replace convergence in probability by almost-sure convergence; but the shape of the argument, moment estimate into probability estimate, is fixed here and reused everywhere.

Exercise 3.2.1

1. Let $X \sim \mathrm{Poi}(\lambda)$, the Poisson law with mean λ, for which $\mathbb{E}[X] = \mathrm{Var}(X) = \lambda$. Use Chebyshev's inequality to bound $\mathbb{P}(X \geq 2\lambda)$, and compare the bound to 1 for large λ.
2. Show that for any random variable X with finite second moment and any $a > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + a) \leq \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + a^2}$$

the *one-sided Chebyshev* (or Cantelli) inequality. *Hint:* for $t \geq 0$ apply Markov to $(X - \mathbb{E}X + t)^2 \geq (a+t)^2$ on the event $\{X - \mathbb{E}X \geq a\}$, then optimize over t.

3. Let X be a positive random variable. Prove that $\mathbb{E}[X]\,\mathbb{E}[1/X] \geq 1$, and determine when equality holds. *Hint:* Cauchy-Schwarz applied to $\sqrt{X}$ and $1/\sqrt{X}$.

4. A random variable X satisfies $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = 1$. Give the best bound Chebyshev provides on $\mathbb{P}(|X| \geq 3)$, and exhibit a distribution for which this Chebyshev bound is attained with equality.

5. For a random variable X with finite moments of all orders and any even integer k, prove the moment tail bound $\mathbb{P}(|X| \geq a) \leq \mathbb{E}[X^k]/a^k$ for $a > 0$, and explain why taking k large can beat Chebyshev's $k = 2$ bound.

6. Let $1 \leq p < r < \infty$. Using Jensen's inequality with the convex function $\varphi(x) = x^{r/p}$ applied to $|X|^p$, prove that $\|X\|_p \leq \|X\|_r$ on a probability space (the nesting of L^p norms).

3.3 L^p Spaces of Random Variables

The inequalities of section 3.2 treat the quantities $\mathbb{E}|X|^p$ one at a time, as bounds on individual random variables. Collected together, these quantities organize the integrable variables into a family of normed spaces, one for each exponent p, and the analytic geometry of these spaces is where a great deal of probability takes place. The passage is a shift in viewpoint: instead of asking how large a single X is, one regards X as a point in a space of functions and measures its size by a norm. Minkowski's inequality is precisely the statement that this size is a norm, and Riesz-Fischer completeness makes the space Banach, so that Cauchy sequences of random variables converge. The exponent $p = 2$ is distinguished: there the norm comes from an inner product, L^2 becomes a Hilbert space, and the inner product is the covariance. Variance is a squared length, uncorrelated variables are orthogonal, and the Pythagorean theorem becomes the additivity of variance. This section builds that dictionary and records the two structural facts about L^p on a probability space that recur throughout the book: the spaces are nested, and L^2 carries the geometry of variance and covariance.

Everything analytic here is inherited from measure theory. What is native to probability is the reading of the L^2 inner product as covariance and the consequences that reading has for sums of independent variables. The house posture, cite the machinery and own the payoff, applies exactly.

The definition packages the norms $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$ introduced alongside Minkowski's inequality into spaces. Two points require care. First, a norm must vanish only at the zero element, yet $\|X\|_p = 0$ says only that $X = 0$ almost surely, not everywhere; the standard remedy is to identify random variables that agree almost surely, so that the "points" of L^p are really equivalence classes. Second, the exponent $p = \infty$ has no integral in its definition and is handled by the essential supremum, the smallest bound that holds almost surely.

Definition 3.3.1: L^p Spaces

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $1 \le p < \infty$. A random variable X is said to be *p-integrable* if $\mathbb{E}|X|^p < \infty$, and for such X one sets

$$\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$$

Declare two random variables *equivalent* if they are equal almost surely. The space $L^p = L^p(\Omega, \mathcal{F}, \mathbb{P})$ is the set of equivalence classes of p-integrable random variables, equipped with $\|\cdot\|_p$.

For $p = \infty$, the *essential supremum* of X is

$$\|X\|_\infty = \operatorname{ess\,sup}|X| = \inf\{\, M \ge 0 : \mathbb{P}(|X| > M) = 0 \,\}$$

and $L^\infty = L^\infty(\Omega, \mathcal{F}, \mathbb{P})$ is the space of equivalence classes of *essentially bounded* random variables, those with $\|X\|_\infty < \infty$.

The identification of almost-surely-equal variables is not a technicality to be tolerated but the correct notion of "the same random variable" for the purposes of expectation: two variables that differ on a null set have the same law, the same expectation, and the same $\mathbb{E}|X|^p$ for every p, so no integral can tell them apart. Working with equivalence classes is what turns the seminorm $\|\cdot\|_p$ (which can vanish on nonzero functions) into a genuine norm. In practice one writes X for both the variable and its class and only invokes the identification when it matters, exactly as one does with L^p in measure theory.

That $\|\cdot\|_p$ is a norm is not automatic. Homogeneity, $\|cX\|_p = |c|\,\|X\|_p$, is immediate, and $\|X\|_p = 0$ forces $X = 0$ as a class by the identification just made. The one substantive axiom is the triangle inequality, and that is exactly Minkowski's inequality proved in section 3.2.

3.3.2 Note: $\|\cdot\|_p$ is a norm by Minkowski For $1 \le p \le \infty$, the triangle inequality $\|X+Y\|_p \le \|X\|_p + \|Y\|_p$ on L^p is precisely Minkowski's inequality theorem 3.2.10 (the case $p = \infty$ following directly from $|X+Y| \le \|X\|_\infty + \|Y\|_\infty$ almost surely). Together with homogeneity and the vanishing property secured by the almost-sure identification, this makes $(L^p, \|\cdot\|_p)$ a normed vector space.

A first concrete instance grounds the definition before the theory develops. On a finite probability space the essential supremum is an ordinary maximum and the L^p norms are weighted power means, so one can see the nesting $\|X\|_p \le \|X\|_q$ of the next results by hand.

Example 3.3.3: L^p Norms on a Finite Space

Let $\Omega = \{1, 2, 3\}$ with $\mathbb{P}(\{k\}) = 1/3$ for each k, and let X take the values $1, 2, 3$. Then $|X|$ is bounded, so $X \in L^p$ for every p, and

$$\|X\|_p = \left(\tfrac{1}{3}\left(1^p + 2^p + 3^p\right)\right)^{1/p}$$

For $p = 1$ this is $\|X\|_1 = (1+2+3)/3 = 2$; for $p = 2$ it is $\|X\|_2 = \sqrt{(1+4+9)/3} = \sqrt{14/3} \approx 2.16$; and as $p \to \infty$ the largest value dominates, giving $\|X\|_\infty = \max\{1, 2, 3\} = 3$. The norms increase with p, and the limit recovers the maximum, both features that persist in general. Because every singleton has positive probability, here "almost surely" means "everywhere" and the essential supremum is the ordinary maximum.

A norm is only as useful as its convergence theory, and the property that makes L^p a workable space is completeness: every Cauchy sequence of p-integrable variables converges in $\|\cdot\|_p$ to a p-integrable limit. This is the Riesz-Fischer theorem. It is a theorem of measure theory, proved there in full for a general measure space, and a probability measure is a measure of total mass one, so the statement is inherited verbatim. Completeness is what lets one define a random variable as an L^p limit of approximations, the mechanism behind L^2 convergence in the laws of large numbers and behind the construction of conditional expectation later in the book.

Theorem 3.3.4: Completeness of L^p (Riesz-Fischer)

For $1 \leq p \leq \infty$, the space $L^p(\Omega, \mathcal{F}, \mathbb{P})$ is complete: every Cauchy sequence in $\|\cdot\|_p$ converges in $\|\cdot\|_p$ to an element of L^p. Hence L^p is a Banach space.
For $p = 2$, the norm is induced by the inner product

$$\langle X, Y \rangle = \mathbb{E}[XY]$$

(defined and finite for $X, Y \in L^2$ by Cauchy-Schwarz), and $\|X\|_2 = \langle X, X\rangle^{1/2}$. With this inner product L^2 is a Hilbert space.

***Proof*:**
Completeness is the Riesz-Fischer theorem for a general measure space, applied to the finite measure $\mathbb{P}$; see [2]. The claim is that a Cauchy sequence has an L^p-convergent subsequence, extracted by choosing indices along which the norms of successive differences are summable and dominating the resulting telescoping series; a Cauchy sequence with a convergent subsequence converges. The full argument is given there and is not reproduced.

For $p = 2$ the pairing $\langle X, Y\rangle = \mathbb{E}[XY]$ is finite for $X, Y \in L^2$ because $\mathbb{E}|XY| \leq \|X\|_2\|Y\|_2$ by Cauchy-Schwarz, the case $p = q = 2$ of theorem 3.2.10. Bilinearity follows from linearity of expectation proposition 3.1.3, symmetry is immediate, and $\langle X, X\rangle = \mathbb{E}[X^2] = \|X\|_2^2 \geq 0$ with equality only for $X = 0$ as a class. Thus $\langle \cdot, \cdot \rangle$ is an inner product inducing $\|\cdot\|_2$, and completeness makes L^2 a Hilbert space, completing the proof.

The Hilbert space structure of L^2 is the reason variance and covariance have a geometry at all. The covariance $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$ is the inner product of the centered variables $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$; the variance $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$ is the squared length of the centered X; and the standard deviation is its length. Correlation is the cosine of the angle between centered variables. Every second-order identity in probability, additivity of variance, the Cauchy-Schwarz bound $|\mathrm{Cov}(X, Y)| \leq \sigma_X\sigma_Y$ on correlation, the least-squares interpretation of conditional expectation, is a theorem of Hilbert-space geometry read through this dictionary. The rest of the section develops the two pieces of that geometry the book needs first.

On a general measure space the L^p spaces for different p are incomparable: $1/x$ near infinity lies in $L^2(\mathbb{R})$ but not $L^1(\mathbb{R})$, while $1/\sqrt{x}$ near zero lies in L^1 but not L^2. On a *probability* space this pathology disappears, and for one reason: the total mass is finite. A variable with a finite high moment automatically has finite lower moments, because raising to a larger power and averaging against a probability measure can only increase the norm. The nesting this produces, $L^q \subseteq L^p$ for $p \leq q$, is used constantly: it is why "X has a finite variance" ($X \in L^2$) already implies "X is integrable" ($X \in L^1$), so that $\mathbb{E}[X]$ makes sense whenever $\mathrm{Var}(X)$ does.

Proposition 3.3.5: Nesting of L^p Spaces

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $1 \le p \le q \le \infty$. Then $L^q \subseteq L^p$, and for every $X \in L^q$,

$$\|X\|_p \le \|X\|_q$$

In particular $L^\infty \subseteq \cdots \subseteq L^2 \subseteq L^1$, and a variable with a finite q-th moment has finite r-th moments for all $r \le q$.

***Proof*:**
Assume first $q < \infty$ and $X \in L^q$, and set $r = q/p \ge 1$. The function $\varphi(t) = t^r$ is convex on $[0, \infty)$, so applying Jensen's inequality theorem 3.2.7 to the nonnegative random variable $|X|^p$ gives

$$(\mathbb{E}|X|^p)^r = \varphi(\mathbb{E}|X|^p) \le \mathbb{E}[\varphi(|X|^p)] = \mathbb{E}|X|^{pr} = \mathbb{E}|X|^q$$

Raising both sides to the power $1/q = 1/(pr)$ yields $(\mathbb{E}|X|^p)^{1/p} \le (\mathbb{E}|X|^q)^{1/q}$, which is $\|X\|_p \le \|X\|_q$. In particular $\|X\|_q < \infty$ forces $\|X\|_p < \infty$, so $X \in L^p$ and $L^q \subseteq L^p$.

For $q = \infty$, note $|X| \le \|X\|_\infty$ almost surely, so $\mathbb{E}|X|^p \le \|X\|_\infty^p\, \mathbb{P}(\Omega) = \|X\|_\infty^p$, using $\mathbb{P}(\Omega) = 1$; taking p-th roots gives $\|X\|_p \le \|X\|_\infty$, completing the proof.

The single step that consumes the finiteness of the mass is the use of Jensen's inequality, which requires a probability measure: it is the identity $\mathbb{P}(\Omega) = 1$ that lets $\mathbb{E}$ act as a genuine average, and it is the same 1 that appears as $\mathbb{P}(\Omega)$ in the $q = \infty$ case. On an infinite measure space the inequality reverses in character and the containment fails, as the examples $1/x$ and $1/\sqrt{x}$ above show. The nesting is therefore not a formal consequence of the definitions but a feature special to finite total mass, and it is one of the structural conveniences that makes probability easier than general measure theory. The exponent inequality $\|X\|_p \le \|X\|_q$ has a name, Lyapunov's inequality, and says the map $p \mapsto \|X\|_p$ is nondecreasing.

An immediate check on a running example illustrates the containment.

Example 3.3.6: A Coin Step Lives in Every L^p

Let X be a single fair ± 1 coin step, $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = 1/2$, the increment of the simple random walk of section 2.3. Then $|X| = 1$ identically, so $\mathbb{E}|X|^p = 1$ and $\|X\|_p = 1$ for every $1 \le p \le \infty$; in particular $X \in L^\infty \subseteq L^q \subseteq L^p$ for all $p \le q$. The nesting is visible: the norm is constant in p because the variable is essentially bounded with $|X|$ taking a single value. Compare a variable that is integrable but not square-integrable, which cannot occur here but can on an infinite measure space, to see that the containment is about finite mass.

The geometry of L^2 pays off first in the behaviour of variance under addition. Two centered variables are orthogonal in L^2 exactly when their covariance vanishes, and vanishing covariance is the definition of being uncorrelated. This identification turns a probabilistic hypothesis (uncorrelatedness) into a geometric one (orthogonality), and the Pythagorean theorem then delivers the additivity of variance for free. The result is Bienaymé's identity, the identity behind every variance computation for sums, and in particular the source of the $\sqrt{n}$ scaling that the central limit theorem later refines.

Proposition 3.3.7: L^2 Geometry and Bienaymé's Identity

Let $X, Y, X_1, \dots, X_n \in L^2(\Omega, \mathcal{F}, \mathbb{P})$.

1. X and Y are *uncorrelated*, meaning $\operatorname{Cov}(X, Y) = 0$, if and only if their centered versions $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$ are orthogonal in L^2, that is $\langle X - \mathbb{E}X, Y - \mathbb{E}Y \rangle = 0$.

2. (*Bienaymé's identity*) If $X_1, \dots, X_n$ are pairwise uncorrelated, then

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i)$$

In particular this holds when $X_1, \dots, X_n$ are independent (definition 2.2.2).

***Proof*:**

1. By definition of the L^2 inner product theorem 3.3.4,

$$\langle X - \mathbb{E}X,\, Y - \mathbb{E}Y \rangle = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \operatorname{Cov}(X, Y)$$

so the covariance is zero precisely when the centered variables are orthogonal. Both variables lie in L^2, so the centered variables do too, and the pairing is finite by Cauchy-Schwarz, so the statement is meaningful.

2. Write $\mu_i = \mathbb{E}[X_i]$ and center: the variable $\sum_i X_i$ has mean $\sum_i \mu_i$ by linearity, so its centered version is $\sum_i (X_i - \mu_i)$. Expanding the square and using linearity of expectation proposition 3.1.3,

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \mathbb{E}\left[\left(\sum_{i=1}^{n} (X_i - \mu_i)\right)^2\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \sum_{i=1}^{n}\sum_{j=1}^{n} \operatorname{Cov}(X_i, X_j)$$

The diagonal terms $i = j$ contribute $\operatorname{Cov}(X_i, X_i) = \operatorname{Var}(X_i)$; the off-diagonal terms $i \neq j$ contribute $\operatorname{Cov}(X_i, X_j) = 0$ by pairwise uncorrelatedness. Hence the double sum collapses to $\sum_{i=1}^{n} \operatorname{Var}(X_i)$.

Finally, if $X_1, \dots, X_n$ are independent, then for $i \neq j$ the product law gives $\mathbb{E}[X_i X_j] = \mathbb{E}[X_i]\mathbb{E}[X_j]$, so $\operatorname{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] = 0$; independent L^2 variables are pairwise uncorrelated, and the identity applies, as we sought to show.

Two features of this result deserve emphasis. First, the hypothesis is only *pairwise* uncorrelatedness, weaker than independence: additivity of variance never sees the higher-order dependence that separates uncorrelated from independent variables, because it is a purely second-order, hence purely Hilbert-space, statement. The converse fails, uncorrelated variables need not be independent, so Bienaymé's identity holds strictly more often than independence. Second, geometrically the identity is the Pythagorean theorem: the centered summands are pairwise orthogonal vectors in L^2, and the squared length of their sum is the sum of their squared lengths. This is the cleanest reason variance adds and covariance does not, and it is the template for the projection arguments that recur in the limit theory.

That projection viewpoint is worth naming, because it returns in force later in the book. In a Hilbert space, given a closed subspace, every vector has a unique nearest point in that subspace, its orthogonal projection, characterized by the residual being orthogonal to the subspace. Conditional expectation is exactly this projection: $\mathbb{E}[X|\mathcal{G}]$ will be constructed as the orthogonal projection of $X \in L^2$ onto the subspace of $\mathcal{G}$-measurable variables, the best least-squares approximation to X using only the information in $\mathcal{G}$. The additivity of variance proved here and the least-squares optimality of conditioning are two faces of the same L^2 geometry; the conditioning face is developed when the machinery of conditional expectation is built.

The running examples make the additivity of variance concrete and expose the scaling law that drives the rest of the book. For a sum of independent identically distributed steps, Bienaymé's identity turns an n-fold sum of variances into a single multiplication by n, so the standard deviation grows like $\sqrt{n}$ while the mean grows like n. This is the fluctuation-versus-drift dichotomy that the laws of large numbers and the central limit theorem sharpen.

Example 3.3.8: Variance of a Sum and the Coin Walk

Let $X_1, X_2, \ldots$ be independent and identically distributed with common variance $\sigma^2 = \mathrm{Var}(X_1) < \infty$, so each $X_i \in L^2$ (proposition 3.3.5), and let $S_n = X_1 + \cdots + X_n$ be the associated random walk of section 2.3.

1. Since the X_i are independent, hence pairwise uncorrelated, Bienaymé's identity proposition 3.3.7 gives

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = n\sigma^2$$

The standard deviation of S_n is $\sigma\sqrt{n}$. If the common mean is $\mu = \mathbb{E}[X_1]$, then $\mathbb{E}[S_n] = n\mu$ by linearity, so the drift grows linearly in n while the fluctuation grows only as $\sqrt{n}$; the ratio of fluctuation to drift is order $1/\sqrt{n}$ and vanishes, the seed of the law of large numbers.

2. For the fair coin walk, each step is $X_i = \pm 1$ with probability $1/2$, so $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] = 1$ (example 3.3.6). Hence

$$\mathbb{E}[S_n] = 0, \qquad \mathrm{Var}(S_n) = n$$

The walk has mean zero and standard deviation $\sqrt{n}$: after n fair steps the typical distance from the origin is of order $\sqrt{n}$, not n. This single computation, $\mathrm{Var}(S_n) = n$, is the quantitative core of the diffusive scaling of the random walk.

3. Normalizing, $S_n/\sqrt{n}$ has variance $\mathrm{Var}(S_n)/n = \sigma^2$ for the general walk and 1 for the coin walk, independent of n. This is why $S_n/\sqrt{n}$, rather than S_n or S_n/n, is the object with a nondegenerate limit: it is the unique power of n that keeps the variance bounded away from 0 and ∞. The central limit theorem identifies that limit as Gaussian.

The three computations trace the logic of the entire limit theory in miniature. Dividing S_n by n kills the variance ($\mathrm{Var}(S_n/n) = \sigma^2/n \to 0$), which by Chebyshev's inequality corollary 3.2.5 forces S_n/n to concentrate at the mean, the weak law. Dividing instead by $\sqrt{n}$ fixes the variance at σ^2, leaving a nondegenerate fluctuation whose distribution is the subject of the central limit theorem. The choice of scaling is dictated entirely by Bienaymé's identity, and Bienaymé's identity is the Pythagorean theorem in L^2.

Exercise 3.3.1

1. Let X be a random variable with $\mathbb{P}(X = 2^n) = 2^{-n}$ for $n = 1, 2, 3, \ldots$. Determine for which $p \in [1, \infty]$ the variable X lies in L^p. Does the conclusion respect the nesting $L^q \subseteq L^p$ of proposition 3.3.5?
2. Let $X, Y \in L^2$. Prove the parallelogram law
$$\|X + Y\|_2^2 + \|X - Y\|_2^2 = 2\|X\|_2^2 + 2\|Y\|_2^2$$
and interpret it as a statement about $\mathrm{Var}(X + Y)$ and $\mathrm{Var}(X - Y)$ when X and Y are centered.
3. Give an example of two random variables X and Y on a finite probability space that are uncorrelated but not independent. Verify that Bienaymé's identity proposition 3.3.7 still gives $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ for your example, and explain why this is consistent with the failure of independence.
4. Let $X \in L^2$ with $\mathrm{Var}(X) > 0$. Using the inner-product structure of L^2, find the constant c minimizing $\mathbb{E}[(X - c)^2]$, and identify the minimum value. Explain how this realizes $\mathbb{E}[X]$ as the orthogonal projection of X onto the subspace of constants.
5. Show that on a probability space the nesting is strict in general: exhibit, on the unit interval $[0, 1]$ with Lebesgue measure (example 1.1.3), a random variable belonging to L^1 but not L^2. Contrast this with example 3.3.6, where the variable lies in every L^p, and identify what feature of the coin step accounts for the difference.
6. Let $S_n = X_1 + \cdots + X_n$ be a sum of pairwise uncorrelated L^2 variables with $\mathrm{Var}(X_i) = \sigma_i^2$. Use Bienaymé's identity together with Chebyshev's inequality corollary 3.2.5 to bound $\mathbb{P}(|S_n - \mathbb{E}S_n| \geq a)$ in terms of $\sum_i \sigma_i^2$. When all $\sigma_i = \sigma$, deduce a bound on $\mathbb{P}(|S_n/n - \mathbb{E}[S_n]/n| \geq \varepsilon)$ and comment on its behaviour as $n \to \infty$.

3.4 Moments and Generating Functions

The variance measures the spread of X about its mean, but it is only the second in an infinite ladder of numerical summaries: the *moments* $\mathbb{E}[X^k]$. Each moment captures a coarser feature of the distribution than the next, and together they often pin down the law of X completely. The trouble is that moments arrive one at a time, each a separate integral, with no visible relation among them. This section introduces a single object, the *moment generating function* $M_X(t) = \mathbb{E}[e^{tX}]$, that packages all of them at once: its derivatives at the origin are the moments, its behaviour under addition of independent variables is a plain product, and its exponential shape converts Markov's inequality into the first tail bounds sharp enough to survive the passage to a limit. This last property, exponential concentration, is the technical seed from which the sharpest results of the book grow.

Throughout, X is a random variable on a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and the change-of-variables formula $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g \, d\mu_X$ from theorem 3.1.2 lets every computation below be carried out against the law μ_X on $\mathbb{R}$.

The ladder of moments is where the section begins. The k-th *moment* is the expectation of X^k; centering X at its mean before raising to the power isolates the shape of the fluctuation from its location, giving the *central* moments; and inserting absolute values gives a quantity that always makes sense as an element of $[0, \infty]$, which is the right object for questions of existence.

Definition 3.4.1: Moments

Let X be a random variable and k a positive integer.

1. The *k-th moment* of X is $\mathbb{E}[X^k]$, defined whenever X^k is integrable, that is whenever $\mathbb{E}|X|^k < \infty$.
2. The *k-th central moment* of X is $\mathbb{E}[(X - \mathbb{E}X)^k]$, defined when X is integrable and $\mathbb{E}|X|^k < \infty$. The second central moment is the variance $\operatorname{Var}(X)$ (definition 3.2.1).
3. The *k-th absolute moment* of X is $\mathbb{E}|X|^k \in [0, \infty]$, which is always defined because $|X|^k \geq 0$.

X is said to *have moments of all orders* (or to have *finite moments of every order*) if $\mathbb{E}|X|^k < \infty$ for every $k \geq 1$.

The absolute moment $\mathbb{E}|X|^k$ is exactly the $p = k$ case of the L^p norm from definition 3.3.1: $\mathbb{E}|X|^k = \|X\|_k^k$. This is the link that organizes the whole ladder. On a probability space the L^p spaces are nested, $L^q \subseteq L^p$ for $p \leq q$, with $\|X\|_p \leq \|X\|_q$ (proposition 3.3.5). Reading that nesting through the identification $\mathbb{E}|X|^p = \|X\|_p^p$ gives the governing principle of moments: existence of a high moment forces existence of every lower one.

Proposition 3.4.2: The Moment Hierarchy

If $\mathbb{E}|X|^k < \infty$ for some integer $k \geq 1$, then $\mathbb{E}|X|^j < \infty$ for every integer $1 \leq j \leq k$, and

$$(\mathbb{E}|X|^j)^{1/j} \leq (\mathbb{E}|X|^k)^{1/k}$$

In particular a variable with a finite k-th moment has all moments and all central moments of order at most k.

***Proof*:**
The hypothesis $\mathbb{E}|X|^k < \infty$ says $X \in L^k$. By the nesting of L^p spaces on a probability space (proposition 3.3.5), $X \in L^j$ with $\|X\|_j \leq \|X\|_k$ for every $1 \leq j \leq k$. Unwinding the norms, $\|X\|_j \leq \|X\|_k$ is precisely $(\mathbb{E}|X|^j)^{1/j} \leq (\mathbb{E}|X|^k)^{1/k}$, and $\|X\|_j < \infty$ is $\mathbb{E}|X|^j < \infty$. Finiteness of the j-th absolute moment gives finiteness of the j-th ordinary moment, since $|X^j| = |X|^j$; and the binomial expansion $(X - \mathbb{E}X)^j = \sum_{i=0}^{j} \binom{j}{i} X^i (-\mathbb{E}X)^{j-i}$ writes the j-th central moment as a finite linear combination of the ordinary moments of orders $0, \ldots, j$, each finite, so the j-th central moment is finite as well. This is what we sought to show.

The hierarchy explains why the variance is so often the first nontrivial quantity one computes: as soon as a variable is square-integrable, its mean exists too, and the two together are the data Chebyshev's inequality consumes. It also warns that the ladder can terminate. A variable may have a mean but no variance, or a finite variance but no third moment; the classic example is a density with a heavy polynomial tail, developed in example 3.4.3 below, where $\mathbb{E}|X|^k = \infty$ past a threshold. Nothing forces the ladder to be infinite, and the moment generating function, defined next, exists exactly when it is infinite and the growth of the moments is controlled.

Example 3.4.3: The Truncated-Tail Ladder

Consider the density $f(x) = \frac{c_\alpha}{(1+|x|)^{\alpha+1}}$ on $\mathbb{R}$, where $\alpha > 0$ and c_α is the normalizing constant making $\int_{\mathbb{R}} f = 1$. A variable X with this law has

$$\mathbb{E}|X|^k = c_\alpha \int_{\mathbb{R}} \frac{|x|^k}{(1+|x|)^{\alpha+1}}\,dx$$

The integrand behaves like $|x|^{k-\alpha-1}$ as $|x| \to \infty$, so the integral converges precisely when $k - \alpha - 1 < -1$, that is when $k < \alpha$. Thus X has finite moments exactly of orders $k < \alpha$ and infinite moments of orders $k \geq \alpha$: taking $\alpha = \frac{5}{2}$, for instance, X has a mean and a variance but no third moment. The hierarchy of proposition 3.4.2 is visible directly here, the set of admissible orders being the initial segment $\{k : k < \alpha\}$. For $\alpha = 1$ this is the Cauchy-type tail, which fails even to have a mean, showing that a perfectly good probability distribution can sit at the very bottom of the ladder.

Collecting the moments one integral at a time is unwieldy. The device that collects them all is the expectation of the exponential e^{tX}, viewed as a function of the real parameter t. Formally expanding $e^{tX} = \sum_k t^k X^k/k!$ and integrating term by term suggests that the moments are the Taylor coefficients of $t \mapsto \mathbb{E}[e^{tX}]$; making that suggestion precise is the content of the theorem after next. First the definition, together with the set of t for which the defining expectation is finite.

Definition 3.4.4: Moment Generating Function

The *moment generating function* (MGF) of a random variable X is the function

$$M_X(t) = \mathbb{E}\big[e^{tX}\big] = \int_{\mathbb{R}} e^{tx}\,d\mu_X(x)$$

defined for those $t \in \mathbb{R}$ at which the expectation is finite. Its *domain* is the set

$$D_X = \{t \in \mathbb{R} : M_X(t) < \infty\}$$

Since $e^{tX} \geq 0$, the expectation $M_X(t) \in (0, \infty]$ is always defined, and D_X is the set on which it is finite. Always $0 \in D_X$, with $M_X(0) = 1$.

The domain D_X is an interval containing 0. If $M_X(t_0) < \infty$ for some $t_0 > 0$, then for $0 \leq t \leq t_0$ one has $e^{tx} \leq e^{t_0 x}$ when $x \geq 0$ and $e^{tx} \leq 1$ when $x < 0$, so $e^{tX} \leq e^{t_0 X} + 1$ and $M_X(t) \leq M_X(t_0) + 1 < \infty$. A symmetric argument handles $t_0 < 0$, and convexity of $t \mapsto e^{tx}$ passes to M_X, so D_X is a (possibly degenerate, possibly infinite) interval about the origin. The case that matters is when this interval is genuinely open around 0: then, as the next result shows, X is controlled to every order at once.

Theorem 3.4.5: The MGF Generates the Moments

Suppose M_X is finite on an open interval $(-\delta, \delta)$ about the origin, for some $\delta > 0$. Then:

1. X has finite moments of every order, and moreover $\mathbb{E}\big[e^{s|X|}\big] < \infty$ for every $0 \leq s < \delta$.
2. M_X is infinitely differentiable on $(-\delta, \delta)$, and its k-th derivative is obtained by differentiating under the expectation,
$$M_X^{(k)}(t) = \mathbb{E}\big[X^k e^{tX}\big] \qquad (t \in (-\delta, \delta))$$
In particular, evaluating at $t = 0$,
$$M_X^{(k)}(0) = \mathbb{E}[X^k]$$
so the k-th moment is the k-th Taylor coefficient of M_X times $k!$.
3. M_X is real-analytic on $(-\delta, \delta)$, with the convergent expansion
$$M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!}\, t^k$$
valid for $|t| < \delta$.

Before the proof, a word on why the open interval is the right hypothesis. A single one-sided finiteness $M_X(t_0) < \infty$ controls the moments only through that one exponential and does not by itself give differentiability at 0; what makes the derivatives exist is that the difference quotients of e^{tX} are dominated by an integrable envelope, and building that envelope needs a little room on both sides of t. The proof produces the envelope explicitly and then lets the dominated convergence theorem do the analytic work.

***Proof*:**
Fix s with $0 \leq s < \delta$ and choose t_0 with $s < t_0 < \delta$. Then
$$e^{s|X|} \leq e^{sX} + e^{-sX} \leq e^{t_0 X} + e^{-t_0 X}$$
holds pointwise, because $e^{s|x|} = \max(e^{sx}, e^{-sx}) \leq e^{sx} + e^{-sx}$ and the map $u \mapsto e^{ux}$ is monotone in u on each half-line, so $e^{sx} \leq e^{t_0 x}$ when $x \geq 0$ and $e^{-sx} \leq e^{-t_0 x}$ when $x < 0$, with the remaining term bounded by $1 \leq e^{t_0 x} + e^{-t_0 x}$ in either case. Taking expectations, $\mathbb{E}[e^{s|X|}] \leq M_X(t_0) + M_X(-t_0) < \infty$ since both $\pm t_0$ lie in $(-\delta, \delta)$. This proves the finiteness in (1).

Step 1: All moments are finite. For each integer $k \geq 0$ and any $s \in (0, \delta)$, the elementary bound $|x|^k \leq C_{k,s}\, e^{s|x|}$ holds with $C_{k,s} = (k/s)^k e^{-k}$, obtained by maximizing $|x|^k e^{-s|x|}$ over x. Hence
$$\mathbb{E}|X|^k \leq C_{k,s}\, \mathbb{E}\big[e^{s|X|}\big] < \infty$$
by the finiteness just established. So X has moments of all orders, completing part (1).

Step 2: Differentiation under the expectation. Fix $t \in (-\delta, \delta)$ and choose $\eta > 0$ small enough that $[t-\eta, t+\eta] \subseteq (-\delta, \delta)$. For a real increment h with $0 < |h| \leq \eta$, write the difference quotient of M_X as an expectation:

$$\frac{M_X(t+h)-M_X(t)}{h} = \mathbb{E}\left[e^{tX}\,\frac{e^{hX}-1}{h}\right]$$

As $h \to 0$ the integrand converges pointwise to Xe^{tX}. To pass the limit inside, dominate the integrand uniformly in h. By the mean value theorem applied to $u \mapsto e^{uX}$ on the interval between t and $t+h$, there is a point ξ (depending on ω and h) strictly between t and $t+h$ with

$$e^{tX}\,\frac{e^{hX}-1}{h} = \frac{e^{(t+h)X}-e^{tX}}{h} = Xe^{\xi X}$$

and $\xi \in [t-\eta, t+\eta]$. Therefore

$$\left|e^{tX}\,\frac{e^{hX}-1}{h}\right| = |X|\,e^{\xi X} \le |X|\big(e^{(t+\eta)X}+e^{(t-\eta)X}\big)$$

where the last step bounds $e^{\xi X}$ by the sum of the two endpoint exponentials (one of the two exceeds $e^{\xi X}$ according to the sign of X). The dominating variable $Y = |X|\big(e^{(t+\eta)X}+e^{(t-\eta)X}\big)$ is integrable: choosing $s \in (0,\delta)$ with $t \pm \eta \in (-\delta+s, \delta-s)$, the factor $e^{(t\pm\eta)X}$ times $|X|$ is bounded by a constant multiple of $e^{(t\pm\eta)X}e^{s|X|} \le e^{(|t|+\eta+s)|X|}$ up to the polynomial factor $|X|$, which Step 1 absorbs, and this last exponent can be kept below δ by taking η, s small. Concretely $Y \le C\,e^{s'|X|}$ for some $s' \in (0,\delta)$ and constant C, so $\mathbb{E}[Y] < \infty$. The dominated convergence theorem of *Measure Theory* [2] now licenses passing the limit through the expectation:

$$M_X'(t) = \lim_{h\to 0}\mathbb{E}\left[e^{tX}\,\frac{e^{hX}-1}{h}\right] = \mathbb{E}\big[Xe^{tX}\big]$$

Thus M_X is differentiable on $(-\delta,\delta)$ with $M_X'(t) = \mathbb{E}[Xe^{tX}]$. Iterating the identical argument on $t \mapsto \mathbb{E}[X^k e^{tX}]$, whose difference quotient is dominated by $|X|^{k+1}\big(e^{(t+\eta)X}+e^{(t-\eta)X}\big)$, again integrable by Step 1, gives $M_X^{(k)}(t) = \mathbb{E}[X^k e^{tX}]$ for all k. Setting $t=0$ yields $M_X^{(k)}(0) = \mathbb{E}[X^k]$, which is part (2).

Step 3: Analyticity. Fix t with $|t| < \delta$ and pick $r > 0$ with $|t| + r < \delta$. For any real h with $|h| \le r$, Taylor's theorem with integral remainder, or the pointwise power series $e^{(t+h)X} = \sum_{k\ge 0}\frac{h^k}{k!}X^k e^{tX}$, gives after taking expectations

$$M_X(t+h) = \sum_{k=0}^{\infty}\frac{h^k}{k!}\,\mathbb{E}\big[X^k e^{tX}\big] = \sum_{k=0}^{\infty}\frac{M_X^{(k)}(t)}{k!}\,h^k$$

provided the interchange of sum and expectation is justified. It is: the partial sums are dominated by $\sum_k \frac{|h|^k}{k!}|X|^k e^{tX} = e^{|h|\,|X|}e^{tX}$, and $\mathbb{E}\big[e^{|h||X|}e^{tX}\big] \le \mathbb{E}\big[e^{(|t|+r)|X|}\big] < \infty$ by part (1), so the dominated convergence theorem allows summation and expectation to be exchanged. Hence M_X agrees with a convergent power series in a neighborhood of every point of $(-\delta,\delta)$, so it is real-analytic there. Taking $t=0$ specializes the expansion to $M_X(h) = \sum_k \frac{\mathbb{E}[X^k]}{k!}h^k$ for $|h| < \delta$, which is part (3) and completes the proof.

The name "moment generating" is now literal: M_X is the exponential generating function of the sequence of moments, and its derivatives at the origin read them off. The theorem also delivers something the moments alone cannot, namely a single analytic object attached to X whose finiteness on an interval already forces the entire moment ladder to be infinite with controlled growth. This is why the hypothesis "M_X finite near 0" recurs: it is the clean sufficient condition under which the moment method works, and it holds for every distribution with an exponentially decaying tail, including all the bounded, Poisson, and Gaussian variables of this book. When it fails, as for the truncated-tail law of example 3.4.3 where $D_X = \{0\}$, the moment method must be replaced by the characteristic function $\mathbb{E}[e^{itX}]$, whose finiteness is automatic and which drives the limit theorems to come.

The property that makes M_X the computational engine of the limit theory is its behaviour under addition of independent variables. Expectation is linear but not multiplicative in general; independence is exactly the hypothesis that restores multiplicativity for products of functions of the separate variables, and the exponential turns a sum in the exponent into a product of exponentials. The two facts combine to turn the MGF of a sum into a product of MGFs.

Proposition 3.4.6: The MGF of a Sum of Independent Variables

Let $X_1, \ldots, X_n$ be independent random variables (definition 2.2.2) and set $S_n = X_1 + \cdots + X_n$. Then for every t at which all the individual MGFs are finite,

$$M_{S_n}(t) = \prod_{i=1}^{n} M_{X_i}(t)$$

In particular $D_{S_n} \supseteq \bigcap_i D_{X_i}$, and if the X_i are identically distributed with common MGF $M(t)$ then $M_{S_n}(t) = M(t)^n$.

***Proof*:**
It suffices to treat $n = 2$; the general case follows by induction, since $X_1 + \cdots + X_n = (X_1 + \cdots + X_{n-1}) + X_n$ and the two summands on the right are independent (a function of $X_1, \ldots, X_{n-1}$ is independent of X_n by definition 2.2.2). For $n = 2$, the variables e^{tX_1} and e^{tX_2} are functions of the independent variables X_1 and X_2, hence themselves independent, and each is nonnegative. The product law for independent nonnegative variables, which is the integrated form of Fubini's theorem on the product law $\mu_{(X_1,X_2)} = \mu_{X_1} \otimes \mu_{X_2}$ (proposition 2.3.2), gives

$$M_{S_2}(t) = \mathbb{E}\big[e^{t(X_1+X_2)}\big] = \mathbb{E}\big[e^{tX_1} e^{tX_2}\big] = \mathbb{E}\big[e^{tX_1}\big]\,\mathbb{E}\big[e^{tX_2}\big] = M_{X_1}(t)\,M_{X_2}(t)$$

the middle equality being the factorization of the expectation of a product of independent variables. If $t \in D_{X_1} \cap D_{X_2}$ then both factors are finite, so $M_{S_2}(t) < \infty$ and $t \in D_{S_2}$, giving the stated inclusion of domains. The identically distributed case is the special case $M_{X_1} = M_{X_2} = M$, completing the proof.

The passage from sums to products is the reason MGFs, and later characteristic functions, dominate the analysis of independent sums. Convolution of laws (definition 2.3.3), which describes the same operation at the level of distributions, is a genuine integral over the simplex and is awkward to iterate; multiplication of MGFs is trivial to iterate. The MGF thus linearizes the sum: it converts the operation "add n independent copies" into "raise to the n-th power", and the analytic behaviour of $M(t)^n$ as $n \to \infty$ is exactly what the laws of large numbers and the central limit theorem extract. This is the concrete sense in which the limit theorems are computations with the MGF.

Multiplicativity has an immediate payoff in the direction of *tail bounds*. Chebyshev's inequality (corollary 3.2.5) controls a tail by the variance and decays only polynomially, like a^{-2}; that decay is too slow to survive summation over the geometric subsequences the sharpest theorems require. The exponential shape of the MGF furnishes a bound that decays exponentially, and it does so by the most economical of arguments: apply Markov's inequality (theorem 3.2.4) not to X but to the monotone transform e^{tX}, then optimize over the free parameter t.

Proposition 3.4.7: The Chernoff Bound

Let X be a random variable and $a \in \mathbb{R}$. For every $t \geq 0$,

$$\mathbb{P}(X \geq a) \leq e^{-ta}\, M_X(t)$$

and consequently

$$\mathbb{P}(X \geq a) \leq \inf_{t\geq 0} e^{-ta} M_X(t)$$

Symmetrically, for every $t \geq 0$, $\mathbb{P}(X \leq -a) \leq e^{-ta} M_X(-t)$.

***Proof*:**
Fix $t \geq 0$. The map $x \mapsto e^{tx}$ is nondecreasing, so the event $\{X \geq a\}$ is contained in $\{e^{tX} \geq e^{ta}\}$, and these two events in fact coincide when $t > 0$. Since $e^{tX} \geq 0$, Markov's inequality (theorem 3.2.4) applied to the nonnegative variable e^{tX} at level e^{ta} gives

$$\mathbb{P}(X \geq a) \leq \mathbb{P}\big(e^{tX} \geq e^{ta}\big) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-ta} M_X(t)$$

This holds for every $t \geq 0$, so the inequality persists after taking the infimum over $t \geq 0$, which is the second display. The lower-tail bound applies the same argument to $-X$, whose MGF is $M_{-X}(t) = M_X(-t)$, completing the proof.

The freedom to choose t is what makes the bound sharp. For a fixed distribution the exponent $-ta + \log M_X(t)$ is a convex function of t, and the optimal t solves $M_X'(t)/M_X(t) = a$; the resulting rate is the Legendre transform of $\log M_X$, an object that reappears as the rate function of large deviation theory. For the running example of coin tossing the optimization is explicit and produces the first exponential concentration bound of the book.

Example 3.4.8: The Subgaussian Tail of the Coin Walk

Let $X_1, X_2, \ldots$ be independent ± 1 steps, each equal to $+1$ or -1 with probability $\frac{1}{2}$ (the symmetric coin walk of section 2.2), and $S_n = X_1 + \cdots + X_n$. Each step has MGF

$$\mathbb{E}\big[e^{tX_i}\big] = \tfrac{1}{2}e^{t} + \tfrac{1}{2}e^{-t} = \cosh t$$

so by proposition 3.4.6 the walk has $M_{S_n}(t) = (\cosh t)^n$. The elementary inequality $\cosh t \leq e^{t^2/2}$, valid for all real t and read off by comparing the power series $\cosh t = \sum_k t^{2k}/(2k)!$ with $e^{t^2/2} = \sum_k t^{2k}/(2^k k!)$ term by term (since $(2k)! \geq 2^k k!$), gives $M_{S_n}(t) \leq e^{nt^2/2}$. The Chernoff bound (proposition 3.4.7) then yields, for every $t \geq 0$ and $a > 0$,

$$\mathbb{P}(S_n \geq a) \leq e^{-ta} e^{nt^2/2} = \exp\big(\tfrac{n}{2}t^2 - at\big)$$

The exponent is minimized at $t = a/n$, where it equals $-a^2/(2n)$. Therefore

$$\mathbb{P}(S_n \geq a) \leq e^{-a^2/(2n)}$$

and by symmetry $\mathbb{P}(|S_n| \geq a) \leq 2e^{-a^2/(2n)}$. This is the *subgaussian* tail: the walk of length n has fluctuations of order $\sqrt{n}$, and the probability of a deviation a decays like that of a Gaussian of variance n. Compare Chebyshev, which gives only $\mathbb{P}(|S_n| \geq a) \leq n/a^2$ from $\mathrm{Var}(S_n) = n$ (example 3.3.8): at the scale $a = \lambda\sqrt{n}$ Chebyshev returns the constant $1/\lambda^2$ while the Chernoff bound returns $2e^{-\lambda^2/2}$, decaying in λ. The exponential decay is what lets these probabilities be summed along the geometric subsequences $n = 2^j$, and that summability is the mechanism behind the upper half of the law of the iterated logarithm developed later in the book.

The bound of example 3.4.8 is the prototype of a *concentration inequality*: a statement that a sum of independent bounded variables sits within a small window of its mean with probability exponentially close to one. The same three moves, dominate the MGF of each summand by a Gaussian MGF, multiply, and optimize the Chernoff exponent, prove the general Azuma-Hoeffding bound taken up with the martingale theory. For now the point is that exponential tails are available the moment an MGF is finite near 0, and that they are strictly stronger than anything the variance alone can give.

The examples below compute the MGFs of the distributions that recur throughout the book and show the multiplicativity of proposition 3.4.6 in action, deriving the binomial MGF from the Bernoulli one and matching the convolution computation of example 2.3.6. The Gaussian MGF $e^{t^2/2}$, appearing here for the first time, is the analytic signature of the normal distribution and the reason the coin walk's tail in example 3.4.8 is called subgaussian.

Example 3.4.9: Moment Generating Functions of the Standard Laws

Each computation evaluates $M_X(t) = \int_{\mathbb{R}} e^{tx}\, d\mu_X(x)$ against the law of X, discrete laws summing against the mass function and continuous laws integrating against the density (theorem 3.1.2).

1. *Bernoulli.* If $X \sim \mathrm{Ber}(p)$ takes the value 1 with probability p and 0 with probability $1-p$, then
$$M_X(t) = (1-p)e^0 + p\, e^t = 1 - p + pe^t$$
finite for all $t \in \mathbb{R}$, so $D_X = \mathbb{R}$. Differentiating, $M_X'(t) = pe^t$ and $M_X''(t) = pe^t$, so $M_X'(0) = p = \mathbb{E}[X]$ and $M_X''(0) = p = \mathbb{E}[X^2]$, whence $\mathrm{Var}(X) = p - p^2 = p(1-p)$, matching the direct computation.

2. *Binomial via multiplicativity.* If $Y \sim \mathrm{Bin}(n, p)$ then $Y = X_1 + \cdots + X_n$ is a sum of n independent $\mathrm{Ber}(p)$ variables (proposition 2.3.4). By proposition 3.4.6,
$$M_Y(t) = \big(1 - p + pe^t\big)^n$$
This recovers the binomial law without a convolution integral: the coefficient of e^{kt} in the binomial expansion of $(1-p+pe^t)^n$ is $\binom{n}{k}p^k(1-p)^{n-k}$, which is exactly the mass function of $\mathrm{Bin}(n,p)$, matching example 2.3.6. Differentiating at 0 gives $\mathbb{E}[Y] = np$.

3. *Poisson.* If $X \sim \mathrm{Poi}(\lambda)$ with mass function $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$, then
$$M_X(t) = \sum_{k=0}^{\infty} e^{tk}\,\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} = e^{-\lambda}e^{\lambda e^t} = \exp\!\big(\lambda(e^t - 1)\big)$$

finite for all $t \in \mathbb{R}$. Then $M_X'(t) = \lambda e^t \exp(\lambda(e^t - 1))$, so $M_X'(0) = \lambda = \mathbb{E}[X]$; a second differentiation gives $\mathbb{E}[X^2] = \lambda + \lambda^2$ and hence $\mathrm{Var}(X) = \lambda$, the equality of mean and variance characteristic of the Poisson law. The product form also shows at once that an independent sum of Poissons is Poisson: $M_{X+X'}(t) = \exp((\lambda + \lambda')(e^t - 1))$.

4. *Standard Gaussian.* If $Z \sim \mathcal{N}(0,1)$ has density $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, then completing the square in the exponent,

$$M_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{tx} e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-t)^2/2}\, dx\; e^{t^2/2} = e^{t^2/2}$$

where the substitution $x \mapsto x + t$ leaves the Gaussian integral equal to 1. Thus $D_Z = \mathbb{R}$ and $M_Z(t) = e^{t^2/2}$. Expanding, $M_Z(t) = \sum_k t^{2k}/(2^k k!)$, so matching Taylor coefficients against $\sum_k \mathbb{E}[Z^k] t^k/k!$ (theorem 3.4.5) gives the moments of the standard normal: the odd moments vanish and $\mathbb{E}[Z^{2k}] = (2k)!/(2^k k!) = (2k-1)!!$, the product of the odd numbers up to $2k - 1$. In particular $\mathbb{E}[Z] = 0$ and $\mathbb{E}[Z^2] = 1$.

The four MGFs above are the ones that reappear whenever an independent sum is analyzed. Their common feature is finiteness on all of $\mathbb{R}$, hence moments of every order and an analytic MGF, so each is a case where the moment method applies without reservation. The Gaussian occupies a distinguished place: its MGF $e^{t^2/2}$ is the fixed point toward which normalized sums are driven, in the precise sense that $M(t/\sqrt{n})^n \to e^{t^2/2}$ for a mean-zero, unit-variance summand with a finite MGF near 0, which is the moment-method preview of the central limit theorem. The characteristic function $\mathbb{E}[e^{itX}]$, which replaces the MGF when the latter fails to exist, carries out that limit in full generality in the chapters on weak convergence to come; the MGF developed here is the transparent special case where every step is a convergent integral rather than a contour argument.

Exercise 3.4.1

1. Let $X \sim \mathrm{Unif}[0,1]$. Compute $\mathbb{E}[X^k]$ for every $k \geq 1$ directly, and then compute the MGF $M_X(t)$ for $t \neq 0$. Verify that $M_X'(0)$ recovers $\mathbb{E}[X] = \frac{1}{2}$, and identify the Taylor coefficients of M_X with the moments you found.

2. Let $X \sim \mathrm{Poi}(\lambda)$. Using the MGF $M_X(t) = \exp(\lambda(e^t - 1))$ from example 3.4.9, compute the third central moment $\mathbb{E}[(X - \lambda)^3]$. (Differentiate to obtain the first three moments, then expand the central moment.)

3. Show that if X has MGF M_X finite near 0, then for any constants $a, b \in \mathbb{R}$ the affine transform $aX + b$ has MGF $M_{aX+b}(t) = e^{bt} M_X(at)$. Deduce that a general Gaussian $Y \sim \mathcal{N}(\mu, \sigma^2)$, obtained as $\mu + \sigma Z$ with $Z \sim \mathcal{N}(0,1)$, has MGF $M_Y(t) = \exp\bigl(\mu t + \frac{1}{2}\sigma^2 t^2\bigr)$.

4. Use the Chernoff bound (proposition 3.4.7) to prove the upper-tail estimate $\mathbb{P}(Y \geq a) \leq e^{-\lambda}(e\lambda/a)^a$ for $Y \sim \mathrm{Poi}(\lambda)$ and $a > \lambda$. (Optimize $e^{-ta}\exp(\lambda(e^t - 1))$ over $t \geq 0$.)

5. Give a random variable $X \geq 0$ whose moments $\mathbb{E}[X^k]$ are all finite but whose MGF domain is $D_X = \{0\}$, that is $M_X(t) = \infty$ for every $t > 0$. (Consider a law with a tail decaying like $e^{-\sqrt{x}}$, for example a density proportional to $e^{-\sqrt{x}}$ on $[1, \infty)$; explain why every moment is finite yet $\mathbb{E}[e^{tX}] = \infty$ for $t > 0$.)

6. Let S_n be the symmetric ± 1 coin walk of example 3.4.8. Using the subgaussian tail $\mathbb{P}(S_n \geq$

$a) \le e^{-a^2/(2n)}$, show that for every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}\big(S_n \ge (1+\varepsilon)\sqrt{2n\log n}\big) < \infty$$

and explain, invoking the first Borel-Cantelli lemma (theorem 1.4.5), what this says about the size of S_n for large n. (This is the summability that seeds the upper bound in the law of the iterated logarithm.)

Part II

Limit Theorems

The apparatus assembled in Part I (a probability space, random variables and their laws, independence, and expectation with its inequalities) now goes to work on the central question of the subject. Given a sequence of independent random variables with a common law, what governs the long-run behaviour of their partial sums $S_n = X_1 + \cdots + X_n$? The answer is a sequence of increasingly precise limit theorems, each sharpening the last.

The progression is one of resolution. The laws of large numbers fix the first-order behaviour: the average S_n/n converges to the mean, almost surely. The central limit theorem resolves the fluctuation around that mean: S_n, recentered and rescaled by $\sqrt{n}$, converges in distribution to a Gaussian, whatever the law of the steps. Between them sits the machinery of weak convergence and characteristic functions that makes the central limit theorem provable, and beyond them the law of the iterated logarithm, which pins the almost-sure envelope of the fluctuations to the exact scale $\sqrt{2n \log \log n}$.

This chapter is the foundation of that progression. Before the averages can be shown to converge, the sums themselves must be controlled: how large the running maximum $\max_{k \le n} |S_k|$ can grow, and whether an infinite series $\sum_n X_n$ of independent terms converges at all. The two tools developed here, Kolmogorov's maximal inequality and the three-series theorem, are the engine on which the strong law and the finer results downstream depend.

4

Sums of Independent Random Variables

The partial sums $S_n = X_1 + \cdots + X_n$ of an independent sequence are the objects the rest of the book studies. Before their averages and fluctuations can be analyzed, two prior questions must be settled: how large the sums can get along the way, and whether, when the terms are small enough in aggregate, the infinite series $\sum_n X_n$ converges at all. Both are answered by controlling the entire trajectory $S_1, S_2, \dots$ at once rather than any single S_n, and the instrument for that control is a maximal inequality.

section 4.1 proves Kolmogorov's maximal inequality, which bounds the probability that the running maximum $\max_{k\leq n} |S_k|$ ever crosses a level by the same variance that bounds a single S_n through Chebyshev, a strengthening obtained essentially for free from independence. Section 4.2 turns this into convergence criteria for series of independent terms, culminating in Kolmogorov's three-series theorem, a complete characterization of the almost-sure convergence of $\sum_n X_n$. Section 4.3 converts series convergence into statements about averages by way of Kronecker's lemma, delivering a strong law of large numbers under a variance condition and connecting the whole circle back to the zero-one law of section 2.4. Throughout, the lesson is that independence promotes control of one sum into control of the whole sequence.

4.1 Kolmogorov's Maximal Inequality

The story of Part I closed with the standard inequalities: Markov's inequality bounds the tail of a nonnegative random variable by its mean, and its second-moment specialization, Chebyshev's inequality (corollary 3.2.5), bounds $\mathbb{P}(|S_n| \geq \lambda)$ by $\mathrm{Var}(S_n)/\lambda^2$. Those inequalities constrain a random variable at a single instant. The sums $S_n = X_1 + \cdots + X_n$ of an independent sequence, however, are not to be studied one at a time: their almost-sure behaviour, the convergence of the series $\sum_n X_n$, and the strong law of large numbers all hinge on what the sequence $S_1, S_2, \ldots$ does over its entire history, not on where it happens to sit at one fixed index. The governing quantity is therefore the running maximum $\max_{k \le n} |S_k|$, and the question is how often the trajectory ever strays far from the origin.

A first attempt is crude but instructive. The event $\{\max_{k\le n} |S_k| \ge \lambda\}$ is the union of the n events $\{|S_k| \ge \lambda\}$, so the union bound gives

$$\mathbb{P}\Big(\max_{k\le n} |S_k| \ge \lambda\Big) \le \sum_{k=1}^{n} \mathbb{P}(|S_k| \ge \lambda) \le \sum_{k=1}^{n} \frac{\operatorname{Var}(S_k)}{\lambda^2}$$

by Chebyshev applied to each S_k. For the coin walk, where $\operatorname{Var}(S_k) = k$, this sums to $n(n+1)/(2\lambda^2)$, a bound of order n^2/λ^2. The single-time Chebyshev bound $\mathbb{P}(|S_n| \ge \lambda) \le n/\lambda^2$ is of order n/λ^2, so the union bound has lost an entire factor of n: controlling the whole trajectory this way costs far more than controlling one endpoint. Kolmogorov's maximal inequality removes that loss completely. It bounds the running maximum by $\operatorname{Var}(S_n)/\lambda^2$, exactly the Chebyshev bound for the single value S_n. The whole history of the walk is controlled at the price of its endpoint, and the mechanism that makes this possible is independence.

The idea of the proof is to look at the trajectory up to the *first* time it crosses the level λ. Splitting the crossing event by its first-passage index makes the constituent events disjoint, and independence lets the remainder of the walk after the crossing be discarded without loss, because it contributes a nonnegative amount that is uncorrelated with what has happened so far.

Theorem 4.1.1: Kolmogorov's Maximal Inequality

Let $X_1, \ldots, X_n$ be independent random variables (definition 2.2.2) with $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[X_i^2] < \infty$ for each i, and set $S_k = X_1 + \cdots + X_k$ for $1 \le k \le n$, with $S_0 = 0$. Then for every $\lambda > 0$,

$$\mathbb{P}\Big(\max_{1\le k\le n} |S_k| \ge \lambda\Big) \le \frac{\operatorname{Var}(S_n)}{\lambda^2} = \frac{1}{\lambda^2}\sum_{i=1}^{n} \operatorname{Var}(X_i)$$

***Proof*:**
Since each X_i has mean zero, $\mathbb{E}[S_k] = 0$ and $\operatorname{Var}(S_k) = \mathbb{E}[S_k^2]$ for every k; the final equality $\operatorname{Var}(S_n) = \sum_{i\le n} \operatorname{Var}(X_i)$ is the additivity of variance for independent summands (proposition 3.3.7). The work is in the first inequality.

Step 1: The first-passage decomposition. For $1 \le k \le n$ let

$$A_k = \{|S_1| < \lambda,\ \ldots,\ |S_{k-1}| < \lambda,\ |S_k| \ge \lambda\}$$

be the event that k is the *first* index at which the walk reaches level λ (for $k = 1$ this reads $A_1 = \{|S_1| \ge \lambda\}$, with no prior constraints). The A_k are pairwise disjoint by construction: at most one index can be the first crossing. Their union is exactly the event that a crossing occurs at all,

$$\bigsqcup_{k=1}^{n} A_k = \Big\{\max_{1\le k\le n} |S_k| \ge \lambda\Big\} \tag{4.1}$$

since the running maximum reaches λ precisely when some $|S_k|$ does, and then there is a first such k.

Step 2: Discarding the future of the walk. Since $S_n^2 \geq 0$ and the A_k are disjoint,

$$\operatorname{Var}(S_n) = \mathbb{E}[S_n^2] \geq \mathbb{E}\Big[S_n^2\,;\,\bigsqcup_k A_k\Big] = \sum_{k=1}^{n} \mathbb{E}[S_n^2\,;\,A_k] \tag{4.2}$$

where $\mathbb{E}[Z\,;\,A]$ abbreviates $\mathbb{E}[Z\mathbb{1}_A]$. On each event A_k split the endpoint at the crossing time by writing $S_n = S_k + (S_n - S_k)$, so that

$$S_n^2 = S_k^2 + 2\,S_k(S_n - S_k) + (S_n - S_k)^2$$

Multiplying by $\mathbb{1}_{A_k}$ and taking expectations gives three terms. The last is nonnegative, $\mathbb{E}[(S_n - S_k)^2\mathbb{1}_{A_k}] \geq 0$, and may be dropped. The middle term vanishes, which is the crux of the argument and where independence enters.

Step 3: The cross term vanishes by independence. The increment $S_n - S_k = X_{k+1} + \cdots + X_n$ is a function of $X_{k+1}, \ldots, X_n$ alone. The product $S_k\mathbb{1}_{A_k}$ is a function of $X_1, \ldots, X_k$ alone, because S_k depends only on those variables and the event A_k is defined by the constraints $|S_1| < \lambda, \ldots, |S_{k-1}| < \lambda, |S_k| \geq \lambda$, each of which involves only $X_1, \ldots, X_k$. The two blocks of variables $\{X_1, \ldots, X_k\}$ and $\{X_{k+1}, \ldots, X_n\}$ are independent, so the random variables $S_k\mathbb{1}_{A_k}$ and $S_n - S_k$ are independent (theorem 2.3.1). Independent integrable variables have factoring expectation, and $\mathbb{E}[S_n - S_k] = \sum_{i>k} \mathbb{E}[X_i] = 0$, so

$$\mathbb{E}[S_k(S_n - S_k)\mathbb{1}_{A_k}] = \mathbb{E}[S_k\mathbb{1}_{A_k}] \cdot \mathbb{E}[S_n - S_k] = \mathbb{E}[S_k\mathbb{1}_{A_k}] \cdot 0 = 0$$

Thus the cross term contributes nothing, and $\mathbb{E}[S_n^2\mathbb{1}_{A_k}] \geq \mathbb{E}[S_k^2\mathbb{1}_{A_k}]$.

Step 4: The crossing threshold. On A_k the walk has by definition reached the level, $|S_k| \geq \lambda$, so $S_k^2 \geq \lambda^2$ pointwise on that event. Hence

$$\mathbb{E}[S_k^2\mathbb{1}_{A_k}] \geq \lambda^2\,\mathbb{E}[\mathbb{1}_{A_k}] = \lambda^2\,\mathbb{P}(A_k)$$

Combining Steps 2 through 4, $\mathbb{E}[S_n^2\mathbb{1}_{A_k}] \geq \lambda^2\mathbb{P}(A_k)$ for each k.

Step 5: Summation. Substituting into (4.2) and using the disjointness (4.1),

$$\operatorname{Var}(S_n) \geq \sum_{k=1}^{n} \mathbb{E}[S_n^2\mathbb{1}_{A_k}] \geq \lambda^2 \sum_{k=1}^{n} \mathbb{P}(A_k) = \lambda^2\,\mathbb{P}\Big(\max_{1\leq k\leq n} |S_k| \geq \lambda\Big)$$

Dividing by λ^2 gives the claimed bound, as we sought to show.

The proof repays a second reading, because its structure is the template for every maximal inequality to come, including Doob's inequalities for martingales in the final chapters. The decomposition on the first-passage index is what turns a statement about the maximum, an object with no additive structure, into a sum over the disjoint events A_k, objects to which linearity of expectation applies. Independence then does the work in Step 3: once the walk has first crossed λ at time k, everything it does afterwards is independent of the crossing and, having mean zero, neither helps nor hurts the second moment on average. That is why $\mathbb{E}[S_n^2\mathbb{1}_{A_k}]$ can be bounded below by $\mathbb{E}[S_k^2\mathbb{1}_{A_k}]$: the future increment $S_n - S_k$ can only add variance, never subtract it. The mean-zero hypothesis is what drives this cancellation; without it the cross term $\mathbb{E}[S_k\mathbb{1}_{A_k}] \cdot \mathbb{E}[S_n - S_k]$ need not vanish, and the whole argument fails (example 4.1.3 and the exercises examine what the hypothesis gives).

There is a way to see the same phenomenon that clarifies its mechanism. The sequence $S_1^2, S_2^2, \ldots$ is not monotone, but the mean-zero independence makes $\mathbb{E}[S_n^2|X_1, \ldots, X_k] \geq S_k^2$ almost surely: conditioning on the first k steps, the expected future square equals the current square plus the variance of the increment $S_n - S_k$, hence is at least the current square. A process with that property is a *submartingale*, and Kolmogorov's inequality is the first instance of a maximal inequality for submartingales, a theme that returns in full generality in the martingale chapters. For now the inequality stands on its own, proved by hand from independence, and it is exactly the tool the next section needs.

The gain over the single-time Chebyshev bound is worth isolating as a statement, because it is the whole point: the maximal inequality controls an entire trajectory for the price of one endpoint.

Proposition 4.1.2: Trajectory Control at the Chebyshev Price

Under the hypotheses of theorem 4.1.1, Chebyshev's inequality (corollary 3.2.5) applied to the single sum S_n gives

$$\mathbb{P}(|S_n| \geq \lambda) \leq \frac{\mathrm{Var}(S_n)}{\lambda^2}$$

and Kolmogorov's maximal inequality gives the same bound for the running maximum,

$$\mathbb{P}\Big(\max_{1\leq k\leq n} |S_k| \geq \lambda\Big) \leq \frac{\mathrm{Var}(S_n)}{\lambda^2}$$

Since $|S_n| \leq \max_{k\leq n} |S_k|$, the left-hand event of the second bound contains that of the first, so the maximal inequality is a strict strengthening of Chebyshev applied to S_n, at no cost in the bound.

***Proof*:**
The first display is corollary 3.2.5 with the random variable S_n, whose mean is $\mathbb{E}[S_n] = \sum_{i\leq n} \mathbb{E}[X_i] = 0$, so that $\mathbb{E}[S_n^2] = \mathrm{Var}(S_n)$; the second is theorem 4.1.1. The containment $\{|S_n| \geq \lambda\} \subseteq \{\max_{k\leq n} |S_k| \geq \lambda\}$ holds because $|S_n|$ is one of the terms in the maximum, so whenever $|S_n| \geq \lambda$ the maximum is at least λ as well. Monotonicity of probability then makes the second probability at least the first, yet both are bounded by the same $\mathrm{Var}(S_n)/\lambda^2$, completing the proof.

The content of proposition 4.1.2 is a statement about what independence gives. For an arbitrary (dependent) sequence one has no such inequality: the union bound of order n^2/λ^2 computed above is essentially the best a general argument gives, and it can be attained. The improvement to $\mathrm{Var}(S_n)/\lambda^2$ is a return on the independence hypothesis, and it is the reason the running maximum of a sum of independent variables can be treated with the same ease as a single sum. Every convergence theorem in the next section is an application of exactly this leverage.

Example 4.1.3: The Coin Walk and Non-Identical Sums

Two instances make the inequality concrete.

1. *The simple random walk.* Let $X_1, X_2, \ldots$ be independent signs taking values ± 1 with probability $\frac{1}{2}$ each, the steps of the coin walk from example 1.3.6, and let $S_n = X_1 + \cdots + X_n$. Each step has $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] = 1$, so $\mathrm{Var}(S_n) = n$ (proposition 3.3.7).

Kolmogorov's inequality gives

$$\mathbb{P}\Big(\max_{1\le k\le n} |S_k| \ge \lambda\Big) \le \frac{n}{\lambda^2}$$

The single-time Chebyshev bound is $\mathbb{P}(|S_n| \ge \lambda) \le n/\lambda^2$, the identical right-hand side. Both are of order n/λ^2, and the maximal inequality delivers the stronger left-hand statement (the maximum, not the endpoint) for free. Taking $\lambda = t\sqrt{n}$ recasts this as

$$\mathbb{P}\Big(\max_{1\le k\le n} |S_k| \ge t\sqrt{n}\Big) \le \frac{1}{t^2}$$

a bound on the probability that the walk ever wanders more than t standard deviations from the origin during its first n steps, uniform in n. By contrast, the union bound of the section opening gives here $n(n+1)/(2\lambda^2)$, of order n^2/λ^2, weaker by a factor of n.

2. *Non-identically-distributed summands.* Independence does not require identical distributions. Let $Y_1, Y_2, \dots$ be independent with Y_i uniform on $\{-i, +i\}$, so $\mathbb{E}[Y_i] = 0$ and $\mathrm{Var}(Y_i) = i^2$. Writing $T_k = Y_1 + \cdots + Y_k$, additivity of variance gives $\mathrm{Var}(T_n) = \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$, of order $n^3/3$. The maximal inequality reads

$$\mathbb{P}\Big(\max_{1\le k\le n} |T_k| \ge \lambda\Big) \le \frac{n(n+1)(2n+1)}{6\lambda^2}$$

The variances grow, so the walk spreads faster, and the bound tracks the total accumulated variance $\mathrm{Var}(T_n)$ rather than a per-step constant. This is exactly the form the convergence theorems of the next section exploit: what enters is $\sum_i \mathrm{Var}(X_i)$, whatever the individual laws.

The uniform-in-n form of the coin-walk bound is the first hint of the section's payoff. The estimate $\mathbb{P}(\max_{k\le n} |S_k| \ge \lambda) \le n/\lambda^2$ holds for every n at once, so letting λ grow with n (as $t\sqrt{n}$) keeps the probability bounded no matter how long the walk runs. This is precisely the kind of control needed to make statements about the infinite trajectory rather than any finite piece of it. Applied not to the whole walk but to the tail increments $\sum_{k>m} X_k$, the same inequality shows the partial sums form an almost-surely Cauchy sequence whenever the variances are summable, and that is the content of the one-series theorem developed next.

The mean-zero hypothesis deserves one more remark, because it is the pivot of the proof and its role is easy to underappreciate. If the summands have a common nonzero mean μ, then $S_n \approx n\mu$ drifts linearly, and the running maximum $\max_{k\le n} |S_k|$ is dominated by this drift rather than by the fluctuations; a variance bound cannot then control the maximum, and indeed the cross term in Step 3 no longer vanishes. The correct object in the presence of drift is the centered walk $S_k - \mathbb{E}[S_k] = \sum_{i\le k}(X_i - \mathbb{E}[X_i])$, to which the theorem applies verbatim. This centering is the routine first move in every application to come: subtract the mean, bound the fluctuations of the centered sum, and read the drift off separately.

Exercise 4.1.1

1. Let $X_1, \dots, X_n$ be independent, mean-zero, and L^2, with partial sums S_k. Prove the one-sided

(upper) maximal inequality

$$\mathbb{P}\Big(\max_{1\le k\le n} S_k \ge \lambda\Big) \le \frac{\mathrm{Var}(S_n)}{\lambda^2 + \mathrm{Var}(S_n)} \qquad (\lambda > 0)$$

(Hint: this is the reverse-tail refinement. Run the first-passage argument on $A_k = \{S_1 < \lambda, \dots, S_{k-1} < \lambda,\, S_k \ge \lambda\}$, but bound $\mathbb{E}[S_n^2 \mathbb{1}_{A_k}]$ below using $(S_k + c)^2$ for a constant $c \ge 0$ to be optimized, exploiting that $S_k \ge \lambda$ on A_k and that $S_n - S_k$ is independent of A_k with mean zero.)

2. Show that Kolmogorov's maximal inequality (theorem 4.1.1) implies Chebyshev's inequality for S_n (corollary 3.2.5) but not conversely, by exhibiting an independent mean-zero sequence for which the maximal bound is strictly stronger than what the union bound over $\{|S_k| \ge \lambda\}$ yields.

3. Where exactly does the mean-zero hypothesis enter the proof of theorem 4.1.1? Give an explicit independent sequence X_1, X_2 with a common nonzero mean for which the inequality $\mathbb{P}(\max_{k\le 2} |S_k| \ge \lambda) \le \mathrm{Var}(S_n)/\lambda^2$ *fails*, and identify which step of the proof breaks.

4. (Etemadi-style maximal inequality without the mean-zero hypothesis.) Let $X_1, \dots, X_n$ be independent, and let S_k be the partial sums (no moment or mean assumption). Prove that for every $\lambda > 0$,
$$\mathbb{P}\Big(\max_{1\le k\le n} |S_k| \ge 3\lambda\Big) \le 3 \max_{1\le k\le n} \mathbb{P}(|S_k| \ge \lambda)$$
(Hint: decompose on the first index k with $|S_k| \ge 3\lambda$; on that event either $|S_n| \ge \lambda$ or $|S_n - S_k| \ge 2\lambda$, and $S_n - S_k$ is independent of A_k.)

5. For the coin walk S_n of example 4.1.3, use the maximal inequality to bound $\mathbb{P}(\max_{k\le n} |S_k| \ge 3\sqrt{n})$, and compare with the single-time Chebyshev bound on $\mathbb{P}(|S_n| \ge 3\sqrt{n})$. Then compute both bounds numerically for $n = 10^4$.

6. Let (X_n) be independent, mean-zero, and L^2 with $\sum_n \mathrm{Var}(X_n) < \infty$. Using theorem 4.1.1 applied to the tail block $X_{m+1}, \dots, X_{m+p}$, show that
$$\mathbb{P}\Big(\max_{1\le p\le N} |S_{m+p} - S_m| \ge \lambda\Big) \le \frac{1}{\lambda^2} \sum_{i>m} \mathrm{Var}(X_i)$$
uniformly in N, and deduce that $\mathbb{P}(\sup_{p\ge 1} |S_{m+p} - S_m| \ge \lambda) \le \lambda^{-2} \sum_{i>m} \mathrm{Var}(X_i)$. (This is the estimate the next section turns into almost-sure convergence.)

4.2 Convergence of Random Series

The maximal inequality of the previous section controls the whole trajectory of the partial sums for a fixed number of terms. The natural next question is what happens in the limit: given an independent sequence (X_n), does the infinite series $\sum_n X_n$ converge? This is a new kind of question, because convergence of the series is a statement about the tail of the sequence, and independence forces a stark dichotomy on such statements. Fix a point ω in the sample space; whether the sequence of partial sums $S_n(\omega) = X_1(\omega) + \cdots + X_n(\omega)$ converges depends only on the terms $X_m(\omega), X_{m+1}(\omega), \dots$ from any index m onward, since altering finitely many terms changes the limit's existence not at all. The event $\{\sum_n X_n$ converges$\}$ therefore lies in the tail σ-algebra $\bigcap_m \sigma(X_m, X_{m+1}, \dots)$ of definition 2.4.1. By Kolmogorov's zero-one law theorem 2.4.3 its probability is 0 or 1: the series either converges almost surely or diverges almost surely, with nothing in between. The whole task of this section is to decide which, and to do so from computable features of the individual laws of the X_n.

The zero-one law is what makes the question tractable. Without it one might fear a series that converges on some middling event of probability $\frac{1}{2}$; the law rules this out in advance and reduces the problem to producing a single sharp criterion. The three main results below build that criterion in stages: first for mean-zero terms with summable variances, then, by a truncation device, for completely general independent terms. The device that bridges the two is worth naming at the outset, since it recurs throughout the limit theory of sums.

Definition 4.2.1: Truncation

Let X be a random variable and $A > 0$ a level. The *truncation* of X at level A is the random variable

$$Y = X\mathbb{1}_{\{|X| \le A\}}$$

which agrees with X where $|X| \le A$ and is set to 0 where $|X| > A$. For an independent sequence (X_n) we write $Y_n = X_n\mathbb{1}_{\{|X_n| \le A\}}$ for the truncation of each term at the common level A.

Truncation replaces a possibly ill-behaved variable, whose expectation or variance may not even exist, by a bounded one for which every moment is finite: $|Y_n| \le A$, so $Y_n \in L^p$ for every p (definition 3.3.1). Two features make it the right tool here. First, the truncated variables inherit independence, because $Y_n = f(X_n)$ for the fixed Borel function $f(x) = x\mathbb{1}_{\{|x| \le A\}}$, and a sequence of functions of independent variables is independent (definition 2.2.2). Second, Y_n and X_n differ only on the event $\{|X_n| > A\}$, so if these events are rare enough, in the sense that $\sum_n \mathbb{P}(|X_n| > A) < \infty$, the two series $\sum_n X_n$ and $\sum_n Y_n$ have the same convergence behaviour by the first Borel-Cantelli lemma. The three-series theorem is precisely the bookkeeping that turns these two observations into a criterion.

To ground the definition immediately, take X uniform on $[-2, 2]$ and truncate at level $A = 1$. Then $Y = X$ on $\{|X| \le 1\}$, an event of probability $\frac{1}{2}$, and $Y = 0$ on $\{|X| > 1\}$. By symmetry $\mathbb{E}[Y] = 0$, while the variance is

$$\mathrm{Var}(Y) = \mathbb{E}[Y^2] = \int_{-1}^{1} x^2 \cdot \tfrac{1}{4}\, dx = \tfrac{1}{4} \cdot \tfrac{2}{3} = \tfrac{1}{6}$$

so truncation has produced a bounded, mean-zero, finite-variance surrogate for X, ready for the variance-based machinery below.

The first substantial result assumes the terms are already centered and have summable variances. Under this hypothesis the maximal inequality does all the work: it shows that the partial sums form an almost surely Cauchy sequence, which in a complete space is the same as convergence.

The mechanism deserves a word before the statement. Convergence of a real sequence is equivalent to its being Cauchy, and being Cauchy means the oscillation $\sup_{k>m} |S_k - S_m|$ over the tail shrinks to 0 as $m \to \infty$. This oscillation is exactly a running maximum of the kind the maximal inequality bounds, applied not to $S_1, \dots, S_n$ but to the block of increments past index m. Summable variances make that bound tend to 0, and the Cauchy criterion converts the analytic bound into almost sure convergence.

Theorem 4.2.2: Kolmogorov's One-Series Theorem

Let (X_n) be independent random variables with $\mathbb{E}[X_n] = 0$ for every n and

$$\sum_{n=1}^{\infty} \operatorname{Var}(X_n) < \infty$$

Then $\sum_n X_n$ converges almost surely.

***Proof*:**
Write $S_n = X_1 + \cdots + X_n$. The strategy is to show that the sequence (S_n) is almost surely Cauchy; since $\mathbb{R}$ is complete, almost sure Cauchy convergence is almost sure convergence.

Step 1: A maximal bound on each tail block. Fix indices $m < m'$ and a level $\varepsilon > 0$. Consider the finite block of increments $X_{m+1}, \dots, X_{m'}$. These are independent, mean zero, and in L^2 by hypothesis, so theorem 4.1.1 applies to their partial sums. The k-th partial sum of the block is $S_{m+k} - S_m$, and its variance is

$$\operatorname{Var}(S_{m+k} - S_m) = \sum_{j=m+1}^{m+k} \operatorname{Var}(X_j)$$

by proposition 3.3.7. Taking $\lambda = \varepsilon$ and the largest block length, the maximal inequality gives

$$\mathbb{P}\Big(\max_{m<k\le m'} |S_k - S_m| \ge \varepsilon\Big) \le \frac{1}{\varepsilon^2} \sum_{j=m+1}^{m'} \operatorname{Var}(X_j) \tag{4.3}$$

The right side of (4.3) is a partial sum of the convergent series $\sum_j \operatorname{Var}(X_j)$, hence a tail that is finite, and it does not depend on the truncation length m' except monotonically.

Step 2: Let the block grow. Fix m and let $m' \to \infty$. The events $\{\max_{m<k\le m'} |S_k - S_m| \ge \varepsilon\}$ increase to $\{\sup_{k>m} |S_k - S_m| \ge \varepsilon\}$, so by continuity from below (proposition 1.1.5),

$$\mathbb{P}\Big(\sup_{k>m} |S_k - S_m| \ge \varepsilon\Big) \le \frac{1}{\varepsilon^2} \sum_{j>m} \operatorname{Var}(X_j) \tag{4.4}$$

the bound in (4.4) being the limit of the right side of (4.3) as $m' \to \infty$.

Step 3: Let the base grow. As $m \to \infty$ the tail sum $\sum_{j>m} \operatorname{Var}(X_j)$ tends to 0, because the full series converges. Hence the right side of (4.4) tends to 0, and so for every fixed $\varepsilon > 0$,

$$\lim_{m\to\infty} \mathbb{P}\Big(\sup_{k>m} |S_k - S_m| \ge \varepsilon\Big) = 0$$

Now define the oscillation of the tail past index m by $D_m = \sup_{k,\ell>m} |S_k - S_\ell|$. By the triangle inequality $D_m \le 2\sup_{k>m} |S_k - S_m|$, so $\mathbb{P}(D_m \ge 2\varepsilon) \le \mathbb{P}(\sup_{k>m} |S_k - S_m| \ge \varepsilon) \to 0$. The sequence D_m is nonincreasing in m, hence converges pointwise to a limit $D_\infty = \inf_m D_m \ge 0$. For each $\varepsilon > 0$ the event $\{D_\infty \ge 2\varepsilon\}$ is contained in $\{D_m \ge 2\varepsilon\}$ for every m, so $\mathbb{P}(D_\infty \ge 2\varepsilon) \le \inf_m \mathbb{P}(D_m \ge 2\varepsilon) = 0$.

Step 4: Conclude. Letting ε run through $1/j$ for $j \in \mathbb{N}$ and taking the union of the null events $\{D_\infty \ge 2/j\}$ shows $\mathbb{P}(D_\infty > 0) = 0$, so $D_\infty = 0$ almost surely. On the event $\{D_\infty = 0\}$ the oscillations $D_m \downarrow 0$, which is exactly the Cauchy criterion for the sequence (S_n). Thus (S_n) is Cauchy almost surely, and by completeness of $\mathbb{R}$ it converges almost surely, completing the proof.

The one-series theorem is the section's main result, and its hypotheses map cleanly onto the two things that can wreck a series of random terms. Mean zero prevents the systematic drift that a nonzero mean would accumulate: a series $\sum_n c$ of a positive constant diverges no matter how the randomness behaves, and centering removes exactly this deterministic part. Summable variance controls the random fluctuation: it forces the increments to become small in L^2 fast enough that their accumulated wobble is finite. When both hold, the series settles down. The converse fails for the mean condition alone but, with the truncation device, an exact converse is available at the level of the full three-series criterion below.

A first consequence is worth extracting, since it converts the theorem into a statement about a general (uncentered) series once the means are summable on their own.

Corollary 4.2.3: Centering a Summable-Variance Series

Let (X_n) be independent with $\sum_n \operatorname{Var}(X_n) < \infty$. Then $\sum_n (X_n - \mathbb{E}[X_n])$ converges almost surely. If in addition $\sum_n \mathbb{E}[X_n]$ converges, then $\sum_n X_n$ converges almost surely.

***Proof*:**
The variables $X_n - \mathbb{E}[X_n]$ are independent (functions of independent variables), have mean zero, and $\operatorname{Var}(X_n - \mathbb{E}[X_n]) = \operatorname{Var}(X_n)$, so their variances are summable. Theorem 4.2.2 gives almost sure convergence of $\sum_n (X_n - \mathbb{E}[X_n])$. Adding the convergent deterministic series $\sum_n \mathbb{E}[X_n]$ term by term yields convergence of $\sum_n X_n$ on the same almost sure event, as we sought to show.

This corollary already contains the sufficiency direction of the general theorem in embryo: truncate, center the truncations, apply the one-series theorem to the centered truncations, add back the deterministic series of truncated means, and finally pass from the truncations to the original terms. The three-series theorem is the statement that these four steps, each of which requires a series to converge, are not only sufficient but necessary.

The truncations Y_n are bounded, hence have finite mean $\mathbb{E}[Y_n]$ and finite variance $\operatorname{Var}(Y_n)$; the events $\{|X_n| > A\}$ where X_n and Y_n disagree have probabilities $\mathbb{P}(|X_n| > A)$. These are exactly the three quantities whose summation the theorem tests. Before stating it, note why three conditions and not fewer are needed: $\sum \mathbb{P}(|X_n| > A)$ handles the large excursions that truncation discards, $\sum \operatorname{Var}(Y_n)$ handles the fluctuation of what remains, and $\sum \mathbb{E}[Y_n]$ handles the drift of what remains. Each addresses a different way the series could fail to converge, and the examples at the end of the section exhibit a series that passes two of the three but fails the third.

Theorem 4.2.4: Kolmogorov's Three-Series Theorem

Let (X_n) be independent random variables. The series $\sum_n X_n$ converges almost surely if and only if, for some (equivalently, every) level $A > 0$, all three of the following series converge, where $Y_n = X_n \mathbb{1}_{\{|X_n| \le A\}}$ is the truncation of definition 4.2.1:

1. $\displaystyle\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > A)$;
2. $\displaystyle\sum_{n=1}^{\infty} \mathbb{E}[Y_n]$;
3. $\displaystyle\sum_{n=1}^{\infty} \mathrm{Var}(Y_n)$.

***Proof*:**
We prove sufficiency in full, then necessity, isolating in necessity the one step that draws on a tool beyond this chapter.

Sufficiency: the three series converge $\Rightarrow \sum_n X_n$ converges a.s. Fix a level $A > 0$ for which (1), (2), (3) hold. The centered truncations $Y_n - \mathbb{E}[Y_n]$ are independent, mean zero, and have variance $\mathrm{Var}(Y_n - \mathbb{E}[Y_n]) = \mathrm{Var}(Y_n)$, which is summable by (3). By theorem 4.2.2,

$$\sum_{n=1}^{\infty} (Y_n - \mathbb{E}[Y_n]) \quad \text{converges almost surely}$$

The series $\sum_n \mathbb{E}[Y_n]$ converges by (2), and it is deterministic, so adding it back gives that $\sum_n Y_n$ converges almost surely. It remains to pass from Y_n to X_n. By (1), $\sum_n \mathbb{P}(|X_n| > A) < \infty$, so the first Borel-Cantelli lemma theorem 1.4.5 gives $\mathbb{P}(|X_n| > A$ for infinitely many $n) = 0$. Thus for almost every ω there is an index $N(\omega)$ with $|X_n(\omega)| \le A$, hence $X_n(\omega) = Y_n(\omega)$, for all $n \ge N(\omega)$. The two series $\sum_n X_n(\omega)$ and $\sum_n Y_n(\omega)$ then differ only in finitely many terms, so they converge or diverge together. Since $\sum_n Y_n$ converges almost surely, so does $\sum_n X_n$.

Necessity: $\sum_n X_n$ converges a.s. $\Rightarrow$ the three series converge. Suppose $\sum_n X_n$ converges almost surely. The argument proceeds in three steps, running the sufficiency logic backward with a symmetrization device to remove the unknown means.

Step 1: The first series converges. If $\sum_n X_n$ converges almost surely then its terms tend to 0 almost surely, so $\mathbb{P}(|X_n| > A$ for infinitely many $n) = 0$. The events $\{|X_n| > A\}$ are independent, being functions of the independent X_n. By the second Borel-Cantelli lemma theorem 1.4.6, if $\sum_n \mathbb{P}(|X_n| > A)$ diverged then these independent events would occur infinitely often with probability 1, a contradiction. Hence $\sum_n \mathbb{P}(|X_n| > A) < \infty$, which is (1). By the first Borel-Cantelli lemma this also shows $X_n = Y_n$ for all large n almost surely, so $\sum_n Y_n$ converges almost surely as well.

Step 2: Symmetrize and read off the variance series. To extract a variance bound without knowing the means $\mathbb{E}[Y_n]$, introduce an independent copy. Let (Y_n') be an independent sequence with the same laws as (Y_n), independent of the whole (Y_n) sequence; such a sequence exists on a suitable product space by theorem 2.3.5 applied to the given laws. Set

$$Y_n^s = Y_n - Y_n'$$

the *symmetrization* of Y_n. The Y_n^s are independent, bounded by $2A$, symmetric (so $\mathbb{E}[Y_n^s] = 0$), and $\mathrm{Var}(Y_n^s) = 2\mathrm{Var}(Y_n)$. Since $\sum_n Y_n$ converges almost surely and $\sum_n Y_n'$ converges almost surely with the same law, their difference $\sum_n Y_n^s$ converges almost surely. The symmetrized terms are mean zero, so their almost sure convergence forces $\sum_n \mathrm{Var}(Y_n^s) < \infty$; this is the one step that runs the maximal-inequality logic in reverse, and we isolate it below. Granting it, $\sum_n \mathrm{Var}(Y_n^s) = 2\sum_n \mathrm{Var}(Y_n) < \infty$, which is (3).

Step 3: The mean series converges. With (3) in hand, the centered truncations $Y_n - \mathbb{E}[Y_n]$ have summable variance, so $\sum_n (Y_n - \mathbb{E}[Y_n])$ converges almost surely by theorem 4.2.2. But $\sum_n Y_n$ converges almost surely by Step 1. Subtracting the two almost surely convergent series leaves the deterministic series $\sum_n \mathbb{E}[Y_n]$, which must therefore converge; this is (2). All three series converge, completing the argument.

The reversed variance bound (Step 2, isolated). The claim used in Step 2 is a converse to the one-series theorem for bounded symmetric terms: if (Z_n) are independent, symmetric, uniformly bounded by C, and $\sum_n Z_n$ converges almost surely, then $\sum_n \mathrm{Var}(Z_n) < \infty$. The proof is a converse maximal inequality of Kolmogorov type: for independent, mean-zero, bounded terms one has a lower bound $\mathbb{P}(\max_{k\le n} |T_k| \ge \lambda) \ge 1 - (\lambda + C)^2/\mathrm{Var}(T_n)$ on the running maximum $T_k = Z_1 + \cdots + Z_k$, which forces $\mathrm{Var}(T_n)$ to stay bounded when the maximum does not run off to infinity. Establishing that lower bound requires the reverse Kolmogorov inequality, a self-contained but separate variance estimate whose derivation is not developed in this chapter; this one lower bound is the sole input to necessity we take on faith, and every other step of the necessity argument has been given in full above.

The three-series theorem is a complete solution to the convergence problem: it translates an analytic question about an infinite random sum into three deterministic numerical series that can be checked by computing means, variances, and tail probabilities of the individual laws. The equivalence "for some A" versus "for every A" is not vacuous. If the three series converge for one level A, then changing to a level $A' > A$ alters each term Y_n only on the event $\{A < |X_n| \le A'\}$, whose probabilities are dominated by $\sum_n \mathbb{P}(|X_n| > A) < \infty$; a finite total change of probability mass can shift the three quantities by a summable amount, so convergence is preserved. Thus the criterion, once satisfied at any level, is satisfied at all, and one is free to choose whichever A is convenient, typically $A = 1$.

The interpretive payoff loops back to where the section began. The three-series theorem does more than decide convergence; combined with the zero-one law it explains the shape of the answer. Because $\{\sum_n X_n \text{ converges}\}$ is a tail event, the answer is never a probability strictly between 0 and 1, and the theorem identifies the deterministic data, three numerical series, on which the binary verdict depends. Convergence of a random series is thus as deterministically decidable as convergence of a numerical one, once the right three numerical series are formed.

The theorems come to life on the random-signs series, the running example of this section, which shows how independence turns a divergent absolute series into a convergent random one.

Example 4.2.5: Random-Signs Series

Let (ε_n) be independent fair signs, $\mathbb{P}(\varepsilon_n = +1) = \mathbb{P}(\varepsilon_n = -1) = \frac{1}{2}$, and let (a_n) be real numbers.

Consider the random series

$$\sum_{n=1}^{\infty} \varepsilon_n a_n$$

obtained by attaching a random sign to each a_n. The task is to determine exactly when it converges almost surely, then read off two instances.

1. *The dichotomy: convergence iff* $\sum_n a_n^2 < \infty$. The terms $X_n = \varepsilon_n a_n$ are independent, and $\mathbb{E}[X_n] = a_n \mathbb{E}[\varepsilon_n] = 0$, so each is mean zero. Their variances are

$$\mathrm{Var}(X_n) = \mathbb{E}[\varepsilon_n^2] a_n^2 = a_n^2$$

since $\varepsilon_n^2 = 1$. If $\sum_n a_n^2 < \infty$ then theorem 4.2.2 applies directly and $\sum_n \varepsilon_n a_n$ converges almost surely. Conversely, if $\sum_n a_n^2 = \infty$ the three-series test at level $A = 1$ must fail, so by theorem 4.2.4 the series diverges almost surely. To see the failure directly, note the terms $\varepsilon_n a_n$ are already symmetric and bounded when the a_n are bounded; the necessity direction of the three-series theorem then forces $\sum_n \mathrm{Var}(X_n) = \sum_n a_n^2 < \infty$ for convergence. Either way,

$$\sum_{n=1}^{\infty} \varepsilon_n a_n \text{ converges almost surely} \iff \sum_{n=1}^{\infty} a_n^2 < \infty$$

2. *Conditional convergence made almost sure:* $\sum_n \varepsilon_n / n$. Take $a_n = 1/n$. Then $\sum_n a_n^2 = \sum_n 1/n^2 = \pi^2/6 < \infty$, so by part (1) the random-signs harmonic series $\sum_n \varepsilon_n/n$ converges almost surely. This is despite the fact that the series converges nowhere absolutely: $\sum_n |\varepsilon_n/n| = \sum_n 1/n = \infty$ for every ω. The randomness of the signs is what saves it, exactly as the alternating signs save the deterministic alternating harmonic series, but here for almost every sign pattern rather than one special one. By contrast $a_n = 1/\sqrt{n}$ gives $\sum_n a_n^2 = \sum_n 1/n = \infty$, so $\sum_n \varepsilon_n/\sqrt{n}$ diverges almost surely.

3. *A series that needs all three conditions.* The random-signs series is too symmetric to exercise the mean series (2), which vanishes identically there. To see all three conditions bite, let (X_n) be independent with

$$X_n = \begin{cases} n & \text{with probability } n^{-2} \\ c_n & \text{with probability } 1 - n^{-2} \end{cases}$$

where $c_n = -n^{-1}/(1-n^{-2})$ (for $n \geq 2$; the $n = 1$ term is almost surely the constant 1 and does not affect the tail) is chosen so that $\mathbb{E}[X_n] = n \cdot n^{-2} + c_n(1 - n^{-2}) = n^{-1} - n^{-1} = 0$. Truncate at $A = 1$: for $n \geq 2$ the large value n exceeds A, so $Y_n = c_n \mathbb{1}_{\{X_n = c_n\}}$, while $Y_n = 0$ on the event $\{X_n = n\}$. Then:

   - $\sum_n \mathbb{P}(|X_n| > 1) = \sum_n \mathbb{P}(X_n = n) = \sum_n n^{-2} < \infty$, so (1) holds;
   - $\mathbb{E}[Y_n] = c_n(1 - n^{-2}) = -n^{-1}$, so $\sum_n \mathbb{E}[Y_n] = -\sum_n n^{-1} = -\infty$ diverges, and (2) fails.

   Because (2) fails, $\sum_n X_n$ diverges almost surely, even though the untruncated terms are mean zero. The point is that the mean-zero balance is maintained only by the rare large excursions $X_n = n$, which truncation removes; once removed, the surviving part Y_n carries a systematic negative drift $-n^{-1}$ that accumulates to $-\infty$. This is a series on which the naive test "the terms are centered, so it should converge" gives the wrong answer, and only the full three-series bookkeeping detects the failure.

The last two parts of the example isolate two of the three conditions in turn. Part (2) is a pure variance phenomenon: only condition (3) is nontrivial, and it decides everything. Part (3) is a pure mean phenomenon: conditions (1) and (3) hold but (2) fails, and the failure comes from a drift that the mean-zero appearance of the untruncated terms hides. Together they show why no two of the three conditions can be dropped, and why truncation, which separates the drift of the bulk from the mass of the excursions, is the device that makes the separation visible.

There is a broader reading. The random-signs dichotomy $\sum_n a_n^2 < \infty$ is the same condition that characterizes membership in the Hilbert space ℓ^2. A random-signs series converges almost surely exactly when its coefficient sequence is square-summable, so the almost-sure convergence of $\sum_n \varepsilon_n a_n$ is a probabilistic statement about the geometry of ℓ^2 built in chapter 3 through the variance inner product (proposition 3.3.7). This is the first place in the book where the L^2 structure of random variables and the convergence of an infinite sum meet, and the meeting point is the summability of variances that both theorems of this section rest on.

Exercise 4.2.1

1. Let (ε_n) be independent fair signs. For which real exponents p does the random-signs series $\sum_n \varepsilon_n n^{-p}$ converge almost surely? For which p does it converge absolutely for almost every ω? Explain the gap between the two ranges.
2. Let (X_n) be independent with $\mathbb{P}(X_n = 2^{-n}) = \mathbb{P}(X_n = -2^{-n}) = \frac{1}{2}$. Show that $\sum_n X_n$ converges almost surely, and identify the smallest and largest possible values of the limit. What is the law of the sum on the coin-tossing interpretation?
3. Let (X_n) be independent with X_n uniform on $[-b_n, b_n]$ for constants $b_n > 0$. Give a necessary and sufficient condition on (b_n) for $\sum_n X_n$ to converge almost surely, and prove both directions using the results of this section.
4. Give an example of an independent mean-zero sequence (X_n) for which $\sum_n X_n$ converges almost surely but $\sum_n \mathrm{Var}(X_n) = \infty$. Explain why this does not contradict theorem 4.2.2, and check your example against the three-series theorem at level $A = 1$.
5. Let (X_n) be independent Poisson variables, $X_n \sim \mathrm{Poi}(\lambda_n)$, with $\sum_n \lambda_n < \infty$. Determine whether $\sum_n (X_n - \lambda_n)$ converges almost surely, and whether $\sum_n X_n$ converges almost surely.
6. Use the zero-one law and the three-series theorem to prove the following: for any independent sequence (X_n), the set of levels $A > 0$ at which the three series all converge is either empty or all of $(0, \infty)$. (This is the "some A iff every A" equivalence; supply the proof the theorem statement only asserts.)

4.3 Application: Strong Laws from Series Convergence

The previous section decided when an infinite series $\sum_n X_n$ of independent terms converges. That is a statement about the tail of the sequence of partial sums, and on its face it says nothing about the averages S_n/n that a law of large numbers concerns. The bridge between the two is a purely deterministic fact: if a weighted series $\sum_n x_n/b_n$ converges and the weights b_n increase to infinity, then the raw partial sums $\sum_{k \le n} x_k$ grow slower than b_n, so their b_n-average tends to zero. This is Kronecker's lemma. Fed the almost-sure convergence of a random series from theorem 4.2.2, it converts that convergence directly into an almost-sure statement about averages. The output is the first strong law of large numbers of the book: under a summable-variance condition, the centered average $(S_n - \mathbb{E}S_n)/n$ tends to zero almost surely.

The logic runs in one direction and is worth fixing in mind before the details. The maximal inequality of section 4.1 controlled the whole trajectory of the partial sums; the one-series theorem of section 4.2 turned that control into almost-sure convergence of a series; and Kronecker's lemma now turns convergence of a series into convergence of averages. Each step trades a statement about one object for a statement about the entire sequence, and independence is what makes the trades possible.

The deterministic core is elementary but not obvious, and it is cleanest to isolate it completely from probability. There are no random variables in the next lemma: $x_1, x_2, \ldots$ is an arbitrary real sequence and $b_1, b_2, \ldots$ is an arbitrary sequence of positive weights increasing to infinity. The claim is that convergence of the weighted series forces the unweighted averages to vanish. The mechanism is Abel summation, which rewrites a partial sum of products as a boundary term minus a sum of differences, exactly as integration by parts does for an integral of a product.

Lemma 4.3.1: Kronecker's Lemma

Let $(x_n)_{n\geq 1}$ be a sequence of real numbers and let $(b_n)_{n\geq 1}$ be a sequence of positive reals with $b_n \uparrow \infty$, that is, $b_1 \leq b_2 \leq \cdots$ and $b_n \to \infty$. If the series

$$\sum_{n=1}^{\infty} \frac{x_n}{b_n}$$

converges, then

$$\frac{1}{b_n} \sum_{k=1}^{n} x_k \longrightarrow 0 \qquad \text{as } n \to \infty$$

***Proof*:**
Write $u_n = \sum_{k=1}^{n} x_k/b_k$ for the partial sums of the convergent series, and set $u_0 = 0$. By hypothesis $u_n \to u$ for some finite limit u. The individual terms are recovered as $x_k/b_k = u_k - u_{k-1}$, so $x_k = b_k(u_k - u_{k-1})$, and the sum to be controlled is

$$\sum_{k=1}^{n} x_k = \sum_{k=1}^{n} b_k\big(u_k - u_{k-1}\big)$$

Step 1: Summation by parts. The point of Abel summation is to move the difference off the factor $u_k - u_{k-1}$ and onto the weights b_k, where it telescopes against the boundary. Compute

$$\sum_{k=1}^{n} b_k\big(u_k - u_{k-1}\big) = \sum_{k=1}^{n} b_k u_k - \sum_{k=1}^{n} b_k u_{k-1} = b_n u_n - \sum_{k=1}^{n-1}\big(b_{k+1} - b_k\big)u_k$$

where the last equality reindexes the second sum by $k \mapsto k+1$ and uses $u_0 = 0$; explicitly, the second sum equals $\sum_{j=1}^{n} b_j u_{j-1} = \sum_{j=2}^{n} b_j u_{j-1} = \sum_{k=1}^{n-1} b_{k+1} u_k$ (dropping the $j = 1$ term, which is $b_1 u_0 = 0$), and subtracting it from $\sum_{k=1}^{n} b_k u_k = b_n u_n + \sum_{k=1}^{n-1} b_k u_k$ leaves

$b_n u_n - \sum_{k=1}^{n-1}(b_{k+1} - b_k)u_k$. Dividing by b_n,

$$\frac{1}{b_n}\sum_{k=1}^{n} x_k = u_n - \frac{1}{b_n}\sum_{k=1}^{n-1}\big(b_{k+1} - b_k\big)u_k \tag{4.5}$$

Step 2: The averaged tail vanishes. The differences $b_{k+1} - b_k$ are nonnegative because b_n is nondecreasing, and they telescope to

$$\sum_{k=1}^{n-1}\big(b_{k+1} - b_k\big) = b_n - b_1$$

So the coefficients $(b_{k+1} - b_k)/b_n$ in the last term of (4.5) are nonnegative and sum to $(b_n - b_1)/b_n \le 1$. The last term is therefore a weighted average of the values $u_1, \ldots, u_{n-1}$ (with a total weight slightly below 1). A weighted average of a convergent sequence, weighted so that the mass escapes to the far end as $n \to \infty$, converges to the same limit as the sequence. Concretely, fix $\varepsilon > 0$ and choose N so that $|u_k - u| < \varepsilon$ for all $k \ge N$. Split

$$\frac{1}{b_n}\sum_{k=1}^{n-1}\big(b_{k+1} - b_k\big)u_k = \frac{1}{b_n}\sum_{k=1}^{N-1}\big(b_{k+1} - b_k\big)u_k + \frac{1}{b_n}\sum_{k=N}^{n-1}\big(b_{k+1} - b_k\big)u_k$$

The first sum is a fixed finite quantity divided by $b_n \to \infty$, so it tends to 0. In the second sum replace each u_k by u at a cost bounded by $\varepsilon \cdot \frac{1}{b_n}\sum_{k=N}^{n-1}(b_{k+1} - b_k) \le \varepsilon$, and $\frac{1}{b_n}\sum_{k=N}^{n-1}(b_{k+1} - b_k) = (b_n - b_N)/b_n \to 1$, so the second sum converges to a limit within ε of u. Hence

$$\limsup_{n\to\infty}\left|\frac{1}{b_n}\sum_{k=1}^{n-1}\big(b_{k+1} - b_k\big)u_k - u\right| \le \varepsilon$$

and since $\varepsilon > 0$ was arbitrary, the weighted average converges to u.

Step 3: Conclusion. In (4.5) the first term u_n converges to u and the subtracted weighted average also converges to u, so their difference converges to $u - u = 0$. Therefore $b_n^{-1}\sum_{k\le n} x_k \to 0$, as we sought to show.

The lemma is a statement about the interplay of two scales. Convergence of $\sum x_n/b_n$ says the terms x_n are small *relative to* the growing weights b_n; the conclusion says their unweighted sum is small relative to the same weights. The intuition is that summability at scale b_n leaves no room for the partial sums to keep pace with b_n. The Cesàro theorem, that averages of a convergent sequence converge to its limit, is the special case $b_n = n$ applied to the sequence of partial sums, and Step 2 is precisely a Cesàro-type argument with the escaping-mass weights $(b_{k+1} - b_k)/b_n$. What the general weights give is the freedom to choose b_n later; for the strong law the choice will be $b_n = n$ and x_n the centered summands.

A concrete instance makes the mechanism visible before it is deployed probabilistically. Take $x_n = (-1)^n$ and $b_n = n$. The weighted series $\sum_n (-1)^n/n$ converges (it is the alternating harmonic series, summing to $-\log 2$), so the lemma predicts $n^{-1}\sum_{k\le n}(-1)^k \to 0$. That is immediate directly, since the partial sums $\sum_{k\le n}(-1)^k$ alternate between -1 and 0 and are therefore bounded, so dividing by n sends them to zero; but the lemma reaches the same conclusion knowing only that the weighted series converges, without inspecting the partial sums at all. That is exactly the leverage needed when the x_n are random and the partial sums cannot be inspected term by term.

Everything is now in place for the first strong law of large numbers. The setup is a sequence of independent random variables that need not be identically distributed; the only hypothesis is that their variances are summable after division by n^2. Under this hypothesis the centered average converges to zero almost surely, which is the strong law: the empirical average of the X_k tracks the average of their means, not just in probability but along almost every sample path. The proof is the assembly the section has been building toward. Divide each centered summand by n to form a series whose terms are independent and mean zero with summable variances; theorem 4.2.2 makes that series converge almost surely; and lemma 4.3.1 with $b_n = n$ converts the convergence into the vanishing of the average.

Theorem 4.3.2: Strong Law under a Variance Condition

Let $(X_n)_{n\geq 1}$ be independent random variables, each with finite variance, and suppose

$$\sum_{n=1}^{\infty} \frac{\mathrm{Var}(X_n)}{n^2} < \infty \tag{4.6}$$

Then, writing $S_n = X_1 + \cdots + X_n$,

$$\frac{S_n - \mathbb{E}S_n}{n} \longrightarrow 0 \qquad \text{almost surely}$$

In particular, if the variances are uniformly bounded, $\sup_n \mathrm{Var}(X_n) < \infty$, then (4.6) holds and the conclusion follows.

***Proof*:**
Set $Y_n = (X_n - \mathbb{E}X_n)/n$. These are independent, being fixed measurable functions of the independent variables X_n (independence is preserved under the affine maps $x \mapsto (x - \mathbb{E}X_n)/n$; see definition 2.2.2). Each has mean zero, since $\mathbb{E}[Y_n] = (\mathbb{E}X_n - \mathbb{E}X_n)/n = 0$ by linearity of expectation (proposition 3.1.3). Its variance is

$$\mathrm{Var}(Y_n) = \mathrm{Var}\left(\frac{X_n - \mathbb{E}X_n}{n}\right) = \frac{\mathrm{Var}(X_n)}{n^2}$$

using $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$ from the definition of variance (definition 3.2.1) with $a = 1/n$ and $b = -\mathbb{E}X_n/n$. By hypothesis (4.6),

$$\sum_{n=1}^{\infty} \mathrm{Var}(Y_n) = \sum_{n=1}^{\infty} \frac{\mathrm{Var}(X_n)}{n^2} < \infty$$

so the sequence (Y_n) satisfies the hypotheses of the one-series theorem: independent, mean zero, with summable variances. By theorem 4.2.2 the series

$$\sum_{n=1}^{\infty} Y_n = \sum_{n=1}^{\infty} \frac{X_n - \mathbb{E}X_n}{n}$$

converges almost surely. Let Ω_0 be the event of full probability on which this convergence holds.

Fix $\omega \in \Omega_0$ and apply lemma 4.3.1 pathwise, with weights $b_n = n$ (which increase to infinity) and terms $x_k = x_k(\omega) = X_k(\omega) - \mathbb{E}X_k$. The weighted series $\sum_k x_k/b_k = \sum_k (X_k(\omega) - \mathbb{E}X_k)/k$ is exactly $\sum_k Y_k(\omega)$, which converges by the choice of ω. Kronecker's lemma therefore gives

$$\frac{1}{n}\sum_{k=1}^{n}\big(X_k(\omega) - \mathbb{E}X_k\big) \longrightarrow 0$$

Since $\sum_{k\le n} \mathbb{E}X_k = \mathbb{E}[S_n]$ by linearity of expectation (proposition 3.1.3), the left side is $(S_n(\omega) - \mathbb{E}S_n)/n$. This holds for every $\omega \in \Omega_0$, an event of probability one, so $(S_n - \mathbb{E}S_n)/n \to 0$ almost surely.

For the final clause, if $\sup_n \operatorname{Var}(X_n) =: C < \infty$ then $\sum_n \operatorname{Var}(X_n)/n^2 \le C\sum_n 1/n^2 < \infty$, so (4.6) holds and the main conclusion applies, completing the proof.

This is the strong law of large numbers in its general-variance form, and it is worth being precise about what it does and does not assume. It requires finite variances and the summability (4.6), but it imposes no identical-distribution hypothesis: the X_n may have wildly different laws, as long as their variances do not grow faster than n^2. What it delivers is convergence of the average of the X_k to the average of their means along almost every path. When the X_n are identically distributed with common mean μ, the centering $\mathbb{E}S_n = n\mu$ and the conclusion reads $S_n/n \to \mu$ almost surely, the form the strong law is usually quoted in. But that identically distributed case does not need finite variance at all: a finite mean suffices, and reaching the strong law under only $\mathbb{E}|X_1| < \infty$ is the business of the next chapter, whose proof builds on the machinery established here. The present theorem is the variance-hypothesis version, powerful enough for most examples and proved entirely from the maximal inequality by way of series convergence.

The gap between this theorem and the sharper identically distributed statement is instructive. The condition (4.6) is a smallness condition on the tails of the summands, expressed through their variances; it is what makes the series $\sum_n Y_n$ converge. When the summands are identically distributed one can trade the second moment for the first by a truncation argument that exploits the common law, which is why the finite-mean strong law lives one chapter downstream rather than here. For now the variance route is the direct one, and it already covers the running examples.

The examples below instantiate the theorem for the book's anchor processes and return to the zero-one law. The first is the coin walk, where the variance is constant and the hypothesis is trivially met; the second is a bounded but non-identically-distributed sequence, showing the theorem's reach beyond the identically distributed world; and the third reads the conclusion through theorem 2.4.3 to explain *why* the limit is a constant, and why that constant is zero.

Example 4.3.3: Strong Laws for Sums of Independent Variables

1. *The coin walk.* Let (X_n) be independent with $\mathbb{P}(X_n = +1) = \mathbb{P}(X_n = -1) = 1/2$, the ± 1 steps of the simple symmetric random walk on $\mathbb{Z}$ from example 1.3.6, and let $S_n = X_1 + \cdots + X_n$ be the position after n steps. Each step has $\mathbb{E}X_n = 0$ and $\operatorname{Var}(X_n) = \mathbb{E}[X_n^2] - (\mathbb{E}X_n)^2 = 1 - 0 = 1$, so the variances are uniformly bounded and (4.6) holds. Since $\mathbb{E}S_n = 0$, theorem 4.3.2 gives

$$\frac{S_n}{n} \longrightarrow 0 \qquad \text{almost surely}$$

The walk wanders, and the maximal inequality showed it can be far from the origin along the way, but its *average displacement per step* settles to zero on almost every path. This is the strong companion of the weak statement that $S_n/n \to 0$ in probability, and it is sharper: it controls the whole tail of the sequence S_n/n at once, not each S_n/n separately.

2. *A bounded non-identically-distributed sequence.* Let (X_n) be independent with

$$\mathbb{P}\big(X_n = +\tfrac{1}{2}\big) = \mathbb{P}\big(X_n = -\tfrac{1}{2}\big) = \frac{1}{2n}, \qquad \mathbb{P}(X_n = 0) = 1 - \frac{1}{n}$$

so each X_n takes values in $\{-\frac{1}{2}, 0, \frac{1}{2}\}$ and is more and more likely to be zero as n grows. Here $\mathbb{E}X_n = 0$ and

$$\mathrm{Var}(X_n) = \mathbb{E}[X_n^2] = \tfrac{1}{4} \cdot \tfrac{1}{n} = \frac{1}{4n}$$

The variances are not bounded below by a positive constant and the law of X_n changes with n, so this sequence is outside the identically distributed regime. Nonetheless $\mathrm{Var}(X_n) = 1/(4n) \le 1/4$ is bounded, so (4.6) holds (indeed $\sum_n \mathrm{Var}(X_n)/n^2 = \frac{1}{4}\sum_n n^{-3} < \infty$), and $\mathbb{E}S_n = 0$, so theorem 4.3.2 again yields $S_n/n \to 0$ almost surely. The theorem's freedom from an identical-distribution hypothesis is what makes it apply.

3. *The limit as an almost-sure constant.* That the limit in both cases is a genuine number, not a random quantity, is a structural fact traceable to the zero-one law. Fix any independent sequence (X_n) satisfying (4.6) with $\mathbb{E}X_n = 0$ for all n, so that $\mathbb{E}S_n = 0$ and the theorem gives $S_n/n \to L$ almost surely for the constant $L = 0$. Consider instead the event

$$E = \Big\{\omega : \lim_{n\to\infty} \tfrac{1}{n} S_n(\omega) \text{ exists and equals } 0\Big\}$$

Changing finitely many of the X_k does not change $\lim_n S_n/n$: if S_n' differs from S_n only through the first m terms, then $(S_n - S_n')/n \to 0$, so the two averages have the same limit. Hence E is a tail event, measurable with respect to the tail σ-algebra $\mathcal{T} = \bigcap_m \sigma(X_m, X_{m+1}, \dots)$ of definition 2.4.1. By Kolmogorov's zero-one law (theorem 2.4.3), $\mathbb{P}(E) \in \{0, 1\}$. The theorem has shown $\mathbb{P}(E) = 1$, which is consistent with the dichotomy and, more to the point, illustrates it: the *value* $\lim_n S_n/n$, being a tail random variable, is almost surely constant, and the strong law identifies the constant as 0. The zero-one law guarantees that a limit of averages, if it exists at all, must be deterministic; the strong law is the theorem that produces the limit and names it.

The third item is the payoff of viewing the strong law alongside the zero-one law. The zero-one law is a soft statement: it says the limit of the averages, being a tail quantity, is almost surely equal to some constant, but it is silent about which constant and about whether the limit exists. The strong law is the hard statement that supplies both, producing the limit and pinning it to $\mathbb{E}S_n/n$. The two results are complementary halves of the same picture, one asserting determinism and the other computing the deterministic value.

Stepping back, the three sections of this chapter form a single argument with a consistent shape. The maximal inequality of section 4.1 was the seed: it controlled an entire trajectory of partial sums with the variance that Chebyshev spends on one sum. Section 4.2 spent that trajectory-level control to prove that a series of independent terms with summable variance converges almost surely. The present section spent *that* convergence, through the deterministic Kronecker's lemma, to prove that the averages converge. The through-line is that independence lets control of a single quantity propagate to control of the whole sequence, and each result in the chapter is one more link in that propagation. The strong law obtained here is the entry point to the next chapter, where the identically distributed case is reached under a weaker hypothesis by exactly the truncation ideas rehearsed in the three-series theorem.

Exercise 4.3.1

1. Verify the necessity of the growth hypothesis in Kronecker's lemma by exhibiting a counterexample when b_n does not tend to infinity. Take $b_n = 1$ for all n and $x_n = (-1)^n/n$. Show that $\sum_n x_n/b_n$ converges yet $b_n^{-1}\sum_{k\le n} x_k$ does not tend to 0. Explain in one sentence which step of the proof of lemma 4.3.1 fails.

2. Let (X_n) be independent with X_n uniformly distributed on the interval $[-n, n]$, so $\mathrm{Var}(X_n) = n^2/3$. Does theorem 4.3.2 apply? Compute $\sum_n \mathrm{Var}(X_n)/n^2$ and state precisely what the theorem does or does not tell you about $(S_n - \mathbb{E}S_n)/n$.

3. Let (X_n) be independent with $\mathbb{P}(X_n = \pm n^\alpha) = 1/2$ for a fixed real α. For which α does theorem 4.3.2 give $S_n/n \to 0$ almost surely? Compute $\mathrm{Var}(X_n)$ and determine the range of α for which (4.6) holds.

4. Suppose (X_n) are independent with $\mathbb{E}X_n = 0$ and $\mathrm{Var}(X_n) = \sigma_n^2$, and let (b_n) be positive with $b_n \uparrow \infty$. Adapt the proof of theorem 4.3.2 to show that if $\sum_n \sigma_n^2/b_n^2 < \infty$ then $b_n^{-1}S_n \to 0$ almost surely. Which choice of b_n recovers the theorem?

5. (The limit is a tail variable.) Let (X_n) be an arbitrary independent sequence, not assumed to satisfy any variance condition. Show that the set $\{\omega : \lim_n S_n(\omega)/n \text{ exists}\}$ is a tail event and that, on this set, the limit $\lim_n S_n/n$ is a tail-measurable random variable. Conclude from theorem 2.4.3 that the limit, when it exists almost surely, is almost surely constant, without using theorem 4.3.2.

6. Give an example of an independent sequence (X_n) for which $b_n^{-1}\sum_{k\le n} X_k \to 0$ almost surely with $b_n = n$, yet $\sum_n X_n$ diverges almost surely. Explain why this is not a contradiction of anything in this section. (Hint: the coin walk.)

5

Laws of Large Numbers

The strong law under a variance condition (theorem 4.3.2) already delivered a first law of large numbers: for independent summands whose variances do not grow too fast, the average S_n/n converges to the mean almost surely. This chapter completes the picture. It separates the two modes of convergence that the phrase "law of large numbers" can name, weak (in probability) and strong (almost sure), establishes each under the natural hypothesis of independent, identically distributed summands with a finite mean, and then puts the strong law to work.

section 5.1 proves the weak law, that $S_n/n \to \mu$ in probability, first by the second-moment method under a finite variance and then in the sharp form that requires only a finite mean. Section 5.2 proves the strong law of large numbers for i.i.d. summands with finite mean by two independent routes: Kolmogorov's, through truncation and the variance-condition law of section 4.3, and Etemadi's, which needs only pairwise independence. Section 5.3 harvests the consequences that make the strong law the most used theorem of applied probability: Monte Carlo integration, the Glivenko-Cantelli theorem on empirical distributions, Bernstein's probabilistic proof of the Weierstrass approximation theorem, and Borel's theorem on normal numbers. Section 5.4 closes with a law of large numbers from another world entirely: the number of distinct prime factors of a random integer concentrates around $\log \log n$, the Hardy-Ramanujan theorem, and its proof is nothing but Chebyshev's inequality applied to the primes.

5.1 The Weak Law of Large Numbers

The strong law under a variance condition (theorem 4.3.2) already produced a statement of the form "the average S_n/n approaches the mean", but it did so in the strongest available sense, almost sure convergence, and it paid for that strength with a variance hypothesis. The weak law asks less and delivers less, and it is the right place to begin. It replaces almost sure convergence by convergence in probability, a weaker mode that speaks only about the distribution of a single S_n/n

at a time rather than about the joint behaviour of the whole sequence along an individual sample path. This weakening gives two things. First, the proof under a finite variance becomes a single line of Chebyshev's inequality, with an explicit rate. Second, and more substantially, the finite-variance hypothesis can be dropped entirely: Khinchin's theorem gives the weak law for any i.i.d. sequence with a finite mean, a generality the second-moment method cannot reach on its own. This section develops both, after first isolating the mode of convergence they are stated in.

Convergence in probability is the first of the several modes of convergence that organize the limit theory of this Part. It sits strictly between convergence in distribution, which the central limit theorem to come is phrased in, and almost sure convergence, the mode of the strong law. Its definition asks that the event on which X_n differs appreciably from its limit becomes negligible.

Definition 5.1.1: Convergence in Probability

Let X and $X_1, X_2, \ldots$ be random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The sequence (X_n) *converges to X in probability*, written $X_n \xrightarrow{\mathbb{P}} X$, if for every $\varepsilon > 0$,

$$\mathbb{P}\big(|X_n - X| > \varepsilon\big) \longrightarrow 0 \qquad \text{as } n \to \infty$$

The definition quantifies over ε from the outside: for each fixed tolerance ε, the probability that X_n misses X by more than that tolerance vanishes in the limit. The order of the quantifiers matters. Convergence in probability does not assert that, for a fixed sample point ω, the numbers $X_n(\omega)$ converge to $X(\omega)$; the exceptional set on which X_n deviates is allowed to move around with n, so long as it shrinks in measure. This is exactly the loophole that separates convergence in probability from almost sure convergence, and it is what makes the weak law weaker than the strong law. Whether the strict inequality $|X_n - X| > \varepsilon$ or the non-strict $\geq \varepsilon$ appears in the definition is immaterial, since running ε over all positive reals recovers the same notion; the strict form is chosen here to match the tail events that appear in the proofs.

A first observation grounds the definition against the almost sure convergence of section 4.3: the stronger mode implies the weaker. This is the implication that lets the strong law, once proved, subsume the weak law, and it is proved by the continuity of probability along monotone sequences.

Proposition 5.1.2: Almost Sure Convergence Implies Convergence in Probability

Let $X, X_1, X_2, \ldots$ be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. If $X_n \to X$ almost surely, then $X_n \xrightarrow{\mathbb{P}} X$.

***Proof*:**
Fix $\varepsilon > 0$. The plan is to bound the single-n deviation event by a tail event whose probability is controlled by continuity of probability. For each n define the tail deviation event

$$E_n = \bigcup_{k \geq n} \{|X_k - X| > \varepsilon\}$$

the event that *some* index from n onward deviates by more than ε. The events E_n decrease, $E_1 \supseteq E_2 \supseteq \cdots$, since shrinking the range of the union can only shrink it. Their intersection $E = \bigcap_n E_n$ is the event that $|X_k - X| > \varepsilon$ for infinitely many k.

We claim $\mathbb{P}(E) = 0$. On the almost sure convergence event $\{X_n \to X\}$, which has probability one, one has $|X_k - X| \leq \varepsilon$ for all sufficiently large k, so $\omega \in \{X_n \to X\}$ forces $\omega \notin E$. Hence

$E \subseteq \{X_n \to X\}^c$, an event of probability zero, and so $\mathbb{P}(E) = 0$. By continuity of probability from above (proposition 1.1.5, part 2), applied to the decreasing sequence $E_n \downarrow E$,

$$\mathbb{P}(E_n) \longrightarrow \mathbb{P}(E) = 0$$

Finally, the single-index event is contained in the tail event, $\{|X_n - X| > \varepsilon\} \subseteq E_n$, so by monotonicity $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{P}(E_n) \to 0$. As $\varepsilon > 0$ was arbitrary, $X_n \xrightarrow{\mathbb{P}} X$, as we sought to show.

The implication is strict: convergence in probability does not imply almost sure convergence. The standard witness is a "moving bump" that sweeps across the sample space with ever smaller support but visits every point infinitely often; its probability of being large tends to zero, yet no single sample path settles down. This example, and the precise sense in which the two modes differ, is developed in section 5.2, where almost sure convergence is defined and the strong law is proved (see also exercise 5.1.1 (5)). For the weak law, convergence in probability is the correct and sufficient target.

The most direct route to the weak law runs through Chebyshev's inequality (corollary 3.2.5). The average $\bar{X}_n = S_n/n$ of n summands with a common mean μ has mean μ as well; if the summands are uncorrelated with a common variance σ^2, then the variance of the average is σ^2/n, which shrinks to zero. Chebyshev converts that shrinkage of the variance directly into a bound on the probability of a deviation, and the weak law falls out with an explicit rate. The hypothesis needed is only pairwise uncorrelatedness, weaker than independence, because the only fact used about the summands is the additivity of variance, which is a second-order (Hilbert-space) statement blind to higher-order dependence.

Theorem 5.1.3: Weak Law of Large Numbers (L^2)

Let $(X_n)_{n\geq 1}$ be square-integrable random variables that are pairwise uncorrelated, with a common mean $\mu = \mathbb{E}[X_i]$ and a common variance $\sigma^2 = \operatorname{Var}(X_i) < \infty$. Write $S_n = X_1 + \cdots + X_n$ and $\bar{X}_n = S_n/n$. Then for every $\varepsilon > 0$,

$$\mathbb{P}\big(|\bar{X}_n - \mu| > \varepsilon\big) \leq \frac{\sigma^2}{n\varepsilon^2} \tag{5.1}$$

and in particular $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$. The same conclusion holds, with the same bound, for an i.i.d. sequence in L^2, since independence implies pairwise uncorrelatedness.

***Proof*:**

The average is centered at μ: by linearity of expectation, $\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] = \mu$. Its variance is computed from Bienaymé's identity. Since the X_i are pairwise uncorrelated, proposition 3.3.7 gives $\operatorname{Var}(S_n) = \sum_{i=1}^n \operatorname{Var}(X_i) = n\sigma^2$, and since $\bar{X}_n = S_n/n$ scales the variance by $1/n^2$,

$$\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\operatorname{Var}(S_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Now apply Chebyshev's inequality (corollary 3.2.5) to $\bar{X}_n$, whose mean is μ, with threshold

$a = \varepsilon$. Because the strict and non-strict deviation events are bounded the same way,

$$\mathbb{P}\big(|\bar{X}_n - \mu| > \varepsilon\big) \le \mathbb{P}\big(|\bar{X}_n - \mu| \ge \varepsilon\big) \le \frac{\operatorname{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}$$

which is the bound (5.1). For each fixed $\varepsilon > 0$ the right-hand side tends to 0 as $n \to \infty$, so $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$. The final assertion holds because independent square-integrable variables are pairwise uncorrelated (proposition 3.3.7), so an i.i.d. L^2 sequence satisfies the hypotheses, completing the proof.

The bound (5.1) is not only a convergence statement but a *rate*: the probability of a deviation of size ε decays like $1/n$, uniformly in the underlying distribution, controlled only by the single number σ^2. This is already enough for many applications. Setting $\varepsilon = \sigma/\sqrt{n} \cdot t$ shows that deviations on the scale $\sigma/\sqrt{n}$ have probability bounded by $1/t^2$, which foreshadows the $\sqrt{n}$ scaling at which the central limit theorem will find a nondegenerate Gaussian limit; the weak law says the fluctuations are $o(1)$, and the fine structure of those fluctuations is precisely what the later chapters resolve. The proof also exposes why the hypothesis is a second moment: Chebyshev needs a variance to bound, and the variance of the average is what carries the $1/n$ decay. Removing the variance hypothesis, as the next theorem does, requires a different idea.

Before that, a concrete instance. The running coin-tossing example is the prototype of the weak law, and the L^2 theorem applies to it directly.

Example 5.1.4: The Weak Law for Fair Coins

Let (ξ_n) be the coordinate variables of the fair-coin-tossing space (example 1.3.6), so the ξ_n are i.i.d. with $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = 0) = 1/2$; here $\xi_n = 1$ records heads on toss n. Each ξ_n has mean $\mu = 1/2$ and variance $\sigma^2 = \mathbb{E}[\xi_n^2] - (\mathbb{E}\xi_n)^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. The fraction of heads in the first n tosses is $\bar{\xi}_n = \frac{1}{n}\sum_{i=1}^n \xi_i$, and theorem 5.1.3 gives, for every $\varepsilon > 0$,

$$\mathbb{P}\big(\big|\bar{\xi}_n - \tfrac{1}{2}\big| > \varepsilon\big) \le \frac{1/4}{n\varepsilon^2} = \frac{1}{4n\varepsilon^2}$$

so $\bar{\xi}_n \xrightarrow{\mathbb{P}} 1/2$: the empirical head frequency converges in probability to $1/2$. Equivalently, in terms of the count $S_n = n\bar{\xi}_n \sim \operatorname{Bin}(n, 1/2)$, whose variance is $\operatorname{Var}(S_n) = n\sigma^2 = n/4$, Chebyshev with threshold $a = \varepsilon n$ gives the same bound directly: $\mathbb{P}(|S_n - n/2| \ge \varepsilon n) \le (n/4)/(\varepsilon n)^2 = 1/(4n\varepsilon^2)$. Taking $\varepsilon = 0.01$ and $n = 10^6$ bounds the probability that the head fraction misses $1/2$ by more than a hundredth by $1/(4 \cdot 10^6 \cdot 10^{-4}) = 1/400 = 0.0025$. This is the assertion that a fair coin's long-run frequency is $1/2$, made quantitative.

The L^2 theorem is sharp about its rate but crude about its hypotheses: it requires a finite variance, and there are integrable random variables, with a perfectly good mean, whose variance is infinite. For an i.i.d. sequence the weak law nonetheless holds under a finite mean alone. This is Khinchin's theorem, and its proof introduces a technique, *truncation*, that recurs throughout the limit theory of sums: replace each X_i by a bounded surrogate Y_i, prove the result for the surrogate by the second-moment method, and control the discarded part separately. The finite mean is exactly the hypothesis under which both pieces can be made to work.

Theorem 5.1.5: Khinchin's Weak Law of Large Numbers (L^1)

Let $(X_n)_{n\geq 1}$ be i.i.d. random variables with $\mathbb{E}|X_1| < \infty$, and let $\mu = \mathbb{E}[X_1]$. Write $\bar{X}_n = \frac{1}{n}(X_1 + \cdots + X_n)$. Then

$$\bar{X}_n \xrightarrow{\mathbb{P}} \mu$$

The idea is to cut each X_i off at level n. The truncated variables are bounded, hence square-integrable, so the L^2 theorem controls their average; the point is that as n grows the truncation level grows with it, so the truncated mean returns to μ and the truncated variance, divided by n, still vanishes. The two error terms that the truncation introduces, the shift in the mean and the event that some X_i was cut, are both controlled by the single hypothesis $\mathbb{E}|X_1| < \infty$ through the tail-integral estimate that a finite mean forces the tail $\int_{|x|>t} |x|\, d\mu_X \to 0$.

***Proof*:**
Write μ_X for the common law of the X_i and let $\varepsilon > 0$ be fixed throughout. The truncation level is the index n itself.

Step 1: Truncation. For each n define the truncated variables

$$Y_{n,i} = X_i\, \mathbb{1}_{\{|X_i|\leq n\}}, \qquad 1 \leq i \leq n$$

and set $T_n = Y_{n,1} + \cdots + Y_{n,n}$ and $\bar{Y}_n = T_n/n$. For fixed n the variables $Y_{n,1}, \ldots, Y_{n,n}$ are i.i.d., being the common measurable function $x \mapsto x\, \mathbb{1}_{\{|x|\leq n\}}$ applied to the i.i.d. X_i; each is bounded by n in absolute value, hence square-integrable. Write

$$\mu_n = \mathbb{E}[Y_{n,1}] = \int_{\{|x|\leq n\}} x\, d\mu_X(x)$$

for their common mean, using the change-of-variables formula (theorem 3.1.2) to express the expectation of the truncated variable as an integral against the law of X_1.

Step 2: The truncated mean returns to μ. Since $\mathbb{E}|X_1| < \infty$, the function x is μ_X-integrable, so by dominated convergence the truncated integral converges to the full one:

$$\mu_n = \int_{\{|x|\leq n\}} x\, d\mu_X(x) \longrightarrow \int_{\mathbb{R}} x\, d\mu_X(x) = \mu$$

the dominating function being $|x| \in L^1(\mu_X)$ and the integrands $x\, \mathbb{1}_{\{|x|\leq n\}} \to x$ pointwise. Consequently $\bar{Y}_n$ is centered near μ: choose N_1 so that $|\mu_n - \mu| < \varepsilon/2$ for all $n \geq N_1$.

Step 3: The truncated variance is $o(n)$. The heart of the argument is that $\mathbb{E}[Y_{n,1}^2] = o(n)$, that is, $\mathbb{E}[Y_{n,1}^2]/n \to 0$. By theorem 3.1.2 applied to $g(x) = x^2\, \mathbb{1}_{\{|x|\leq n\}}$,

$$\mathbb{E}[Y_{n,1}^2] = \int_{\{|x|\leq n\}} x^2\, d\mu_X(x) = \int_{\{|x|\leq n\}} |x| \cdot |x|\, d\mu_X(x) \leq n \int_{\{|x|\leq n\}} |x|\, d\mu_X(x) \leq n\, \mathbb{E}|X_1|$$

where the first inequality bounds one factor $|x| \leq n$ on the domain of integration. This crude bound gives $\mathbb{E}[Y_{n,1}^2] \leq n\, \mathbb{E}|X_1|$, which is $O(n)$ but not yet $o(n)$; the improvement uses that a finite mean makes the tail contribution small. Split the integral at a level M with $0 < M \leq n$:

$$\frac{\mathbb{E}[Y_{n,1}^2]}{n} = \frac{1}{n} \int_{\{|x|\leq M\}} x^2\, d\mu_X(x) + \frac{1}{n} \int_{\{M<|x|\leq n\}} x^2\, d\mu_X(x)$$

The first integral is at most M^2, so the first term is at most $M^2/n \to 0$ as $n \to \infty$ with M fixed. In the second integral $|x| \le n$ on the domain, so $x^2 = |x| \cdot |x| \le n|x|$ and

$$\frac{1}{n}\int_{\{M<|x|\le n\}} x^2\, d\mu_X(x) \le \int_{\{M<|x|\le n\}} |x|\, d\mu_X(x) \le \int_{\{|x|>M\}} |x|\, d\mu_X(x)$$

Given $\eta > 0$, integrability of $|x|$ lets us fix M so large that this last tail integral is below $\eta/2$; then choosing n large enough that $M^2/n < \eta/2$ makes $\mathbb{E}[Y_{n,1}^2]/n < \eta$. As $\eta > 0$ was arbitrary, $\mathbb{E}[Y_{n,1}^2]/n \to 0$, which is the claim.

Now estimate the variance of the truncated average. The $Y_{n,i}$ are, for fixed n, i.i.d., hence pairwise uncorrelated, so Bienaymé's identity (proposition 3.3.7) gives $\operatorname{Var}(T_n) = n \operatorname{Var}(Y_{n,1})$, and therefore

$$\operatorname{Var}(\bar{Y}_n) = \frac{1}{n^2}\operatorname{Var}(T_n) = \frac{\operatorname{Var}(Y_{n,1})}{n} \le \frac{\mathbb{E}[Y_{n,1}^2]}{n} \longrightarrow 0$$

using $\operatorname{Var}(Y_{n,1}) \le \mathbb{E}[Y_{n,1}^2]$ and Step 3.

Step 4: The truncated average converges in probability to μ. For $n \ge N_1$, combine the centering of Step 2 with Chebyshev's inequality (corollary 3.2.5). If $|\bar{Y}_n - \mu| > \varepsilon$ then, since $|\mu_n - \mu| < \varepsilon/2$, the triangle inequality forces $|\bar{Y}_n - \mu_n| > \varepsilon/2$, where $\mu_n = \mathbb{E}[\bar{Y}_n]$. Hence

$$\mathbb{P}\big(|\bar{Y}_n - \mu| > \varepsilon\big) \le \mathbb{P}\big(|\bar{Y}_n - \mathbb{E}\bar{Y}_n| > \varepsilon/2\big) \le \frac{\operatorname{Var}(\bar{Y}_n)}{(\varepsilon/2)^2} = \frac{4\operatorname{Var}(\bar{Y}_n)}{\varepsilon^2} \longrightarrow 0$$

by the variance estimate of Step 3. So $\bar{Y}_n \xrightarrow{\mathbb{P}} \mu$.

Step 5: Removing the truncation. It remains to pass from $\bar{Y}_n$ back to $\bar{X}_n$. The two averages agree unless some summand was truncated, that is, on the complement of the event

$$D_n = \bigcup_{i=1}^{n}\{X_i \ne Y_{n,i}\} = \bigcup_{i=1}^{n}\{|X_i| > n\}$$

By the union bound and the fact that the X_i are identically distributed,

$$\mathbb{P}(D_n) \le \sum_{i=1}^{n} \mathbb{P}(|X_i| > n) = n\,\mathbb{P}(|X_1| > n)$$

Now $n\,\mathbb{P}(|X_1| > n) \to 0$ because $\mathbb{E}|X_1| < \infty$. Indeed, by Markov's tail formula for the mean (or directly),

$$n\,\mathbb{P}(|X_1| > n) = \int_{\{|X_1|>n\}} n\, d\mathbb{P} \le \int_{\{|X_1|>n\}} |X_1|\, d\mathbb{P} \longrightarrow 0$$

the convergence again by dominated convergence, since $|X_1| \in L^1$ and the sets $\{|X_1| > n\}$ shrink to a null set. On D_n^c every summand satisfies $X_i = Y_{n,i}$, so $\bar{X}_n = \bar{Y}_n$ there, and therefore

$$\mathbb{P}\big(|\bar{X}_n - \mu| > \varepsilon\big) \le \mathbb{P}\big(|\bar{Y}_n - \mu| > \varepsilon\big) + \mathbb{P}(D_n)$$

Both terms on the right tend to 0, the first by Step 4 and the second by the tail estimate just established. As $\varepsilon > 0$ was arbitrary, $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$, completing the proof.

Khinchin's theorem is the definitive weak law: for i.i.d. summands it asks for a finite mean and nothing more, exactly the hypothesis under which the target quantity $\mu = \mathbb{E}[X_1]$ is even defined. The truncation argument is worth absorbing on its own terms. It is the standard mechanism for converting an L^2-only tool, here the pairing of Chebyshev with the additivity of variance, into an L^1 statement, and it will reappear in the strong law (section 5.2) and in the study of sums of independent variables. Two estimates carried the proof, and both are readings of the single fact $\mathbb{E}|X_1| < \infty$: the truncated second moment is $o(n)$, which makes the variance of the average vanish, and the discard probability $n\,\mathbb{P}(|X_1| > n)$ is $o(1)$, which makes the truncation asymptotically invisible. The tail integral $\int_{|x|>t} |x|\, d\mu_X$ tending to zero is what powers both.

The gap between the two theorems is genuine, not a defect of the proofs. There exist integrable random variables with infinite variance, and for such variables Khinchin's theorem applies while the L^2 theorem does not. The next example exhibits one, showing that the extra generality of Khinchin's theorem is used.

Example 5.1.6: Two Weak Laws: Finite versus Infinite Variance

1. *The coin average.* As in example 5.1.4, an i.i.d. fair-coin sequence (ξ_n) has $\mu = 1/2$ and $\sigma^2 = 1/4$, so both weak laws apply and $\bar{\xi}_n \xrightarrow{\mathbb{P}} 1/2$. This is the finite-variance case, where the L^2 theorem is already enough and supplies the rate $\mathbb{P}(|\bar{\xi}_n - 1/2| > \varepsilon) \le 1/(4n\varepsilon^2)$.

2. *An integrable law with infinite variance.* Let X have the symmetric density

$$f(x) = \frac{1}{|x|^3}\, \mathbb{1}_{\{|x|\ge 1\}}$$

on $\mathbb{R}$. This is a probability density: by symmetry and the substitution on $[1,\infty)$,

$$\int_{\mathbb{R}} f(x)\, dx = 2\int_1^\infty x^{-3}\, dx = 2\left[-\tfrac{1}{2}x^{-2}\right]_1^\infty = 1$$

The first absolute moment is finite,

$$\mathbb{E}|X| = 2\int_1^\infty x \cdot x^{-3}\, dx = 2\int_1^\infty x^{-2}\, dx = 2\left[-x^{-1}\right]_1^\infty = 2 < \infty$$

so X is integrable, and by symmetry of f its mean is $\mu = \mathbb{E}[X] = 0$. The second moment, however, diverges:

$$\mathbb{E}[X^2] = 2\int_1^\infty x^2 \cdot x^{-3}\, dx = 2\int_1^\infty \frac{dx}{x} = 2\lim_{R\to\infty} \log R = \infty$$

so $\mathrm{Var}(X) = \infty$. For an i.i.d. sequence (X_n) with this law, theorem 5.1.3 does not apply, its finite-variance hypothesis failing, yet theorem 5.1.5 does apply, since $\mathbb{E}|X_1| = 2 < \infty$, and gives $\bar{X}_n \xrightarrow{\mathbb{P}} 0$. The average of these heavy-tailed but integrable variables still concentrates at the mean; only the second-moment rate is lost. The tail here decays like $\mathbb{P}(|X| > t) = t^{-2}$, borderline for a finite variance, exactly the regime the truncation argument was built to handle.

The two theorems mark the two hypotheses under which an empirical average is governed by its mean. A finite variance is a statement about the spread of a single trial and yields the quantitative $1/n$ rate; a finite mean is the minimal statement under which the mean is defined at all and yields only qualitative convergence, but for i.i.d. data it yields it in full. The distinction previews the central theme of the Part: the strength of a limit theorem is gave with the moments assumed. The weak law leaves two questions open. It says each individual average $\bar{X}_n$ is likely close to μ, but says nothing about whether a fixed sample path $n \mapsto \bar{X}_n(\omega)$ converges, this is the almost sure statement of the strong law, taken up next, and it does not resolve the $\sqrt{n}$-scale fluctuations of $\bar{X}_n$ around μ, which the central limit theory later in this Part will identify as Gaussian.

Exercise 5.1.1

1. Let (X_n) be i.i.d. with $\mathbb{E}[X_1] = \mu$ and $\operatorname{Var}(X_1) = \sigma^2 < \infty$. Using the rate (5.1), find, for given $\varepsilon > 0$ and $\delta > 0$, an explicit n guaranteeing $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \delta$. Evaluate it for a fair die (values $1, \ldots, 6$ equally likely) with $\varepsilon = 0.1$ and $\delta = 0.05$.

2. Show that the pairwise-uncorrelated hypothesis in theorem 5.1.3 is strictly weaker than the i.i.d. hypothesis, and cannot be dropped to "identically distributed" alone: exhibit identically distributed but perfectly correlated square-integrable variables $X_1 = X_2 = \cdots = X$ (a single nonconstant X repeated) for which $\bar{X}_n = X$ does not converge in probability to a constant.

3. Let (X_n) be i.i.d. with the Cauchy density $f(x) = \frac{1}{\pi(1+x^2)}$. Show $\mathbb{E}|X_1| = \infty$, so neither weak law of this section applies. (In fact $\bar{X}_n$ has the same Cauchy law as X_1 for every n, so $\bar{X}_n$ does not converge in probability to any constant; you need not prove this last claim, which uses characteristic functions from a later chapter.)

4. *Weak law without identical distributions.* Let (X_n) be independent with $\mathbb{E}[X_n] = \mu$ for all n and $\operatorname{Var}(X_n) \leq C$ for a constant C independent of n. Show that $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ with the rate $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq C/(n\varepsilon^2)$. Where in the proof of theorem 5.1.3 is the common-variance hypothesis used?

5. *Convergence in probability, examined.* On $([0,1], \mathcal{B}, \lambda)$ with λ Lebesgue measure, for each n write $n = 2^k + j$ with $0 \leq j < 2^k$ and set $X_n = \mathbb{1}_{[j2^{-k},\,(j+1)2^{-k}]}$. Show that $X_n \xrightarrow{\mathbb{P}} 0$ but that for *no* sample point ω does $X_n(\omega) \to 0$. (This is the moving-bump example alluded to after proposition 5.1.2; it shows the converse implication there fails.)

6. Let $X_n \xrightarrow{\mathbb{P}} X$ and $Y_n \xrightarrow{\mathbb{P}} Y$. Show $X_n + Y_n \xrightarrow{\mathbb{P}} X + Y$. (Hint: if $|X_n + Y_n - X - Y| > \varepsilon$ then $|X_n - X| > \varepsilon/2$ or $|Y_n - Y| > \varepsilon/2$.) Show also that if $g\colon \mathbb{R} \to \mathbb{R}$ is uniformly continuous and $X_n \xrightarrow{\mathbb{P}} X$, then $g(X_n) \xrightarrow{\mathbb{P}} g(X)$.

5.2 The Strong Law of Large Numbers

The weak law of section 5.1 says that the sample average $\bar{X}_n$ is close to the mean μ with high probability once n is large: for each fixed n the event $\{|\bar{X}_n - \mu| > \varepsilon\}$ is unlikely. What it does not say is that a single realization of the sequence $\bar{X}_1, \bar{X}_2, \ldots$ settles down to μ. A gambler watching the running average over one infinite run of the process is not asking about the marginal distribution of $\bar{X}_n$ for each separate n; the gambler is asking whether *this* path converges. That is a statement about the joint behaviour of the whole tail of the sequence at once, and it is a strictly stronger requirement.

This section proves that under the same hypothesis of an integrable, identically distributed sequence, almost every path does converge to μ, the *strong* law of large numbers, and it gives two independent proofs of it, Kolmogorov's and Etemadi's, each illuminating a different mechanism.

The right notion of convergence for "every path settles down" is convergence along almost every sample point, which is the pointwise convergence of measure theory read in probabilistic language.

Definition 5.2.1: Almost Sure Convergence

Let $(X_n)_{n\geq 1}$ and X be random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The sequence (X_n) *converges almost surely* to X, written $X_n \to X$ almost surely (abbreviated a.s.), if

$$\mathbb{P}\big(\{\omega \in \Omega \colon X_n(\omega) \to X(\omega) \text{ as } n \to \infty\}\big) = 1$$

that is, if the set of sample points along which the numerical sequence $X_n(\omega)$ fails to converge to $X(\omega)$ is a $\mathbb{P}$-null event.

The event inside the probability is the set of ω for which the ordinary real sequence $\big(X_n(\omega)\big)_n$ converges to the number $X(\omega)$; its measurability is a standard fact of measure theory, since it can be written as the countable combination $\bigcap_{k\geq 1}\bigcup_{m\geq 1}\bigcap_{n\geq m}\{|X_n - X| \leq 1/k\}$ of events (see [2]). Almost sure convergence asks for this event to carry full probability: apart from a null set of exceptional paths, every realization of the process converges in the elementary sense of calculus. This is a much more demanding requirement than the weak law's, which controls only each X_n in isolation. The distinction between controlling the marginals and controlling the paths is the entire content of the difference between the weak and strong laws, and the first task is to check that the strong notion really is stronger.

Proposition 5.2.2: Almost Sure Convergence Implies Convergence in Probability

If $X_n \to X$ almost surely, then $X_n \to^{\mathbb{P}} X$ (definition 5.1.1).

***Proof*:**
Fix $\varepsilon > 0$ and, for $m \geq 1$, set

$$B_m = \bigcup_{n\geq m} \{|X_n - X| > \varepsilon\}$$

the event that some deviation past index m exceeds ε. The sets B_m decrease as m grows, since enlarging m removes terms from the union. We claim their intersection $\bigcap_m B_m$ is contained in the exceptional null set of non-convergence. If $\omega \in \bigcap_m B_m$, then for every m there is some $n \geq m$ with $|X_n(\omega) - X(\omega)| > \varepsilon$, so the inequality $|X_n(\omega) - X(\omega)| > \varepsilon$ holds for infinitely many n, and the sequence $X_n(\omega)$ cannot converge to $X(\omega)$. By hypothesis the set of such ω is null, so $\mathbb{P}(\bigcap_m B_m) = 0$. Continuity of probability along the decreasing sequence (B_m) (proposition 1.1.5), applicable since $\mathbb{P}(B_1) \leq 1 < \infty$, gives

$$\lim_{m\to\infty} \mathbb{P}(B_m) = \mathbb{P}\Big(\bigcap_m B_m\Big) = 0$$

Since $\{|X_m - X| > \varepsilon\} \subseteq B_m$, monotonicity of $\mathbb{P}$ yields $\mathbb{P}(|X_m - X| > \varepsilon) \leq \mathbb{P}(B_m) \to 0$. As $\varepsilon > 0$ was arbitrary, this is exactly convergence in probability, as we sought to show.

The proof isolates the mechanism cleanly: almost sure convergence controls the entire tail $\{|X_n - X| > \varepsilon \text{ for some } n \geq m\}$ and forces its probability to zero, whereas convergence in probability controls only the single slice $\{|X_m - X| > \varepsilon\}$. Because the slice sits inside the tail, the strong statement implies the weak one, and the gap between them is precisely the gap between a supremum over $n \geq m$ and a single term. The converse fails: a sequence can converge in probability while almost every path oscillates forever. The standard witness is the moving-bump or typewriter sequence, in which a bump of shrinking width marches across $[0, 1]$ and returns again and again; it is constructed in full in example 5.2.4, and the exercises at the end of the section ask the reader to verify both halves of its behaviour. The lesson is that convergence in probability is strictly weaker, and that the strong law is a strictly stronger theorem than the weak law it refines.

The variance-condition strong law of section 4.3 (theorem 4.3.2) already gives an almost sure law of large numbers, but it pays for it with a finite-variance hypothesis. For an identically distributed sequence that is too strong: the average $\bar{X}_n$ ought to converge to μ as soon as μ exists, that is, as soon as $\mathbb{E}|X_1| < \infty$, whether or not the second moment is finite. Kolmogorov's theorem removes the variance hypothesis entirely, at the cost of the identical-distribution assumption that lets a single truncation level be tuned to the tail of the common law. It is the definitive form of the strong law.

Theorem 5.2.3: Kolmogorov's Strong Law of Large Numbers

Let $(X_n)_{n\geq 1}$ be independent and identically distributed random variables with $\mathbb{E}|X_1| < \infty$, and set $\mu = \mathbb{E}[X_1]$. Then, writing $S_n = X_1 + \cdots + X_n$ and $\bar{X}_n = S_n/n$,

$$\bar{X}_n \longrightarrow \mu \qquad \text{almost surely}$$

The main device of the proof is truncation, in the same spirit as definition 4.2.1, but with a level that grows with the index. The single integrability hypothesis $\mathbb{E}|X_1| < \infty$ is spent in two places: once through the first Borel-Cantelli lemma to show that truncating at level n changes only finitely many terms along almost every path, and once through a series estimate to show that the truncated variables, though not identically distributed, satisfy the variance condition of theorem 4.3.2. Two elementary facts about the tail of an integrable variable drive both estimates, and recording them before the proof keeps its four steps unencumbered. For a nonnegative random variable Z one has the layer-cake identity $\mathbb{E}[Z] = \int_0^\infty \mathbb{P}(Z > t)\, dt$, and specialized to $Z = |X_1|$ and summed over integer thresholds it gives

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n) \leq \int_0^\infty \mathbb{P}(|X_1| > t)\, dt = \mathbb{E}|X_1| < \infty \tag{5.2}$$

the inequality because the sum is a lower Riemann sum for the decreasing integrand. The second fact is the elementary tail bound on the sum of inverse squares,

$$\sum_{n\geq k} \frac{1}{n^2} \leq \frac{1}{k^2} + \int_k^\infty \frac{dt}{t^2} = \frac{1}{k^2} + \frac{1}{k} \leq \frac{2}{k} \qquad (k \geq 1) \tag{5.3}$$

again comparing the sum to an integral. Both (5.2) and (5.3) are pure calculus; the mathematics of the theorem is in how they combine.

***Proof*:**
Write F for the common distribution function of the X_n and let $\mu = \mathbb{E}[X_1]$. Define the truncated variables

$$Y_n = X_n \mathbb{1}_{\{|X_n|\leq n\}}$$

truncating the n-th variable at its own index. The proof proceeds in four steps: replace X_n by Y_n up to finitely many terms, verify the variance condition for the Y_n, run the variance-condition strong law on the centered Y_n, and identify the limit of the centerings with μ.

Step 1: $X_n = Y_n$ eventually, almost surely. The events on which truncation changes the n-th term are $\{X_n \neq Y_n\} = \{|X_n| > n\}$. Since the X_n are identically distributed, $\mathbb{P}(|X_n| > n) = \mathbb{P}(|X_1| > n)$, and summing over n and invoking (5.2),

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n) \leq \mathbb{E}|X_1| < \infty$$

By the first Borel-Cantelli lemma (theorem 1.4.5), which needs no independence, the event $\{X_n \neq Y_n \text{ infinitely often}\}$ has probability zero. Hence for almost every ω there is a finite index $N(\omega)$ with $X_n(\omega) = Y_n(\omega)$ for all $n \geq N(\omega)$. The two partial sums $\sum_{k\leq n} X_k$ and $\sum_{k\leq n} Y_k$ then differ, for such ω, by a fixed finite quantity for all large n, so their averages have the same limit:

$$\frac{1}{n}\sum_{k=1}^{n}(X_k - Y_k) \longrightarrow 0 \qquad \text{almost surely}$$

since the numerator is eventually constant in n. It therefore suffices to prove $\frac{1}{n}\sum_{k\leq n} Y_k \to \mu$ almost surely; the passage back to $\bar{X}_n$ is free.

Step 2: the variance condition for the Y_n. The $Y_n = X_n \mathbb{1}_{\{|X_n|\leq n\}}$ are independent, being measurable functions of the independent X_n (definition 2.2.2), and each is bounded, hence in L^2 with a finite variance. Bound the variance by the second moment, $\operatorname{Var}(Y_n) \leq \mathbb{E}[Y_n^2]$, and compute the second moment from the common law. Using that the X_n are identically distributed and the change-of-variables formula (theorem 3.1.2),

$$\mathbb{E}[Y_n^2] = \mathbb{E}\big[X_n^2 \mathbb{1}_{\{|X_n|\leq n\}}\big] = \mathbb{E}\big[X_1^2 \mathbb{1}_{\{|X_1|\leq n\}}\big] = \int_{\{|x|\leq n\}} x^2 \, dF(x)$$

Now sum $\mathbb{E}[Y_n^2]/n^2$ over n and exchange the order of summation and integration (both are sums of nonnegative terms, so Tonelli's theorem applies, see [2]):

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}[Y_n^2]}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} \int_{\{|x|\leq n\}} x^2 \, dF(x) = \int_{\mathbb{R}} x^2 \Bigg(\sum_{n\geq|x|} \frac{1}{n^2}\Bigg) dF(x)$$

where the inner sum runs over those n with $|x| \leq n$, that is $n \geq |x|$, so it starts at the smallest integer $k = \lceil |x| \rceil \geq |x|$. For $|x| \geq 1$ the tail bound (5.3) with $k = \lceil |x| \rceil \geq |x|$ gives $\sum_{n\geq|x|} 1/n^2 \leq 2/\lceil |x| \rceil \leq 2/|x|$, so $x^2 \sum_{n\geq|x|} 1/n^2 \leq 2|x|$; for $|x| < 1$ the entire sum $\sum_{n\geq 1} 1/n^2 = \pi^2/6$ is bounded and $x^2 \cdot \pi^2/6 \leq \pi^2/6$. In either range $x^2 \sum_{n\geq|x|} 1/n^2 \leq 2|x| + \pi^2/6$, a bound integrable against dF because $\mathbb{E}|X_1| < \infty$. Hence

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}[Y_n^2]}{n^2} \leq \int_{\mathbb{R}} \big(2|x| + \tfrac{\pi^2}{6}\big) \, dF(x) = 2\,\mathbb{E}|X_1| + \frac{\pi^2}{6} < \infty$$

In particular $\sum_n \operatorname{Var}(Y_n)/n^2 \leq \sum_n \mathbb{E}[Y_n^2]/n^2 < \infty$, so the sequence (Y_n) satisfies the variance condition (4.6) of theorem 4.3.2.

Step 3: the strong law for the truncated variables. The Y_n are independent with finite variances satisfying the summability of Step 2, so theorem 4.3.2 applies and gives

$$\frac{1}{n}\sum_{k=1}^{n}\big(Y_k - \mathbb{E}[Y_k]\big) \longrightarrow 0 \qquad \text{almost surely}$$

It remains to show that the Cesàro average of the centerings $\mathbb{E}[Y_k]$ converges to μ; combined with the last display this will give $\frac{1}{n}\sum_{k\le n} Y_k \to \mu$ almost surely.

Step 4: $\frac{1}{n}\sum_{k\le n}\mathbb{E}[Y_k] \to \mu$. By the change-of-variables formula and identical distribution,

$$\mathbb{E}[Y_k] = \mathbb{E}\big[X_k \mathbb{1}_{\{|X_k|\le k\}}\big] = \mathbb{E}\big[X_1 \mathbb{1}_{\{|X_1|\le k\}}\big]$$

As $k \to \infty$ the integrands $X_1\mathbb{1}_{\{|X_1|\le k\}}$ converge pointwise to X_1 and are dominated by the integrable variable $|X_1|$, so dominated convergence ([2]) gives $\mathbb{E}[Y_k] \to \mathbb{E}[X_1] = \mu$. A convergent sequence has the same limit for its Cesàro averages: if $a_k \to \mu$ then $\frac{1}{n}\sum_{k\le n} a_k \to \mu$. Applying this to $a_k = \mathbb{E}[Y_k]$,

$$\frac{1}{n}\sum_{k=1}^{n}\mathbb{E}[Y_k] \longrightarrow \mu$$

Adding this to the centered convergence of Step 3,

$$\frac{1}{n}\sum_{k=1}^{n} Y_k = \frac{1}{n}\sum_{k=1}^{n}\big(Y_k - \mathbb{E}[Y_k]\big) + \frac{1}{n}\sum_{k=1}^{n}\mathbb{E}[Y_k] \longrightarrow 0 + \mu = \mu \qquad \text{almost surely}$$

By Step 1 the average of the X_k has the same almost sure limit as the average of the Y_k, so $\bar{X}_n \to \mu$ almost surely, completing the proof.

The proof is a machine with four interlocking parts, and each part uses the integrability hypothesis exactly once. Truncation at the moving level n is the device that makes an identically distributed sequence with only a first moment behave, for the purposes of the variance-condition law, like a sequence with controlled second moments: the higher the index, the more of the tail is retained, and the retained second moment $\mathbb{E}[Y_n^2]$ is allowed to grow, but slowly enough that $\sum \mathbb{E}[Y_n^2]/n^2$ still converges. The Borel-Cantelli step guarantees that this truncation is invisible in the limit, since it disturbs only finitely many terms of almost every path. The estimate (5.3) is what converts a first moment into the summability of second moments over n; it is the exact point at which $\mathbb{E}|X_1| < \infty$, and nothing more, is enough. The statement that emerges is sharp in its hypothesis: the sample average of an i.i.d. sequence converges almost surely to the mean precisely when the mean exists.

The connection to the zero-one law bears making explicit. The limit $\lim_n \bar{X}_n$, when it exists, is a tail-measurable random variable: changing finitely many of the X_k does not change it, so it is measurable with respect to the tail σ-algebra of definition 2.4.1. Kolmogorov's zero-one law (theorem 2.4.3) then forces it to be almost surely constant, so the *existence* of an almost sure limit already implies the limit is a deterministic number; the strong law's contribution is to name that number, μ. This is the same reading applied to the coin walk in example 4.3.3, now in the sharp identically distributed setting.

Example 5.2.4: The Strong Law for Coin Tossing and Beyond

1. *The coin average converges almost surely.* Let (X_n) be i.i.d. with $\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = 0) = 1/2$, the head indicators of the fair coin-tossing space (example 1.3.6), so $\bar{X}_n$ is the fraction of heads in the first n tosses. Here $\mathbb{E}|X_1| = \mathbb{E}[X_1] = 1/2 < \infty$, so Kolmogorov's strong law (theorem 5.2.3) applies and gives

$$\bar{X}_n \longrightarrow \frac{1}{2} \qquad \text{almost surely}$$

Along almost every infinite sequence of tosses the head fraction settles to $1/2$. This is the assertion that fair coins obey the frequentist interpretation of probability path by path, not merely in the weak sense that $\bar{X}_n$ is close to $1/2$ for each large n: the set of sequences whose head fraction fails to converge to $1/2$, or converges to anything else, has probability zero. The centered version ± 1 walk of example 4.3.3 is the same statement shifted, $S_n/n \to 0$.

2. *In probability but not almost surely: the typewriter sequence.* To see that proposition 5.2.2 has no converse, work on the unit interval $\Omega = [0, 1]$ with Lebesgue measure (example 1.1.3). Index the dyadic intervals block by block: for each $m \geq 0$ and each $0 \leq j < 2^m$, let

$$I_{m,j} = \left[\tfrac{j}{2^m}, \tfrac{j+1}{2^m}\right]$$

and list these in a single sequence, first the one interval of level $m = 0$, then the two of level $m = 1$, then the four of level $m = 2$, and so on, calling the resulting indicators $X_1, X_2, X_3, \ldots$, so that $X_n = \mathbb{1}_{I_{m,j}}$ for the appropriate (m, j). As $n \to \infty$ the level m tends to infinity, and the width of the supporting interval is $\mathbb{P}(X_n = 1) = 2^{-m} \to 0$. Hence for any $\varepsilon \in (0, 1)$,

$$\mathbb{P}(|X_n - 0| > \varepsilon) = \mathbb{P}(X_n = 1) = 2^{-m} \longrightarrow 0$$

so $X_n \to^{\mathbb{P}} 0$: the sequence converges to 0 in probability. But it converges almost nowhere. Fix any $\omega \in [0, 1]$. Within each level m exactly one interval $I_{m,j}$ contains ω, so among the 2^m indicators of level m exactly one equals 1 at ω and the rest equal 0. Thus $X_n(\omega) = 1$ for infinitely many n (one per level) and $X_n(\omega) = 0$ for infinitely many n, and the numerical sequence $X_n(\omega)$ oscillates between 0 and 1 forever without converging. The bump of height one sweeps across $[0, 1]$ again and again, ever narrower, hitting every point infinitely often; the marginal probability of a hit shrinks to zero, but the path never settles. This is the promised witness that convergence in probability is strictly weaker than almost sure convergence.

Kolmogorov's route reaches the strong law through the convergence of a random series, the machinery of section 4.2, and it leans on independence in the form the one-series theorem requires. A second proof, due to Etemadi, reaches the same conclusion by a different road that dispenses with the full force of independence: it needs only that the variables be *pairwise* independent, and it never invokes a series theorem, working instead with a geometric subsequence and monotonicity. The two proofs are worth having side by side, since each teaches a technique that recurs, series convergence in one case, subsequence-plus-monotonicity in the other. Etemadi's theorem is also strictly more general in its independence hypothesis, since pairwise independence is weaker than the mutual independence Kolmogorov's proof uses.

Theorem 5.2.5: Etemadi's Strong Law of Large Numbers

Let $(X_n)_{n\geq 1}$ be identically distributed and *pairwise* independent random variables with $\mathbb{E}|X_1| < \infty$, and set $\mu = \mathbb{E}[X_1]$. Then $\bar{X}_n \to \mu$ almost surely.

The reduction to nonnegative summands is the first move, and it is what makes the monotonicity argument available: if the X_n are nonnegative then so are the partial sums S_n, and a nondecreasing sequence is easy to control between the terms of a sparse subsequence. Along a geometric subsequence $n_k = \lfloor \alpha^k \rfloor$ the variance estimate needed for Chebyshev is summable because the n_k grow geometrically, so the first Borel-Cantelli lemma pins $\bar{X}_{n_k}$ to μ; the gaps between consecutive n_k are then filled by squeezing S_n between S_{n_k} and $S_{n_{k+1}}$, and letting $\alpha \downarrow 1$ closes the squeeze.

***Proof Alternative proof*:**
We give the proof for nonnegative X_n, then reduce the general case to it.

Reduction to $X_n \geq 0$. Suppose the theorem is known for nonnegative summands. For general integrable X_n, decompose $X_n = X_n^+ - X_n^-$ into positive and negative parts. The sequences (X_n^+) and (X_n^-) are each identically distributed (as measurable functions of the identically distributed X_n) and pairwise independent (for $i \neq j$ the variables X_i^+, X_j^+ are functions of the independent pair X_i, X_j, hence independent, and likewise for the negative parts), and each is nonnegative and integrable since $\mathbb{E}[X_1^\pm] \leq \mathbb{E}|X_1| < \infty$. The nonnegative case gives $\frac{1}{n}\sum_{k\leq n} X_k^+ \to \mathbb{E}[X_1^+]$ and $\frac{1}{n}\sum_{k\leq n} X_k^- \to \mathbb{E}[X_1^-]$ almost surely, and subtracting,

$$\bar{X}_n = \frac{1}{n}\sum_{k=1}^{n}(X_k^+ - X_k^-) \longrightarrow \mathbb{E}[X_1^+] - \mathbb{E}[X_1^-] = \mathbb{E}[X_1] = \mu$$

almost surely. So it suffices to treat $X_n \geq 0$, which we now assume.

Step 1: truncation and its harmlessness. Set $Y_n = X_n \mathbb{1}_{\{X_n \leq n\}}$ and $T_n = Y_1 + \cdots + Y_n$. Exactly as in Step 1 of the proof of theorem 5.2.3, identical distribution and (5.2) give

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) = \sum_{n=1}^{\infty} \mathbb{P}(X_1 > n) \leq \mathbb{E}|X_1| < \infty$$

so by the first Borel-Cantelli lemma (theorem 1.4.5) almost surely $X_n = Y_n$ for all large n, and hence $\frac{1}{n}(S_n - T_n) \to 0$ almost surely. It therefore suffices to prove $T_n/n \to \mu$ almost surely.

Step 2: the truncated means converge to μ. As in Step 4 of theorem 5.2.3, $\mathbb{E}[Y_n] = \mathbb{E}[X_1 \mathbb{1}_{\{X_1\leq n\}}] \to \mathbb{E}[X_1] = \mu$ by dominated convergence, so the Cesàro averages satisfy $\frac{1}{n}\mathbb{E}[T_n] = \frac{1}{n}\sum_{k\leq n}\mathbb{E}[Y_k] \to \mu$. It thus suffices to show $\frac{1}{n}\big(T_n - \mathbb{E}[T_n]\big) \to 0$ almost surely.

Step 3: control along a geometric subsequence. Fix $\alpha > 1$ and set $n_k = \lfloor \alpha^k \rfloor$. The Y_j are pairwise independent (functions of the pairwise independent X_j), hence pairwise uncorrelated, so Bienaymé's identity (proposition 3.3.7) gives $\mathrm{Var}(T_n) = \sum_{j=1}^{n}\mathrm{Var}(Y_j)$. For each $\varepsilon > 0$, Chebyshev's inequality (corollary 3.2.5) applied to T_{n_k} bounds the deviation of the subsequence, and summing over k,

$$\sum_{k=1}^{\infty}\mathbb{P}\Big(\big|T_{n_k} - \mathbb{E}[T_{n_k}]\big| > \varepsilon n_k\Big) \leq \sum_{k=1}^{\infty}\frac{\mathrm{Var}(T_{n_k})}{\varepsilon^2 n_k^2} = \frac{1}{\varepsilon^2}\sum_{k=1}^{\infty}\frac{1}{n_k^2}\sum_{j=1}^{n_k}\mathrm{Var}(Y_j)$$

Interchange the order of summation over j and k (nonnegative terms):

$$\sum_{k=1}^{\infty} \frac{1}{n_k^2} \sum_{j=1}^{n_k} \operatorname{Var}(Y_j) = \sum_{j=1}^{\infty} \operatorname{Var}(Y_j) \sum_{k\colon n_k \geq j} \frac{1}{n_k^2}$$

The inner sum is controlled by the geometric growth of the n_k. Since $n_k = \lfloor \alpha^k \rfloor \geq \alpha^k/2$ for all $k \geq 1$, one has $n_k^{-2} \leq 4\alpha^{-2k}$, and summing the geometric tail over those k with $n_k \geq j$ (equivalently $\alpha^k \gtrsim j$) gives a bound of the form

$$\sum_{k\colon n_k \geq j} \frac{1}{n_k^2} \leq \frac{C_\alpha}{j^2}$$

for a constant C_α depending only on α: the largest term is $\asymp j^{-2}$ and the remaining terms decrease geometrically, so their sum is a bounded multiple of the largest. Substituting and using $\operatorname{Var}(Y_j) \leq \mathbb{E}[Y_j^2]$ together with the estimate $\sum_j \mathbb{E}[Y_j^2]/j^2 \leq 2\mathbb{E}|X_1| + \pi^2/6 < \infty$ from Step 2 of the proof of theorem 5.2.3 (identical computation), the whole double sum is finite:

$$\sum_{k=1}^{\infty} \mathbb{P}\Big(\big|T_{n_k} - \mathbb{E}[T_{n_k}]\big| > \varepsilon n_k\Big) \leq \frac{C_\alpha}{\varepsilon^2} \sum_{j=1}^{\infty} \frac{\operatorname{Var}(Y_j)}{j^2} < \infty$$

By the first Borel-Cantelli lemma (theorem 1.4.5), for each fixed $\varepsilon > 0$ almost surely only finitely many of the events $\{|T_{n_k} - \mathbb{E}[T_{n_k}]| > \varepsilon n_k\}$ occur. Taking $\varepsilon = 1/i$ over $i \in \mathbb{N}$ and intersecting the corresponding probability-one events, almost surely

$$\frac{T_{n_k} - \mathbb{E}[T_{n_k}]}{n_k} \longrightarrow 0 \qquad \text{as } k \to \infty$$

and since $\mathbb{E}[T_{n_k}]/n_k \to \mu$ by Step 2, this reads $T_{n_k}/n_k \to \mu$ almost surely.

Step 4: filling the gaps by monotonicity. The convergence so far is only along the sparse subsequence (n_k). Here the nonnegativity of the Y_j pays off: the partial sums T_n are nondecreasing in n, since each $Y_j \geq 0$. For any n choose k with $n_k \leq n < n_{k+1}$; monotonicity of T_n then sandwiches

$$\frac{T_{n_k}}{n_{k+1}} \leq \frac{T_n}{n} \leq \frac{T_{n_{k+1}}}{n_k}$$

Rewrite each outer term to isolate a converging factor. Because $n_{k+1}/n_k \to \alpha$ as $k \to \infty$ (the floors of consecutive powers of α), the left bound is

$$\frac{T_{n_k}}{n_{k+1}} = \frac{T_{n_k}}{n_k} \cdot \frac{n_k}{n_{k+1}} \longrightarrow \mu \cdot \frac{1}{\alpha}$$

and the right bound is

$$\frac{T_{n_{k+1}}}{n_k} = \frac{T_{n_{k+1}}}{n_{k+1}} \cdot \frac{n_{k+1}}{n_k} \longrightarrow \mu \cdot \alpha$$

both almost surely, using $T_{n_k}/n_k \to \mu$ from Step 3. As $n \to \infty$ the index k also tends to infinity, so almost surely

$$\frac{\mu}{\alpha} \leq \liminf_{n\to\infty} \frac{T_n}{n} \leq \limsup_{n\to\infty} \frac{T_n}{n} \leq \alpha\mu$$

This holds for every fixed $\alpha > 1$ on a probability-one event; taking a sequence $\alpha \downarrow 1$, say $\alpha = 1 + 1/i$, and intersecting the countably many probability-one events squeezes the liminf and limsup together at μ. Hence $T_n/n \to \mu$ almost surely. Combined with Step 1, $\bar{X}_n = S_n/n \to \mu$ almost surely, as we sought to show.

The two proofs isolate two hypotheses that Kolmogorov's argument bundles together. Kolmogorov uses mutual independence, because the one-series theorem behind theorem 4.3.2 needs the truncated summands to be a mutually independent sequence; in exchange it never has to fill gaps, since the series machinery delivers convergence of the full sequence directly. Etemadi uses only pairwise independence, which is all Bienaymé's identity for the variance of a sum requires (proposition 3.3.7); the price is that Chebyshev along the sparse geometric subsequence controls only $\bar{X}_{n_k}$, and the intervening $\bar{X}_n$ must be recovered by the monotonicity of the partial sums, which is why the nonnegative reduction is required by that route. That pairwise independence suffices is a strengthening with content: pairwise independent sequences that are not mutually independent exist (the parity triple of example 2.2.6 illustrates the pairwise-versus-mutual distinction), and Etemadi's theorem still governs their averages. Both proofs, though, share the same skeleton, truncate at level n, discard the overflow by Borel-Cantelli, and identify the limit of the truncated means by dominated convergence, so the difference is entirely in how the fluctuation of the truncated sum is tamed.

The strong law also closes a loop with the zero-one law that began the chapter's independence theory. For an i.i.d. sequence the average $\bar{X}_n$ has, when it converges, a tail-measurable limit, so theorem 2.4.3 guarantees the limit is deterministic before either strong law is invoked; the theorems of this section supply the value of that deterministic constant and, more, the fact that the limit exists at all. The weak law, the two strong laws, and the zero-one law thus form a single circle of ideas: independence forces the tail to be trivial, triviality forces any limit to be constant, and integrability forces that constant to be the mean.

Exercise 5.2.1

1. Give an example of a sequence (X_n) of random variables and a random variable X with $X_n \to^{\mathbb{P}} X$ but for which $X_n \to X$ almost surely fails, and prove both claims. (You may use the typewriter construction of example 5.2.4 or produce your own.)

2. For the typewriter sequence of example 5.2.4, show directly that $\mathbb{E}[X_n] \to 0$ and that $\mathbb{E}[(X_n - 0)^2] \to 0$, so that $X_n \to 0$ in L^2 as well, while $X_n(\omega)$ fails to converge for *every* $\omega \in [0, 1]$. Conclude that convergence in L^2, like convergence in probability, does not imply almost sure convergence.

3. Prove the following partial converse to proposition 5.2.2: if $X_n \to^{\mathbb{P}} X$, then there is a subsequence (X_{n_k}) with $X_{n_k} \to X$ almost surely. (Choose n_k so that $\mathbb{P}(|X_{n_k} - X| > 1/k) \leq 2^{-k}$ and apply the first Borel-Cantelli lemma, theorem 1.4.5.)

4. Let (X_n) be i.i.d. with the standard Cauchy distribution, whose density is $\frac{1}{\pi(1+x^2)}$. Show that $\mathbb{E}|X_1| = \infty$, so that neither strong law applies, and explain in one or two sentences why the average $\bar{X}_n$ does *not* converge to a constant. (You are not asked to prove the non-convergence; identify which hypothesis of theorem 5.2.3 fails and why the conclusion should be expected to fail with it.)

5. In Kolmogorov's proof the truncation level was the index n itself. Suppose instead one truncates at a level c_n with $c_n \to \infty$. Identify a growth condition on (c_n) under which Step 1 (the Borel-Cantelli step) still gives $\sum_n \mathbb{P}(|X_1| > c_n) < \infty$ for every integrable X_1, and explain why $c_n = n$ is, up to constants, the natural borderline choice. (Hint: compare with (5.2).)

6. Etemadi's theorem assumes only pairwise independence. Explain precisely which step of Kolmogorov's proof of theorem 5.2.3 breaks if the X_n are only pairwise independent, and which step of Etemadi's proof of theorem 5.2.5 uses nonnegativity in a way Kolmogorov's does

not.

5.3 Applications of the Laws of Large Numbers

The strong law of large numbers (theorem 5.2.3) is easy to state and, once proved, easy to underestimate. Its content is a single sentence: the empirical average of an i.i.d. sample converges to the mean along almost every sample path. Yet that sentence is the mechanism behind a randomized method of numerical integration whose accuracy does not degrade with dimension, behind the theorem that a histogram built from data eventually looks like the distribution it was drawn from, behind a probabilistic construction of the polynomials that approximate any continuous function, and behind the fact that almost every real number has perfectly balanced binary digits. This section collects these four consequences. They are not a miscellany. Each one converts the asymptotic certainty the strong law provides into a statement in a neighbouring field, and together they are the reason the law of large numbers, rather than being one theorem among many, is the bridge along which probability reaches statistics and analysis.

The common move in all four is to recognize some quantity of interest as an expectation, $\mathbb{E}[g(X_1)]$ for a well-chosen g and a well-chosen distribution, and then let the strong law replace the expectation by a computable empirical average. Monte Carlo integration makes the move openly and by design; the other three make it in disguise, where the "sample" is a grid of quantiles, a binomial count, or the binary digits of a random real. Seeing the disguise is the whole art.

Numerical integration in one variable is a solved problem: the trapezoid rule, Simpson's rule, and Gaussian quadrature all compute $\int_0^1 g$ to high accuracy with a handful of function evaluations. The difficulty appears in high dimension. A grid-based rule that places m points along each of d axes uses m^d points, and to keep a fixed accuracy as d grows the cost explodes exponentially. This is the curse of dimensionality, and it makes deterministic quadrature useless for the integrals that arise in statistical physics, mathematical finance, and Bayesian inference, where d is routinely in the hundreds. The strong law suggests an escape. Rather than choosing evaluation points on a grid, choose them at random, and estimate the integral by the average of the integrand over the random points. The error of such an estimate, as the next result shows, is governed by the variance of the integrand and the number of samples alone, with no reference whatsoever to the dimension.

Proposition 5.3.1: Monte Carlo Integration

Let $(U_i)_{i\geq 1}$ be i.i.d. random variables uniformly distributed on $[0,1]$, and let $g\colon [0,1] \to \mathbb{R}$ be Borel measurable and integrable, $\int_0^1 |g| < \infty$. Then

$$\frac{1}{n}\sum_{i=1}^{n} g(U_i) \longrightarrow \int_0^1 g(x)\,dx \qquad \text{almost surely}$$

If moreover $g \in L^2[0,1]$, write $I = \int_0^1 g$ and $\sigma^2(g) = \int_0^1 \big(g(x) - I\big)^2\,dx = \operatorname{Var}\big(g(U_1)\big)$. Then the estimator $\widehat{I}_n = \frac{1}{n}\sum_{i=1}^{n} g(U_i)$ is unbiased, $\mathbb{E}[\widehat{I}_n] = I$, and its root-mean-square error is

$$\sqrt{\mathbb{E}\big[(\widehat{I}_n - I)^2\big]} = \frac{\sigma(g)}{\sqrt{n}}$$

***Proof*:**
Set $Y_i = g(U_i)$. The Y_i are i.i.d.: each is the fixed measurable function g applied to U_i, and measurable functions of independent variables are independent (definition 2.2.2), while identical distribution of the U_i forces identical distribution of the $g(U_i)$. Their common mean is the integral: by the change-of-variables formula for expectation (theorem 3.1.2), pushing U_1 forward through g and using that the law of U_1 is Lebesgue measure on $[0, 1]$,

$$\mathbb{E}[Y_1] = \mathbb{E}[g(U_1)] = \int_0^1 g(x)\,dx = I$$

which is finite because $\int_0^1 |g| < \infty$. The sequence (Y_i) is therefore i.i.d. with a finite mean, so theorem 5.2.3 applies and gives

$$\frac{1}{n}\sum_{i=1}^{n} g(U_i) = \frac{1}{n}\sum_{i=1}^{n} Y_i \longrightarrow \mathbb{E}[Y_1] = I \qquad \text{almost surely}$$

which is the first claim.

For the error, assume $g \in L^2$, so $\mathbb{E}[Y_1^2] = \int_0^1 g^2 < \infty$ and Y_1 has a finite variance $\sigma^2(g) = \operatorname{Var}(Y_1) = \mathbb{E}[(Y_1 - I)^2]$ (definition 3.2.1), the second expression in the statement following again from theorem 3.1.2. Unbiasedness is linearity of expectation: $\mathbb{E}[\widehat{I}_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Y_i] = I$. For the mean-square error, the Y_i are independent, hence uncorrelated, so variance is additive (proposition 3.3.7):

$$\mathbb{E}\big[(\widehat{I}_n - I)^2\big] = \operatorname{Var}(\widehat{I}_n) = \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}(Y_i) = \frac{n\sigma^2(g)}{n^2} = \frac{\sigma^2(g)}{n}$$

Taking square roots gives the root-mean-square error $\sigma(g)/\sqrt{n}$, completing the proof.

The content of the second half is the $\sqrt{n}$ rate and its independence from dimension. The error $\sigma(g)/\sqrt{n}$ involves only the variance of the integrand and the sample size; it contains no factor that grows with the number of variables. An integral over the d-dimensional cube $[0, 1]^d$ is handled by exactly the same estimator, sampling i.i.d. points $\mathbf{U}_i \sim \operatorname{Unif}[0, 1]^d$ and averaging $g(\mathbf{U}_i)$, and the proof is unchanged because the strong law and the variance identity never saw the dimension. The price is the slow rate: halving the error requires quadrupling the sample, since $1/\sqrt{n}$ improves only as the square root. For a smooth one-dimensional integrand this is far worse than Simpson's rule. But Simpson's rule in d dimensions costs m^d points for a comparable per-axis resolution m, whereas Monte Carlo pays $\sigma(g)/\sqrt{n}$ regardless of d. There is a crossover dimension beyond which random sampling wins, and for the high-dimensional integrals of real applications the crossover is passed early. This is the trade at the heart of the method: a dimension-blind but slow rate defeats a dimension-cursed but fast one once the dimension is large.

Example 5.3.2: Estimating an Integral by Sampling

Consider $I = \int_0^1 e^x\,dx = e - 1 \approx 1.71828$, an integral one can do in closed form precisely so the estimator can be checked. The integrand is $g(x) = e^x$, with

$$\mathbb{E}[g(U_1)^2] = \int_0^1 e^{2x}\,dx = \frac{e^2 - 1}{2} \approx 3.19453$$

so the variance of a single sample is

$$\sigma^2(g) = \int_0^1 e^{2x}\,dx - \left(\int_0^1 e^x\,dx\right)^2 = \frac{e^2-1}{2} - (e-1)^2 \approx 3.19453 - 2.95249 = 0.24204$$

giving $\sigma(g) \approx 0.492$. The Monte Carlo estimator $\widehat{I}_n = \frac{1}{n}\sum_{i=1}^n e^{U_i}$ then has root-mean-square error $\sigma(g)/\sqrt{n} \approx 0.492/\sqrt{n}$. To pin the estimate to within 10^{-3} of the true value in root-mean-square, one needs $0.492/\sqrt{n} \leq 10^{-3}$, that is $n \geq (492)^2 \approx 2.4 \times 10^5$ samples. The slowness is on display: a quarter of a million evaluations for three-digit accuracy, where Simpson's rule would need a handful. The point of the example is not that Monte Carlo is efficient here, on a one-dimensional smooth integrand it is not, but that the same $0.492/\sqrt{n}$ bound would hold verbatim for $\int_{[0,1]^{100}} e^{x_1}\,d\mathbf{x}$ with a hundred variables, where no deterministic grid rule is feasible at all.

The Monte Carlo estimate is the strong law used as intended: an expectation is what one wants, and the empirical average is what one computes. The three results that follow use the same engine on quantities that are expectations only after a change of viewpoint. The first is the theorem that legitimizes the very idea of estimating a distribution from a sample.

Suppose $X_1, X_2, \dots$ are i.i.d. draws from an unknown distribution with distribution function F. From the sample one forms the empirical distribution function

$$F_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{X_i \leq x\}}$$

the fraction of the first n observations that fall at or below x. This is the staircase a statistician sees: it is the data's own estimate of F. For each fixed x the strong law immediately gives $F_n(x) \to F(x)$, since the indicators $\mathbb{1}_{\{X_i \leq x\}}$ are i.i.d. Bernoulli variables with mean $\mathbb{P}(X_1 \leq x) = F(x)$. What is not immediate, and what makes the estimate trustworthy, is that the convergence is *uniform* in x: the entire graph of F_n converges to the entire graph of F, so that with probability one no point of the distribution is eventually mis-estimated by more than a vanishing amount. The uniformity is the Glivenko-Cantelli theorem, sometimes called the fundamental theorem of statistics because it is what guarantees that a large sample determines the underlying law.

Theorem 5.3.3: Glivenko-Cantelli

Let $(X_i)_{i\geq 1}$ be i.i.d. real random variables with distribution function $F(x) = \mathbb{P}(X_1 \leq x)$, and let

$$F_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{X_i \leq x\}}$$

be the empirical distribution function of the first n observations. Then

$$\|F_n - F\|_\infty = \sup_{x\in\mathbb{R}} \big|F_n(x) - F(x)\big| \longrightarrow 0 \qquad \text{almost surely}$$

***Proof*:**
The argument has two ingredients: the pointwise strong law, applied at finitely many points at once, and the monotonicity of both F and F_n, which lets control at finitely many points be upgraded to control everywhere.

Step 1: Pointwise convergence at a value and at a left limit. Fix $x \in \mathbb{R}$. The random variables

$\mathbb{1}_{\{X_i \le x\}}$, $i \ge 1$, are i.i.d. (measurable functions of the i.i.d. X_i; definition 2.2.2) and bounded, hence integrable, with mean $\mathbb{E}[\mathbb{1}_{\{X_1 \le x\}}] = \mathbb{P}(X_1 \le x) = F(x)$. By theorem 5.2.3,

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \le x\}} \longrightarrow F(x) \qquad \text{almost surely}$$

The same argument applied to the indicators $\mathbb{1}_{\{X_i < x\}}$ gives $F_n(x^-) \to F(x^-)$ almost surely, where $F_n(x^-) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i < x\}}$ is the left limit of the staircase F_n at x and $F(x^-) = \mathbb{P}(X_1 < x)$ is the left limit of F. (Right continuity makes $F(x)$ itself the right limit, so the value and the left limit are the two quantities the jumps of F are built from.)

Step 2: A finite grid of quantiles. Fix an integer $m \ge 1$. For each $j = 1, \dots, m-1$ define the quantile

$$x_j = \inf\{x \in \mathbb{R} : F(x) \ge j/m\}$$

the smallest point at which F reaches the level j/m, with the conventions $x_0 = -\infty$ and $x_m = +\infty$. These $m-1$ points partition $\mathbb{R}$ into the intervals (x_{j-1}, x_j), $j = 1, \dots, m$, and by the definition of x_j as an infimum, together with the right continuity and monotonicity of F,

$$F(x_j) \ge \frac{j}{m} \qquad \text{and} \qquad F(x_j^-) \le \frac{j}{m} \tag{5.4}$$

for $1 \le j \le m-1$. The first inequality holds because $F(x_j) \ge j/m$ by right continuity applied to the infimum; the second because any $x < x_j$ has $F(x) < j/m$, and $F(x_j^-)$ is the supremum of these values. Consequently consecutive grid values differ by at most $1/m$ in a controlled way: $F(x_j^-) - F(x_{j-1}) \le j/m - (j-1)/m = 1/m$ for each j, so F increases by no more than $1/m$ across the interior of each interval (x_{j-1}, x_j).

Step 3: A full-probability event on which the grid is controlled. By Step 1 there is, for each of the finitely many grid points $x_1, \dots, x_{m-1}$, an event of probability one on which $F_n(x_j) \to F(x_j)$ and $F_n(x_j^-) \to F(x_j^-)$. The intersection Ω_m of these finitely many full-probability events again has probability one, and on Ω_m there is a random index $N = N(m, \omega)$ such that for all $n \ge N$ and all $j = 1, \dots, m-1$,

$$\left|F_n(x_j) - F(x_j)\right| \le \frac{1}{m} \qquad \text{and} \qquad \left|F_n(x_j^-) - F(x_j^-)\right| \le \frac{1}{m} \tag{5.5}$$

Fix $\omega \in \Omega_m$ and $n \ge N$ for the remainder of the step; all quantities below are evaluated at this ω.

Step 4: Monotonicity upgrades the grid to all of $\mathbb{R}$. Let $x \in \mathbb{R}$ be arbitrary and let j be the index with $x \in [x_{j-1}, x_j)$ (taking $x_{j-1} = -\infty$ or $x_j = +\infty$ at the ends, where the bounds below are trivial since F and F_n agree with 0 or 1 in the limit). Both F and F_n are nondecreasing, so

$$F_n(x_{j-1}) \le F_n(x) \le F_n(x_j^-) \qquad \text{and} \qquad F(x_{j-1}) \le F(x) \le F(x_j^-)$$

Subtracting the second chain from the first and using the extreme members,

$$F_n(x) - F(x) \le F_n(x_j^-) - F(x_{j-1}) \le \left[F(x_j^-) + \tfrac{1}{m}\right] - F(x_{j-1}) \le \tfrac{1}{m} + \tfrac{1}{m} = \tfrac{2}{m}$$

where the middle inequality is (5.5) and the last uses $F(x_j^-) - F(x_{j-1}) \le 1/m$ from Step 2. Symmetrically,

$$F_n(x) - F(x) \ge F_n(x_{j-1}) - F(x_j^-) \ge \left[F(x_{j-1}) - \tfrac{1}{m}\right] - F(x_j^-) \ge -\tfrac{2}{m}$$

Thus $|F_n(x) - F(x)| \le 2/m$ for every $x \in \mathbb{R}$ simultaneously, so $\|F_n - F\|_\infty \le 2/m$ for all $n \ge N(m, \omega)$.

Step 5: Let the mesh shrink. Steps 3 and 4 show that on Ω_m, $\limsup_n \|F_n - F\|_\infty \le 2/m$. Let $\Omega_\infty = \bigcap_{m \ge 1} \Omega_m$, a countable intersection of full-probability events, so $\mathbb{P}(\Omega_\infty) = 1$ (proposition 1.1.5). On Ω_∞ the bound $\limsup_n \|F_n - F\|_\infty \le 2/m$ holds for every m, hence $\limsup_n \|F_n - F\|_\infty = 0$, that is $\|F_n - F\|_\infty \to 0$. This holds on an event of probability one, as we sought to show.

The theorem is the reason inferential statistics works. A single realization of the sample produces the function F_n, and Glivenko-Cantelli certifies that this observed function is, for large n, uniformly close to the unknown F that generated the data. Every functional of F that is continuous in the uniform metric (medians, quantiles, and moments computed against dF_n among them) is thereby consistently estimated by the corresponding functional of F_n. The uniformity is what distinguishes the theorem from the pointwise strong law: pointwise convergence would allow the worst discrepancy $\sup_x |F_n(x) - F(x)|$ to persist by migrating to ever-new locations x, and the monotonicity argument of Steps 2 to 4 is exactly what forbids that migration. The quantitative refinement, that $\sqrt{n}\,\|F_n - F\|_\infty$ has a nondegenerate limiting distribution (the Kolmogorov-Smirnov statistic), belongs to the theory of weak convergence developed in the chapters ahead; the qualitative fact that the discrepancy vanishes at all is already a theorem about almost-sure limits, and it is the strong law that delivers it.

The next application turns the strong law, or rather its quantitative cousin Chebyshev's inequality, into a construction in classical analysis. Weierstrass's approximation theorem asserts that every continuous function on a closed interval is a uniform limit of polynomials. Bernstein gave a proof that is entirely probabilistic: he wrote down an explicit sequence of polynomials, the Bernstein polynomials, and recognized each one as an expectation of the function evaluated at a binomial average. The law of large numbers, in the concrete form of binomial concentration, then forces the approximation. The proof is a model of the section's theme, an analytic statement proved by reading a polynomial as an expectation.

The starting observation is that the binomial distribution is exactly the law of a scaled coin-average. If $S_n \sim \mathrm{Bin}(n, x)$ counts the successes in n independent trials each of success probability x, then $S_n = \sum_{i=1}^n \mathbb{1}_{\{U_i \le x\}}$ for i.i.d. $U_i \sim \mathrm{Unif}[0,1]$, so S_n/n is the empirical frequency of success and, by the law of large numbers, concentrates near x. The Bernstein polynomial is what one gets by averaging the target function f against this concentrating distribution.

Theorem 5.3.4: Weierstrass Approximation via Bernstein Polynomials

Let $f\colon [0,1] \to \mathbb{R}$ be continuous. For each $n \ge 1$ define the n-th Bernstein polynomial of f,

$$B_n(f)(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k} \qquad (x \in [0,1])$$

Then $B_n(f) \to f$ uniformly on $[0,1]$:

$$\sup_{x \in [0,1]} \bigl|B_n(f)(x) - f(x)\bigr| \longrightarrow 0 \qquad \text{as } n \to \infty$$

In particular, every continuous function on $[0,1]$ is a uniform limit of polynomials.

***Proof*:**
Fix $x \in [0,1]$ and let S_n be a random variable with the binomial distribution $\mathrm{Bin}(n,x)$, that is $\mathbb{P}(S_n = k) = \binom{n}{k} x^k (1-x)^{n-k}$ for $k = 0, \ldots, n$; this is the law of a sum of n i.i.d. $\mathrm{Bernoulli}(x)$ variables (example 2.3.6). The Bernstein polynomial is then precisely the expectation of f evaluated at the binomial average S_n/n:

$$\mathbb{E}\left[f\left(\frac{S_n}{n}\right)\right] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \mathbb{P}(S_n = k) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k} = B_n(f)(x) \tag{5.6}$$

using theorem 3.1.2 for the first equality. Since $f(x)$ is a constant and S_n/n has mean x, the error to control is

$$B_n(f)(x) - f(x) = \mathbb{E}\left[f\left(\frac{S_n}{n}\right) - f(x)\right] \tag{5.7}$$

Step 1: The variance of the binomial average. The sum S_n has mean $\mathbb{E}[S_n] = nx$ and variance $\mathrm{Var}(S_n) = nx(1-x)$ (the variance of a sum of n independent $\mathrm{Bernoulli}(x)$ variables, each of variance $x(1-x)$; definition 3.2.1). Hence the average S_n/n has mean x and variance

$$\mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{x(1-x)}{n} \leq \frac{1}{4n} \tag{5.8}$$

where the bound uses $x(1-x) \leq 1/4$ for $x \in [0,1]$, attained at $x = 1/2$. The crucial feature of (5.8) is that the bound $1/(4n)$ is independent of x; this is what will make the approximation uniform.

Step 2: Uniform continuity fixes the split. Since f is continuous on the compact interval $[0,1]$, it is uniformly continuous and bounded; write $M = \sup_{[0,1]} |f| < \infty$. Fix $\varepsilon > 0$. Uniform continuity provides a $\delta > 0$ such that $|f(u) - f(v)| < \varepsilon$ whenever $|u - v| < \delta$ with $u, v \in [0,1]$. Split the expectation in (5.7) according to whether the binomial average is within δ of x:

$$\left|B_n(f)(x) - f(x)\right| \leq \mathbb{E}\Big[\Big|f\big(\tfrac{S_n}{n}\big) - f(x)\Big|\Big] = \mathbb{E}\big[\cdots ; \big|\tfrac{S_n}{n} - x\big| < \delta\big] + \mathbb{E}\big[\cdots ; \big|\tfrac{S_n}{n} - x\big| \geq \delta\big]$$

where the semicolon denotes expectation restricted to the indicated event and $\cdots$ abbreviates $|f(S_n/n) - f(x)|$.

Step 3: Bound each piece. On the event $\{|S_n/n - x| < \delta\}$ uniform continuity gives $|f(S_n/n) - f(x)| < \varepsilon$, so the first term is at most ε. On the complementary event the integrand is at most $2M$, so the second term is at most $2M \cdot \mathbb{P}(|S_n/n - x| \geq \delta)$. Chebyshev's inequality (corollary 3.2.5), applied to S_n/n with its mean x and the variance bound (5.8), controls this probability:

$$\mathbb{P}\left(\left|\frac{S_n}{n} - x\right| \geq \delta\right) \leq \frac{\mathrm{Var}(S_n/n)}{\delta^2} \leq \frac{1}{4n\delta^2}$$

Combining the two pieces,

$$\left|B_n(f)(x) - f(x)\right| \leq \varepsilon + \frac{2M}{4n\delta^2} = \varepsilon + \frac{M}{2n\delta^2} \tag{5.9}$$

Step 4: Uniformity and passage to the limit. The bound (5.9) holds for every $x \in [0,1]$ with the same ε, δ, and M, because the variance bound $1/(4n)$ of Step 1 is uniform in x and δ was chosen from uniform continuity, not pointwise. Taking the supremum over x,

$$\sup_{x \in [0,1]} \left|B_n(f)(x) - f(x)\right| \leq \varepsilon + \frac{M}{2n\delta^2}$$

Letting $n \to \infty$ sends the second term to 0, so $\limsup_n \sup_x |B_n(f)(x) - f(x)| \le \varepsilon$. Since $\varepsilon > 0$ was arbitrary, the supremum tends to 0, which is uniform convergence. As each $B_n(f)$ is a polynomial, every continuous f on $[0,1]$ is a uniform limit of polynomials, completing the proof.

The proof is the law of large numbers dressed as approximation theory. The binomial average S_n/n concentrates on its mean x (Step 1 is exactly the variance computation behind the weak law), so the distribution against which f is averaged in (5.6) collapses onto the single point x, and the average $\mathbb{E}[f(S_n/n)]$ converges to $f(x)$. Two features make the convergence uniform rather than pointwise, and both were flagged in the proof: the variance bound $x(1-x) \le 1/4$ is independent of x, and the modulus of continuity δ comes from uniform, not pointwise, continuity of f on the compact interval. The Bernstein polynomials are, beyond this existence proof, a usable approximation scheme, and their probabilistic origin is not incidental: it is why they reproduce constants and linear functions exactly and why they never overshoot the range of f, since $B_n(f)(x)$ is an average of values of f.

Example 5.3.5: Bernstein Approximation of a Quadratic

Take $f(x) = x^2$ on $[0,1]$, where the Bernstein polynomials can be computed in closed form and the convergence watched directly. By (5.6), $B_n(f)(x) = \mathbb{E}[(S_n/n)^2]$ with $S_n \sim \mathrm{Bin}(n,x)$. Using $\mathbb{E}[(S_n/n)^2] = \mathrm{Var}(S_n/n) + (\mathbb{E}[S_n/n])^2$ and the moments from Step 1 of the proof,

$$B_n(x^2) = \frac{x(1-x)}{n} + x^2 = x^2 + \frac{x - x^2}{n}$$

The Bernstein polynomial of x^2 is thus x^2 plus a correction of order $1/n$, and $\sup_{x\in[0,1]} |B_n(x^2) - x^2| = \sup_x \frac{x-x^2}{n} = \frac{1}{4n}$, the supremum attained at $x = 1/2$. The error decays like $1/n$ and vanishes uniformly, in accord with the theorem. The example also shows that Bernstein polynomials do *not* reproduce x^2 exactly, unlike constants ($B_n(1) = 1$) and linear functions ($B_n(x) = x$, since $\mathbb{E}[S_n/n] = x$): the approximation is exact only up to degree one, which is the price of its probabilistic robustness.

The final application returns to the running coin-tossing space and reads the strong law as a statement about the decimal, or rather binary, expansions of real numbers. Borel proved in 1909 that almost every real number in $[0,1]$ is *normal*: in its binary expansion the digit 1 appears with limiting frequency exactly $1/2$, as does the digit 0. The proof is a single application of the strong law once the binary digits of a uniformly random real are i.i.d. fair coin flips. This identification, between a uniform variable on the interval and an infinite sequence of independent bits, is the same one that underlies the construction of the coin-tossing space, and here it pays off as a theorem about the arithmetic of almost every number.

Theorem 5.3.6: Borel's Normal Number Theorem

For Lebesgue-almost-every $x \in [0,1]$, writing $x = \sum_{n=1}^{\infty} d_n(x) 2^{-n}$ for the binary expansion with digits $d_n(x) \in \{0,1\}$, the frequency of the digit 1 among the first n digits converges to $1/2$:

$$\frac{1}{n}\sum_{k=1}^{n} d_k(x) \longrightarrow \frac{1}{2} \qquad \text{for Lebesgue-almost-every } x \in [0,1]$$

Such an x is called *normal to base* 2.

***Proof*:**
Model x as a uniform random variable. Let $X \sim \mathrm{Unif}[0,1]$ be the identity coordinate on the probability space $([0,1], \mathcal{B}, \lambda)$ of example 1.1.3, where λ is Lebesgue measure, so that "Lebesgue-almost-every x" is exactly "almost surely" for this X. Its binary digits are the functions

$$d_n(X) = \lfloor 2^n X \rfloor - 2\lfloor 2^{n-1} X \rfloor \in \{0,1\}$$

(a dyadically rational X, of which there are only countably many and hence a Lebesgue-null set, has two expansions; discard this null set, which does not affect an almost-sure statement).

Step 1: The digits are i.i.d. fair coin flips. The event $\{d_1(X) = e_1, \ldots, d_n(X) = e_n\}$ for prescribed bits $e_1, \ldots, e_n \in \{0,1\}$ is exactly the dyadic interval

$$\left[\sum_{k=1}^{n} e_k 2^{-k},\ \sum_{k=1}^{n} e_k 2^{-k} + 2^{-n}\right)$$

of length 2^{-n}, so under Lebesgue measure

$$\mathbb{P}\big(d_1(X) = e_1, \ldots, d_n(X) = e_n\big) = 2^{-n} = \prod_{k=1}^{n} \frac{1}{2} = \prod_{k=1}^{n} \mathbb{P}\big(d_k(X) = e_k\big)$$

the last equality because each single-digit event $\{d_k(X) = e_k\}$ is a union of 2^{k-1} dyadic intervals of length 2^{-k}, of total measure $2^{k-1} \cdot 2^{-k} = 1/2$. The product formula over all choices of bits says that $(d_n(X))_{n \geq 1}$ is a sequence of independent random variables, each uniform on $\{0,1\}$, that is i.i.d. Bernoulli$(1/2)$: an infinite sequence of fair coin flips, the coin-tossing space of example 1.3.6 realized inside the unit interval.

Step 2: Apply the strong law. The digits $d_1(X), d_2(X), \ldots$ are i.i.d. with $\mathbb{E}[d_1(X)] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = \frac{1}{2}$, finite. By theorem 5.2.3,

$$\frac{1}{n} \sum_{k=1}^{n} d_k(X) \longrightarrow \mathbb{E}[d_1(X)] = \frac{1}{2} \qquad \text{almost surely}$$

Translating back through the identification of "almost surely" with "Lebesgue-almost-every x", the digit frequency converges to $1/2$ for almost every $x \in [0,1]$, which is the assertion, completing the proof.

The theorem is a clean instance of the strong law converting a probabilistic certainty into an arithmetic one. The set of non-normal numbers is nonempty and even uncountable, for instance every number whose binary expansion is eventually all zeros, or the number $0.010101\ldots_2$ whose digit-1 frequency is $1/2$ but which fails a stronger normality (that every finite block appears with its Bernoulli frequency), or numbers engineered to have digit frequency $1/3$. What Borel's theorem says is that this exceptional set, however large in cardinality, is negligible in measure: a real number drawn at random is normal with probability one.

Two extensions are immediate from the same proof and worth recording, though the details are left as exercises. First, the base is irrelevant: for any integer base $b \geq 2$, the base-b digits of a uniform X are i.i.d. uniform on $\{0, 1, \ldots, b-1\}$ by the identical dyadic-interval computation with intervals of length b^{-n}, so almost every x has each base-b digit appearing with frequency $1/b$; this is *simple normality* to base b. Second, full normality, that every finite block of length ℓ appears with its expected frequency $b^{-\ell}$, follows by applying the strong law to the overlapping-block indicators, and a number normal to every base simultaneously is called *absolutely normal*. Borel's theorem, in the form that almost every real is absolutely normal, is the assertion that this most stringent balance is the generic behaviour, even though exhibiting a single explicit absolutely normal number is difficult and no "natural" constant such as π or $\sqrt{2}$ is known to be normal to any base.

Standing back, the four applications trace the reach of a single theorem across three subjects. In numerical analysis the strong law is the guarantee behind Monte Carlo integration, and it is the source of a method whose error ignores dimension. In statistics it is the Glivenko-Cantelli theorem, the assurance that a sample determines the distribution it was drawn from, on which the entire enterprise of estimation rests. In analysis it is Bernstein's constructive proof of the Weierstrass theorem, a polynomial approximation of any continuous function read off from binomial concentration. In number theory it is Borel's theorem that almost every real number is digit-balanced. The strong law is, in each case, the statement that averages of independent samples are asymptotically deterministic; what changes is only what the samples are and what quantity their average computes. This is the sense in which the law of large numbers is the bridge from probability to the quantitative sciences: it is the theorem that lets a random average stand in for a deterministic quantity, and once that substitution is licensed, the quantity can be an integral, a distribution function, a value of a continuous function, or a digit frequency, and the same theorem covers them all.

Exercise 5.3.1

1. (Variance reduction by antithetic sampling.) Let $g\colon [0,1] \to \mathbb{R}$ be in L^2 and consider, alongside the plain Monte Carlo estimator of proposition 5.3.1, the antithetic estimator $\widetilde{I}_n = \frac{1}{n}\sum_{i=1}^n \frac{g(U_i)+g(1-U_i)}{2}$ for i.i.d. $U_i \sim \mathrm{Unif}[0,1]$. Show that $\widetilde{I}_n$ is unbiased for $\int_0^1 g$, and that its per-sample variance is $\frac{1}{4}\mathrm{Var}\big(g(U_1)+g(1-U_1)\big)$. Deduce that if g is monotone then antithetic sampling never increases the variance, and compute the antithetic variance for $g(x) = x$ to show it can be zero.

2. (Sharpness of Glivenko-Cantelli's uniformity.) Let X_i be i.i.d. $\mathrm{Unif}[0,1]$, so $F(x) = x$ on $[0,1]$. For fixed n, describe the graph of F_n and explain why $\sup_x |F_n(x) - F(x)|$ is always attained at (or infinitesimally to the left of) one of the observed data points $X_{(1)} \le \cdots \le X_{(n)}$. Using this, express $\|F_n - F\|_\infty$ in terms of the order statistics $X_{(k)}$.

3. (Bernstein reproduces low-degree polynomials.) Using (5.6) and the binomial moments, show that $B_n(1) = 1$ and $B_n(x) = x$ for every n, so constants and linear functions are fixed by the Bernstein operator. Then compute $B_n(x^3)$ in closed form and identify the order in n of $\sup_x |B_n(x^3) - x^3|$.

4. (Base-b normality.) Adapt the proof of theorem 5.3.6 to an arbitrary integer base $b \ge 2$: show that the base-b digits of a uniform $X \in [0,1]$ are i.i.d. uniform on $\{0,1,\dots,b-1\}$, and conclude that for almost every x each base-b digit has limiting frequency $1/b$. Where in the argument is the value of b used?

5. (The non-normal numbers are uncountable.) Exhibit an uncountable set of $x \in [0,1]$ that are not simply normal to base 2, and reconcile this with theorem 5.3.6. (Hint: force the digit frequency to a value other than $1/2$ by prescribing digits along a fixed positive-density subsequence and letting the rest range freely.)

6. (Monte Carlo beats the grid in high dimension.) Fix an accuracy target $\eta > 0$ and a bounded integrand g on $[0,1]^d$ with variance σ^2. Compare the number of function evaluations needed

to reach root-mean-square (or worst-case) error η for (i) Monte Carlo integration and (ii) a product midpoint rule using m points per axis, whose error scales like C/m^2 for twice-differentiable g. For which dimensions d does Monte Carlo require fewer evaluations? Explain qualitatively why the crossover dimension is small in practice.

5.4 Application: The Normal Order of $\omega(n)$

The laws of large numbers of the preceding sections concern a sequence of independent trials laid out in time: coins flipped one after another, samples drawn one after another. This section applies the same machinery to an object that has nothing to do with time and, on its face, nothing to do with chance: a single integer n and the count of its distinct prime factors. The passage from one world to the other is a change of viewpoint that Mark Kac made famous with the slogan that "the primes play a game of chance". Fix a large bound N, draw an integer n uniformly at random from $\{1, \ldots, N\}$, and ask how many distinct primes divide it. Divisibility by 2 is decided by a coin that comes up heads with probability close to $1/2$ (half of the integers up to N are even); divisibility by 3, by an almost independent coin with probability close to $1/3$; and so on through the primes. The number of distinct prime factors is then a sum of these nearly independent indicator coins, and the theory of sums of independent random variables should have something to say about it. It does: it says that the count is almost always close to $\log \log n$, a growth so slow that it is nearly constant across the whole range of everyday integers. This is the Hardy-Ramanujan theorem, and the argument that proves it is Chebyshev's inequality (corollary 3.2.5) applied to a carefully computed variance, the second-moment method that Turán isolated as the heart of the matter.

The one non-probabilistic ingredient is a fact about the primes, Mertens' theorem, which supplies the $\log \log$ scale. It is imported from analytic number theory; everything probabilistic, the moment computations and the concentration argument, is carried out in full here.

The model is the uniform measure on a finite set of integers, with the divisibility events as its basic random variables.

Definition 5.4.1: The Random Integer and Its Prime Divisors

Fix an integer $N \geq 2$, and let n be uniformly distributed on $\{1, \ldots, N\}$, that is, n is a random variable on the probability space $\{1, \ldots, N\}$ with $\mathbb{P}(n = k) = 1/N$ for each k (a scaled counting measure, the finite analogue of example 1.1.3). For each prime p define the *divisibility indicator*

$$D_p \colon \{1, \ldots, N\} \to \{0, 1\}, \qquad D_p(n) = \mathbb{1}_{\{p \mid n\}}$$

the random variable that is 1 when p divides n and 0 otherwise. The *number of distinct prime factors* of n is

$$\omega(n) = \sum_{p \leq N} D_p(n) = \#\{p \text{ prime} : p \mid n\}$$

the sum running over all primes $p \leq N$ (no prime exceeding N can divide an integer $n \leq N$, so the restriction $p \leq N$ loses nothing).

Two conventions deserve emphasis. First, ω counts *distinct* primes, without multiplicity: $\omega(12) =$

$\omega(2^2 \cdot 3) = 2$, not 3. The count with multiplicity, usually written $\Omega(n)$, differs from $\omega(n)$ only by the contribution of the higher prime powers, and that contribution is bounded on average, so the two functions have the same normal order; the argument below is cleanest for ω. Second, $\omega(n)$ is a deterministic arithmetic function of n, an ordinary object of number theory. Randomness enters only through the choice of n: once N is fixed and n is drawn uniformly, ω becomes a random variable, and its distribution encodes how prime factorizations are distributed across the integers up to N. The whole strategy is to study this random variable by the tools of chapter 4 and the present chapter, treating the D_p as the summands.

Example 5.4.2: Small Values of ω

Take $N = 12$ and n uniform on $\{1, \ldots, 12\}$. The primes $p \leq 12$ are $2, 3, 5, 7, 11$. The values of ω are

n	1	2	3	4	5	6	7	8	9	10	11	12
$\omega(n)$	0	1	1	1	1	2	1	1	1	2	1	2

Only $n = 1$ has $\omega = 0$; the primes and prime powers $(2, 3, 4, 5, 7, 8, 9, 11)$ have $\omega = 1$; the numbers $6 = 2 \cdot 3$, $10 = 2 \cdot 5$, $12 = 2^2 \cdot 3$ have $\omega = 2$. Even at this tiny scale the count clusters: no integer up to 12 has three distinct prime factors, since the smallest such integer is $2 \cdot 3 \cdot 5 = 30$. The concentration that the theorem below makes precise is already visible as this reluctance of ω to spread out.

The individual indicator D_p is a Bernoulli variable whose success probability is, to a small error, $1/p$. Making the "almost independent" picture precise means computing the first two moments of the D_p: the means, which will feed the expected value of ω, and the pairwise products, which control the covariances and hence the variance. Both computations reduce to counting the multiples of an integer in $\{1, \ldots, N\}$, and that count is a floor.

Proposition 5.4.3: Moments of the Divisibility Indicators

Let n be uniform on $\{1, \ldots, N\}$ and let D_p be as in definition 5.4.1. Then for every prime $p \leq N$,

$$\mathbb{E}[D_p] = \frac{\lfloor N/p \rfloor}{N} = \frac{1}{p} + O\left(\frac{1}{N}\right)$$

and for distinct primes p, q,

$$\mathbb{E}[D_p D_q] = \frac{\lfloor N/pq \rfloor}{N} = \begin{cases} \frac{1}{pq} + O\left(\frac{1}{N}\right) & \text{if } pq \leq N \\ 0 & \text{if } pq > N \end{cases}$$

Consequently, for distinct primes p, q,

$$\operatorname{Cov}(D_p, D_q) = \mathbb{E}[D_p D_q] - \mathbb{E}[D_p]\mathbb{E}[D_q] = O\left(\frac{1}{N}\right)$$

with the implied constant absolute (independent of p, q, N).

***Proof*:**
Step 1: The single-prime mean. The event $\{D_p = 1\}$ is exactly the set of multiples of p in $\{1, \dots, N\}$, namely $p, 2p, \dots, \lfloor N/p \rfloor p$, of which there are $\lfloor N/p \rfloor$. Since D_p is an indicator, $\mathbb{E}[D_p] = \mathbb{P}(p \big| n) = \lfloor N/p \rfloor / N$ by the change-of-variables formula for expectation (theorem 3.1.2), applied to the two-valued D_p. Writing $N/p = \lfloor N/p \rfloor + \{N/p\}$ with the fractional part $\{N/p\} \in [0, 1)$ gives

$$\frac{\lfloor N/p \rfloor}{N} = \frac{N/p - \{N/p\}}{N} = \frac{1}{p} - \frac{\{N/p\}}{N} = \frac{1}{p} + O\left(\frac{1}{N}\right)$$

since $0 \le \{N/p\} < 1$ bounds the error by $1/N$ in absolute value.

Step 2: The two-prime product. For distinct primes p, q the product $D_p D_q$ is the indicator of $\{p \big| n\} \cap \{q \big| n\}$, and since p, q are distinct primes, $p \big| n$ and $q \big| n$ together hold if and only if $pq \big| n$ (coprimality of p and q). Thus $D_p D_q = D_{pq}$ as functions, an indicator of the multiples of pq, and the count of those in $\{1, \dots, N\}$ is $\lfloor N/pq \rfloor$. Hence $\mathbb{E}[D_p D_q] = \lfloor N/pq \rfloor / N$. If $pq > N$ then no $n \le N$ is a multiple of pq, so $\lfloor N/pq \rfloor = 0$ and the expectation vanishes, recording that no integer up to N can be divisible by two distinct primes whose product already exceeds N. If instead $pq \le N$, the same floor estimate as in Step 1 (with pq in place of p) gives $\mathbb{E}[D_p D_q] = 1/(pq) + O(1/N)$.

Step 3: The covariance. Fix distinct primes p, q. When $pq \le N$, combine the two estimates:

$$\operatorname{Cov}(D_p, D_q) = \mathbb{E}[D_p D_q] - \mathbb{E}[D_p]\mathbb{E}[D_q] = \left(\tfrac{1}{pq} + O(\tfrac{1}{N})\right) - \left(\tfrac{1}{p} + O(\tfrac{1}{N})\right)\left(\tfrac{1}{q} + O(\tfrac{1}{N})\right)$$

The product of the two means is $1/(pq) + O(1/N)$, because each factor is at most 1 and one of them carries an $O(1/N)$ error, so the leading terms $1/(pq)$ cancel and what remains is $O(1/N)$. When $pq > N$, we have $\mathbb{E}[D_p D_q] = 0$, so

$$\operatorname{Cov}(D_p, D_q) = -\mathbb{E}[D_p]\mathbb{E}[D_q] = -\frac{\lfloor N/p \rfloor}{N} \cdot \frac{\lfloor N/q \rfloor}{N}$$

and here $\mathbb{E}[D_p] = \lfloor N/p \rfloor / N \le 1/p$ and likewise for q, so $|\operatorname{Cov}(D_p, D_q)| \le 1/(pq) < 1/N$ using $pq > N$. In both cases the covariance is $O(1/N)$ with an absolute implied constant, completing the proof.

proposition 5.4.3 is the quantitative form of the "game of chance" picture. If the events $\{p \big| n\}$ were exactly independent, then $\mathbb{E}[D_p D_q]$ would equal $\mathbb{E}[D_p]\mathbb{E}[D_q] = 1/(pq)$ exactly and every covariance would vanish; the divisibility of n by p and by q are *almost* independent, failing only by the $O(1/N)$ arithmetic error that the floor introduces, and, for large primes with $pq > N$, by an outright negative correlation forced by the finiteness of the range. This near-independence is precisely what the second-moment method needs: a variance that behaves as if the summands were independent, up to errors that sum to something negligible. The next result carries out that summation.

Before stating the theorem, the analytic input must be named. The prime numbers are not distributed by chance in reality, and the only ingredient of the argument that reaches outside probability is a statement about how the reciprocals of the primes accumulate. This is Mertens' theorem, one of the central results of elementary analytic number theory, which is proved in full in that book and imported here.

5.4.4 Note: Mertens' Theorem (Imported) For $x \geq 2$, the primes satisfy

$$\sum_{p \leq x} \frac{1}{p} = \log\log x + M + O\left(\frac{1}{\log x}\right), \qquad \sum_{p \leq x} \frac{1}{p^2} = C + O\left(\frac{1}{x \log x}\right)$$

where M is the Mertens constant and $C = \sum_p p^{-2} < \infty$. In particular $\sum_{p \leq x} 1/p = \log\log x + O(1)$ and $\sum_{p \leq x} 1/p^2 = O(1)$, the second sum being bounded because $\sum_p p^{-2} \leq \sum_{k \geq 2} k^{-2} < \infty$ converges. These are proved in *Analytic Number Theory* [1]; here they enter as the source of the $\log\log$ scale and are used as black boxes.

Two auxiliary counts about primes are used in the error analysis below, and both are cited to the same source rather than developed here, since they are standard analytic number theory whose full proofs (the prime number theorem, or the weaker Chebyshev bounds that suffice) would be a lengthy detour with no probabilistic content. The first is that the number of primes up to x, $\pi(x) = \#\{p \leq x\}$, satisfies $\pi(x) = O(x/\log x)$, so in particular $\pi(N) = O(N/\log N) = o(N)$. The second is the tail estimate needed to discard the large primes: $\sum_{p \leq N,\, pq > N}$-type sums are controlled by $\sum_{p \leq N} 1/p = \log\log N + O(1)$ together with $\sum_p 1/p^2 = O(1)$. These are the delicate prime-sum inputs, cited to *Analytic Number Theory* [1] with the note that each is a standard-but-lengthy estimate, so that the probabilistic skeleton of the argument stands clear.

With the moments computed and the analytic input named, the mean and variance of ω fall out by linearity, and Chebyshev's inequality converts the small variance into the concentration statement.

Theorem 5.4.5: Hardy-Ramanujan: The Normal Order of ω

Let n be uniform on $\{1, \ldots, N\}$ and $\omega(n) = \sum_{p \leq N} D_p(n)$ its number of distinct prime factors (definition 5.4.1). Then

$$\mathbb{E}[\omega] = \log\log N + O(1), \qquad \mathrm{Var}(\omega) = \log\log N + O(1)$$

Consequently, for any function ψ with $\psi(N) \to \infty$ as $N \to \infty$,

$$\frac{1}{N}\#\left\{ n \leq N : \left|\omega(n) - \log\log N\right| > \psi(N)\sqrt{\log\log N} \right\} \longrightarrow 0$$

Equivalently, $(\omega - \log\log N)/\sqrt{\log\log N} \to 0$ in probability (definition 5.1.1). In the language of number theory, $\omega(n)$ has *normal order* $\log\log n$: for almost all n (all but a density-zero set), $\omega(n)$ is asymptotic to $\log\log n$.

***Proof*:**
Step 1: The mean. By linearity of expectation (proposition 3.1.3) and the single-prime mean of proposition 5.4.3,

$$\mathbb{E}[\omega] = \sum_{p \leq N} \mathbb{E}[D_p] = \sum_{p \leq N} \left(\frac{1}{p} + O\left(\frac{1}{N}\right)\right) = \sum_{p \leq N} \frac{1}{p} + O\left(\frac{\pi(N)}{N}\right)$$

The error is a sum of $\pi(N)$ terms each $O(1/N)$, hence $O(\pi(N)/N) = O(1)$, since $\pi(N) =$

$O(N/\log N) = o(N)$. The main term is Mertens' sum, $\sum_{p\le N} 1/p = \log\log N + O(1)$. Combining,

$$\mathbb{E}[\omega] = \log\log N + O(1)$$

as claimed.

Step 2: The variance as a sum over pairs. Since ω is a finite sum of the D_p, its variance expands into single-prime variances and pairwise covariances (the multivariate Bienaymé identity of proposition 3.3.7, itself linearity of expectation applied to $(\omega - \mathbb{E}\omega)^2$):

$$\mathrm{Var}(\omega) = \sum_{p\le N} \mathrm{Var}(D_p) + \sum_{\substack{p,q\le N\\ p\ne q}} \mathrm{Cov}(D_p, D_q)$$

The two sums are handled separately: the diagonal gives the main term, the off-diagonal is shown to be $O(1)$.

Step 3: The diagonal (single-prime variances). Each D_p is a $\{0,1\}$-valued indicator, so $D_p^2 = D_p$ and, by definition 3.2.1,

$$\mathrm{Var}(D_p) = \mathbb{E}[D_p^2] - \mathbb{E}[D_p]^2 = \mathbb{E}[D_p] - \mathbb{E}[D_p]^2 = \mathbb{E}[D_p]\big(1 - \mathbb{E}[D_p]\big)$$

Using $\mathbb{E}[D_p] = 1/p + O(1/N)$ from proposition 5.4.3, and summing,

$$\sum_{p\le N} \mathrm{Var}(D_p) = \sum_{p\le N}\left(\frac{1}{p} - \frac{1}{p^2}\right) + O\left(\frac{\pi(N)}{N}\right) = \sum_{p\le N}\frac{1}{p} - \sum_{p\le N}\frac{1}{p^2} + O(1)$$

Here the cross terms from $\mathbb{E}[D_p](1-\mathbb{E}[D_p]) = (1/p + O(1/N))(1 - 1/p + O(1/N))$ produce, beyond $1/p - 1/p^2$, only further $O(1/N)$ contributions, and $\pi(N)$ of them sum to $O(1)$ as in Step 1. By Mertens' theorem the first sum is $\log\log N + O(1)$ and the second is $O(1)$, so

$$\sum_{p\le N} \mathrm{Var}(D_p) = \log\log N + O(1)$$

Step 4: The off-diagonal (covariances) is $O(1)$. The number of ordered pairs of distinct primes is at most $\pi(N)^2$, and each covariance is $O(1/N)$ by proposition 5.4.3, so a crude bound gives $O(\pi(N)^2/N)$, which need not be $O(1)$; a finer accounting is required, and it is here that the split between small and large primes matters. Separate the pairs by whether $pq \le N$ or $pq > N$.

For pairs with $pq \le N$, the covariance is a difference of two floor estimates. From Step 3 of proposition 5.4.3, when $pq \le N$,

$$\mathrm{Cov}(D_p, D_q) = \frac{\lfloor N/pq\rfloor}{N} - \frac{\lfloor N/p\rfloor}{N}\frac{\lfloor N/q\rfloor}{N}$$

Writing $\lfloor N/m\rfloor/N = 1/m - \theta_m/N$ with $\theta_m = \{N/m\} \in [0,1)$ for $m \in \{p, q, pq\}$, the leading $1/(pq)$ terms cancel and each remaining term is $O(1/N)$ times a factor bounded by 1; hence $|\mathrm{Cov}(D_p, D_q)| \le c/N$ for an absolute constant c. To sum this over the pairs with $pq \le N$, note that the map $(p,q) \mapsto pq$ sends distinct ordered pairs to squarefree integers $m \le N$, each hit at most twice (by (p,q) and (q,p)), so the number of such pairs is at most $2N$. Therefore

$$\Bigg|\sum_{\substack{p\ne q\\ pq\le N}} \mathrm{Cov}(D_p, D_q)\Bigg| \le \frac{c}{N}\cdot \#\{(p,q) : p\ne q,\ pq\le N\} \le \frac{c}{N}\cdot 2N = 2c$$

an $O(1)$ contribution from the small-product pairs.

For pairs with $pq > N$, the covariance is negative and equals $-\mathbb{E}[D_p]\mathbb{E}[D_q]$, so its absolute value is $\mathbb{E}[D_p]\mathbb{E}[D_q] \le (1/p)(1/q)$ by the mean bound $\mathbb{E}[D_p] \le 1/p$, and hence

$$\sum_{\substack{p\neq q\\ pq>N}} \bigl|\mathrm{Cov}(D_p, D_q)\bigr| \le \sum_{\substack{p\neq q\\ pq>N}} \frac{1}{pq}$$

That this tail sum is $O(1)$ is the one step of the argument that lies in analytic number theory rather than probability. A crude bound already comes from Mertens' theorem: the constraint $pq > N$ forces $\max(p,q) > \sqrt{N}$, so the sum is at most $2\bigl(\sum_{\sqrt{N}<p\le N} 1/p\bigr)\bigl(\sum_{p\le N} 1/q\bigr)$, and $\sum_{\sqrt{N}<p\le N} 1/p = \log\log N - \log\log\sqrt{N} + O(1) = \log 2 + O(1) = O(1)$ since $\log\log\sqrt{N} = \log(\frac{1}{2}\log N) = \log\log N - \log 2$, leaving $O(\log\log N)$. The sharpening to the full $O(1)$ bound is Turán's estimate for this prime double-sum, a standard-but-lengthy computation with no probabilistic content, cited to *Analytic Number Theory* [1]. Granting it,

$$\sum_{\substack{p\neq q\\ pq>N}} \mathrm{Cov}(D_p, D_q) = O(1)$$

Step 5: Assembling the variance. Combining Steps 3 and 4,

$$\mathrm{Var}(\omega) = \underbrace{\sum_{p\le N} \mathrm{Var}(D_p)}_{\log\log N + O(1)} + \underbrace{\sum_{p\neq q} \mathrm{Cov}(D_p, D_q)}_{O(1)} = \log\log N + O(1)$$

Step 6: Chebyshev's inequality (Turán's argument). With the mean and variance in hand, the concentration is immediate. Write $\lambda = \log\log N$, so $\mathbb{E}[\omega] = \lambda + O(1)$ and $\mathrm{Var}(\omega) = \lambda + O(1)$. Fix $\psi(N) \to \infty$. Chebyshev's inequality (corollary 3.2.5) applied to ω about its own mean gives, for any $t > 0$,

$$\mathbb{P}\bigl(|\omega - \mathbb{E}\omega| > t\bigr) \le \frac{\mathrm{Var}(\omega)}{t^2}$$

Because $\mathbb{E}[\omega] = \lambda + O(1)$, there is a constant A with $|\mathbb{E}\omega - \lambda| \le A$ for all large N. For the event $|\omega - \lambda| > \psi(N)\sqrt{\lambda}$ to occur with $\psi(N)\sqrt{\lambda} > 2A$ (which holds for all large N), the triangle inequality forces $|\omega - \mathbb{E}\omega| > \psi(N)\sqrt{\lambda} - A > \frac{1}{2}\psi(N)\sqrt{\lambda}$. Hence

$$\mathbb{P}\bigl(|\omega - \lambda| > \psi(N)\sqrt{\lambda}\bigr) \le \mathbb{P}\bigl(|\omega - \mathbb{E}\omega| > \tfrac{1}{2}\psi(N)\sqrt{\lambda}\bigr) \le \frac{\mathrm{Var}(\omega)}{\frac{1}{4}\psi(N)^2\lambda} = \frac{\lambda + O(1)}{\frac{1}{4}\psi(N)^2\lambda}$$

The right side is $4(1+O(1/\lambda))/\psi(N)^2 \to 0$ as $N \to \infty$, since $\psi(N) \to \infty$ and $\lambda = \log\log N \to \infty$. Because $\mathbb{P}(\,\cdot\,)$ on the uniform space is exactly $(1/N)\#\{\cdot\}$, this is the stated density statement, and it says precisely that $(\omega - \lambda)/\sqrt{\lambda} \to 0$ in probability (definition 5.1.1), completing the proof.

The theorem says that the graph of ω is, for almost all n, pinned to the curve $\log\log n$ within a window of width $\sqrt{\log\log n}$, a window vanishingly narrow relative to the height on the logarithmic double scale. The phrase *normal order* names exactly this phenomenon: an arithmetic function f has normal order g when $f(n) \sim g(n)$ for all n outside a set of density zero, and Hardy and Ramanujan's discovery was that ω, despite its wild pointwise behaviour, has the very regular normal order $\log\log n$.

The proof deserves a second look for what it does and does not use. It does not use any hard theorem about primes beyond Mertens' estimate for $\sum 1/p$; it does not use independence of the D_p, only their near-independence quantified as a summable covariance; and its concentration step is nothing more than Chebyshev's inequality, which uses only the first two moments. That the first two moments suffice is Turán's insight, and it is the reason this argument, the *second-moment method*, became a template far beyond number theory: whenever a quantity of interest is a sum of many weakly correlated indicators, its mean and variance alone may already pin it down.

A word on the scale itself, because $\log\log n$ is deceptive. It grows without bound, so a "typical" integer has more and more prime factors as $n \to \infty$, but it grows so slowly that across the entire range of integers a human being will ever write down the count is essentially a small constant. The next example makes this concrete and then contrasts the normal order with the extreme values that it averages over.

Example 5.4.6: The Normal Order Numerically, and the Extremes It Hides

1. *The typical count near a million.* For n near 10^6, the normal order is

$$\log\log 10^6 = \log(6\log 10) = \log(6\cdot 2.3026) = \log(13.816) \approx 2.626$$

so a typical integer near a million has about 2.6, that is roughly three, distinct prime factors. Even at $n = 10^{12}$ the count is only $\log\log 10^{12} = \log(12\cdot 2.3026) = \log(27.63) \approx 3.32$, still about three; and at $n = 10^{80}$, an integer with as many digits as there are atoms in the observable universe, one finds $\log\log 10^{80} = \log(80\cdot 2.3026) = \log(184.2) \approx 5.2$. The number of distinct prime factors of a random integer is, for all practical purposes, a single-digit number.

2. *The extremes the average washes out.* The normal order is a statement about almost all n; the exceptional n can be as far from $\log\log n$ as arithmetic allows. At the low end, every prime power $n = p^k$ has $\omega(n) = 1$, no matter how large n is: infinitely many integers have exactly one distinct prime factor, sitting far below $\log\log n$. At the high end, the *primorial* $n = p_1 p_2 \cdots p_k$ (the product of the first k primes) has $\omega(n) = k$ by construction. Since the k-th prime is about $k\log k$, the primorial has size roughly $\exp(\sum_{j\le k}\log p_j)$, and by the prime number theorem $\log n = \sum_{j\le k}\log p_j \sim p_k \sim k\log k$, so $k \sim \log n/\log\log n$. Thus the primorial achieves

$$\omega(n) = k \sim \frac{\log n}{\log\log n}$$

the maximal order of ω, which outgrows the normal order $\log\log n$ by the factor $\log n/(\log\log n)^2 \to \infty$. Both the prime powers and the primorials form density-zero sets, exactly the exceptional sets that the Hardy-Ramanujan theorem discards, and their existence is why the theorem must be phrased in terms of density rather than as a bound valid for every n.

The two halves of example 5.4.6 frame what the normal order is and is not. It is not a bound: ω ranges from 1 (on the prime powers) up to about $\log n/\log\log n$ (on the primorials), a spread that grows without limit. It is a statement about the overwhelming majority: strip away a set of density zero, and on everything that remains $\omega(n)$ hugs $\log\log n$. This is the same structure as every law of large numbers in this chapter. There, the average $\bar{X}_n$ could in principle land anywhere in the range of the X_i, but the set of outcomes on which it strays far from μ has probability tending to zero; here, $\omega(n)$ could be as small as 1 or as large as $\log n/\log\log n$, but the set of integers on which it strays far from $\log\log n$ has density tending to zero. Theorem 5.4.5 is, in the precise sense of definition 5.1.1, a weak law of large numbers, with the uniform measure on $\{1,\ldots,N\}$ playing the role of the probability space and the nearly-independent divisibility indicators playing the role of the i.i.d. summands. The mean $\log\log N$ replaces μ, and the normalization $\sqrt{\log\log N}$, the square root of the variance, is exactly the scale on which a weak law lives.

Naming this a weak law invites the question that a weak law always invites: what does the fluctuation look like on the scale $\sqrt{\log\log N}$, once the crude Chebyshev bound is sharpened? For sums of i.i.d. variables the answer is the central limit theorem, and the analogy extends far enough that the same answer holds here. The centred and rescaled count $(\omega(n)-\log\log N)/\sqrt{\log\log N}$ does more than tend to zero in the loose sense of Chebyshev, which leaves its shape unspecified: its distribution across $\{1,\ldots,N\}$ converges to the standard Gaussian. The number of distinct prime factors of a random integer is asymptotically normal, with mean and variance both $\log\log N$. This is the Erdős-Kac theorem, the central-limit refinement of Hardy-Ramanujan, and proving it requires replacing the second moment by all the moments, or equivalently by a characteristic-function argument, the machinery developed for the central limit theorem later in this book. The primes will play their game of chance one more time, and the next time the bell curve will appear.

Exercise 5.4.1

1. Compute $\mathbb{E}[\omega]$ for $N=12$ directly from the table in example 5.4.2, and separately from the formula $\mathbb{E}[\omega]=\sum_{p\le 12}\lfloor 12/p\rfloor/12$. Confirm the two agree, and compare the result to the asymptotic prediction $\log\log 12$.

2. Let D be the indicator of an event of probability r. Show $\operatorname{Var}(D)=r(1-r)$, and deduce that $\operatorname{Var}(D)\le 1/4$ with equality exactly when $r=1/2$. Explain in one sentence why this bound is why the diagonal sum $\sum_p \operatorname{Var}(D_p)$ in theorem 5.4.5 is dominated by the small primes.

3. Let $\Omega(n)$ count prime factors *with* multiplicity, so $\Omega(n)=\sum_{p^k\le N}\mathbb{1}_{\{p^k|n\}}$ (summing over prime powers). Show that $0\le\Omega(n)-\omega(n)$ and that

$$\mathbb{E}[\Omega-\omega]=\sum_p\sum_{k\ge 2}\frac{\lfloor N/p^k\rfloor}{N}\le\sum_p\sum_{k\ge 2}\frac{1}{p^k}=\sum_p\frac{1}{p(p-1)}$$

   Conclude that $\mathbb{E}[\Omega-\omega]=O(1)$, so Ω has the same normal order $\log\log n$ as ω.

4. Turán's original argument bounds the sum $\sum_{n\le N}(\omega(n)-\log\log N)^2$ directly. Show that this sum equals $N\cdot\mathbb{E}[(\omega-\log\log N)^2]$ and that, by theorem 5.4.5, it is $O(N\log\log N)$. Deduce the density statement of theorem 5.4.5 from this bound, thereby recovering the theorem from the "sum of squares" formulation.

5. This exercise probes what the growth condition $\psi(N)\to\infty$ gives, and what the second-moment method can and cannot say about dropping it.

   1. *The prime powers give only a density-zero exception.* On every prime power $n=p^k$ one has $\omega(n)=1$, so $|\omega(n)-\log\log N|=\log\log N-1$ exceeds any fixed constant c once N is large. Show, however, that the prime powers up to N number $\pi(N)+\pi(N^{1/2})+\cdots$ and hence have density $O(1/\log N)\to 0$. Conclude that this construction does *not* contradict theorem 5.4.5 for a constant threshold: it exhibits an exceptional set, but only a density-zero one, exactly the kind the theorem is allowed to discard.

2. *Why a constant threshold is nonetheless expected to fail.* Explain heuristically why one still expects a *constant* threshold $\psi(N) \equiv c$ to capture a non-vanishing fraction of the integers, so that some growth $\psi(N) \to \infty$ really is needed. The reason lies beyond the second moment: the true fluctuations of ω are of order $\sqrt{\log \log N}$, and their limiting shape is Gaussian (the Erdős-Kac refinement foreshadowed above). A Gaussian of standard deviation $\sqrt{\log \log N} \to \infty$ places mass bounded away from zero outside any fixed window of width $2c$. Note explicitly that Chebyshev's inequality and the variance computation of theorem 5.4.5 give only an *upper* bound on the exceptional set and so cannot by themselves force this spread; the necessity of growth is a forward-looking claim, not one deducible from the second moment alone.

6. For a squarefree integer $n = p_1 \cdots p_r$ one has $\omega(n) = \Omega(n) = r$. Using that the density of squarefree integers is $6/\pi^2$, argue informally why restricting the Hardy-Ramanujan theorem to squarefree n changes neither the normal order nor its proof, and identify the one step of the proof of theorem 5.4.5 that this restriction would simplify.

6

Weak Convergence and Characteristic Functions

The weak law and the central limit theorem are both statements about the distribution of a random variable in a limit, but of different kinds. The weak law says a distribution collapses to a point; the central limit theorem says a rescaled distribution approaches a fixed shape, the Gaussian. Making the second kind of statement precise requires a notion of convergence for probability distributions themselves, one that captures "the law of X_n approaches the law of X" without demanding that the variables share a sample space or converge pointwise. That notion is weak convergence, and this chapter develops it together with its principal analytic tool, the characteristic function.

section 6.1 defines convergence in distribution and characterizes it several equivalent ways through the portmanteau theorem. Section 6.2 addresses compactness: Prokhorov's theorem identifies tightness as the condition under which every sequence of laws has a weakly convergent subsequence, the probabilistic Bolzano-Weierstrass. Section 6.3 introduces the characteristic function $\varphi_X(t) = \mathbb{E}[e^{itX}]$, the Fourier transform of a law, and proves the inversion theorem that recovers the law from it. section 6.4 ties the threads together in Lévy's continuity theorem: pointwise convergence of characteristic functions is equivalent to weak convergence, the engine that drives the central limit theorem in the chapter to come.

6.1 Convergence in Distribution and the Portmanteau Theorem

The modes of convergence studied so far, almost sure convergence and convergence in probability (definition 5.2.1, definition 5.1.1), compare random variables pointwise on a common sample space: they ask whether $X_n(\omega)$ is close to $X(\omega)$ for most ω. The central limit theorem to come asks a different question. It concerns the normalized sum $S_n/\sqrt{n}$, which does not settle down to any fixed random variable at all; what stabilizes is its *distribution*, which approaches the Gaussian shape. To state such a limit precisely one needs a notion of convergence that compares the *laws* of the variables, not their sample-space realizations, and that makes sense even when the X_n live on different probability spaces. That notion is weak convergence, the subject of this section, and the portmanteau theorem is the tool that makes it flexible: it repackages a single definition into five interchangeable characterizations, each convenient for a different purpose.

The first task is to decide what "the law of X_n approaches the law of X" should mean. A law on $\mathbb{R}$ is determined by its distribution function (theorem 1.3.3), so a first guess is to require $F_n(x) \to F(x)$ for every x. This is almost right, but a single example shows it requires too much. Let X_n be the point mass at $1/n$, so X_n takes the value $1/n$ with probability one, and let X be the point mass at 0. Every reasonable reading of "the law of X_n approaches the law of X" should call these convergent: the mass sits at $1/n$, which slides to 0. Yet the distribution functions are

$$F_n(x) = \begin{cases} 0 & x < 1/n \\ 1 & x \geq 1/n \end{cases} \qquad F(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

and at the point $x = 0$ they disagree in the limit: $F_n(0) = 0$ for every n because $1/n > 0$, whereas $F(0) = 1$. So $F_n(0) \not\to F(0)$. The failure happens precisely at $x = 0$, which is the one point where F jumps, that is, its one discontinuity. Away from 0 the convergence is fine: for any $x > 0$ one has $1/n < x$ eventually, so $F_n(x) = 1 = F(x)$, and for $x < 0$ both are 0. The lesson is that a jump in F can be approached from the wrong side, and the value of F_n exactly at the jump is an artifact of where the approaching mass happens to sit. Requiring agreement only at the continuity points of F discards this artifact while keeping everything that matters.

Definition 6.1.1: Weak Convergence (Convergence in Distribution)

Let $\mu, \mu_1, \mu_2, \ldots$ be probability laws on $\mathbb{R}$ with distribution functions $F, F_1, F_2, \ldots$ (definition 1.3.1). The sequence μ_n *converges weakly* to μ, written $\mu_n \Rightarrow \mu$, if

$$F_n(x) \longrightarrow F(x) \qquad \text{for every continuity point } x \text{ of } F$$

If X_n and X are random variables (possibly on different probability spaces) with laws μ_{X_n} and μ_X (definition 2.1.3), then X_n *converges in distribution* to X, written $X_n \Rightarrow X$, if $\mu_{X_n} \Rightarrow \mu_X$; equivalently, if $F_{X_n}(x) \to F_X(x)$ at every continuity point x of F_X.

Two features of the definition deserve emphasis at once. First, convergence in distribution is a statement about laws, not about the variables themselves: writing $X_n \Rightarrow X$ is shorthand for $\mu_{X_n} \Rightarrow \mu_X$, and the actual random variables serve only to name the laws. Two sequences with the same laws converge or fail to converge together, and the limit X is determined only up to its law. This is exactly why the notion suits limit theorems, where the interesting content is the distribution and the sample space is incidental. Second, because F is nondecreasing it has at most countably many discontinuities (definition 1.3.1), so "every continuity point" still means all but a countable set; the definition constrains F_n on a dense set of points and the monotonicity of the F_n propagates the constraint. The continuity points of the limit are what one checks, and the point-mass example above shows why the exceptional set cannot be dropped.

A worked instance grounds the definition and previews the running example of the chapter. The discrete law $\mathrm{Poi}(\lambda)$ arises as a limit of binomials, a fact already visible at the level of mass functions;

the full characteristic-function proof is deferred to the continuity theorem later in this chapter.

Example 6.1.2: Weak Convergence of Laws on $\mathbb{R}$
Three instances isolate the moving parts of the definition.

1. *A shifted variable:* $X + 1/n \Rightarrow X$. Let X be any random variable with distribution function F, and set $X_n = X + 1/n$. Then $F_n(x) = \mathbb{P}(X + 1/n \le x) = F(x - 1/n)$. Fix a continuity point x of F. Since F is right-continuous and nondecreasing, and $x - 1/n \uparrow x$ from below, the left limit gives $F(x - 1/n) \to F(x^-)$; but x is a continuity point, so $F(x^-) = F(x)$, and hence $F_n(x) \to F(x)$. Thus $X_n \Rightarrow X$. Translating the mass by a vanishing amount does not change the limiting law, which is the content the definition is built to record.

2. *Binomial to Poisson:* $\mathrm{Bin}(n, \lambda/n) \Rightarrow \mathrm{Poi}(\lambda)$. Fix $\lambda > 0$ and let $\mu_n = \mathrm{Bin}(n, \lambda/n)$, the law of the number of successes in n independent trials each of success probability $p_n = \lambda/n$. For a fixed nonnegative integer k the mass at k is
$$\mu_n(\{k\}) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1 - \frac{\lambda}{n}\right)^{n-k} = \frac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \frac{\lambda^k}{k!} \cdot \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$
As $n \to \infty$ the first factor tends to 1 (it is a product of k terms each tending to 1), the middle factor is constant, the fourth factor tends to 1, and $(1 - \lambda/n)^n \to e^{-\lambda}$. Hence
$$\mu_n(\{k\}) \longrightarrow e^{-\lambda}\frac{\lambda^k}{k!} = \mathrm{Poi}(\lambda)(\{k\})$$
for each k. Convergence of the masses at every integer forces convergence of the distribution functions at every continuity point of the Poisson distribution function: for a noninteger x with $m = \lfloor x \rfloor$,
$$F_n(x) = \sum_{k=0}^{m} \mu_n(\{k\}) \longrightarrow \sum_{k=0}^{m} e^{-\lambda}\frac{\lambda^k}{k!} = F(x)$$
a finite sum of convergent terms, and the noninteger points are exactly the continuity points of the Poisson F. Therefore $\mathrm{Bin}(n, \lambda/n) \Rightarrow \mathrm{Poi}(\lambda)$. The characteristic-function route, which is shorter and dispenses with the arithmetic, is given in section 6.4; the convolution structure of the binomial as a sum of Bernoullis is recorded in example 2.3.6.

3. *Failure at a discontinuity: $F_n \to F$ need not hold at a jump.* Let X_n be the point mass at $1/n$ and X the point mass at 0, as in the discussion preceding the definition. Then $X_n \Rightarrow X$ holds, yet $F_n(0) = 0$ for every n while $F(0) = 1$, so $F_n(0) \not\to F(0)$. The single point of failure is the jump of F, which is not a continuity point, so the definition is satisfied. Had one instead used $X_n = -1/n$, the mass would approach 0 from below and one would find $F_n(0) = 1 = F(0)$; the value of F_n at the jump depends on the side of approach and carries no information about the limiting law.

The definition through distribution functions is concrete but rigid: it is tied to the order structure of $\mathbb{R}$ and to the somewhat awkward continuity-point proviso. For proofs one wants characterizations that are coordinate-free and that interact well with functions and sets. The portmanteau theorem supplies them. Its name (a portmanteau is a suitcase with two hinged compartments) signals that a single notion of convergence has several equivalent faces packed together; one opens whichever compartment fits the problem at hand. The most important face for the sequel is the first: convergence of expectations $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ against all bounded continuous test functions f. This is the definition that generalizes verbatim from $\mathbb{R}$ to any metric space and that connects weak convergence to the analytic machinery of characteristic functions, since $x \mapsto e^{itx}$ is bounded and continuous.

Theorem 6.1.3: Portmanteau Theorem

Let $\mu, \mu_1, \mu_2, \dots$ be probability laws on $\mathbb{R}$, realized by random variables $X, X_1, X_2, \dots$ with $\mu = \mu_X$ and $\mu_n = \mu_{X_n}$. The following are equivalent.

1. $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for every bounded continuous function $f\colon \mathbb{R} \to \mathbb{R}$.
2. $\limsup_n \mathbb{P}(X_n \in C) \le \mathbb{P}(X \in C)$ for every closed set $C \subseteq \mathbb{R}$.
3. $\liminf_n \mathbb{P}(X_n \in G) \ge \mathbb{P}(X \in G)$ for every open set $G \subseteq \mathbb{R}$.
4. $\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$ for every Borel set A with $\mu(\partial A) = 0$ (a *μ-continuity set*).
5. $X_n \Rightarrow X$ in the sense of definition 6.1.1: $F_n(x) \to F(x)$ at every continuity point x of F.

Each clause reads a limit of laws through a different aperture: bounded continuous averages in (1), mass that cannot suddenly appear on closed sets in (2), mass that cannot suddenly vanish from open sets in (3), exact convergence on sets whose boundary the limit ignores in (4), and the concrete cumulative distribution function in (5). The asymmetry between (2) and (3) is not an accident of the proof but a feature of the phenomenon: mass can pile up onto the boundary of a closed set in the limit (raising $\mathbb{P}(X \in C)$ above the limsup of $\mathbb{P}(X_n \in C)$ would be forbidden, but the reverse leak is allowed), and dually mass can leak off the boundary of an open set. Clause (4) says these two effects cancel exactly on any set whose boundary is invisible to μ.

***Proof*:**
The argument is a cycle: (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) $\Rightarrow$ (1). Throughout, write $\mathbb{E}[f(X_n)] = \int_{\mathbb{R}} f\, d\mu_n$ and $\mathbb{E}[f(X)] = \int_{\mathbb{R}} f\, d\mu$, which is the change-of-variables identity of theorem 3.1.2 applied to f; the expectations are of bounded measurable functions of the X_n and X, hence finite (definition 3.1.1).

Step 1: (1) $\Rightarrow$ (2). Fix a closed set C. The device is to approximate the indicator $\mathbb{1}_C$ from above by bounded continuous functions. For a point x write $d(x, C) = \inf_{y \in C} |x - y|$ for its distance to C; this is a 1-Lipschitz, hence continuous, function of x, and since C is closed, $d(x, C) = 0$ if and only if $x \in C$. For each integer $k \ge 1$ define

$$f_k(x) = \big(1 - k\, d(x, C)\big)^+ = \max\big(0,\ 1 - k\, d(x, C)\big)$$

Each f_k is continuous and takes values in $[0, 1]$, so it is bounded and continuous. On C one has $d(x, C) = 0$, so $f_k = 1$; thus $\mathbb{1}_C \le f_k$. Moreover f_k decreases pointwise to $\mathbb{1}_C$ as $k \to \infty$: if $x \in C$ then $f_k(x) = 1 = \mathbb{1}_C(x)$ for all k, and if $x \notin C$ then $d(x, C) > 0$, so $1 - k\, d(x, C) < 0$ for large k and $f_k(x) = 0 = \mathbb{1}_C(x)$ eventually. Now for each fixed k,

$$\limsup_n \mathbb{P}(X_n \in C) = \limsup_n \mathbb{E}[\mathbb{1}_C(X_n)] \le \limsup_n \mathbb{E}[f_k(X_n)] = \mathbb{E}[f_k(X)]$$

where the inequality uses $\mathbb{1}_C \le f_k$ and monotonicity of expectation (proposition 3.1.3), and the last equality is (1) applied to the bounded continuous f_k. Letting $k \to \infty$ and applying the dominated convergence theorem ([2]) with the constant dominating function 1, which is μ-integrable since μ is a probability measure, gives $\mathbb{E}[f_k(X)] \to \mathbb{E}[\mathbb{1}_C(X)] = \mathbb{P}(X \in C)$. Therefore $\limsup_n \mathbb{P}(X_n \in C) \le \mathbb{P}(X \in C)$, proving (2).

Step 2: (2) ⇔ (3). Closed and open sets are complementary, and this exchange converts a lim sup bound into a lim inf bound. If G is open then $C = G^c$ is closed, and $\mathbb{P}(X_n \in G) = 1 - \mathbb{P}(X_n \in C)$. Taking lim inf and using $\liminf(1 - a_n) = 1 - \limsup a_n$,

$$\liminf_n \mathbb{P}(X_n \in G) = 1 - \limsup_n \mathbb{P}(X_n \in C) \ge 1 - \mathbb{P}(X \in C) = \mathbb{P}(X \in G)$$

where the inequality is (2) for C. This proves (2) ⇒ (3); the reverse implication is identical with the roles of open and closed exchanged, so (2) and (3) are equivalent. In particular (1) ⇒ (3) as well.

Step 3: (2) and (3) ⇒ (4). Let A be a Borel set with $\mu(\partial A) = 0$. Write A° for its interior (open) and $\overline{A}$ for its closure (closed), so $A^\circ \subseteq A \subseteq \overline{A}$ and $\overline{A} \setminus A^\circ = \partial A$. The hypothesis $\mu(\partial A) = 0$ says the interior and closure carry the same μ-mass:

$$\mathbb{P}(X \in A^\circ) = \mathbb{P}(X \in \overline{A}) = \mathbb{P}(X \in A)$$

the outer equalities because $\mu(\overline{A}) - \mu(A^\circ) = \mu(\partial A) = 0$, and A is squeezed between them. Now chain the two inequalities using $A^\circ \subseteq A \subseteq \overline{A}$:

$$\mathbb{P}(X \in A) = \mathbb{P}(X \in A^\circ) \overset{(3)}{\le} \liminf_n \mathbb{P}(X_n \in A^\circ) \le \liminf_n \mathbb{P}(X_n \in A) \le \limsup_n \mathbb{P}(X_n \in A) \le \limsup_n \mathbb{P}(X_n \in \overline{A}) \overset{(2)}{\le} \mathbb{P}(X \in$$

The two ends of the chain coincide, so every inequality is an equality; in particular $\liminf_n \mathbb{P}(X_n \in A) = \limsup_n \mathbb{P}(X_n \in A) = \mathbb{P}(X \in A)$, which is exactly $\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$. This proves (4).

Step 4: (4) ⇒ (5). Fix a continuity point x of F and apply (4) to the half-line $A = (-\infty, x]$. Its boundary is the single point $\partial A = \{x\}$, and $\mu(\{x\}) = F(x) - F(x^-) = 0$ precisely because x is a continuity point of F. So A is a μ-continuity set, and (4) gives

$$F_n(x) = \mathbb{P}(X_n \le x) = \mathbb{P}(X_n \in A) \longrightarrow \mathbb{P}(X \in A) = \mathbb{P}(X \le x) = F(x)$$

Since x was an arbitrary continuity point, this is (5).

Step 5: (5) ⇒ (1). This is the substantive direction: from convergence of the cumulative distribution functions at continuity points one must recover convergence of $\mathbb{E}[f(X_n)]$ for *every* bounded continuous f. Fix such an f, say $|f| \le M$, and fix $\varepsilon > 0$.

First localize. The set D of continuity points of F is dense in $\mathbb{R}$ (its complement is countable), and $F(x) \to 0$ as $x \to -\infty$, $F(x) \to 1$ as $x \to +\infty$. Choose continuity points $a < b$ in D with $F(a) < \varepsilon$ and $F(b) > 1 - \varepsilon$, so that

$$\mathbb{P}(X \notin (a, b]) = F(a) + \big(1 - F(b)\big) < 2\varepsilon$$

Because a, b are continuity points, (5) gives $F_n(a) \to F(a)$ and $F_n(b) \to F(b)$, so for all large n,

$$\mathbb{P}(X_n \notin (a, b]) = F_n(a) + \big(1 - F_n(b)\big) < 3\varepsilon$$

Thus, up to mass 3ε for the X_n and 2ε for X, all of the distribution lives in the compact interval $[a, b]$.

Next approximate f on $[a,b]$ by a step function whose jumps sit at continuity points. On the compact interval $[a,b]$ the function f is uniformly continuous, so choose $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ whenever $|x-y| < \delta$. Since D is dense, pick a partition $a = t_0 < t_1 < \cdots < t_m = b$ with every $t_j \in D$ and each $t_j - t_{j-1} < \delta$. Define the step function

$$g = \sum_{j=1}^{m} f(t_j)\, \mathbb{1}_{(t_{j-1}, t_j]}$$

which is constant on each $(t_{j-1}, t_j]$ and vanishes off $(a,b]$. By the choice of δ, on $(a,b]$ one has $|f - g| \le \varepsilon$ pointwise (any $x \in (t_{j-1}, t_j]$ satisfies $|x - t_j| < \delta$, so $|f(x) - f(t_j)| < \varepsilon$). Now estimate, for the law μ,

$$\big|\mathbb{E}[f(X)] - \mathbb{E}[g(X)]\big| \le \mathbb{E}\big[|f(X) - g(X)|\, \mathbb{1}_{X \in (a,b]}\big] + \mathbb{E}\big[|f(X) - g(X)|\, \mathbb{1}_{X \notin (a,b]}\big] \le \varepsilon + 2M \cdot 2\varepsilon$$

using $|f-g| \le \varepsilon$ on $(a,b]$ and $|f-g| \le |f| + |g| \le 2M$ off it, together with $\mathbb{P}(X \notin (a,b]) < 2\varepsilon$. The identical estimate for μ_n, using $\mathbb{P}(X_n \notin (a,b]) < 3\varepsilon$, gives $|\mathbb{E}[f(X_n)] - \mathbb{E}[g(X_n)]| \le \varepsilon + 2M \cdot 3\varepsilon$ for all large n.

Finally pass $\mathbb{E}[g(X_n)]$ to the limit exactly. Each expectation of the step function is a finite sum,

$$\mathbb{E}[g(X_n)] = \sum_{j=1}^{m} f(t_j)\, \mathbb{P}\big(X_n \in (t_{j-1}, t_j]\big) = \sum_{j=1}^{m} f(t_j)\, \big(F_n(t_j) - F_n(t_{j-1})\big)$$

and every t_j is a continuity point of F, so $F_n(t_j) \to F(t_j)$ by (5). The sum has finitely many terms, so

$$\mathbb{E}[g(X_n)] \longrightarrow \sum_{j=1}^{m} f(t_j)\big(F(t_j) - F(t_{j-1})\big) = \mathbb{E}[g(X)]$$

Combining the three estimates by the triangle inequality, for all large n,

$$\big|\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]\big| \le \big|\mathbb{E}[f(X_n)] - \mathbb{E}[g(X_n)]\big| + \big|\mathbb{E}[g(X_n)] - \mathbb{E}[g(X)]\big| + \big|\mathbb{E}[g(X)] - \mathbb{E}[f(X)]\big|$$

The outer two terms are at most $\varepsilon(1 + 6M)$ and $\varepsilon(1 + 4M)$ respectively, and the middle term is below ε for large n. Hence $\limsup_n |\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \le \varepsilon(3 + 10M)$. Since $\varepsilon > 0$ was arbitrary, the limsup is 0, that is, $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$. This establishes (5) $\Rightarrow$ (1) and closes the cycle, completing the proof.

The proof rewards a second look at where the continuity-point restriction and the one-sided inequalities came from. Clause (5), the cumulative distribution function definition, is the concrete input, and it is used in exactly two places: to build a step function whose breakpoints avoid the jumps of F, and to pass that step function to the limit. The jumps of F are the only obstruction to naively equating F_n and F, and the argument routes around them by choosing partition points in the dense set of continuity points. This is the same phenomenon the point-mass example flagged, now handled once and for all. In the sequel it is almost always clause (1) that gets used, because it turns a question about laws into a question about the numbers $\mathbb{E}[f(X_n)]$, and because the complex exponentials $f(x) = e^{itx}$ that define the characteristic function are themselves bounded and continuous.

With the equivalences in hand, one can locate weak convergence among the modes of convergence developed in chapter 5. Convergence in probability compares variables on a common sample space; weak convergence compares only their laws. Every argument that pins mass on a common space should therefore imply the weaker statement about laws, and the next proposition confirms this. The converse fails in general, but it holds in the one degenerate case that recurs throughout the limit theory: convergence to a constant.

Proposition 6.1.4: Convergence in Probability versus in Distribution

Let $X, X_1, X_2, \dots$ be random variables on a common probability space.

1. If $X_n \to X$ in probability (definition 5.1.1), then $X_n \Rightarrow X$.
2. If $X_n \Rightarrow c$ for a constant $c \in \mathbb{R}$ (that is, the limit law is the point mass at c), then $X_n \to c$ in probability.

Convergence in distribution is strictly weaker than convergence in probability in general: (1) has no converse unless the limit is constant.

***Proof*:**
We prove the two implications and then exhibit the failure of the converse.

Part (1): convergence in probability implies weak convergence. We verify clause (2) of the portmanteau theorem (theorem 6.1.3), which is equivalent to $X_n \Rightarrow X$. Fix a closed set C and $\varepsilon > 0$, and let $C^\varepsilon = \{x : d(x, C) \le \varepsilon\}$ be its closed ε-neighbourhood. If $X_n \in C$ but $X \notin C^\varepsilon$, then $|X_n - X| \ge d(X, C) > \varepsilon$, since X lies at distance more than ε from C while $X_n \in C$. Contrapositively,

$$\{X_n \in C\} \subseteq \{X \in C^\varepsilon\} \cup \{|X_n - X| > \varepsilon\}$$

so, taking probabilities and using subadditivity,

$$\mathbb{P}(X_n \in C) \le \mathbb{P}(X \in C^\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$$

The second term tends to 0 as $n \to \infty$ by convergence in probability, so $\limsup_n \mathbb{P}(X_n \in C) \le \mathbb{P}(X \in C^\varepsilon)$. Now let $\varepsilon \downarrow 0$: the sets C^ε decrease to $\bigcap_k C^{1/k} = \overline{C} = C$ (the intersection of all closed neighbourhoods of a closed set is the set itself), so by continuity of the measure along decreasing sequences (proposition 1.1.5), $\mathbb{P}(X \in C^\varepsilon) \downarrow \mathbb{P}(X \in C)$. Therefore $\limsup_n \mathbb{P}(X_n \in C) \le \mathbb{P}(X \in C)$, which is clause (2). By the portmanteau theorem, $X_n \Rightarrow X$.

Part (2): weak convergence to a constant implies convergence in probability. Suppose $X_n \Rightarrow c$, so the limit law is the point mass δ_c with distribution function $F(x) = \mathbb{1}_{[c,\infty)}(x)$, which is continuous at every $x \neq c$. Fix $\varepsilon > 0$. The points $c - \varepsilon$ and $c + \varepsilon$ are continuity points of F, so by definition 6.1.1,

$$F_n(c - \varepsilon) \to F(c - \varepsilon) = 0, \qquad F_n(c + \varepsilon) \to F(c + \varepsilon) = 1$$

Now

$$\mathbb{P}(|X_n - c| > \varepsilon) \le \mathbb{P}(X_n \le c - \varepsilon) + \mathbb{P}(X_n > c + \varepsilon) = F_n(c - \varepsilon) + \big(1 - F_n(c + \varepsilon)\big)$$

and the right side tends to $0 + (1 - 1) = 0$. Hence $\mathbb{P}(|X_n - c| > \varepsilon) \to 0$ for every $\varepsilon > 0$, that is, $X_n \to c$ in probability.

Failure of the converse. Let X be a symmetric random variable that is not almost surely constant, for instance $X = \pm 1$ each with probability $\frac{1}{2}$, and set $X_n = (-1)^n X$. Since X is symmetric, $-X$

has the same law as X, so every X_n has the same law as X; trivially $X_n \Rightarrow X$ (the distribution functions are identical for all n). But $X_n - X$ equals 0 when n is even and $-2X = \pm 2$ when n is odd, so $\mathbb{P}(|X_n - X| > 1) = 1$ for odd n, and X_n does not converge to X in probability. Thus convergence in distribution does not imply convergence in probability, completing the proof.

Part (1) is the reason weak convergence sits at the bottom of the hierarchy of modes of convergence: almost sure convergence implies convergence in probability (chapter 5), which by the proposition implies convergence in distribution, and each implication is strict in general. The failure example is instructive about *why* the converse fails. The variables $X_n = (-1)^n X$ all carry the identical law, so from the standpoint of laws nothing is moving at all, and yet on the sample space they oscillate maximally. Convergence in distribution simply cannot see this oscillation, because it never looks at the joint behaviour of X_n and X, only at their marginal laws one at a time. Part (2) is the escape hatch that the limit theory relies on: when the target is a constant there is no joint behaviour to lose track of, since the point mass δ_c has no spread, and the two notions coincide. This is precisely the situation of the weak law of large numbers, where $\overline{X}_n \to \mu$, and it is why that theorem may be read interchangeably as a statement in probability or in distribution.

A closing word on why this is the right notion for the theorems ahead. The limit theorems of this Part are assertions about the shape of a distribution in the large, not about the fate of any particular sample point. The weak law says the law of the empirical average concentrates at a point; the central limit theorem to come says the law of the normalized fluctuation approaches the Gaussian shape, whatever the underlying distribution of the summands. Neither statement would even typecheck under almost sure or L^p convergence, since the normalized sums $S_n/\sqrt{n}$ do not converge pointwise or in L^p to anything. What converges is the law, and weak convergence is exactly the topology on laws in which these statements are true. That the same convergence can be tested through continuous averages, through closed and open sets, or through the cumulative distribution function, is what makes it usable: one proves the shape limit by whichever face of the portmanteau theorem the problem hands over most readily, and in the central limit theorem that face will be the characteristic function.

Exercise 6.1.1

1. Let U_n be uniform on $\{1/n, 2/n, \dots, n/n\}$ (the equally spaced grid of n points in $(0, 1]$). Show that $U_n \Rightarrow U$, where U is uniform on $[0, 1]$ (example 1.1.3), by computing $F_n(x)$ and taking the limit at each $x \in (0, 1)$.
2. Suppose $X_n \Rightarrow X$ and let $a > 0$ and $b \in \mathbb{R}$ be constants. Using clause (1) of the portmanteau theorem (theorem 6.1.3), show that $aX_n + b \Rightarrow aX + b$.
3. *Continuous mapping.* Let $g\colon \mathbb{R} \to \mathbb{R}$ be continuous and bounded, and suppose $X_n \Rightarrow X$. Show directly from theorem 6.1.3(1) that $g(X_n) \Rightarrow g(X)$. (Hint: if f is bounded continuous, so is $f \circ g$.)
4. Give an example of random variables with $X_n \Rightarrow X$ but $\mathbb{E}[X_n] \not\to \mathbb{E}[X]$. Explain why this does not contradict clause (1) of theorem 6.1.3. (Hint: the test function $f(x) = x$ is continuous but not bounded.)
5. Let $X_1, X_2, \dots$ be i.i.d. uniform on $[0, 1]$ and set $M_n = \max(X_1, \dots, X_n)$. Show that $n(1 - M_n) \Rightarrow E$, where E has the exponential law with $\mathbb{P}(E > t) = e^{-t}$ for $t \geq 0$, by computing the distribution function of $n(1 - M_n)$ and passing to the limit.

6. Let $\mu_n \Rightarrow \mu$ and let $A = (a, b]$ with $a < b$. Show that if $\mu(\{a\}) = \mu(\{b\}) = 0$ then $\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$, and give an example with $\mu(\{b\}) > 0$ where this convergence fails.

6.2 Tightness and Prokhorov's Theorem

Weak convergence, defined in section 6.1, is a mode of convergence, and every mode of convergence raises the same structural question: when does a sequence have a convergent subsequence? For sequences of real numbers the answer is the Bolzano-Weierstrass theorem, and its hypothesis is boundedness. A sequence of laws can fail to have a weak limit for a reason that has no analogue for a single real number: the mass can leak out to infinity. The laws $\mu_n = \delta_n$, unit point masses at the integer n, have distribution functions $F_n(x) = \mathbb{1}_{[n,\infty)}(x)$ that converge pointwise to the constant 0; the limit is not a distribution function at all, because all of the probability has escaped past every finite window. No subsequence does any better. The obstruction is that the family $\{\delta_n\}$ spreads its mass out with no uniform control on how far it travels.

Tightness is the condition that rules this out. It is the exact analogue of boundedness for a family of probability measures: it requires a single compact set that captures all but ε of the mass of every member of the family simultaneously. Prokhorov's theorem then says that tightness is precisely the compactness hypothesis one wants, the probabilistic Bolzano-Weierstrass. The route to it runs through Helly's selection theorem, a compactness statement for distribution functions that costs nothing beyond monotonicity and a diagonal argument, and tightness is exactly what upgrades the possibly-defective limit Helly produces into a genuine probability law.

Definition 6.2.1: Tightness

A family $\{\mu_i\}_{i\in I}$ of probability laws on $\mathbb{R}$ is *tight* if for every $\varepsilon > 0$ there is a finite $M > 0$ such that

$$\mu_i\big([-M, M]\big) > 1 - \varepsilon \qquad \text{for every } i \in I$$

A sequence $(\mu_n)_{n\geq 1}$ of laws is tight if the family $\{\mu_n\}$ is, and a sequence of random variables (X_n) is tight if the family of their laws $\{\mu_{X_n}\}$ is.

The force of the definition is in the two words "every" and the single M: one threshold M, depending on ε but not on i, must work for all members of the family at once. Dropping either quantifier trivializes the notion. For a single law μ, tightness is automatic: since $\mu\big([-M, M]\big) \to 1$ as $M \to \infty$ by continuity from below (proposition 1.1.5), some M suffices, and likewise any finite family is tight because one may take the maximum of finitely many thresholds. The content is entirely in controlling an infinite family uniformly. Equivalently, in terms of distribution functions (definition 1.3.1), tightness of $\{\mu_i\}$ says that for every $\varepsilon > 0$ there is M with $F_i(-M) < \varepsilon$ and $F_i(M) > 1 - \varepsilon$ for all i, so the tails of all the F_i are pushed down and up uniformly.

Example 6.2.2: The Uniform Family and a Failure

Let μ_i be the law of a uniform random variable on $[-i, i]$ for $i \in \{1, 2, 3\}$. For $\varepsilon > 0$ choose $M = 3$; then $\mu_i\big([-3, 3]\big) = 1 > 1 - \varepsilon$ for each of the three, so this finite family is tight, as every finite family must be. The interesting behaviour appears only in the limit of infinitely many. Take instead $\mu_n = \delta_n$, the point mass at n, for all $n \geq 1$. Given any M, every index $n > M$ has $\mu_n\big([-M, M]\big) = 0$, so no M can make $\mu_n\big([-M, M]\big) > 1 - \varepsilon$ hold for all n once $\varepsilon < 1$. The family

$\{\delta_n\}$ is not tight, and this is the analytic content of the escape of mass observed above.

The path to Prokhorov's theorem begins with a compactness statement that makes no reference to tightness at all and holds for every sequence of distribution functions. The price for this generality is that the limit it produces may fail to be a distribution function: mass may have escaped, and the limit may be *defective*, or *sub-probability*, meaning it rises from a value at least 0 to a value at most 1 but need not span the full unit interval.

Theorem 6.2.3: Helly's Selection Theorem

Let $(F_n)_{n\geq 1}$ be a sequence of distribution functions on $\mathbb{R}$. Then there is a subsequence (F_{n_k}) and a nondecreasing, right-continuous function $F\colon \mathbb{R} \to [0,1]$ such that

$$F_{n_k}(x) \longrightarrow F(x) \qquad \text{at every continuity point } x \text{ of } F$$

The limit F satisfies $0 \leq F(x) \leq 1$ but need not be a distribution function: its limits $F(-\infty) = \lim_{x\to-\infty} F(x)$ and $F(+\infty) = \lim_{x\to+\infty} F(x)$ may satisfy $F(+\infty) - F(-\infty) < 1$, so F may be defective.

***Proof*:**
The argument has two stages: a diagonal extraction that forces convergence on the rationals, followed by a right-continuous regularization that produces a function on all of $\mathbb{R}$.

Step 1: Diagonal extraction on the rationals. Enumerate the rationals as $\mathbb{Q} = \{q_1, q_2, q_3, \dots\}$. For each fixed j the sequence of real numbers $\big(F_n(q_j)\big)_{n\geq 1}$ lies in the compact interval $[0,1]$, so by the Bolzano-Weierstrass theorem it has a convergent subsequence. Extract these one coordinate at a time and diagonalize. Concretely, choose an infinite index set $N_1 \subseteq \mathbb{N}$ along which $F_n(q_1)$ converges; then choose an infinite $N_2 \subseteq N_1$ along which $F_n(q_2)$ converges (still convergent at q_1, since $N_2 \subseteq N_1$); continue, obtaining nested infinite sets $N_1 \supseteq N_2 \supseteq N_3 \supseteq \cdots$ with $F_n(q_j)$ convergent along N_j. Let n_k be the k-th element of N_k (the diagonal). For each fixed j, all but finitely many of the n_k lie in N_j, so $\big(F_{n_k}(q_j)\big)_k$ converges. Define

$$G(q) = \lim_{k\to\infty} F_{n_k}(q) \qquad (q \in \mathbb{Q})$$

a function $G\colon \mathbb{Q} \to [0,1]$. Because each F_{n_k} is nondecreasing, G is nondecreasing on $\mathbb{Q}$: if $q < q'$ then $F_{n_k}(q) \leq F_{n_k}(q')$ for every k, and passing to the limit preserves the inequality.

Step 2: Right-continuous regularization. The function G is defined only on $\mathbb{Q}$ and need not be right-continuous there. Define $F\colon \mathbb{R} \to [0,1]$ by

$$F(x) = \inf\{\, G(q) \,\big|\, q \in \mathbb{Q},\ q > x \,\}$$

Since G takes values in $[0,1]$, so does F. Monotonicity of F is immediate from monotonicity of G: if $x < x'$ then the infimum defining $F(x)$ is over a superset of the rationals used for $F(x')$, hence $F(x) \leq F(x')$. We check that F is right-continuous. Fix x and $\varepsilon > 0$. By definition of the infimum there is a rational $q > x$ with $G(q) < F(x) + \varepsilon$. For any y with $x \leq y < q$, every rational exceeding y that is at most q still lies above y, so $F(y) \leq G(q) < F(x) + \varepsilon$; combined with $F(y) \geq F(x)$ from monotonicity, this gives $F(x) \leq F(y) < F(x) + \varepsilon$ for all $y \in [x, q)$. Hence $F(y) \to F(x)$ as $y \downarrow x$, so F is right-continuous.

Step 3: Convergence at continuity points. Let x be a continuity point of F. Choose rationals r, s with $r < x < s$; monotonicity of the F_{n_k} gives

$$F_{n_k}(r) \le F_{n_k}(x) \le F_{n_k}(s)$$

Passing to the limit in k and using $F_{n_k}(r) \to G(r)$, $F_{n_k}(s) \to G(s)$,

$$G(r) \le \liminf_k F_{n_k}(x) \le \limsup_k F_{n_k}(x) \le G(s)$$

Now let $r \uparrow x$ and $s \downarrow x$ through the rationals. As $s \downarrow x$ the values $G(s)$ decrease to $F(x)$ by the very definition of F. As $r \uparrow x$, the values $G(r)$ increase to $F(x^-)$, the left limit of F at x; here one uses that for rationals $r < x$ one has $G(r) \le F(r) \le F(x^-)$, and that $F(x^-)$ is the supremum of F over points below x, which the $G(r)$ approach. Because x is a continuity point of F, $F(x^-) = F(x)$, so both outer bounds converge to $F(x)$, forcing

$$\lim_{k\to\infty} F_{n_k}(x) = F(x)$$

Thus F_{n_k} converges to F at every continuity point of F.

Step 4: F may be defective. Nothing in the construction controls the behaviour of F at $\pm\infty$. The point masses δ_n have $F_n(x) = \mathbb{1}_{[n,\infty)}(x)$, and along any subsequence $F_{n_k}(x) \to 0$ for every fixed x, giving $F \equiv 0$, for which $F(+\infty) - F(-\infty) = 0 < 1$. So F is defective in general, completing the proof.

Helly's theorem is a soft compactness statement, purchased entirely with monotonicity and boundedness in $[0, 1]$; it is the reason a limiting distribution function exists to talk about at all. What it cannot guarantee is that the limit carries full mass. The convergence $F_{n_k}(x) \to F(x)$ is exactly the convergence appearing in the definition of weak convergence (definition 6.1.1), pointwise at continuity points of the limit, and this notion of convergence to a possibly-defective limit is called *vague convergence*. Prokhorov's theorem is the statement that tightness is exactly what promotes vague convergence to weak convergence: it rules out the escape of mass to infinity, so the vague limit is a genuine distribution function and the convergence is weak.

Theorem 6.2.4: Prokhorov's Theorem on $\mathbb{R}$

A family $\{\mu_i\}_{i\in I}$ of probability laws on $\mathbb{R}$ is tight if and only if it is relatively compact for weak convergence, meaning that every sequence drawn from the family has a subsequence converging weakly to a probability law (the limit need not belong to the family).

***Proof*:**
Both implications are proved. The forward direction (tightness implies relative compactness) is where Helly's theorem does its work; the converse extracts tightness from compactness by contradiction.

Step 1: Tightness implies relative compactness. Let (μ_n) be any sequence drawn from the tight family, with distribution functions F_n. By Helly's selection theorem (theorem 6.2.3) there is a subsequence (F_{n_k}) and a nondecreasing, right-continuous $F\colon \mathbb{R} \to [0, 1]$ with $F_{n_k}(x) \to F(x)$ at every continuity point of F. It remains to show that tightness forces $F(+\infty) - F(-\infty) = 1$, so that F is a genuine distribution function.

Fix $\varepsilon > 0$. By tightness there is $M_0 > 0$ with $\mu_n\big([-M_0, M_0]\big) > 1-\varepsilon$ for all n. The discontinuity points of the monotone F are countable, so choose continuity points $M > M_0$ and $-M < -M_0$ of F; then $[-M_0, M_0] \subseteq (-M, M]$, hence $F_n(M) - F_n(-M) = \mu_n\big((-M, M]\big) \geq \mu_n\big([-M_0, M_0]\big) > 1-\varepsilon$ for all n. Then $F_{n_k}(M) \to F(M)$ and $F_{n_k}(-M) \to F(-M)$, and passing to the limit,

$$F(M) - F(-M) = \lim_k \big(F_{n_k}(M) - F_{n_k}(-M)\big) \geq 1 - \varepsilon$$

Therefore $F(+\infty) - F(-\infty) \geq F(M) - F(-M) \geq 1 - \varepsilon$. Since $\varepsilon > 0$ was arbitrary and $F(+\infty) - F(-\infty) \leq 1$ always, we obtain $F(+\infty) - F(-\infty) = 1$. Since F takes values in $[0, 1]$, the bounds $F(-\infty) \geq 0$ and $F(+\infty) \leq 1$ together with $F(+\infty) - F(-\infty) = 1$ force $F(-\infty) = 0$ and $F(+\infty) = 1$. Thus F is a nondecreasing, right-continuous function increasing from 0 to 1, hence a distribution function, and by the correspondence between distribution functions and Borel probability laws (theorem 1.3.3) it is the distribution function of a probability law μ. The convergence $F_{n_k}(x) \to F(x)$ at every continuity point of F is exactly $\mu_{n_k} \Rightarrow \mu$ (definition 6.1.1). Thus every sequence from the family has a weakly convergent subsequence, which is relative compactness.

Step 2: Relative compactness implies tightness. Suppose the family is relatively compact but, for contradiction, not tight. Failure of tightness means there is $\varepsilon_0 > 0$ such that for every $M > 0$ some member μ of the family has $\mu\big([-M, M]\big) \leq 1 - \varepsilon_0$. Applying this with $M = 1, 2, 3, \ldots$ produces a sequence (μ_n) from the family with

$$\mu_n\big([-n, n]\big) \leq 1 - \varepsilon_0 \qquad \text{for every } n \geq 1$$

By relative compactness there is a subsequence $\mu_{n_k} \Rightarrow \mu$ for some probability law μ. Fix any $R > 0$ whose endpoints $\pm R$ are continuity points of the limiting distribution function F of μ. The open interval $(-R, R)$ is an open set, so by the portmanteau theorem (theorem 6.1.3, the open-set inequality),

$$\mu\big((-R, R)\big) \leq \liminf_k \mu_{n_k}\big((-R, R)\big)$$

For every k with $n_k \geq R$ we have $(-R, R) \subseteq [-n_k, n_k]$, hence $\mu_{n_k}\big((-R, R)\big) \leq \mu_{n_k}\big([-n_k, n_k]\big) \leq 1 - \varepsilon_0$. Taking the liminf, $\mu\big((-R, R)\big) \leq 1 - \varepsilon_0$. This holds for every such R, and letting $R \to \infty$ gives, by continuity from below (proposition 1.1.5), $\mu(\mathbb{R}) \leq 1 - \varepsilon_0 < 1$, contradicting that μ is a probability law with $\mu(\mathbb{R}) = 1$. Hence the family must be tight, completing the proof.

Prokhorov's theorem supplies the compactness for weak convergence. In one direction it manufactures subsequential limits and is used exactly as Bolzano-Weierstrass is used in analysis: to prove a sequence of laws converges, one first secures a convergent subsequence and then shows every such subsequential limit is the same, whence the whole sequence converges. Lévy's continuity theorem, in the section to come, is built on precisely this pattern, extracting a subsequential limit by tightness and identifying it by its characteristic function. In the other direction the theorem certifies that tightness is not only sufficient for compactness but necessary: a relatively compact family has nowhere for its mass to escape to, so it must have been tight to begin with. The two implications together make "tight" and "relatively compact" interchangeable, and it is the qualitative hypothesis, tightness, that is checkable in practice.

The remaining task is to make tightness checkable. Verifying the definition directly requires uniform control of tail probabilities, which is awkward. A single uniform moment bound delivers exactly that control through Markov's inequality, and this is the criterion used in nearly every application, including the proof of the central limit theorem to come.

Proposition 6.2.5: Moment Bound Implies Tightness

Let $\{\mu_i\}_{i\in I}$ be a family of probability laws on $\mathbb{R}$, and let X_i denote a random variable with law μ_i. If there is $p > 0$ and a finite constant C with

$$\sup_{i\in I} \mathbb{E}\big[|X_i|^p\big] \le C < \infty$$

then the family $\{\mu_i\}$ is tight.

***Proof*:**
Fix $\varepsilon > 0$. Since $t \mapsto t^p$ is strictly increasing on $[0,\infty)$, the event $\{|X_i| > M\}$ equals $\{|X_i|^p > M^p\}$, and Markov's inequality applied to the nonnegative random variable $|X_i|^p$ (theorem 3.2.4) gives, for every $M > 0$ and every i,

$$\mu_i\big(\{x : |x| > M\}\big) = \mathbb{P}\big(|X_i|^p > M^p\big) \le \frac{\mathbb{E}\big[|X_i|^p\big]}{M^p} \le \frac{C}{M^p}$$

The final bound is uniform in i. Choose M large enough that $C/M^p < \varepsilon$, that is $M > (C/\varepsilon)^{1/p}$. Then for every i,

$$\mu_i\big([-M,M]\big) = 1 - \mu_i\big(\{x : |x| > M\}\big) \ge 1 - \frac{C}{M^p} > 1 - \varepsilon$$

Since the same M works for all i, the family is tight, as we sought to show.

The proposition converts a quantitative uniform integrability condition, a bound on a single moment, into the qualitative compactness condition of tightness, and the conversion is Markov's inequality with nothing else added. The value of p is immaterial to the conclusion, so the mildest available bound suffices: a uniform bound on the first absolute moments $\mathbb{E}|X_i|$, or on the second moments (variances plus squared means), each implies tightness. This is the criterion invoked whenever a sequence of normalized sums is shown to be tight, and it is why control of a single moment is enough to launch a subsequence argument.

Example 6.2.6: Gaussian Families and an Escaping Sequence
Two families make the roles of boundedness and escape concrete.

1. *Gaussians with varying variance.* Let $\mu_n = \mathcal{N}(0,\sigma_n^2)$ be the centered Gaussian law with variance σ_n^2. The family $\{\mu_n\}$ is tight if and only if the sequence (σ_n) is bounded.

   Suppose first $\sigma_n \le \sigma$ for all n. If $X_n \sim \mathcal{N}(0,\sigma_n^2)$ then $\mathbb{E}[X_n^2] = \sigma_n^2 \le \sigma^2$, a uniform second-moment bound, so tightness follows from proposition 6.2.5 with $p = 2$ and $C = \sigma^2$. Explicitly, if $Z \sim \mathcal{N}(0,1)$ then X_n has the law of $\sigma_n Z$, and for any M,

   $$\mu_n\big([-M,M]\big) = \mathbb{P}\big(|Z| \le M/\sigma_n\big) \ge \mathbb{P}\big(|Z| \le M/\sigma\big)$$

   which exceeds $1-\varepsilon$ once M is large, uniformly in n.

   Conversely, suppose (σ_n) is unbounded, so $\sigma_{n_k} \to \infty$ along some subsequence. For any fixed M,

   $$\mu_{n_k}\big([-M,M]\big) = \mathbb{P}\big(|Z| \le M/\sigma_{n_k}\big) \longrightarrow \mathbb{P}(|Z| \le 0) = 0$$

as $k \to \infty$, since $M/\sigma_{n_k} \to 0$. Thus no fixed M keeps $\mu_{n_k}\big([-M, M]\big)$ above $1-\varepsilon$ for all large k once $\varepsilon < 1$, and the family is not tight. The mass of a Gaussian with growing variance spreads out and thins over every bounded window.

2. *Mass escaping to infinity.* The sequence of point masses $\mu_n = \delta_n$ from example 6.2.2 is not tight, as shown there: no window $[-M, M]$ retains the mass of δ_n for $n > M$. Its distribution functions $F_n = \mathbb{1}_{[n,\infty)}$ converge vaguely to $F \equiv 0$, the wholly defective limit, so no subsequence converges weakly to a probability law, consistent with Prokhorov's theorem (theorem 6.2.4) in its contrapositive form. A moment bound cannot hold here: $\mathbb{E}_{\mu_n}|X| = n \to \infty$, so proposition 6.2.5 does not apply, exactly as it should not.

The two cases isolate the mechanism. In the Gaussian family the mass stays centered but its spread is the free parameter, and boundedness of the spread is tightness. In the point masses the spread is zero but the location runs off, and the escape of location is the failure of tightness. Prokhorov's theorem sees only the mass in bounded windows, so it treats both spreading and translation as the same phenomenon: mass no longer confined to any compact set.

Exercise 6.2.1

1. Show directly from the definition that a family consisting of a single law μ on $\mathbb{R}$ is tight, and that any finite family $\{\mu_1, \dots, \mu_m\}$ is tight. Which step uses that the family is finite?
2. Suppose $\{\mu_i\}_{i\in I}$ and $\{\nu_j\}_{j\in J}$ are both tight families of laws on $\mathbb{R}$. Show that the combined family $\{\mu_i\}_{i\in I} \cup \{\nu_j\}_{j\in J}$ is tight. Deduce that a sequence (μ_n) with $\mu_n \Rightarrow \mu$ is tight (use that the tail of a convergent sequence is controlled by its limit, and Helly or the definition for the finitely many early terms).
3. Let X be a random variable with law μ and let μ_n be the law of $X + c_n$ for a real sequence (c_n). Prove that $\{\mu_n\}$ is tight if and only if (c_n) is bounded.
4. Give a sequence of laws (μ_n) on $\mathbb{R}$ that converges vaguely to a defective (nonzero) subprobability limit, so that some of the mass, but not all, escapes to infinity. Compute the total mass of the vague limit and confirm the sequence is not tight.
5. The moment criterion (proposition 6.2.5) is sufficient but not necessary for tightness. Construct a tight sequence (μ_n) of laws with $\mathbb{E}_{\mu_n}|X| \to \infty$, showing that tightness does not force any uniform moment bound.
6. Let (X_n) be random variables with $\sup_n \mathbb{E}[X_n^2] < \infty$, and suppose every weakly convergent subsequence of (μ_{X_n}) has the same limit μ. Prove that $\mu_{X_n} \Rightarrow \mu$. (This is the standard use of Prokhorov: compactness plus a unique subsequential limit gives full convergence.)

6.3 Characteristic Functions

The portmanteau theorem (theorem 6.1.3) shows what weak convergence means, and Prokhorov's theorem (theorem 6.2.4) shows when a sequence of laws has a convergent subsequence, but neither gives a practical way to *compute* a weak limit. Testing $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ against every bounded continuous f is not something one does with a pencil. What is needed is a single transform of a law, a function of one real variable, whose convergence already forces weak convergence and which turns the operation that dominates probability, addition of independent random variables, into something manageable. The moment generating function of definition 3.4.4 is one candidate, but it fails for the heavy-tailed laws that matter most, since $\mathbb{E}[e^{tX}]$ can be infinite for every $t \neq 0$. Replacing the real exponent tX by the imaginary exponent itX cures this at a stroke: $|e^{itX}| = 1$, so the expectation always exists. This is the characteristic function, and it is the analytic instrument around which the rest of the chapter, and the central limit theorem to come, is built.

Before defining it we record that expectation extends to complex-valued random variables in the only sensible way. A complex-valued random variable is a map $Z = U + iV$ with U, V real random variables, and one sets

$$\mathbb{E}[Z] = \mathbb{E}[U] + i\,\mathbb{E}[V]$$

whenever $\mathbb{E}|U|$ and $\mathbb{E}|V|$ are finite. Linearity over $\mathbb{C}$ follows from linearity over $\mathbb{R}$ applied to real and imaginary parts, and the fundamental inequality $|\mathbb{E}[Z]| \leq \mathbb{E}|Z|$ persists: writing $\mathbb{E}[Z] = re^{i\theta}$ in polar form,

$$|\mathbb{E}[Z]| = r = \mathbb{E}[e^{-i\theta}Z] = \mathbb{E}\big[\mathrm{Re}(e^{-i\theta}Z)\big] \leq \mathbb{E}\big|e^{-i\theta}Z\big| = \mathbb{E}|Z|$$

where the middle equality holds because the left side is real, so the imaginary part of $\mathbb{E}[e^{-i\theta}Z]$ vanishes, and $\mathrm{Re}(e^{-i\theta}Z) \leq |e^{-i\theta}Z|$ pointwise.

Definition 6.3.1: Characteristic Function

Let X be a real-valued random variable with law μ_X (definition 2.1.3). Its *characteristic function* is the function $\varphi_X : \mathbb{R} \to \mathbb{C}$ defined by

$$\varphi_X(t) = \mathbb{E}\big[e^{itX}\big] = \int_{\mathbb{R}} e^{itx}\, d\mu_X(x)$$

the Fourier transform of the law μ_X. The second equality is the change of variables formula (theorem 3.1.2) applied to the bounded continuous function $x \mapsto e^{itx} = \cos(tx) + i\sin(tx)$.

The characteristic function depends on X only through its law, so laws with equal characteristic functions cannot be distinguished by this transform. The inversion theorem below will show the converse, that φ_X determines μ_X completely, which is what makes the transform useful rather than convenient. For now the integral is guaranteed to converge because the integrand has modulus one, and this is the decisive advantage over the moment generating function: φ_X exists for *every* law, with no integrability hypothesis whatsoever.

The most basic properties fall out of the definition and the elementary estimate $|e^{itx}| = 1$.

Proposition 6.3.2: Elementary Properties of the Characteristic Function

For any real-valued random variable X the characteristic function φ_X satisfies:

1. $\varphi_X(0) = 1$;
2. $|\varphi_X(t)| \leq 1$ for every $t \in \mathbb{R}$;
3. φ_X is uniformly continuous on $\mathbb{R}$;
4. $\varphi_{-X}(t) = \overline{\varphi_X(t)}$, and in particular φ_X is real-valued if and only if X and $-X$ have the same law.

***Proof*:**
For (1), $\varphi_X(0) = \mathbb{E}[e^0] = \mathbb{E}[1] = 1$. For (2), the modulus inequality for complex expectations gives

$$|\varphi_X(t)| = \left|\mathbb{E}[e^{itX}]\right| \leq \mathbb{E}\left|e^{itX}\right| = \mathbb{E}[1] = 1$$

For (3), fix $t \in \mathbb{R}$ and $h \in \mathbb{R}$. Then

$$|\varphi_X(t+h) - \varphi_X(t)| = \left|\mathbb{E}[e^{itX}(e^{ihX} - 1)]\right| \leq \mathbb{E}\left|e^{ihX} - 1\right|$$

The bound on the right does not depend on t, which is what will give *uniform* continuity. The integrand $|e^{ihX} - 1|$ is bounded by 2 and tends to 0 pointwise as $h \to 0$, so dominated convergence ([2], with dominating function the constant 2, which is integrable since μ_X is a probability measure) gives $\mathbb{E}|e^{ihX} - 1| \to 0$ as $h \to 0$. Because this bound is independent of t, for every $\varepsilon > 0$ there is $\delta > 0$ with $|\varphi_X(t+h) - \varphi_X(t)| < \varepsilon$ for all t whenever $|h| < \delta$, which is uniform continuity.

For (4), $\varphi_{-X}(t) = \mathbb{E}[e^{-itX}] = \mathbb{E}[\overline{e^{itX}}]$, and since $\mathbb{E}[\bar{Z}] = \overline{\mathbb{E}[Z]}$ for a complex-valued Z (conjugation commutes with the real and imaginary parts), this equals $\overline{\varphi_X(t)}$. Now φ_X is real-valued exactly when $\varphi_X = \overline{\varphi_X} = \varphi_{-X}$; by the uniqueness half of the inversion theorem (corollary 6.3.6, proved below and resting on theorem 6.3.5 rather than on this property, so the reference is not circular) this holds if and only if X and $-X$ share a law, completing the proof.

Property (2) says the characteristic function lives in the closed unit disk, with its maximum modulus attained at the origin. Property (3) is worth pausing on: no regularity of the law is assumed, yet its Fourier transform is automatically uniformly continuous. This is the general principle that the Fourier transform trades decay for smoothness, here in its crudest form, and it is why φ_X is a far more docile object than μ_X itself, which may have atoms, singular parts, and every other pathology. Property (4) identifies the symmetric laws, those invariant under $x \mapsto -x$, as exactly the ones with real characteristic function.

Example 6.3.3: The Characteristic Function of a Point Mass
Let $X = c$ almost surely, so $\mu_X = \delta_c$ is the point mass at c. Then

$$\varphi_X(t) = \mathbb{E}[e^{itc}] = e^{itc}$$

a pure phase of modulus one, so $|\varphi_X(t)| = 1$ for all t. This is the extreme opposite of a spread-out law: the modulus never decays, reflecting that no averaging takes place. When $c = 0$ the law is δ_0 and $\varphi_X \equiv 1$, the constant function, which by inversion below is the unique law with this property.

The definition and its elementary properties in hand, the transform becomes useful through three structural rules: its behaviour under affine changes of the variable, under addition of independent variables, and under differentiation, where it encodes the moments of X.

Proposition 6.3.4: Structural Properties of the Characteristic Function

Let X, Y be real-valued random variables.

1. *Affine transformations.* For real constants a, b,

$$\varphi_{aX+b}(t) = e^{itb}\varphi_X(at)$$

2. *Independent sums.* If X and Y are independent (definition 2.2.2), then

$$\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$$

3. *Moments and derivatives.* If $\mathbb{E}|X|^k < \infty$ for some integer $k \geq 1$, then φ_X is k times continuously differentiable, with

$$\varphi_X^{(j)}(t) = \mathbb{E}\big[(iX)^j e^{itX}\big] \qquad (0 \leq j \leq k)$$

and in particular

$$\varphi_X^{(j)}(0) = i^j\,\mathbb{E}[X^j]$$

***Proof*:**
We prove the three parts in turn.

1. By definition and linearity of the exponent,

$$\varphi_{aX+b}(t) = \mathbb{E}\big[e^{it(aX+b)}\big] = \mathbb{E}\big[e^{itb}e^{i(ta)X}\big] = e^{itb}\,\mathbb{E}\big[e^{i(at)X}\big] = e^{itb}\varphi_X(at)$$

the constant e^{itb} pulling out of the expectation because it does not depend on ω.

2. Since X and Y are independent, so are the bounded complex-valued random variables e^{itX} and e^{itY} (a measurable function of X is independent of a measurable function of Y). Independence factorizes the expectation of a product, and applying this to the real and imaginary parts separately,

$$\varphi_{X+Y}(t) = \mathbb{E}\big[e^{itX}e^{itY}\big] = \mathbb{E}\big[e^{itX}\big]\,\mathbb{E}\big[e^{itY}\big] = \varphi_X(t)\varphi_Y(t)$$

Equivalently, the law of $X + Y$ is the convolution $\mu_X * \mu_Y$ (proposition 2.3.4), and the Fourier transform carries convolution to pointwise product, which is the same statement read through definition 6.3.1.

3. Consider the difference quotient of the integrand at a point t:

$$\frac{e^{i(t+h)X} - e^{itX}}{h} = e^{itX} \cdot \frac{e^{ihX} - 1}{h}$$

As $h \to 0$ the right side converges pointwise to $iX\,e^{itX}$. To differentiate under the integral sign we need a dominating function independent of h. The elementary bound $|e^{is} - 1| \leq |s|$

for real s gives
$$\left|\frac{e^{ihX}-1}{h}\right| \le \frac{|hX|}{|h|} = |X|$$
so the difference quotients are dominated by $|X|$, which is integrable by hypothesis for $k \ge 1$. Dominated convergence ([2]) then justifies passing the limit inside the expectation:

$$\varphi_X'(t) = \lim_{h\to 0} \mathbb{E}\left[e^{itX}\frac{e^{ihX}-1}{h}\right] = \mathbb{E}\big[iX\,e^{itX}\big]$$

This is the case $j=1$. For $1 \le j \le k$ the identical argument applies with the integrand $(iX)^{j-1}e^{itX}$ and dominating function $|X|^j$, which is integrable because $\mathbb{E}|X|^j < \infty$ for $j \le k$ by Jensen's inequality (theorem 3.2.7); induction on j then gives $\varphi_X^{(j)}(t) = \mathbb{E}[(iX)^j e^{itX}]$. Continuity of $\varphi_X^{(j)}$ follows exactly as in proposition 6.3.2(3), since $t \mapsto \mathbb{E}[(iX)^j e^{itX}]$ is uniformly continuous by the same dominated-convergence estimate with dominator $|X|^j$. Setting $t=0$ gives $\varphi_X^{(j)}(0) = \mathbb{E}[(iX)^j] = i^j\mathbb{E}[X^j]$, as we sought to show.

Part (2) is the reason characteristic functions dominate the study of sums. Independence of the summands, which is awkward to exploit at the level of laws because it forces a convolution integral, becomes ordinary multiplication of scalar functions. The characteristic function of a sum of n independent variables is the product of the n characteristic functions, and the analysis of such a product is elementary complex analysis rather than an n-fold convolution. This is precisely the mechanism by which the central limit theorem, an assertion about a sum of many independent variables, will reduce to the pointwise limit of a product. Part (3) makes the characteristic function a moment-generating device in the literal sense: its Taylor coefficients at the origin are the moments of X, scaled by powers of i. This mirrors theorem 3.4.5 for the moment generating function but without any integrability hypothesis beyond the moment in question, and it is the local expansion that the central limit theorem exploits, where $\varphi(t) = 1 - \frac{1}{2}\sigma^2 t^2 + o(t^2)$ near the origin controls the Gaussian limit.

The section now arrives at the theorem that justifies the whole enterprise: the characteristic function is not only *a* transform of the law but an *invertible* one. Knowing φ at every t is knowing μ on every interval, hence on every Borel set. The formula is a contour-free Fourier inversion, and its proof is a controlled application of Fubini's theorem together with one classical improper integral.

Theorem 6.3.5: The Inversion Theorem

Let X have law μ, distribution function F, and characteristic function φ. If $a < b$ are continuity points of F, then

$$\mu\big((a,b)\big) = \lim_{T\to\infty} \frac{1}{2\pi}\int_{-T}^{T} \frac{e^{-ita}-e^{-itb}}{it}\,\varphi(t)\,dt$$

***Proof*:**
Write I_T for the integral $\frac{1}{2\pi}\int_{-T}^{T}\frac{e^{-ita}-e^{-itb}}{it}\varphi(t)\,dt$.

Step 1: Insert the definition of φ and apply Fubini. Expanding $\varphi(t) = \int_{\mathbb{R}} e^{itx}\, d\mu(x)$ gives

$$I_T = \frac{1}{2\pi} \int_{-T}^{T} \int_{\mathbb{R}} \frac{e^{-ita} - e^{-itb}}{it} e^{itx}\, d\mu(x)\, dt$$

The integrand is bounded in modulus uniformly on $[-T, T] \times \mathbb{R}$: the numerator satisfies

$$\left| \frac{e^{-ita} - e^{-itb}}{it} \right| = \left| \frac{1}{it} \int_a^b (-it) e^{-its}\, ds \right| = \left| \int_a^b e^{-its}\, ds \right| \le b - a$$

so the modulus of the full integrand is at most $b - a$, and integrating this bound over the finite rectangle $[-T, T] \times \mathbb{R}$ against $dt\, d\mu$ gives the finite quantity $2T(b - a)$. Fubini's theorem ([2]) therefore permits exchanging the order of integration:

$$I_T = \frac{1}{2\pi} \int_{\mathbb{R}} \left(\int_{-T}^{T} \frac{e^{it(x-a)} - e^{it(x-b)}}{it}\, dt \right) d\mu(x)$$

Step 2: Evaluate the inner integral via the Dirichlet integral. Fix x and abbreviate the inner integral as $J_T(x)$. Splitting the two exponentials and expanding via Euler's formula, the even (cosine) parts of $\frac{e^{it(x-a)}}{it}$ produce odd integrands in t that vanish over the symmetric interval, leaving the sine parts:

$$J_T(x) = \int_{-T}^{T} \frac{\sin(t(x-a))}{t}\, dt - \int_{-T}^{T} \frac{\sin(t(x-b))}{t}\, dt$$

The classical Dirichlet integral, computed in [2], states that for real θ,

$$\int_{-T}^{T} \frac{\sin(t\theta)}{t}\, dt = 2 \int_0^T \frac{\sin(t\theta)}{t}\, dt \xrightarrow[T\to\infty]{} \pi \operatorname{sgn}(\theta)$$

where $\operatorname{sgn}(\theta)$ is $+1, 0, -1$ according as θ is positive, zero, or negative, and the convergence is bounded uniformly in T and θ (the partial integrals $\int_0^T \frac{\sin u}{u}\, du$ are uniformly bounded). Consequently

$$J_T(x) \xrightarrow[T\to\infty]{} \pi\big(\operatorname{sgn}(x-a) - \operatorname{sgn}(x-b)\big) =: 2\pi\, g(x)$$

Reading off the sign function shows $g(x) = 0$ for $x < a$ or $x > b$, $g(x) = 1$ for $a < x < b$, and $g(x) = \frac{1}{2}$ at the endpoints $x = a$ and $x = b$.

Step 3: Pass to the limit under the outer integral. The functions $x \mapsto J_T(x)$ are uniformly bounded in T and x by the uniform bound on the Dirichlet partial integrals, and they converge pointwise to $2\pi g(x)$. Dominated convergence ([2]), with the constant dominator supplied by that uniform bound, gives

$$\lim_{T\to\infty} I_T = \frac{1}{2\pi} \int_{\mathbb{R}} 2\pi\, g(x)\, d\mu(x) = \int_{\mathbb{R}} g(x)\, d\mu(x) = \mu\big((a, b)\big) + \tfrac{1}{2}\mu(\{a\}) + \tfrac{1}{2}\mu(\{b\})$$

Because a and b are continuity points of F, the law μ has no atom at either, so $\mu(\{a\}) = \mu(\{b\}) = 0$ and the two endpoint terms vanish. The limit is therefore $\mu((a, b))$, completing the proof.

The inversion formula recovers the mass $\mu((a,b))$ of every interval whose endpoints are continuity points of F from the characteristic function alone. The continuity-point restriction is the same one that pervades weak convergence: at an atom of μ the endpoint contributes half its mass, exactly the value $g = \frac{1}{2}$ computed in Step 2, which is why one restricts to continuity points where this ambiguity disappears. Since the continuity points of F are dense (a monotone function has at most countably many jumps), the intervals (a,b) with such endpoints generate the Borel σ-algebra, and a probability measure is determined by its values on a generating π-system of such intervals. The immediate consequence is the uniqueness that the whole method rests on.

Corollary 6.3.6: Characteristic Functions Determine Laws

Two real-valued random variables have the same law if and only if they have the same characteristic function. Equivalently, the map $\mu \mapsto \varphi_\mu$ from probability laws on $\mathbb{R}$ to characteristic functions is injective.

***Proof*:**
If X and Y have the same law then $\varphi_X = \varphi_Y$ directly from definition 6.3.1, since φ depends on the variable only through its law. Conversely, suppose $\varphi_X = \varphi_Y$, and write μ, ν for the two laws and F, G for their distribution functions. By the inversion theorem (theorem 6.3.5), $\mu((a,b)) = \nu((a,b))$ for all $a < b$ that are common continuity points of F and G. The set of common continuity points is dense, since each of F, G has at most countably many discontinuities. For any x that is a common continuity point, choosing a sequence of common continuity points $a_n \to -\infty$ gives

$$F(x) = \lim_{n\to\infty} \mu\big((a_n, x)\big) = \lim_{n\to\infty} \nu\big((a_n, x)\big) = G(x)$$

using $\mu((a_n, x)) = F(x^-) - F(a_n) \to F(x)$ as $a_n \to -\infty$, where $F(x^-) = F(x)$ because x is a continuity point. Thus $F = G$ on a dense set, and both being right-continuous, $F = G$ everywhere. By the correspondence between distribution functions and laws (theorem 1.3.3), $\mu = \nu$, so X and Y have the same law, as we sought to show.

Uniqueness is the property that converts the transform into a proof technique. To show two laws agree, it now suffices to show their characteristic functions agree, and characteristic functions are concrete functions one can compute and compare. The next section will strengthen this static uniqueness into a dynamic one: not only does the limit of the characteristic functions determine the limit law, but pointwise convergence of the φ_n forces weak convergence of the μ_n. First, the transform must be made concrete on the laws that recur throughout the book.

Example 6.3.7: Characteristic Functions of the Standard Laws

The characteristic function φ of the running laws of the book is computed below.

1. *Bernoulli.* If $X \sim \mathrm{Ber}(p)$, taking value 1 with probability p and 0 with probability $1-p$, then
$$\varphi_X(t) = \mathbb{E}[e^{itX}] = (1-p)e^{it\cdot 0} + p\, e^{it\cdot 1} = 1 - p + pe^{it}$$

2. *Poisson.* If $X \sim \mathrm{Poi}(\lambda)$ with $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$, then summing the exponential series,
$$\varphi_X(t) = \sum_{k=0}^{\infty} e^{itk}\,\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^{it})^k}{k!} = e^{-\lambda}e^{\lambda e^{it}} = e^{\lambda(e^{it}-1)}$$

3. *Standard Gaussian.* Let $X \sim \mathcal{N}(0,1)$ with density $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Then $\mathbb{E}|X| < \infty$, so by proposition 6.3.4(3), φ is differentiable with

$$\varphi'(t) = \mathbb{E}[iX\, e^{itX}] = \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} x\, e^{itx} e^{-x^2/2}\, dx$$

Integrating by parts, with $u = e^{itx}$ and $dv = xe^{-x^2/2}\, dx$ so that $v = -e^{-x^2/2}$ and the boundary terms vanish,

$$\varphi'(t) = \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} (it) e^{itx} e^{-x^2/2}\, dx = -t \cdot \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{itx} e^{-x^2/2}\, dx = -t\, \varphi(t)$$

The first-order linear equation $\varphi'(t) = -t\varphi(t)$ with $\varphi(0) = 1$ has the unique solution

$$\varphi_X(t) = e^{-t^2/2}$$

the Gaussian being, up to scaling, its own Fourier transform.

4. *Uniform on* $[-1,1]$. If $X \sim \mathrm{Unif}[-1,1]$ with density $\frac{1}{2}$ on $[-1,1]$, then for $t \neq 0$,

$$\varphi_X(t) = \frac{1}{2} \int_{-1}^{1} e^{itx}\, dx = \frac{1}{2} \cdot \frac{e^{it} - e^{-it}}{it} = \frac{\sin t}{t}$$

and $\varphi_X(0) = 1$, consistent with $\sin t/t \to 1$ as $t \to 0$. This φ is real (the law is symmetric) and, unlike the previous examples, changes sign and decays only like $1/t$, reflecting the discontinuity of the density at the endpoints.

5. *Cauchy.* The standard Cauchy law has density $\frac{1}{\pi(1+x^2)}$ and no finite mean, so no moment method applies. A contour integration (or the inversion of the previous example read backwards, since $e^{-|t|}$ and $\frac{1}{\pi(1+x^2)}$ are a Fourier pair, [2]) gives

$$\varphi_X(t) = \int_{\mathbb{R}} \frac{e^{itx}}{\pi(1+x^2)}\, dx = e^{-|t|}$$

The corner at $t = 0$, where φ is continuous but not differentiable, is consistent with the missing first moment: were $\mathbb{E}|X| < \infty$, proposition 6.3.4(3) would make φ differentiable with $\varphi'(0) = i\mathbb{E}[X]$, so a non-differentiable φ cannot come from a law with a finite mean. (The reverse implication, that a differentiable φ forces a finite mean, is false in general, so the corner is a symptom of the absent moment rather than a proof of it.)

6. *Binomial, from the product rule.* A variable $X \sim \mathrm{Bin}(n,p)$ is a sum $X = Y_1 + \cdots + Y_n$ of n independent $\mathrm{Ber}(p)$ variables (example 2.3.6). By the independent-sum rule proposition 6.3.4(2) and part (1) above,

$$\varphi_X(t) = \prod_{j=1}^{n} \varphi_{Y_j}(t) = \left(1 - p + pe^{it}\right)^n$$

recovering the binomial characteristic function from the Bernoulli one without touching the binomial coefficients.

These computations exhibit the dictionary between the shape of a law and the behaviour of its characteristic function. The Gaussian, the smoothest law, has the characteristic function of fastest decay, $e^{-t^2/2}$, decaying faster than any power; the uniform law, with a jump in its density, decays like $1/t$; and the Cauchy law, with no mean at all, has a characteristic function with a corner at the origin. The rule that emerges, and that the moment property proposition 6.3.4(3) makes precise, is that smoothness of φ near the origin measures the number of finite moments of X, while decay of φ at infinity measures the smoothness of the law. The Cauchy computation also illustrates a point of principle: the characteristic function exists and is perfectly well-behaved even when every moment is infinite and the moment generating function is defined only at $t = 0$, which is exactly the failure that motivated passing from the real exponent to the imaginary one.

The picture to carry forward is that the characteristic function is the analytic transform that linearizes probability. Convolution of laws, the operation that governs sums of independent random variables and that is intractable to iterate directly, becomes pointwise multiplication of characteristic functions; the law is recovered from the transform by inversion; and, as the next section shows, weak convergence of laws is detected by pointwise convergence of transforms. These three facts, factorization, inversion, and continuity, are the reason the characteristic function, and not the distribution function or the moment generating function, is the instrument of choice for the limit theory of sums, and they are what make the central limit theorem a short computation once they are in place.

Exercise 6.3.1

1. Show that a real-valued random variable X has a characteristic function φ_X that is real-valued for all t if and only if X and $-X$ have the same law (a *symmetric* law). Deduce that for any X the function $t \mapsto \mathrm{Re}\varphi_X(t)$ is the characteristic function of the symmetrized variable $X - X'$, where X' is an independent copy of X, up to the correct normalization; identify the correct statement and prove it.

2. Compute the characteristic function of the exponential law with density e^{-x} on $[0, \infty)$. Verify that $\varphi(0) = 1$ and that $\varphi'(0) = i\,\mathbb{E}[X] = i$, consistent with proposition 6.3.4(3).

3. Using proposition 6.3.4 and example 6.3.7, show that if $X \sim \mathcal{N}(0, \sigma^2)$ then $\varphi_X(t) = e^{-\sigma^2 t^2/2}$, and that the sum of two independent Gaussians $\mathcal{N}(0, \sigma_1^2)$ and $\mathcal{N}(0, \sigma_2^2)$ is $\mathcal{N}(0, \sigma_1^2 + \sigma_2^2)$. Explain how the characteristic function makes this stability under addition immediate.

4. Show that $|\varphi_X(t)| = 1$ for some $t \neq 0$ if and only if X is supported on a lattice $b + \frac{2\pi}{t}\mathbb{Z}$ for some $b \in \mathbb{R}$ (that is, X takes values in an arithmetic progression with common difference $2\pi/|t|$). This characterizes the boundary case of the inequality $|\varphi_X(t)| \leq 1$.

5. Suppose φ_X is integrable, $\int_{\mathbb{R}} |\varphi_X(t)|\,dt < \infty$. Assuming the density-inversion formula $f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-itx}\varphi_X(t)\,dt$ (the pointwise Fourier inversion of an integrable transform, [2]), show that X has a bounded continuous density f. Apply this to the standard Gaussian to recover its density from $e^{-t^2/2}$.

6.4 Lévy's Continuity Theorem

The inversion theorem (theorem 6.3.5) and its corollary (corollary 6.3.6) settle a static question: the characteristic function pins down the law, so two distributions with the same φ are equal. But the reason to care about characteristic functions in a book about limit theorems is dynamic. A limit theorem asserts that a sequence of laws μ_n approaches a target law μ, and the characteristic function turns each law into an ordinary function on $\mathbb{R}$, a function that behaves multiplicatively under the one operation that dominates the subject, convolution (proposition 6.3.4). If convergence of laws could be detected by convergence of these functions, then a limit theorem would reduce to a pointwise limit of an explicit integral, the kind of computation one can carry out by hand. Lévy's continuity theorem provides exactly this dictionary: weak convergence of laws is the same thing as pointwise convergence of their characteristic functions, provided the limiting function is continuous at the origin. It is the result that makes the characteristic function the working tool of the central limit theorem.

One direction of the correspondence is immediate. If $\mu_n \Rightarrow \mu$, then for each fixed t the function $x \mapsto e^{itx}$ is bounded and continuous, so its integral against μ_n converges to its integral against μ by the very definition of weak convergence in the portmanteau form (theorem 6.1.3). The content of the theorem is the converse, and the converse is where the hypothesis at the origin earns its place. Pointwise convergence of the φ_n alone is not enough: the sequence of laws could leak its mass off to $\pm\infty$, in which case the limit function is a characteristic function of nothing. The single quantitative estimate that controls this leakage is a tail bound expressing the mass a law places far from the origin in terms of the behaviour of its characteristic function near the origin. That estimate is the technical heart of the proof, so it is isolated first.

Lemma 6.4.1: Tail Bound from the Characteristic Function

Let μ be a probability law on $\mathbb{R}$ with characteristic function φ. For every $\delta > 0$,

$$\mu\left(\left\{x \middle| |x| \geq \tfrac{2}{\delta}\right\}\right) \leq \frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}\varphi(t)\big)\,dt$$

***Proof*:**
The idea is to integrate $1-\varphi(t)$ over a small symmetric interval and read the result as an average of $1 - e^{itx}$ against μ, whose real part is a nonnegative function of x that is bounded below away from the origin.

Write $\varphi(t) = \int_{\mathbb{R}} e^{itx}\,d\mu(x)$ (definition 6.3.1). Fubini's theorem applies to the bounded integrand $1 - e^{itx}$ on the finite measure space $[-\delta, \delta] \times \mathbb{R}$ (see [2]), so

$$\frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \varphi(t)\big)\,dt = \frac{1}{\delta}\int_{\mathbb{R}}\left(\int_{-\delta}^{\delta}\big(1 - e^{itx}\big)\,dt\right)d\mu(x)$$

The inner integral is elementary. For $x \neq 0$,

$$\int_{-\delta}^{\delta}\big(1 - e^{itx}\big)\,dt = 2\delta - \frac{e^{i\delta x} - e^{-i\delta x}}{ix} = 2\delta - \frac{2\sin(\delta x)}{x} = 2\delta\left(1 - \frac{\sin(\delta x)}{\delta x}\right)$$

and the value 0 at $x = 0$ agrees with the limit of the right side. Taking real parts (the left side of the displayed Fubini identity has real part $\frac{1}{\delta}\int_{-\delta}^{\delta}(1 - \mathrm{Re}\varphi(t))\,dt$) gives

$$\frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}\varphi(t)\big)\,dt = 2\int_{\mathbb{R}}\left(1 - \frac{\sin(\delta x)}{\delta x}\right)d\mu(x)$$

with the convention that the integrand equals 0 at $x = 0$. The function $u \mapsto 1 - \frac{\sin u}{u}$ is nonnegative for all real u, so the integrand is nonnegative, and the integral only decreases when restricted to the region $|x| \geq 2/\delta$. On that region $|\delta x| \geq 2$, and since $|\sin(\delta x)| \leq 1$,

$$1 - \frac{\sin(\delta x)}{\delta x} \geq 1 - \frac{1}{|\delta x|} \geq 1 - \frac{1}{2} = \frac{1}{2}$$

Restricting the integral to $\{|x| \geq 2/\delta\}$ and bounding the integrand below by $\frac{1}{2}$ there,

$$\frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}\varphi(t)\big)\,dt \geq 2\int_{\{|x|\geq 2/\delta\}}\left(1 - \frac{\sin(\delta x)}{\delta x}\right)d\mu(x) \geq 2\cdot\frac{1}{2}\,\mu\big(\{|x| \geq \tfrac{2}{\delta}\}\big)$$

which is the claimed inequality, as we sought to show.

The lemma is a translation device. The left side measures how much of a law's mass sits at distance $2/\delta$ or more from the origin, a purely spatial quantity; the right side measures how far the characteristic function departs from its value $\varphi(0) = 1$ over the small interval $[-\delta, \delta]$, a purely frequency-side quantity. Because φ is continuous with $\varphi(0) = 1$, the right side is small for small δ, and the bound converts this into control of the tail. When the same estimate is applied uniformly across a whole sequence μ_n whose characteristic functions converge to a function continuous at 0, it produces tightness (definition 6.2.1), and tightness is what Prokhorov's theorem (theorem 6.2.4) needs. With the lemma in hand the main theorem follows by assembling pieces already built in this chapter.

Theorem 6.4.2: Lévy's Continuity Theorem

Let $\mu, \mu_1, \mu_2, \ldots$ be probability laws on $\mathbb{R}$ with characteristic functions $\varphi, \varphi_1, \varphi_2, \ldots$.

1. If $\mu_n \Rightarrow \mu$, then $\varphi_n(t) \to \varphi(t)$ for every $t \in \mathbb{R}$.
2. Conversely, suppose $\varphi_n(t) \to g(t)$ for every $t \in \mathbb{R}$, where the limit function g is continuous at 0. Then g is the characteristic function of a probability law μ, and $\mu_n \Rightarrow \mu$.

***Proof*:**
The two parts are proved separately; the first is a one-line consequence of the portmanteau theorem, and the second occupies the bulk of the argument.

1. Fix $t \in \mathbb{R}$. The function $x \mapsto e^{itx} = \cos(tx) + i\sin(tx)$ is bounded (modulus 1) and continuous, so both its real and imaginary parts are bounded continuous functions of x. By the portmanteau theorem (theorem 6.1.3, in the form $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for bounded continuous f) applied to $\cos(t\cdot)$ and $\sin(t\cdot)$,

$$\varphi_n(t) = \int_{\mathbb{R}} e^{itx}\,d\mu_n(x) \longrightarrow \int_{\mathbb{R}} e^{itx}\,d\mu(x) = \varphi(t)$$

   Since t was arbitrary, $\varphi_n \to \varphi$ pointwise on $\mathbb{R}$.

2. Assume $\varphi_n(t) \to g(t)$ for all t, with g continuous at 0. The argument proceeds in three steps: first tightness, then extraction of a subsequential limit, then identification of that limit and promotion to convergence of the full sequence.

   Step 1: The family $\{\mu_n\}$ is tight. Each φ_n has $\varphi_n(0) = 1$, so $g(0) = \lim_n \varphi_n(0) = 1$. Fix $\varepsilon > 0$. Because g is continuous at 0 with $g(0) = 1$, and $1 - \mathrm{Re}g$ vanishes at 0, there is

$\delta > 0$ small enough that

$$\frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}g(t)\big)\,dt < \frac{\varepsilon}{2}$$

Indeed, $|1 - \mathrm{Re}g(t)| \leq |1 - g(t)|$ can be made smaller than $\varepsilon/4$ for $|t| \leq \delta$ by choosing δ small, and then the average over $[-\delta, \delta]$ of length 2δ is at most $\frac{1}{\delta} \cdot 2\delta \cdot \frac{\varepsilon}{4} = \frac{\varepsilon}{2}$. The integrands $1 - \mathrm{Re}\varphi_n(t)$ are bounded in modulus by 2 on the fixed compact interval $[-\delta, \delta]$ and converge pointwise to $1 - \mathrm{Re}g(t)$, so by dominated convergence (see [2])

$$\frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}\varphi_n(t)\big)\,dt \longrightarrow \frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}g(t)\big)\,dt < \frac{\varepsilon}{2}$$

Hence there is N with $\frac{1}{\delta}\int_{-\delta}^{\delta}(1 - \mathrm{Re}\varphi_n(t))\,dt < \varepsilon$ for all $n \geq N$. Applying the tail bound (lemma 6.4.1) to each such μ_n,

$$\mu_n\big(\{|x| \geq \tfrac{2}{\delta}\}\big) \leq \frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \mathrm{Re}\varphi_n(t)\big)\,dt < \varepsilon \qquad (n \geq N)$$

Setting $M_1 = 2/\delta$, this reads $\mu_n([-M_1, M_1]) > 1 - \varepsilon$ for all $n \geq N$. The finitely many remaining laws $\mu_1, \ldots, \mu_{N-1}$ are each a single probability measure on $\mathbb{R}$, so for each there is a radius with $\mu_j([-M_j, M_j]) > 1 - \varepsilon$; taking $M = \max\{M_1, \ldots, M_{N-1}\}$ gives $\mu_n([-M, M]) > 1 - \varepsilon$ for *every* n. As $\varepsilon > 0$ was arbitrary, $\{\mu_n\}$ is tight (definition 6.2.1).

Step 2: Every weakly convergent subsequence has the same limit. By Prokhorov's theorem (theorem 6.2.4), tightness makes $\{\mu_n\}$ relatively compact for weak convergence: every subsequence has a further subsequence converging weakly to some probability law. Let $\mu_{n_k} \Rightarrow \nu$ be any such weakly convergent subsequence, with ν a probability law. By part (i) applied to this subsequence, its characteristic functions converge to that of ν: $\varphi_{n_k}(t) \to \varphi_\nu(t)$ for every t. But $\varphi_{n_k}(t) \to g(t)$ as well, since the full sequence converges to g. A limit is unique, so $\varphi_\nu(t) = g(t)$ for every t. In particular the same function g arises for *every* weakly convergent subsequence, and $g = \varphi_\nu$ is the characteristic function of a genuine probability law. Writing $\mu := \nu$ for this common limit, the uniqueness corollary of the inversion theorem (corollary 6.3.6) shows that any two subsequential limits, both having characteristic function g, are equal to μ. This proves g is a characteristic function, establishing the first assertion of part (ii).

Step 3: The full sequence converges to μ. It remains to upgrade "every convergent subsequence has limit μ" to $\mu_n \Rightarrow \mu$. Suppose not. Then there is a bounded continuous function f and an $\eta > 0$ with $\left|\int f\,d\mu_n - \int f\,d\mu\right| \geq \eta$ for infinitely many n, say along a subsequence (μ_{n_k}). This subsequence is itself tight (a subfamily of a tight family), so by Prokhorov it has a further subsequence $(\mu_{n_{k_j}})$ converging weakly to some probability law, which by Step 2 must be μ. Weak convergence gives $\int f\,d\mu_{n_{k_j}} \to \int f\,d\mu$, contradicting $\left|\int f\,d\mu_{n_{k_j}} - \int f\,d\mu\right| \geq \eta$ for all j. Hence $\int f\,d\mu_n \to \int f\,d\mu$ for every bounded continuous f, which is weak convergence $\mu_n \Rightarrow \mu$ by the portmanteau theorem (theorem 6.1.3), completing the proof.

The pattern of Step 3, a "subsequence of a subsequence" argument, recurs throughout probability and deserves a name to itself: to prove a sequence converges to x, it suffices to show that every subsequence has a further subsequence converging to the fixed limit x. Tightness supplies the convergent-subsequence half; uniqueness of the limit, forced here by corollary 6.3.6, supplies the identification half; the two together deliver convergence of the whole sequence. The role of the continuity hypothesis on g is worth isolating. It is used in exactly one place, Step 1, to guarantee that the mass of μ_n does not escape to infinity. Without it the conclusion fails: if $\mu_n = \mathcal{N}(0, n)$ is centered Gaussian with variance n, then $\varphi_n(t) = e^{-nt^2/2} \to \mathbb{1}_{\{t=0\}}$, whose limit g equals 1 at $t = 0$ and 0 elsewhere, discontinuous at the origin. Here the laws spread out and place no mass on any bounded interval in the limit, so there is no limiting probability law, and indeed g is not the characteristic function of one.

The theorem is stated as a characterization, but its working use is one-directional, and it is that direction, the converse, that the next chapter runs on. To prove that a sequence of laws converges weakly to a target, it suffices to compute characteristic functions, take a pointwise limit, and check that the limit is a recognizable characteristic function. This reduces a statement about the shapes of distributions to a statement about pointwise limits of explicit integrals in its own right.

Corollary 6.4.3: The Characteristic-Function Method

Let μ be a probability law on $\mathbb{R}$ with characteristic function φ_μ, and let $\mu_1, \mu_2, \ldots$ be probability laws with characteristic functions φ_n. If $\varphi_n(t) \to \varphi_\mu(t)$ for every $t \in \mathbb{R}$, then $\mu_n \Rightarrow \mu$. Equivalently, for random variables, if $\varphi_{X_n}(t) \to \varphi_X(t)$ for every t, then $X_n \Rightarrow X$.

***Proof*:**
The limit function $g = \varphi_\mu$ is the characteristic function of the probability law μ, hence is continuous everywhere and in particular at 0 (definition 6.3.1). By part (ii) of Lévy's continuity theorem (theorem 6.4.2), μ_n converges weakly to the law with characteristic function g, which is μ by uniqueness (corollary 6.3.6), completing the proof.

The corollary is the form of the theorem one invokes. When the target law is known in advance, its characteristic function is known, so its continuity is not in question, and the hypothesis reduces to the pointwise limit $\varphi_n \to \varphi_\mu$. This is a computation, and often a short one, because the characteristic function of a sum of independent variables factors (proposition 6.3.4). Two computations make the method concrete: the first proves a convergence already stated in this chapter, and the second is a preview of the central limit theorem to come, carried out here in the special case where the machinery is cleanest.

Example 6.4.4: Poisson Approximation and the Gaussian Preview

1. *Binomial to Poisson.* Fix $\lambda > 0$ and let $\mu_n = \mathrm{Bin}(n, \lambda/n)$, the law of a sum of n independent $\mathrm{Ber}(\lambda/n)$ variables. A single $\mathrm{Ber}(p)$ variable has characteristic function $\mathbb{E}[e^{itX}] = (1-p) + pe^{it} = 1 + p(e^{it} - 1)$, and by independence the characteristic function of the binomial is the n-th power (proposition 6.3.4),

$$\varphi_n(t) = \left(1 + \frac{\lambda}{n}(e^{it} - 1)\right)^n$$

Fix t and set $z_n = \lambda(e^{it} - 1)$, a constant independent of n. The standard limit $(1 + z/n)^n \to e^z$ (valid for complex z, since $n \log(1 + z_n/n) = n\big(\frac{z_n}{n} + O(n^{-2})\big) \to z_n$) gives

$$\varphi_n(t) = \left(1 + \frac{z_n}{n}\right)^n \longrightarrow e^{z_n} = e^{\lambda(e^{it} - 1)}$$

The right side is the characteristic function of $\mathrm{Poi}(\lambda)$ (example 6.3.7). By the characteristic-function method (corollary 6.4.3), $\mathrm{Bin}(n, \lambda/n) \Rightarrow \mathrm{Poi}(\lambda)$, recovering the convergence stated in example 6.1.2 and the underlying computation of example 2.3.6. The characteristic-function proof is shorter than a direct comparison of the probability mass functions, and it makes transparent why the Poisson appears: the binomial's characteristic function is a power that exponentiates in the limit.

2. *The Gaussian preview.* Let $X_1, X_2, \dots$ be independent and identically distributed with $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = 1$, and set $S_n = X_1 + \cdots + X_n$. Consider the normalized sum $S_n/\sqrt{n}$. Write $\psi(t) = \varphi_{X_1}(t) = \mathbb{E}[e^{itX_1}]$ for the common characteristic function of the summands. Because $\mathbb{E}[X_1^2] = 1 < \infty$, ψ is twice continuously differentiable with $\psi(0) = 1$, $\psi'(0) = i\mathbb{E}[X_1] = 0$, and $\psi''(0) = i^2\mathbb{E}[X_1^2] = -1$ (proposition 6.3.4). A second-order Taylor expansion at 0 therefore reads

$$\psi(t) = 1 - \frac{t^2}{2} + o(t^2) \qquad (t \to 0)$$

The independence and identical distribution of the summands make the characteristic function of the sum factor (proposition 6.3.4), and scaling by $1/\sqrt{n}$ rescales the argument (proposition 6.3.4 again), so

$$\varphi_{S_n/\sqrt{n}}(t) = \left(\psi\left(\frac{t}{\sqrt{n}}\right)\right)^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n$$

where the error is $o(1/n)$ for each fixed t because $\psi(t/\sqrt{n}) - 1 = -\frac{t^2}{2n} + o(t^2/n)$. The same complex $(1 + z/n)^n \to e^z$ limit, now with $z_n \to -t^2/2$, gives

$$\varphi_{S_n/\sqrt{n}}(t) \longrightarrow e^{-t^2/2}$$

for every t. The right side is the characteristic function of the standard Gaussian $\mathcal{N}(0,1)$ (example 6.3.7), which is continuous, so the characteristic-function method (corollary 6.4.3) yields $S_n/\sqrt{n} \Rightarrow \mathcal{N}(0,1)$. This is the central limit theorem for independent identically distributed summands with finite variance, obtained here from a Taylor expansion and a single scalar limit. The chapter to come develops this argument in full and extends it well beyond the identically distributed case.

The two examples exhibit the same three-move choreography, and it is worth naming the moves because every application of the method repeats them. First, factor: the characteristic function of a sum of independent variables is a product, and for identically distributed summands a power, so the problem passes from a sum to a single characteristic function raised to a power. Second, expand: the behaviour of that single characteristic function near $t = 0$, governed by the summand's first two moments, determines the exponent. Third, exponentiate: the power $(1 + z_n/n)^n$ converges to e^z, and the limiting exponent identifies the target law. The Poisson case and the Gaussian case differ only in which law sits at the end, and the difference is entirely a matter of what the summands look like, not of the method. The characteristic function is the object that makes convolution multiplication and therefore turns the analysis of sums into the analysis of products, and Lévy's theorem is the guarantee that the resulting pointwise limits mean what one wants them to mean about distributions. Everything the central limit theorem needs is now in place: a notion of weak convergence, a criterion for compactness of laws through tightness and Prokhorov, an invertible transform, and the continuity theorem tying transform-convergence to weak convergence. The next chapter turns the crank.

Exercise 6.4.1

1. Let X_n be uniform on $\{-n, n\}$, that is $\mathbb{P}(X_n = n) = \mathbb{P}(X_n = -n) = \frac{1}{2}$. Compute $\varphi_{X_n}(t)$ and its pointwise limit $g(t)$ as $n \to \infty$. Show that g is discontinuous at 0, and explain, in terms of Lévy's continuity theorem, why (X_n) does not converge in distribution to any random variable.

2. Let $X_1, X_2, \ldots$ be independent and identically distributed with common characteristic function ψ, and let $S_n = X_1 + \cdots + X_n$. Suppose $\mathbb{E}[X_1] = m$. Using the characteristic-function method, prove the weak law of large numbers in the form $S_n/n \Rightarrow \delta_m$ (the point mass at m), and reconcile this with convergence in probability using proposition 6.1.4.

3. The standard Cauchy law has characteristic function $\varphi(t) = e^{-|t|}$ (example 6.3.7). Let $X_1, \ldots, X_n$ be independent standard Cauchy variables and let $\bar{X}_n = (X_1 + \cdots + X_n)/n$ be their average. Compute $\varphi_{\bar{X}_n}(t)$ and identify the limiting law of $\bar{X}_n$. Contrast the result with the weak law of large numbers and explain which hypothesis of the weak law fails.

4. Let $Y_\lambda \sim \mathrm{Poi}(\lambda)$ and set $Z_\lambda = (Y_\lambda - \lambda)/\sqrt{\lambda}$. Using the characteristic function $\varphi_{Y_\lambda}(t) = e^{\lambda(e^{it}-1)}$ (example 6.3.7), show that $Z_\lambda \Rightarrow \mathcal{N}(0,1)$ as $\lambda \to \infty$ along the reals. Interpret this as a central-limit statement for the Poisson family.

5. Fix a probability law μ with characteristic function φ, and suppose $|1 - \mathrm{Re}\varphi(t)| \leq Ct^2$ for all $|t| \leq \delta_0$, for some constants $C, \delta_0 > 0$. Using the tail bound (lemma 6.4.1), prove that $\mu(\{|x| \geq M\}) \leq \dfrac{8C}{3M^2}$ for all $M \geq 2/\delta_0$. State what this bound says about a family of laws $\{\mu_i\}$ whose characteristic functions all satisfy $|1 - \mathrm{Re}\varphi_i(t)| \leq Ct^2$ on $|t| \leq \delta_0$ with the same constants.

7

The Central Limit Theorem

The laws of large numbers said the average S_n/n settles to the mean. The central limit theorem describes what remains after that settling: the fluctuation $S_n - n\mu$, rescaled by $\sqrt{n}$, does not vanish but organizes itself into a fixed shape, the Gaussian, whatever the distribution of the individual summands. This universality is why the bell curve is ubiquitous, and Lévy's continuity theorem of chapter 6 makes it provable in a few lines: the characteristic function of the normalized sum converges to $e^{-t^2/2}$, the characteristic function of the standard normal.

section 7.1 proves the classical Lindeberg-Lévy theorem for i.i.d. summands with finite variance. Section 7.2 extends it to triangular arrays under the Lindeberg condition, the sharp hypothesis for the asymptotic normality of sums of independent but not identically distributed variables. section 7.3 develops Gaussian random vectors and the multivariate central limit theorem through the Cramér-Wold device. Section 7.4 records the quantitative refinement, the Berry-Esseen bound on the rate of convergence, and the delta method that transports asymptotic normality through a smooth function. Section 7.5 closes the number-theoretic thread opened in section 5.4: the number of distinct prime factors of a random integer is asymptotically Gaussian, the Erdős-Kac theorem, the central-limit refinement of the Hardy-Ramanujan law.

7.1 The Classical Central Limit Theorem

The laws of large numbers pinned down the first-order behaviour of a sum of independent variables: S_n/n converges to the mean μ, so the fluctuation $S_n - n\mu$ is $o(n)$. It does not, however, vanish. The central limit theorem is the statement that once this fluctuation is rescaled by exactly $\sqrt{n}$, it settles into a fixed distribution, and that distribution is always the same Gaussian bell, no matter what law the individual summands carry beyond their mean and variance. This universality is the reason the normal distribution is the one that appears whenever a quantity is built up from many small independent contributions, and it is what this section makes precise for independent and identically distributed summands with finite variance.

The proof is short because the machinery of the previous chapter was built for it. A characteristic function turns a sum into a product, and Lévy's continuity theorem (theorem 6.4.2) turns pointwise convergence of characteristic functions into weak convergence of the underlying laws. The whole argument reduces to a Taylor expansion of a single characteristic function near the origin, raised to the n. Before stating the theorem the target of that convergence must be named, so this section opens with the normal law itself.

Chapter 6 met the standard normal in passing, as the law whose characteristic function is $e^{-t^2/2}$ (example 6.3.7). The general normal law is obtained from it by shifting and scaling.

Definition 7.1.1: The Normal Distribution

For $\mu \in \mathbb{R}$ and $\sigma^2 \in (0, \infty)$, the *normal law* $\mathcal{N}(\mu, \sigma^2)$ is the probability measure on $\mathbb{R}$ with density

$$\varphi_{\mu,\sigma^2}(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

with respect to Lebesgue measure. A random variable X with this law is called *normal* (or *Gaussian*) with mean μ and variance σ^2, written $X \sim \mathcal{N}(\mu, \sigma^2)$. The case $\mu = 0$, $\sigma^2 = 1$ is the *standard normal*; its density is written $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and its cumulative distribution function

$$\Phi(x) = \int_{-\infty}^{x} \varphi(t)\, dt$$

is the *standard normal cdf.* By convention $\mathcal{N}(\mu, 0)$ denotes the point mass δ_μ, the degenerate normal of zero variance.

That φ_{μ,σ^2} integrates to one is the Gaussian integral $\int_{\mathbb{R}} e^{-x^2/2}\, dx = \sqrt{2\pi}$, and the substitution $x = \mu + \sigma z$ shows every member of the family is a genuine density with the advertised mean and variance: if $Z \sim \mathcal{N}(0, 1)$ then $\mu + \sigma Z$ has density φ_{μ,σ^2}, mean μ, and variance σ^2. The family is therefore a single shape, the standard bell φ, translated to sit at μ and dilated to have spread σ. This closure under affine maps is what makes the two-parameter family the correct object: any limit built from centred and scaled sums will be normal, and the scaling only sets μ and σ^2.

The characteristic function records the same closure algebraically and is the form in which the normal law enters every proof in this chapter.

Proposition 7.1.2: Characteristic Function of the Normal Law

If $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma^2 \geq 0$, then

$$\varphi_X(t) = \exp\big(i\mu t - \tfrac{1}{2}\sigma^2 t^2\big)$$

for all $t \in \mathbb{R}$.

***Proof*:**
For the standard normal, example 6.3.7 of chapter 6 computed $\varphi_Z(t) = e^{-t^2/2}$. Writing a general normal as $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$ and using the transformation rule $\varphi_{a+bX}(t) = e^{iat}\varphi_X(bt)$

from the properties of characteristic functions (proposition 6.3.4), we obtain

$$\varphi_X(t) = e^{i\mu t}\,\varphi_Z(\sigma t) = e^{i\mu t}\,e^{-\sigma^2 t^2/2} = \exp\bigl(i\mu t - \tfrac{1}{2}\sigma^2 t^2\bigr)$$

The degenerate case $\sigma^2 = 0$ gives $\varphi_X(t) = e^{i\mu t}$, the characteristic function of the point mass δ_μ, which is the stated formula with $\sigma^2 = 0$, completing the proof.

The characteristic function is again of the form (exponential of $i\mu t$ minus a quadratic in t), and the exponent is exactly log of the moment generating function evaluated at it: a normal variable has $\mathbb{E}[e^{sX}] = e^{\mu s + \sigma^2 s^2/2}$ (definition 3.4.4), and setting $s = it$ recovers the display above. The single most useful consequence for what follows is that the family is closed under convolution: if $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then $\varphi_{X+Y} = \varphi_X \varphi_Y = \exp(i(\mu_1+\mu_2)t - \frac{1}{2}(\sigma_1^2+\sigma_2^2)t^2)$, so $X + Y \sim \mathcal{N}(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$. A sum of independent normals is normal, with means and variances adding. This is the finite-n case of the theorem that a sum of independent variables of *any* law, suitably normalized, becomes normal in the limit.

With the limit law in hand the theorem can be stated. The natural object is the *standardized* sum: given i.i.d. summands with mean μ and variance σ^2, the sum $S_n = X_1 + \cdots + X_n$ has mean $n\mu$ and variance $n\sigma^2$, so subtracting the mean and dividing by the standard deviation $\sigma\sqrt{n}$ produces a variable with mean 0 and variance 1 for every n. The central limit theorem asserts that this standardized sum converges in distribution to the standard normal.

Theorem 7.1.3: The Lindeberg-Lévy Central Limit Theorem

Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with mean $\mu = \mathbb{E}[X_1]$ and variance $\sigma^2 = \mathrm{Var}(X_1) \in (0, \infty)$. Set $S_n = X_1 + \cdots + X_n$. Then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \;\Rightarrow\; \mathcal{N}(0,1)$$

as $n \to \infty$; that is, for every $x \in \mathbb{R}$,

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \le x\right) \longrightarrow \Phi(x)$$

Before the proof, three remarks fix its meaning. The convergence is in distribution (definition 6.1.1), the weakest of the modes: it is a statement about the laws of the standardized sums, not about the variables themselves, which need not converge in any pointwise sense. The equivalence of the two displayed lines is the fact, recorded with weak convergence on $\mathbb{R}$, that convergence in distribution to a law with continuous cdf is equivalent to pointwise convergence of the cdfs at every point; here the limiting cdf Φ is continuous, so convergence holds at *every* x, with no exceptional set. Finally, the hypothesis is minimal: only a finite second moment is assumed, and the requirement $\sigma^2 > 0$ excludes the degenerate case of a constant summand, for which $S_n - n\mu$ is identically zero.

***Proof*:**
The strategy is to compute the characteristic function of the standardized sum, show it converges pointwise to $e^{-t^2/2}$, and invoke Lévy's continuity theorem.

Step 1: Reduce to mean zero and unit variance. Set $Y_i = (X_i - \mu)/\sigma$. The Y_i are again i.i.d., with

$\mathbb{E}[Y_1] = 0$ and $\operatorname{Var}(Y_1) = \mathbb{E}[Y_1^2] = 1$. The standardized sum is unchanged under this reduction:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma} = \frac{Y_1 + \cdots + Y_n}{\sqrt{n}}$$

Writing $T_n = Y_1 + \cdots + Y_n$, it therefore suffices to prove $T_n/\sqrt{n} \Rightarrow \mathcal{N}(0,1)$ when the summands have mean 0 and variance 1.

Step 2: The characteristic function of the standardized sum. Let $\varphi(t) = \varphi_{Y_1}(t) = \mathbb{E}[e^{itY_1}]$ be the common characteristic function of the Y_i. Because the Y_i are independent, the characteristic function of a sum is the product of the characteristic functions, and because they are identically distributed this product is a power:

$$\varphi_{T_n}(t) = \varphi(t)^n$$

The map $y \mapsto y/\sqrt{n}$ rescales the argument, so by the transformation rule (proposition 6.3.4),

$$\varphi_{T_n/\sqrt{n}}(t) = \varphi_{T_n}\left(\frac{t}{\sqrt{n}}\right) = \varphi\left(\frac{t}{\sqrt{n}}\right)^n \tag{7.1}$$

Everything now rests on the behaviour of φ near the origin, since the argument $t/\sqrt{n} \to 0$ for each fixed t.

Step 3: Taylor expansion of φ at the origin. Because Y_1 has a finite second moment, its characteristic function is twice continuously differentiable, with

$$\varphi'(0) = i\,\mathbb{E}[Y_1] = 0, \qquad \varphi''(0) = -\,\mathbb{E}[Y_1^2] = -1$$

by the differentiation-under-the-integral properties of characteristic functions of variables with finite moments (proposition 6.3.4). A second-order Taylor expansion of φ about 0, valid whenever the second moment is finite, gives

$$\varphi(s) = \varphi(0) + \varphi'(0)\,s + \tfrac{1}{2}\varphi''(0)\,s^2 + o(s^2) = 1 - \tfrac{1}{2}s^2 + o(s^2) \qquad (s \to 0) \tag{7.2}$$

Here $\varphi(0) = 1$ because $\varphi(0) = \mathbb{E}[e^0] = 1$, and the error term $o(s^2)$ is the standard Peano remainder, legitimate precisely because $\varphi''(0)$ exists.

Step 4: Raise to the n and pass to the limit. Substitute $s = t/\sqrt{n}$ into (7.2). Fixing t and letting $n \to \infty$, the argument $t/\sqrt{n} \to 0$, so

$$\varphi\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)$$

where the remainder is $o(1/n)$ because $o(s^2) = o(t^2/n) = o(1/n)$ for fixed t. By (7.1),

$$\varphi_{T_n/\sqrt{n}}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \tag{7.3}$$

This has the form $(1 + z_n/n)^n$ with $z_n = -t^2/2 + o(1) \to -t^2/2$. The elementary lemma that $(1 + z_n/n)^n \to e^z$ whenever $z_n \to z$ (lemma 7.1.4 below) applies, giving

$$\varphi_{T_n/\sqrt{n}}(t) \longrightarrow e^{-t^2/2}$$

for every $t \in \mathbb{R}$.

Step 5: Conclude by Lévy continuity. The pointwise limit $e^{-t^2/2}$ is the characteristic function of $\mathcal{N}(0,1)$ (proposition 7.1.2), and it is continuous at $t = 0$. Lévy's continuity theorem (theorem 6.4.2), in the form recorded for verifying weak convergence by characteristic functions (corollary 6.4.3), concludes that

$$\frac{T_n}{\sqrt{n}} \Rightarrow \mathcal{N}(0,1)$$

By Step 1 this is exactly the assertion for the original standardized sum, completing the proof.

The single analytic fact used at the end of Step 4 deserves to be isolated, because the naive real-variable version $(1 + z/n)^n \to e^z$ requires care when z_n is complex and only converges rather than being fixed.

Lemma 7.1.4: Convergence to the Exponential

Let (z_n) be a sequence of complex numbers with $z_n \to z \in \mathbb{C}$. Then

$$\left(1 + \frac{z_n}{n}\right)^n \longrightarrow e^z$$

***Proof*:**
Two facts about complex numbers suffice. First, for complex numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$ all of modulus at most some $R \geq 1$,

$$\left|\prod_{k=1}^{n} a_k - \prod_{k=1}^{n} b_k\right| \leq R^{n-1} \sum_{k=1}^{n} |a_k - b_k|$$

a telescoping estimate proved by induction on n: writing the difference of products as $\sum_k \big(\prod_{j<k} a_j\big)(a_k - b_k)\big(\prod_{j>k} b_j\big)$ and bounding each of the $n-1$ retained factors by R. Second, for $|w| \leq 1$,

$$|e^w - (1 + w)| \leq |w|^2$$

obtained by comparing the power series term by term, since $e^w - 1 - w = \sum_{k\geq 2} w^k/k!$ and $\sum_{k\geq 2} |w|^k/k! \leq |w|^2 \sum_{k\geq 2} 1/k! \leq |w|^2$.
Fix z and choose N so that $|z_n| \leq |z| + 1 =: M$ for all $n \geq N$. For $n \geq \max(N, M)$ the numbers $w_n := z_n/n$ satisfy $|w_n| \leq M/n \leq 1$, so both estimates apply with $a_k = 1 + w_n$, $b_k = e^{w_n}$ (all n factors equal). These factors are not of modulus at most 1 when $\mathrm{Re} w_n > 0$, but they are uniformly bounded: $|1 + w_n| \leq 1 + |w_n| \leq e^{|w_n|} \leq e^{M/n}$ and $|e^{w_n}| = e^{\mathrm{Re} w_n} \leq e^{|w_n|} \leq e^{M/n}$, so both lie within $R := e^{M/n}$, and hence $R^{n-1} \leq e^{M(n-1)/n} \leq e^M$. The first estimate with this R and then the second give

$$\left|\left(1 + \frac{z_n}{n}\right)^n - e^{z_n}\right| = |(1 + w_n)^n - (e^{w_n})^n| \leq R^{n-1}\, n\, |(1+w_n) - e^{w_n}| \leq e^M\, n\, |w_n|^2 = e^M \frac{|z_n|^2}{n} \leq \frac{e^M M^2}{n}$$

which tends to 0. Since $e^{z_n} \to e^z$ by continuity of the exponential, the triangle inequality yields $(1 + z_n/n)^n \to e^z$, as we sought to show.

The theorem is worth reading against the law of large numbers it refines. The weak law (theorem 5.1.5) says $S_n/n \to \mu$ in probability, which in the standardized variable is the statement $(S_n - n\mu)/n \to 0$: dividing the fluctuation by n crushes it to a point. The central limit theorem inspects the same fluctuation under the gentler scaling $\sqrt{n}$ and finds it neither collapses nor diverges but converges to a nondegenerate law. The two theorems describe the same sum at two resolutions, and $\sqrt{n}$ is the exact borderline between them: any faster growth than $\sqrt{n}$ in the denominator sends the standardized sum to 0, any slower sends it to infinity in probability. The Gaussian is the fixed point that lives precisely at the $\sqrt{n}$ scale.

Two features of the proof carry all the content. The reduction to mean zero and unit variance shows the limit cannot remember anything about the summand law beyond μ and σ^2: after standardizing, every problem looks identical through second order. The Taylor expansion (7.2) shows why: the characteristic function near the origin is determined, to the order that survives the limit, entirely by $\varphi''(0) = -\mathbb{E}[Y_1^2]$, and the third and higher moments live in the $o(s^2)$ remainder that the limit discards. This is the analytic form of universality: the Gaussian is universal because the second-order Taylor coefficient is all that survives raising $\varphi(t/\sqrt{n})$ to the n.

The oldest instance of the theorem, predating the general statement by two centuries, is the coin-tossing case. Recall the fair coin-tossing space (example 1.3.6), where independent Bernoulli trials model repeated flips. Counting successes gives a binomial variable, and its normalization was analyzed by de Moivre for the fair coin and by Laplace in general.

Corollary 7.1.5: The de Moivre-Laplace Theorem

Fix $p \in (0, 1)$ and let $S_n \sim \mathrm{Bin}(n, p)$, the number of successes in n independent trials each of success probability p. Then

$$\frac{S_n - np}{\sqrt{np(1-p)}} \Rightarrow \mathcal{N}(0, 1)$$

as $n \to \infty$; equivalently, for all $a < b$,

$$\mathbb{P}\left(a \le \frac{S_n - np}{\sqrt{np(1-p)}} \le b\right) \longrightarrow \Phi(b) - \Phi(a)$$

***Proof*:**
A binomial variable $S_n \sim \mathrm{Bin}(n, p)$ is the sum $X_1 + \cdots + X_n$ of i.i.d. Bernoulli variables $X_i \sim \mathrm{Ber}(p)$, each taking value 1 with probability p and 0 with probability $1 - p$ (the successes on the coin-tossing space, example 1.3.6). These have mean $\mu = p$ and variance $\sigma^2 = p(1-p) \in (0, \infty)$, so the Lindeberg-Lévy theorem (theorem 7.1.3) applies verbatim with $n\mu = np$ and $\sigma\sqrt{n} = \sqrt{np(1-p)}$, giving the first display. The second follows from the first because Φ is continuous, so the interval probabilities of the limit are obtained by differencing the cdf at the endpoints, as we sought to show.

The corollary is the concrete face of the theorem, and it is the source of the normal approximation to the binomial used throughout applied probability: for large n,

$$\mathbb{P}(S_n \le k) \approx \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right)$$

The approximation is a genuine asymptotic statement, not an identity, and its quality is governed by how far S_n is from its mean in units of the standard deviation; the rate at which the two sides agree, uniformly in k, is quantified by the Berry-Esseen bound to come later in this chapter. In practice the approximation is sharpened by a *continuity correction*, replacing k by $k + \frac{1}{2}$ to account for the integer-valued S_n being approximated by a continuous law, but the asymptotic content is exactly corollary 7.1.5.

The corollary also explains the historical order of discovery. De Moivre found the $p = 1/2$ case in the 1730s while approximating the central binomial coefficients $\binom{2n}{n}$, well before the notion of a probability density was available; Laplace extended it to general p. The general central limit theorem (theorem 7.1.3) is the twentieth-century statement that the special role of the binomial is an illusion: any i.i.d. sum with finite variance obeys the same law, and the coin flip is one instance among all of them.

The theorem is best absorbed by watching it act on specific sums. The following example carries the two anchors of this chapter, a discrete binomial and a continuous exponential sum, and pauses on the universality that makes them share a limit.

Example 7.1.6: Normal Approximation in Practice

1. *The binomial, numerically.* Let $S_{100} \sim \mathrm{Bin}(100, \frac{1}{2})$, the number of heads in 100 fair flips. Its mean is $np = 50$ and its standard deviation is $\sqrt{np(1-p)} = \sqrt{25} = 5$. To estimate the probability of at least 60 heads, standardize and apply corollary 7.1.5:
$$\mathbb{P}(S_{100} \geq 60) = \mathbb{P}\left(\frac{S_{100} - 50}{5} \geq 2\right) \approx 1 - \Phi(2) \approx 0.0228$$
With the continuity correction the threshold 60 becomes 59.5, giving $(59.5 - 50)/5 = 1.9$ and $1 - \Phi(1.9) \approx 0.0287$; the exact binomial probability is 0.0284, so the corrected normal value is accurate to three decimals already at $n = 100$. The bell curve centred at 50 with spread 5 is a faithful stand-in for the discrete histogram of S_{100}.

2. *A continuous example: sums of exponentials.* Let $X_1, X_2, \ldots$ be i.i.d. exponential with rate 1, so each has density e^{-x} on $(0, \infty)$, mean $\mu = 1$, and variance $\sigma^2 = 1$. The sum $S_n = X_1 + \cdots + X_n$ is Gamma-distributed with shape n and rate 1, an exactly known law skewed to the right. The Lindeberg-Lévy theorem (theorem 7.1.3) nonetheless gives
$$\frac{S_n - n}{\sqrt{n}} \Rightarrow \mathcal{N}(0,1)$$
For $n = 100$ this says S_{100}, which has mean 100 and standard deviation 10, is well approximated by $\mathcal{N}(100, 100)$; for instance $\mathbb{P}(S_{100} \leq 110) \approx \Phi(1) \approx 0.841$. The standardized density $\sqrt{n}\,\varphi_{S_n}(n + \sqrt{n}\,x)$ is visibly skewed for small n and relaxes onto the symmetric bell $\varphi(x)$ as n grows: the initial asymmetry of the exponential is a third-moment effect, and (7.2) shows the third moment is precisely what the limit forgets.

3. *Universality across laws.* The two limits above are numerically identical objects: the standardized binomial and the standardized exponential sum both converge to $\mathcal{N}(0,1)$, with no trace of the Bernoulli or the exponential surviving. This is the content of theorem 7.1.3: the limit depends on the summand law only through μ and σ^2, both of which are standardized away. Replace the summands by uniforms on $[0,1]$, by Poisson counts, by any law with finite positive variance, and the standardized sum converges to the same standard normal. The summand law sets the *rate* of convergence, but not the *limit.*

The three parts point at the same phenomenon from complementary sides, and it is worth stating the interpretation plainly, since it is the reason the central limit theorem occupies the place it does. A quantity produced by accumulating many small, independent contributions, none of which dominates the total, is approximately normal, and the approximation improves as the number of contributions grows. Measurement errors, the total displacement of a random walk, the aggregate of many independent shocks in a financial or physical system: each is a sum of the kind the theorem governs, and each is therefore Gaussian to the extent that its constituents are numerous, independent, and individually negligible.

The universality has a sharp mechanism, already visible in the proof, and it is not a vague appeal to averaging. The characteristic function of a standardized summand agrees with that of the standard normal to second order at the origin, $\varphi(s) = 1 - s^2/2 + o(s^2)$, and raising it to the n-th power at argument $t/\sqrt{n}$ magnifies exactly the second-order coefficient while sending the higher-order remainder to zero. The Gaussian is the unique law whose characteristic function is $e^{-t^2/2}$, and this is the only law that can survive the operation because it is the only one with no structure beyond the second-order term. Every finite-variance summand is, through the lens of $\sqrt{n}$-scaling and this Taylor expansion, indistinguishable from a Gaussian; the bell curve is universal not because all randomness looks the same, but because the operation of summing and rescaling erases everything except the variance. This is the same fixed-point phenomenon that will reappear in the next section, where the Lindeberg-Feller theorem identifies the exact condition under which independent but non-identically-distributed summands are still individually negligible enough for the Gaussian limit to hold.

Exercise 7.1.1

1. Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_1] = 3$ and $\mathrm{Var}(X_1) = 4$, and let $S_n = X_1 + \cdots + X_n$. For large n, find the approximate distribution of S_n (its mean and variance), and use the central limit theorem to estimate $\mathbb{P}(S_{100} > 320)$ in terms of Φ.

2. Let $U_1, \ldots, U_n$ be i.i.d. uniform on $[0, 1]$. Compute $\mu = \mathbb{E}[U_1]$ and $\sigma^2 = \mathrm{Var}(U_1)$, write down the standardized sum, and state the central limit theorem for it. For $n = 12$, explain why $U_1 + \cdots + U_{12} - 6$ is a convenient approximate standard normal (this is a classical method for generating pseudo-Gaussian samples).

3. Show directly from the characteristic function that a sum of two independent normal variables is normal: if $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, prove $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Which theorem from chapter 6 guarantees that the characteristic function determines the law?

4. The central limit theorem uses the scaling $\sqrt{n}$. Show that this scaling is forced: with X_i i.i.d. mean 0, variance $\sigma^2 > 0$, prove that $(S_n)/n^\alpha$ converges in probability to 0 if $\alpha > 1/2$ and diverges (in the sense that $\mathbb{P}(|S_n|/n^\alpha > M) \to 1$ for every M) if $\alpha < 1/2$. Only $\alpha = 1/2$ produces a nondegenerate limit.

5. Give an example of an i.i.d. sequence for which the standardized sum $(S_n - n\mu)/\sqrt{n}$ does *not* converge to a normal law, and identify which hypothesis of theorem 7.1.3 fails. (Hint: consider a law with infinite variance, such as one with density proportional to $|x|^{-3}$ for large $|x|$.)

6. Let $N_\lambda \sim \mathrm{Poi}(\lambda)$ be Poisson with mean λ. Using that N_λ has the law of a sum of λ (integer) independent $\mathrm{Poi}(1)$ variables, show that $(N_\lambda - \lambda)/\sqrt{\lambda} \Rightarrow \mathcal{N}(0, 1)$ as $\lambda \to \infty$ through the integers. Interpret this as the normal approximation to the Poisson for large λ.

7.2 The Lindeberg-Feller Theorem

The classical central limit theorem (theorem 7.1.3) rests on two hypotheses about the summands: they are independent, and they are identically distributed. Independence is indispensable, the mechanism by which the characteristic function of a sum factors into a product. The identical distribution is not. What the classical proof used was that each summand $X_i/(\sigma\sqrt{n})$ is small, of order $1/\sqrt{n}$, so that the local expansion of a single characteristic function, applied n times, controls the whole product. Identical distribution guaranteed this smallness uniformly, but any arrangement in which no single summand dominates the sum should serve equally well. This section replaces the identical-distribution hypothesis with the sharp condition, due to Lindeberg, that makes the Gaussian limit persist, and records Feller's theorem that the condition is not only sufficient but necessary once one asks for the individual terms to be negligible.

The natural setting is no longer a single sequence of variables but a doubly indexed family, one row of variables for each n, whose row sums are the objects whose limit is sought. A sum of n independent measurements with different precisions, a weighted average $\sum_k a_{n,k}Y_k$ in which the weights change with n, or the standardized sum $S_n/(\sigma\sqrt{n})$ of the classical theorem itself, in which the n-th row is the collection $X_1/(\sigma\sqrt{n}), \dots, X_n/(\sigma\sqrt{n})$, are all instances of one structure. Abstracting it isolates the two features that drive the limit, row-wise independence and the smallness of each entry, from the accidental feature of a common distribution.

Definition 7.2.1: Triangular Array

A *triangular array* of random variables is a family $\{X_{n,k} : n \geq 1,\ 1 \leq k \leq k_n\}$, where $(k_n)_{n\geq 1}$ is a sequence of positive integers, such that for each fixed n the variables $X_{n,1}, \dots, X_{n,k_n}$ in the n-th row are independent (definition 2.2.2). The array is called *centered* if $\mathbb{E}[X_{n,k}] = 0$ for all n, k. Its *row sums* are

$$S_n = \sum_{k=1}^{k_n} X_{n,k}$$

Write $\sigma_{n,k}^2 = \mathrm{Var}(X_{n,k})$ (definition 3.2.1) and $s_n^2 = \sum_{k=1}^{k_n} \sigma_{n,k}^2 = \mathrm{Var}(S_n)$ for the variance of the n-th row sum. Throughout this section the array is centered and satisfies $s_n^2 \to \sigma^2$ for some finite $\sigma^2 > 0$, the variance of the target Gaussian.

Independence is imposed only *within* each row; nothing is assumed across rows, and indeed the rows will typically be built from a common source in a way that makes them highly dependent. The name records the mental picture of the classical case, where the n-th row has n entries and stacking the rows fills a triangle, though k_n need not equal n and the array need not literally be triangular. The only structural require is that each row is a finite independent family whose variances sum to something converging to σ^2; the theorem below asks when the row sums converge to $\mathcal{N}(0, \sigma^2)$ (definition 7.1.1).

Example 7.2.2: The Classical Sum as a Triangular Array

Let $Y_1, Y_2, \dots$ be i.i.d. with mean 0 and variance $\sigma^2 \in (0, \infty)$, the setting of the classical theorem. Set $k_n = n$ and

$$X_{n,k} = \frac{Y_k}{\sqrt{n}} \qquad (1 \leq k \leq n)$$

The n-th row is independent because $Y_1, \dots, Y_n$ are, so this is a centered triangular array. Each entry has variance $\sigma_{n,k}^2 = \sigma^2/n$, and the row-sum variance is

$$s_n^2 = \sum_{k=1}^{n} \frac{\sigma^2}{n} = \sigma^2$$

exactly, for every n, so $s_n^2 \to \sigma^2$ holds trivially. The row sum is $S_n = n^{-1/2} \sum_{k=1}^{n} Y_k$, the normalized sum of the classical theorem, and its target limit is $\mathcal{N}(0, \sigma^2)$. This is the array whose limit theorem 7.1.3 already computes, and it is the one against which the hypotheses below should be tested.

The reduction that entries are individually small must now be made precise, and made precise in a way that survives non-identical distributions. In the array of example 7.2.2 each entry has variance $\sigma^2/n \to 0$, so the largest entry variance $\max_k \sigma_{n,k}^2$ tends to zero: no single summand carries a nonvanishing fraction of the total variance. This is the smallness the classical proof exploited. But controlling variances is not enough for the characteristic-function expansion, which needs the entries to be small not just in their second moment but in a truncated sense that also handles the tail. The condition that delivers exactly this is Lindeberg's.

Definition 7.2.3: The Lindeberg Condition

A centered triangular array $\{X_{n,k}\}$ satisfies the *Lindeberg condition* if for every $\varepsilon > 0$,

$$L_n(\varepsilon) = \sum_{k=1}^{k_n} \mathbb{E}\big[X_{n,k}^2 \, \mathbb{1}_{\{|X_{n,k}| > \varepsilon\}}\big] \xrightarrow[n\to\infty]{} 0$$

The quantity $L_n(\varepsilon)$ is the *Lindeberg function* of the array.

The Lindeberg function isolates, for each row, the part of the total second moment $\sum_k \mathbb{E}[X_{n,k}^2]$ that comes from entries exceeding the threshold ε. Because the array is centered, $\mathbb{E}[X_{n,k}^2] = \sigma_{n,k}^2$, so $\sum_k \mathbb{E}[X_{n,k}^2] = s_n^2 \to \sigma^2$ is bounded, and the whole second moment is finite; the condition asks that the fraction of it living above any fixed level ε vanishes as n grows. In the running array of example 7.2.2, where $X_{n,k} = Y_k/\sqrt{n}$, the event $\{|X_{n,k}| > \varepsilon\}$ is $\{|Y_k| > \varepsilon\sqrt{n}\}$, and

$$L_n(\varepsilon) = \sum_{k=1}^{n} \frac{1}{n} \mathbb{E}\big[Y_k^2 \, \mathbb{1}_{\{|Y_k| > \varepsilon\sqrt{n}\}}\big] = \mathbb{E}\big[Y_1^2 \, \mathbb{1}_{\{|Y_1| > \varepsilon\sqrt{n}\}}\big]$$

using that the Y_k are identically distributed, and this tends to 0 by dominated convergence, since Y_1^2 is integrable and $\mathbb{1}_{\{|Y_1| > \varepsilon\sqrt{n}\}} \to 0$ pointwise. The Lindeberg condition thus holds automatically in the i.i.d. case with finite variance, which is why the classical theorem is a special case of the one below.

The proper reading of the condition is as *uniform asymptotic negligibility* measured in the second moment. To see this, observe that for any $\varepsilon > 0$,

$$\max_{1\le k\le k_n} \sigma_{n,k}^2 = \max_k \mathbb{E}[X_{n,k}^2] \le \varepsilon^2 + \max_k \mathbb{E}\big[X_{n,k}^2 \, \mathbb{1}_{\{|X_{n,k}| > \varepsilon\}}\big] \le \varepsilon^2 + L_n(\varepsilon)$$

where the first inequality splits $\mathbb{E}[X_{n,k}^2]$ according to whether $|X_{n,k}|$ exceeds ε and bounds the small part by ε^2. Letting $n \to \infty$ and then $\varepsilon \to 0$ shows that the Lindeberg condition forces

$$\max_{1\le k\le k_n} \sigma_{n,k}^2 \xrightarrow[n\to\infty]{} 0 \tag{7.4}$$

the requirement that no single summand contributes a nonvanishing share of the variance. The condition (7.4) is exactly *uniform asymptotic negligibility*: the largest entry, measured by variance, becomes negligible uniformly across the row. Lindeberg's condition is strictly stronger, controlling not just the largest variance but the tail contribution to the second moment, and this extra control is what the proof will convert into the Gaussian limit.

7.2.4 Note: Negligibility Is Necessary for a Gaussian Limit Uniform asymptotic negligibility (7.4) is not a technical convenience but a constraint on when a limit can be Gaussian. If one summand carries a fixed positive fraction of the variance, its own law leaves a permanent imprint on the limit, which can then be any infinitely divisible law, not only the Gaussian. The degenerate array in example 7.2.10(2) exhibits this failure of negligibility. Feller's converse, stated below, shows that under negligibility the Gaussian limit and the Lindeberg condition are the same thing.

Everything is now in place for the central result of the section. The theorem asserts that the Lindeberg condition alone, with no assumption of a common distribution, forces the row sums to be asymptotically Gaussian, and that the condition is sharp in the presence of negligibility.

Theorem 7.2.5: The Lindeberg-Feller Central Limit Theorem

Let $\{X_{n,k} \colon 1 \le k \le k_n\}$ be a centered triangular array with $s_n^2 = \sum_k \operatorname{Var}(X_{n,k}) \to \sigma^2 \in (0, \infty)$.

1. *Sufficiency (Lindeberg).* If the array satisfies the Lindeberg condition (definition 7.2.3), then the row sums converge in distribution to a centered Gaussian:

$$S_n = \sum_{k=1}^{k_n} X_{n,k} \;\Rightarrow\; \mathcal{N}(0, \sigma^2)$$

2. *Necessity (Feller).* Conversely, if the row sums satisfy $S_n \Rightarrow \mathcal{N}(0, \sigma^2)$ *and* the array is uniformly asymptotically negligible in the sense of (7.4), that is $\max_k \sigma_{n,k}^2 \to 0$, then the array satisfies the Lindeberg condition.

Part (1) is the extension of the classical theorem that this section exists to prove, and its proof occupies the rest of the technical discussion. Part (2) is Feller's contribution, which converts sufficiency into an equivalence: once the individual terms are required to be negligible, asymptotic normality holds *exactly* when Lindeberg's condition does. The proof of sufficiency is a refinement of the classical characteristic-function argument, with three moving parts, a telescoping estimate that compares a product of characteristic functions to a product of Gaussian surrogates, a Taylor bound controlled uniformly across a row, and the Lindeberg truncation that ties the error to $L_n(\varepsilon)$. The necessity direction is more delicate and is discussed after the sufficiency proof.

Two elementary lemmas carry the analytic weight and are isolated so that the main argument reads cleanly. The first is a product-comparison inequality: a product of numbers in the closed unit disk changes by at most the sum of the changes of its factors. It is the multiplicative analogue of the triangle inequality and is what lets the row sum be compared to its Gaussian surrogate factor by factor.

Lemma 7.2.6: Product Comparison in the Unit Disk

Let $z_1, \dots, z_m$ and $w_1, \dots, w_m$ be complex numbers with $|z_k| \le 1$ and $|w_k| \le 1$ for every k. Then

$$\left| \prod_{k=1}^{m} z_k - \prod_{k=1}^{m} w_k \right| \le \sum_{k=1}^{m} |z_k - w_k|$$

***Proof*:**
The proof is by a telescoping sum. Write the difference of products as a sum of terms in each of which a single factor is switched from w to z:

$$\prod_{k=1}^{m} z_k - \prod_{k=1}^{m} w_k = \sum_{j=1}^{m} \left(\prod_{k<j} z_k \right) (z_j - w_j) \left(\prod_{k>j} w_k \right)$$

Each successive term cancels the trailing part of its predecessor, leaving only the two extreme products. Taking moduli and using $|z_k| \le 1$ and $|w_k| \le 1$, each factored product has modulus at most 1, so the j-th term is bounded by $|z_j - w_j|$. Summing over j gives the claim, completing the proof.

The second lemma bounds how far a complex exponential deviates from its second-order Taylor polynomial, in a form that will be applied to $e^{itX_{n,k}}$ and that captures the two regimes the Lindeberg truncation distinguishes: for a small argument the cubic remainder controls the error, for a large argument the quadratic terms themselves do.

Lemma 7.2.7: Taylor Bound for the Complex Exponential

For every real x and every integer $n \ge 0$,

$$\left| e^{ix} - \sum_{j=0}^{n} \frac{(ix)^j}{j!} \right| \le \min\left(\frac{|x|^{n+1}}{(n+1)!}, \ \frac{2|x|^n}{n!} \right)$$

In particular, for $n = 2$,

$$\left| e^{ix} - 1 - ix + \tfrac{1}{2}x^2 \right| \le \min\left(\tfrac{1}{6}|x|^3, \ x^2 \right)$$

***Proof*:**
Let $R_n(x) = e^{ix} - \sum_{j=0}^{n} (ix)^j / j!$ be the remainder. The exact integral form of Taylor's theorem, obtained by repeated integration by parts starting from $e^{ix} - 1 = i \int_0^x e^{is}\, ds$, gives

$$R_n(x) = \frac{i^{n+1}}{n!} \int_0^x (x-s)^n e^{is}\, ds$$

Bounding $|e^{is}| = 1$ and integrating the polynomial yields

$$|R_n(x)| \le \frac{1}{n!} \left| \int_0^x |x-s|^n\, ds \right| = \frac{|x|^{n+1}}{(n+1)!}$$

the first term of the minimum. For the second, integrate by parts once more: the identity $R_n(x) = R_{n-1}(x) - (ix)^n/n!$ together with the same integral representation for R_{n-1} gives

$$R_n(x) = \frac{i^n}{(n-1)!}\int_0^x (x-s)^{n-1}\big(e^{is}-1\big)\,ds$$

and $|e^{is}-1| \le 2$ bounds this by $\frac{2}{(n-1)!}\cdot\frac{|x|^n}{n} = \frac{2|x|^n}{n!}$, the second term of the minimum. Taking the smaller of the two bounds gives the general inequality, and specializing to $n=2$ gives $\frac{1}{6}|x|^3$ and $\frac{2}{2!}x^2 = x^2$, as we sought to show.

With both lemmas in hand the sufficiency proof is a matter of assembling the pieces. The strategy is to compare the characteristic function of S_n, which is the product $\prod_k \varphi_{n,k}(t)$ of the entry characteristic functions (definition 6.3.1), with the product $\prod_k e^{-\sigma_{n,k}^2 t^2/2}$ of Gaussian surrogates, which equals $e^{-s_n^2t^2/2} \to e^{-\sigma^2t^2/2}$, the characteristic function of $\mathcal{N}(0,\sigma^2)$. The product-comparison lemma reduces the difference of the two products to a sum of the differences $|\varphi_{n,k}(t) - e^{-\sigma_{n,k}^2t^2/2}|$, and the Taylor bound, split by the Lindeberg truncation, shows that this sum vanishes. Lévy's continuity theorem (theorem 6.4.2) then converts the convergence of characteristic functions into the asserted weak convergence.

***Proof of theorem 7.2.5(1), sufficiency*:**

Fix $t \in \mathbb{R}$; the target is $\varphi_{S_n}(t) \to e^{-\sigma^2t^2/2}$. Write $\varphi_{n,k}(t) = \mathbb{E}[e^{itX_{n,k}}]$ for the characteristic function of the (n,k) entry. By independence within the row and the product rule for characteristic functions of independent sums (proposition 6.3.4),

$$\varphi_{S_n}(t) = \prod_{k=1}^{k_n}\varphi_{n,k}(t)$$

Introduce the Gaussian surrogate factors $\psi_{n,k}(t) = e^{-\sigma_{n,k}^2t^2/2}$, each the characteristic function of a $\mathcal{N}(0,\sigma_{n,k}^2)$ variable, so that

$$\prod_{k=1}^{k_n}\psi_{n,k}(t) = \exp\left(-\tfrac{1}{2}t^2\sum_{k=1}^{k_n}\sigma_{n,k}^2\right) = e^{-s_n^2t^2/2} \xrightarrow[n\to\infty]{} e^{-\sigma^2t^2/2}$$

by the hypothesis $s_n^2 \to \sigma^2$. It therefore suffices to show that the two products have the same limit, that is,

$$\Delta_n(t) := \left|\prod_{k=1}^{k_n}\varphi_{n,k}(t) - \prod_{k=1}^{k_n}\psi_{n,k}(t)\right| \xrightarrow[n\to\infty]{} 0$$

Step 1: Reduce to a sum of factorwise differences. Both $\varphi_{n,k}(t)$ and $\psi_{n,k}(t)$ have modulus at most 1: the first because it is a characteristic function (proposition 6.3.4), the second because $\sigma_{n,k}^2t^2 \ge 0$. The product-comparison lemma (lemma 7.2.6) applies and gives

$$\Delta_n(t) \le \sum_{k=1}^{k_n}\big|\varphi_{n,k}(t) - \psi_{n,k}(t)\big|$$

The problem is now to bound each factorwise difference and sum the bounds.

Step 2: Bound each factor by its second-order deviation. Fix k and abbreviate $X = X_{n,k}$, $\sigma_k^2 = \sigma_{n,k}^2$. Since X is centered, $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = \sigma_k^2$, so the second-order Taylor expansion of e^{itX} has expectation

$$\mathbb{E}\big[1 + itX - \tfrac{1}{2}t^2X^2\big] = 1 - \tfrac{1}{2}\sigma_k^2 t^2$$

Applying the Taylor bound (lemma 7.2.7) with $x = tX$ inside the expectation, together with the modulus inequality for expectations,

$$\big|\varphi_{n,k}(t) - \big(1 - \tfrac{1}{2}\sigma_k^2 t^2\big)\big| = \big|\mathbb{E}[e^{itX} - 1 - itX + \tfrac{1}{2}t^2X^2]\big| \le \mathbb{E}\big[\min\big(\tfrac{1}{6}|t|^3|X|^3,\ t^2X^2\big)\big]$$

Separately, the Gaussian surrogate deviates from the same quadratic by a controlled amount: since $|e^{-u} - (1-u)| \le \frac{1}{2}u^2$ for $u \ge 0$, taking $u = \frac{1}{2}\sigma_k^2 t^2$ gives

$$\big|\psi_{n,k}(t) - \big(1 - \tfrac{1}{2}\sigma_k^2 t^2\big)\big| \le \tfrac{1}{2}\left(\tfrac{1}{2}\sigma_k^2 t^2\right)^2 = \tfrac{1}{8}\sigma_k^4 t^4$$

Combining the two by the triangle inequality,

$$\big|\varphi_{n,k}(t) - \psi_{n,k}(t)\big| \le \mathbb{E}\big[\min\big(\tfrac{1}{6}|t|^3|X|^3,\ t^2X^2\big)\big] + \tfrac{1}{8}t^4\sigma_k^4 \tag{7.5}$$

Step 3: Sum over the row and truncate at ε. Sum (7.5) over k. The second, quartic term is negligible: using the uniform bound $\sigma_k^2 \le \max_j \sigma_{n,j}^2$ on one factor,

$$\sum_{k=1}^{k_n} \tfrac{1}{8}t^4\sigma_k^4 \le \tfrac{1}{8}t^4\Big(\max_{1\le j\le k_n} \sigma_{n,j}^2\Big)\sum_{k=1}^{k_n}\sigma_k^2 = \tfrac{1}{8}t^4 s_n^2 \max_j \sigma_{n,j}^2$$

which tends to 0 because $s_n^2 \to \sigma^2$ is bounded and $\max_j \sigma_{n,j}^2 \to 0$ by (7.4), itself a consequence of the Lindeberg condition. For the main term, split each expectation at the threshold ε, using the small bound $\frac{1}{6}|t|^3|X|^3$ on $\{|X| \le \varepsilon\}$ and the large bound t^2X^2 on $\{|X| > \varepsilon\}$:

$$\mathbb{E}\big[\min\big(\tfrac{1}{6}|t|^3|X|^3,\ t^2X^2\big)\big] \le \tfrac{1}{6}|t|^3\,\mathbb{E}\big[|X|^3\mathbb{1}_{\{|X|\le\varepsilon\}}\big] + t^2\,\mathbb{E}\big[X^2\mathbb{1}_{\{|X|>\varepsilon\}}\big]$$

On $\{|X| \le \varepsilon\}$ one has $|X|^3 \le \varepsilon X^2$, so the first piece is at most $\frac{1}{6}|t|^3\varepsilon\,\mathbb{E}[X^2]$. Summing over k and using $\sum_k \mathbb{E}[X_{n,k}^2] = s_n^2$ and the definition of the Lindeberg function,

$$\sum_{k=1}^{k_n}\mathbb{E}\big[\min\big(\tfrac{1}{6}|t|^3|X_{n,k}|^3,\ t^2X_{n,k}^2\big)\big] \le \tfrac{1}{6}|t|^3\varepsilon\, s_n^2 + t^2L_n(\varepsilon)$$

Step 4: Send $n \to \infty$, then $\varepsilon \to 0$. Collecting the bounds from Step 3,

$$\Delta_n(t) \le \tfrac{1}{6}|t|^3\varepsilon\, s_n^2 + t^2L_n(\varepsilon) + \tfrac{1}{8}t^4 s_n^2 \max_j \sigma_{n,j}^2$$

Fix $\varepsilon > 0$ and let $n \to \infty$. The third term vanishes as shown, and $L_n(\varepsilon) \to 0$ by the Lindeberg condition, so

$$\limsup_{n\to\infty} \Delta_n(t) \le \tfrac{1}{6}|t|^3\varepsilon\,\sigma^2$$

using $s_n^2 \to \sigma^2$. The left side does not depend on ε, so letting $\varepsilon \to 0$ forces $\limsup_n \Delta_n(t) = 0$, hence $\Delta_n(t) \to 0$. Therefore

$$\varphi_{S_n}(t) = \prod_k \varphi_{n,k}(t) \xrightarrow[n\to\infty]{} e^{-\sigma^2t^2/2}$$

for every t. This is the characteristic function of $\mathcal{N}(0,\sigma^2)$ (definition 7.1.1), so by Lévy's continuity theorem (theorem 6.4.2) the row sums converge weakly, $S_n \Rightarrow \mathcal{N}(0,\sigma^2)$, completing the proof of sufficiency.

The main device of the proof is the truncation in Step 3, where the Lindeberg function appears exactly once, absorbing the entire contribution of the large entries. The small entries are handled by the crude bound $|X|^3 \le \varepsilon X^2$, which trades one power of X for the threshold ε and so contributes a term proportional to ε that can be made arbitrarily small; the large entries contribute $L_n(\varepsilon)$, which the hypothesis sends to zero. This is the mechanism by which controlling $\sum_k \mathbb{E}[\min(|X_{n,k}|^2, |X_{n,k}|^3)]$, the combined small-and-large moment, forces the Gaussian limit, and it is why the condition is stated in terms of the second moment of the large part rather than a third moment of everything.

The necessity direction, part (2), reverses this reasoning and is the reason Lindeberg's condition is called sharp rather than only sufficient. A full proof runs through the same characteristic-function comparison read backwards, and is recorded for completeness rather than for a new idea.

7.2.8 Key Ideas: The Feller Converse Suppose $S_n \Rightarrow \mathcal{N}(0,\sigma^2)$ and $\max_k \sigma_{n,k}^2 \to 0$. The negligibility hypothesis makes each factor $\varphi_{n,k}(t)$ uniformly close to 1 across the row, so that $\log \varphi_{n,k}(t)$ is well defined and $\sum_k (\varphi_{n,k}(t) - 1)$ has the same limit as $\log \varphi_{S_n}(t) \to -\sigma^2 t^2/2$. Taking real parts and using $1 - \mathrm{Re}\varphi_{n,k}(t) = \mathbb{E}[1 - \cos(tX_{n,k})]$ turns the statement into $\sum_k \mathbb{E}[1 - \cos(tX_{n,k})] \to \frac{1}{2}\sigma^2 t^2$ for every t. The elementary inequalities $1 - \cos u \ge \frac{u^2}{2} - \frac{u^4}{24}$ for the small part and $1 - \cos u \le 2$ for the large part, applied after integrating the identity over a small t-interval to average out the oscillation, isolate $\sum_k \mathbb{E}[X_{n,k}^2 \mathbb{1}_{\{|X_{n,k}|>\varepsilon\}}]$ and show it must vanish. The argument uses only the characteristic-function machinery already developed; the full computation is carried out in Durrett, *Probability: Theory and Examples*, and in Billingsley, *Probability and Measure*.

Together the two parts pin down the exact frontier of the central limit theorem for independent summands. Under the standing negligibility of individual terms, a sum of independent variables is asymptotically Gaussian if and only if the Lindeberg function vanishes; there is no gap between the sufficient and the necessary condition. The moral, which the next corollary and examples make operational, is that the central limit theorem is not really about identical distributions or even about moments beyond the second, but about the single requirement that the total variance be spread thinly across many summands, none of which dominates.

The Lindeberg function can be awkward to estimate directly, since it involves a truncated second moment for every ε. A more convenient sufficient condition, requiring only a single moment slightly above the second, is due to Lyapunov and is the form in which the theorem is most often applied.

Corollary 7.2.9: Lyapunov's Condition

Let $\{X_{n,k}\}$ be a centered triangular array with $s_n^2 \to \sigma^2 \in (0,\infty)$. If there exists $\delta > 0$ such that

$$\sum_{k=1}^{k_n} \mathbb{E}\left[|X_{n,k}|^{2+\delta}\right] \xrightarrow[n\to\infty]{} 0$$

then the array satisfies the Lindeberg condition, and hence $S_n \Rightarrow \mathcal{N}(0,\sigma^2)$.

***Proof*:**
Fix $\varepsilon > 0$. On the event $\{|X_{n,k}| > \varepsilon\}$ one has $1 < |X_{n,k}|^\delta/\varepsilon^\delta$, so $X_{n,k}^2 \mathbb{1}_{\{|X_{n,k}|>\varepsilon\}} \le \varepsilon^{-\delta}|X_{n,k}|^{2+\delta}$

pointwise. Taking expectations and summing over the row,

$$L_n(\varepsilon) = \sum_{k=1}^{k_n} \mathbb{E}\big[X_{n,k}^2 \mathbb{1}_{\{|X_{n,k}|>\varepsilon\}}\big] \le \frac{1}{\varepsilon^\delta} \sum_{k=1}^{k_n} \mathbb{E}\big[|X_{n,k}|^{2+\delta}\big]$$

The right side tends to 0 by hypothesis, and $\varepsilon > 0$ is arbitrary, so $L_n(\varepsilon) \to 0$ for every ε, which is the Lindeberg condition. The weak convergence then follows from theorem 7.2.5(1), completing the proof.

Lyapunov's condition trades the family of truncated second moments for a single absolute moment of order $2+\delta$, most commonly $\delta = 1$, giving the third absolute moment $\sum_k \mathbb{E}|X_{n,k}|^3 \to 0$. The bound is Markov's inequality (theorem 3.2.4) in disguise: raising the indicator threshold to a power δ is exactly the trick that estimates a tail second moment by a higher absolute moment. Whenever a uniform bound on third moments is available, and it usually is when the summands are bounded or have well-controlled tails, Lyapunov delivers the central limit theorem with no truncation argument at all.

The two examples below exhibit the theorem from both sides. The first is a weighted sum of independent variables in which Lyapunov's condition is straightforward to check and the Gaussian limit follows; it is the prototype for the number-theoretic application of the next sections and for any regression or averaging problem with non-uniform weights. The second is a cautionary array in which a single term retains a fixed share of the variance, negligibility fails, and the limit ceases to be universal, its shape recording the law of the persistent summand, showing that the hypotheses cannot be dropped.

Example 7.2.10: Lindeberg and Its Failure

1. *A weighted independent sum.* Let $Y_1, Y_2, \dots$ be i.i.d. with $\mathbb{E}[Y_k] = 0$, $\mathbb{E}[Y_k^2] = 1$, and finite third moment $\mathbb{E}|Y_k|^3 = \rho < \infty$ (for instance Y_k a centered bounded variable, or standard Gaussian). Let $(a_{n,k})$ be real weights, and set

$$X_{n,k} = a_{n,k} Y_k \qquad (1 \le k \le n)$$

so the row sum is the weighted sum $S_n = \sum_{k=1}^n a_{n,k} Y_k$. The entries of each row are independent, and centered since $\mathbb{E}[Y_k] = 0$, so this is a centered triangular array with $\sigma_{n,k}^2 = a_{n,k}^2$ and $s_n^2 = \sum_k a_{n,k}^2$. Assume the weights are normalized so that $s_n^2 \to \sigma^2 \in (0,\infty)$, and that they are individually negligible in the strong sense

$$\max_{1\le k\le n} |a_{n,k}| \xrightarrow[n\to\infty]{} 0$$

Lyapunov's condition with $\delta = 1$ holds: since $\mathbb{E}|X_{n,k}|^3 = |a_{n,k}|^3 \rho$,

$$\sum_{k=1}^n \mathbb{E}|X_{n,k}|^3 = \rho \sum_{k=1}^n |a_{n,k}|^3 \le \rho\Big(\max_j |a_{n,j}|\Big) \sum_{k=1}^n a_{n,k}^2 = \rho\, s_n^2 \max_j |a_{n,j}| \to 0$$

because $s_n^2 \to \sigma^2$ is bounded and $\max_j |a_{n,j}| \to 0$. By corollary 7.2.9 the weighted sum is asymptotically Gaussian,

$$\sum_{k=1}^n a_{n,k} Y_k \;\Rightarrow\; \mathcal{N}(0,\sigma^2)$$

The concrete case $a_{n,k} = 1/\sqrt{n}$ recovers the classical theorem (example 7.2.2), while non-uniform weights, for instance $a_{n,k} \propto k$ suitably normalized, are covered with no extra work as long as no single weight dominates.

2. *A single dominant term.* Now let the weights concentrate on one coordinate. Take $Y_1, Y_2, \dots$ i.i.d. $\mathcal{N}(0,1)$ and set

$$X_{n,1} = Y_1, \qquad X_{n,k} = \frac{Y_k}{\sqrt{n}} \quad (2 \le k \le n)$$

The row is independent, centered, and has variances $\sigma_{n,1}^2 = 1$ and $\sigma_{n,k}^2 = 1/n$ for $k \ge 2$, so

$$s_n^2 = 1 + \sum_{k=2}^{n} \frac{1}{n} = 1 + \frac{n-1}{n} \xrightarrow[n\to\infty]{} 2$$

giving the finite target $\sigma^2 = 2$. The first entry, however, keeps a fixed variance $\sigma_{n,1}^2 = 1$, so

$$\max_{1 \le k \le n} \sigma_{n,k}^2 = 1 \not\to 0$$

and uniform asymptotic negligibility (7.4) fails. The Lindeberg condition fails with it: for $0 < \varepsilon < 1$,

$$L_n(\varepsilon) \ge \mathbb{E}\big[Y_1^2 \mathbb{1}_{\{|Y_1| > \varepsilon\}}\big] =: c_\varepsilon > 0$$

a fixed positive constant that does not decay, since the offending first term never shrinks. And the limit is not standard for the reason the theorem predicts: the row sum is $S_n = Y_1 + n^{-1/2} \sum_{k=2}^{n} Y_k$, the sum of an independent $\mathcal{N}(0,1)$ and a term converging in distribution to $\mathcal{N}(0,1)$, so $S_n \Rightarrow \mathcal{N}(0,2)$ does hold here by direct computation, but the *shape* of the limit is dictated by the surviving first summand rather than being washed out into universality. Perturb Y_1 to any centered non-Gaussian variable Z with variance 1, keeping the tail $n^{-1/2} \sum_{k \ge 2} Y_k \Rightarrow \mathcal{N}(0,1)$, and the limit becomes the convolution $Z * \mathcal{N}(0,1)$, which is not Gaussian. This is the failure that negligibility exists to exclude: when one term carries a nonvanishing share of the variance, its own law imprints on the limit and the Gaussian universality of the central limit theorem is lost.

The contrast between the two arrays is the whole content of the section in miniature. In the first, the variance is spread across many comparable summands, each individually negligible, and the limit is Gaussian regardless of the common law of the Y_k; the Gaussian shape is emergent, a property of the summation rather than of any summand. In the second, one summand refuses to become negligible, and its law survives into the limit; the central limit theorem does not apply, not because the sum fails to converge, but because what it converges to is no longer universal. Lindeberg's condition is the exact dividing line between these two regimes, and Feller's converse guarantees that there is no third possibility hiding between them: under negligibility, either the Lindeberg function vanishes and the limit is Gaussian, or it does not and the limit records the persistent summand.

Exercise 7.2.1

1. Let $p_1, p_2, \dots \in (0,1)$ and let $B_k \sim \mathrm{Ber}(p_k)$ be independent, so B_k has mean p_k and variance $p_k(1-p_k)$. Set $X_{n,k} = (B_k - p_k)/s_n$ for $1 \le k \le n$, where $s_n^2 = \sum_{k=1}^{n} p_k(1-p_k)$. Show that if $s_n^2 \to \infty$ then this centered array satisfies the Lindeberg condition, and conclude that

$\left(\sum_{k=1}^{n}(B_k - p_k)\right)/s_n \Rightarrow \mathcal{N}(0,1)$. (This is the central limit theorem for sums of independent indicators with varying success probabilities.)

2. Suppose a centered triangular array is *uniformly bounded*: there is a constant M with $|X_{n,k}| \le M/\sqrt{n}$ for all $k \le n$, and $s_n^2 \to \sigma^2 \in (0,\infty)$. Show that the Lyapunov condition holds with $\delta = 1$, and hence $S_n \Rightarrow \mathcal{N}(0,\sigma^2)$.

3. In example 7.2.10(1) take $Y_k \sim \mathcal{N}(0,1)$ and weights $a_{n,k} = c_n k$ for $1 \le k \le n$, where c_n is chosen so that $s_n^2 = c_n^2 \sum_{k=1}^{n} k^2 = 1$. Compute c_n asymptotically, verify that $\max_k |a_{n,k}| \to 0$, and conclude that $\sum_{k=1}^{n} a_{n,k} Y_k \Rightarrow \mathcal{N}(0,1)$.

4. Show by example that uniform asymptotic negligibility (7.4) is strictly weaker than the Lindeberg condition: construct a centered triangular array with $s_n^2 \to \sigma^2 \in (0,\infty)$ and $\max_k \sigma_{n,k}^2 \to 0$ for which the Lindeberg condition nonetheless fails. (Hint: let each row consist of a growing number of large but individually negligible entries whose combined truncated second moment above a fixed level does not vanish.)

5. Give a centered triangular array that satisfies the Lindeberg condition but for which the Lyapunov condition fails for every $\delta > 0$. Conclude that Lindeberg is strictly more general than Lyapunov. (Hint: in each row, let one entry have a heavy enough tail that all its absolute moments of order > 2 blow up, while its truncated second moment above any fixed level still vanishes as $n \to \infty$.)

7.3 Gaussian Vectors and the Multivariate CLT

The classical central limit theorem (theorem 7.1.3) describes the fluctuation of a single scalar average. But the objects of interest in practice are rarely one number at a time. A statistician estimates a mean and a variance together; a physicist tracks several correlated observables; a portfolio has many assets whose returns fluctuate jointly. In each case the natural question is not how one average fluctuates but how a whole vector of averages fluctuates, and whether the components move independently or in concert. This section answers that question. The limit object is the multivariate normal law, a Gaussian on $\mathbb{R}^d$ whose shape encodes not only the spread of each coordinate but every pairwise correlation, and the theorem that produces it, the multivariate central limit theorem, follows from the one-dimensional case by a device that reduces all of $\mathbb{R}^d$ to its scalar projections: the Cramér-Wold device.

The strategy is worth stating at the outset, because it recurs throughout multivariate probability. A random vector in $\mathbb{R}^d$ is a complicated object, but its one-dimensional projections $\langle \theta, X \rangle$, taken over all directions θ, are scalar random variables to which the entire one-dimensional theory already applies. If those projections determine the vector (they do, via characteristic functions), then a statement about the vector can be proved one direction at a time. The multivariate normal is defined so that all of its projections are ordinary normals; the Cramér-Wold device says convergence of all projections is convergence of the vector; and the multivariate central limit theorem is then just the scalar theorem applied in every direction at once.

Throughout, $\langle x, y \rangle = \sum_{i=1}^{d} x_i y_i$ is the standard inner product on $\mathbb{R}^d$, vectors are columns, and for a random vector $X = (X_1, \dots, X_d)$ with integrable coordinates the mean vector is $\mu = \mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_d])$ and, when the coordinates are square-integrable, the covariance matrix is the

$d \times d$ matrix

$$\Sigma = \mathrm{Cov}(X) = \mathbb{E}\big[(X-\mu)(X-\mu)^{\mathsf{T}}\big], \qquad \Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$$

The diagonal entries $\Sigma_{ii} = \mathrm{Var}(X_i)$ record the individual variances (definition 3.2.1) and the off-diagonal entries record the correlations. Two algebraic facts about Σ are used constantly. First, Σ is symmetric, since $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$. Second, Σ is positive semidefinite: for any $\theta \in \mathbb{R}^d$,

$$\langle \theta, \Sigma\theta \rangle = \mathbb{E}\big[\langle \theta, X-\mu\rangle^2\big] = \mathrm{Var}\big(\langle \theta, X\rangle\big) \geq 0$$

because it is the variance of the scalar random variable $\langle \theta, X\rangle$. This computation also names the quantity that will be the variance in every projected central limit theorem below: the variance of the linear combination $\langle \theta, X\rangle$ is exactly $\langle \theta, \Sigma\theta\rangle$.

Before defining the multivariate normal, one point of motivation. The one-dimensional normal was pinned down (definition 7.1.1) by two parameters, a mean and a variance, and its characteristic function $e^{i\mu t - \sigma^2 t^2/2}$ displayed them cleanly. The multivariate object should be pinned down by a mean vector and the full covariance matrix, and the cleanest route to it, both for the definition and for every proof, is again the characteristic function. The characteristic function of a random vector X is the function $\varphi_X \colon \mathbb{R}^d \to \mathbb{C}$ given by $\varphi_X(t) = \mathbb{E}[e^{i\langle t, X\rangle}]$, the direct analogue of the scalar case (definition 6.3.1); as there, it determines the law and it linearizes independence. Defining the multivariate normal by its characteristic function makes the parameters manifest and makes the projection property, the technical engine of the section, a one-line computation.

Definition 7.3.1: Multivariate Normal Distribution

Let $\mu \in \mathbb{R}^d$ and let Σ be a symmetric positive semidefinite $d \times d$ matrix. A random vector X taking values in $\mathbb{R}^d$ has the *multivariate normal* (or *Gaussian*) *law* $\mathcal{N}(\mu, \Sigma)$ if its characteristic function is

$$\varphi_X(t) = \exp\big(i\langle \mu, t\rangle - \tfrac{1}{2}\langle t, \Sigma t\rangle\big), \qquad t \in \mathbb{R}^d$$

A random vector is called *Gaussian* if its law is $\mathcal{N}(\mu, \Sigma)$ for some such μ and Σ.

That this formula does define a genuine probability law, for every symmetric positive semidefinite Σ, is part of the content and is verified below by exhibiting $\mathcal{N}(\mu, \Sigma)$ as an explicit linear image of a vector of independent standard normals. The definition through the characteristic function is preferred because it covers the degenerate case where Σ is singular on the same footing as the nondegenerate case, and a singular covariance is not a pathology to be excluded: it is exactly what appears when the coordinates of X satisfy a linear relation, as they do, for instance, when the last coordinate is the sum of the others. A definition through a density would have to treat that case separately, since a law supported on a proper subspace of $\mathbb{R}^d$ has no density with respect to Lebesgue measure on $\mathbb{R}^d$.

The single most useful feature of the definition is that the class of Gaussian vectors is characterized by its projections, and this is what makes the Cramér-Wold reduction possible. The following proposition records the equivalence and, in passing, computes the law of every linear image of a Gaussian vector.

Proposition 7.3.2: Linear Combinations of a Gaussian Vector

Let X be a random vector in $\mathbb{R}^d$. The following are equivalent.

1. $X \sim \mathcal{N}(\mu, \Sigma)$ for some $\mu \in \mathbb{R}^d$ and symmetric positive semidefinite Σ.
2. For every $\theta \in \mathbb{R}^d$, the scalar random variable $\langle \theta, X \rangle$ has a univariate normal law.

In that case $\langle \theta, X \rangle \sim \mathcal{N}\big(\langle \theta, \mu \rangle, \langle \theta, \Sigma\theta \rangle\big)$ for every θ, and more generally $AX + b \sim \mathcal{N}(A\mu + b,\ A\Sigma A^\mathsf{T})$ for every $m \times d$ matrix A and $b \in \mathbb{R}^m$.

***Proof*:**
From (i) to the projection formula. Suppose $X \sim \mathcal{N}(\mu, \Sigma)$. Fix $\theta \in \mathbb{R}^d$ and let $Y = \langle \theta, X \rangle$. The characteristic function of the scalar Y, evaluated at $s \in \mathbb{R}$, is

$$\varphi_Y(s) = \mathbb{E}\big[e^{isY}\big] = \mathbb{E}\big[e^{i\langle s\theta, X\rangle}\big] = \varphi_X(s\theta) = \exp\big(i\langle \mu, s\theta\rangle - \tfrac{1}{2}\langle s\theta, \Sigma\, s\theta\rangle\big)$$

Pulling the scalar s out of each bracket, $\langle \mu, s\theta\rangle = s\langle \theta, \mu\rangle$ and $\langle s\theta, \Sigma\, s\theta\rangle = s^2\langle \theta, \Sigma\theta\rangle$, so

$$\varphi_Y(s) = \exp\big(is\langle \theta, \mu\rangle - \tfrac{1}{2}s^2\langle \theta, \Sigma\theta\rangle\big)$$

This is the characteristic function of the univariate normal law $\mathcal{N}\big(\langle \theta, \mu\rangle, \langle \theta, \Sigma\theta\rangle\big)$ (definition 7.1.1), so by uniqueness of the characteristic function (corollary 6.3.6) Y has that law. In particular $\langle \theta, X\rangle$ is univariate normal for every θ, which is (ii).

From (ii) to (i). Suppose every $\langle \theta, X\rangle$ is univariate normal. Each coordinate $X_i = \langle e_i, X\rangle$ is then in particular integrable and square-integrable (a univariate normal has moments of all orders), so the mean vector $\mu = \mathbb{E}[X]$ and covariance matrix $\Sigma = \mathrm{Cov}(X)$ exist; Σ is symmetric positive semidefinite as noted above. Fix $\theta \in \mathbb{R}^d$. The scalar $\langle \theta, X\rangle$ is normal by hypothesis, and by linearity of expectation (proposition 3.1.3) its mean is $\mathbb{E}[\langle \theta, X\rangle] = \langle \theta, \mu\rangle$ and its variance is $\mathrm{Var}(\langle \theta, X\rangle) = \langle \theta, \Sigma\theta\rangle$ by the computation opening this section. Hence $\langle \theta, X\rangle \sim \mathcal{N}(\langle \theta, \mu\rangle, \langle \theta, \Sigma\theta\rangle)$, and its scalar characteristic function at $s = 1$ is

$$\mathbb{E}\big[e^{i\langle \theta, X\rangle}\big] = \exp\big(i\langle \theta, \mu\rangle - \tfrac{1}{2}\langle \theta, \Sigma\theta\rangle\big)$$

The left side is precisely $\varphi_X(\theta)$, and since θ was arbitrary this identifies φ_X with the characteristic function of $\mathcal{N}(\mu, \Sigma)$. Thus $X \sim \mathcal{N}(\mu, \Sigma)$, which is (i).

The affine image. For the last claim, let A be $m \times d$ and $b \in \mathbb{R}^m$, and put $Z = AX + b$. For $t \in \mathbb{R}^m$,

$$\varphi_Z(t) = \mathbb{E}\big[e^{i\langle t, AX+b\rangle}\big] = e^{i\langle t, b\rangle}\,\mathbb{E}\big[e^{i\langle A^\mathsf{T}t, X\rangle}\big] = e^{i\langle t, b\rangle}\,\varphi_X(A^\mathsf{T}t)$$

using $\langle t, AX\rangle = \langle A^\mathsf{T}t, X\rangle$. Substituting the Gaussian characteristic function and collecting terms,

$$\varphi_Z(t) = \exp\big(i\langle t, b\rangle + i\langle \mu, A^\mathsf{T}t\rangle - \tfrac{1}{2}\langle A^\mathsf{T}t, \Sigma A^\mathsf{T}t\rangle\big) = \exp\big(i\langle A\mu + b, t\rangle - \tfrac{1}{2}\langle t, A\Sigma A^\mathsf{T}t\rangle\big)$$

the second equality by $\langle \mu, A^\mathsf{T}t\rangle = \langle A\mu, t\rangle$ and $\langle A^\mathsf{T}t, \Sigma A^\mathsf{T}t\rangle = \langle t, A\Sigma A^\mathsf{T}t\rangle$. This is the characteristic function of $\mathcal{N}(A\mu + b, A\Sigma A^\mathsf{T})$, completing the proof.

The equivalence in proposition 7.3.2 is the working definition of a Gaussian vector that most probabilists carry in their heads: a vector is Gaussian precisely when every linear combination of its coordinates is a scalar Gaussian. This is more than a convenient reformulation. It shows that Gaussianity of a vector is not a statement about each coordinate separately (each X_i being normal is far weaker) but a joint condition on all directions at once, and it is the reason the class is closed under every linear map: an affine image of a Gaussian is Gaussian, with mean and covariance transforming by the visible rules. The affine-image formula $A\Sigma A^\mathsf{T}$ is the multivariate analogue of the scalar rule that scaling by a multiplies the variance by a^2.

The definition through characteristic functions leaves open the concrete question of what $\mathcal{N}(\mu, \Sigma)$ looks like as a distribution on $\mathbb{R}^d$. When Σ is nonsingular the law has a density, and it is the exact multivariate generalization of the bell curve.

Proposition 7.3.3: The Gaussian Density

Let $\mu \in \mathbb{R}^d$ and let Σ be symmetric positive definite (hence nonsingular). Then $\mathcal{N}(\mu, \Sigma)$ is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^d$, with density

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{d/2}\sqrt{\det \Sigma}} \exp\bigl(-\tfrac{1}{2}\langle x-\mu, \Sigma^{-1}(x-\mu)\rangle\bigr), \qquad x \in \mathbb{R}^d$$

When Σ is singular the law is supported on the affine subspace $\mu + \operatorname{range}(\Sigma)$ and has no density on $\mathbb{R}^d$.

***Proof*:**
Since Σ is symmetric positive definite, the spectral theorem furnishes an orthogonal matrix Q and a diagonal matrix $D = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$ with all $\lambda_i > 0$ such that $\Sigma = QDQ^\mathsf{T}$. Write $\Sigma^{1/2} = QD^{1/2}Q^\mathsf{T}$, a symmetric positive definite square root with $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$. Let $G = (G_1, \dots, G_d)$ be a vector of independent standard normal variables; the characteristic function of G factors as $\varphi_G(t) = \prod_{k=1}^d e^{-t_k^2/2} = e^{-\langle t,t\rangle/2}$ by independence (theorem 2.3.1), so $G \sim \mathcal{N}(0, I)$. Set $X = \mu + \Sigma^{1/2}G$. By the affine-image formula of proposition 7.3.2,

$$X \sim \mathcal{N}\bigl(\mu,\ \Sigma^{1/2}I(\Sigma^{1/2})^\mathsf{T}\bigr) = \mathcal{N}(\mu, \Sigma)$$

so X realizes the law in question, and it remains to compute its density by the change-of-variables formula for the invertible affine map $g \mapsto \mu + \Sigma^{1/2}g$. The vector G has density $\prod_k (2\pi)^{-1/2}e^{-g_k^2/2} = (2\pi)^{-d/2}e^{-\langle g,g\rangle/2}$ on $\mathbb{R}^d$. The map $x = \mu + \Sigma^{1/2}g$ has inverse $g = \Sigma^{-1/2}(x-\mu)$ and Jacobian determinant $\det \Sigma^{1/2} = \sqrt{\det \Sigma}$, so the density of X at x is

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{d/2}\sqrt{\det \Sigma}} \exp\Bigl(-\tfrac{1}{2}\bigl\langle \Sigma^{-1/2}(x-\mu), \Sigma^{-1/2}(x-\mu)\bigr\rangle\Bigr)$$

Since $\Sigma^{-1/2}$ is symmetric, $\langle \Sigma^{-1/2}(x-\mu), \Sigma^{-1/2}(x-\mu)\rangle = \langle x-\mu, \Sigma^{-1}(x-\mu)\rangle$, giving the stated density.

When Σ is singular, let $r = \operatorname{rank}\Sigma < d$ and choose a unit vector θ in the kernel of Σ. Then $\operatorname{Var}(\langle \theta, X-\mu\rangle) = \langle \theta, \Sigma\theta\rangle = 0$, so the scalar $\langle \theta, X\rangle$ equals its mean $\langle \theta, \mu\rangle$ almost surely; X therefore lies almost surely in the hyperplane $\{x\colon \langle \theta, x\rangle = \langle \theta, \mu\rangle\}$, and intersecting over a basis of $\ker \Sigma$ confines X to the affine subspace $\mu + \operatorname{range}(\Sigma)$, which has Lebesgue measure zero in $\mathbb{R}^d$. A law giving full mass to a Lebesgue-null set has no density on $\mathbb{R}^d$, as we sought to show.

The construction $X = \mu + \Sigma^{1/2}G$ in the proof is worth isolating, because it gives the concrete picture behind the definition: every Gaussian vector is an affine image of a vector of independent standard normals. The matrix $\Sigma^{1/2}$ stretches the round, uncorrelated cloud of the standard Gaussian along the eigendirections of Σ by the factors $\sqrt{\lambda_i}$, and the shift by μ centers it. Correlation between coordinates is nothing but a rotation of this stretched cloud away from the coordinate axes. When one of the λ_i is zero, the cloud is flattened completely in that direction, which is the geometric meaning of a singular covariance and of the degenerate case handled above. The density, where it exists, has level sets $\{x\colon \langle x-\mu, \Sigma^{-1}(x-\mu)\rangle = c\}$ that are ellipsoids centered at μ with axes along the eigenvectors of Σ, the contours of constant probability.

Example 7.3.4: The Bivariate Normal
Take $d = 2$, mean $\mu = 0$, and covariance

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \qquad \sigma_1, \sigma_2 > 0, \quad \rho \in [-1, 1]$$

the covariance of a pair (X_1, X_2) with variances σ_1^2, σ_2^2 and correlation coefficient $\rho = \operatorname{Cov}(X_1, X_2)/(\sigma_1\sigma_2)$. The matrix Σ is positive semidefinite exactly when $\det\Sigma = \sigma_1^2\sigma_2^2(1-\rho^2) \geq 0$, that is when $|\rho| \leq 1$, and positive definite when $|\rho| < 1$. For $|\rho| < 1$, $\det\Sigma = \sigma_1^2\sigma_2^2(1-\rho^2)$ and

$$\Sigma^{-1} = \frac{1}{1-\rho^2}\begin{pmatrix} 1/\sigma_1^2 & -\rho/(\sigma_1\sigma_2) \\ -\rho/(\sigma_1\sigma_2) & 1/\sigma_2^2 \end{pmatrix}$$

so the density of proposition 7.3.3 is the familiar

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{x_1^2}{\sigma_1^2} - \frac{2\rho x_1 x_2}{\sigma_1\sigma_2} + \frac{x_2^2}{\sigma_2^2}\right]\right)$$

Two features are visible in this formula. First, setting $\rho = 0$ makes the exponent split as $-x_1^2/(2\sigma_1^2) - x_2^2/(2\sigma_2^2)$ and the density factors as a product $f(x_1, x_2) = f_1(x_1)f_2(x_2)$ of two univariate normal densities; for jointly Gaussian coordinates, zero correlation is therefore the same as independence (theorem 2.3.1). This equivalence is special to the Gaussian and fails for general random variables. Second, as $\rho \to \pm 1$ the constant $\det\Sigma \to 0$, the ellipsoidal level sets collapse onto the line $x_2 = \pm(\sigma_2/\sigma_1)x_1$, and the density degenerates: the boundary correlation $\rho = \pm 1$ forces $X_2 = \pm(\sigma_2/\sigma_1)X_1$ almost surely, a singular Gaussian with no density, exactly the degenerate case of proposition 7.3.3.

The projection property built into proposition 7.3.2 suggests a general principle: if projections of a Gaussian vector determine it, perhaps projections of any vector determine its distribution, and, more usefully, its convergence in distribution. This is the Cramér-Wold device, and it is what turns the one-dimensional central limit theorem into the d-dimensional one. First the mode of convergence must be fixed. A sequence of random vectors X_n in $\mathbb{R}^d$ *converges in distribution* to X, written $X_n \Rightarrow X$, if $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for every bounded continuous $f\colon \mathbb{R}^d \to \mathbb{R}$, the exact transcription of the scalar definition (definition 6.1.1) with $\mathbb{R}$ replaced by $\mathbb{R}^d$. As on the line, Lévy's continuity theorem holds in this setting: $X_n \Rightarrow X$ if and only if $\varphi_{X_n}(t) \to \varphi_X(t)$ for every $t \in \mathbb{R}^d$, with the same proof, the multivariate inversion and tail estimates replacing their scalar versions (theorem 6.4.2). The Cramér-Wold device reduces this vector criterion to the scalar one already in hand.

Theorem 7.3.5: The Cramér-Wold Device

Let X_n and X be random vectors in $\mathbb{R}^d$. Then $X_n \Rightarrow X$ in $\mathbb{R}^d$ if and only if

$$\langle \theta, X_n \rangle \Rightarrow \langle \theta, X \rangle \quad \text{in } \mathbb{R}, \qquad \text{for every } \theta \in \mathbb{R}^d$$

***Proof*:**
The whole argument rests on one identity relating the characteristic function of a vector to the characteristic functions of its scalar projections: for any random vector Y in $\mathbb{R}^d$ and any $\theta \in \mathbb{R}^d$, the characteristic function of the scalar $\langle \theta, Y \rangle$, evaluated at $s \in \mathbb{R}$, is

$$\varphi_{\langle \theta, Y \rangle}(s) = \mathbb{E}\big[e^{is\langle \theta, Y \rangle}\big] = \mathbb{E}\big[e^{i\langle s\theta, Y \rangle}\big] = \varphi_Y(s\theta) \tag{7.6}$$

In particular, taking $s = 1$, the scalar characteristic function of the projection in direction θ, evaluated at 1, equals the vector characteristic function φ_Y evaluated at the point θ. This is the bridge across which the two Lévy continuity theorems, scalar and vector, are made to talk to each other.

Necessity. Suppose $X_n \Rightarrow X$ in $\mathbb{R}^d$. Fix $\theta \in \mathbb{R}^d$ and let $g_\theta \colon \mathbb{R}^d \to \mathbb{R}$ be the linear map $g_\theta(x) = \langle \theta, x \rangle$, which is continuous. For any bounded continuous $h \colon \mathbb{R} \to \mathbb{R}$ the composite $h \circ g_\theta \colon \mathbb{R}^d \to \mathbb{R}$ is bounded and continuous, so weak convergence of the vectors gives

$$\mathbb{E}\big[h(\langle \theta, X_n \rangle)\big] = \mathbb{E}\big[(h \circ g_\theta)(X_n)\big] \longrightarrow \mathbb{E}\big[(h \circ g_\theta)(X)\big] = \mathbb{E}\big[h(\langle \theta, X \rangle)\big]$$

Since h was an arbitrary bounded continuous function on $\mathbb{R}$, this is exactly the statement $\langle \theta, X_n \rangle \Rightarrow \langle \theta, X \rangle$. (This direction is the continuous-mapping principle: a continuous image of a weakly convergent sequence converges weakly.)

Sufficiency. Suppose $\langle \theta, X_n \rangle \Rightarrow \langle \theta, X \rangle$ in $\mathbb{R}$ for every $\theta \in \mathbb{R}^d$. Fix a point $t \in \mathbb{R}^d$; the goal is $\varphi_{X_n}(t) \to \varphi_X(t)$, after which the vector Lévy continuity theorem finishes the argument. Apply the scalar direction of Lévy's continuity theorem (theorem 6.4.2, part (i)) to the convergence $\langle t, X_n \rangle \Rightarrow \langle t, X \rangle$ in the single direction $\theta = t$: it yields

$$\varphi_{\langle t, X_n \rangle}(s) \longrightarrow \varphi_{\langle t, X \rangle}(s) \qquad \text{for every } s \in \mathbb{R}$$

Evaluate this scalar convergence at the single value $s = 1$ and invoke the identity (7.6) on both sides, with $Y = X_n$ on the left and $Y = X$ on the right:

$$\varphi_{X_n}(t) = \varphi_{\langle t, X_n \rangle}(1) \longrightarrow \varphi_{\langle t, X \rangle}(1) = \varphi_X(t)$$

Since $t \in \mathbb{R}^d$ was arbitrary, $\varphi_{X_n} \to \varphi_X$ pointwise on $\mathbb{R}^d$. The limit φ_X is a characteristic function, hence continuous at the origin, so the vector form of Lévy's continuity theorem (theorem 6.4.2) gives $X_n \Rightarrow X$, completing the proof.

The device earns its name by trading a hard problem for an easy one. Deciding weak convergence in $\mathbb{R}^d$ directly means controlling $\mathbb{E}[f(X_n)]$ for every bounded continuous f on $\mathbb{R}^d$, a large and unwieldy class of test functions. The Cramér-Wold device replaces this with the requirement that a single family of scalar sequences, the projections in each direction θ, converge, and each of those is a one-dimensional problem to which the entire scalar theory applies. The one point that must not be overlooked is the quantifier: it is not enough that finitely many projections converge, nor that the coordinate projections $\langle e_i, X_n \rangle = X_{n,i}$ converge individually. Convergence of the coordinate marginals says nothing about the joint law, whereas convergence of *all* projections determines it. This is the vector analogue of the warning in example 7.3.4 that Gaussianity of each coordinate is weaker than Gaussianity of the vector.

With the device in hand the multivariate central limit theorem is immediate. The hypotheses are the exact vector translation of the scalar theorem: independent identically distributed summands, now vector-valued, with a finite mean vector and a finite covariance matrix. The conclusion is that the rescaled deviation of the sample mean from the true mean is asymptotically Gaussian, with the limiting covariance being precisely the common covariance of the summands.

Theorem 7.3.6: The Multivariate Central Limit Theorem

Let $X_1, X_2, \dots$ be independent and identically distributed random vectors in $\mathbb{R}^d$ with common mean vector $\mu = \mathbb{E}[X_1] \in \mathbb{R}^d$ and finite covariance matrix $\Sigma = \mathrm{Cov}(X_1)$. Write $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ for the sample mean. Then

$$\sqrt{n}\,(\bar{X}_n - \mu) \implies \mathcal{N}(0, \Sigma) \qquad \text{in } \mathbb{R}^d$$

***Proof*:**
The strategy is to project onto an arbitrary direction θ, recognize the projection as a scalar i.i.d. sum, apply the classical central limit theorem there, and reassemble by the Cramér-Wold device.

Step 1: Reduce to one dimension. Fix $\theta \in \mathbb{R}^d$ and set $V_n = \sqrt{n}\,(\bar{X}_n - \mu)$. The scalar projection is

$$\langle \theta, V_n \rangle = \sqrt{n}\,\langle \theta, \bar{X}_n - \mu \rangle = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \langle \theta, X_i \rangle - \langle \theta, \mu \rangle \right)$$

Introduce the scalar random variables $Y_i = \langle \theta, X_i \rangle$. Because the vectors $X_1, X_2, \dots$ are i.i.d. and Y_i is a fixed measurable function (a fixed linear functional) of X_i, the sequence $Y_1, Y_2, \dots$ is itself i.i.d. (theorem 2.3.1). In this notation, with $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^n Y_i$,

$$\langle \theta, V_n \rangle = \sqrt{n}\,(\bar{Y}_n - \nu), \qquad \nu = \mathbb{E}[Y_1] = \langle \theta, \mu \rangle$$

so the projected quantity is exactly the standardized sample mean of the scalar i.i.d. sequence (Y_i).

Step 2: Compute the scalar mean and variance. By linearity of expectation (proposition 3.1.3), $\mathbb{E}[Y_1] = \langle \theta, \mu \rangle = \nu$. The variance is

$$\tau^2 := \mathrm{Var}(Y_1) = \mathrm{Var}\big(\langle \theta, X_1 \rangle\big) = \langle \theta, \Sigma\theta \rangle$$

by the identity opening this section, which expresses the variance of a projection through the covariance matrix. Both are finite because Σ is finite.

Step 3: Apply the classical CLT and reassemble. If $\tau^2 > 0$, the classical central limit theorem

(theorem 7.1.3) applied to the i.i.d. scalar sequence (Y_i) gives

$$\langle \theta, V_n \rangle = \sqrt{n}\,(\bar{Y}_n - \nu) \implies \mathcal{N}(0, \tau^2) = \mathcal{N}\big(0, \langle \theta, \Sigma\theta \rangle\big)$$

If instead $\tau^2 = \langle \theta, \Sigma\theta \rangle = 0$, then $Y_i = \langle \theta, X_i \rangle = \nu$ almost surely (a variable of variance zero is almost surely its mean), so $\langle \theta, V_n \rangle = \sqrt{n}\,(\bar{Y}_n - \nu) = 0$ almost surely for every n, and hence $\langle \theta, V_n \rangle \Rightarrow 0$, which is convergence to $\mathcal{N}(0,0)$, the point mass at 0. In both cases the limit is $\mathcal{N}(0, \langle \theta, \Sigma\theta \rangle)$.

Now let $Z \sim \mathcal{N}(0, \Sigma)$. Its projection in direction θ is $\langle \theta, Z \rangle \sim \mathcal{N}(0, \langle \theta, \Sigma\theta \rangle)$ by proposition 7.3.2, which is exactly the limit just identified for $\langle \theta, V_n \rangle$. Therefore

$$\langle \theta, V_n \rangle \implies \langle \theta, Z \rangle \qquad \text{for every } \theta \in \mathbb{R}^d$$

By the Cramér-Wold device (theorem 7.3.5), $V_n \Rightarrow Z$, that is $\sqrt{n}\,(\bar{X}_n - \mu) \Rightarrow \mathcal{N}(0, \Sigma)$, completing the proof.

The theorem says that the joint fluctuations of several averages are asymptotically Gaussian, and it identifies the limiting shape precisely: the covariance matrix of the limit is the covariance matrix of a single summand, unchanged. The role of $\sqrt{n}$ is the same as in one dimension, the rescaling that keeps the fluctuation from either vanishing or exploding, and the role of Σ is to record how the coordinate fluctuations are coupled. If Σ is diagonal the coordinates of the limit are independent (example 7.3.4), and the multivariate theorem reduces to d separate scalar theorems; the content beyond the scalar case is entirely in the off-diagonal entries, which are present exactly when the summand coordinates are correlated. The proof also makes plain why no new idea is needed: the Cramér-Wold device converts a d-dimensional limit theorem into a one-parameter family of scalar limit theorems, and the classical theorem, applied in every direction, supplies each of them.

The covariance matrix deserves a last word of interpretation, because it is the one piece of data the multivariate theorem contributes beyond the scalar case, and reading it correctly is the whole skill of applying the theorem. The limit $\mathcal{N}(0, \Sigma)$ is a cloud of probability centered at the origin, and Σ is its shape. Its eigenvectors are the principal axes of the cloud, the directions along which the fluctuations are uncorrelated, and its eigenvalues are the variances along those axes, the squared lengths of the corresponding semi-axes of the one-standard-deviation ellipsoid. A large eigenvalue is a direction of large joint fluctuation; a small one, a direction in which the averages move almost in lockstep; a zero eigenvalue, an exact linear constraint among the limiting fluctuations that holds with probability one. Correlation between two coordinates tilts the ellipsoid away from the coordinate axes, and the sign of the correlation is the direction of the tilt. All of the geometry of the limit, and all of the statistical content of knowing how the averages co-vary, is packaged in this single matrix.

Example 7.3.7: Joint Limit of the Sample Mean and Sample Second Moment

A single scalar sample produces two natural statistics at once, its average and its average square, and the multivariate central limit theorem describes their joint fluctuation. Let $\xi_1, \xi_2, \dots$ be i.i.d. real random variables with a finite fourth moment, and consider the $\mathbb{R}^2$-valued summands

$$X_i = \begin{pmatrix} \xi_i \\ \xi_i^2 \end{pmatrix}$$

which are i.i.d. vectors because the ξ_i are (theorem 2.3.1). Write $m_k = \mathbb{E}[\xi_1^k]$ for the k-th moment, finite for $k \le 4$ by hypothesis. The sample mean of the vectors is

$$\bar{X}_n = \begin{pmatrix} \frac{1}{n}\sum_i \xi_i \\ \frac{1}{n}\sum_i \xi_i^2 \end{pmatrix}, \qquad \mu = \mathbb{E}[X_1] = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$$

the first coordinate the sample mean and the second the sample second moment. The covariance matrix $\Sigma = \mathrm{Cov}(X_1)$ has entries computed from the moments:

$$\Sigma = \begin{pmatrix} \mathrm{Var}(\xi_1) & \mathrm{Cov}(\xi_1, \xi_1^2) \\ \mathrm{Cov}(\xi_1^2, \xi_1) & \mathrm{Var}(\xi_1^2) \end{pmatrix} = \begin{pmatrix} m_2 - m_1^2 & m_3 - m_1 m_2 \\ m_3 - m_1 m_2 & m_4 - m_2^2 \end{pmatrix}$$

using $\mathrm{Cov}(\xi_1, \xi_1^2) = \mathbb{E}[\xi_1^3] - \mathbb{E}[\xi_1]\mathbb{E}[\xi_1^2] = m_3 - m_1 m_2$ and $\mathrm{Var}(\xi_1^2) = \mathbb{E}[\xi_1^4] - \mathbb{E}[\xi_1^2]^2 = m_4 - m_2^2$. By theorem 7.3.6,

$$\sqrt{n}\,(\bar{X}_n - \mu) = \sqrt{n}\begin{pmatrix} \frac{1}{n}\sum_i \xi_i - m_1 \\ \frac{1}{n}\sum_i \xi_i^2 - m_2 \end{pmatrix} \implies \mathcal{N}(0, \Sigma)$$

The off-diagonal entry $m_3 - m_1 m_2$ is the asymptotic covariance between the fluctuation of the sample mean and that of the sample second moment; it vanishes precisely when $m_3 = m_1 m_2$, in which case the two statistics fluctuate asymptotically independently. To make this concrete, take $\xi_i \sim \mathcal{N}(0,1)$, so $m_1 = 0$, $m_2 = 1$, $m_3 = 0$, $m_4 = 3$. Then

$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \sqrt{n}\begin{pmatrix} \bar{\xi}_n \\ \frac{1}{n}\sum_i \xi_i^2 - 1 \end{pmatrix} \implies \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\right)$$

The zero off-diagonal (a consequence of $m_3 = 0$ for the symmetric standard normal) says the sample mean and the sample variance of a Gaussian sample are asymptotically uncorrelated, and by example 7.3.4 for jointly Gaussian limits this is asymptotic independence. This asymptotic independence of the sample mean and sample variance is the asymptotic form of an exact independence that holds for the Gaussian at every finite n, and it is the reason the t-statistic of classical statistics has a limiting normal law.

Exercise 7.3.1

1. Let $X = (X_1, X_2)$ be a random vector such that each coordinate X_1, X_2 is a standard normal, but $X_2 = \varepsilon X_1$ where $\varepsilon \in \{-1, +1\}$ is independent of X_1 with $\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}$. Show that X_1 and X_2 are each normal, that $\mathrm{Cov}(X_1, X_2) = 0$, yet X is *not* a Gaussian vector. (Hint: compute the law of $X_1 + X_2$ and check it against proposition 7.3.2.)

2. Let $X \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^d$ with Σ nonsingular. Using the affine-image formula of proposition 7.3.2, show that the standardized vector $\Sigma^{-1/2}(X - \mu)$ has law $\mathcal{N}(0, I)$, and deduce that $\langle X - \mu, \Sigma^{-1}(X - \mu)\rangle$ has a chi-squared law with d degrees of freedom (that is, the law of $\sum_{k=1}^d G_k^2$ for independent standard normals G_k).

3. Fix probabilities $p_1, \ldots, p_d > 0$ summing to 1. Roll a d-sided die n times independently and let $N_n = (N_{n,1}, \ldots, N_{n,d})$ record the counts of each face, so $N_n = \sum_{i=1}^n X_i$ where X_i is the indicator vector e_j when the i-th roll shows face j. Compute $\mu = \mathbb{E}[X_1]$ and $\Sigma = \mathrm{Cov}(X_1)$, and write down the limiting law of $\frac{1}{\sqrt{n}}(N_n - n\mu)$. Explain why the limiting covariance matrix is singular, and identify the linear constraint responsible.

4. Let $Z \sim \mathcal{N}(0, I)$ in $\mathbb{R}^d$ and let Q be any orthogonal $d \times d$ matrix. Show that QZ has the

same law as Z. (This rotational invariance, Maxwell's theorem, characterizes the standard Gaussian among all laws on $\mathbb{R}^d$ with independent identically distributed coordinates; you need only the affine-image formula.)

5. Let $\xi_1, \xi_2, \dots$ be i.i.d. real with mean 0, variance σ^2, and finite fourth moment, and suppose additionally $\mathbb{E}[\xi_1^3] = 0$. For the vector summands $X_i = (\xi_i, \xi_i^2)^\top$ of example 7.3.7, show directly from the theorem that the sample mean $\frac{1}{n}\sum \xi_i$ and the sample second moment $\frac{1}{n}\sum \xi_i^2$ have asymptotically independent fluctuations, and give the two limiting variances in terms of the moments.

6. Give an example of two sequences of random vectors X_n, X_n' in $\mathbb{R}^2$ such that each coordinate marginal converges, $X_{n,1} \Rightarrow X_1$ and $X_{n,2} \Rightarrow X_2$, with the same coordinate limits for both sequences, yet X_n and X_n' converge to different joint laws. Explain how this is consistent with the Cramér-Wold device (theorem 7.3.5), which requires convergence of *all* projections, not only the two coordinate ones.

7.4 Refinements: Berry-Esseen and the Delta Method

The central limit theorem of section 7.1 is a statement about a limit: the standardized sum converges in distribution to a Gaussian. Two questions immediately press on this qualitative fact. First, how fast is the convergence? A limit theorem is only useful in practice if the approximation it licenses is good at the sample sizes one meets, and for that one needs a bound on the error $\sup_x |F_n(x) - \Phi(x)|$ at finite n, not just its vanishing in the limit. The Berry-Esseen theorem supplies exactly this, and it does so with the sharp rate $1/\sqrt{n}$. Second, the central limit theorem describes the fluctuations of a sum, but the quantities of interest in statistics are almost never sums: they are functions of sums, sample variances, ratios, correlation coefficients, maximum-likelihood estimators. The delta method is the device that pushes asymptotic normality through a smooth transformation, turning the central limit theorem for T_n into a central limit theorem for $g(T_n)$. Both refinements rest on a single elementary tool, Slutsky's theorem, which governs how weak convergence interacts with convergence to a constant, and it is with Slutsky's theorem that this section begins.

Weak convergence is not, on its own, well-behaved under addition and multiplication: if $X_n \Rightarrow X$ and $Y_n \Rightarrow Y$ then it is false in general that $X_n + Y_n \Rightarrow X + Y$, because the joint behaviour of X_n and Y_n is not determined by their marginals. The obstruction disappears when one of the two sequences converges not merely in distribution but to a *constant*. Convergence to a constant is a strong form of convergence in probability (definition 5.1.1), and it pins the second sequence down tightly enough that it can be combined with the first. This is the content of Slutsky's theorem, and it is what licenses every "and therefore" that transports a limit through an auxiliary term converging to a constant.

Proposition 7.4.1: Slutsky's Theorem

Let X_n, Y_n be random variables on a common probability space with $X_n \Rightarrow X$, and let $c \in \mathbb{R}$ be a constant with $Y_n \to c$ in probability. Then

1. $X_n + Y_n \Rightarrow X + c$, and
2. $X_n Y_n \Rightarrow cX$.

***Proof*:**
The proof of both parts runs through the portmanteau theorem (theorem 6.1.3), which characterizes weak convergence by the convergence of expectations of bounded Lipschitz functions, together with the fact (proposition 6.1.4) that convergence in probability to a constant is enough to control the perturbation each auxiliary term contributes. We treat the two claims separately.

1. *The sum.* Fix a bounded Lipschitz function $f\colon \mathbb{R} \to \mathbb{R}$, say $|f| \le M$ everywhere and $|f(a) - f(b)| \le L|a-b|$ for all a, b. By the portmanteau theorem it suffices to show $\mathbb{E}[f(X_n + Y_n)] \to \mathbb{E}[f(X + c)]$. Decompose the difference as

   $$\mathbb{E}[f(X_n+Y_n)]-\mathbb{E}[f(X+c)] = \big(\mathbb{E}[f(X_n+Y_n)]-\mathbb{E}[f(X_n+c)]\big)+\big(\mathbb{E}[f(X_n+c)]-\mathbb{E}[f(X+c)]\big)$$

   The second bracket tends to 0: the map $x \mapsto f(x+c)$ is again bounded and Lipschitz, so $X_n \Rightarrow X$ gives $\mathbb{E}[f(X_n + c)] \to \mathbb{E}[f(X + c)]$ directly by the portmanteau theorem. For the first bracket, fix $\varepsilon > 0$ and split according to whether Y_n is close to c. On the event $\{|Y_n - c| \le \varepsilon\}$ the Lipschitz bound gives $|f(X_n + Y_n) - f(X_n + c)| \le L\varepsilon$; on the complementary event the crude bound $|f(X_n + Y_n) - f(X_n + c)| \le 2M$ applies. Hence

   $$\big|\mathbb{E}[f(X_n + Y_n)] - \mathbb{E}[f(X_n + c)]\big| \le L\varepsilon + 2M\,\mathbb{P}(|Y_n - c| > \varepsilon)$$

   Since $Y_n \to c$ in probability, $\mathbb{P}(|Y_n-c| > \varepsilon) \to 0$, so $\limsup_n \big|\mathbb{E}[f(X_n+Y_n)]-\mathbb{E}[f(X_n+c)]\big| \le L\varepsilon$. As $\varepsilon > 0$ was arbitrary the first bracket tends to 0, and combining the two brackets gives $\mathbb{E}[f(X_n + Y_n)] \to \mathbb{E}[f(X + c)]$. By the portmanteau theorem, $X_n + Y_n \Rightarrow X + c$.

2. *The product.* The argument mirrors the previous one, the only new point being that multiplication by X_n is not bounded, so the splitting must also control the size of X_n. Fix a bounded Lipschitz $f\colon \mathbb{R} \to \mathbb{R}$ with $|f| \le M$ and Lipschitz constant L. Write

   $$\mathbb{E}[f(X_nY_n)] - \mathbb{E}[f(cX)] = \big(\mathbb{E}[f(X_nY_n)] - \mathbb{E}[f(cX_n)]\big) + \big(\mathbb{E}[f(cX_n)] - \mathbb{E}[f(cX)]\big)$$

   The second bracket tends to 0 because $x \mapsto f(cx)$ is bounded and Lipschitz and $X_n \Rightarrow X$. For the first bracket, fix $\varepsilon > 0$ and, in addition, a real $A > 0$; the extra parameter A caps how large X_n is allowed to be. On the event $\{|Y_n - c| \le \varepsilon\} \cap \{|X_n| \le A\}$ the Lipschitz bound gives

   $$|f(X_nY_n) - f(cX_n)| \le L\,|X_n|\,|Y_n - c| \le LA\varepsilon$$

   while off this event $|f(X_nY_n) - f(cX_n)| \le 2M$ as before. Therefore

   $$\big|\mathbb{E}[f(X_nY_n)] - \mathbb{E}[f(cX_n)]\big| \le LA\varepsilon + 2M\,\mathbb{P}(|Y_n - c| > \varepsilon) + 2M\,\mathbb{P}(|X_n| > A)$$

   Let $n \to \infty$. The middle term vanishes since $Y_n \to c$ in probability. For the last term, $X_n \Rightarrow X$ implies tightness of $\{X_n\}$ (definition 6.2.1): given $\delta > 0$ one may choose A, a continuity point of the law of X, so large that $\limsup_n \mathbb{P}(|X_n| > A) \le \delta$. Thus

   $$\limsup_n \big|\mathbb{E}[f(X_nY_n)] - \mathbb{E}[f(cX_n)]\big| \le LA\varepsilon + 2M\delta$$

   Now send $\varepsilon \to 0$ with A (hence δ) fixed, then $\delta \to 0$; the right-hand side tends to 0. Hence $\mathbb{E}[f(X_nY_n)] \to \mathbb{E}[f(cX)]$, and the portmanteau theorem gives $X_nY_n \Rightarrow cX$, completing the proof.

The hypothesis that c is a constant is doing all the work, and it cannot be dropped. If $Y_n \Rightarrow Y$ for a random limit Y, the conclusion fails, because the sum $X_n + Y_n$ depends on the joint law of (X_n, Y_n), which the two marginal convergences leave undetermined. The concrete failure is easy to exhibit: let $X_n \equiv Z$ and $Y_n \equiv -Z$ for a single standard normal Z, so $X_n \Rightarrow Z$ and $Y_n \Rightarrow -Z$, both standard normal; yet $X_n + Y_n \equiv 0$, whose limit is the constant 0, not the $\mathcal{N}(0, 2)$ one would get by naively adding independent copies. Slutsky's theorem sidesteps this by demanding that one summand collapse onto a point, where there is no joint-law ambiguity left to exploit. In the applications to come, Y_n is typically an estimated scaling factor (a sample standard deviation, a ratio of sample moments) that converges by the law of large numbers to its deterministic target, and Slutsky's theorem is what lets one replace the random scaling by its limit inside a weak-convergence statement.

A first use of Slutsky's theorem records the practical fact that studentizing a sum by an *estimated* standard deviation does not disturb the central limit theorem. It is the reason the t-statistic of elementary statistics is asymptotically standard normal.

Example 7.4.2: Studentizing the Central Limit Theorem

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$, and write $\bar{X}_n = S_n/n$ for the sample mean. The central limit theorem (theorem 7.1.3) gives

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0, 1)$$

In practice σ is unknown and is replaced by the sample standard deviation s_n, where $s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$. The weak law of large numbers (theorem 5.1.5) applied to X_i^2 and $\bar{X}_n \to \mu$ give $s_n^2 \to \sigma^2$ in probability, whence $\sigma/s_n \to 1$ in probability. Writing

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{s_n} = \underbrace{\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}}_{\Rightarrow \mathcal{N}(0,1)} \cdot \underbrace{\frac{\sigma}{s_n}}_{\to 1 \text{ in prob.}}$$

the product form of Slutsky's theorem (proposition 7.4.1) applies with $c = 1$, and the studentized statistic satisfies

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{s_n} \Rightarrow \mathcal{N}(0, 1)$$

The estimated normalization is asymptotically invisible: replacing σ by s_n leaves the limiting distribution standard normal.

With Slutsky's theorem in hand, the two refinements can be stated. The first is quantitative. The central limit theorem gives the shape of the limit but says nothing about how close F_n is to Φ for a fixed n. For that one needs a bound, uniform in x, on the discrepancy between the true distribution function of the standardized sum and the Gaussian one. The Berry-Esseen theorem provides such a bound, and its form is the one a scaling heuristic predicts: the error decays like $1/\sqrt{n}$, and the constant in front is governed by the third absolute moment, which measures how far the summands are from symmetric and light-tailed. The appearance of $1/\sqrt{n}$ is the sharp rate, and no assumption beyond a finite third moment is needed to obtain it.

Theorem 7.4.3: Berry-Esseen Theorem

Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ, variance $\sigma^2 \in (0, \infty)$, and finite third absolute central moment

$$\rho = \mathbb{E}|X_1 - \mu|^3 < \infty$$

Let F_n denote the distribution function of the standardized sum $(S_n - n\mu)/(\sigma\sqrt{n})$, and let Φ be the standard normal distribution function. Then there is a universal constant C, independent of n and of the common law of the X_i, such that

$$\sup_{x\in\mathbb{R}} \big|F_n(x) - \Phi(x)\big| \le \frac{C\rho}{\sigma^3\sqrt{n}}$$

for every $n \ge 1$.

A full proof of the Berry-Esseen theorem is a long Fourier-analytic smoothing estimate: one bounds the distance between F_n and Φ by the distance between their characteristic functions through the Berry-Esseen smoothing inequality, then controls the characteristic-function difference on a growing frequency window using the third-moment Taylor remainder. The argument is technical but introduces no new probabilistic idea beyond the characteristic-function expansion already used to prove the central limit theorem itself; it is for this reason omitted here. The standard references give it in full: Feller's *An Introduction to Probability Theory and Its Applications*, Volume II, and Petrov's *Sums of Independent Random Variables* both carry the complete proof, the latter with the sharp modern constants.

The dimensionless quantity controlling the bound is ρ/σ^3, the standardized third absolute moment, which is invariant under the affine rescalings $X \mapsto aX + b$ that the standardization already removes; only the shape of the summand's law, not its location or scale, affects the rate. The theorem is often quoted in the equivalent form $\sup_x |F_n(x) - \Phi(x)| \le C\,\mathbb{E}|Y_1|^3/\sqrt{n}$ where $Y_i = (X_i - \mu)/\sigma$ are the standardized summands, since $\mathbb{E}|Y_1|^3 = \rho/\sigma^3$. The best known value of the universal constant is close to $C = 0.4748$ (Shevtsova), and the constant cannot be taken below $(\sqrt{10}+3)/(6\sqrt{2\pi}) \approx 0.4097$ (Esseen's lower bound); the exact optimal C is still open. For the purposes of this book the existence of *some* finite universal C is what matters: it converts the asymptotic statement of the central limit theorem into a finite-sample guarantee.

The $1/\sqrt{n}$ rate is both a promise and a warning. It promises that the Gaussian approximation improves steadily with the sample size, but it warns that the improvement is slow: to halve the worst-case error one must quadruple the number of summands. It also identifies the regime where the approximation is poor. When the summand law is heavily skewed, ρ/σ^3 is large, and the bound degrades accordingly; the central limit theorem still holds in the limit, but the approach to the Gaussian is delayed. This is exactly the situation for a biased coin near $p = 0$ or $p = 1$, as the worked example below makes quantitative.

The second refinement addresses transformation. Statistical estimators are functions of sample averages, and one wants the limiting distribution of $g(T_n)$ from that of T_n. When g is differentiable at the limit point θ, its behaviour near θ is governed by its linearization $g(t) \approx g(\theta) + g'(\theta)(t-\theta)$, and this linear approximation is faithful precisely on the $1/\sqrt{n}$ scale over which T_n fluctuates around θ. The delta method makes this precise: the fluctuations of $g(T_n)$ are, to leading order, the fluctuations of T_n magnified by the factor $g'(\theta)$, so the limiting variance is scaled by $g'(\theta)^2$.

Theorem 7.4.4: The Delta Method

Let $T_1, T_2, \dots$ be random variables and $\theta \in \mathbb{R}$ a constant such that

$$\sqrt{n}\,(T_n - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$$

for some $\sigma^2 \geq 0$. Let $g\colon \mathbb{R} \to \mathbb{R}$ be a function differentiable at θ with $g'(\theta) \neq 0$. Then

$$\sqrt{n}\,\big(g(T_n) - g(\theta)\big) \Rightarrow \mathcal{N}\big(0,\, g'(\theta)^2\sigma^2\big)$$

***Proof*:**
The strategy is to replace g by its first-order Taylor expansion at θ and to show that the remainder, once multiplied by $\sqrt{n}$, vanishes in probability, so that Slutsky's theorem converts the limit for T_n into the limit for $g(T_n)$.

Step 1: The Taylor expansion with a controlled remainder. Differentiability of g at θ means there is a function $r\colon \mathbb{R} \to \mathbb{R}$ with

$$g(t) = g(\theta) + g'(\theta)(t - \theta) + r(t)(t - \theta), \qquad r(t) \to 0 \text{ as } t \to \theta$$

where $r(\theta)$ is set to 0 so that r is continuous at θ; this is exactly the statement that $g'(\theta)$ is the derivative. Applying this with $t = T_n$ and multiplying by $\sqrt{n}$,

$$\sqrt{n}\,\big(g(T_n) - g(\theta)\big) = g'(\theta)\,\sqrt{n}\,(T_n - \theta) + r(T_n) \cdot \sqrt{n}\,(T_n - \theta) \tag{7.7}$$

Step 2: The linear term converges. By hypothesis $\sqrt{n}\,(T_n - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$. Multiplying by the constant $g'(\theta)$ is a continuous map, so by the mapping property of weak convergence (equivalently, by the product form of Slutsky's theorem (proposition 7.4.1) with the constant sequence $Y_n \equiv g'(\theta)$),

$$g'(\theta)\,\sqrt{n}\,(T_n - \theta) \Rightarrow \mathcal{N}\big(0,\, g'(\theta)^2\sigma^2\big)$$

using that if $W \sim \mathcal{N}(0, \sigma^2)$ then $aW \sim \mathcal{N}(0, a^2\sigma^2)$.

Step 3: The remainder term is negligible. It remains to show the second summand in (7.7) tends to 0 in probability. First, $T_n \to \theta$ in probability: since $\sqrt{n}\,(T_n - \theta)$ converges in distribution it is a tight sequence, so $T_n - \theta = (\sqrt{n}\,(T_n - \theta))/\sqrt{n} \to 0$ in probability (a tight sequence divided by $\sqrt{n} \to \infty$ collapses to 0). Because r is continuous at θ with $r(\theta) = 0$, the continuous mapping theorem for convergence in probability gives $r(T_n) \to 0$ in probability. Meanwhile $\sqrt{n}\,(T_n - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$, so the product form of Slutsky's theorem (proposition 7.4.1), now with $X_n = \sqrt{n}\,(T_n - \theta)$ and $Y_n = r(T_n) \to c = 0$, yields

$$r(T_n) \cdot \sqrt{n}\,(T_n - \theta) \Rightarrow 0 \cdot \mathcal{N}(0, \sigma^2) = 0$$

that is, the remainder converges in distribution to the constant 0, which is the same as convergence to 0 in probability.

Step 4: Assemble via Slutsky. Return to (7.7). The linear term converges in distribution to $\mathcal{N}(0, g'(\theta)^2\sigma^2)$ by Step 2, and the remainder term converges to the constant 0 in probability by Step 3. The sum form of Slutsky's theorem (proposition 7.4.1) then gives

$$\sqrt{n}\,\big(g(T_n) - g(\theta)\big) \Rightarrow \mathcal{N}\big(0,\, g'(\theta)^2\sigma^2\big) + 0 = \mathcal{N}\big(0,\, g'(\theta)^2\sigma^2\big)$$

which is the claim, as we sought to show.

The delta method is the single most-used tool of large-sample statistics, and its mechanism is worth restating in words: on the $1/\sqrt{n}$ window over which T_n lives, a differentiable g looks linear, and a linear map sends a Gaussian to a Gaussian with variance scaled by the square of the slope. The condition $g'(\theta) \neq 0$ is essential to the stated form: at a critical point the linear term vanishes and the leading fluctuation is governed by the next nonzero derivative, so the correct scaling is no longer $\sqrt{n}$ but n, and the limit is no longer Gaussian. If $g'(\theta) = 0$ but $g''(\theta) \neq 0$, the same Taylor argument at second order gives $n\,(g(T_n) - g(\theta)) \Rightarrow \frac{1}{2}g''(\theta)\sigma^2\chi_1^2$, a chi-square law rather than a normal one, because the square of a Gaussian is chi-square. The worked examples below use only the first-order form, where $g'(\theta) \neq 0$.

One reading of the delta method is that it identifies which transformations *stabilize* the variance. If the asymptotic variance of T_n depends on the unknown parameter θ, say $\sigma^2 = v(\theta)$, then the transformed estimator $g(T_n)$ has asymptotic variance $g'(\theta)^2 v(\theta)$, and choosing g with $g'(\theta) = 1/\sqrt{v(\theta)}$ makes this identically 1, free of θ. Such a g is a *variance-stabilizing transformation*, obtained by integrating $1/\sqrt{v(\theta)}$; the second worked example carries this out for a sample proportion and recovers the classical arcsine transform.

Example 7.4.5: Berry-Esseen and Variance Stabilization

1. *Berry-Esseen for the centered coin sum.* Let $X_1, \ldots, X_n$ be i.i.d. $\mathrm{Ber}(p)$ with $0 < p < 1$, so $S_n \sim \mathrm{Bin}(n,p)$. Here $\mu = p$, $\sigma^2 = p(1-p)$, and the third absolute central moment is

$$\rho = \mathbb{E}|X_1 - p|^3 = p(1-p)^3 + (1-p)p^3 = p(1-p)\big(p^2 + (1-p)^2\big)$$

using $|X_1 - p| = 1 - p$ with probability p and $|X_1 - p| = p$ with probability $1 - p$. The Berry-Esseen ratio is therefore

$$\frac{\rho}{\sigma^3} = \frac{p(1-p)\big(p^2 + (1-p)^2\big)}{\big(p(1-p)\big)^{3/2}} = \frac{p^2 + (1-p)^2}{\sqrt{p(1-p)}}$$

and theorem 7.4.3 gives, with the working constant $C = 0.4748$,

$$\sup_x \big|F_n(x) - \Phi(x)\big| \leq \frac{0.4748}{\sqrt{n}} \cdot \frac{p^2 + (1-p)^2}{\sqrt{p(1-p)}}$$

For a fair coin $p = \frac{1}{2}$ the ratio is $(\frac{1}{4} + \frac{1}{4})/\sqrt{1/4} = 1$, so the bound reads $\sup_x |F_n(x) - \Phi(x)| \leq 0.4748/\sqrt{n}$; at $n = 100$ this is about 0.047, and at $n = 2500$ about 0.0095, the characteristic slow $1/\sqrt{n}$ decay. For a skewed coin $p = 0.05$ the ratio is $(0.0025 + 0.9025)/\sqrt{0.0475} \approx 4.15$, roughly four times larger, so the same accuracy requires roughly sixteen times as many tosses. This is the de Moivre-Laplace approximation (corollary 7.1.5) equipped with an explicit error bar, and it quantifies the folklore that the normal approximation to the binomial is trustworthy only once $np(1-p)$ is comfortably large.

2. *The arcsine variance-stabilizing transform.* Let $\hat{p}_n = S_n/n$ be the sample proportion from n i.i.d. $\mathrm{Ber}(p)$ trials, so $\mathbb{E}\hat{p}_n = p$ and, by the central limit theorem,

$$\sqrt{n}\,(\hat{p}_n - p) \Rightarrow \mathcal{N}\big(0,\, p(1-p)\big)$$

The asymptotic variance $v(p) = p(1-p)$ depends on the unknown p, which is inconvenient: a confidence interval built directly on $\hat{p}_n$ has a width that varies with the parameter being estimated. To remove this dependence, seek g with $g'(p) = 1/\sqrt{p(1-p)}$. Integrating,

$$g(p) = \int \frac{dp}{\sqrt{p(1-p)}} = 2 \arcsin \sqrt{p}$$

as one checks by differentiating: with $u = \sqrt{p}$, $\frac{d}{dp} 2 \arcsin \sqrt{p} = 2 \cdot \frac{1}{\sqrt{1-p}} \cdot \frac{1}{2\sqrt{p}} = 1/\sqrt{p(1-p)}$. Applying the delta method (theorem 7.4.4) with this g and $\sigma^2 = p(1-p)$,

$$\sqrt{n}\left(2 \arcsin \sqrt{\hat{p}_n} - 2 \arcsin \sqrt{p}\right) \Rightarrow \mathcal{N}\Big(0,\ \tfrac{1}{p(1-p)} \cdot p(1-p)\Big) = \mathcal{N}(0,1)$$

The limiting variance is 1 for every $p \in (0,1)$: the transformed estimator has an asymptotic distribution free of the unknown parameter. This is the arcsine (or angular) transform, the classical device for putting proportions on a scale where a fixed-width confidence interval is valid uniformly in p, and it is a direct payoff of the delta method read as a variance-stabilization principle.

Taken together, the two refinements sit on opposite sides of the central limit theorem. Berry-Esseen looks *inward*, sharpening the limit into a finite-sample statement: it answers "how good is the Gaussian approximation at this n?" with a bound exhibiting the slow $1/\sqrt{n}$ rate and the penalty for skewness. The delta method looks *outward*, extending the reach of the limit: it answers "what is the limiting law of a smooth function of my estimator?" by linearizing and scaling the variance. The first is the quantitative refinement of the central limit theorem as a theorem about numbers; the second is the structural refinement that makes it the organizing principle of asymptotic statistics, where every standard-error formula, confidence interval, and Wald test descends from a delta-method variance computation resting on Slutsky's theorem. The number-theoretic thread resumes in section 7.5, where the central limit theorem is turned on the prime factorization of a random integer.

Exercise 7.4.1

1. Let $Z_n \Rightarrow \mathcal{N}(0,1)$ and let $a_n \to a$, $b_n \to b$ be sequences of *constants* (viewed as degenerate random variables). Prove that $a_n Z_n + b_n \Rightarrow \mathcal{N}(b, a^2)$. Identify precisely where each part of Slutsky's theorem (proposition 7.4.1) is used, and explain why the constancy of a_n, b_n matters.

2. Give an explicit example of random variables with $X_n \Rightarrow X$ and $Y_n \Rightarrow Y$ (both limits nondegenerate) for which $X_n + Y_n$ does *not* converge to $X + Y$ in distribution. Explain in one sentence why Slutsky's theorem does not apply.

3. Let $X_1, X_2, \ldots$ be i.i.d. with mean μ, variance $\sigma^2 \in (0,\infty)$, and finite fourth moment. Show that $\sqrt{n}\,(s_n^2 - \sigma^2) \Rightarrow \mathcal{N}(0, \mu_4 - \sigma^4)$, where $s_n^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ and $\mu_4 = \mathbb{E}(X_1 - \mu)^4$. (Hint: reduce to the average of $(X_i - \mu)^2$, apply the central limit theorem to it, and dispose of the $\bar{X}_n - \mu$ correction with Slutsky's theorem.)

4. Let $\hat{\lambda}_n = \bar{X}_n$ be the sample mean of i.i.d. positive random variables with mean $\lambda > 0$ and variance $\tau^2 \in (0,\infty)$. Using the delta method, find the asymptotic distribution of $\sqrt{n}\,(\log \hat{\lambda}_n - \log \lambda)$. For which one-parameter family is the resulting asymptotic variance free of λ, so that the log transform stabilizes the variance?

5. Using the Berry-Esseen bound with working constant $C = 0.4748$, find the smallest n guaranteeing $\sup_x |F_n(x) - \Phi(x)| \le 0.01$ for a fair-coin sum ($p = \frac{1}{2}$). Compare with the n

needed for a coin with $p = 0.1$, and comment on the ratio.

6. Suppose $\sqrt{n}\,(T_n - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$ and g is twice differentiable at θ with $g'(\theta) = 0$ but $g''(\theta) \neq 0$. Show that $n\,(g(T_n) - g(\theta)) \Rightarrow \frac{1}{2} g''(\theta)\sigma^2\chi_1^2$, where χ_1^2 is the law of the square of a standard normal. (This is the second-order delta method; explain why the scaling changes from $\sqrt{n}$ to n.)

7.5 Application: The Erdős-Kac Theorem

The Hardy-Ramanujan theorem of section 5.4 said that the number of distinct prime factors of a random integer concentrates: for n drawn uniformly from $\{1, \ldots, N\}$, the count $\omega(n)$ sits within a window of width $\sqrt{\log\log N}$ around $\log\log N$ for all but a density-zero set of integers. That is a law of large numbers, and it was proved by exactly the tool a law of large numbers is proved by, Chebyshev's inequality applied to a small variance. The present chapter has taught us to ask a sharper question of any such concentration statement. Once the crude second-moment bound has pinned a quantity to its mean, what is the shape of the residual fluctuation on the scale set by the standard deviation? For sums of independent variables the answer is the central limit theorem, and the divisibility indicators D_p of definition 5.4.1, though not independent, are correlated so weakly that the same answer survives: every moment of the standardized count matches the corresponding moment of an independent sum of coins in the limit, and that is enough to force the Gaussian. The centred and rescaled count $(\omega(n) - \log\log N)/\sqrt{\log\log N}$ does more than tend to zero: its distribution across $\{1, \ldots, N\}$ converges to the standard Gaussian. This is the Erdős-Kac theorem, and it is the founding result of probabilistic number theory, the discovery that the primes-as-coin-flips heuristic is not a mere analogy but obeys the central limit theorem on the nose.

The proof has two ideas, and the interplay between them is the whole content. The first is that the Lindeberg-Feller theorem (theorem 7.2.5) does deliver a Gaussian, but only for a triangular array whose rows are *independent*; its proof factors a characteristic function over the summands, and that factorization fails outright for the D_p, which are dependent. So Lindeberg-Feller cannot be applied to the divisibility indicators themselves. It is applied instead to an *independent model*, a row of independent Bernoulli coins Y_p with the same biases $\mathbb{P}(Y_p = 1) = 1/p$, where the hypothesis is met exactly and the conclusion follows. The second idea discharges the dependence: the true standardized count and the independent model are shown to have the *same limit* by the method of moments, comparing every moment of one to the corresponding moment of the other and showing the difference vanishes. This is where the near-independence of the D_p does its work, and where it must do more than control a single variance. A moment of order k is a sum over k-tuples of primes, and the deviation of a mixed moment $\mathbb{E}[\prod D_{p_i}]$ from the product $\prod \mathbb{E}[D_{p_i}]$ that independence would give is exactly the kind of prime-tuple correlation that proposition 5.4.3 quantifies; that every such deviation is of lower order, tuple by tuple, is the arithmetic input, imported from analytic number theory as section 5.4 imported Mertens. Because the Gaussian is determined by its moments, matching all of them forces the same limit.

The strategy is therefore to split ω at a threshold prime p_0, throw away the contribution of the large primes above p_0 using a mean bound, apply Lindeberg-Feller to the independent model on the small primes below p_0 to identify its limit as Gaussian, and then transfer that limit to the true count by matching moments. The one non-probabilistic ingredient, as in section 5.4, is Mertens' theorem and the associated prime-sum and prime-tuple estimates: these are imported from analytic number theory. Everything probabilistic, the Lindeberg verification for the independent model, the moment comparison, and the truncation bound, is done here.

Before the theorem, the truncation must be set up precisely, since the choice of threshold is what makes both halves of the argument work at once. The threshold is placed at

$$p_0 = N^{1/\log\log N}$$

a prime cutoff that grows with N but far more slowly than N itself. Two facts about this choice drive the proof, and both are consequences of Mertens' theorem. First, the small primes up to p_0 already carry the whole variance to leading order: by Mertens' theorem,

$$\sum_{p\le p_0}\frac{1}{p} = \log\log p_0+O(1) = \log\Big(\frac{\log N}{\log\log N}\Big)+O(1) = \log\log N-\log\log\log N+O(1) = \log\log N+O(\log\log\log N)$$

so $\sum_{p\le p_0} 1/p = \log\log N\,(1+o(1))$, the small primes accounting for the full $\log\log N$ to leading order. Second, the large primes above p_0 contribute a mean that is of lower order: again by Mertens,

$$\sum_{p_0<p\le N}\frac{1}{p} = \big(\log\log N + O(1)\big) - \big(\log\log p_0 + O(1)\big) = \log\log\log N + O(1)$$

which is $o(\sqrt{\log\log N})$, since $\log\log\log N$ grows far more slowly than $\sqrt{\log\log N}$. The large-prime part therefore has a mean too small to matter on the central-limit scale, and a variance no larger than its mean, so a Chebyshev bound will erase it. These two computations are recorded once here so that the proof can invoke them cleanly.

Theorem 7.5.1: The Erdős-Kac Theorem

Let n be uniform on $\{1,\dots,N\}$ and let $\omega(n) = \sum_{p\le N} D_p(n)$ be its number of distinct prime factors (definition 5.4.1). Then, as $N\to\infty$,

$$\frac{\omega(n)-\log\log N}{\sqrt{\log\log N}} \Rightarrow \mathcal{N}(0,1)$$

that is, for every real x,

$$\frac{1}{N}\,\#\Big\{\,1\le n\le N\big|\omega(n)\le \log\log N + x\sqrt{\log\log N}\,\Big\} \longrightarrow \Phi(x)$$

where Φ is the standard normal distribution function. The number of distinct prime factors of a random integer is asymptotically Gaussian, with mean and variance both $\log\log N$.

***Proof*:**
Write $\lambda = \log\log N$ throughout, so the claim is that $(\omega-\lambda)/\sqrt{\lambda}\Rightarrow\mathcal{N}(0,1)$. Set the threshold $p_0 = N^{1/\lambda}$ and split the count at p_0:

$$\omega = \omega_{\le p_0}+\omega_{>p_0}, \qquad \omega_{\le p_0} = \sum_{p\le p_0} D_p, \qquad \omega_{>p_0} = \sum_{p_0<p\le N} D_p$$

The proof has three movements: the large-prime tail is shown negligible on the scale $\sqrt{\lambda}$; the standardized small-prime part is shown asymptotically normal by comparison with an independent model, to which Lindeberg-Feller applies with its independence hypothesis met;

and the two are recombined by Slutsky's theorem.

Step 1: The large-prime tail is negligible. The variable $\omega_{>p_0} = \sum_{p_0<p\le N} D_p$ is a sum of nonnegative indicators, so it suffices to bound its expectation: a nonnegative random variable that is small in mean is small in probability. By linearity of expectation (proposition 3.1.3) and the single-prime mean of proposition 5.4.3,

$$\mathbb{E}[\omega_{>p_0}] = \sum_{p_0<p\le N} \mathbb{E}[D_p] = \sum_{p_0<p\le N} \left(\frac{1}{p} + O\Big(\frac{1}{N}\Big)\right) = \sum_{p_0<p\le N} \frac{1}{p} + O\Big(\frac{\pi(N)}{N}\Big)$$

The error is $\pi(N)$ terms of size $O(1/N)$, hence $O(\pi(N)/N) = o(1)$ since $\pi(N) = O(N/\log N)$, so $\pi(N)/N = O(1/\log N) = o(1)$. The main term is the large-prime reciprocal sum computed above from Mertens' theorem, namely

$$\sum_{p_0<p\le N} \frac{1}{p} = \log\log\log N + O(1)$$

Therefore $\mathbb{E}[\omega_{>p_0}] = \log\log\log N + O(1)$. Dividing by $\sqrt{\lambda} = \sqrt{\log\log N}$,

$$\mathbb{E}\left[\frac{\omega_{>p_0}}{\sqrt{\lambda}}\right] = \frac{\log\log\log N + O(1)}{\sqrt{\log\log N}} \longrightarrow 0$$

since $\log\log\log N = o(\sqrt{\log\log N})$. Because $\omega_{>p_0} \ge 0$, Markov's inequality (theorem 3.2.4) turns this mean bound into a probability bound: for every $\varepsilon > 0$,

$$\mathbb{P}\left(\frac{\omega_{>p_0}}{\sqrt{\lambda}} > \varepsilon\right) \le \frac{1}{\varepsilon}\,\mathbb{E}\left[\frac{\omega_{>p_0}}{\sqrt{\lambda}}\right] \longrightarrow 0$$

Hence $\omega_{>p_0}/\sqrt{\lambda} \to 0$ in probability (definition 5.1.1). (A second-moment bound via Chebyshev's inequality, corollary 3.2.5, gives the same conclusion, and is what is needed if one prefers to center $\omega_{>p_0}$; here the first moment already suffices because the tail is a sum of nonnegative terms.) The large primes have been discarded.

Step 2: Standardizing the small-prime part and its independent model. Attention now turns to $\omega_{\le p_0} = \sum_{p\le p_0} D_p$. Index the primes $p \le p_0$ as $p_1 < p_2 < \cdots < p_{k_N}$, where $k_N = \pi(p_0)$ is the number of primes below the threshold, center each indicator, and set

$$X_{N,j} = \frac{D_{p_j} - \mathbb{E}[D_{p_j}]}{\sqrt{\lambda}}, \qquad 1 \le j \le k_N, \qquad S_N := \sum_{j=1}^{k_N} X_{N,j} = \frac{\omega_{\le p_0} - \mathbb{E}[\omega_{\le p_0}]}{\sqrt{\lambda}}$$

so S_N is the standardized small-prime count about its own mean. The variables $D_{p_1}, \ldots, D_{p_{k_N}}$ are *not* independent, so $\{X_{N,j}\}$ is not a triangular array in the sense of definition 7.2.1 and theorem 7.2.5 may not be applied to it directly. Alongside it, define the *independent model*: for each prime $p \le p_0$ let Y_p be a Bernoulli variable with $\mathbb{P}(Y_p = 1) = 1/p$, with the Y_p mutually independent, and set

$$\widetilde{X}_{N,j} = \frac{Y_{p_j} - 1/p_j}{\sqrt{\lambda}}, \qquad \widetilde{S}_N := \sum_{j=1}^{k_N} \widetilde{X}_{N,j}$$

The Y_p share the biases $1/p$ of the divisibility events but are mutually independent, so $\{\widetilde{X}_{N,j}\}$ *is* a triangular array, and Lindeberg-Feller does apply to it. The plan is to prove $\widetilde{S}_N \Rightarrow \mathcal{N}(0,1)$

for the model in Step 3, and then to show in Step 4 that S_N has the same limit by matching all moments.

Step 3: The independent model is asymptotically normal. The array $\{\widetilde{X}_{N,j}\}$ has independent, mean-zero rows, so its two Lindeberg-Feller hypotheses can be checked directly. For the row variance, the Y_p are independent, so variances add with no covariance terms:

$$\sum_{j=1}^{k_N} \mathrm{Var}(\widetilde{X}_{N,j}) = \frac{1}{\lambda} \sum_{p \le p_0} \mathrm{Var}(Y_p) = \frac{1}{\lambda} \sum_{p \le p_0} \left(\frac{1}{p} - \frac{1}{p^2}\right)$$

using $Y_p^2 = Y_p$, hence $\mathrm{Var}(Y_p) = 1/p - 1/p^2$. By the small-prime Mertens sum computed above, $\sum_{p \le p_0} 1/p = \log\log N + O(\log\log\log N)$, while $\sum_{p \le p_0} 1/p^2 \le \sum_p 1/p^2 = O(1)$ is bounded, so

$$\sum_{j=1}^{k_N} \mathrm{Var}(\widetilde{X}_{N,j}) = \frac{\lambda + O(\log\log\log N)}{\lambda} \longrightarrow 1$$

the first hypothesis, with limiting variance $\sigma^2 = 1$. For the Lindeberg condition the array is uniformly small in the strongest possible sense: each $Y_p \in \{0,1\}$ gives $|Y_p - 1/p| \le 1$, so

$$|\widetilde{X}_{N,j}| \le \frac{1}{\sqrt{\lambda}} = \frac{1}{\sqrt{\log\log N}} =: b_N$$

uniformly in j, with $b_N \to 0$. Fixing $\varepsilon > 0$, once N is large enough that $b_N < \varepsilon$ the event $\{|\widetilde{X}_{N,j}| > \varepsilon\}$ is empty for every j, so the Lindeberg sum (definition 7.2.3) is exactly zero for all large N and the Lindeberg condition holds. By the Lindeberg-Feller theorem (theorem 7.2.5), therefore,

$$\widetilde{S}_N \Rightarrow \mathcal{N}(0,1)$$

Here independence is real and both hypotheses hold, so the theorem applies directly. What remains is to move its conclusion from the model to the count.

Step 4: Matching moments transfers the limit from the model to the count. The variables S_N and $\widetilde{S}_N$ have the same variance to leading order, but that alone cannot equate their limiting laws; the transfer is made moment by moment. Fix an integer $k \ge 1$. Expanding the k-th power of a row sum gives, in both cases, a sum over ordered k-tuples $(p_{i_1}, \dots, p_{i_k})$ of small primes of a mixed moment of the centred indicators, divided by $\lambda^{k/2}$:

$$\mathbb{E}[S_N^k] = \frac{1}{\lambda^{k/2}} \sum_{i_1,\dots,i_k} \mathbb{E}\Big[\textstyle\prod_{r=1}^k \left(D_{p_{i_r}} - \mathbb{E}[D_{p_{i_r}}]\right)\Big], \qquad \mathbb{E}[\widetilde{S}_N^k] = \tfrac{1}{\lambda^{k/2}} \textstyle\sum_{i_1,\dots,i_k} \mathbb{E}\Big[\prod_{r=1}^k \left(Y_{p_{i_r}} - 1/p_{i_r}\right)\Big]$$

Compare the two term by term. For a tuple involving m *distinct* primes, the model factor is a product over the distinct primes of independent centred Bernoulli moments, computable exactly; the true factor is the corresponding *joint* centred moment of the D_p. These agree up to the discrepancy between the joint moment $\mathbb{E}[\prod D_p]$ over the distinct primes and the product $\prod \mathbb{E}[D_p]$ that independence would give, which is exactly the prime-tuple correlation that the moment estimates of proposition 5.4.3 control: for a squarefree product over any fixed set of distinct primes, $\mathbb{E}[\prod_p D_p] = \prod_p (1/p) + O(1/N)$, with the same $O(1/N)$ inclusion-exclusion error that produced the single-prime and pairwise estimates, now for a product of boundedly many indicators. The accounting that turns these per-tuple agreements into agreement of the two moments is exactly the prime-tuple bookkeeping used for the variance in theorem 5.4.5, carried one order at a time and carried out in full in *Analytic Number Theory* [1]. Two features of the centred moments organize the count. First, because the factors are centred, any prime that appears in a tuple exactly once contributes a mean-zero factor: in the model it is independent of the rest and kills the term outright, and for the true count it survives only through its $O(1/N)$ correlation with the other factors. So the tuples that carry a nonzero main term are exactly those in which every distinct prime appears with multiplicity at least 2; a tuple with m distinct primes then has $2m \le k$, that is $m \le k/2$. Second, over the small primes a distinct prime appearing with multiplicity $a \ge 2$ contributes the centred-Bernoulli moment $\mathbb{E}[(D_p - \mathbb{E}[D_p])^a] = 1/p + O(1/p^2)$, and summing $1/p$ over $p \le p_0$ gives $\sum_{p \le p_0} 1/p = \lambda\,(1 + o(1))$ by Mertens, one factor of λ for each distinct prime. A configuration with m distinct primes therefore sums to $\Theta(\lambda^m)$ before standardizing, hence to $\Theta(\lambda^{m-k/2})$ after division by $\lambda^{k/2}$. This is $o(1)$ exactly when $m < k/2$, so every tuple with fewer than $k/2$ distinct primes is subdominant, while the fully paired tuples with $m = k/2$ (each of the $k/2$ primes appearing with multiplicity exactly 2) sit at the top order λ^0. The correlation corrections separating the true moment from the model, each of relative size $O(1/N)$ against a paired term, also vanish once summed over the $k_N = \pi(p_0)$ small primes, since $k_N \le p_0 = N^{1/\lambda}$ is far fewer than N. Only the diagonal combinatorics of the fully paired tuples survive at the top order, and those are identical for the two arrays because they depend only on the shared variances $\mathrm{Var}(D_p) = \mathrm{Var}(Y_p) + O(1/N)$. Consequently, for every fixed k,

$$\mathbb{E}[S_N^k] - \mathbb{E}[\widetilde{S}_N^k] \longrightarrow 0, \qquad \text{and both converge to} \qquad \mu_k := \mathbb{E}[Z^k],\ Z \sim \mathcal{N}(0,1)$$

the common limit being the k-th Gaussian moment. That $\mathbb{E}[\widetilde{S}_N^k] \to \mu_k$ is the same tuple computation run for the model alone, where the top-order diagonal combinatorics reproduce exactly the Wick pairing that yields the Gaussian moments μ_k (this is the moment form of Step 3's conclusion, and holds because the bounded independent summands make every moment converge along with the distribution). The standard normal law is determined by its moments, so convergence of every moment $\mathbb{E}[S_N^k] \to \mu_k$ forces $S_N \Rightarrow \mathcal{N}(0,1)$ (the method of moments; the Gaussian moment problem is determinate by the Carleman criterion, a standard measure-theoretic fact, [2]). The dependence of the D_p has now been discharged in full, not assumed away: it enters only through correlation corrections that are of strictly lower order at every moment.

Step 5: Recentering the small-prime mean. The variable S_N is centred at $\mathbb{E}[\omega_{\le p_0}]$, but the theorem centres at $\lambda = \log\log N$. These differ by a lower-order amount. From $\mathbb{E}[\omega_{\le p_0}] = \sum_{p \le p_0} \mathbb{E}[D_p] = \sum_{p \le p_0} 1/p + O(1)$ and the small-prime Mertens sum,

$$\mathbb{E}[\omega_{\le p_0}] = \log\log N + O(\log\log\log N) = \lambda + O(\log\log\log N)$$

so the centering discrepancy is

$$c_N := \frac{\mathbb{E}[\omega_{\le p_0}] - \lambda}{\sqrt{\lambda}} = \frac{O(\log\log\log N)}{\sqrt{\log\log N}} \longrightarrow 0$$

a deterministic sequence tending to zero. Writing

$$\frac{\omega_{\le p_0} - \lambda}{\sqrt{\lambda}} = \frac{\omega_{\le p_0} - \mathbb{E}[\omega_{\le p_0}]}{\sqrt{\lambda}} + c_N = S_N + c_N$$

and applying Slutsky's theorem (proposition 7.4.1) with the convergent constant sequence $c_N \to 0$ gives $(\omega_{\le p_0} - \lambda)/\sqrt{\lambda} \Rightarrow \mathcal{N}(0,1)$.

Step 6: Reassembly by Slutsky. Finally, restore the large primes. Since

$$\frac{\omega-\lambda}{\sqrt{\lambda}}=\frac{\omega_{\le p_0}-\lambda}{\sqrt{\lambda}}+\frac{\omega_{>p_0}}{\sqrt{\lambda}}$$

and the first term converges weakly to $\mathcal{N}(0,1)$ by Step 5 while the second converges to 0 in probability by Step 1, Slutsky's theorem (proposition 7.4.1) yields

$$\frac{\omega-\lambda}{\sqrt{\lambda}}\Rightarrow\mathcal{N}(0,1)$$

Unwinding the standard-normal distribution function, this is exactly the density statement $\frac{1}{N}\#\{n\le N\big|\omega(n)\le\lambda+x\sqrt{\lambda}\}\to\Phi(x)$ at every x, since Φ is continuous and weak convergence is convergence of distribution functions at continuity points (definition 6.1.1). This is the claim, completing the proof.

Three features of the argument deserve comment, because each is where the number-theoretic and probabilistic strands cross. The truncation at $p_0=N^{1/\log\log N}$ is chosen to make both halves work simultaneously: high enough that the small primes below it carry the full variance $\log\log N$ to leading order, so the standardized small-prime sum has the right scale, yet low enough that the large primes above it contribute a mean of only $\log\log\log N=o(\sqrt{\log\log N})$, so their omission costs nothing on the central-limit scale. The Lindeberg-Feller theorem is applied not to the divisibility indicators, which are dependent and so fall outside its hypotheses, but to an independent model of Bernoulli coins with the same biases $1/p$; there the Lindeberg condition is verified by the crudest possible observation, that each centred coin is bounded by 1 and hence each array entry is bounded by $1/\sqrt{\log\log N}\to0$, the "no single prime dominates" uniform smallness the central limit theorem needs. The near-independence of the D_p then enters in exactly one place, the moment comparison of Step 4: it is what makes the true count and the independent model share every limiting moment, because the correlation corrections to each mixed moment are of strictly lower order in λ. That the corrections are lower-order at every moment, not just that the covariances are summable at second order, is what the argument needs and what the prime-tuple estimates supply; controlling the second moment alone, as in the Hardy-Ramanujan law, fixes the scale of the fluctuation but not its Gaussian shape.

The relationship between the two theorems is the relationship between a law of large numbers and its central limit theorem, transported from sums of i.i.d. variables to the arithmetic setting. Theorem 5.4.5 is the law of large numbers for ω: it says the count concentrates at $\log\log N$ on the scale $\sqrt{\log\log N}$, using only the first two moments and Chebyshev's inequality. theorem 7.5.1 is its central limit theorem: it resolves the residual fluctuation on that same scale into a definite shape, the Gaussian, using all the moments in the guise of the Lindeberg-Feller theorem. The mean $\log\log N$ plays the role of $n\mu$, the normalization $\sqrt{\log\log N}$ plays the role of $\sigma\sqrt{n}$, and the divisibility indicators play the role of the i.i.d. summands, near-independent rather than independent but close enough that the covariance corrections wash out. The historical sequence mirrors the logical one: Hardy and Ramanujan proved the concentration in 1917 by a variance computation, and Erdős and Kac proved the Gaussian fluctuation in 1940 by importing the central limit theorem, in the very form developed in this chapter.

Example 7.5.2: $\omega(n)$ Near a Billion, and Concentration versus Fluctuation

1. *The Gaussian approximation near $N = 10^9$.* The scale parameter is

$$\lambda = \log\log 10^9 = \log(9\log 10) = \log(9 \cdot 2.3026) = \log(20.72) \approx 3.03$$

so theorem 7.5.1 says that for a random integer near a billion, the number of distinct prime factors is approximately $\mathcal{N}(3.03, 3.03)$: a Gaussian centred near 3 with standard deviation $\sqrt{3.03} \approx 1.74$. Concretely, the fraction of integers $n \leq 10^9$ with $\omega(n)$ between $\lambda - \sqrt{\lambda} \approx 1.3$ and $\lambda + \sqrt{\lambda} \approx 4.8$ (a one-standard-deviation window) is approximately $\Phi(1) - \Phi(-1) \approx 0.68$, and the fraction with $\omega(n) \geq \lambda + 2\sqrt{\lambda} \approx 6.5$, that is with seven or more distinct prime factors, is approximately $1 - \Phi(2) \approx 0.023$. About two percent of the integers near a billion have seven or more distinct prime factors, and this the Gaussian predicts without any prime counting, from the single number $\lambda \approx 3$. The empirical distribution of $\omega(n)$ over $n \leq 10^9$ does track a bell curve of these parameters, the discreteness of the integer values notwithstanding.

2. *Concentration is the law of large numbers, fluctuation is its central limit theorem.* It is instructive to place the two theorems side by side on the same axis. Theorem 5.4.5 asserts that the *standardized* count $(\omega(n) - \lambda)/\sqrt{\lambda}$ tends to 0 in probability: for any window growing to infinity, however slowly, almost all n eventually fall inside it. That is a statement of concentration, and it is silent about the shape of the distribution within the window. theorem 7.5.1 asserts that the *same* standardized count converges in distribution to $\mathcal{N}(0,1)$: not only does the fluctuation stay of order $\sqrt{\lambda}$, its histogram, suitably rescaled, is the bell curve. The first is the arithmetic analogue of $\bar{X}_n \to \mu$; the second is the arithmetic analogue of $\sqrt{n}(\bar{X}_n - \mu) \Rightarrow \mathcal{N}(0, \sigma^2)$. A concrete consequence visible only at the finer resolution: the median of $\omega(n)$ near N and its mean both sit at λ, but the Gaussian shape further predicts that the interquartile spread of ω is about $1.35\sqrt{\lambda}$, a prediction the concentration theorem cannot make because it fixes only the location, not the profile.

The two parts of example 7.5.2 exhibit the gap that Erdős-Kac fills. The Hardy-Ramanujan law locates ω; the Erdős-Kac theorem draws its profile. Everything that a central limit theorem adds to a law of large numbers, a definite limiting shape, quantitative tail probabilities, the passage from a location statement to a distributional one, is added here to the count of prime factors, and the object that provides the shape is the same Gaussian that governs coin tossing and the symmetric random walk.

That the bell curve should govern the prime factorizations of the integers is the content of the slogan, made precise, that "the primes play a game of chance". The number of distinct prime factors of a random integer is built from the divisibility indicators D_p, which behave like independent coins of biases $1/2, 1/3, 1/5, \ldots$, and any sum of many small independent contributions, none dominant, is asymptotically Gaussian by the central limit theorem. The arithmetic obstacle, that the D_p are not exactly independent, is precisely the obstacle the moment estimates of proposition 5.4.3 were built to overcome, and it dissolves once the count and an independent sum of coins share every limiting moment, their prime-tuple correlations being of strictly lower order at each order. The Gaussian, determined by its moments, is then forced for the count exactly as the central limit theorem forces it for the coins. Erdős and Kac's theorem is the capstone of the number-theoretic thread of this book, opened at the Hardy-Ramanujan law of section 5.4 and closed here: the recognition that a purely arithmetic quantity, the count of prime divisors, obeys the same universal Gaussian law as the sums of independent random variables that this chapter is about. The theorem opened the field of probabilistic number theory, in which arithmetic functions are studied as random variables and their distributions extracted by the limit theorems of probability; the additive functions of an integer, of which ω is the prototype, are the natural objects there, and the Erdős-Kac theorem and its refinements (culminating in the Kubilius-Shapiro theory of general additive functions) are its central results.

Exercise 7.5.1

1. Verify the large-prime mean estimate at the heart of Step 1. Using Mertens' theorem in the form $\sum_{p\le x} 1/p = \log\log x + O(1)$ and the threshold $p_0 = N^{1/\log\log N}$, show that $\log\log p_0 = \log\log N - \log\log\log N$ and hence that $\sum_{p_0<p\le N} 1/p = \log\log\log N + O(1)$. Conclude that this is $o(\sqrt{\log\log N})$, which is what makes the large-prime part negligible on the central-limit scale.

2. The proof succeeds because the threshold $p_0 = N^{1/\log\log N}$ is chosen so that both the small-prime variance is $\log\log N\,(1+o(1))$ and the large-prime mean is $o(\sqrt{\log\log N})$. Investigate what goes wrong at the extremes.

    1. Suppose one truncated at $p_0' = N^{1/2}$ instead. Show that the large-prime mean $\sum_{N^{1/2}<p\le N} 1/p$ is then $O(1)$, so the tail is even smaller; but explain why this threshold is nonetheless legitimate for the tail bound. Which half of the argument, if any, would this larger truncation threaten?
    2. Suppose instead one truncated at a fixed prime $p_0'' = P$ independent of N. Show that the small-prime variance $\sum_{p\le P} 1/p$ is then a constant, not $\log\log N\,(1+o(1))$, so the standardization by $\sqrt{\log\log N}$ would drive the row variance to 0 and Lindeberg-Feller would give a degenerate limit. Conclude that p_0 must grow with N.

3. Reprove that the Lindeberg condition holds for the independent-model array $\widetilde{X}_{N,j} = (Y_{p_j} - 1/p_j)/\sqrt{\log\log N}$ of Step 2, this time via Lyapunov's condition (corollary 7.2.9) with $\delta = 1$, and compare the two routes. Specifically, show that $\sum_j \mathbb{E}|\widetilde{X}_{N,j}|^3 \le b_N \sum_j \mathbb{E}\widetilde{X}_{N,j}^2$ where $b_N = 1/\sqrt{\log\log N}$, and deduce that the Lyapunov sum is $O(b_N) \to 0$. Which route is shorter here, and why does the boundedness of the summands make them equivalent?

4. Use the Gaussian approximation of theorem 7.5.1 to estimate, for n uniform on $\{1,\dots,10^{18}\}$, the proportion of integers with at least 8 distinct prime factors. Compute $\lambda = \log\log 10^{18}$, standardize the threshold $\omega = 8$, and express the answer as $1-\Phi(x)$ for the appropriate x. Then explain in one sentence why the answer is only an approximation, referring to the two sources of error (the discreteness of ω and the finite-N rate of convergence).

5. Let $\Omega(n)$ count prime factors with multiplicity, as in exercise 5.4.1 (3). Granting from that exercise that $\mathbb{E}[\Omega-\omega] = O(1)$ and that $\mathrm{Var}(\Omega-\omega) = O(1)$, prove that Ω satisfies the same central limit theorem:
$$\frac{\Omega(n) - \log\log N}{\sqrt{\log\log N}} \Rightarrow \mathcal{N}(0,1)$$
(Hint: write $(\Omega-\lambda)/\sqrt{\lambda} = (\omega-\lambda)/\sqrt{\lambda} + (\Omega-\omega)/\sqrt{\lambda}$, bound the second term in L^2, and apply Slutsky's theorem as in Step 6.)

6. An *additive* arithmetic function is one with $f(mn) = f(m) + f(n)$ whenever $\gcd(m,n) = 1$; the count ω is the special case $f(p^k) = 1$. For a bounded additive f with $f(p^k) = g(p)$

depending only on p, define $A(N) = \sum_{p \le N} g(p)/p$ and $B(N)^2 = \sum_{p \le N} g(p)^2/p$. State, by direct analogy with the proof of theorem 7.5.1, the hypothesis on g under which one expects $(f(n) - A(N))/B(N) \Rightarrow \mathcal{N}(0,1)$, and identify which step of the proof (Step 1 truncation, Step 3 model normality, or Step 4 moment comparison) each part of the hypothesis controls. (This is the Erdős-Kac theorem for general additive functions; a full proof is developed in the Kubilius-Shapiro theory of probabilistic number theory.)

8

The Law of the Iterated Logarithm

The strong law of large numbers pinned the average S_n/n to the mean, and the central limit theorem resolved the fluctuation $S_n/\sqrt{n}$ into a Gaussian. Between the almost-sure statement of the one and the distributional statement of the other lies a question neither answers: how large do the fluctuations of S_n get, almost surely, along a single trajectory? The central limit theorem says S_n is of order $\sqrt{n}$ in distribution, but for a fixed path the $\limsup$ of $S_n/\sqrt{n}$ is $+\infty$, the Gaussian scale being exceeded infinitely often. The exact almost-sure envelope is finer, and it is the subject of this chapter, the law of the iterated logarithm, which locates it precisely at $\sqrt{2n\log\log n}$.

The result is the sharpest description of the fluctuations of a random walk, and its proof is the culmination of the machinery built in this Part. section 8.1 proves the upper half, that $\limsup_n S_n/\sqrt{2n\log\log n} \le 1$ almost surely, by a first Borel-Cantelli argument (theorem 1.4.5) along a geometric subsequence of times, with the subgaussian tail bound of proposition 3.4.7 controlling the exceptional probabilities and a maximal inequality in the manner of theorem 4.1.1 bridging the gaps between subsequence times. section 8.2 proves the matching lower half, that the $\limsup$ is at least 1, by a second Borel-Cantelli argument (theorem 1.4.6) on the independent increments over disjoint blocks. Section 8.3 reads the theorem on the binary digits of a random real, sharpening Borel's normal-number theorem (theorem 5.3.6) from a statement about digit frequencies into one about the exact size of their fluctuations.

8.1 The Upper Bound

The chapter opening located the almost-sure envelope of a random walk at $\sqrt{2n\log\log n}$; this section proves the upper half of that claim, that the walk stays below $(1+\varepsilon)\sqrt{2n\log\log n}$ from some point on, almost surely. The obstacle is that the central limit theorem (theorem 7.1.3) is a distributional statement: it controls a single time n, telling us $S_n/\sqrt{n}$ is approximately standard normal, but says nothing about the whole trajectory $n \mapsto S_n$ on one sample path. A single time n can never yield an almost-sure envelope, because the walk gets many chances to be large as n ranges over all of $\mathbb{N}$, and a bound holding with high probability at each fixed n may still fail infinitely often. Two devices bridge the gap. A geometric subsequence $n_k = \lfloor c^k \rfloor$ reduces the uncountably many constraints to a countable, summable family, so the first Borel-Cantelli lemma (theorem 1.4.5) can be applied. An exponential maximal inequality controls the walk over the entire block of times between two consecutive subsequence points at once, upgrading the pointwise tail bound to a statement about the running maximum. This is the mechanism that makes the whole argument possible, and it is where we begin.

We first fix the normalizer and the standing hypotheses for the chapter.

Definition 8.1.1: The Iterated-Logarithm Normalizer

For $n \geq 3$ the *iterated-logarithm normalizer* is

$$b_n = \sqrt{2n \log \log n}$$

the restriction $n \geq 3$ ensuring $\log \log n > 0$ so that b_n is a positive real number. Throughout this chapter the steps $X_1, X_2, \ldots$ are independent and identically distributed (definition 2.2.2) with

$$\mathbb{E}[X_i] = 0, \qquad \operatorname{Var}(X_i) = 1$$

(definition 3.2.1), and *subgaussian*: there is a constant, here normalized to the variance, for which

$$\mathbb{E}\big[e^{tX_i}\big] \leq e^{t^2/2} \qquad \text{for all } t \in \mathbb{R} \tag{8.1}$$

The partial sums are $S_n = X_1 + \cdots + X_n$, with $S_0 = 0$.

The subgaussian hypothesis (8.1) is the single analytic input the whole chapter runs on. It says the moment generating function (definition 3.4.4) of each step is dominated by that of a standard Gaussian, so the steps have tails no heavier than Gaussian tails. The bound is not an extra assumption piled on top of the variance normalization but a real restriction on the steps: it holds automatically for bounded steps, and in particular for the running example of this chapter, but fails for heavy-tailed variables. Restricting to subgaussian steps is what lets the exponential estimates below close; the general finite-variance law of the iterated logarithm (Hartman-Wintner), which drops (8.1) and keeps only $\operatorname{Var}(X_i) = 1$, is stated in section 8.2 together with the reason its proof needs a truncation argument beyond the tools developed here.

The reason ± 1 steps satisfy (8.1) is a one-line computation, which grounds the definition immediately.

Example 8.1.2: Subgaussianity of ± 1 Steps

Let X take the values $+1$ and -1 each with probability $\frac{1}{2}$, the step of the simple symmetric random walk (example 1.3.6). Then $\mathbb{E}[X] = 0$ and $\operatorname{Var}(X) = \mathbb{E}[X^2] = 1$, so the mean and variance normalizations hold. For the subgaussian bound, expand the exponential moment as a series using $X^{2k} = 1$ and $X^{2k+1} = X$:

$$\mathbb{E}\big[e^{tX}\big] = \frac{e^t + e^{-t}}{2} = \cosh t = \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!}$$

Compare this term by term with the series for $e^{t^2/2}$,

$$e^{t^2/2} = \sum_{k=0}^{\infty} \frac{(t^2/2)^k}{k!} = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k\, k!}$$

Since $(2k)! = (2k)(2k-1)\cdots(k+1)\cdot k! \geq 2^k\, k!$ (the product of the k factors from $k+1$ to $2k$ is at least 2^k, each factor at least 2 once $k \geq 1$, and the two sides agree at $k = 0, 1$), one has $\frac{1}{(2k)!} \leq \frac{1}{2^k k!}$ for every k, hence $\cosh t \leq e^{t^2/2}$ for all t. Thus (8.1) holds, and the ± 1 walk falls under the hypotheses of definition 8.1.1.

Every bounded mean-zero variable is subgaussian for the same reason, by a comparison of the same shape (Hoeffding's lemma), so the hypotheses of the chapter cover all bounded steps and not only the ± 1 case; the binary-digit walk of section 8.3 is the other instance we will need.

The heart of the section is the passage from a tail bound at one time to a tail bound on the running maximum. For a single time, the Chernoff bound (proposition 3.4.7) applied to (8.1) already gives a Gaussian tail: for $\lambda > 0$,

$$\mathbb{P}(S_n \geq \lambda) \leq \inf_{t \geq 0} e^{-t\lambda} \mathbb{E}\big[e^{tS_n}\big] \leq \inf_{t \geq 0} e^{-t\lambda + t^2 n/2} = e^{-\lambda^2/(2n)} \tag{8.2}$$

the middle inequality using independence to factor $\mathbb{E}[e^{tS_n}] = \prod_{i \leq n} \mathbb{E}[e^{tX_i}] \leq e^{t^2 n/2}$, and the optimum occurring at $t = \lambda/n$. What the law of the iterated logarithm needs is the same bound with S_n replaced by the maximum $\max_{k \leq n} S_k$, so that a single estimate controls the walk over an entire block of times rather than at one endpoint. The next proposition supplies exactly this, at no cost in the exponent.

Proposition 8.1.3: Exponential Maximal Inequality

Let $X_1, \ldots, X_n$ be independent mean-zero subgaussian variables satisfying (8.1), with partial sums $S_k = X_1 + \cdots + X_k$. Then for every $\lambda > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} S_k \geq \lambda\right) \leq e^{-\lambda^2/(2n)}$$

***Proof*:**
Fix $t > 0$; it will be optimized at the end. The idea is that the exponential e^{tS_k} turns the additive walk into a nonnegative submartingale, to which a first-passage decomposition in the manner of Kolmogorov's maximal inequality (theorem 4.1.1) applies.

Step 1: The exponential process is a submartingale. Let $\mathcal{F}_k = \sigma(X_1, \ldots, X_k)$ and $M_k = e^{tS_k}$. Since X_{k+1} is independent of $\mathcal{F}_k$ and $\mathbb{E}[e^{tX_{k+1}}] \geq e^{t\mathbb{E}[X_{k+1}]} = 1$ by Jensen's inequality applied to the convex function $x \mapsto e^{tx}$,

$$\mathbb{E}[M_{k+1} | \mathcal{F}_k] = M_k\, \mathbb{E}\big[e^{tX_{k+1}}\big] \geq M_k$$

so (M_k) is a nonnegative submartingale. Applying (8.1) to each factor gives the terminal control

$$\mathbb{E}[M_n] = \prod_{i=1}^{n} \mathbb{E}\big[e^{tX_i}\big] \leq e^{t^2 n/2} \tag{8.3}$$

Step 2: First-passage decomposition. Let $a = e^{t\lambda} > 0$. Because $x \mapsto e^{tx}$ is increasing, the event $\{\max_{k\le n} S_k \ge \lambda\}$ coincides with $\{\max_{k\le n} M_k \ge a\}$. Let τ be the first time the running maximum reaches a,

$$\tau = \min\{k \le n \colon M_k \ge a\}$$

with the convention $\tau = \infty$ if the level is never reached, and decompose the event according to the value of τ:

$$\left\{\max_{k\le n} M_k \ge a\right\} = \bigsqcup_{j=1}^{n} \{\tau = j\}$$

a disjoint union because τ takes exactly one value on the event. On $\{\tau = j\}$ one has $M_j \ge a$, and this event lies in $\mathcal{F}_j$ (it is determined by $X_1, \dots, X_j$). The submartingale property, iterated from time j to time n, gives $\mathbb{E}[M_n | \mathcal{F}_j] \ge M_j$, so

$$\mathbb{E}\big[M_n\, \mathbb{1}_{\{\tau=j\}}\big] = \mathbb{E}\big[\mathbb{E}[M_n|\mathcal{F}_j]\, \mathbb{1}_{\{\tau=j\}}\big] \ge \mathbb{E}\big[M_j\, \mathbb{1}_{\{\tau=j\}}\big] \ge a\, \mathbb{P}(\tau = j)$$

the first equality because $\{\tau = j\} \in \mathcal{F}_j$. Summing over $j = 1, \dots, n$ and using that the events $\{\tau = j\}$ are disjoint subsets of the sample space,

$$a\, \mathbb{P}\left(\max_{k\le n} M_k \ge a\right) = \sum_{j=1}^{n} a\, \mathbb{P}(\tau = j) \le \sum_{j=1}^{n} \mathbb{E}\big[M_n\, \mathbb{1}_{\{\tau=j\}}\big] \le \mathbb{E}[M_n]$$

the last step because the indicators sum to at most 1 and $M_n \ge 0$. This is the submartingale form of Kolmogorov's maximal inequality: the running maximum of M_k is controlled by the terminal expectation $\mathbb{E}[M_n]$.

Step 3: Optimize. Dividing by $a = e^{t\lambda}$ and inserting the bound (8.3) on $\mathbb{E}[M_n]$,

$$\mathbb{P}\left(\max_{k\le n} S_k \ge \lambda\right) = \mathbb{P}\left(\max_{k\le n} M_k \ge a\right) \le \frac{\mathbb{E}[M_n]}{a} \le e^{-t\lambda + t^2 n/2}$$

This holds for every $t > 0$. The exponent $-t\lambda + t^2 n/2$ is a quadratic in t minimized at $t = \lambda/n$, where it takes the value $-\lambda^2/(2n)$. Substituting gives $\mathbb{P}(\max_{k\le n} S_k \ge \lambda) \le e^{-\lambda^2/(2n)}$, as we sought to show.

The inequality is the exact Gaussian tail (8.2) but for the running maximum rather than the terminal value, with no loss in the exponent. This is the feature that makes it usable: the price of controlling the entire path $S_1, \dots, S_n$ instead of the single point S_n is nothing at all. The mechanism is that the exponential e^{tS_k} converts the walk into a submartingale, and a submartingale's running maximum is governed by its terminal expectation through the first-passage decomposition, exactly as in Kolmogorov's inequality (theorem 4.1.1) for the second moment. Here the second moment is replaced by the exponential moment, which is what produces a Gaussian rather than a polynomial tail. Without this upgrade the subsequence argument below would control the walk only at the times n_k and leave the blocks between them uncontrolled; with it, a single term of the Borel-Cantelli sum handles a whole block.

Everything is now in place for the upper half of the law of the iterated logarithm. The strategy is to choose a geometric subsequence of times, bound the probability that the walk exceeds $(1+\varepsilon)b_{n_k}$ anywhere in the k-th block using proposition 8.1.3, check that these probabilities sum, and conclude by the first Borel-Cantelli lemma that overshoots stop happening. The subgaussian tail is what makes the sum converge: the exponent $(1+\varepsilon)^2 \log\log n_k$ turns into a power $k^{-(1+\varepsilon)^2}$ of the index, and $(1+\varepsilon)^2 > 1$ is exactly the summability threshold.

Theorem 8.1.4: Upper Bound in the Law of the Iterated Logarithm

Under the standing hypotheses of definition 8.1.1 (i.i.d. mean-zero variance-one subgaussian steps),

$$\limsup_{n\to\infty} \frac{S_n}{b_n} \le 1 \qquad \text{almost surely}$$

where $b_n = \sqrt{2n \log\log n}$.

***Proof*:**
Fix $\varepsilon > 0$ and a ratio $c > 1$; both will be sent to their limits at the end. Set $n_k = \lfloor c^k \rfloor$, an increasing sequence of times with $n_k \to \infty$ and $n_k/n_{k-1} \to c$ as $k \to \infty$. The plan is to show that the walk exceeds the level $(1+\varepsilon)b_{n_k}$ somewhere in the block of times $(n_{k-1}, n_k]$ only finitely often.

Step 1: The exceptional events and their probabilities. For k large enough that $n_{k-1} \ge 3$, define

$$A_k = \left\{ \max_{n_{k-1} < n \le n_k} S_n \ge (1+\varepsilon) b_{n_k} \right\}$$

Enlarging the maximum to run over all of $1 \le n \le n_k$ only increases the probability, so proposition 8.1.3 with $\lambda = (1+\varepsilon)b_{n_k}$ and block length n_k gives

$$\mathbb{P}(A_k) \le \mathbb{P}\left(\max_{n \le n_k} S_n \ge (1+\varepsilon) b_{n_k} \right) \le \exp\left(-\frac{(1+\varepsilon)^2 b_{n_k}^2}{2n_k} \right)$$

Now $b_{n_k}^2 = 2n_k \log\log n_k$, so the ratio $b_{n_k}^2/(2n_k) = \log\log n_k$, and the bound simplifies to

$$\mathbb{P}(A_k) \le \exp\big(-(1+\varepsilon)^2 \log\log n_k \big) = \big(\log n_k\big)^{-(1+\varepsilon)^2} \tag{8.4}$$

Step 2: Summability. As $k \to \infty$ one has $n_k = \lfloor c^k \rfloor$, hence $\log n_k = k \log c + O(1)$, and so $\log n_k \ge \frac{1}{2} k \log c$ for all large k. Feeding this into (8.4),

$$\mathbb{P}(A_k) \le \big(\log n_k\big)^{-(1+\varepsilon)^2} \le \big(\tfrac{1}{2} k \log c\big)^{-(1+\varepsilon)^2}$$

which is a constant multiple of $k^{-(1+\varepsilon)^2}$. Because $(1+\varepsilon)^2 > 1$, the series $\sum_k k^{-(1+\varepsilon)^2}$ converges, and therefore

$$\sum_k \mathbb{P}(A_k) < \infty$$

This is the one place the subgaussian tail is spent: the squared exponent $(1+\varepsilon)^2$ is what clears the summability threshold 1, and a polynomial (Chebyshev, corollary 3.2.5) tail would not.

Step 3: Borel-Cantelli and the block bound. By the first Borel-Cantelli lemma (theorem 1.4.5), $\mathbb{P}(A_k$ infinitely often$) = 0$; that is (definition 1.4.1), almost surely only finitely many A_k occur. Fix a sample point outside this null set. Then there is a (path-dependent) index K such that for all $k > K$ and all n with $n_{k-1} < n \le n_k$,

$$S_n < (1+\varepsilon) b_{n_k} \tag{8.5}$$

Every integer $n > n_K$ lies in exactly one block $(n_{k-1}, n_k]$, so (8.5) controls S_n for all large n, with the normalizer evaluated at the block endpoint n_k rather than at n itself.

Step 4: Compare b_{n_k} with b_n and take limits. For n in the block $(n_{k-1}, n_k]$ the normalizer b_{n_k} exceeds b_n only by the block ratio. Writing b_{n_k}/b_n and using $b_m^2 = 2m \log\log m$,

$$\frac{b_{n_k}^2}{b_n^2} = \frac{n_k}{n} \cdot \frac{\log\log n_k}{\log\log n} \le \frac{n_k}{n_{k-1}} \cdot \frac{\log\log n_k}{\log\log n_{k-1}}$$

As $k \to \infty$ the first factor tends to c and the second tends to 1 (since $\log\log n_k / \log\log n_{k-1} \to 1$, the double logarithm growing slower than any ratio of the underlying times). Hence $\limsup_k b_{n_k}/b_n \le \sqrt{c}$ uniformly over n in the k-th block. Combining with (8.5), for every large n,

$$\frac{S_n}{b_n} < (1+\varepsilon)\frac{b_{n_k}}{b_n}$$

and taking $\limsup_{n\to\infty}$,

$$\limsup_{n\to\infty} \frac{S_n}{b_n} \le (1+\varepsilon)\sqrt{c}$$

almost surely. This bound holds for every $\varepsilon > 0$ and every $c > 1$ simultaneously outside a single null set (take a countable sequence $\varepsilon_j \downarrow 0$, $c_j \downarrow 1$ and union the null sets). Letting $\varepsilon \downarrow 0$ and $c \downarrow 1$ yields $\limsup_n S_n/b_n \le 1$ almost surely, completing the proof.

The theorem says that from some finite time onward, no matter how the path has behaved, the walk never again rises above $(1+\varepsilon)\sqrt{2n \log\log n}$. This is a true almost-sure ceiling on a single trajectory, not a statement about the distribution at a fixed time, and it is strictly finer than the central limit theorem: the Gaussian scale $\sqrt{n}$ is exceeded infinitely often, but the slightly larger scale $\sqrt{2n \log\log n}$ is not. The extra factor $\sqrt{2\log\log n}$ over the CLT scale is precisely the correction that separates "large in distribution" from "large along the whole path". Two features of the proof deserve emphasis. First, the subsequence $n_k = \lfloor c^k \rfloor$ is geometric because only a geometric spacing makes the tail probabilities summable while keeping the blocks short enough that b_{n_k} and b_n agree in the limit; an arithmetic subsequence would fail on both counts. Second, the maximal inequality is what lets one term of the sum cover a whole block: without it the argument would bound S_{n_k} at the subsequence times but say nothing about the overshoots in between, and the theorem is a statement about all n, not just the n_k.

The matching lower bound, that the lim sup is at least 1, is proved in section 8.2 and requires a different engine, the second Borel-Cantelli lemma applied to independent blocks; together the two halves give the law of the iterated logarithm proper.

The upper bound for the running example is the concrete statement to carry forward.

Example 8.1.5: The Upper Envelope of the ± 1 Walk

Let S_n be the simple symmetric random walk on $\mathbb{Z}$, the partial sums of the ± 1 steps of example 1.3.6. By example 8.1.2 these steps satisfy the standing hypotheses of definition 8.1.1, so theorem 8.1.4 applies verbatim: for every $\varepsilon > 0$, almost surely

$$S_n \le (1+\varepsilon)\sqrt{2n \log\log n} \qquad \text{for all sufficiently large } n$$

Contrast the three scales the reader has now met for this same walk. The strong law (theorem 5.2.3) gives $S_n/n \to 0$, so S_n is of smaller order than n. The central limit theorem (theorem 7.1.3) gives $S_n/\sqrt{n} \Rightarrow \mathcal{N}(0,1)$, fixing the distributional scale at $\sqrt{n}$; but along a fixed path $\limsup_n S_n/\sqrt{n} =$

$+\infty$, the $\sqrt{n}$ ceiling being breached infinitely often. The law of the iterated logarithm interposes the exact almost-sure scale $\sqrt{2n\log\log n}$: at $n = 10^6$ the CLT scale is $\sqrt{n} = 1000$, while the iterated-logarithm envelope is $\sqrt{2n\log\log n} = \sqrt{2\cdot 10^6 \cdot \log\log 10^6} \approx \sqrt{2\cdot 10^6 \cdot 2.62} \approx 2290$, larger by the factor $\sqrt{2\log\log n} \approx 2.29$. The double logarithm grows so slowly that this factor is never large in practice, yet it is exactly the correction the almost-sure envelope requires.

Exercise 8.1.1

1. Prove the one-point Gaussian tail bound (8.2) directly from the Chernoff bound (proposition 3.4.7) and the subgaussian hypothesis (8.1), showing that the infimum over $t \geq 0$ is attained at $t = \lambda/n$. Explain in one sentence why proposition 8.1.3 is strictly stronger than this.
2. Suppose the steps satisfy the more general subgaussian bound $\mathbb{E}[e^{tX_i}] \leq e^{\sigma^2 t^2/2}$ with variance proxy σ^2 (not necessarily 1). Redo the proof of proposition 8.1.3 to show $\mathbb{P}(\max_{k\leq n} S_k \geq \lambda) \leq e^{-\lambda^2/(2\sigma^2 n)}$, and identify the value of σ^2 for which the normalizer b_n in theorem 8.1.4 must be replaced by $\sqrt{2\sigma^2 n\log\log n}$.
3. In Step 2 of theorem 8.1.4 the summability of $\sum_k \mathbb{P}(A_k)$ used $(1+\varepsilon)^2 > 1$. Show by exhibiting the divergent series that if one tried to run the argument at level b_{n_k} itself (that is, with $\varepsilon = 0$), the first Borel-Cantelli lemma would give no conclusion. What does this failure at $\varepsilon = 0$ suggest about the correct value of $\limsup_n S_n/b_n$?
4. Explain why the subsequence $n_k = \lfloor c^k \rfloor$ cannot be replaced by an arithmetic subsequence $n_k = k$. Address both failures: the tail sum $\sum_k \mathbb{P}(\max_{n\leq k} S_n \geq (1+\varepsilon)b_k)$, and the comparison of b_{n_k} with b_n across a block. (Consider what $\sum_k (\log k)^{-(1+\varepsilon)^2}$ does.)
5. Deduce from theorem 8.1.4 that $\limsup_n |S_n|/b_n \leq 1$ almost surely. (Apply the theorem to both S_n and $-S_n$; note that $-X_i$ satisfies the same standing hypotheses.)
6. Let X_i be i.i.d. standard Gaussian rather than ± 1. Verify that the Gaussian satisfies (8.1) with equality, so theorem 8.1.4 applies. Why does this show that the upper bound cannot be improved beyond a constant by strengthening the tail hypothesis?

8.2 The Lower Bound and the Law of the Iterated Logarithm

The upper bound of section 8.1 confined the walk beneath the envelope $b_n = \sqrt{2n\log\log n}$: for every $\varepsilon > 0$, the walk eventually stays below $(1+\varepsilon)b_n$ almost surely. That half of the story used only that large deviations are rare, packaged through the first Borel-Cantelli lemma. It says nothing about whether the walk ever comes close to the envelope. A random walk that oscillated only within $\frac{1}{2}b_n$ would satisfy the upper bound just as well, and nothing seen so far rules it out.

This section closes the gap from below: the walk does approach the envelope, infinitely often. The mechanism is the other half of the Borel-Cantelli dichotomy first met in section 1.4. Where the upper bound summed probabilities and concluded that an unwanted event happens only finitely often, the lower bound will exhibit events with a *divergent* probability sum on which the walk is large, and independence will force those events to occur infinitely often. The two lemmas are thus the two structural halves of the whole theorem: convergence of a probability series for the ceiling, divergence of one (plus independence) for the floor. What makes divergence usable is that the walk's increments over disjoint blocks of time are independent, so the second Borel-Cantelli lemma (theorem 1.4.6) applies to them directly.

To run that argument, a large-deviation estimate is needed in the opposite direction to the maximal inequality of section 8.1: a lower bound on the probability that a block sum is large, matching the Gaussian tail $e^{-\lambda^2/(2n)}$ to leading exponential order. This is the content of the first result below.

The exponential maximal inequality $\mathbb{P}(\max_{k\le n} S_k \ge \lambda) \le e^{-\lambda^2/(2n)}$ of section 8.1 is an *upper* bound on the tail, and it is exactly the size of the standard Gaussian tail. For the lower bound to have any chance of being sharp, the true tail $\mathbb{P}(S_n \ge \lambda)$ must be at least as large as the Gaussian tail as well, at least to leading order in the exponent. That the two Gaussian bounds pinch together is precisely what the central limit theorem provides in the relevant range of λ. The next proposition records the lower half.

Proposition 8.2.1: Matching Lower Tail for Block Sums

Let $X_1, X_2, \dots$ be i.i.d. with mean 0, variance 1, and subgaussian with $\mathbb{E}[e^{tX_1}] \le e^{t^2/2}$, and set $S_n = X_1 + \cdots + X_n$. Then, for any sequence $\lambda_n \to \infty$ with $\lambda_n = o(n^{2/3})$,

$$\mathbb{P}(S_n \ge \lambda_n) \ge \exp\left(-(1+o(1))\,\frac{\lambda_n^2}{2n}\right) \tag{8.6}$$

as $n \to \infty$, where the $o(1)$ is a deterministic sequence tending to 0 that depends only on the law of X_1 and on the sequence (λ_n) through the ratio $\lambda_n/\sqrt{n}$.

***Proof*:**
Write $x_n = \lambda_n/\sqrt{n}$, so that $\mathbb{P}(S_n \ge \lambda_n) = \mathbb{P}(S_n/\sqrt{n} \ge x_n)$. The range $\lambda_n = o(n^{2/3})$ is exactly $x_n = o(n^{1/6})$; the normalized threshold x_n may grow, but slowly relative to $\sqrt{n}$. This is the *moderate deviation* regime, the window in which the central limit theorem still governs the tail.

Step 1: The Gaussian target. Let Φ denote the standard normal distribution function and $\bar{\Phi} = 1 - \Phi$ its tail. The elementary lower bound for the Gaussian tail states that for $x > 0$,

$$\bar{\Phi}(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\,du \ge \frac{1}{\sqrt{2\pi}}\left(\frac{1}{x} - \frac{1}{x^3}\right) e^{-x^2/2} \tag{8.7}$$

obtained from the identity $\frac{d}{du}\big(-(u^{-1} - u^{-3})e^{-u^2/2}\big) = (1 - 3u^{-4})e^{-u^2/2} \le e^{-u^2/2}$: integrating both sides over $[x,\infty)$ and evaluating the exact left side gives $(x^{-1} - x^{-3})e^{-x^2/2} \le \int_x^\infty e^{-u^2/2}\,du$. Taking logarithms in (8.7) and using $\log(1/x - 1/x^3) = -\log x + o(1)$ gives

$$\log \bar{\Phi}(x) = -\frac{x^2}{2} - \log x + O(1) = -(1+o(1))\frac{x^2}{2} \tag{8.8}$$

as $x \to \infty$, since $\log x = o(x^2)$. Thus, to leading order in the exponent, the standard Gaussian tail at height x_n is $\exp(-(1+o(1))x_n^2/2)$, which is exactly the right-hand side of (8.6) with $x_n^2 = \lambda_n^2/n$.

Step 2: Transfer to S_n through the central limit theorem. By the central limit theorem (theorem 7.1.3), $S_n/\sqrt{n}$ converges in distribution to a standard normal. For a *fixed* threshold x

this gives $\mathbb{P}(S_n/\sqrt{n} \geq x) \to \bar{\Phi}(x)$, which combined with (8.8) already yields (8.6) whenever $\lambda_n/\sqrt{n}$ stays bounded. The point of the moderate-deviation range $x_n = o(n^{1/6})$ is that the same comparison survives when the threshold is allowed to grow: there is a deterministic $\delta_n \to 0$ with

$$\mathbb{P}\left(\frac{S_n}{\sqrt{n}} \geq x_n\right) \geq \bar{\Phi}(x_n)\, e^{-\delta_n x_n^2} \tag{8.9}$$

for all large n. This is the one estimate whose full proof lies beyond the tools assembled in this chapter: it is a *moderate-deviation* statement, a quantitative refinement of the central limit theorem asserting that the relative error in the Gaussian approximation to the tail stays negligible in the exponent throughout the window $x_n = o(n^{1/6})$. Under the subgaussian (indeed finite exponential moment) hypothesis in force here, this is the Cramer-type moderate-deviation theorem; a statement and proof, via the tilted (exponentially changed) measure, is given by Petrov, *Sums of Independent Random Variables*, Chapter VIII, and by Stout, *Almost Sure Convergence*, in the course of the law of the iterated logarithm itself. The mechanism is worth naming even without the full proof: tilting the law of X_1 by the factor $e^{tX}/\mathbb{E}[e^{tX}]$ with $t = x_n/\sqrt{n}$ recenters the sum so that the threshold λ_n becomes the new mean, turning a tail probability into a central one to which the central limit theorem applies directly, and the subgaussian bound controls the tilt.

Step 3: Assemble. Combining (8.9) with the logarithmic form (8.8) at $x = x_n$,

$$\log \mathbb{P}(S_n \geq \lambda_n) \geq \log \bar{\Phi}(x_n) - \delta_n x_n^2 = -(1 + o(1))\frac{x_n^2}{2} - \delta_n x_n^2 = -(1 + o(1))\frac{x_n^2}{2}$$

because $\delta_n \to 0$ folds into the $o(1)$. Exponentiating and rewriting $x_n^2 = \lambda_n^2/n$ gives (8.6), as we sought to show.

The estimate (8.6) is the mirror image of the maximal inequality $\mathbb{P}(\max_{k \leq n} S_k \geq \lambda) \leq e^{-\lambda^2/(2n)}$ from section 8.1. There the Gaussian tail was an upper bound on the whole maximum; here it is a lower bound on a single sum. Together they pin the logarithm of the tail to $-\lambda^2/(2n)$ up to a factor $1 + o(1)$, which is the precision the law of the iterated logarithm requires. The restriction $\lambda = o(n^{2/3})$ is not an artifact: at the LIL scale $\lambda \asymp b_n = \sqrt{2n \log \log n}$ one has $x_n = \sqrt{2 \log \log n}$, which grows far more slowly than $n^{1/6}$, so the walk's fluctuations sit comfortably inside the moderate-deviation window where the Gaussian approximation is trustworthy in the exponent.

The strategy for the lower half of the theorem is now visible. Chop time into consecutive blocks along a geometric sequence $n_k = \lfloor c^k \rfloor$, so that the block sum $B_k = S_{n_k} - S_{n_{k-1}}$ has variance $n_k - n_{k-1}$. If c is large this variance is close to n_k, so a block sum of size $(1 - \varepsilon)b_{n_k}$ is a moderate deviation of just under the full envelope, and proposition 8.2.1 gives it a probability large enough that, summed over k, the series diverges. The blocks being over disjoint time intervals, the B_k are independent (definition 2.2.2), and the second Borel-Cantelli lemma converts the divergent sum into: infinitely many blocks are large. A large block forces S_{n_k} itself to be large, provided the accumulated sum $S_{n_{k-1}}$ at the start of the block does not cancel it, and the upper bound of section 8.1 controls exactly that.

Theorem 8.2.2: The Lower Bound

Let $X_1, X_2, \dots$ be i.i.d. with mean 0, variance 1, and subgaussian with $\mathbb{E}[e^{tX_1}] \le e^{t^2/2}$, and let $S_n = X_1 + \cdots + X_n$ with normalizer $b_n = \sqrt{2n \log\log n}$ from definition 8.1.1. Then

$$\limsup_{n\to\infty} \frac{S_n}{b_n} \ge 1 \qquad \text{almost surely}$$

***Proof*:**
Fix $\varepsilon \in (0,1)$ and a large real $c > 1$, and set $n_k = \lfloor c^k \rfloor$ for $k \ge 1$. The two parameters will be sent to their limits ($\varepsilon \downarrow 0$ and $c \to \infty$) only at the very end; for now they are fixed. Define the block increments

$$B_k = S_{n_k} - S_{n_{k-1}} = \sum_{i=n_{k-1}+1}^{n_k} X_i$$

for $k \ge 2$. Since the B_k are sums over disjoint index sets of the independent sequence (X_i), they are themselves independent (definition 2.2.2), and B_k is a sum of

$$m_k := n_k - n_{k-1} = (1 + o(1))\, n_k \left(1 - \frac{1}{c}\right)$$

i.i.d. copies of X_1, so $\mathrm{Var}(B_k) = m_k$ and B_k has the same subgaussian law as the partial sum S_{m_k}. The whole argument compares B_k at the threshold $(1-\varepsilon)b_{n_k}$.

Step 1: The block events have divergent probability. Consider the events

$$E_k = \{B_k \ge (1-\varepsilon)\, b_{n_k}\} \qquad (k \ge 2)$$

Apply proposition 8.2.1 to B_k, a sum of m_k steps, at the threshold $\lambda = (1-\varepsilon)b_{n_k}$. First check that λ lies in the moderate-deviation window $\lambda = o(m_k^{2/3})$: since $b_{n_k} = \sqrt{2n_k \log\log n_k}$ and $m_k \asymp n_k$, the ratio $\lambda/m_k^{2/3} \asymp n_k^{1/2}(\log\log n_k)^{1/2}/n_k^{2/3} = n_k^{-1/6}(\log\log n_k)^{1/2} \to 0$. Hence (8.6) applies, giving

$$\mathbb{P}(E_k) \ge \exp\left(-(1+o(1))\frac{(1-\varepsilon)^2 b_{n_k}^2}{2m_k}\right)$$

Substitute $b_{n_k}^2 = 2n_k \log\log n_k$ and $m_k = (1+o(1))n_k(1-1/c)$:

$$\frac{(1-\varepsilon)^2 b_{n_k}^2}{2m_k} = (1+o(1))\,\frac{(1-\varepsilon)^2}{1-1/c}\,\log\log n_k$$

Write $\theta := \dfrac{(1-\varepsilon)^2}{1-1/c}$ for the constant governing the exponent. Because $n_k = \lfloor c^k \rfloor$, we have $\log n_k = (1+o(1))k \log c$ and hence $\log\log n_k = (1+o(1))\log k$, so

$$\mathbb{P}(E_k) \ge \exp\big(-(1+o(1))\,\theta\,\log\log n_k\big) = \exp\big(-(1+o(1))\,\theta\,\log k\big) = k^{-(1+o(1))\theta}$$

The argument now requires $\theta < 1$. For fixed ε small, $(1-\varepsilon)^2 < 1$, and $1 - 1/c \to 1$ as $c \to \infty$; so for c large enough (depending on ε) one has $\theta < 1$. With $\theta < 1$ the exponent $-(1+o(1))\theta$ exceeds

-1 for all large k, and the series $\sum_k k^{-(1+o(1))\theta}$ diverges by comparison with the harmonic series. Therefore

$$\sum_{k\geq 2} \mathbb{P}(E_k) = \infty \tag{8.10}$$

Step 2: The blocks are large infinitely often. The events E_k are defined on the independent random variables B_k, so they are mutually independent. By the second Borel-Cantelli lemma (theorem 1.4.6), the divergence (8.10) forces

$$\mathbb{P}(E_k \text{ occurs for infinitely many } k) = 1$$

That is, almost surely $B_k \geq (1-\varepsilon)b_{n_k}$ for infinitely many k.

Step 3: Transfer from the block B_k to the partial sum S_{n_k}. The block sum splits as $S_{n_k} = B_k + S_{n_{k-1}}$, so on the event E_k,

$$\frac{S_{n_k}}{b_{n_k}} \geq (1-\varepsilon) - \frac{|S_{n_{k-1}}|}{b_{n_k}} \tag{8.11}$$

It remains to control the correction $|S_{n_{k-1}}|/b_{n_k}$. Apply the upper bound (theorem 8.1.4) both to the walk S_n and to its reflection $-S_n$ (whose steps $-X_i$ satisfy the same hypotheses): almost surely $\limsup_n S_n/b_n \leq 1$ and $\limsup_n(-S_n)/b_n \leq 1$, hence $\limsup_n |S_n|/b_n \leq 1$. In particular, almost surely $|S_{n_{k-1}}| \leq 2b_{n_{k-1}}$ for all large k. The ratio of consecutive normalizers is

$$\frac{b_{n_{k-1}}}{b_{n_k}} = \sqrt{\frac{n_{k-1}\log\log n_{k-1}}{n_k \log\log n_k}} = (1+o(1))\sqrt{\frac{n_{k-1}}{n_k}} = (1+o(1))\frac{1}{\sqrt{c}} \tag{8.12}$$

since $n_{k-1}/n_k \to 1/c$ and $\log\log n_{k-1}/\log\log n_k \to 1$. Therefore, almost surely for all large k,

$$\frac{|S_{n_{k-1}}|}{b_{n_k}} \leq 2\,\frac{b_{n_{k-1}}}{b_{n_k}} = (1+o(1))\frac{2}{\sqrt{c}}$$

Step 4: Combine and take limits. Fix a sample point in the almost-sure intersection of the two events (infinitely many E_k occur, and $|S_{n_{k-1}}| \leq 2b_{n_{k-1}}$ eventually). Along the infinite subsequence of k on which E_k holds, (8.11) and Step 3 give

$$\frac{S_{n_k}}{b_{n_k}} \geq (1-\varepsilon) - (1+o(1))\frac{2}{\sqrt{c}}$$

infinitely often, and hence

$$\limsup_{n\to\infty} \frac{S_n}{b_n} \geq \limsup_{k\to\infty} \frac{S_{n_k}}{b_{n_k}} \geq (1-\varepsilon) - \frac{2}{\sqrt{c}} \qquad \text{almost surely}$$

This holds almost surely for every fixed admissible pair (ε, c). Take a countable sequence of pairs with $\varepsilon \downarrow 0$ and $c \to \infty$ (say $\varepsilon = 1/j$ and $c = j^2$, keeping $\theta < 1$ for all large j); the countable intersection of the corresponding almost-sure events still has probability 1, and on it

$$\limsup_{n\to\infty} \frac{S_n}{b_n} \geq \sup_j \left((1-1/j) - \frac{2}{j} \right) = 1$$

completing the proof.

The proof is the exact structural counterpart of the upper bound. There the events $A_k = \{\max_n S_n \geq (1+\varepsilon)b_{n_k}\}$ had a *summable* probability series and the first Borel-Cantelli lemma made them stop; here the events $E_k = \{B_k \geq (1-\varepsilon)b_{n_k}\}$ have a *divergent* series and the second Borel-Cantelli lemma makes them recur. The difference in the hypotheses is independence: the first lemma needs none, but the second requires it, and it is available precisely because the blocks B_k live over disjoint stretches of time. This is why the argument is forced to work with block increments rather than the partial sums S_{n_k} directly, which are strongly dependent. The geometric spacing $n_k = \lfloor c^k \rfloor$ does double duty: it makes consecutive normalizers differ by the fixed factor $\sqrt{c}$ (so the leftover $S_{n_{k-1}}$ is a controllable $2/\sqrt{c}$ fraction of b_{n_k}), and it makes $\log\log n_k$ grow like $\log k$, which is what turns the exponent into a power of k and lets the divergence be read off. Sending $c \to \infty$ shrinks the correction $2/\sqrt{c}$ to zero at the cost of nothing, because the divergence in Step 1 held for every large c.

The two bounds meet. Theorem 8.1.4 caps the lim sup at 1 and theorem 8.2.2 floors it at 1, so the lim sup equals 1 exactly. Reflecting the walk turns the statement about the lim sup into one about the lim inf, and the full theorem follows. It is the sharpest almost-sure description of the size of a random walk, and the promised resolution of the foreshadowing in box 1.4.8.

Theorem 8.2.3: The Law of the Iterated Logarithm

Let $X_1, X_2, \ldots$ be i.i.d. with mean 0, variance 1, and subgaussian with $\mathbb{E}[e^{tX_1}] \leq e^{t^2/2}$, and let $S_n = X_1 + \cdots + X_n$. Then, with $b_n = \sqrt{2n \log\log n}$,

$$\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n\log\log n}} = 1 \qquad \text{and} \qquad \liminf_{n\to\infty} \frac{S_n}{\sqrt{2n\log\log n}} = -1$$

almost surely. Consequently, for every $\varepsilon > 0$, almost surely $|S_n| \leq (1+\varepsilon)b_n$ for all large n, while $S_n \geq (1-\varepsilon)b_n$ and $S_n \leq -(1-\varepsilon)b_n$ each occur for infinitely many n.

***Proof*:**
The lim sup statement is the combination of the two preceding results: theorem 8.1.4 gives $\limsup_n S_n/b_n \leq 1$ and theorem 8.2.2 gives $\limsup_n S_n/b_n \geq 1$, so the two are equal to 1 almost surely. For the lim inf, apply the lim sup statement to the walk $\widetilde{S}_n = -S_n$, whose steps $-X_i$ are again i.i.d. with mean 0, variance 1, and the same subgaussian bound $\mathbb{E}[e^{t(-X_1)}] \leq e^{t^2/2}$ (the bound is symmetric in t). Thus $\limsup_n(-S_n)/b_n = 1$ almost surely, and since $\liminf_n S_n/b_n = -\limsup_n(-S_n)/b_n$, the lim inf equals -1 almost surely.

The two displayed consequences follow at once: $\limsup_n S_n/b_n = 1$ means $S_n \geq (1-\varepsilon)b_n$ infinitely often but $S_n \leq (1+\varepsilon)b_n$ eventually, and symmetrically for the lim inf; intersecting the four almost-sure events gives the stated envelope, as we sought to show.

The theorem describes a walk that forever brushes against a slowly widening envelope. The $\pm(1+\varepsilon)b_n$ curves are eventual barriers the walk never crosses, yet the $\pm(1-\varepsilon)b_n$ curves are touched again and again, from both sides. What the theorem as proven does *not* decide is the behavior at the exact curve $\pm b_n$: knowing only that $\limsup_n S_n/b_n = 1$ leaves open whether b_n itself is crossed infinitely often, since a lim sup equal to 1 is compatible both with the exact envelope being exceeded infinitely often and with its never being reached. The envelope is sharp in the limiting sense that any factor $1-\varepsilon$ below it is exceeded infinitely often and any factor $1+\varepsilon$ above it is eventually respected; the finer question of the exact curve is governed by a more delicate integral test lying beyond this chapter (the outcome, for the envelope $b_n = \sqrt{2n \log\log n}$, is that b_n is in fact crossed infinitely often). The doubly iterated logarithm in the normalizer sets the scale: $\log\log n$ grows so slowly that at $n = 10^6$ it is only about 2.6, and the correction it applies to the central-limit scale $\sqrt{n}$ is a factor near $\sqrt{2 \cdot 2.6} \approx 2.3$. The random walk exceeds its typical size $\sqrt{n}$ by this modest but unbounded factor infinitely often, and never by more.

The subgaussian hypothesis is doing real work here, but only in the lower bound and only through the moderate-deviation estimate proposition 8.2.1; the upper bound of section 8.1 used the subgaussian tail directly. It is a theorem of Hartman and Wintner that the same conclusion holds under the weakest possible hypothesis, finite variance alone.

Theorem 8.2.4: Hartman-Wintner

Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = \sigma^2 \in (0, \infty)$, and let $S_n = X_1 + \cdots + X_n$. Then

$$\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n \log\log n}} = \sigma \qquad \text{and} \qquad \liminf_{n\to\infty} \frac{S_n}{\sqrt{2n \log\log n}} = -\sigma$$

almost surely.

***Proof Key ideas*:**
The general finite-variance case is not proved here: it requires a truncation and approximation argument beyond the tools of this chapter. The steps X_i are truncated at a level growing with n, the moderate-deviation lower bound is recovered for the truncated (now bounded) variables while the contribution of the discarded tails is shown to be negligible on the b_n scale, and a Kolmogorov-type exponential inequality replaces the subgaussian bound throughout. The complete argument is in Stout, *Almost Sure Convergence*, and in Kallenberg, *Foundations of Modern Probability*. The subgaussian case proved above (theorem 8.2.3) is the $\sigma = 1$ instance and covers the running examples in full.

Normalizing by σ recovers the variance-one statement, so theorem 8.2.4 extends theorem 8.2.3 rather than restates it: it drops the exponential-moment hypothesis entirely, asking only what the central limit theorem itself asks. The finite second moment is also necessary: if $\mathbb{E}[X_1^2] = \infty$ then $\limsup_n S_n/b_n = +\infty$ almost surely, so no envelope of order b_n exists (this is the Strassen-Feller converse, again beyond the present scope). The law of the iterated logarithm is in this sense the exact almost-sure companion of the central limit theorem: both live on finite variance, one describing the distributional scale $\sqrt{n}$ and the other the almost-sure envelope $\sqrt{2n \log\log n}$.

The running example of the whole Part now receives its sharpest reading. The simple symmetric random walk is the case of ± 1 steps, and it satisfies every hypothesis of theorem 8.2.3.

Example 8.2.5: The Envelope of the Simple Symmetric Walk
Let $X_i = \pm 1$ with probability $\frac{1}{2}$ each, independent (example 1.3.6), and $S_n = X_1 + \cdots + X_n$ the position of the simple symmetric random walk after n steps. The steps have mean 0 and variance $\mathbb{E}[X_1^2] = 1$, and they are subgaussian with the sharp constant: for every real t,

$$\mathbb{E}[e^{tX_1}] = \frac{e^t + e^{-t}}{2} = \cosh t \le e^{t^2/2}$$

the inequality following termwise from $\cosh t = \sum_{k\ge 0} t^{2k}/(2k)!$ and $e^{t^2/2} = \sum_{k\ge 0} t^{2k}/(2^k k!)$ with

$(2k)! \geq 2^k k!$. Thus theorem 8.2.3 applies verbatim:

$$\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n\log\log n}} = 1 \qquad \text{and} \qquad \liminf_{n\to\infty} \frac{S_n}{\sqrt{2n\log\log n}} = -1 \qquad \text{almost surely}$$

The walk oscillates between the two curves $\pm\sqrt{2n\log\log n}$: for every $\varepsilon > 0$ it eventually stays inside $\pm(1+\varepsilon)\sqrt{2n\log\log n}$, and it exceeds $(1-\varepsilon)\sqrt{2n\log\log n}$ in the positive direction infinitely often and $-(1-\varepsilon)\sqrt{2n\log\log n}$ in the negative direction infinitely often.

A concrete consequence sharpens the strong law. The strong law of large numbers (theorem 5.2.3) gives $S_n/n \to 0$ almost surely, so the walk's *average* step tends to 0. The law of the iterated logarithm quantifies the residual: since $S_n = (1+o(1))\sqrt{2n\log\log n}$ infinitely often, the average S_n/n is as large as $(1+o(1))\sqrt{2\log\log n/n}$ infinitely often, decaying to 0 but never faster than this precise rate along the record-setting times.

One feature of theorem 8.2.3 deserves comment: the lim sup is a *deterministic* number, 1, even though S_n is random. That the lim sup takes a constant value with probability one, rather than a value that varies from path to path, is a structural fact, forced before any estimate is made.

Proposition 8.2.6: The Limsup Is Almost Surely Constant

For any i.i.d. sequence (X_i) with the normalizer $b_n = \sqrt{2n\log\log n}$, the random variable

$$L := \limsup_{n\to\infty} \frac{S_n}{b_n}$$

is measurable with respect to the tail σ-algebra (definition 2.4.1) of (X_i), and is therefore almost surely equal to a constant in $[-\infty, +\infty]$.

***Proof*:**
Fix $m \geq 1$. Changing the first m steps changes every partial sum $S_n = X_1 + \cdots + X_n$ by the fixed amount $D = X_1 + \cdots + X_m$, so for $n > m$,

$$\frac{S_n}{b_n} = \frac{D + (X_{m+1} + \cdots + X_n)}{b_n}$$

Since $b_n \to \infty$, the term $D/b_n \to 0$, and hence

$$L = \limsup_{n\to\infty} \frac{S_n}{b_n} = \limsup_{n\to\infty} \frac{X_{m+1} + \cdots + X_n}{b_n}$$

The right-hand side is a function of $(X_{m+1}, X_{m+2}, \dots)$ alone, so L is measurable with respect to $\sigma(X_{m+1}, X_{m+2}, \dots)$. As m was arbitrary, L lies in the intersection $\bigcap_m \sigma(X_{m+1}, X_{m+2}, \dots)$, which is the tail σ-algebra. By Kolmogorov's zero-one law (theorem 2.4.3), every tail event has probability 0 or 1, so the distribution of L is concentrated at a single point: L equals a constant almost surely, as we sought to show.

This is the structural reason the theorem *can* assert an equality. Without the zero-one law one could only hope for a statement about the distribution of L; with it, L is a constant, and the whole content of the law of the iterated logarithm is the evaluation of that constant. The upper bound identifies it as at most 1 and the lower bound as at least 1, and the zero-one law guarantees there is a single number to identify in the first place. The same remark explains why the four events in theorem 8.2.3 (exceeding $(1-\varepsilon)b_n$ or staying below $(1+\varepsilon)b_n$, in each direction) are all zero-one events: each is a tail event by the same first-m-steps argument, so the theorem's probabilities were always going to be 0 or 1, never anything in between.

Exercise 8.2.1

1. In the proof of theorem 8.2.2, the block sum B_k is compared to the threshold $(1-\varepsilon)b_{n_k}$ using proposition 8.2.1, which requires the threshold to lie in the moderate-deviation window $\lambda = o(m_k^{2/3})$ where $m_k = n_k - n_{k-1}$. Verify this in detail: show that $(1-\varepsilon)b_{n_k}/m_k^{2/3} \to 0$ as $k \to \infty$, and identify the exact power of n_k that governs the decay.

2. Suppose in the lower-bound proof one keeps ε fixed but does *not* send $c \to \infty$, taking instead a fixed c with $\theta = (1-\varepsilon)^2/(1-1/c) \geq 1$. Show that the divergence (8.10) may fail, so the argument breaks. Precisely: for which θ does $\sum_k k^{-\theta}$ converge, and what does this say about the necessity of taking c large?

3. The $\liminf$ half of theorem 8.2.3 was obtained by reflecting the walk. Give the direct argument instead: state the analogue of theorem 8.2.2 for the events $\{B_k \leq -(1-\varepsilon)b_{n_k}\}$ and explain, in one or two sentences, why every step of the lower-bound proof transfers without change. Where is the symmetry of the subgaussian bound $\mathbb{E}[e^{tX}] \leq e^{t^2/2}$ used?

4. Equation (8.12) claims $b_{n_{k-1}}/b_{n_k} = (1+o(1))c^{-1/2}$ for $n_k = \lfloor c^k \rfloor$. Prove it carefully, including the claim that $\log\log n_{k-1}/\log\log n_k \to 1$. Then explain why the factor 2 in the bound $|S_{n_{k-1}}| \leq 2b_{n_{k-1}}$ (rather than the sharper $(1+\varepsilon)b_{n_{k-1}}$) costs nothing in the final limit.

5. By proposition 8.2.6, $L = \limsup_n S_n/b_n$ is almost surely a constant $\ell \in [-\infty, +\infty]$ for *any* i.i.d. sequence. Show that if the X_i are additionally symmetric (i.e. X_i and $-X_i$ have the same law), then $\liminf_n S_n/b_n = -\ell$ almost surely. Deduce that the LIL envelope for a symmetric sequence is automatically symmetric about 0, and say why this does not by itself compute ℓ.

6. A bounded mean-zero variance-one step need not satisfy the normalized subgaussian bound $\mathbb{E}[e^{tX}] \leq e^{t^2/2}$ with constant 1. Give an example of a bounded, mean-zero, variance-one random variable X and a value of t for which $\mathbb{E}[e^{tX}] > e^{t^2/2}$. Explain why this does not obstruct applying theorem 8.2.4 to such X, and why the exponential maximal inequality of section 8.1 would need a different constant.

8.3 Application: Fluctuations in the Digits of a Random Real

The law of the iterated logarithm was proved for an abstract sequence of subgaussian steps, and its primary illustration was the ± 1 coin walk. That same walk has a second incarnation that turns the theorem into a statement about a single real number rather than a sequence of coin tosses: the binary digits of a point drawn uniformly from the unit interval. A number $x \in [0,1]$ carries an infinite string of 0s and 1s, and asking how the count of 1s among the first n digits behaves is the same as asking how the partial sums of a fair coin walk behave. This section reads the law of the iterated logarithm through that dictionary and, in doing so, sharpens a theorem already met in section 5.3: Borel's normal-number theorem (theorem 5.3.6), which says the digit frequency tends to $\frac{1}{2}$. Borel fixes the frequency; the law of the iterated logarithm fixes exactly how far the running count strays from its expected value along the way.

Fix $x \in [0,1]$ and write its binary expansion

$$x = \sum_{i=1}^{\infty} \frac{d_i(x)}{2^i} \qquad d_i(x) \in \{0,1\}$$

where, for the countably many dyadic rationals, the expansion terminating in 0s is chosen (the ambiguity is a Lebesgue-null set and affects nothing that follows). The map $x \mapsto (d_1(x), d_2(x), \dots)$ is the coordinate map of the coin-tossing model. The content that makes it a probabilistic object is recorded in example 1.1.3: under Lebesgue measure on $[0,1]$, which is the law of a uniform random variable, the digit functions $d_1, d_2, \dots$ form an i.i.d. sequence of $\mathrm{Ber}(1/2)$ random variables. This is the same fact that underlies Borel's theorem, and it is the hinge of the entire section: it lets a question about one random real become a question about a sum of independent ± 1 steps, to which the law of the iterated logarithm applies verbatim.

To measure how the digits deviate from a perfectly balanced string, center each digit at its mean $\frac{1}{2}$ and accumulate.

Definition 8.3.1: Centered Digit Count

For $x \in [0,1]$ with binary digits $d_i(x) \in \{0,1\}$, the *centered digit count* of the first n digits is

$$V_n(x) = \sum_{i=1}^{n} \Big(d_i(x) - \tfrac{1}{2} \Big)$$

Equivalently, if $N_n(x) = \sum_{i \le n} d_i(x)$ counts the 1s among the first n digits, then $V_n = N_n - \frac{n}{2}$, the excess of the observed 1-count over the balanced value $n/2$.

The quantity V_n is a running scoreboard for the tug-of-war between 0s and 1s: it increases by $\frac{1}{2}$ at each 1 and decreases by $\frac{1}{2}$ at each 0. A string with exactly half 1s among its first n digits scores $V_n = 0$; a surplus of k ones scores $V_n = k$. Because each centered digit $d_i - \frac{1}{2}$ takes the values $\pm\frac{1}{2}$ with equal probability, it has mean 0 and variance $\frac{1}{4}$, so V_n is a sum of n i.i.d. mean-zero steps of variance $\frac{1}{4}$. This is precisely the coin walk again, rescaled by a factor of two.

Example 8.3.2: The First Few Digits of a Specific Real

Take $x = \frac{11}{16} = 0.1011_2$, whose binary digits are $d_1 = 1, d_2 = 0, d_3 = 1, d_4 = 1$ and then all 0s. The centered digit count reads

$$V_1 = \tfrac{1}{2}, \quad V_2 = 0, \quad V_3 = \tfrac{1}{2}, \quad V_4 = 1$$

and $V_n = 1 - \frac{n-4}{2}$ for $n \ge 4$, drifting downward once the digits become all 0. This particular x is a dyadic rational, hence eventually constant in its digits and not typical of a uniform draw; it is displayed only to exhibit the bookkeeping. For a generic x the digits do not terminate, and V_n executes an unending fair walk whose size is the subject of this section.

The connection to the ± 1 walk is exact, not approximate. Recording it as an identity before the main theorem makes the rescaling in the proof transparent. Let

$$S_n = \sum_{i=1}^{n} (2d_i - 1)$$

be the partial sums of the steps $2d_i - 1 \in \{-1, +1\}$. Since $2d_i - 1 = 2(d_i - \frac{1}{2})$, summing gives $S_n = 2V_n$: the coin walk is exactly twice the centered digit count. The steps $2d_i - 1$ are i.i.d., take values ± 1 with probability $\frac{1}{2}$ each, have mean 0 and variance 1, and are subgaussian with $\mathbb{E}[e^{t(2d_i-1)}] = \cosh t \le e^{t^2/2}$, so they satisfy every hypothesis of the law of the iterated logarithm as proved in section 8.2.

Theorem 8.3.3: Law of the Iterated Logarithm for Binary Digits

Let x be uniform on $[0, 1]$, let $d_i(x) \in \{0, 1\}$ be its binary digits, and let $V_n = \sum_{i \le n} \big(d_i - \frac{1}{2}\big)$ be the centered digit count of definition 8.3.1. Then, almost surely,

$$\limsup_{n\to\infty} \frac{V_n}{\sqrt{\frac{1}{2}\, n \log\log n}} = 1 \qquad \text{and} \qquad \liminf_{n\to\infty} \frac{V_n}{\sqrt{\frac{1}{2}\, n \log\log n}} = -1$$

***Proof*:**
By example 1.1.3, applied via Borel's normal-number setup (theorem 5.3.6), the digits $d_1, d_2, \ldots$ are i.i.d. Ber(1/2) random variables on the probability space $([0, 1], \mathcal{B}, \lambda)$. Set $X_i = 2d_i - 1$. The X_i are then i.i.d., take the values ± 1 with probability $\frac{1}{2}$ each, and therefore have $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = 1$; they are subgaussian with $\mathbb{E}[e^{tX_i}] = \cosh t \le e^{t^2/2}$. These are exactly the standing hypotheses under which the law of the iterated logarithm (theorem 8.2.3) was established, so writing $S_n = \sum_{i \le n} X_i$ and $b_n = \sqrt{2n \log\log n}$ for the normalizer of definition 8.1.1, we have almost surely

$$\limsup_{n\to\infty} \frac{S_n}{b_n} = 1 \qquad \text{and} \qquad \liminf_{n\to\infty} \frac{S_n}{b_n} = -1$$

It remains to translate this from S_n to V_n. Since $X_i = 2d_i - 1 = 2\big(d_i - \frac{1}{2}\big)$, summation gives $S_n = 2V_n$, so $V_n = \frac{1}{2} S_n$. Dividing the coin-walk normalizer accordingly,

$$\frac{V_n}{\sqrt{\frac{1}{2}\, n \log\log n}} = \frac{\frac{1}{2} S_n}{\frac{1}{\sqrt{2}} \sqrt{n \log\log n}} = \frac{\frac{1}{2} S_n}{\frac{1}{2} \sqrt{2n \log\log n}} = \frac{S_n}{b_n}$$

the middle equality using $\sqrt{\frac{1}{2}\, n \log\log n} = \frac{1}{\sqrt{2}} \sqrt{n \log\log n} = \frac{1}{2} \sqrt{2n \log\log n} = \frac{1}{2} b_n$. The rescaled digit count is thus literally equal to S_n / b_n for every n, so its lim sup is 1 and its lim inf is -1 almost surely, as we sought to show.

The scale $\sqrt{\frac{1}{2}\, n \log\log n}$ that appears here, rather than the coin walk's $\sqrt{2n \log\log n}$, is the signature of the variance $\frac{1}{4}$ of a centered digit. The general form of the law of the iterated logarithm attaches the envelope $\sqrt{2\sigma^2 n \log\log n}$ to a sum of i.i.d. steps of variance σ^2; with $\sigma^2 = \frac{1}{4}$ this reads $\sqrt{2 \cdot \frac{1}{4} \cdot n \log\log n} = \sqrt{\frac{1}{2}\, n \log\log n}$, exactly the normalizer above. The factor of two between the two envelopes is nothing more than the factor of two between a step of size 1 and a step of size $\frac{1}{2}$.

The force of theorem 8.3.3 is clearest when set beside Borel's normal-number theorem, which it refines. Borel (theorem 5.3.6) asserts that for almost every x the digit frequency N_n/n tends to $\frac{1}{2}$, equivalently $V_n/n \to 0$ almost surely: the excess 1-count is $o(n)$, a law of large numbers for the digits. This says the surplus of 1s over 0s is negligible on the scale of n, but says nothing about its actual size. The law of the iterated logarithm supplies that size. It states that the surplus is not only $o(n)$ but is pinned, along almost every trajectory, to the envelope

$$V_n = \pm\big(1 + o(1)\big)\sqrt{\tfrac{1}{2}\, n \log\log n} \quad \text{infinitely often}$$

and never appreciably exceeds it: for every $\varepsilon > 0$, eventually $|V_n| \le (1+\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$, while the value $(1-\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$ is surpassed by V_n (and by $-V_n$) for infinitely many n. Borel centers the running count at $n/2$; the law of the iterated logarithm fixes the exact amplitude of its wandering about that center.

8.3.4 Remark: A Frequency Theorem Versus a Fluctuation Theorem The contrast between theorem 5.3.6 and theorem 8.3.3 is the contrast between a first-order and a second-order statement about the same sequence. Borel controls the mean of the empirical digit distribution and concludes convergence of frequencies; the law of the iterated logarithm controls the fluctuation of the running count around that mean and concludes an exact almost-sure envelope. A normal number, in Borel's sense, is one whose digit frequencies are all correct; the law of the iterated logarithm shows that passing that test still leaves the running 1-count free to oscillate, and quantifies by exactly how much.

To make the envelope concrete, it helps to read off what it says about a prefix of a definite length. The next result records the one-sided consequence and then attaches a number to it.

Corollary 8.3.5: Infinitely Many Digit-Rich Prefixes

Fix $\varepsilon \in (0,1)$. For almost every $x \in [0,1]$, there are infinitely many n for which the number of 1s among the first n binary digits of x exceeds

$$\frac{n}{2} + (1-\varepsilon)\sqrt{\tfrac{1}{2}\, n \log\log n}$$

and, symmetrically, infinitely many n for which it falls below $\frac{n}{2} - (1-\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$.

***Proof*:**

By theorem 8.3.3, $\limsup_n V_n/\sqrt{\frac{1}{2}\, n \log\log n} = 1$ almost surely, so for almost every x and every $\varepsilon \in (0,1)$ the ratio exceeds $1-\varepsilon$ for infinitely many n, that is, $V_n > (1-\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$ infinitely often. Since $N_n = \frac{n}{2} + V_n$, this is the first claim. The second follows from $\liminf_n V_n/\sqrt{\frac{1}{2}\, n \log\log n} = -1$ in the same way, completing the proof.

Example 8.3.6: The Fluctuation Scale at a Million Digits
Take $n = 10^6$, so the balanced count is $n/2 = 500{,}000$. The fluctuation scale is

$$\sqrt{\tfrac{1}{2}\, n \log\log n} = \sqrt{\tfrac{1}{2} \cdot 10^6 \cdot \log\log 10^6}$$

Here $\log 10^6 = 6 \log 10 \approx 13.8$ and $\log(13.8) \approx 2.63$, so $\log\log 10^6 \approx 2.63$. Substituting,

$$\sqrt{\tfrac{1}{2} \cdot 10^6 \cdot 2.63} = \sqrt{1.32 \times 10^6} \approx 1146$$

For almost every x, then, the value 1146 is the height of the envelope at this one length: it marks the band $500{,}000 \pm 1146$ that the 1-count of the first million digits is measured against. The infinitely-often statement is about n ranging to infinity: for infinitely many lengths n the 1-count of the first n digits lands near $\frac{n}{2} + (1-\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$ and, for infinitely many others, near $\frac{n}{2} - (1-\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$, while a count outside the band of half-width about $(1+\varepsilon)\sqrt{\frac{1}{2}\, n \log\log n}$ does not occur for large n. A normal number's running 1-count wanders about eleven hundred above and below half a million and, past the transient, never appreciably further. Set against the central limit theorem's distributional scale $\sqrt{\frac{1}{4} n} = \sqrt{2.5 \times 10^5} \approx 500$ for a single fixed n, the extra factor $\sqrt{2 \log\log n} \approx 2.3$ is the price of asking for the almost-sure envelope of the whole trajectory rather than the typical size at one instant.

The example exposes how slowly the iterated logarithm grows and therefore how tight the envelope is. Between a million digits and a trillion, $\log\log n$ climbs only from about 2.6 to about 3.3, so the $\log\log$ correction factor $\sqrt{\frac{1}{2} \log\log n}$ that multiplies $\sqrt{n}$ barely moves, from about 1.14 to about 1.28, a rise of some twelve percent even as n multiplies by 10^6. The fluctuation scale itself is not nearly so quiet: dominated by the $\sqrt{n}$ factor, it grows about a thousandfold over this range, from roughly 1146 to roughly 1.29 million. It is the correction relative to $\sqrt{n}$, not the scale in absolute terms, that stays almost fixed, and this near-constancy of the $\log\log$ correction is what makes the law of the iterated logarithm a tight envelope rather than a loose bound: the running count is held to within a whisker, on the logarithmic scale, of $\sqrt{n}$.

This reading of the digits closes the arc of Part II. The three limit theorems of this Part describe the same sum S_n of independent steps at three successively finer resolutions. The strong law of large numbers (theorem 5.2.3) fixes the location: $S_n/n \to 0$, the average settles at the mean. The central limit theorem (theorem 7.1.3) fixes the distributional scale: $S_n/\sqrt{n}$ converges in law to a Gaussian, so a typical fluctuation is of order $\sqrt{n}$. Neither, though, describes a single trajectory's extremes, and there the two answers seem to conflict, since $S_n/\sqrt{n}$ has $\limsup = +\infty$ almost surely even as it is Gaussian in distribution. The law of the iterated logarithm (theorem 8.2.3) resolves the tension by fixing the exact almost-sure envelope: the extreme excursions of the path sit precisely at $\pm\sqrt{2n \log\log n}$, larger than the distributional scale by the slowly-growing factor $\sqrt{2 \log\log n}$ and no larger. Mean, then distributional scale, then almost-sure envelope: each theorem answers the question its predecessor left open, and the digits of a random real display all three at once.

The story does not end with discrete time. The coin walk S_n, suitably interpolated and rescaled, converges to Brownian motion, and Brownian motion carries its own law of the iterated logarithm, describing the fine oscillation of a continuous path both at large times and, in a local form, near any fixed instant. The continuous-time envelope is $\sqrt{2t \log\log(1/t)}$ near $t = 0$, the direct analogue of the theorem proved here, and its proof rests on the same pair of Borel-Cantelli arguments over a geometric sequence of scales. Pursuing it requires the continuous-time framework that lies past the boundary of this book; it is one of the first destinations pointed to in the closing chapter.

Exercise 8.3.1

1. Verify directly that the centered digit steps $d_i - \frac{1}{2}$ have variance $\frac{1}{4}$, and deduce from the general envelope $\sqrt{2\sigma^2 n \log\log n}$ for i.i.d. steps of variance σ^2 that the correct normalizer for V_n is $\sqrt{\frac{1}{2}\, n \log\log n}$. Then explain in one sentence why the coin walk $S_n = 2V_n$ carries the larger envelope $\sqrt{2n \log\log n}$.

2. Borel's normal-number theorem (theorem 5.3.6) and the law of the iterated logarithm for digits (theorem 8.3.3) both hold for almost every x. Show that the two almost-sure events are compatible by exhibiting, for a hypothetical x obeying theorem 8.3.3, the limit of V_n/n; conclude that theorem 8.3.3 implies the frequency statement $N_n/n \to \frac{1}{2}$ of Borel's theorem, so the law of the iterated logarithm is strictly stronger.

3. A binary string is called *k-balanced up to n* if $|N_n - \frac{n}{2}| \le k$. Using theorem 8.3.3, decide for which fixed constants k (independent of n) it is true that almost every x is k-balanced up to n for all sufficiently large n, and for which it fails.

4. Let x be uniform on $[0, 1]$ and let $e_i(x) \in \{0, 1, \dots, 9\}$ be its decimal digits, which are i.i.d. uniform on $\{0, \dots, 9\}$. Fix a target digit, say 7, and let $W_n = \sum_{i \le n} \big(\mathbb{1}_{\{e_i = 7\}} - \frac{1}{10}\big)$ be the centered count of 7s. State the law of the iterated logarithm for W_n, identifying the correct normalizer, and justify it by computing the variance of a single centered indicator.

5. Explain why the event $\{\limsup_n V_n/\sqrt{\frac{1}{2}\, n \log\log n} = 1\}$ has probability either 0 or 1 before any envelope is computed, by identifying it as a tail event of the i.i.d. digit sequence and invoking Kolmogorov's zero-one law (theorem 2.4.3). Why does this not by itself determine that the value is 1?

6. A student reasons: "The central limit theorem says $V_n/\sqrt{n/4}$ is approximately standard normal, and a standard normal rarely exceeds 3, so V_n should essentially never exceed $3\sqrt{n}/2$." Explain precisely where this reasoning breaks down as a claim about a single trajectory, and reconcile it with theorem 8.3.3.

Part III

Conditioning and Martingales

The limit theorems of Part II analyzed a fixed body of randomness all at once. This final Part introduces time. Its subject is how an estimate should be revised as information accumulates, and the processes whose defining feature is that they cannot be improved upon by such revision.

The organizing tool is conditional expectation, the best estimate of a random variable given partial information encoded as a sub-σ-algebra. chapter 9 constructs it through the Radon-Nikodym theorem and develops its calculus, culminating in its interpretation as an orthogonal projection. On that foundation chapter 10 builds the theory of martingales, the discrete-time processes whose conditional increments vanish, and proves the two results that make them indispensable: the optional stopping theorem, that a fair game stopped at a cleverly chosen time is still fair, and the martingale convergence theorem, that a martingale bounded in size must settle to a limit. The chapter closes with the applications, from Lévy's zero-one law to a first concentration inequality, that show why martingales are the natural language for information revealed over time.

This Part is also the gateway to continuous time. The discrete martingale theory developed here is the prerequisite for the stochastic processes of the sequel, and the conditional expectation of chapter 9 is the object on which the edifice of stochastic calculus rests.

9

Conditional Expectation

Expectation collapses a random variable to a single number, its average over all of the randomness. Conditional expectation is the refinement that averages over only part of it. Given a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ representing the information available, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the best estimate of X that can be formed from that information: itself a random variable, measurable with respect to $\mathcal{G}$, agreeing with X in average over every event of $\mathcal{G}$. It is the central construction of the second half of probability, the object on which martingales, Markov processes, and stochastic calculus are all built.

section 9.1 constructs $\mathbb{E}[X \mid \mathcal{G}]$ through the Radon-Nikodym theorem and fixes its defining property. Section 9.2 develops the calculus: linearity, the tower property, taking out what is known, and the conditional forms of Jensen's inequality and the convergence theorems. section 9.3 recasts conditional expectation on L^2 as orthogonal projection onto the subspace of $\mathcal{G}$-measurable variables, the geometry that makes it the best mean-square predictor. Section 9.4 closes with regular conditional distributions, which package the conditional law of one variable given another into a genuine probability measure and recover the elementary conditional densities.

9.1 Conditioning on a Sigma-Algebra

The expectation $\mathbb{E}[X]$ of an integrable random variable is a single number: the average of X against the probability measure $\mathbb{P}$, taken over every outcome at once. It answers the question "what is X on average, knowing nothing about which outcome occurred?" In practice one usually knows something. A gambler has seen the first ten cards; a statistician has observed a covariate; a process has run for a while and its history is on the table. The information available is exactly a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, the collection of events whose occurrence or non-occurrence can be decided from what is known. Conditional expectation is the answer to the refined question: "what is X on average, given the information in $\mathcal{G}$?"

The point that distinguishes the measure-theoretic treatment from a first course is that this answer is not a number. If the information $\mathcal{G}$ is itself random, the conditional average of X depends on the outcome, and so it is a random variable. Its value at ω is the average of X over the smallest event of $\mathcal{G}$ that is known to contain ω. Two requires pin it down. First, it must be computable from the information: it is $\mathcal{G}$-measurable. Second, it must average correctly on every event one can decide: for each $A \in \mathcal{G}$, the average of the estimate over A equals the average of X over A. The fact, and the content of this section, is that these two requirements determine a random variable uniquely up to a null set, and that the Radon-Nikodym theorem manufactures it.

The elementary case makes the two requirements concrete before the general definition abstracts them. Suppose $\mathcal{G}$ is generated by a finite partition $\Omega = G_1 \cup \cdots \cup G_m$ into events of positive probability. To know the information in $\mathcal{G}$ is to know which block G_i the outcome fell into, and nothing finer. The natural estimate of X is then constant on each block, equal to the elementary conditional average

$$\frac{\mathbb{E}[X\mathbb{1}_{G_i}]}{\mathbb{P}(G_i)}$$

of X over that block. Writing this estimate as a random variable gives a $\mathcal{G}$-measurable function, and one checks that its integral over each G_i returns $\mathbb{E}[X\mathbb{1}_{G_i}]$, so it averages correctly on the generators, hence on all of $\mathcal{G}$. The general definition keeps precisely these two properties and drops the partition, which is what allows $\mathcal{G}$ to be infinite.

Definition 9.1.1: Conditional Expectation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let X be an integrable random variable (definition 3.1.1), and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. A *conditional expectation of X given $\mathcal{G}$* is a random variable Z satisfying:

1. Z is $\mathcal{G}$-measurable and integrable;
2. for every $A \in \mathcal{G}$,
$$\int_A Z \, d\mathbb{P} = \int_A X \, d\mathbb{P} \tag{9.1}$$

Any two such random variables agree almost surely (theorem 9.1.2), so the conditional expectation is unique up to a $\mathbb{P}$-null set; the almost-surely-determined equivalence class is written $\mathbb{E}[X|\mathcal{G}]$. An individual Z satisfying (1) and (2) is called a *version* of $\mathbb{E}[X|\mathcal{G}]$.

Every symbol in the definition carries weight, and it is worth reading each against the informal picture. Condition (1), $\mathcal{G}$-measurability, is the requirement that the estimate be computable from the available information: no version of $\mathbb{E}[X|\mathcal{G}]$ may distinguish two outcomes that $\mathcal{G}$ cannot tell apart. Condition (2) is the averaging requirement. It does not ask that Z equal X, which is impossible when X is not $\mathcal{G}$-measurable; it asks only that Z and X have the same integral over every event whose membership can be decided from $\mathcal{G}$. The defining identity (9.1) is used throughout the chapter: to check that a candidate is (a version of) the conditional expectation, one verifies that it is $\mathcal{G}$-measurable and integrates like X over every $A \in \mathcal{G}$. Finally, the object is defined only up to a null set. Statements about $\mathbb{E}[X|\mathcal{G}]$ therefore carry an almost-sure qualifier, and identities between conditional expectations are identities between equivalence classes, holding $\mathbb{P}$-almost surely. This is not a technicality to be waved away; it is the reason every property proved in the next section is stated "a.s."

The definition asserts the existence and near-uniqueness of Z but does not yet supply it. Uniqueness is elementary and self-contained. Existence is where the machinery of the prerequisite volume enters: the defining identity (9.1) says exactly that Z is a Radon-Nikodym density of the measure $A \mapsto \int_A X\, d\mathbb{P}$ with respect to $\mathbb{P}$, both regarded as measures on the smaller σ-algebra $\mathcal{G}$. This is the mechanism that turns an abstract averaging requirement into an actual random variable.

Theorem 9.1.2: Existence and Uniqueness of Conditional Expectation

Let X be integrable and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra. Then a conditional expectation $\mathbb{E}[X|\mathcal{G}]$ exists, and any two versions agree almost surely.

***Proof*:**
The two claims are independent, and the argument treats them in turn: uniqueness directly from the defining identity, existence via Radon-Nikodym.

Step 1: Uniqueness. Suppose Z and Z' both satisfy conditions (1) and (2) of definition 9.1.1. Both are $\mathcal{G}$-measurable, so the set

$$A = \{Z > Z'\} = \{\omega : Z(\omega) > Z'(\omega)\}$$

belongs to $\mathcal{G}$. Applying the defining identity (9.1) to each of Z and Z' on this set and subtracting gives

$$\int_A (Z - Z')\, d\mathbb{P} = \int_A Z\, d\mathbb{P} - \int_A Z'\, d\mathbb{P} = \int_A X\, d\mathbb{P} - \int_A X\, d\mathbb{P} = 0$$

On A the integrand $Z - Z'$ is strictly positive, so a nonnegative random variable with zero integral vanishes almost everywhere on A; this forces $\mathbb{P}(A) = 0$ (proposition 3.1.3). Hence $\mathbb{P}(Z > Z') = 0$, and by the same argument with the roles of Z and Z' exchanged, $\mathbb{P}(Z' > Z) = 0$. Therefore $Z = Z'$ almost surely, which is the asserted uniqueness.

Step 2: Existence for $X \geq 0$. Assume first that $X \geq 0$. Define a set function on the smaller σ-algebra $\mathcal{G}$ by

$$\nu(A) = \int_A X\, d\mathbb{P}, \qquad A \in \mathcal{G} \tag{9.2}$$

That ν is a measure on $(\Omega, \mathcal{G})$ is immediate: $\nu(\emptyset) = 0$, and countable additivity over disjoint $A_1, A_2, \dots \in \mathcal{G}$ follows from the monotone convergence theorem applied to the partial sums $X\mathbb{1}_{A_1 \cup \dots \cup A_n} \uparrow X\mathbb{1}_{\bigcup_k A_k}$. It is finite because $\nu(\Omega) = \mathbb{E}[X] < \infty$ by integrability. Moreover ν is absolutely continuous with respect to the restriction $\mathbb{P}|_{\mathcal{G}}$: if $A \in \mathcal{G}$ has $\mathbb{P}(A) = 0$, then $\int_A X\, d\mathbb{P} = 0$, so $\nu(A) = 0$. The Radon-Nikodym theorem, proved on the space $(\Omega, \mathcal{G}, \mathbb{P}|_{\mathcal{G}})$ [2], now supplies a nonnegative $\mathcal{G}$-measurable density

$$Z = \frac{d\nu}{d\mathbb{P}|_{\mathcal{G}}}$$

with $\nu(A) = \int_A Z\, d\mathbb{P}|_{\mathcal{G}} = \int_A Z\, d\mathbb{P}$ for every $A \in \mathcal{G}$ (the last equality because Z is $\mathcal{G}$-measurable, so its integral over a $\mathcal{G}$-set is computed identically against $\mathbb{P}$ and against $\mathbb{P}|_{\mathcal{G}}$). Comparing with (9.2) gives $\int_A Z\, d\mathbb{P} = \int_A X\, d\mathbb{P}$ for all $A \in \mathcal{G}$, which is exactly (9.1); taking $A = \Omega$ shows $\mathbb{E}[Z] = \mathbb{E}[X] < \infty$, so Z is integrable. Thus Z is a version of $\mathbb{E}[X|\mathcal{G}]$.

Step 3: Existence for general integrable X. Decompose $X = X^+ - X^-$ into its positive and negative parts. Both X^+ and X^- are nonnegative and integrable ($\mathbb{E}[X^{\pm}] \leq \mathbb{E}[|X|] < \infty$), so Step 2 furnishes nonnegative $\mathcal{G}$-measurable integrable versions Z^+ of $\mathbb{E}[X^+ \middle| \mathcal{G}]$ and Z^- of $\mathbb{E}[X^- \middle| \mathcal{G}]$. Set $Z = Z^+ - Z^-$. Then Z is $\mathcal{G}$-measurable and integrable, and for every $A \in \mathcal{G}$, by linearity of the integral,

$$\int_A Z\,d\mathbb{P} = \int_A Z^+\,d\mathbb{P} - \int_A Z^-\,d\mathbb{P} = \int_A X^+\,d\mathbb{P} - \int_A X^-\,d\mathbb{P} = \int_A X\,d\mathbb{P}$$

so Z satisfies the defining identity and is a version of $\mathbb{E}[X \middle| \mathcal{G}]$. Existence therefore holds for every integrable X, completing the proof.

The proof exposes what conditional expectation *is* in three registers, and each is worth carrying forward. Analytically, $\mathbb{E}[X \middle| \mathcal{G}]$ is a Radon-Nikodym derivative, the density of the "mass distribution" $\nu(A) = \int_A X\,d\mathbb{P}$ once that distribution is coarsened to the σ-algebra $\mathcal{G}$; all the calculus of the next section is, at bottom, the calculus of such densities. Structurally, existence is not a probabilistic fact at all but a measure-theoretic one, imported wholesale from the prerequisite volume, and this is the sense in which conditional expectation "costs nothing new" once Radon-Nikodym is in hand. Operationally, the uniqueness argument gives the standard tool for identifying a conditional expectation in practice: to prove $\mathbb{E}[X \middle| \mathcal{G}] = Z$ a.s. for a proposed $\mathcal{G}$-measurable Z, it suffices to verify the single identity (9.1) on all $A \in \mathcal{G}$, and by the same $\{Z > Z'\} \in \mathcal{G}$ device this is often reduced to a generating class. Every property in section 9.2 is proved this way.

One reading of the construction deserves to be named, because it is the reading that governs the whole chapter. Passing from X to $\mathbb{E}[X \middle| \mathcal{G}]$ replaces X by the coarsest object that is indistinguishable from X at the resolution of $\mathcal{G}$ and preserves all $\mathcal{G}$-averages. When $\mathcal{G} = \{\emptyset, \Omega\}$ is trivial, the only $\mathcal{G}$-measurable variables are constants, and (9.1) on $A = \Omega$ forces that constant to be $\mathbb{E}[X]$: with no information, the best estimate is the unconditional mean, so $\mathbb{E}[X \middle| \{\emptyset, \Omega\}] = \mathbb{E}[X]$ a.s. At the other extreme, when $\mathcal{G} = \mathcal{F}$ contains all the information, X itself is $\mathcal{G}$-measurable and satisfies (9.1) trivially, so $\mathbb{E}[X \middle| \mathcal{F}] = X$ a.s. Conditioning interpolates between knowing nothing and knowing everything.

Two pieces of standard notation and one specialization of the definition organize the rest of the chapter. Conditioning an indicator produces a conditional probability, and conditioning on a random variable means conditioning on the information that variable carries.

Definition 9.1.3: Conditional Probability Given a Sigma-Algebra

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. For an event $A \in \mathcal{F}$, the *conditional probability of A given $\mathcal{G}$* is the conditional expectation of its indicator,

$$\mathbb{P}(A \middle| \mathcal{G}) = \mathbb{E}[\mathbb{1}_A \middle| \mathcal{G}]$$

a random variable determined up to a null set. For a random variable Y, or a finite family $Y_1, \ldots, Y_k$, conditioning on Y means conditioning on the σ-algebra it generates (definition 2.1.5):

$$\mathbb{E}[X \middle| Y] = \mathbb{E}[X \middle| \sigma(Y)], \qquad \mathbb{P}(A \middle| Y) = \mathbb{P}(A \middle| \sigma(Y))$$

Because $0 \le \mathbb{1}_A \le 1$, monotonicity of the conditional expectation (proved in the next section) gives $0 \le \mathbb{P}(A \middle| \mathcal{G}) \le 1$ almost surely, so $\mathbb{P}(A \middle| \mathcal{G})$ is a $[0,1]$-valued random variable, as its name promises. Taking $A = \Omega$ in the defining identity gives $\mathbb{E}[\mathbb{P}(A \middle| \mathcal{G})] = \mathbb{P}(A)$, which is the conditional form of the law of total probability: averaging the conditional probability over the information recovers the unconditional probability. One subtlety, deferred to section 9.4, is worth flagging now. For each fixed A the object $\mathbb{P}(A|\mathcal{G})$ is defined only up to a null set, and there is no reason a priori that these separate choices, made for uncountably many A, can be reconciled into a map $A \mapsto \mathbb{P}(A|\mathcal{G})(\omega)$ that is a genuine probability measure for almost every ω. That they can, on well-behaved spaces, is the theory of regular conditional distributions.

The specialization $\mathbb{E}[X|Y]$ is the form the reader will most often meet. Since $\mathbb{E}[X|Y]$ is $\sigma(Y)$-measurable, the Doob-Dynkin lemma (lemma 2.1.6) says it is a measurable function of Y: there is a Borel function g with $\mathbb{E}[X|Y] = g(Y)$ almost surely. The function g is what elementary probability calls the regression function $y \mapsto \mathbb{E}[X|Y = y]$, and the next example computes it in the two cases where it is transparent.

Example 9.1.4: Conditional Expectation on Partitions and Discrete Variables

Three computations ground the definition, from the fully elementary to the continuous case whose proof is deferred.

1. *A finite partition.* Let $\mathcal{G} = \sigma(G_1, \ldots, G_m)$ be generated by a partition of Ω into events G_i with $\mathbb{P}(G_i) > 0$. Every $\mathcal{G}$-measurable random variable is constant on each block G_i, so the candidate
$$Z = \sum_{i=1}^{m} \frac{\mathbb{E}[X\mathbb{1}_{G_i}]}{\mathbb{P}(G_i)}\, \mathbb{1}_{G_i} \tag{9.3}$$
is $\mathcal{G}$-measurable. To verify (9.1) it is enough to check it on a generating class, and the generators G_j (together with $\emptyset$) form one. On G_j the sum (9.3) collapses to its j-th term because the blocks are disjoint, so
$$\int_{G_j} Z\, d\mathbb{P} = \frac{\mathbb{E}[X\mathbb{1}_{G_j}]}{\mathbb{P}(G_j)}\, \mathbb{P}(G_j) = \mathbb{E}[X\mathbb{1}_{G_j}] = \int_{G_j} X\, d\mathbb{P}$$
Every $A \in \mathcal{G}$ is a disjoint union of blocks, so additivity of the integral extends the identity from the G_j to all of $\mathcal{G}$. Thus $\mathbb{E}[X|\mathcal{G}] = Z$ a.s.: the conditional expectation is the block-wise elementary conditional average, and the first-course notion is recovered exactly.

2. *A discrete random variable.* Let Y take countably many values $\{y_1, y_2, \ldots\}$, each with $\mathbb{P}(Y = y_k) > 0$. The events $G_k = \{Y = y_k\}$ partition Ω and generate $\sigma(Y)$, so (9.3) applies (its countable version follows by monotone convergence exactly as in Step 2 of theorem 9.1.2) and gives $\mathbb{E}[X|Y] = g(Y)$ a.s. with
$$g(y_k) = \frac{\mathbb{E}[X\mathbb{1}_{\{Y=y_k\}}]}{\mathbb{P}(Y = y_k)} = \mathbb{E}[X|Y = y_k]$$
the elementary conditional expectation of X given the event $\{Y = y_k\}$. For a concrete instance, roll two fair dice with faces X_1, X_2 and let $Y = X_1 + X_2$ be the sum. Given $Y = y$, symmetry makes X_1 uniform on the $\{(a, b) : a + b = y\}$ pairs, and averaging gives $\mathbb{E}[X_1|Y] = Y/2$: knowing the total, the best estimate of one die is half the total.

3. *A pair with a joint density.* Suppose (X, Y) has a joint density $f(x, y)$ with marginal $f_Y(y) = \int f(x, y)\, dx$. Then, at the values y with $f_Y(y) > 0$, the regression function is the integral of x against the conditional density $f(x, y)/f_Y(y)$,
$$\mathbb{E}[X|Y = y] = \int_{\mathbb{R}} x\, \frac{f(x, y)}{f_Y(y)}\, dx$$

> recovering the elementary conditional expectation from a first course. This is the continuous analogue of the discrete formula in (2), with the sum over atoms replaced by an integral against the conditional density. The verification that this formula defines a version of $\mathbb{E}[X|Y]$, and its clean statement through the disintegration of the joint law, are the business of section 9.4; we defer to proposition 9.4.5.

The three cases fall on a single spectrum. Conditioning on a finite partition is the whole of elementary conditional probability; conditioning on a discrete variable is the same computation with countably many blocks; conditioning on a variable with a density is its continuous limit, where blocks shrink to points and the block-average becomes an integral against the conditional density. What the abstract definition gives is a construction that survives when there is no partition and no density at all, where the Radon-Nikodym density is the only handle available, and where the estimate can still fail to be given by any explicit formula. That generality is what the martingales of the next chapter require.

Two readings of $\mathbb{E}[X|\mathcal{G}]$ will recur, and both are already visible. The first is the reading emphasized so far: $\mathbb{E}[X|\mathcal{G}]$ is a *coarser average*, the random variable obtained by replacing X with its average over the finest cells $\mathcal{G}$ can resolve, preserving every $\mathcal{G}$-average in the process. The second reading, developed fully in section 9.3, is that among all $\mathcal{G}$-measurable variables $\mathbb{E}[X|\mathcal{G}]$ is the *best estimate* of X: when X is square-integrable it is the one minimizing the mean-square error $\mathbb{E}[(X-Z)^2]$, the orthogonal projection of X onto the space $L^2(\mathcal{G})$ of $\mathcal{G}$-measurable square-integrable variables. The two readings are the same object seen twice, once as an average and once as a projection, and the interplay between them is the source of most of the chapter's computations. Throughout, the almost-sure caveat stands: $\mathbb{E}[X|\mathcal{G}]$ names an equivalence class, every identity involving it holds only up to a null set, and the phrase "a.s." on each such identity is a statement about the theory, not a disclaimer to be forgotten.

Exercise 9.1.1

1. Let X be integrable. Using only the defining identity (9.1), prove directly that $\mathbb{E}[X|\{\emptyset,\Omega\}] = \mathbb{E}[X]$ a.s. and $\mathbb{E}[X|\mathcal{F}] = X$ a.s., identifying which σ-measurability require does the work in each case.
2. Let c be a constant and $\mathcal{G} \subseteq \mathcal{F}$ any sub-σ-algebra. Show that $\mathbb{E}[c|\mathcal{G}] = c$ a.s. More generally, show that if X is itself $\mathcal{G}$-measurable and integrable then $\mathbb{E}[X|\mathcal{G}] = X$ a.s.
3. Let $\mathcal{G} = \sigma(G_1, \ldots, G_m)$ be generated by a partition into events of positive probability, and let $A \in \mathcal{F}$. Compute $\mathbb{P}(A|\mathcal{G})$ explicitly as a random variable, and verify that $\mathbb{E}[\mathbb{P}(A|\mathcal{G})] = \mathbb{P}(A)$.
4. Roll two fair dice with faces X_1, X_2, and let $Y = \max(X_1, X_2)$. Compute $g(y) = \mathbb{E}[X_1|Y = y]$ for each $y \in \{1, \ldots, 6\}$, and hence write $\mathbb{E}[X_1|Y]$ as a function of Y.
5. Let Z be $\mathcal{G}$-measurable and integrable, and suppose $\int_A Z\,d\mathbb{P} \geq \int_A X\,d\mathbb{P}$ for every $A \in \mathcal{G}$. Prove that $Z \geq \mathbb{E}[X|\mathcal{G}]$ a.s. (This one-sided version of the uniqueness argument is the template for the monotonicity results of the next section.)
6. Let (X, Y) be uniform on the triangle $\{(x, y) : 0 < x < y < 1\}$, so the joint density is $f(x, y) = 2$ there and 0 elsewhere. Using case (3) of example 9.1.4, compute the conditional density $f(x, y)/f_Y(y)$ and the regression function $\mathbb{E}[X|Y = y]$, and identify the resulting random variable $\mathbb{E}[X|Y]$.

9.2 Properties of Conditional Expectation

The construction in section 9.1 produces $\mathbb{E}[X|\mathcal{G}]$ from the Radon-Nikodym theorem, but it is not the density that one computes with in practice. Almost every manipulation of conditional expectation rests instead on a short list of structural properties, each of which follows directly from the defining property of definition 9.1.1: that $\mathbb{E}[X|\mathcal{G}]$ is the almost-surely-unique $\mathcal{G}$-measurable integrable variable Z with

$$\int_A Z\,d\mathbb{P} = \int_A X\,d\mathbb{P} \qquad \text{for every } A \in \mathcal{G} \tag{9.4}$$

This section assembles that list. The strategy never changes: to prove that a given $\mathcal{G}$-measurable variable equals $\mathbb{E}[X|\mathcal{G}]$, one checks (9.4) for it on an arbitrary $A \in \mathcal{G}$; uniqueness then does the rest. Linearity, monotonicity, the tower property, taking out a known factor, and the conditional forms of Jensen's inequality and the convergence theorems all fall to this single method. The payoff is a calculus that lets one compute with $\mathbb{E}[\,\cdot\,|\mathcal{G}]$ as fluently as with the ordinary expectation, and it is this calculus that the martingale theory of the next chapter runs on.

Throughout, all random variables are assumed integrable unless stated otherwise, $\mathcal{G} \subseteq \mathcal{F}$ is a sub-σ-algebra, and every identity between conditional expectations holds only almost surely, since $\mathbb{E}[X|\mathcal{G}]$ is defined only up to a $\mathbb{P}$-null set. Rather than repeat "a.s." inside every displayed line, the qualifier is stated once with each result and understood throughout its proof.

The first group of properties records the ways in which $\mathbb{E}[\,\cdot\,|\mathcal{G}]$ behaves like the ordinary expectation and the two extreme cases in which it degenerates: when X already lives in $\mathcal{G}$ (all its information is available, so conditioning changes nothing) and when X is independent of $\mathcal{G}$ (none of its information is available, so conditioning returns the constant $\mathbb{E}[X]$). These are the identities one reaches for first in any computation.

Proposition 9.2.1: Basic Properties of Conditional Expectation

Let X, Y be integrable and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra.

1. *(Linearity.)* For $a, b \in \mathbb{R}$,
$$\mathbb{E}[aX + bY|\mathcal{G}] = a\,\mathbb{E}[X|\mathcal{G}] + b\,\mathbb{E}[Y|\mathcal{G}] \quad \text{a.s.}$$
2. *(Monotonicity.)* If $X \le Y$ a.s., then $\mathbb{E}[X|\mathcal{G}] \le \mathbb{E}[Y|\mathcal{G}]$ a.s. In particular $|\mathbb{E}[X|\mathcal{G}]| \le \mathbb{E}[|X||\mathcal{G}]$ a.s.
3. *(Total expectation.)* $\mathbb{E}\big[\mathbb{E}[X|\mathcal{G}]\big] = \mathbb{E}[X]$.
4. *(Measurability.)* If X is $\mathcal{G}$-measurable, then $\mathbb{E}[X|\mathcal{G}] = X$ a.s.
5. *(Independence.)* If X is independent of $\mathcal{G}$ (definition 2.2.2), then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$ a.s.

***Proof*:**
Each part is proved by exhibiting a $\mathcal{G}$-measurable integrable variable satisfying the defining identity (9.4) and invoking the uniqueness in theorem 9.1.2. Write $Z_X = \mathbb{E}[X|\mathcal{G}]$ and $Z_Y = \mathbb{E}[Y|\mathcal{G}]$ for the versions fixed by the construction.

1. *Linearity.* The variable $aZ_X + bZ_Y$ is $\mathcal{G}$-measurable and integrable, and for every $A \in \mathcal{G}$ the additivity and homogeneity of the ordinary integral (proposition 3.1.3) give

$$\int_A (aZ_X + bZ_Y)\, d\mathbb{P} = a \int_A Z_X\, d\mathbb{P} + b \int_A Z_Y\, d\mathbb{P} = a \int_A X\, d\mathbb{P} + b \int_A Y\, d\mathbb{P} = \int_A (aX + bY)\, d\mathbb{P}$$

using (9.4) for Z_X and Z_Y at the middle step. Thus $aZ_X + bZ_Y$ satisfies the defining identity for $aX + bY$, and by uniqueness it equals $\mathbb{E}[aX + bY \big| \mathcal{G}]$ a.s.

2. *Monotonicity.* Suppose $X \le Y$ a.s. and consider the $\mathcal{G}$-measurable event $A = \{Z_X > Z_Y\} = \{Z_X - Z_Y > 0\} \in \mathcal{G}$. By the defining identity applied to X and to Y,

$$\int_A (Z_X - Z_Y)\, d\mathbb{P} = \int_A X\, d\mathbb{P} - \int_A Y\, d\mathbb{P} = \int_A (X - Y)\, d\mathbb{P} \le 0$$

since $X - Y \le 0$ a.s. But on A the integrand $Z_X - Z_Y$ is strictly positive, so a nonnegative integrand over A can have a nonpositive integral only if $\mathbb{P}(A) = 0$. Hence $Z_X \le Z_Y$ a.s. For the absolute-value bound, apply this to the pair of inequalities $-|X| \le X \le |X|$ together with linearity, giving $-\mathbb{E}[|X| \big| \mathcal{G}] \le Z_X \le \mathbb{E}[|X| \big| \mathcal{G}]$ a.s.

3. *Total expectation.* Take $A = \Omega \in \mathcal{G}$ in (9.4):

$$\mathbb{E}\big[\mathbb{E}[X \big| \mathcal{G}]\big] = \int_\Omega Z_X\, d\mathbb{P} = \int_\Omega X\, d\mathbb{P} = \mathbb{E}[X]$$

4. *Measurability.* If X is itself $\mathcal{G}$-measurable, then X is a $\mathcal{G}$-measurable integrable variable, and it trivially satisfies $\int_A X\, d\mathbb{P} = \int_A X\, d\mathbb{P}$ for every $A \in \mathcal{G}$. By uniqueness $\mathbb{E}[X \big| \mathcal{G}] = X$ a.s.

5. *Independence.* The constant $\mathbb{E}[X]$ is $\mathcal{G}$-measurable and integrable. Fix $A \in \mathcal{G}$. Since X is independent of $\mathcal{G}$, it is independent of the indicator $\mathbb{1}_A$, so $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[X]\,\mathbb{E}[\mathbb{1}_A] = \mathbb{E}[X]\,\mathbb{P}(A)$ by the product rule for independent variables (theorem 2.3.1). Therefore

$$\int_A X\, d\mathbb{P} = \mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[X]\,\mathbb{P}(A) = \int_A \mathbb{E}[X]\, d\mathbb{P}$$

so the constant $\mathbb{E}[X]$ satisfies the defining identity for X, and by uniqueness $\mathbb{E}[X \big| \mathcal{G}] = \mathbb{E}[X]$ a.s., as we sought to show.

Parts (4) and (5) are the two poles of conditioning, and reading them together clarifies what $\mathbb{E}[\,\cdot \big| \mathcal{G}]$ measures. A $\mathcal{G}$-measurable variable is one whose value is already determined by the information in $\mathcal{G}$, so there is nothing left to average and conditioning is the identity. A variable independent of $\mathcal{G}$ shares no information with it, so knowing $\mathcal{G}$ tells one nothing about X and the best estimate reverts to the unconditional mean $\mathbb{E}[X]$. Every case in between interpolates these two, and part (3), total expectation, says that averaging out the remaining randomness in $\mathbb{E}[X \big| \mathcal{G}]$ recovers the full mean. This last identity is used constantly: it lets one compute an ordinary expectation by first conditioning on a convenient σ-algebra and then averaging, the "condition and average" pattern behind almost every second-moment calculation in the sequel.

Monotonicity deserves separate emphasis. It says conditional expectation is a positive, order-preserving operator: it never turns a smaller variable into a larger conditional estimate. Combined with linearity, it makes $\mathbb{E}[\,\cdot\,|\mathcal{G}]$ a positive linear operator on L^1 that fixes the constants, which is exactly the abstract profile of an averaging operator. The conditional Jensen inequality below is the precise sense in which this operator is an average.

The next property is the one that makes iterated conditioning tractable. Information accumulates through a chain of ever-finer σ-algebras, and one wants to know what happens when a conditional estimate is itself conditioned on coarser information. The answer is that the coarser conditioning wins outright: conditioning on $\mathcal{G}$ and then on the smaller $\mathcal{H}$ is the same as conditioning on $\mathcal{H}$ from the start. This is the tower property, and it is the single most frequently invoked identity in the martingale theory that follows.

Proposition 9.2.2: Tower Property

Let $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ be nested sub-σ-algebras and X integrable. Then

$$\mathbb{E}\big[\mathbb{E}[X|\mathcal{G}] \mid \mathcal{H}\big] = \mathbb{E}[X|\mathcal{H}] \quad \text{a.s.}$$

Symmetrically, $\mathbb{E}\big[\mathbb{E}[X|\mathcal{H}]|\mathcal{G}\big] = \mathbb{E}[X|\mathcal{H}]$ a.s.: conditioning a coarser estimate on finer information leaves it unchanged.

***Proof*:**
Write $Z = \mathbb{E}[X|\mathcal{G}]$; it is integrable, and the outer conditional expectation $\mathbb{E}[Z|\mathcal{H}]$ is a well-defined $\mathcal{H}$-measurable integrable variable. To identify it with $\mathbb{E}[X|\mathcal{H}]$ it suffices, by uniqueness (theorem 9.1.2), to check that $\mathbb{E}[Z|\mathcal{H}]$ satisfies the defining identity (9.4) for $\mathbb{E}[X|\mathcal{H}]$: namely that its integral over every $A \in \mathcal{H}$ equals $\int_A X\, d\mathbb{P}$.

Fix $A \in \mathcal{H}$. Since $\mathcal{H} \subseteq \mathcal{G}$, we also have $A \in \mathcal{G}$, and this containment is the whole point. Applying the defining identity of $\mathbb{E}[Z|\mathcal{H}]$ (with respect to $\mathcal{H}$) and then the defining identity of $Z = \mathbb{E}[X|\mathcal{G}]$ (with respect to $\mathcal{G}$, using $A \in \mathcal{G}$),

$$\int_A \mathbb{E}[Z|\mathcal{H}]\, d\mathbb{P} = \int_A Z\, d\mathbb{P} = \int_A \mathbb{E}[X|\mathcal{G}]\, d\mathbb{P} = \int_A X\, d\mathbb{P}$$

where the last equality is (9.4) for $\mathbb{E}[X|\mathcal{G}]$ applied to the $\mathcal{G}$-set A. Thus $\mathbb{E}[Z|\mathcal{H}]$ is a $\mathcal{H}$-measurable integrable variable whose integral over every $A \in \mathcal{H}$ agrees with that of X, so by uniqueness $\mathbb{E}[Z|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$ a.s.

For the symmetric statement, $\mathbb{E}[X|\mathcal{H}]$ is $\mathcal{H}$-measurable, and since $\mathcal{H} \subseteq \mathcal{G}$ it is $\mathcal{G}$-measurable as well. Part (4) of proposition 9.2.1 then gives $\mathbb{E}[\mathbb{E}[X|\mathcal{H}]|\mathcal{G}] = \mathbb{E}[X|\mathcal{H}]$ a.s., completing the proof.

The tower property is best read as a statement about coarsening. The chain $\mathcal{H} \subseteq \mathcal{G}$ describes two levels of information, the finer $\mathcal{G}$ and the coarser $\mathcal{H}$. Forming $\mathbb{E}[X|\mathcal{G}]$ discards the information outside $\mathcal{G}$; conditioning the result on $\mathcal{H}$ discards still more. Because the second step only removes information, and information already absent from $\mathcal{H}$ was never in play, the composite is the same as conditioning on $\mathcal{H}$ directly. The smaller σ-algebra always controls the outcome. The total-expectation identity of proposition 9.2.1(3) is the special case $\mathcal{H} = \{\emptyset, \Omega\}$, where conditioning on the trivial σ-algebra is just taking the mean. The chapter's martingale sequel will use the tower property in exactly this form, applied to the filtration $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$ of accumulating information.

The third property expresses the sense in which $\mathcal{G}$-measurable variables behave like constants once one has conditioned on $\mathcal{G}$. If a factor Y is already known given $\mathcal{G}$, it should pass outside the conditional expectation untouched, exactly as a genuine constant does. This is "taking out what is known," and it is the conditional analogue of the elementary fact that constants factor out of an integral.

Proposition 9.2.3: Taking Out What Is Known

Let Y be $\mathcal{G}$-measurable, and suppose X and YX are both integrable. Then

$$\mathbb{E}[YX \big| \mathcal{G}] = Y\,\mathbb{E}[X \big| \mathcal{G}] \quad \text{a.s.}$$

The proof runs the standard machine of measure theory: verify the claim for indicators of $\mathcal{G}$-sets, extend by linearity to $\mathcal{G}$-measurable simple functions, pass to nonnegative Y by monotone convergence, and finally split a general Y into its positive and negative parts. The shape of a general $\mathcal{G}$-measurable Y that this ladder climbs is exactly the one supplied by the standard simple-function approximation [2]: every nonnegative $\mathcal{G}$-measurable variable is an increasing pointwise limit of $\mathcal{G}$-measurable simple functions.

***Proof*:**
Write $Z = \mathbb{E}[X \big| \mathcal{G}]$. The claim is that YZ is a version of $\mathbb{E}[YX \big| \mathcal{G}]$, which by uniqueness means YZ is $\mathcal{G}$-measurable, YX is integrable (assumed), and

$$\int_A YZ \, d\mathbb{P} = \int_A YX \, d\mathbb{P} \qquad \text{for every } A \in \mathcal{G} \tag{9.5}$$

Since Y and Z are both $\mathcal{G}$-measurable, so is their product, and it suffices to verify (9.5). We build up Y in four stages.

Step 1: Indicators. Suppose first $Y = \mathbb{1}_B$ for some $B \in \mathcal{G}$. For $A \in \mathcal{G}$ we have $A \cap B \in \mathcal{G}$, and $\mathbb{1}_B \mathbb{1}_A = \mathbb{1}_{A \cap B}$, so by the defining identity of Z applied to the $\mathcal{G}$-set $A \cap B$,

$$\int_A \mathbb{1}_B Z \, d\mathbb{P} = \int_{A \cap B} Z \, d\mathbb{P} = \int_{A \cap B} X \, d\mathbb{P} = \int_A \mathbb{1}_B X \, d\mathbb{P}$$

which is (9.5) for $Y = \mathbb{1}_B$.

Step 2: Simple functions. If $Y = \sum_{i=1}^n c_i \mathbb{1}_{B_i}$ with $B_i \in \mathcal{G}$ and $c_i \in \mathbb{R}$, then by linearity of the integral and Step 1 applied to each term,

$$\int_A YZ \, d\mathbb{P} = \sum_{i=1}^n c_i \int_A \mathbb{1}_{B_i} Z \, d\mathbb{P} = \sum_{i=1}^n c_i \int_A \mathbb{1}_{B_i} X \, d\mathbb{P} = \int_A YX \, d\mathbb{P}$$

so (9.5) holds for every $\mathcal{G}$-measurable simple Y.

Step 3: Nonnegative Y. Assume $Y \geq 0$ and, for now, $X \geq 0$ as well, so that $Z \geq 0$ a.s. by monotonicity (proposition 9.2.1(2)). By the standard simple-function approximation [2] choose $\mathcal{G}$-measurable simple functions $0 \leq Y_1 \leq Y_2 \leq \cdots$ with $Y_k \uparrow Y$ pointwise. Then $Y_k Z \uparrow YZ$ and $Y_k X \uparrow YX$ pointwise, both increasing sequences of nonnegative variables. Applying the monotone convergence theorem inside each integral and Step 2 to each Y_k,

$$\int_A YZ \, d\mathbb{P} = \lim_{k \to \infty} \int_A Y_k Z \, d\mathbb{P} = \lim_{k \to \infty} \int_A Y_k X \, d\mathbb{P} = \int_A YX \, d\mathbb{P}$$

Since YX is integrable, the common value is finite, and (9.5) holds for nonnegative Y and nonnegative X.

Step 4: General X and Y. For general integrable X write $X = X^+ - X^-$; both $X^\pm \geq 0$ are integrable and $Z = \mathbb{E}[X^+ \big| \mathcal{G}] - \mathbb{E}[X^- \big| \mathcal{G}]$ a.s. by linearity. Likewise write $Y = Y^+ - Y^-$ with $Y^\pm \geq 0$ $\mathcal{G}$-measurable. Each of the four products $Y^\pm X^\pm$ is nonnegative with a nonnegative factor, so Step 3 applies to each, and expanding $YX = (Y^+ - Y^-)(X^+ - X^-)$ and YZ correspondingly and adding the four identities (each term integrable because $|YX| \leq |Y|(X^+ + X^-)$ has finite integral by hypothesis) yields (9.5) for the general case. By uniqueness $\mathbb{E}[YX \big| \mathcal{G}] = YZ = Y\,\mathbb{E}[X \big| \mathcal{G}]$ a.s., completing the proof.

The rule "taking out what is known" is the conditional counterpart of pulling a constant out of an integral, with $\mathcal{G}$-measurability playing the role of being constant. Once one has conditioned on $\mathcal{G}$, any $\mathcal{G}$-measurable factor is frozen, no longer random from the perspective of $\mathcal{G}$, and so it factors out. In tandem with the tower property this is the main step of the martingale calculus: a typical step conditions on the information $\mathcal{F}_n$ available at time n, pulls out the $\mathcal{F}_n$-measurable quantities accumulated so far, and averages only the new increment. The integrability hypothesis on YX cannot be dropped: without it the product need not be conditionable at all, and the identity has no meaning.

Jensen's inequality (theorem 3.2.7) says that a convex function commutes with expectation only up to an inequality: $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$. Because conditional expectation is itself an averaging operator, the same inequality should hold conditionally, with the constant $\mathbb{E}[X]$ replaced by the random variable $\mathbb{E}[X \big| \mathcal{G}]$. This is the conditional Jensen inequality, and it is the engine behind the fact that convex functions of martingales are submartingales, a structural fact used throughout the next chapter.

Proposition 9.2.4: Conditional Jensen Inequality

Let $\varphi \colon \mathbb{R} \to \mathbb{R}$ be convex, and let X be integrable with $\varphi(X)$ integrable. Then

$$\varphi\big(\mathbb{E}[X \big| \mathcal{G}]\big) \leq \mathbb{E}[\varphi(X) \big| \mathcal{G}] \quad \text{a.s.}$$

***Proof*:**
The idea is to replace φ by its supporting lines and condition each linear bound. A convex φ on $\mathbb{R}$ is the supremum of a countable family of affine functions lying below it. Concretely, at each rational q the convex function has a supporting line: there is a slope $m(q) \in \mathbb{R}$ with

$$\varphi(x) \geq \varphi(q) + m(q)(x - q) \qquad \text{for all } x \in \mathbb{R} \tag{9.6}$$

Because φ is continuous and the affine minorants at rational points already touch φ on a dense set, taking the supremum over $q \in \mathbb{Q}$ recovers φ exactly:

$$\varphi(x) = \sup_{q \in \mathbb{Q}} \big(\varphi(q) + m(q)(x - q)\big) \qquad \text{for all } x \in \mathbb{R} \tag{9.7}$$

Fix a rational q and abbreviate $a_q = \varphi(q) - m(q)q$ and $b_q = m(q)$, so the right-hand side of (9.6) is $a_q + b_q x$. Substituting $x = X$ in (9.6) gives $\varphi(X) \geq a_q + b_q X$ pointwise. Both sides are integrable, so conditioning on $\mathcal{G}$ and using monotonicity and linearity (proposition 9.2.1),

$$\mathbb{E}[\varphi(X) \big| \mathcal{G}] \geq \mathbb{E}[a_q + b_q X \big| \mathcal{G}] = a_q + b_q\,\mathbb{E}[X \big| \mathcal{G}] \quad \text{a.s.}$$

Here a_q passes through as a constant and $b_q\,\mathbb{E}[X|\mathcal{G}]$ by linearity. Let N_q be the null set off which this inequality holds, and let $N = \bigcup_{q\in\mathbb{Q}} N_q$, a countable union of null sets, hence null. Off N the inequality

$$\mathbb{E}[\varphi(X)|\mathcal{G}] \geq a_q + b_q\,\mathbb{E}[X|\mathcal{G}] = \varphi(q) + m(q)\big(\mathbb{E}[X|\mathcal{G}] - q\big)$$

holds simultaneously for *every* rational q.

Taking the supremum over $q \in \mathbb{Q}$ on the right and applying the representation (9.7) with $x = \mathbb{E}[X|\mathcal{G}](\omega)$ for each fixed $\omega \notin N$,

$$\mathbb{E}[\varphi(X)|\mathcal{G}] \geq \sup_{q\in\mathbb{Q}}\Big(\varphi(q) + m(q)\big(\mathbb{E}[X|\mathcal{G}] - q\big)\Big) = \varphi\big(\mathbb{E}[X|\mathcal{G}]\big)$$

off the null set N. This is the claimed inequality a.s., as we sought to show.

The countable family of supporting lines is what makes the argument work: a single supporting line would give the inequality only where its slope is the right one, but conditioning is a random operation and $\mathbb{E}[X|\mathcal{G}]$ ranges over an interval, so one needs a line for every possible value at once. Handling countably many rational slopes controls the exceptional null sets, and density of $\mathbb{Q}$ recovers the full convex function in the limit. Two special cases are worth recording immediately: $\varphi(x) = |x|$ recovers the bound $|\mathbb{E}[X|\mathcal{G}]| \leq \mathbb{E}[|X||\mathcal{G}]$ already noted in proposition 9.2.1(2), and $\varphi(x) = x^2$ gives $\mathbb{E}[X|\mathcal{G}]^2 \leq \mathbb{E}[X^2|\mathcal{G}]$, the conditional form of the inequality $\mathbb{E}[X]^2 \leq \mathbb{E}[X^2]$ that underlies the conditional variance of the next section. Applied to the convex map $x \mapsto |x|^p$ for $p \geq 1$, it also shows that $\mathbb{E}[\,\cdot\,|\mathcal{G}]$ is a contraction on L^p: taking expectations of $|\mathbb{E}[X|\mathcal{G}]|^p \leq \mathbb{E}[|X|^p|\mathcal{G}]$ and using total expectation gives $\mathbb{E}[|\mathbb{E}[X|\mathcal{G}]|^p] \leq \mathbb{E}[|X|^p]$, so conditioning never increases the L^p norm.

The final structural fact is that all three of the standard convergence theorems for the ordinary integral, monotone convergence, Fatou's lemma, and dominated convergence, have conditional versions. They are proved by the same principle that has governed the whole section: each is obtained by applying the corresponding ordinary theorem inside the defining integral identity.

Proposition 9.2.5: Conditional Convergence Theorems

Let $\mathcal{G} \subseteq \mathcal{F}$ and let X_n be random variables.

1. *(Conditional monotone convergence.)* If $0 \leq X_n \uparrow X$ a.s. with X integrable, then $\mathbb{E}[X_n|\mathcal{G}] \uparrow \mathbb{E}[X|\mathcal{G}]$ a.s.
2. *(Conditional Fatou.)* If $X_n \geq 0$ a.s. and $\liminf_n X_n$ is integrable, then
$$\mathbb{E}\Big[\liminf_{n\to\infty} X_n \;\Big|\; \mathcal{G}\Big] \leq \liminf_{n\to\infty} \mathbb{E}[X_n|\mathcal{G}] \quad \text{a.s.}$$
3. *(Conditional dominated convergence.)* If $X_n \to X$ a.s. and $|X_n| \leq W$ a.s. for all n with W integrable, then $\mathbb{E}[X_n|\mathcal{G}] \to \mathbb{E}[X|\mathcal{G}]$ a.s.

***Proof*:**

1. *Monotone convergence.* By monotonicity (proposition 9.2.1(2)) the sequence $Z_n = \mathbb{E}[X_n|\mathcal{G}]$ is a.s. nondecreasing, so $Z = \lim_n Z_n$ exists a.s. in $[0, \infty]$ and is $\mathcal{G}$-measurable. For $A \in \mathcal{G}$,

two applications of the ordinary monotone convergence theorem, together with the defining identity for each Z_n, give

$$\int_A Z\,d\mathbb{P} = \lim_{n\to\infty}\int_A Z_n\,d\mathbb{P} = \lim_{n\to\infty}\int_A X_n\,d\mathbb{P} = \int_A X\,d\mathbb{P}$$

the first and last limits by monotone convergence applied to $Z_n \uparrow Z$ and $X_n \uparrow X$ on A. In particular $\int_\Omega Z\,d\mathbb{P} = \mathbb{E}[X] < \infty$, so Z is integrable, hence finite a.s., and its integral over every $A \in \mathcal{G}$ agrees with that of X. By uniqueness $Z = \mathbb{E}[X|\mathcal{G}]$ a.s.

2. *Fatou.* Set $Y_n = \inf_{k\geq n} X_k$, so $0 \leq Y_n \uparrow \liminf_n X_n$ a.s. Since $Y_n \leq X_k$ for every $k \geq n$, monotonicity gives $\mathbb{E}[Y_n|\mathcal{G}] \leq \mathbb{E}[X_k|\mathcal{G}]$ a.s. for all $k \geq n$, hence $\mathbb{E}[Y_n|\mathcal{G}] \leq \inf_{k\geq n}\mathbb{E}[X_k|\mathcal{G}]$ a.s. Letting $n \to \infty$ and using part (1) on the increasing sequence $Y_n \uparrow \liminf_n X_n$,

$$\mathbb{E}\Big[\liminf_{n\to\infty} X_n \,\Big|\, \mathcal{G}\Big] = \lim_{n\to\infty}\mathbb{E}[Y_n|\mathcal{G}] \leq \lim_{n\to\infty}\inf_{k\geq n}\mathbb{E}[X_k|\mathcal{G}] = \liminf_{n\to\infty}\mathbb{E}[X_n|\mathcal{G}] \quad \text{a.s.}$$

3. *Dominated convergence.* The variables $W + X_n \geq 0$ and $W - X_n \geq 0$ are nonnegative a.s. Applying conditional Fatou (part 2) to $W + X_n$, whose liminf is $W + X$,

$$\mathbb{E}[W|\mathcal{G}] + \mathbb{E}[X|\mathcal{G}] \leq \liminf_{n\to\infty}\big(\mathbb{E}[W|\mathcal{G}] + \mathbb{E}[X_n|\mathcal{G}]\big) = \mathbb{E}[W|\mathcal{G}] + \liminf_{n\to\infty}\mathbb{E}[X_n|\mathcal{G}]$$

and since $\mathbb{E}[W|\mathcal{G}]$ is finite a.s. it cancels, giving $\mathbb{E}[X|\mathcal{G}] \leq \liminf_n \mathbb{E}[X_n|\mathcal{G}]$ a.s. Applying the same to $W - X_n$, whose liminf is $W - X$, gives $\mathbb{E}[X|\mathcal{G}] \geq \limsup_n \mathbb{E}[X_n|\mathcal{G}]$ a.s. Together $\limsup_n \mathbb{E}[X_n|\mathcal{G}] \leq \mathbb{E}[X|\mathcal{G}] \leq \liminf_n \mathbb{E}[X_n|\mathcal{G}]$ a.s., forcing the limit to exist and equal $\mathbb{E}[X|\mathcal{G}]$ a.s., completing the proof.

These conditional limit theorems are exactly what one expects: conditional expectation, being an average, respects monotone and dominated passage to the limit just as the ordinary integral does. They are the tools that let one interchange a limit with a conditioning operation, which is needed at every point where a conditional expectation is defined as, or manipulated through, a limit, most immediately in the closure argument for the L^2-projection of the next section and in the martingale convergence theorems beyond it.

With the calculus assembled, the following worked computations show it in action on the grounding examples of this chapter.

Example 9.2.6: Computing with the Conditional Calculus

1. *The tower property collapsing a two-stage average.* Let $(X_1, X_2, \dots)$ be an i.i.d. sequence of fair coin tosses on the coin-tossing space (example 1.3.6, theorem 2.3.5), with $X_i \in \{0,1\}$ and $\mathbb{P}(X_i = 1) = \frac{1}{2}$, and let $S_n = X_1 + \cdots + X_n$. Fix $m < n$ and put $\mathcal{G} = \sigma(X_1, \dots, X_m)$. Since S_m is $\mathcal{G}$-measurable and $X_{m+1}, \dots, X_n$ are independent of $\mathcal{G}$, the measurability rule (proposition 9.2.1(4)) on the $\mathcal{G}$-measurable part and the independence rule (proposition 9.2.1(5)) on the rest give

$$\mathbb{E}[S_n|\mathcal{G}] = S_m + \sum_{i=m+1}^{n}\mathbb{E}[X_i|\mathcal{G}] = S_m + (n-m)\cdot\tfrac{1}{2}$$

Now condition on the coarser $\mathcal{H} = \sigma(X_1, \dots, X_\ell)$ with $\ell < m$. The tower property (proposition 9.2.2) predicts $\mathbb{E}[\mathbb{E}[S_n|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[S_n|\mathcal{H}]$, and both sides evaluate to $S_\ell + (n-\ell)\cdot\frac{1}{2}$: the left side because $\mathbb{E}[S_m|\mathcal{H}] = S_\ell + (m-\ell)\cdot\frac{1}{2}$ by the same argument, and the constant $(n-m)/2$ passes through, while the right side is the one-step computation. The two-stage average has collapsed to a single-stage one.

2. *Taking out a known factor.* Let Y be $\mathcal{G}$-measurable and bounded, and let X be integrable. Then YX is integrable and proposition 9.2.3 gives $\mathbb{E}[YX|\mathcal{G}] = Y\,\mathbb{E}[X|\mathcal{G}]$. Concretely, if $Y = \mathbb{1}_G$ for a set $G \in \mathcal{G}$, this reads $\mathbb{E}[\mathbb{1}_G X|\mathcal{G}] = \mathbb{1}_G\,\mathbb{E}[X|\mathcal{G}]$: conditioning commutes with restriction to a $\mathcal{G}$-event. Taking $\mathcal{G} = \sigma(Y)$ for a discrete Y with $\mathbb{E}[X|Y] = g(Y)$, this recovers the elementary formula $\mathbb{E}[Yh(Y)X|Y] = Yh(Y)\,g(Y)$ for any function h, the known factor $Yh(Y)$ coming straight out.

3. *Conditional Jensen and the second moment.* Applying proposition 9.2.4 to the convex map $\varphi(x) = x^2$ yields
$$\mathbb{E}[X|\mathcal{G}]^2 \le \mathbb{E}[X^2|\mathcal{G}] \quad \text{a.s.}$$
whenever $X \in L^2$. Taking ordinary expectations of both sides and using total expectation (proposition 9.2.1(3)) recovers the familiar $\mathbb{E}[X]^2 = \mathbb{E}\big[\mathbb{E}[X|\mathcal{G}]\big]^2 \le \mathbb{E}[X^2]$, but the conditional statement is strictly finer: it holds pointwise (a.s.) as an inequality between random variables, and the gap $\mathbb{E}[X^2|\mathcal{G}] - \mathbb{E}[X|\mathcal{G}]^2$ is precisely the conditional variance developed in the next section. For a concrete instance, take $X = S_n$ in part (1) and $\mathcal{G} = \sigma(X_1, \ldots, X_m)$: then $\mathbb{E}[S_n|\mathcal{G}]^2 = \big(S_m + (n-m)/2\big)^2$, while $\mathbb{E}[S_n^2|\mathcal{G}]$ exceeds it by the variance $(n-m)/4$ of the independent future increments, so the inequality is strict on the event that future tosses remain random.

Taken together, the properties of this section portray $\mathbb{E}[\,\cdot|\mathcal{G}]$ as a positive, order-preserving, averaging operator on L^1 that fixes every $\mathcal{G}$-measurable variable and reverts to the unconditional mean on variables independent of $\mathcal{G}$. Linearity and monotonicity make it a positive linear map; conditional Jensen makes precise that it is an *average* and not only a positive map, since it moves convex functions in the direction an average must; the tower property organizes iterated conditioning by the coarser σ-algebra; and taking out what is known says that $\mathcal{G}$-measurable variables behave as scalars under it. The next section sharpens this operator picture on L^2, where these same properties become the geometry of orthogonal projection and $\mathbb{E}[X|\mathcal{G}]$ emerges as the best mean-square predictor of X from the information in $\mathcal{G}$.

Exercise 9.2.1

1. Let X be integrable and $\mathcal{G} \subseteq \mathcal{F}$. Using only the defining property (9.4) and proposition 9.2.1, show that if $c \in \mathbb{R}$ is a constant then $\mathbb{E}[c|\mathcal{G}] = c$ a.s., and deduce that $\mathbb{E}[X + c|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}] + c$ a.s.

2. Let Y be a bounded $\mathcal{G}$-measurable variable and X integrable. Prove that
$$\mathbb{E}\big[Y\big(X - \mathbb{E}[X|\mathcal{G}]\big)\big] = 0$$
Interpret this as an orthogonality statement: the residual $X - \mathbb{E}[X|\mathcal{G}]$ is uncorrelated with every bounded $\mathcal{G}$-measurable variable. (This is the seed of the projection picture of the next section.)

3. Let $\mathcal{H} \subseteq \mathcal{G}$ and let X be integrable. Working directly from the defining property (9.4), give a one-line derivation of the tower property $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$ a.s. Deduce that the residual

satisfies $\mathbb{E}\big[(X - \mathbb{E}[X|\mathcal{G}])\big|\mathcal{H}\big] = 0$ a.s.

4. Let X be a fair coin toss, $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = \frac{1}{2}$, and let $\mathcal{G} = \{\emptyset, \Omega\}$ be trivial. Compute both sides of the conditional Jensen inequality for $\varphi(x) = x^2$ and confirm the inequality is strict. Then exhibit a $\mathcal{G}$ for which it becomes an equality, and explain in one sentence what distinguishes the two cases.

5. Using conditional Jensen, prove that for any $p \geq 1$ and $X \in L^p$,

$$\big\|\mathbb{E}[X|\mathcal{G}]\big\|_p \leq \|X\|_p$$

so that conditioning is a contraction on L^p (definition 3.3.1). Where does the argument use the integrability of $|X|^p$?

6. Give an example of nonnegative random variables X_n with $X_n \to 0$ a.s. but $\mathbb{E}[X_n|\mathcal{G}] \not\to 0$ a.s. for some $\mathcal{G}$, showing that conditional dominated convergence requires the dominating variable W. (Hint: adapt the standard sliding-mass counterexample and take $\mathcal{G}$ trivial, so conditional expectation is ordinary expectation.)

9.3 Conditional Expectation as Projection

The defining property of $\mathbb{E}[X|\mathcal{G}]$ from section 9.1 is an averaging property: the conditional expectation agrees with X in integral over every event of $\mathcal{G}$. That property is what makes the Radon-Nikodym construction work, but it says nothing directly about $\mathbb{E}[X|\mathcal{G}]$ being a good *estimate* of X. This section supplies the estimation reading, and it is the reading that governs every applied use of conditioning. Restrict attention to variables with a finite second moment. Among all $\mathcal{G}$-measurable functions, which one is closest to X in mean square, that is, which $\mathcal{G}$-measurable Z minimizes the average squared error $\mathbb{E}[(X - Z)^2]$? The answer is $\mathbb{E}[X|\mathcal{G}]$, and the reason is geometry: on L^2 the conditional expectation is nothing but the orthogonal projection of X onto the subspace of $\mathcal{G}$-measurable variables. The averaging property and the least-squares property are the same fact seen from two sides, and the bridge between them is the Hilbert space structure of L^2 recorded in theorem 3.3.4.

The picture to carry is the elementary one from finite-dimensional geometry. Given a plane through the origin in $\mathbb{R}^3$ and a point X off the plane, the point of the plane nearest to X is the foot of the perpendicular dropped from X; the error vector from that foot to X is orthogonal to the plane. Replace $\mathbb{R}^3$ by the Hilbert space $L^2(\mathcal{F})$ of square-integrable random variables with inner product $\langle X, Y\rangle = \mathbb{E}[XY]$, and replace the plane by the subspace $L^2(\mathcal{G})$ of those square-integrable variables that are $\mathcal{G}$-measurable. The nearest point of $L^2(\mathcal{G})$ to X is $\mathbb{E}[X|\mathcal{G}]$, and the error $X - \mathbb{E}[X|\mathcal{G}]$ is orthogonal to every $\mathcal{G}$-measurable variable. Everything in this section is the translation of that one image into the language of conditioning.

To make the statement precise, write $L^2(\mathcal{G})$ for the set of (equivalence classes of) random variables Z that are $\mathcal{G}$-measurable and satisfy $\mathbb{E}[Z^2] < \infty$, a subset of the space $L^2(\mathcal{F}) = L^2(\Omega, \mathcal{F}, \mathbb{P})$ from definition 3.3.1. The first thing to check is that this subset is a *closed* subspace, for only then does the projection exist; a nearest point onto a non-closed set need not be attained. That closedness is where the $\mathcal{G}$-measurability enters, and it is the technical heart of the whole construction.

Lemma 9.3.1: $L^2(\mathcal{G})$ is a Closed Subspace

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra of a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$L^2(\mathcal{G}) = \{Z \in L^2(\Omega, \mathcal{F}, \mathbb{P}) \colon Z \text{ is } \mathcal{G}\text{-measurable}\}$$

is a closed linear subspace of the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$. In fact $L^2(\mathcal{G})$ is, with the restricted inner product, exactly the Hilbert space $L^2(\Omega, \mathcal{G}, \mathbb{P}|_{\mathcal{G}})$ of square-integrable variables on the smaller measurable space.

***Proof*:**
Linearity is immediate: a linear combination $aZ_1 + bZ_2$ of $\mathcal{G}$-measurable variables is $\mathcal{G}$-measurable, and it lies in L^2 because L^2 is a vector space (theorem 3.3.4). The subspace also contains the constants, since a constant is measurable with respect to every σ-algebra and is square-integrable on a probability space.

For closedness, suppose $Z_n \in L^2(\mathcal{G})$ and $Z_n \to Z$ in the $L^2(\mathcal{F})$ norm; the claim is that $Z \in L^2(\mathcal{G})$. Convergence in L^2 gives a subsequence Z_{n_k} converging to Z almost surely: an L^2-convergent sequence is L^1-convergent on a probability space by the nesting $L^2 \subseteq L^1$, and an L^1-convergent sequence has an almost-surely convergent subsequence, both facts recorded in theorem 3.3.4 and its surrounding development. Set $Z' = \limsup_k Z_{n_k}$, a genuinely $\mathcal{G}$-measurable function everywhere on Ω, since the limsup of a sequence of $\mathcal{G}$-measurable functions is $\mathcal{G}$-measurable. On the event where Z_{n_k} converges, which has probability one, Z' equals that limit, so $Z' = Z$ almost surely. Elements of L^2 are equivalence classes modulo null sets, so Z and Z' represent the same class; the class of Z thus has the $\mathcal{G}$-measurable representative Z'. Hence $Z \in L^2(\mathcal{G})$, and the subspace is closed.

The identification with $L^2(\Omega, \mathcal{G}, \mathbb{P}|_{\mathcal{G}})$ is a matter of unwinding definitions: a $\mathcal{G}$-measurable function is exactly a measurable function on $(\Omega, \mathcal{G})$, and its L^2 norm computed against $\mathbb{P}$ agrees with the norm computed against the restriction $\mathbb{P}|_{\mathcal{G}}$, since the two measures assign the same value to every set of $\mathcal{G}$. A closed subspace of a Hilbert space is itself a Hilbert space, completing the proof.

The lemma is the only place where measurability does real work; once $L^2(\mathcal{G})$ is known to be a closed subspace, the rest is Hilbert-space geometry that applies to any closed subspace of any Hilbert space. The closedness is a consequence of $\mathcal{G}$-measurability, not a formality: the set of variables satisfying some non-σ-algebra constraint, say "Z is a polynomial in a fixed variable of degree at most d", is a subspace but the argument that its L^2 limits stay inside would fail without the measurability structure. With the closed subspace in hand, the projection theorem for Hilbert spaces gives a unique nearest point, and the entire content of the main theorem is that this nearest point is the conditional expectation already constructed.

Before stating the theorem, it is worth isolating the geometric fact it rests on, because the proof uses it twice and it is the source of both the existence of the best predictor and its characterization by orthogonality. The statement is the Hilbert projection theorem specialized to L^2; the proof is short enough, and central enough to conditioning, that it is given in full rather than cited.

Proposition 9.3.2: Orthogonal Projection onto a Closed Subspace

Let H be a Hilbert space, $M \subseteq H$ a closed subspace, and $X \in H$. Then there is a unique $Y \in M$ minimizing $\|X - Z\|$ over $Z \in M$, and Y is characterized among elements of M by the orthogonality condition

$$\langle X - Y, W\rangle = 0 \qquad \text{for every } W \in M$$

The map $X \mapsto Y$, written $P_M X$, is linear, satisfies $P_M X = X$ for $X \in M$, and obeys the Pythagorean decomposition $\|X\|^2 = \|P_M X\|^2 + \|X - P_M X\|^2$.

***Proof*:**
Existence. Let $d = \inf_{Z \in M} \|X - Z\|$ and choose $Z_n \in M$ with $\|X - Z_n\| \to d$. The parallelogram law, valid in any inner product space, applied to the vectors $X - Z_n$ and $X - Z_m$ gives

$$\|Z_n - Z_m\|^2 = 2\|X - Z_n\|^2 + 2\|X - Z_m\|^2 - 4\left\|X - \tfrac{Z_n + Z_m}{2}\right\|^2$$

Since M is a subspace, the midpoint $\frac{1}{2}(Z_n + Z_m)$ lies in M, so the last norm is at least d. Letting $n, m \to \infty$, the first two terms tend to $2d^2$ each and the subtracted term is at least d^2, so $\|Z_n - Z_m\|^2 \le 4d^2 - 4d^2 = 0$ in the limit; (Z_n) is Cauchy. By completeness of H it converges to some Y, and by closedness of M the limit Y lies in M; continuity of the norm gives $\|X - Y\| = d$, so Y attains the infimum.

Orthogonality characterizes the minimizer. Suppose $Y \in M$ attains the minimum and fix $W \in M$. For real t the vector $Y + tW$ lies in M, so the function

$$\varphi(t) = \|X - Y - tW\|^2 = \|X - Y\|^2 - 2t\,\langle X - Y, W\rangle + t^2\|W\|^2$$

is minimized at $t = 0$. Its derivative at 0 is $-2\langle X - Y, W\rangle$, which must vanish, giving $\langle X - Y, W\rangle = 0$ for every $W \in M$. Conversely, if $Y \in M$ satisfies this orthogonality condition, then for any $Z \in M$ the difference $W = Y - Z \in M$ is orthogonal to $X - Y$, and Pythagoras gives

$$\|X - Z\|^2 = \|(X - Y) + (Y - Z)\|^2 = \|X - Y\|^2 + \|Y - Z\|^2 \ge \|X - Y\|^2$$

so Y is a minimizer. In particular the minimizer is unique, since the displayed inequality is strict unless $Z = Y$.

Linearity and the Pythagorean identity. If $Y_1 = P_M X_1$ and $Y_2 = P_M X_2$, then $aY_1 + bY_2 \in M$ and, for every $W \in M$, $\langle (aX_1 + bX_2) - (aY_1 + bY_2), W\rangle = a\langle X_1 - Y_1, W\rangle + b\langle X_2 - Y_2, W\rangle = 0$, so by the characterization $aY_1 + bY_2 = P_M(aX_1 + bX_2)$. For $X \in M$ the orthogonality holds trivially with $Y = X$. Finally, taking $W = P_M X \in M$ in the orthogonality condition makes $X - P_M X$ and $P_M X$ orthogonal, and Pythagoras applied to $X = (X - P_M X) + P_M X$ gives the stated decomposition, completing the proof.

The proposition is the geometry; the theorem is its probabilistic content. The projection $P_{L^2(\mathcal{G})}X$ is characterized by orthogonality to every $\mathcal{G}$-measurable variable, and orthogonality against the *indicators* of $\mathcal{G}$-sets is precisely the defining averaging property of the conditional expectation. That is the identification, and it is exact: no almost-sure slack beyond the usual one, no hypothesis beyond $X \in L^2$.

Theorem 9.3.3: Conditional Expectation as Orthogonal Projection

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra, and $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Then:

1. $\mathbb{E}[X|\mathcal{G}]$ lies in $L^2(\mathcal{G})$, and it is the orthogonal projection of X onto the closed subspace $L^2(\mathcal{G})$; that is,
$$\mathbb{E}[X|\mathcal{G}] = P_{L^2(\mathcal{G})}X$$
2. The residual $X - \mathbb{E}[X|\mathcal{G}]$ is orthogonal to $L^2(\mathcal{G})$: for every $\mathcal{G}$-measurable $W \in L^2$,
$$\mathbb{E}\big[(X - \mathbb{E}[X|\mathcal{G}])\,W\big] = 0$$
3. $\mathbb{E}[X|\mathcal{G}]$ is the best mean-square predictor of X by a $\mathcal{G}$-measurable variable: among all $\mathcal{G}$-measurable $Z \in L^2$,
$$\mathbb{E}\big[(X - \mathbb{E}[X|\mathcal{G}])^2\big] = \min_{Z \in L^2(\mathcal{G})} \mathbb{E}\big[(X - Z)^2\big]$$
and the minimizer is almost-surely unique.

***Proof*:**
Write $Y = P_{L^2(\mathcal{G})}X$ for the orthogonal projection supplied by proposition 9.3.2, applied to the Hilbert space $H = L^2(\mathcal{F})$ and the closed subspace $M = L^2(\mathcal{G})$ of lemma 9.3.1. The crux is that Y satisfies the defining property of $\mathbb{E}[X|\mathcal{G}]$ from definition 9.1.1. By construction Y is $\mathcal{G}$-measurable and square-integrable, hence integrable on a probability space. Fix $A \in \mathcal{G}$. Its indicator $\mathbb{1}_A$ is $\mathcal{G}$-measurable and bounded, hence lies in $L^2(\mathcal{G})$, so the orthogonality characterization in proposition 9.3.2 applied to $W = \mathbb{1}_A$ gives

$$\mathbb{E}\big[(X - Y)\mathbb{1}_A\big] = \langle X - Y, \mathbb{1}_A \rangle = 0$$

that is $\int_A Y\, d\mathbb{P} = \int_A X\, d\mathbb{P}$. This holds for every $A \in \mathcal{G}$, so Y is a $\mathcal{G}$-measurable integrable variable with the same integral as X over every $\mathcal{G}$-set. By the almost-sure uniqueness clause of definition 9.1.1 (equivalently the uniqueness in the existence theorem for conditional expectation), $Y = \mathbb{E}[X|\mathcal{G}]$ almost surely. With this identification the three parts follow.

1. The displayed identification $Y = \mathbb{E}[X|\mathcal{G}]$ is precisely $\mathbb{E}[X|\mathcal{G}] = P_{L^2(\mathcal{G})}X$, and since $Y \in M = L^2(\mathcal{G})$ by construction, the conditional expectation lies in $L^2(\mathcal{G})$.
2. The residual $X - \mathbb{E}[X|\mathcal{G}]$ equals $X - Y$, which by proposition 9.3.2 is orthogonal to every element of $M = L^2(\mathcal{G})$. Thus for every $\mathcal{G}$-measurable $W \in L^2$,
$$\mathbb{E}\big[(X - \mathbb{E}[X|\mathcal{G}])\,W\big] = \langle X - Y, W \rangle = 0$$
so the residual has conditional mean zero and is uncorrelated with every $\mathcal{G}$-measurable statistic.
3. By the minimization clause of proposition 9.3.2, Y minimizes $\|X - Z\|^2 = \mathbb{E}[(X - Z)^2]$ over $Z \in M = L^2(\mathcal{G})$, and the minimizer is unique in L^2, that is almost-surely unique. Since $Y = \mathbb{E}[X|\mathcal{G}]$, this is the best-mean-square-predictor claim, completing the proof.

The theorem is the geometric heart of estimation, and the vocabulary it licenses is worth pausing on. Reading $L^2(\mathcal{G})$ as the collection of predictions that can be computed from the information in $\mathcal{G}$, the conditional expectation is the *prediction of least squared error*. The residual $X - \mathbb{E}[X|\mathcal{G}]$ is the part of X that the information in $\mathcal{G}$ cannot explain; part (2) says this unexplained part is orthogonal to everything $\mathcal{G}$ can produce, which is the statistician's statement that the optimal predictor extracts every feature of X that $\mathcal{G}$ can express, leaving a residual uncorrelated with all of them. The three parts are one fact: minimizing squared error, projecting orthogonally, and averaging over $\mathcal{G}$-sets all name the same operation.

Two remarks sharpen the scope. First, the L^2 hypothesis is what gives the geometry; for integrable X that is not square-integrable the conditional expectation still exists by the Radon-Nikodym construction of section 9.1, but it is no longer literally a projection, because L^1 is not a Hilbert space and has no inner product to project along. The projection theorem is therefore a statement about the well-behaved L^2 variables, extended to L^1 afterward by approximation and the order and continuity properties of section 9.2. Second, the orthogonal-projection identity gives, at no cost, a short proof of the L^2 contraction property: since $X = P_{L^2(\mathcal{G})}X + (X - P_{L^2(\mathcal{G})}X)$ is an orthogonal decomposition, the Pythagorean identity yields

$$\mathbb{E}[X^2] = \mathbb{E}\big[\mathbb{E}[X|\mathcal{G}]^2\big] + \mathbb{E}\big[(X - \mathbb{E}[X|\mathcal{G}])^2\big]$$

so in particular $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]^2] \le \mathbb{E}[X^2]$; conditioning never increases the second moment. Conditioning is a contraction on L^2, and the defect $\mathbb{E}[X^2] - \mathbb{E}[\mathbb{E}[X|\mathcal{G}]^2]$ is exactly the mean squared prediction error.

That last display is the identity from which the law of total variance falls out, and it is the second boxed result of the section. The idea of decomposing the variability of X into a within-$\mathcal{G}$ part and a between-$\mathcal{G}$ part is one of the most heavily used facts in statistics, appearing under the names analysis of variance, the bias-variance tradeoff, and Rao-Blackwellization. To state it cleanly, the conditional variance is defined by mimicking the definition of variance with conditional in place of ordinary expectation.

Proposition 9.3.4: Conditional Variance and the Law of Total Variance

Let $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra. Define the *conditional variance* of X given $\mathcal{G}$ by

$$\mathrm{Var}(X|\mathcal{G}) = \mathbb{E}\big[(X - \mathbb{E}[X|\mathcal{G}])^2\big|\mathcal{G}\big]$$

a nonnegative $\mathcal{G}$-measurable random variable. Then

$$\mathrm{Var}(X|\mathcal{G}) = \mathbb{E}[X^2|\mathcal{G}] - \mathbb{E}[X|\mathcal{G}]^2 \quad \text{a.s.}$$

and the *law of total variance* holds:

$$\mathrm{Var}(X) = \mathbb{E}\big[\mathrm{Var}(X|\mathcal{G})\big] + \mathrm{Var}\big(\mathbb{E}[X|\mathcal{G}]\big)$$

***Proof*:**
Throughout, write $M = \mathbb{E}[X|\mathcal{G}]$, a $\mathcal{G}$-measurable variable in L^2 by theorem 9.3.3, so that M^2 and XM are integrable (the latter by Cauchy-Schwarz, both in L^2).

Step 1: the computational form. Expand the square inside the conditional expectation and use conditional linearity from proposition 9.2.1:

$$\operatorname{Var}(X|\mathcal{G}) = \mathbb{E}\big[X^2 - 2XM + M^2\big|\mathcal{G}\big] = \mathbb{E}[X^2|\mathcal{G}] - 2\,\mathbb{E}[XM|\mathcal{G}] + \mathbb{E}[M^2|\mathcal{G}]$$

Because M is $\mathcal{G}$-measurable, taking out what is known (proposition 9.2.3) gives $\mathbb{E}[XM|\mathcal{G}] = M\,\mathbb{E}[X|\mathcal{G}] = M^2$ and $\mathbb{E}[M^2|\mathcal{G}] = M^2$. Substituting,

$$\operatorname{Var}(X|\mathcal{G}) = \mathbb{E}[X^2|\mathcal{G}] - 2M^2 + M^2 = \mathbb{E}[X^2|\mathcal{G}] - M^2$$

which is the claimed identity $\mathbb{E}[X^2|\mathcal{G}] - \mathbb{E}[X|\mathcal{G}]^2$. In particular $\operatorname{Var}(X|\mathcal{G}) \geq 0$ almost surely, being a conditional expectation of the nonnegative variable $(X-M)^2$.

Step 2: take ordinary expectations of the two pieces. Apply the total-expectation identity $\mathbb{E}[\mathbb{E}[\,\cdot\,|\mathcal{G}]] = \mathbb{E}[\,\cdot\,]$ from proposition 9.2.1. From Step 1,

$$\mathbb{E}\big[\operatorname{Var}(X|\mathcal{G})\big] = \mathbb{E}\big[\mathbb{E}[X^2|\mathcal{G}]\big] - \mathbb{E}[M^2] = \mathbb{E}[X^2] - \mathbb{E}[M^2]$$

For the second summand, $M = \mathbb{E}[X|\mathcal{G}]$ has the same mean as X, since $\mathbb{E}[M] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[X]$ by the same identity; writing $\mu = \mathbb{E}[X]$,

$$\operatorname{Var}(M) = \mathbb{E}[M^2] - (\mathbb{E}[M])^2 = \mathbb{E}[M^2] - \mu^2$$

Step 3: add. Combining the two lines,

$$\mathbb{E}\big[\operatorname{Var}(X|\mathcal{G})\big] + \operatorname{Var}\big(\mathbb{E}[X|\mathcal{G}]\big) = \big(\mathbb{E}[X^2] - \mathbb{E}[M^2]\big) + \big(\mathbb{E}[M^2] - \mu^2\big) = \mathbb{E}[X^2] - \mu^2 = \operatorname{Var}(X)$$

the middle terms cancelling, which is the law of total variance, as we sought to show.

The decomposition splits the total variability of X into two orthogonal contributions. The first, $\mathbb{E}[\operatorname{Var}(X|\mathcal{G})]$, is the average residual variability that remains after $\mathcal{G}$ is known; it is the part $\mathcal{G}$ cannot explain, and it equals the mean squared prediction error $\mathbb{E}[(X - \mathbb{E}[X|\mathcal{G}])^2]$ appearing in theorem 9.3.3. The second, $\operatorname{Var}(\mathbb{E}[X|\mathcal{G}])$, is the variability of the prediction itself as the information varies; it is the part of the variance that $\mathcal{G}$ *explains*. The law of total variance is thus the Pythagorean identity of the previous discussion, taken in expectation: total squared length equals explained plus unexplained, exactly as $\mathbb{E}[X^2] = \mathbb{E}[M^2] + \mathbb{E}[(X-M)^2]$ after centering. Two extreme cases anchor it. If X is itself $\mathcal{G}$-measurable then $\mathbb{E}[X|\mathcal{G}] = X$, the conditional variance is zero, and all variance is explained; if X is independent of $\mathcal{G}$ then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$ is constant, the explained part is zero, and all variance is residual.

With the theory in place, the three worked examples make it operational: the best mean-square predictor of one variable from another, the total-variance bookkeeping for a mixture, and the contrast that shows why the optimal predictor is generally a curved function of the data rather than a straight line.

Example 9.3.5: Best Mean-Square Prediction, Mixtures, and Linear versus Nonlinear Predictors

1. *The best predictor of X from Y.* Let (X,Y) be a pair of square-integrable variables and condition on $\mathcal{G} = \sigma(Y)$. By the Doob-Dynkin lemma (lemma 2.1.6), every $\sigma(Y)$-measurable variable is a Borel function $g(Y)$, so $L^2(\sigma(Y)) = \{g(Y) \colon \mathbb{E}[g(Y)^2] < \infty\}$. Theorem 9.3.3 then says that among all functions g of the data, the one minimizing the mean squared error

$\mathbb{E}[(X - g(Y))^2]$ is the *regression function*

$$g^\star(y) = \mathbb{E}[X \big| Y = y] \quad \text{so that} \quad \mathbb{E}[X \big| Y] = g^\star(Y)$$

This is the precise sense in which conditional expectation is "the" predictor: no measurable function of Y beats the regression function in mean square. Take a concrete case: $Y \sim \mathrm{Unif}(0,1)$ and, given $Y = y$, let X be uniform on $[0, y]$, so $\mathbb{E}[X \big| Y = y] = y/2$. The best predictor is $\mathbb{E}[X \big| Y] = Y/2$, and its mean squared error is $\mathbb{E}[(X - Y/2)^2] = \mathbb{E}[\mathrm{Var}(X \big| Y)]$. Since the conditional variance of a $\mathrm{Unif}(0, y)$ variable is $y^2/12$, this error is $\mathbb{E}[Y^2/12] = \frac{1}{12}\int_0^1 y^2\, dy = \frac{1}{36}$.

2. *Total-variance decomposition for a two-component mixture.* Let a coin with $\mathbb{P}(N = 1) = p$, $\mathbb{P}(N = 0) = 1 - p$ select which of two populations a measurement comes from: given $N = 1$ the observation X has mean μ_1 and variance σ_1^2, and given $N = 0$ it has mean μ_0 and variance σ_0^2. Take $\mathcal{G} = \sigma(N)$. Then $\mathbb{E}[X \big| \mathcal{G}]$ equals μ_1 on $\{N = 1\}$ and μ_0 on $\{N = 0\}$, and $\mathrm{Var}(X \big| \mathcal{G})$ equals σ_1^2 and σ_0^2 respectively. The law of total variance (proposition 9.3.4) reads

$$\mathrm{Var}(X) = \underbrace{\left(p\sigma_1^2 + (1-p)\sigma_0^2\right)}_{\mathbb{E}[\mathrm{Var}(X \big| \mathcal{G})]} + \underbrace{p(1-p)(\mu_1 - \mu_0)^2}_{\mathrm{Var}(\mathbb{E}[X \big| \mathcal{G}])}$$

the first term the average within-population variance, the second the between-population spread of the group means. The between-group term is the variance of a two-valued variable taking μ_1 with probability p and μ_0 with probability $1 - p$, namely $p(1-p)(\mu_1 - \mu_0)^2$. This is the analysis-of-variance decomposition: separated group means inflate variance through the second term even when each group is individually tight.

3. *Linear versus nonlinear prediction.* The best predictor $\mathbb{E}[X \big| Y]$ is generally a curved function of Y, and it coincides with the best *linear* predictor $a + bY$ only in special cases. Let $Y \sim \mathrm{Unif}(-1, 1)$ and $X = Y^2$. The best predictor is exact, $\mathbb{E}[X \big| Y] = Y^2$, with zero error, since X is a function of Y. The best linear predictor $a + bY$ minimizes $\mathbb{E}[(Y^2 - a - bY)^2]$; by orthogonality of the residual to 1 and to Y, the coefficients solve $\mathbb{E}[Y^2 - a - bY] = 0$ and $\mathbb{E}[(Y^2 - a - bY)Y] = 0$. With $\mathbb{E}[Y] = \mathbb{E}[Y^3] = 0$, $\mathbb{E}[Y^2] = 1/3$, $\mathbb{E}[Y^4] = 1/5$, these give $a = 1/3$ and $b = 0$, so the best linear predictor is the constant $1/3$, with mean squared error $\mathbb{E}[(Y^2 - 1/3)^2] = \mathbb{E}[Y^4] - 1/9 = 1/5 - 1/9 = 4/45$. The nonlinear predictor is perfect and the linear one is useless, because X and Y are uncorrelated ($\mathrm{Cov}(X, Y) = \mathbb{E}[Y^3] = 0$) yet functionally dependent. The linear predictor sees only the covariance (proposition 3.3.7, definition 3.2.1); the conditional expectation sees the full joint law. The two agree exactly when $\mathbb{E}[X \big| Y]$ happens to be affine in Y, as for jointly Gaussian (X, Y), developed in section 9.4.

The third example is the one to keep in mind whenever "correlation" is used as a proxy for "dependence". Zero correlation means the best straight-line predictor is flat, not that Y carries no information about X; the conditional expectation is the object that captures the full predictive content, and it reduces to a linear rule only under the extra structure, most notably joint normality, that makes the regression function affine. This gap between the linear predictor read off the covariance and the true conditional expectation is exactly the gap between second-order and full distributional information.

The projection viewpoint is the geometry the rest of Part III runs on. A martingale is a sequence (X_n) in which each term is the conditional-expectation projection of every later term onto the information available at time n; the tower property of section 9.2 is then the statement that projecting first onto a large information set and then onto a smaller one is the same as projecting directly onto the smaller, the nesting of orthogonal projections. Doob's L^2-convergence theory, when it arrives, is the assertion that a bounded increasing chain of these projections converges in L^2, which is a Hilbert-space statement about nested closed subspaces. The least-squares reading of conditioning established here is thus not an aside but the lens through which the martingale convergence theorems become geometrically inevitable.

Exercise 9.3.1

1. Let $X, Y \in L^2$ and let $Z = \mathbb{E}[X \big| Y]$ be the best mean-square predictor of X from Y. Show directly, without invoking the projection theorem, that for every Borel g with $g(Y) \in L^2$ one has $\mathbb{E}[(X-Z)^2] \le \mathbb{E}[(X-g(Y))^2]$, with equality only if $g(Y) = Z$ almost surely. (Expand $(X-g(Y))^2 = ((X-Z)+(Z-g(Y)))^2$ and use taking-out-what-is-known.)
2. Interpret conditioning as a linear operator $T\colon L^2(\mathcal{F}) \to L^2(\mathcal{F})$, $TX = \mathbb{E}[X \big| \mathcal{G}]$. Using theorem 9.3.3 and the tower property, show T is idempotent ($T^2 = T$) and self-adjoint ($\mathbb{E}[(TX)Y] = \mathbb{E}[X(TY)]$ for all $X, Y \in L^2$), the two algebraic hallmarks of an orthogonal projection.
3. A random number $N \sim \mathrm{Poi}(\lambda)$ of customers arrive, and each independently spends an amount with mean μ and variance τ^2; let S be the total spend. Compute $\mathbb{E}[S]$, then use the law of total variance (proposition 9.3.4) conditioning on N to compute $\mathrm{Var}(S)$.
4. Let $Y \sim \mathcal{N}(0,1)$ and $X = Y^2$. Find $\mathbb{E}[X \big| Y]$ and the best linear predictor of X from Y, and compute the mean squared error of each. Explain in one sentence why the two differ.
5. Deduce the L^2 contraction inequality $\mathbb{E}[\mathbb{E}[X \big| \mathcal{G}]^2] \le \mathbb{E}[X^2]$ from theorem 9.3.3 in two ways: first from the Pythagorean identity, and second as the case $\varphi(x) = x^2$ of conditional Jensen followed by the tower property. Identify the quantity by which the two sides differ.
6. Let $\mathcal{G} = \sigma(A)$ for a single event A with $0 < \mathbb{P}(A) < 1$. Describe $L^2(\mathcal{G})$ explicitly as a subspace of L^2, compute its dimension, and verify by hand that the projection of X onto it is $\frac{\mathbb{E}[X\mathbb{1}_A]}{\mathbb{P}(A)}\mathbb{1}_A + \frac{\mathbb{E}[X\mathbb{1}_{A^c}]}{\mathbb{P}(A^c)}\mathbb{1}_{A^c}$, matching the finite-partition formula for $\mathbb{E}[X \big| \mathcal{G}]$.

9.4 Regular Conditional Distributions

The conditional expectation of the previous three sections attaches to each integrable X a single $\mathcal{G}$-measurable random variable $\mathbb{E}[X \big| \mathcal{G}]$. This is enough to average any one variable over the information in $\mathcal{G}$, but it does not yet package that information the way an unconditional probability space does. Unconditionally, all averages $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g\, d\mu_X$ flow from a single object, the law μ_X of X (definition 2.1.3); one measure on the line generates every expectation at once. The question this section answers is whether the conditional averages can be organized the same way: is there, for each outcome ω, a probability measure $\kappa(\omega, \cdot)$ on $\mathbb{R}$ so that

$$\mathbb{E}[g(X) \big| \mathcal{G}](\omega) = \int_{\mathbb{R}} g(x)\, \kappa(\omega, dx)$$

simultaneously for all bounded Borel g? Such a κ would be a bona fide conditional law of X, a measure that varies measurably with the information and from which every conditional average is recovered as an integral.

The obstruction is that $\mathbb{E}[X|\mathcal{G}]$ is defined only up to a null set, and the null set depends on X. For a fixed Borel B the map $\mathbb{P}(X \in B|\mathcal{G})$ is a well-defined class of random variables, but if one chooses a version for each B separately, there is no reason the resulting numbers $B \mapsto \mathbb{P}(X \in B|\mathcal{G})(\omega)$ should be countably additive at a fixed ω: countable additivity involves uncountably many exceptional null sets, one per countable family of disjoint sets, and an uncountable union of null sets need not be null. The content of this section is that on the real line one can choose the versions coherently, so that the exceptional sets collapse to a single null set and what survives is, outcome by outcome, a bona fide probability measure. This regular conditional distribution is the disintegration of the joint law, and it is the object on which the Markov property and the transition kernels of stochastic processes are built.

The object must be named precisely before it can be built. A regular conditional distribution is a map of two arguments, an outcome and a Borel set, that is a version of the conditional probability in its second argument and a genuine measure in its first.

Definition 9.4.1: Regular Conditional Distribution

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra, and $X \colon \Omega \to \mathbb{R}$ a random variable. A *regular conditional distribution* of X given $\mathcal{G}$ is a map

$$\kappa \colon \Omega \times \mathcal{B}(\mathbb{R}) \to [0, 1]$$

satisfying:

1. for each fixed $\omega \in \Omega$, the set function $B \mapsto \kappa(\omega, B)$ is a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$;
2. for each fixed Borel set $B \in \mathcal{B}(\mathbb{R})$, the function $\omega \mapsto \kappa(\omega, B)$ is $\mathcal{G}$-measurable and is a version of $\mathbb{P}(X \in B|\mathcal{G})$, that is

$$\kappa(\omega, B) = \mathbb{P}(X \in B|\mathcal{G})(\omega) \qquad \text{for } \mathbb{P}\text{-a.e. } \omega$$

A map $\kappa \colon \Omega \times \mathcal{B}(\mathbb{R}) \to [0, 1]$ satisfying (1) and the $\mathcal{G}$-measurability in (2) but not tied to any particular X is called a *probability kernel* from $(\Omega, \mathcal{G})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

The two conditions pull in opposite directions, and their tension is exactly the difficulty. Condition (1) fixes ω and asks for additivity across the σ-algebra $\mathcal{B}(\mathbb{R})$; condition (2) fixes B and asks for a measurable version of a conditional probability. Individually each is routine: condition (2) alone is discharged by definition 9.1.3, which defines $\mathbb{P}(X \in B|\mathcal{G}) = \mathbb{E}[\mathbb{1}_{\{X \in B\}}|\mathcal{G}]$ for every B separately. The regularity requirement is that a *single* function of two variables meet both at once, so that the collection $\{\kappa(\omega, \cdot)\}_\omega$ is a measurably-indexed family of laws rather than an incoherent list of numbers. When such a κ exists it is unique up to a $\mathbb{P}$-null set of outcomes: two regular conditional distributions agree as measures for a.e. ω, since they agree on the rationals' half-lines off a null set and a measure on $\mathbb{R}$ is determined by its values on $\{(-\infty, q]|q \in \mathbb{Q}\}$ (corollary 1.2.6).

Before proving existence, it helps to see the object in the case where everything is finite and the tension of the definition dissolves.

Example 9.4.2: Conditioning on a Finite Partition

Let $\mathcal{G} = \sigma(G_1, \ldots, G_m)$ be generated by a finite partition of Ω into events of positive probability, as in example 9.1.4. A $\mathcal{G}$-measurable function is constant on each atom G_i, so any candidate kernel must have the form $\kappa(\omega, B) = k_i(B)$ for $\omega \in G_i$, with k_i a measure on $\mathbb{R}$. Condition (2) forces, for each B,

$$k_i(B) = \mathbb{P}(X \in B \big| \mathcal{G})(\omega) = \frac{\mathbb{P}(\{X \in B\} \cap G_i)}{\mathbb{P}(G_i)} \qquad (\omega \in G_i)$$

which is exactly the elementary conditional probability $\mathbb{P}(X \in B \big| G_i)$. For fixed i the map $B \mapsto \mathbb{P}(X \in B \big| G_i)$ is a probability measure, namely the law of X restricted and renormalized to the event G_i, so condition (1) holds with no further work. Setting

$$\kappa(\omega, \cdot) = \mathbb{P}(X \in \cdot \big| G_i) \qquad \text{for } \omega \in G_i$$

gives a regular conditional distribution. Integrating g against it on the atom G_i reproduces the conditional average of example 9.1.4: for $\omega \in G_i$,

$$\int_{\mathbb{R}} g \, d\kappa(\omega, \cdot) = \int_{\mathbb{R}} g \, d\mathbb{P}(X \in \cdot \big| G_i) = \frac{\mathbb{E}[g(X) \mathbb{1}_{G_i}]}{\mathbb{P}(G_i)}$$

which is the value at ω of the piecewise formula $\mathbb{E}[g(X) \big| \mathcal{G}] = \sum_i \frac{\mathbb{E}[g(X)\mathbb{1}_{G_i}]}{\mathbb{P}(G_i)} \mathbb{1}_{G_i}$. Here the exceptional null set is empty: the partition is finite, so there is no uncountable union of null sets to control, and the kernel exists outright.

The finite case makes the general strategy visible. In it, the kernel was assembled from the finitely many conditional laws $\mathbb{P}(X \in \cdot \big| G_i)$, and additivity was free because only finitely many exceptional events were in play. For general $\mathcal{G}$ the atoms are gone, but the same idea works if the measures $\kappa(\omega, \cdot)$ are built from their distribution functions along a *countable* skeleton, the rationals, so that only countably many null sets arise and their union is again null. This is the content of the existence theorem.

The construction begins from the conditional distribution functions. For each rational q set

$$F(\omega, q) = \mathbb{P}(X \leq q \big| \mathcal{G})(\omega) = \mathbb{E}[\mathbb{1}_{\{X \leq q\}} \big| \mathcal{G}](\omega)$$

choosing one version of each conditional expectation. The monotonicity, limit, and right-continuity properties that a distribution function must have (definition 1.3.1) hold here as *a.s.* identities inherited from the corresponding properties of X, one identity per rational or pair of rationals. Because there are only countably many such identities, the null sets on which they fail have null union, and off that single null set $q \mapsto F(\omega, q)$ is a distribution function on $\mathbb{Q}$ that extends to one on $\mathbb{R}$. The distribution-function correspondence (theorem 1.3.3) then converts each into a probability measure, and those measures are the kernel.

Theorem 9.4.3: Existence of Regular Conditional Distributions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra, and $X\colon \Omega \to \mathbb{R}$ a random variable. Then a regular conditional distribution κ of X given $\mathcal{G}$ exists, and it is unique up to a $\mathbb{P}$-null set of outcomes. Moreover, for every bounded Borel function $g\colon \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}[g(X) \big| \mathcal{G}](\omega) = \int_{\mathbb{R}} g(x)\, \kappa(\omega, dx) \qquad \text{for } \mathbb{P}\text{-a.e. } \omega \tag{9.8}$$

and the same identity holds for every Borel $g \geq 0$ and every g with $g(X)$ integrable.

***Proof*:**
The argument constructs the conditional distribution functions along the rationals, repairs them on a single null set into bona fide distribution functions, passes to measures by the correspondence of theorem 1.3.3, and verifies the two defining conditions and the integral formula.

Step 1: The rational skeleton. For each $q \in \mathbb{Q}$ fix a version of the conditional probability and write

$$F(\omega, q) = \mathbb{P}(X \leq q \big| \mathcal{G})(\omega) = \mathbb{E}[\mathbb{1}_{\{X \leq q\}} \big| \mathcal{G}](\omega)$$

Each $F(\cdot, q)$ is $\mathcal{G}$-measurable and takes values in $[0, 1]$ a.s., since $0 \leq \mathbb{1}_{\{X \leq q\}} \leq 1$ and conditional expectation is monotone (proposition 9.2.1). Three families of a.s. identities hold:

1. *Monotonicity.* If $q < q'$ are rationals then $\mathbb{1}_{\{X \leq q\}} \leq \mathbb{1}_{\{X \leq q'\}}$, so monotonicity of conditional expectation gives $F(\omega, q) \leq F(\omega, q')$ for a.e. ω, one null set $N_{q,q'}$ per rational pair.
2. *Right limits.* For fixed q, as rationals $q_n \downarrow q$ the events $\{X \leq q_n\}$ decrease to $\{X \leq q\}$, so $\mathbb{1}_{\{X \leq q_n\}} \downarrow \mathbb{1}_{\{X \leq q\}}$; the conditional dominated convergence theorem (proposition 9.2.5) gives $F(\omega, q_n) \downarrow F(\omega, q)$ for a.e. ω, one null set N_q per rational.
3. *Total mass.* As integers $k \to +\infty$ the events $\{X \leq k\}$ increase to Ω and as $k \to -\infty$ they decrease to $\emptyset$, so conditional monotone and dominated convergence give $F(\omega, k) \to 1$ and $F(\omega, -k) \to 0$ for a.e. ω, on a null set N_∞.

There are countably many rationals and rational pairs, so the union

$$N = N_\infty \cup \bigcup_q N_q \cup \bigcup_{q < q'} N_{q,q'}$$

is a $\mathbb{P}$-null set. Off N the function $q \mapsto F(\omega, q)$ is nondecreasing, right-continuous along rational sequences, and runs from 0 to 1.

Step 2: Repair on the null set and extend to $\mathbb{R}$. Fix a reference probability measure, say the standard normal law $\mathcal{N}(0, 1)$ with distribution function Φ. Define, for every real x,

$$G(\omega, x) = \begin{cases} \inf\{F(\omega, q) \big| q \in \mathbb{Q},\ q > x\} & \omega \notin N \\ \Phi(x) & \omega \in N \end{cases}$$

For $\omega \notin N$ the infimum over rationals to the right of x produces the right-continuous extension of the rational skeleton: $G(\omega, \cdot)$ is nondecreasing because $F(\omega, \cdot)$ is monotone on $\mathbb{Q}$; it is right-continuous in x because an infimum of the tail of a monotone function is right-continuous; and

$G(\omega, x) \to 1$ as $x \to +\infty$, $G(\omega, x) \to 0$ as $x \to -\infty$ by the total-mass identity. Thus $x \mapsto G(\omega, x)$ is a genuine distribution function for *every* ω: for $\omega \notin N$ by the repair just described, for $\omega \in N$ because Φ is one. Note that at rational $x = q$ the infimum recovers $F(\omega, q)$ itself, by right-continuity of the skeleton, so G extends F off N.

Step 3: Pass to measures. By the distribution-function correspondence (theorem 1.3.3), each $G(\omega, \cdot)$ determines a unique Borel probability measure $\kappa(\omega, \cdot)$ on $\mathbb{R}$ with

$$\kappa(\omega, (-\infty, x]) = G(\omega, x)$$

This defines the map $\kappa \colon \Omega \times \mathcal{B}(\mathbb{R}) \to [0, 1]$, and condition (1) of definition 9.4.1 holds by construction. It remains to check the measurability and version property, condition (2).

Step 4: Measurability in ω for every Borel B. The class

$$\mathcal{D} = \{B \in \mathcal{B}(\mathbb{R}) \big| \omega \mapsto \kappa(\omega, B) \text{ is } \mathcal{G}\text{-measurable and equals } \mathbb{P}(X \in B \big| \mathcal{G}) \text{ a.s.}\}$$

contains every half-line $(-\infty, x]$ with x rational: there $\kappa(\omega, (-\infty, q]) = F(\omega, q) = \mathbb{P}(X \le q \big| \mathcal{G})(\omega)$ off N, which is $\mathcal{G}$-measurable and is the required version. The half-lines $\{(-\infty, q] \big| q \in \mathbb{Q}\}$ form a π-system generating $\mathcal{B}(\mathbb{R})$. The class $\mathcal{D}$ is a λ-system: it contains $\mathbb{R}$ (total mass 1 on both sides); it is closed under proper differences, since for $B \subseteq C$ in $\mathcal{D}$ the measure and the conditional probability are both additive, giving $\kappa(\omega, C \setminus B) = \kappa(\omega, C) - \kappa(\omega, B)$ and $\mathbb{P}(X \in C \setminus B \big| \mathcal{G}) = \mathbb{P}(X \in C \big| \mathcal{G}) - \mathbb{P}(X \in B \big| \mathcal{G})$ a.s.; and it is closed under increasing unions, since for $B_n \uparrow B$ both $\kappa(\omega, B_n) \uparrow \kappa(\omega, B)$ (measure continuity, condition (1)) and $\mathbb{P}(X \in B_n \big| \mathcal{G}) \uparrow \mathbb{P}(X \in B \big| \mathcal{G})$ a.s. (conditional monotone convergence, proposition 9.2.5). By the π-λ theorem behind corollary 1.2.6, $\mathcal{D} = \mathcal{B}(\mathbb{R})$, so condition (2) holds for every Borel B. This establishes that κ is a regular conditional distribution.

Step 5: The integral formula. Identity (9.8) follows by the standard extension through simple functions. For an indicator $g = \mathbb{1}_B$ it is exactly condition (2): $\int_{\mathbb{R}} \mathbb{1}_B \, d\kappa(\omega, \cdot) = \kappa(\omega, B) = \mathbb{P}(X \in B \big| \mathcal{G})(\omega) = \mathbb{E}[\mathbb{1}_B(X) \big| \mathcal{G}](\omega)$ a.s. By linearity of the integral and of conditional expectation (proposition 9.2.1) the identity extends to nonnegative simple g. For general Borel $g \ge 0$ take simple $g_n \uparrow g$; the left side converges by the monotone convergence theorem for the measure $\kappa(\omega, \cdot)$, the right side by conditional monotone convergence, and the two limits agree a.s. Finiteness of the countably many exceptional null sets, one per g_n, keeps the union null. Writing a bounded Borel g as $g^+ - g^-$ gives the identity for bounded g, and the same decomposition handles any g with $g(X)$ integrable, since then both sides are finite a.s.

Uniqueness. If κ' is a second regular conditional distribution, then for each rational q the versions $\kappa(\cdot, (-\infty, q])$ and $\kappa'(\cdot, (-\infty, q])$ both equal $\mathbb{P}(X \le q \big| \mathcal{G})$ a.s., hence agree off a null set N_q. Off the null union $\bigcup_q N_q$ the measures $\kappa(\omega, \cdot)$ and $\kappa'(\omega, \cdot)$ agree on the generating π-system of rational half-lines, so by corollary 1.2.6 they agree as measures, completing the proof.

The theorem asserts that on the real line the conditional law always exists, and identity (9.8) is the payoff. It says $\mathbb{E}[g(X) \big| \mathcal{G}]$ is computed by integrating g against a single measure that depends only on ω, not on g; one object $\kappa(\omega, \cdot)$ serves every g at once, exactly as the law μ_X served every unconditional average. The construction used nothing about $\mathbb{R}$ beyond the distribution-function correspondence and the countability of the rationals, which is why the statement is really a statement about *standard Borel spaces*, the measurable spaces Borel-isomorphic to a Borel subset of $\mathbb{R}$. On such a space every random element admits a regular conditional distribution, by transporting the construction across the isomorphism; the measure-theoretic assembly of this general disintegration is developed in [2], and it is what licenses conditioning of the vector-valued and process-valued random elements that appear once time enters. The real line is the case the rest of this book needs.

9.4.4 Warning: Regularity Can Fail Without Standard Borel Structure Existence is not automatic in full generality. On a probability space whose target σ-algebra is not standard Borel, a regular conditional distribution can fail to exist: there are versions of $\mathbb{P}(X \in B \big| \mathcal{G})$ for every B, but no way to glue them into a measure for a.e. ω. The line's saving grace is the countable generating π-system used in the proof, which is exactly the standard-Borel property. This is why the theory of conditional distributions in this book, and the transition kernels of the stochastic processes it feeds, are always stated on standard Borel spaces ([2]).

The most-used instance is a pair of variables with a joint density. There the kernel is not only abstract: it has a formula, the elementary conditional density of a first course, and identity (9.8) becomes the familiar integral against that density. This is the point at which the measure-theoretic machinery lands back on the calculation every student meets before seeing a σ-algebra.

Proposition 9.4.5: Conditional Density and Disintegration

Let (X, Y) be a pair of real random variables whose joint law has a density $f(x, y)$ with respect to Lebesgue measure on $\mathbb{R}^2$, so that $\mathbb{P}((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$. Let

$$f_Y(y) = \int_{\mathbb{R}} f(x, y)\, dx$$

be the marginal density of Y (proposition 2.3.2). On the set $\{y \big| f_Y(y) > 0\}$ define the *conditional density*

$$f_{X\big|Y}(x\big|y) = \frac{f(x, y)}{f_Y(y)} \tag{9.9}$$

Then a regular conditional distribution of X given Y is given by $\kappa(\omega, B) = \int_B f_{X\big|Y}(x\big|Y(\omega))\, dx$ on $\{f_Y(Y) > 0\}$, and for every bounded Borel g,

$$\mathbb{E}[g(X)\big|Y] = \int_{\mathbb{R}} g(x)\, f_{X\big|Y}(x\big|Y)\, dx \qquad \text{a.s.} \tag{9.10}$$

Writing $\mathbb{E}[g(X)\big|Y = y]$ for the value of this $\sigma(Y)$-measurable variable on $\{Y = y\}$, one has $\mathbb{E}[g(X)\big|Y = y] = \int_{\mathbb{R}} g(x)\, f_{X\big|Y}(x\big|y)\, dx$ for f_Y-a.e. y.

***Proof*:**
Both sides of (9.10) are $\sigma(Y)$-measurable: the right side is $\sigma(Y)$-measurable, being $\psi(Y)$ for the Borel function $\psi(y) = \int_{\mathbb{R}} g(x) f_{X\big|Y}(x\big|y)\, dx$. By the defining property of conditional expectation (definition 9.1.1) and the Doob-Dynkin lemma again, it suffices to check that for every bounded Borel h,

$$\mathbb{E}[g(X)\, h(Y)] = \mathbb{E}\left[h(Y) \int_{\mathbb{R}} g(x) f_{X\big|Y}(x\big|Y)\, dx\right] \tag{9.11}$$

since the events of $\sigma(Y)$ are exactly $\{Y \in B\}$ and testing against $h = \mathbb{1}_B$ recovers the integral condition $\int_{\{Y \in B\}}$. Compute the left side by the change-of-variables formula (theorem 3.1.2) against the joint density and then Fubini's theorem (proposition 2.3.2):

$$\mathbb{E}[g(X)h(Y)] = \iint_{\mathbb{R}^2} g(x)h(y)f(x,y)\,dx\,dy = \int_{\mathbb{R}} h(y)\left(\int_{\mathbb{R}} g(x)f(x,y)\,dx\right)dy$$

On $\{f_Y(y) > 0\}$ multiply and divide by $f_Y(y)$ inside the inner integral, writing $f(x,y) = f_{X|Y}(x|y)\,f_Y(y)$ by (9.9); on $\{f_Y(y) = 0\}$ the density $f(x,y)$ vanishes for Lebesgue-a.e. x, since $\int_{\mathbb{R}} f(x,y)\,dx = f_Y(y) = 0$, so that y-slice contributes nothing. Hence

$$\mathbb{E}[g(X)h(Y)] = \int_{\mathbb{R}} h(y)\left(\int_{\mathbb{R}} g(x)f_{X|Y}(x|y)\,dx\right)f_Y(y)\,dy = \mathbb{E}\left[h(Y)\int_{\mathbb{R}} g(x)f_{X|Y}(x|Y)\,dx\right]$$

the last equality again by change of variables against the marginal law of Y, whose density is f_Y. This is (9.11), so (9.10) holds. Taking $g = \mathbb{1}_B$ shows that $B \mapsto \int_B f_{X|Y}(x|Y)\,dx$ is a version of $\mathbb{P}(X \in B|Y)$, and it is a probability measure in B for each outcome with $f_Y(Y) > 0$, so it is a regular conditional distribution of X given Y; the pointwise statement for $\mathbb{E}[g(X)|Y = y]$ is the evaluation of (9.10) on the level set $\{Y = y\}$, valid for f_Y-a.e. y, as we sought to show.

Formula (9.9) is the disintegration of the joint law: it factors $f(x,y) = f_{X|Y}(x|y)\,f_Y(y)$ into the marginal of Y times a family of conditional laws of X, one for each value y. Read from right to left it says the pair (X,Y) is generated by first drawing Y from f_Y and then drawing X from the conditional density $f_{X|Y}(\cdot|Y)$, which is precisely how a statistician thinks of a hierarchical model.

The proposition also closes the loop promised in example 9.1.4: the abstract $\mathbb{E}[X|Y]$ built from the Radon-Nikodym theorem in section 9.1 agrees, in the density case, with the elementary integral $\int x f_{X|Y}(x|y)\,dx$ of a first probability course. Nothing was lost in the abstraction, and something was gained: the same κ exists when no density does, for discrete pairs, for mixtures, for singular joint laws, wherever theorem 9.4.3 applies.

Three settings show the kernel in action and are worth carrying forward: a discrete pair conditioned on its sum, a continuous pair conditioned on its sum, and the bivariate normal, whose conditional law is the linear-regression model at the heart of every mean-square prediction.

Example 9.4.6: Conditional Laws: Sums and the Bivariate Normal

1. *Two Poissons given their sum.* Let $X \sim \text{Poi}(a)$ and $Y \sim \text{Poi}(b)$ be independent, and condition X on the sum $S = X + Y$. For integers $0 \le k \le n$, using independence and that $S \sim \text{Poi}(a+b)$ (as computed in example 2.3.6),

$$\mathbb{P}(X = k|S = n) = \frac{\mathbb{P}(X = k)\,\mathbb{P}(Y = n-k)}{\mathbb{P}(S = n)} = \frac{\dfrac{a^k e^{-a}}{k!} \cdot \dfrac{b^{n-k}e^{-b}}{(n-k)!}}{\dfrac{(a+b)^n e^{-(a+b)}}{n!}} = \binom{n}{k} p^k (1-p)^{n-k}$$

with $p = a/(a+b)$. The conditional law of X given $S = n$ is $\text{Bin}(n,p)$: once the total count n is fixed, each of the n occurrences is independently an X-occurrence with probability $p = a/(a+b)$. The regular conditional distribution is $\kappa(\omega,\cdot) = \text{Bin}(S(\omega),p)$, a probability measure for each ω and a version of $\mathbb{P}(X \in \cdot \big| S)$ by the computation above. Consequently $\mathbb{E}[X \big| S] = \sum_{k=0}^{S} k\binom{S}{k}p^k(1-p)^{S-k} = pS = \frac{a}{a+b}\, S$, the sum split in proportion to the means.

2. *Two uniforms given their sum.* Let $X, Y \sim \mathrm{Unif}[0,1]$ be independent, with joint density $f(x,y) = 1$ on the unit square. Condition X on $S = X + Y$. The pair (X, S) has density $f_{X,S}(x,s) = 1$ on the parallelogram $\{0 \le x \le 1,\ 0 \le s - x \le 1\}$, so the marginal of S is the triangular density
$$f_S(s) = \int_{\mathbb{R}} f_{X,S}(x,s)\,dx = \begin{cases} s & 0 \le s \le 1 \\ 2-s & 1 \le s \le 2 \end{cases}$$
the length of the x-slice. By (9.9) the conditional density of X given $S = s$ is $f_{X\big|S}(x\big|s) = 1/f_S(s)$ on the slice, that is X given $S = s$ is uniform on the interval $[\max(0, s-1), \min(1, s)]$. For $0 \le s \le 1$ this is $\mathrm{Unif}[0, s]$ and $\mathbb{E}[X\big|S = s] = s/2$; by symmetry $\mathbb{E}[X\big|S = s] = s/2$ for all $s \in [0, 2]$, so $\mathbb{E}[X\big|S] = S/2$, again the symmetric split.

3. *The bivariate normal.* Let (X, Y) be jointly Gaussian with means μ_X, μ_Y, variances σ_X^2, σ_Y^2, and correlation $\rho \in (-1, 1)$. The conditional density is Gaussian: completing the square in the joint density $f(x, y)$ against the marginal f_Y gives, for each y,
$$X\big|Y = y \ \sim\ \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y),\ \sigma_X^2(1-\rho^2)\right)$$
The conditional mean is linear in y, giving the linear regression
$$\mathbb{E}[X\big|Y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(Y - \mu_Y)$$
and in the centered, equal-variance case ($\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y$) this is $\mathbb{E}[X\big|Y] = \rho Y$. The conditional variance $\sigma_X^2(1-\rho^2)$ is constant in y and is strictly smaller than the unconditional variance σ_X^2 whenever $\rho \ne 0$: observing Y reduces the mean-square uncertainty in X by the factor $1-\rho^2$. This is the exact realization of the best-mean-square predictor of theorem 9.3.3: for the Gaussian pair the optimal predictor is the linear one, and the residual variance $\sigma_X^2(1-\rho^2)$ is the value of the total-variance identity $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X\big|Y)] + \mathrm{Var}(\mathbb{E}[X\big|Y])$ from proposition 9.3.4, since here $\mathrm{Var}(\mathbb{E}[X\big|Y]) = \rho^2\sigma_X^2$.

The three computations share a shape: the regular conditional distribution turns a joint law into a rule that maps an observed value of one variable to a probability measure on the other, and the conditional expectation is the mean of that measure. This is the germ of the entire theory to come. A Markov chain is nothing but a sequence in which the conditional law of the next state given the whole past is a regular conditional distribution depending only on the present state, a transition kernel $\kappa(x, \cdot)$; the disintegration of this section is what makes that sentence a definition rather than a wish. In continuous time the same kernels become the transition semigroups of diffusions, and identity (9.8) is the bridge that lets one compute expectations of a process by integrating against its one-step laws. The conditional expectation of section 9.1 gave the average; the regular conditional distribution gives the law, and it is the law that the martingales of the next chapter and the stochastic processes beyond it are built to track.

Exercise 9.4.1

1. Let $N \sim \mathrm{Poi}(\lambda)$ and, given $N = n$, let X be the number of heads in n independent tosses of a coin with heads-probability p, so that $X \big| N = n \sim \mathrm{Bin}(n, p)$. Identify the regular conditional distribution of X given N, and use (9.8) to compute $\mathbb{E}[X \big| N]$. Then find the unconditional law of X.

2. Let (X, Y) have joint density $f(x, y) = 2$ on the triangle $\{0 < x < y < 1\}$ and 0 elsewhere. Compute the marginal f_Y, the conditional density $f_{X \big| Y}(x \big| y)$, and $\mathbb{E}[X \big| Y]$. Verify that $\mathbb{E}[\mathbb{E}[X \big| Y]] = \mathbb{E}[X]$.

3. Let $X, Y \sim \mathrm{Unif}[0, 1]$ be independent and let $M = \max(X, Y)$. Show that the conditional law of X given $M = m$ is $\frac{1}{2}\delta_m + \frac{1}{2}\,\mathrm{Unif}[0, m]$ (an atom of mass $1/2$ at m and a uniform part), and compute $\mathbb{E}[X \big| M]$. Explain why this kernel has an atom even though the joint law of (X, Y) has none.

4. In the proof of theorem 9.4.3, explain precisely where the countability of $\mathbb{Q}$ is used, and give a one-sentence account of why the same construction cannot be run over an uncountable dense set of thresholds.

5. For the centered bivariate normal with $\sigma_X = \sigma_Y = 1$ and correlation ρ, use the conditional law of part (3) of example 9.4.6 to compute $\mathbb{E}[X^2 \big| Y]$ and verify the law of total variance $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \big| Y)] + \mathrm{Var}(\mathbb{E}[X \big| Y])$ (proposition 9.3.4) numerically at $\rho = 3/5$.

6. Let $X \sim \mathrm{Unif}[0, 1]$ and, given $X = x$, let Y be exponential with rate $1/x$, so that $Y \big| X = x$ has density $\frac{1}{x} e^{-y/x}$ for $y > 0$. Write the joint density $f(x, y)$, and use it to find the conditional density $f_{X \big| Y}(x \big| y)$. (You need not evaluate the marginal $f_Y(y)$ in closed form; express $f_{X \big| Y}$ as a ratio.)

10

Discrete-Time Martingales

A martingale is the mathematical model of a fair game: a sequence of random variables whose expected next value, given everything observed so far, is exactly the present value. Conditional expectation, built in chapter 9, is the tool needed to state this, and martingales are its first and most consequential application. The defining equation $\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n$ constrains a process so tightly that two theorems of great generality follow from it alone: a fair game stopped at a cleverly chosen time is still fair, and a martingale that stays bounded in size must converge.

section 10.1 sets up filtrations, martingales, and stopping times, and the martingale transform that formalizes the impossibility of beating a fair game. section 10.2 proves the optional stopping theorem and reads it on the gambler's ruin. Section 10.3 establishes Doob's maximal, L^p, and upcrossing inequalities, the analytic engine of the theory. section 10.4 proves the martingale convergence theorem, that an L^1-bounded martingale converges almost surely, with its uniformly integrable strengthening to L^1 convergence. Section 10.5 collects the payoffs: Lévy's zero-one law, the Radon-Nikodym theorem recovered as a martingale limit, the Galton-Watson branching process, and a first martingale concentration inequality. Together they show why the martingale is the natural language for information revealed over time, and close the book's development of the discrete theory at the threshold of continuous time.

10.1 Filtrations, Martingales, and Stopping Times

Conditional expectation, built in chapter 9, answers the question "what is X on average, given the information in a σ-algebra $\mathcal{G}$?" In every application of interest that information is not fixed once and for all: it accumulates. A gambler sees one more card each round; a stock price records one more day; a random walk takes one more step. At each time n there is a record of what has been revealed so far, and this record only grows. The object that tracks this growing record is a filtration, an increasing sequence of sub-σ-algebras, and it is the stage on which the entire theory of this chapter is played out. A process is *adapted* to the filtration when, at each time, its present value is already part of the record, and a *martingale* is an adapted process whose expected next value, given the record so far, is exactly its present value. This is the mathematical model of a fair game.

The power of the definition is that it constrains a process by a single conditional identity, and yet from that identity alone flow the structural theorems of the chapter. This section establishes the vocabulary and proves the first of those theorems: that a fair game remains fair no matter how a gambler bets on it or when the gambler decides to quit. The formalization of "betting" and "quitting" is the martingale transform and the stopping time, and the conclusion, that you cannot beat a fair game, is the seed from which the optional stopping theorem of the next section grows.

Definition 10.1.1: Filtration and Adapted Process

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space (definition 1.1.1). A *filtration* is an increasing sequence $(\mathcal{F}_n)_{n\geq 0}$ of sub-σ-algebras of $\mathcal{F}$,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}$$

A sequence of random variables $(X_n)_{n\geq 0}$ is *adapted* to $(\mathcal{F}_n)$ if X_n is $\mathcal{F}_n$-measurable for every n. Given any sequence (X_n), its *natural filtration* is

$$\mathcal{F}_n = \sigma(X_0, X_1, \ldots, X_n)$$

the σ-algebra generated by the process up to time n (definition 2.1.5); a process is always adapted to its own natural filtration.

The word to keep in mind is *information*. The σ-algebra $\mathcal{F}_n$ is the collection of events whose occurrence can be decided by an observer who has watched the process through time n; the containment $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ is the statement that information is never lost, only added. To say (X_n) is adapted is to say that X_n is knowable at time n, and no earlier: its value is determined by what has been observed. The natural filtration is the smallest filtration making (X_n) adapted, the one in which the only information available is the process itself. In practice a filtration is often larger than the natural one, carrying side information (an auxiliary source of randomness, a signal observed alongside the process), and the theory is written for a general $(\mathcal{F}_n)$ precisely so that this extra information is allowed. The tail σ-algebra of definition 2.4.1 sits at the opposite end from a filtration: it records what remains knowable after *discarding* any finite prefix of the process, whereas $\mathcal{F}_n$ records what becomes knowable by *keeping* the prefix up to time n.

Example 10.1.2: The Filtration of Coin Tossing

Take the fair-coin-tossing space (example 1.3.6), on which the coordinate variables $\xi_1, \xi_2, \ldots$ are independent with $\mathbb{P}(\xi_k = +1) = \mathbb{P}(\xi_k = -1) = \frac{1}{2}$, and set $S_0 = 0$, $S_n = \xi_1 + \cdots + \xi_n$, the position of the simple symmetric random walk after n steps. The natural filtration is

$$\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$$

the information in the first n tosses. An event lies in $\mathcal{F}_n$ exactly when it can be decided from those tosses: $\{S_3 = 1\} \in \mathcal{F}_3$ and $\{\xi_2 = +1\} \in \mathcal{F}_2 \subseteq \mathcal{F}_3$, whereas $\{\xi_4 = +1\}$ does not lie in $\mathcal{F}_3$, since the fourth toss is not yet seen. Because $S_n = S_{n-1} + \xi_n$ is a function of $\xi_1, \ldots, \xi_n$, the walk (S_n) is adapted to $(\mathcal{F}_n)$, and $\sigma(S_0, \ldots, S_n) = \sigma(\xi_1, \ldots, \xi_n)$ since the partial sums and the increments determine each other. This filtration, and this walk, are the running example against which every definition of the chapter is read.

With the stage set, the central object can be defined. A martingale is an adapted, integrable process whose one-step conditional forecast is its current value. The inequality variants, submartingale and supermartingale, model games biased toward and against the gambler, and they are not a mere technical afterthought: the convergence and inequality theorems of this chapter are proved for submartingales, of which martingales are the boundary case, so the wider class is worth carrying.

Definition 10.1.3: Martingale, Submartingale, Supermartingale

Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration. An adapted process $(M_n)_{n\geq 0}$ with each M_n integrable (definition 3.1.1) is a *martingale* relative to $(\mathcal{F}_n)$ if

$$\mathbb{E}[M_{n+1} | \mathcal{F}_n] = M_n \quad \text{a.s. for every } n \geq 0$$

where the conditional expectation is taken in the sense of definition 9.1.1. It is a *submartingale* if instead $\mathbb{E}[M_{n+1}|\mathcal{F}_n] \geq M_n$ a.s. for every n, and a *supermartingale* if $\mathbb{E}[M_{n+1}|\mathcal{F}_n] \leq M_n$ a.s. for every n. When the filtration is not named, $(\mathcal{F}_n)$ is understood to be the natural filtration of (M_n).

Three conditions do the work, and each is essential. Adaptedness makes M_n part of the record at time n, so that conditioning on $\mathcal{F}_n$ leaves M_n untouched; integrability makes the conditional expectation defined at all; and the identity $\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n$ is the fairness of the game. Read the identity as a forecast: the best estimate of tomorrow's fortune, given everything known today, is today's fortune. A martingale drifts neither up nor down on average, a submartingale drifts up (a game favorable to the player, since "sub" names the present value sitting *below* the forecast), and a supermartingale drifts down. The mnemonic is worth fixing once, because the prefixes run against the intuition that "super" should mean "better": a supermartingale is the gambler's enemy, its expected increments nonpositive, and it is precisely the class for which one expects almost-sure convergence to a finite limit, since a process that can only lose on average cannot run off to $+\infty$. A process is a martingale exactly when it is both a sub and a supermartingale, and (M_n) is a supermartingale if and only if $(-M_n)$ is a submartingale, so the two one-sided notions are mirror images and every theorem about one dualizes to the other.

The forecast identity extends from one step to many. Because the filtration is increasing, conditioning on the earlier $\mathcal{F}_m$ and using the tower property collapses any number of steps at once, so a martingale forecasts every future value, not just the next.

Proposition 10.1.4: Basic Properties of Martingales

Let (M_n) be adapted and integrable relative to $(\mathcal{F}_n)$.

1. If (M_n) is a martingale then $\mathbb{E}[M_n|\mathcal{F}_m] = M_m$ a.s. for all $n \geq m$, and in particular $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for every n; the expectation is constant. For a submartingale the equality $\mathbb{E}[M_n|\mathcal{F}_m] = M_m$ is replaced by $\mathbb{E}[M_n|\mathcal{F}_m] \geq M_m$ a.s. for $n \geq m$, so $\mathbb{E}[M_n]$ is nondecreasing in n; for a supermartingale the inequality reverses and $\mathbb{E}[M_n]$ is nonincreasing.

2. If (M_n) is a martingale and $\varphi\colon \mathbb{R} \to \mathbb{R}$ is convex with each $\varphi(M_n)$ integrable, then $(\varphi(M_n))$ is a submartingale. If (M_n) is a submartingale and φ is convex and *nondecreasing* (with $\varphi(M_n)$ integrable), then $(\varphi(M_n))$ is again a submartingale.

3. Let $X_1, X_2, \ldots$ be independent, integrable, and mean zero (definition 2.2.2), let $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ with $\mathcal{F}_0 = \{\emptyset, \Omega\}$, and set $M_0 = 0$, $M_n = X_1 + \cdots + X_n$. Then (M_n) is a martingale.

***Proof*:**
Part (1): the multi-step forecast. Fix $n \geq m$. Iterating the one-step identity and the tower property (proposition 9.2.2) collapses the forecast to time m. Concretely, for a martingale,

$$\mathbb{E}[M_n|\mathcal{F}_m] = \mathbb{E}[\mathbb{E}[M_n|\mathcal{F}_{n-1}]|\mathcal{F}_m] = \mathbb{E}[M_{n-1}|\mathcal{F}_m]$$

where the first equality is the tower property (valid because $\mathcal{F}_m \subseteq \mathcal{F}_{n-1}$) and the second uses $\mathbb{E}[M_n|\mathcal{F}_{n-1}] = M_{n-1}$. Repeating the step $n-m$ times reduces the index to m and yields $\mathbb{E}[M_n|\mathcal{F}_m] = M_m$ a.s. Taking $m = 0$ and then expectations, and using $\mathbb{E}[\mathbb{E}[M_n|\mathcal{F}_0]] = \mathbb{E}[M_n]$ together with $\mathbb{E}[M_0|\mathcal{F}_0] = M_0$, gives $\mathbb{E}[M_n] = \mathbb{E}[M_0]$. For a submartingale each conditioning step replaces the equality by $\mathbb{E}[M_n|\mathcal{F}_{n-1}] \geq M_{n-1}$; monotonicity of conditional expectation (proposition 9.2.1) preserves the inequality through the tower, so $\mathbb{E}[M_n|\mathcal{F}_m] \geq M_m$ a.s., and taking expectations gives $\mathbb{E}[M_n] \geq \mathbb{E}[M_m]$ for $n \geq m$. The supermartingale case is the same with inequalities reversed.

Part (2): convex functions. Suppose (M_n) is a martingale and φ is convex with $\varphi(M_n)$ integrable. The conditional Jensen inequality (proposition 9.2.4) applied to M_{n+1} given $\mathcal{F}_n$ gives

$$\mathbb{E}[\varphi(M_{n+1})|\mathcal{F}_n] \geq \varphi\big(\mathbb{E}[M_{n+1}|\mathcal{F}_n]\big) = \varphi(M_n)$$

the equality because $\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n$. Since $\varphi(M_n)$ is $\mathcal{F}_n$-measurable and each $\varphi(M_n)$ is integrable, $(\varphi(M_n))$ is adapted and integrable, and the displayed inequality is exactly the submartingale property. If instead (M_n) is a submartingale and φ is convex and nondecreasing, conditional Jensen still gives $\mathbb{E}[\varphi(M_{n+1})|\mathcal{F}_n] \geq \varphi(\mathbb{E}[M_{n+1}|\mathcal{F}_n])$, and now $\mathbb{E}[M_{n+1}|\mathcal{F}_n] \geq M_n$ together with the monotonicity of φ gives $\varphi(\mathbb{E}[M_{n+1}|\mathcal{F}_n]) \geq \varphi(M_n)$; chaining the two inequalities yields the submartingale property for $(\varphi(M_n))$.

Part (3): sums of independent variables. Each $M_n = X_1 + \cdots + X_n$ is a sum of integrable variables, hence integrable, and is $\mathcal{F}_n$-measurable, so (M_n) is adapted and integrable. For the forecast, write $M_{n+1} = M_n + X_{n+1}$ and use linearity of the conditional expectation:

$$\mathbb{E}[M_{n+1}|\mathcal{F}_n] = \mathbb{E}[M_n|\mathcal{F}_n] + \mathbb{E}[X_{n+1}|\mathcal{F}_n]$$

Now M_n is $\mathcal{F}_n$-measurable, so $\mathbb{E}[M_n|\mathcal{F}_n] = M_n$ (proposition 9.2.1). And X_{n+1} is independent of $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, so conditioning discards no usable information: $\mathbb{E}[X_{n+1}|\mathcal{F}_n] = \mathbb{E}[X_{n+1}] = 0$ (proposition 9.2.1, the independence case). Hence $\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n$ a.s., and (M_n) is a martingale, completing the proof.

Each part earns its place in the sequel. Part (1) is the fact that a martingale's mean is a conserved quantity, the single most-used consequence of the definition and the reason the optional stopping theorem of the next section reads as a conservation law. Part (2) is the machine that manufactures submartingales from martingales: applying it to $\varphi(x) = |x|$, or $\varphi(x) = x^2$ when the martingale is square-integrable, shows that $|M_n|$ and M_n^2 are submartingales, and these are exactly the processes to which Doob's inequalities will be applied. Part (3) is the first genuine class of examples and the bridge back to the limit theory: the partial sums of an independent mean-zero sequence, the very object whose limit theorems occupied the laws of large numbers and the central limit theorem, form a martingale. The martingale property is a structural fact about such sums that neither independence nor the limit theorems see directly, and much of the chapter can be read as extracting new consequences from it.

A word on the conserved mean before turning to stopping. That $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for every fixed n is elementary; the depth of the theory is that this conservation survives when the fixed time n is replaced by a random time τ chosen by an observer who watches the process and decides when to stop. Whether it survives, and under what conditions, is the content of optional stopping, and the first step is to say precisely which random times an observer is allowed to choose.

Definition 10.1.5: Stopping Time and the Stopped Process

Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration. A random variable $\tau\colon \Omega \to \{0, 1, 2, \dots\} \cup \{\infty\}$ is a *stopping time* if

$$\{\tau = n\} \in \mathcal{F}_n \quad \text{for every } n \geq 0$$

equivalently, if $\{\tau \leq n\} \in \mathcal{F}_n$ for every n. Given a process (M_n), the *stopped process* is $(M_{n\wedge\tau})_{n\geq 0}$, where $n \wedge \tau = \min(n, \tau)$; on the event $\{\tau = k\}$ it equals M_k for all $n \geq k$ and equals M_n for $n < k$. The *stopping-time σ-algebra* is

$$\mathcal{F}_\tau = \{A \in \mathcal{F} \colon A \cap \{\tau = n\} \in \mathcal{F}_n \text{ for every } n \geq 0\}$$

the collection of events whose occurrence is decidable by the time τ has arrived.

The defining requirement $\{\tau = n\} \in \mathcal{F}_n$ is the mathematical form of a rule that does not peek into the future. To decide at time n whether to stop, the observer may consult only what has been observed by time n; the decision $\{\tau = n\}$ must therefore be an event in $\mathcal{F}_n$. The equivalence with $\{\tau \leq n\} \in \mathcal{F}_n$ is immediate, since $\{\tau \leq n\} = \bigcup_{k\leq n}\{\tau = k\}$ and each $\{\tau = k\} \in \mathcal{F}_k \subseteq \mathcal{F}_n$, and conversely $\{\tau = n\} = \{\tau \leq n\} \setminus \{\tau \leq n-1\}$. The stopped process freezes (M_n) the instant τ occurs: it runs along with M_n until time τ, then holds its value forever. The σ-algebra $\mathcal{F}_\tau$ is the information available at the random time τ, the correct generalization of $\mathcal{F}_n$ to a random stopping index; when $\tau \equiv n$ is constant it reduces to $\mathcal{F}_n$, as one checks directly from the definition.

The distinction between rules that look only backward and rules that peek forward is best seen on a pair of examples, and the difference is exactly the difference between a stopping time and its illegitimate cousin.

Example 10.1.6: Hitting Times and Last-Exit Times

Let (X_n) be adapted to $(\mathcal{F}_n)$ and let B be a Borel set.

1. *The first hitting time is a stopping time.* Define $\tau_B = \inf\{n \geq 0 \colon X_n \in B\}$, the first time the process enters B (with $\inf \emptyset = \infty$ if B is never entered). Then
$$\{\tau_B = n\} = \{X_0 \notin B\} \cap \cdots \cap \{X_{n-1} \notin B\} \cap \{X_n \in B\}$$
and each event on the right is $\mathcal{F}_n$-measurable because $X_0, \ldots, X_n$ are all $\mathcal{F}_n$-measurable; hence $\{\tau_B = n\} \in \mathcal{F}_n$ and τ_B is a stopping time. Deciding that the process enters B for the first time at step n requires only the first n observations, so the rule looks only backward.

2. *The last exit time is not a stopping time.* Suppose the process is run over a finite horizon $0, \ldots, N$ and set $\lambda_B = \sup\{n \leq N \colon X_n \in B\}$, the last time the process is in B. To decide at time n that this is the *last* visit to B, one must know that $X_{n+1}, \ldots, X_N$ never return to B, information not contained in $\mathcal{F}_n$. Concretely, for the coin walk of example 10.1.2 with $B = \{0\}$ and $N = 2$, the event $\{\lambda_{\{0\}} = 0\} = \{S_1 \neq 0, S_2 \neq 0\}$ depends on ξ_1, ξ_2 and so is not in $\mathcal{F}_0 = \{\emptyset, \Omega\}$. The last-exit rule peeks into the future, and it is exactly this kind of rule that optional stopping forbids.

The two examples are the moral of the section in miniature. A hitting time is knowable the moment it occurs; a last-exit time is knowable only in hindsight, and no gambler can act on it. The next construction makes the informal claim "you cannot beat a fair game" precise by encoding a betting strategy as a process and showing that no such strategy can turn a fair game into a favorable one. A strategy must commit to a stake for round n before the outcome of round n is seen, using only information available at time $n-1$; a process with this property is called predictable.

Definition 10.1.7: Predictable Process and Martingale Transform

A process $(H_n)_{n\geq 1}$ is *predictable* with respect to $(\mathcal{F}_n)$ if H_n is $\mathcal{F}_{n-1}$-measurable for every $n \geq 1$. Given a predictable (H_n) and an adapted process $(M_n)_{n\geq 0}$, the *martingale transform* of M by H is the process $((H \cdot M)_n)_{n\geq 0}$ defined by $(H \cdot M)_0 = 0$ and
$$(H \cdot M)_n = \sum_{k=1}^{n} H_k \,(M_k - M_{k-1})$$
the accumulated gain of a gambler who, entering round k with knowledge $\mathcal{F}_{k-1}$, places the stake H_k on the increment $M_k - M_{k-1}$.

The transform is the discrete stochastic integral of the strategy H against the game M, the exact discrete-time forerunner of the Itô integral that opens the continuous theory. Each increment $H_k(M_k - M_{k-1})$ is the gambler's winnings in round k: the stake H_k, fixed in advance on the strength of $\mathcal{F}_{k-1}$, multiplied by the fresh increment $M_k - M_{k-1}$ of the game. Predictability is the no-clairvoyance condition, that the stake cannot depend on the very increment it is about to be multiplied by. The next result is the formal statement that a fair game cannot be beaten: whatever bounded strategy the gambler adopts, the transform of a martingale is again a martingale, so the gambler's fortune remains a fair game with constant expectation.

Proposition 10.1.8: Transforms Preserve the Martingale Property

Let $(H_n)_{n\geq 1}$ be predictable and bounded, say $|H_n| \leq c_n$ for constants c_n, and let (M_n) be adapted with each M_n integrable.

1. If (M_n) is a martingale, then $(H \cdot M)_n$ is a martingale (with $(H \cdot M)_0 = 0$).
2. If (M_n) is a submartingale and $H_n \geq 0$ for every n, then $(H \cdot M)_n$ is a submartingale; if (M_n) is a supermartingale and $H_n \geq 0$, it is a supermartingale.

In particular, for any stopping time τ the stopped process satisfies

$$M_{n\wedge\tau} = M_0 + (\mathbb{1}_{\{k\leq\tau\}} \cdot M)_n \tag{10.1}$$

so if (M_n) is a martingale then $(M_{n\wedge\tau})$ is a martingale, and if (M_n) is a sub- or supermartingale then $(M_{n\wedge\tau})$ is a sub- or supermartingale respectively.

***Proof*:**
Step 1: adaptedness and integrability. Each $(H \cdot M)_n$ is a finite sum of terms $H_k(M_k - M_{k-1})$ with H_k being $\mathcal{F}_{k-1} \subseteq \mathcal{F}_n$-measurable and $M_k - M_{k-1}$ being $\mathcal{F}_k \subseteq \mathcal{F}_n$-measurable, so $(H \cdot M)_n$ is $\mathcal{F}_n$-measurable, hence the transform is adapted. For integrability, the bound $|H_k| \leq c_k$ gives $\mathbb{E}|H_k(M_k - M_{k-1})| \leq c_k(\mathbb{E}|M_k| + \mathbb{E}|M_{k-1}|) < \infty$, so each $(H \cdot M)_n$ is integrable as a finite sum of integrable terms.

Step 2: the one-step forecast. The increment of the transform is $(H \cdot M)_{n+1} - (H \cdot M)_n = H_{n+1}(M_{n+1} - M_n)$. Condition on $\mathcal{F}_n$. Since H_{n+1} is $\mathcal{F}_n$-measurable and bounded, it may be taken outside the conditional expectation (proposition 9.2.1, taking out what is known):

$$\mathbb{E}\big[H_{n+1}(M_{n+1} - M_n)\big|\mathcal{F}_n\big] = H_{n+1}\,\mathbb{E}[M_{n+1} - M_n|\mathcal{F}_n] = H_{n+1}\big(\mathbb{E}[M_{n+1}|\mathcal{F}_n] - M_n\big)$$

the last equality because M_n is $\mathcal{F}_n$-measurable. If (M_n) is a martingale the bracket vanishes, so $\mathbb{E}[(H \cdot M)_{n+1}|\mathcal{F}_n] = (H \cdot M)_n$ and the transform is a martingale, proving (1). If (M_n) is a submartingale the bracket $\mathbb{E}[M_{n+1}|\mathcal{F}_n] - M_n$ is ≥ 0 a.s., and multiplying a nonnegative quantity by $H_{n+1} \geq 0$ preserves the sign, so $\mathbb{E}[(H \cdot M)_{n+1}|\mathcal{F}_n] \geq (H \cdot M)_n$ and the transform is a submartingale; the supermartingale case reverses the inequality, proving (2).

Step 3: the stopped process. Take $H_k = \mathbb{1}_{\{k\leq\tau\}}$. This is predictable, because $\{k \leq \tau\} = \{\tau \leq k-1\}^c \in \mathcal{F}_{k-1}$ by the stopping-time property (definition 10.1.5), and it is bounded by 1 and nonnegative. Its transform telescopes: for $n \geq 1$,

$$(\mathbb{1}_{\{k\leq\tau\}} \cdot M)_n = \sum_{k=1}^{n} \mathbb{1}_{\{k\leq\tau\}}(M_k - M_{k-1}) = \sum_{k=1}^{n\wedge\tau}(M_k - M_{k-1}) = M_{n\wedge\tau} - M_0$$

since $\mathbb{1}_{\{k\leq\tau\}} = 1$ exactly for $k \leq \tau$ and the surviving sum telescopes from M_0 to $M_{n\wedge\tau}$. This is (10.1). Applying parts (1) and (2) to this predictable nonnegative bounded H shows that $(M_{n\wedge\tau})$ inherits the martingale, submartingale, or supermartingale property of (M_n), as we sought to show.

The proposition is the chapter's first structural theorem, and its reading is the one promised at the outset: no bounded gambling strategy converts a fair game into a favorable one. The gambler chooses stakes H_k by any rule using only past information, and the fortune $(H \cdot M)_n$ is still a martingale, still with mean $\mathbb{E}[(H \cdot M)_n] = 0$. Stopping is the special strategy "bet one unit until time τ, then quit," encoded by $H_k = \mathbb{1}_{\{k \le \tau\}}$, and (10.1) exhibits the stopped process as exactly such a transform. The immediate payoff is the identity of means at every fixed horizon: taking expectations in the stopped-process martingale gives $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$ for every n. This is a genuine conservation law, but it holds at $n \wedge \tau$, a bounded time, not yet at τ itself; passing to $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ requires letting $n \to \infty$ under a hypothesis controlling the tail, and supplying that hypothesis is the whole content of the optional stopping theorem in the next section. For a submartingale the same computation gives only $\mathbb{E}[M_{n\wedge\tau}] \ge \mathbb{E}[M_0]$: a favorable game stopped early can only help.

What (10.1) does *not* say bears recording. It does not say $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ in general, and it must not, for the unrestricted claim is false. The doubling strategy on a fair coin walk S, stopped at the first time the gambler is ahead, produces a fortune $(H \cdot S)$ with $\mathbb{E}[(H \cdot S)_\tau] = 1 \ne 0 = (H \cdot S)_0$; the catch is that the required stakes $H_k = 2^{k-1}$ are unbounded, so the boundedness hypothesis of proposition 10.1.8 fails and the transform need no longer be a martingale. The first-win time here is almost surely finite, indeed with finite expectation, so it is the unbounded stakes and not the stopping time that break the conclusion. The stopped-process identity is exact at every bounded horizon and only conditionally at the random time τ, and holding these two apart is the discipline the theory imposes.

The section closes with the three martingales that recur throughout the chapter, all built from the coin walk, and a fourth obtained by stopping. They are the concrete ballast for every abstract theorem to come.

Example 10.1.9: The Random-Walk Martingales

Let (S_n) be the simple symmetric random walk of example 10.1.2, with increments $\xi_k = \pm 1$ each with probability $\frac{1}{2}$, independent, and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$.

1. *The walk itself.* The increments ξ_k are independent, integrable, and mean zero, so by proposition 10.1.4(3) the walk (S_n) is a martingale: $\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[\xi_{n+1}] = S_n$ a.s. A fair coin game is a fair game.

2. *The compensated square.* Set $M_n = S_n^2 - n$. Then M_n is integrable and $\mathcal{F}_n$-measurable, and using $S_{n+1} = S_n + \xi_{n+1}$ with $\xi_{n+1}^2 = 1$,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = \mathbb{E}[S_n^2 + 2S_n\xi_{n+1} + \xi_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 2S_n\mathbb{E}[\xi_{n+1}] + 1 = S_n^2 + 1$$

since S_n is $\mathcal{F}_n$-measurable and ξ_{n+1} is independent of $\mathcal{F}_n$ with mean zero. Hence $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = S_n^2 + 1 - (n+1) = S_n^2 - n = M_n$, so $(S_n^2 - n)$ is a martingale. The subtracted n is the "compensator" that cancels the deterministic growth $\mathbb{E}[S_n^2] = n$ of the variance, and it is the simplest instance of the Doob decomposition to come.

3. *The exponential martingale.* Fix $\theta \in \mathbb{R}$ and set

$$M_n = \frac{e^{\theta S_n}}{(\cosh\theta)^n}$$

Then M_n is positive and integrable, and since $e^{\theta S_{n+1}} = e^{\theta S_n} e^{\theta \xi_{n+1}}$ with $\mathbb{E}[e^{\theta\xi_{n+1}}] = \frac{1}{2}(e^\theta + e^{-\theta}) = \cosh\theta$,

$$\mathbb{E}[e^{\theta S_{n+1}} \mid \mathcal{F}_n] = e^{\theta S_n}\, \mathbb{E}[e^{\theta\xi_{n+1}} \mid \mathcal{F}_n] = e^{\theta S_n} \cosh\theta$$

using independence of ξ_{n+1} from $\mathcal{F}_n$. Dividing by $(\cosh\theta)^{n+1}$ gives $\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n$, so $(e^{\theta S_n}/(\cosh\theta)^n)$ is a martingale for every θ. This one-parameter family is the discrete analogue of the exponential martingale of Brownian motion and the source of the exponential bounds of section 10.5.

4. *The walk stopped at a hitting time.* Let $a < 0 < b$ be integers and let $\tau = \inf\{n \colon S_n \in \{a, b\}\}$ be the first time the walk hits a or b, a stopping time by example 10.1.6. By proposition 10.1.8, the stopped walk $(S_{n\wedge\tau})$ is a martingale, so $\mathbb{E}[S_{n\wedge\tau}] = 0$ for every n. The stopped walk stays in the bounded range $[a, b]$, so this identity survives the limit $n \to \infty$, and the two hitting probabilities can be read off from $\mathbb{E}[S_\tau] = 0$. The computation is carried out in full in the next section as the gambler's ruin; it is recorded here as the prototype of a martingale argument, a conserved mean evaluated at a hitting time.

These four processes are the fixed points of reference for the chapter. The walk (S_n) is the fair game; the compensated square $S_n^2 - n$ measures accumulated variance and will yield the expected duration of the gambler's ruin; the exponential family $e^{\theta S_n}/(\cosh\theta)^n$ is the generating-function martingale behind concentration inequalities; and the stopped walk is the template that turns conservation of the mean into a hitting probability. Each recurs in a later section, so a reader who has these four fully in hand has, in miniature, the whole of the chapter's technique.

Exercise 10.1.1

1. Let $(X_n)_{n\geq 0}$ be any process and $(\mathcal{F}_n)$ any filtration to which it is adapted. Show that the natural filtration $\mathcal{G}_n = \sigma(X_0, \ldots, X_n)$ satisfies $\mathcal{G}_n \subseteq \mathcal{F}_n$ for every n, so that the natural filtration is the smallest filtration making (X_n) adapted. Deduce that if (X_n) is a martingale relative to some filtration $(\mathcal{F}_n)$, then it is also a martingale relative to its natural filtration.

2. Let $Y_1, Y_2, \ldots$ be independent, integrable, and satisfy $\mathbb{E}[Y_k] = 1$ for every k. Set $M_0 = 1$ and $M_n = Y_1 Y_2 \cdots Y_n$, with $\mathcal{F}_n = \sigma(Y_1, \ldots, Y_n)$. Prove that (M_n) is a martingale. (This is the multiplicative analogue of proposition 10.1.4(3); the exponential martingale of example 10.1.9 is the special case $Y_k = e^{\theta\xi_k}/\cosh\theta$.)

3. Let (M_n) be a martingale. Using proposition 10.1.4(2), show that $(|M_n|)$ and, when $M_n \in L^2$ for all n, (M_n^2) are submartingales. Give an example of a martingale for which (M_n^2) is a *strict* submartingale, that is $\mathbb{E}[M_{n+1}^2|\mathcal{F}_n] > M_n^2$ with positive probability.

4. Let σ, τ be stopping times for a filtration $(\mathcal{F}_n)$. Show that $\sigma\wedge\tau = \min(\sigma, \tau)$, $\sigma\vee\tau = \max(\sigma, \tau)$, and $\sigma + \tau$ are all stopping times. Show also that a deterministic constant $\tau \equiv m$ is a stopping time and that $\mathcal{F}_\tau = \mathcal{F}_m$ in that case.

5. Let (S_n) be the coin walk and let τ be a stopping time with $\tau \leq N$ almost surely for a constant N. Using (10.1), prove directly that $\mathbb{E}[S_\tau] = 0$ and $\mathbb{E}[S_\tau^2] = \mathbb{E}[\tau]$. (These are the bounded cases of Wald's identities, proved in general in the next section.)

6. Let (H_n) be predictable and bounded and (M_n) a martingale. Show that if $(H \cdot M)$ has $\mathbb{E}[(H \cdot M)_n] = 0$ for all n (as the proposition guarantees), then no choice of bounded predictable H can make $\mathbb{E}[(H \cdot M)_n] > 0$. Contrast this with the doubling strategy: explain in one or two sentences which hypothesis of proposition 10.1.8 the doubling strategy violates.

10.2 Optional Stopping

The martingale property $\mathbb{E}[M_{n+1} | \mathcal{F}_n] = M_n$ says that a fair game is fair at each deterministic step: whatever the history, the expected fortune after the next play equals the present fortune, so $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for every fixed n (proposition 10.1.4). A gambler, however, does not decide in advance to quit after exactly n plays. The gambler quits when something happens, when the fortune first hits a target, when the money runs out, when a streak ends. The stopping rule is itself random, adapted to the game as it unfolds, and the natural question is whether fairness survives the substitution of a random quitting time τ for the fixed horizon n. Is $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$? If yes, no strategy that chooses only *when* to stop can convert a fair game into a favorable one: one cannot beat a fair game by clever quitting any more than by clever betting.

The stopped process $(M_{n\wedge\tau})$ is already a martingale (proposition 10.1.8), which encodes exactly the impossibility of beating the game up to any finite deterministic horizon: $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$ for every n. The delicate passage is the limit $n \to \infty$. One wants to let the deterministic clock run out and read off $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$, and the entire content of the optional stopping theorem is the set of hypotheses under which this interchange of limit and expectation is legitimate. Without some such hypothesis it can fail, and fail concretely, as a gambler with unlimited credit and unlimited patience will discover.

Theorem 10.2.1: Optional Stopping Theorem

Let $(M_n)_{n\geq 0}$ be a martingale relative to $(\mathcal{F}_n)$ (definition 10.1.3) and let τ be a stopping time (definition 10.1.5). Then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$$

provided any one of the following conditions holds:

1. τ is bounded: $\tau \leq N$ almost surely for some constant $N \in \mathbb{N}$;
2. $\tau < \infty$ almost surely and the stopped process $(M_{n\wedge\tau})_{n\geq 0}$ is uniformly integrable (in particular, if it is bounded: $|M_{n\wedge\tau}| \leq C$ a.s. for a constant C);
3. $\mathbb{E}[\tau] < \infty$ and the increments are bounded: there is a constant K with $|M_{n+1} - M_n| \leq K$ almost surely for every n.

If instead (M_n) is a submartingale, then $\mathbb{E}[M_\tau] \geq \mathbb{E}[M_0]$ under any of (1)-(3) (and $\leq$ for a supermartingale).

Before the proof, it is worth naming what each hypothesis controls, because the three are three ways of solving the *same* problem. In every case the stopped process $M_{n\wedge\tau}$ converges pointwise to M_τ as $n \to \infty$ (once $\tau < \infty$ a.s., eventually $n \wedge \tau = \tau$), and one already knows $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$. The only issue is whether $\mathbb{E}[M_{n\wedge\tau}] \to \mathbb{E}[M_\tau]$, that is, whether the convergence is strong enough to pass to the limit under the expectation. Boundedness of τ removes the limit entirely: the game has ended by time N. Uniform integrability of the stopped process is precisely the condition, from the theory of integration, that licenses interchanging limit and expectation for an almost-surely convergent sequence. The third hypothesis is the one used for concrete random walks: bounded increments plus a finite-mean stopping time produce an integrable dominating function, so dominated convergence applies. The three hypotheses are not competing; they are the same theorem instrumented for the three situations one meets in practice.

***Proof*:**
The submartingale claim is proved alongside the martingale claim: the stopped process is a martingale (resp. submartingale) by proposition 10.1.8, and each equality below becomes the corresponding inequality when "martingale" is weakened to "submartingale". The argument is given for the martingale case; the modification is indicated at the one point where the direction of the inequality matters.

Step 1: The bounded case (1). By proposition 10.1.8 the stopped process $(M_{n\wedge\tau})_{n\geq 0}$ is a martingale, and every martingale has constant mean (proposition 10.1.4): $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$ for all n. If $\tau \leq N$ almost surely, then at time $n = N$ one has $N \wedge \tau = \tau$ almost surely, so

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_{N\wedge\tau}] = \mathbb{E}[M_0]$$

which is the assertion. (For a submartingale $(M_{n\wedge\tau})$ has nondecreasing mean, so $\mathbb{E}[M_\tau] = \mathbb{E}[M_{N\wedge\tau}] \geq \mathbb{E}[M_{0\wedge\tau}] = \mathbb{E}[M_0]$; this is the sole place the inequality direction is used.)

Step 2: Reduction of (2) and (3) to a limit. In both remaining cases $\tau < \infty$ almost surely, so for almost every ω there is an n with $\tau(\omega) \leq n$, whence $M_{n\wedge\tau}(\omega) = M_\tau(\omega)$ for all large n. Thus

$$M_{n\wedge\tau} \longrightarrow M_\tau \qquad \text{almost surely as } n \to \infty \tag{10.2}$$

Step 1 (with N replaced by the running index n) gives $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$ for every n. The theorem will follow once

$$\mathbb{E}[M_\tau] = \lim_{n\to\infty} \mathbb{E}[M_{n\wedge\tau}] \tag{10.3}$$

is established, since the right side is the constant $\mathbb{E}[M_0]$. What remains, in each of (2) and (3), is to justify moving the limit (10.2) inside the expectation to obtain (10.3). For a submartingale Step 1 gives instead $\mathbb{E}[M_{n\wedge\tau}] \geq \mathbb{E}[M_0]$ for every n, and the interchange (10.3) then yields $\mathbb{E}[M_\tau] = \lim_n \mathbb{E}[M_{n\wedge\tau}] \geq \mathbb{E}[M_0]$: the inequality survives the limit because the limit passages below all establish $\mathbb{E}[M_{n\wedge\tau}] \to \mathbb{E}[M_\tau]$, and a limit of numbers each $\geq \mathbb{E}[M_0]$ is $\geq \mathbb{E}[M_0]$.

Step 3: The uniformly integrable case (2). A sequence that converges almost surely and is uniformly integrable converges in L^1, and L^1 convergence carries the expectations along: $\mathbb{E}[M_{n\wedge\tau}] \to \mathbb{E}[M_\tau]$ [2]. This is exactly (10.3). The parenthetical special case is immediate: if $|M_{n\wedge\tau}| \leq C$ almost surely, the constant C dominates the whole sequence, and a sequence dominated by a fixed integrable variable is uniformly integrable. (A bounded, almost-surely convergent sequence also falls to the bounded convergence theorem directly, without invoking uniform integrability.)

Step 4: The bounded-increments case (3). The point is to produce an integrable dominating variable and invoke dominated convergence. Write the stopped process as a telescoping sum of increments,

$$M_{n\wedge\tau} = M_0 + \sum_{k=1}^{n\wedge\tau}(M_k - M_{k-1})$$

Bounding each increment by K and the number of terms by τ,

$$|M_{n\wedge\tau}| \leq |M_0| + \sum_{k=1}^{n\wedge\tau} |M_k - M_{k-1}| \leq |M_0| + K\tau =: Y \tag{10.4}$$

The dominating variable $Y = |M_0| + K\tau$ does not depend on n, and it is integrable: $\mathbb{E}[Y] = \mathbb{E}|M_0| + K\,\mathbb{E}[\tau] < \infty$, since M_0 is integrable (part of the definition of a martingale) and $\mathbb{E}[\tau] < \infty$

by hypothesis. The bound (10.4) holds for every n, so the dominated convergence theorem [2] applies to the almost-sure limit (10.2) and yields $\mathbb{E}[M_{n\wedge\tau}] \to \mathbb{E}[M_\tau]$, which is (10.3). In all three cases $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$, completing the proof.

The theorem is a licence, and its value lies as much in when it may be applied as in what it concludes. Read positively, it says a stopping rule cannot manufacture an edge: for any τ meeting one of the three conditions, the expected fortune at the stopping time equals the starting fortune, so the game is fair however one chooses to quit. Read as a warning, it says the conditions cannot be dropped. The single most common error in applying optional stopping is to write $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ for a stopping time that is finite almost surely but satisfies none of (1)-(3), and the next example shows this equality is then false, not just unproven.

Example 10.2.2: The Random Walk Reaching $+1$

Let (S_n) be the simple symmetric random walk started at $S_0 = 0$, so $S_n = X_1 + \cdots + X_n$ with the X_i independent and $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$ (example 1.3.6). Then (S_n) is a martingale (example 10.1.9). Let

$$\tau = \inf\{n \geq 0 \big| S_n = 1\}$$

be the first time the walk reaches $+1$. It is a standard fact, which the gambler's ruin computation below will confirm as a limiting case (example 10.2.4), that the symmetric walk is recurrent and hits every integer, so $\tau < \infty$ almost surely. At the stopping time the walk sits, by definition, at height 1:

$$S_\tau = 1 \quad \text{almost surely,} \qquad \text{hence} \qquad \mathbb{E}[S_\tau] = 1$$

But $\mathbb{E}[S_0] = 0$. So $\mathbb{E}[S_\tau] = 1 \neq 0 = \mathbb{E}[S_0]$: optional stopping fails outright.

Nothing is wrong with the theorem; every one of its hypotheses fails here. The time τ is unbounded, ruling out (1). The stopped process $S_{n\wedge\tau}$ is not uniformly integrable, ruling out (2): before hitting $+1$ the walk can wander arbitrarily far negative, and indeed $\mathbb{E}[\tau] = \infty$, so the bounded-increments criterion (3) also fails despite $|S_{n+1} - S_n| = 1$. The failure of (3) is the clean diagnosis. Increments are bounded, yet $\mathbb{E}[\tau] = \infty$, and it is exactly the infinitude of $\mathbb{E}[\tau]$ that destroys the integrable dominating variable $|M_0| + K\tau$ built in Step 4 of the proof. The reader who has met the recurrent symmetric walk before will recognize $\mathbb{E}[\tau] = \infty$ as the statement that the walk hits $+1$ with probability one but takes, on average, infinitely long to do it: the mean return time of recurrent one-dimensional walk is infinite. A gambler playing this strategy is guaranteed to end one dollar ahead, but only by risking an unbounded loss over an unbounded time, and the "profit" $\mathbb{E}[S_\tau] - \mathbb{E}[S_0] = 1$ is an accounting artifact of ignoring that risk.

This example is the reason the theorem is stated with hypotheses rather than as a bare identity, and it fixes the discipline for every application to come: to invoke optional stopping one must first certify one of (1)-(3), and for random walks the certification is almost always (3), reduced to the two facts $\mathbb{E}[\tau] < \infty$ and bounded increments. The next result turns the theorem into a computational tool by identifying, for sums of independent variables, exactly the quantity that optional stopping controls.

Corollary 10.2.3: Wald's Identities

Let $X_1, X_2, \dots$ be independent and identically distributed (definition 2.2.2, theorem 2.3.5), and set $S_n = X_1 + \cdots + X_n$ with $S_0 = 0$. Let τ be a stopping time for the natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ with $\mathbb{E}[\tau] < \infty$.

1. *(First identity.)* If X_1 is integrable with mean $\mu = \mathbb{E}[X_1]$, then S_τ is integrable and
$$\mathbb{E}[S_\tau] = \mu\,\mathbb{E}[\tau] \tag{10.5}$$
2. *(Second identity.)* If in addition X_1 is square-integrable with variance $\sigma^2 = \mathrm{Var}(X_1)$, then
$$\mathbb{E}\big[(S_\tau - \mu\tau)^2\big] = \sigma^2\,\mathbb{E}[\tau] \tag{10.6}$$

The first identity is the intuition of the "fair game" made quantitative for a biased game: if each play changes the fortune by μ on average and one plays a random number τ of times, the expected total change is μ times the expected number of plays, exactly as if τ were a constant. The delicate content is that this holds even though τ is chosen adaptively, in response to the very increments being summed, so a naive "$\mathbb{E}[S_\tau] = \mathbb{E}[\sum_{k\le\tau} X_k] = \mathbb{E}[\tau]\mathbb{E}[X_1]$" is not a valid factorization (τ and the X_k are dependent). The second identity does the same for the fluctuation: after centring by the drift, the mean-square size of S_τ is the per-step variance times the expected number of steps, the stopped-time analogue of $\mathrm{Var}(S_n) = n\sigma^2$. Both are proved by feeding a martingale into optional stopping.

***Proof*:**
The finiteness hypothesis $\mathbb{E}[\tau] < \infty$ will feed criterion (3) of theorem 10.2.1 in each part, but the increments of the relevant martingales are not bounded by a constant unless the X_i are. A truncation argument handles the general case; the structure of the proof is to verify the identity on a bounded stopping time and then remove the bound.

Step 1: The centred sum is a martingale. Put $M_n = S_n - n\mu$. Since the X_i are i.i.d. with mean μ, the increments $X_i - \mu$ are independent, mean zero, and integrable, so (M_n) is a martingale for $(\mathcal{F}_n)$ (proposition 10.1.4), with $M_0 = 0$.

Step 2: The first identity for bounded τ. Suppose first $\tau \le N$ almost surely. By the bounded case of optional stopping (theorem 10.2.1(1)), $\mathbb{E}[M_\tau] = \mathbb{E}[M_0] = 0$, that is $\mathbb{E}[S_\tau - \mu\tau] = 0$. Since $\tau \le N$ makes τ integrable and $|S_\tau| \le \sum_{k\le N} |X_k|$ integrable, the expectation splits, giving $\mathbb{E}[S_\tau] = \mu\,\mathbb{E}[\tau]$. This is (10.5) for bounded τ.

Step 3: Removing the bound. Apply Step 2 to the truncated stopping times $\tau \wedge N$, which are bounded and increase to τ:
$$\mathbb{E}[S_{\tau\wedge N}] = \mu\,\mathbb{E}[\tau \wedge N] \tag{10.7}$$
As $N \to \infty$, monotone convergence gives $\mathbb{E}[\tau\wedge N] \uparrow \mathbb{E}[\tau] < \infty$, so the right side converges to $\mu\,\mathbb{E}[\tau]$. For the left side, dominate: by the same telescoping bound as in the proof of theorem 10.2.1,
$$|S_{\tau\wedge N}| \le \sum_{k=1}^{\tau\wedge N} |X_k| \le \sum_{k=1}^{\tau} |X_k| =: W$$

and W is integrable because, by the independence of $\{\tau \geq k\} \in \mathcal{F}_{k-1}$ from X_k,

$$\mathbb{E}[W] = \mathbb{E}\Big[\sum_{k=1}^{\infty} |X_k|\, \mathbb{1}_{\{\tau \geq k\}}\Big] = \sum_{k=1}^{\infty} \mathbb{E}|X_k|\, \mathbb{P}(\tau \geq k) = \mathbb{E}|X_1| \sum_{k=1}^{\infty} \mathbb{P}(\tau \geq k) = \mathbb{E}|X_1|\, \mathbb{E}[\tau] < \infty \tag{10.8}$$

using $\sum_{k \geq 1} \mathbb{P}(\tau \geq k) = \mathbb{E}[\tau]$ for the integer-valued $\tau \geq 0$. Since $S_{\tau \wedge N} \to S_\tau$ almost surely (as $\tau < \infty$) and is dominated by the integrable W, dominated convergence [2] gives $\mathbb{E}[S_{\tau \wedge N}] \to \mathbb{E}[S_\tau]$, and S_τ is integrable ($|S_\tau| \leq W$). Passing to the limit in (10.7) yields $\mathbb{E}[S_\tau] = \mu\, \mathbb{E}[\tau]$, which is (10.5).

Step 4: The second identity. Assume now $\sigma^2 = \mathrm{Var}(X_1) < \infty$. The process

$$N_n = M_n^2 - n\sigma^2 = (S_n - n\mu)^2 - n\sigma^2$$

is a martingale for $(\mathcal{F}_n)$: writing $D_k = X_k - \mu$ (mean zero, variance σ^2, independent), $M_n = \sum_{k \leq n} D_k$, and

$$\mathbb{E}[M_{n+1}^2 \big| \mathcal{F}_n] = \mathbb{E}[(M_n + D_{n+1})^2 \big| \mathcal{F}_n] = M_n^2 + 2M_n \mathbb{E}[D_{n+1}] + \mathbb{E}[D_{n+1}^2] = M_n^2 + \sigma^2$$

where $\mathbb{E}[D_{n+1} \big| \mathcal{F}_n] = \mathbb{E}[D_{n+1}] = 0$ and $\mathbb{E}[D_{n+1}^2 \big| \mathcal{F}_n] = \sigma^2$ by independence (proposition 9.2.1). Hence $\mathbb{E}[N_{n+1} \big| \mathcal{F}_n] = N_n$ with $N_0 = 0$. For bounded $\tau \leq N$, optional stopping (theorem 10.2.1(1)) gives $\mathbb{E}[N_\tau] = 0$, that is

$$\mathbb{E}[(S_\tau - \mu\tau)^2] = \mathbb{E}[M_\tau^2] = \sigma^2\, \mathbb{E}[\tau] \tag{10.9}$$

To remove the bound, apply (10.9) to $\tau \wedge N$ and let $N \to \infty$. The right side tends to $\sigma^2 \mathbb{E}[\tau]$ by monotone convergence. For the left side one needs $\mathbb{E}[M_{\tau \wedge N}^2] \to \mathbb{E}[M_\tau^2]$, and the point is that L^2-boundedness of the sequence $(M_{\tau \wedge N})$ is *not* enough: it makes $(M_{\tau \wedge N})$ uniformly integrable and hence L^1-convergent, which controls the first moments but not the second. What is true, and what does the job, is that $M_{\tau \wedge N} \to M_\tau$ in L^2, and L^2 convergence carries the squared norms along.

To see the L^2 convergence, use that the stopped martingale has orthogonal increments. For $N < N'$ the increment $M_{\tau \wedge N'} - M_{\tau \wedge N} = \sum_{k=N+1}^{N'} D_k\, \mathbb{1}_{\{\tau \geq k\}}$ is a sum of terms each orthogonal to $\mathcal{F}_{\tau \wedge N}$, so $M_{\tau \wedge N} \perp (M_{\tau \wedge N'} - M_{\tau \wedge N})$ in L^2 and, by (10.9) applied at N and at N',

$$\mathbb{E}\big[(M_{\tau \wedge N'} - M_{\tau \wedge N})^2\big] = \mathbb{E}[M_{\tau \wedge N'}^2] - \mathbb{E}[M_{\tau \wedge N}^2] = \sigma^2\big(\mathbb{E}[\tau \wedge N'] - \mathbb{E}[\tau \wedge N]\big)$$

Since $\mathbb{E}[\tau \wedge N] \uparrow \mathbb{E}[\tau] < \infty$, the right side tends to 0 as $N, N' \to \infty$, so $(M_{\tau \wedge N})$ is Cauchy in L^2 and therefore converges in L^2 to some limit. That limit agrees with the almost-sure limit M_τ of (10.2) (an L^2-convergent sequence has a subsequence converging almost surely to its L^2-limit [2]), so $M_{\tau \wedge N} \to M_\tau$ in L^2; in particular $M_\tau \in L^2$. Convergence in L^2 forces convergence of the norms, $\mathbb{E}[M_{\tau \wedge N}^2] \to \mathbb{E}[M_\tau^2]$. Passing to the limit in (10.9) gives (10.6), as we sought to show.

Wald's first identity already settles a question elementary probability leaves awkward: the expected value of a sum of a random number of independent terms, when the number of terms may itself react to the terms. The classical statement, that a gambler who bets one dollar per round on a fair game and quits at an integrable-mean rule breaks even, is the case $\mu = 0$; a positive μ scales the expected winnings by the expected number of rounds. The example the whole chapter has been circling is the one where these identities compute everything explicitly: the walk pinned between two absorbing barriers.

Example 10.2.4: Gambler's Ruin for the Symmetric Walk

A gambler starts with a dollars and plays a fair game, winning or losing one dollar each round independently with probability $\frac{1}{2}$, until either going broke (reaching 0) or reaching a target of N dollars, where $0 \le a \le N$. Model the fortune by the symmetric walk $S_n = a + X_1 + \cdots + X_n$ with $\mathbb{P}(X_i = \pm 1) = \frac{1}{2}$ (example 1.3.6), and let

$$\tau = \inf\{n \ge 0 \big| S_n = 0 \text{ or } S_n = N\}$$

be the first time it hits a barrier. Three quantities are wanted: the probability of ruin, the probability of reaching the target, and the expected duration of play.

The stopping time has finite mean. Confine the walk to $\{0, 1, \ldots, N\}$. From any interior state, the N consecutive steps $+1$ carry the walk to N, so from every starting point the walk is absorbed within any block of N steps with probability at least 2^{-N}. Splitting time into blocks of length N and using independence, $\mathbb{P}(\tau > kN) \le (1 - 2^{-N})^k$, which decays geometrically; hence $\tau < \infty$ almost surely and $\mathbb{E}[\tau] < \infty$ (a geometric tail has all moments finite).

Absorption probabilities from the martingale S_n. The walk (S_n) is a martingale, its increments are bounded by 1, and $\mathbb{E}[\tau] < \infty$, so optional stopping applies in form (3) (theorem 10.2.1):

$$\mathbb{E}[S_\tau] = \mathbb{E}[S_0] = a$$

Now $S_\tau \in \{0, N\}$, taking the value N with the win probability $p_a := \mathbb{P}(S_\tau = N)$ and 0 otherwise. Therefore $\mathbb{E}[S_\tau] = N\, p_a$, and equating to a gives

$$p_a = \mathbb{P}(\text{reach } N) = \frac{a}{N}, \qquad \mathbb{P}(\text{ruin}) = 1 - p_a = \frac{N-a}{N} \tag{10.10}$$

The win probability is the fraction of the way the gambler has already come. This also justifies the recurrence claim of example 10.2.2: letting $N \to \infty$ with $a = 1$ fixed, the probability the walk started at 1 reaches 0 before N is $1 - 1/N \to 1$, so from any positive level the symmetric walk returns to 0 almost surely.

Expected duration from the martingale $S_n^2 - n$. By example 10.1.9 the process $M_n = S_n^2 - n$ is a martingale (the increment computation $\mathbb{E}[S_{n+1}^2 - S_n^2 | \mathcal{F}_n] = 1$ uses only $\mathrm{Var}(X_i) = 1$ and is independent of the starting point, so $S_n^2 - n$ is a martingale for any $S_0 = a$; this is the $\sigma^2 = 1$ case of the martingale N_n in the proof of corollary 10.2.3). Its increments $S_{n+1}^2 - S_n^2 - 1 = 2S_n X_{n+1}$ are bounded in absolute value by $2N$ on the confined walk, and $\mathbb{E}[\tau] < \infty$, so optional stopping (form (3)) gives $\mathbb{E}[S_\tau^2 - \tau] = \mathbb{E}[S_0^2 - 0] = a^2$, that is

$$\mathbb{E}[\tau] = \mathbb{E}[S_\tau^2] - a^2$$

Since $S_\tau \in \{0, N\}$ with $S_\tau = N$ having probability a/N by (10.10), $\mathbb{E}[S_\tau^2] = N^2 \cdot (a/N) = Na$, and therefore

$$\mathbb{E}[\tau] = Na - a^2 = a(N-a) \tag{10.11}$$

The expected number of rounds is the product of the two distances to the barriers. A gambler starting at the midpoint $a = N/2$ of a game to N plays for an expected $N^2/4$ rounds, which grows quadratically in the target: the fair game is fair, but slow.

The symmetric case is clean because S_n and $S_n^2 - n$ are both martingales, and absorption at a barrier is exactly the event that fixes S_τ. When the game is biased the linear martingale S_n is replaced by an exponential one, and the same optional-stopping computation produces the classical asymmetric formulas. The device is the exponential martingale of example 10.1.9, specialized to the value of the parameter that makes the increment mean one.

Example 10.2.5: Gambler's Ruin for the Biased Walk

Now let each round be won with probability p and lost with probability $q = 1 - p$, where $p \neq \frac{1}{2}$, so $\mathbb{P}(X_i = +1) = p$ and $\mathbb{P}(X_i = -1) = q$. The fortune $S_n = a + X_1 + \cdots + X_n$ is no longer a martingale (its mean drifts by $\mu = p - q \neq 0$ per step), but the exponentiated walk is. Set $r = q/p$ and

$$M_n = r^{S_n} = \left(\frac{q}{p}\right)^{S_n}$$

Then (M_n) is a martingale: given $\mathcal{F}_n$, the factor $r^{X_{n+1}}$ is independent of the past, and

$$\mathbb{E}[r^{X_{n+1}}] = p \cdot r^{+1} + q \cdot r^{-1} = p \cdot \frac{q}{p} + q \cdot \frac{p}{q} = q + p = 1$$

so $\mathbb{E}[M_{n+1} \big| \mathcal{F}_n] = M_n \, \mathbb{E}[r^{X_{n+1}}] = M_n$. (This is example 10.1.9 with $e^\theta = r$; the choice $r = q/p$ is exactly the θ making $\mathbb{E}[e^{\theta X}] = 1$.) On the confined walk $M_n = r^{S_n}$ is bounded (its exponent lies in $[0, N]$), so the stopped process is bounded and optional stopping applies in form (2):

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0] = r^a$$

With $S_\tau \in \{0, N\}$ and win probability $p_a = \mathbb{P}(S_\tau = N)$, the left side is $r^0(1 - p_a) + r^N p_a = 1 + (r^N - 1)p_a$. Equating,

$$p_a = \frac{r^a - 1}{r^N - 1} = \frac{(q/p)^a - 1}{(q/p)^N - 1} \tag{10.12}$$

which recovers (10.10) in the limit $r \to 1$ (both sides tend to a/N by l'Hôpital in r). The expected duration is obtained from the martingale $S_n - n\mu$ (proposition 10.1.4): optional stopping in form (3) (bounded increments, $\mathbb{E}[\tau] < \infty$ by the same block argument as before) gives $\mathbb{E}[S_\tau - \mu\tau] = a$, hence $\mathbb{E}[\tau] = (\mathbb{E}[S_\tau] - a)/\mu$ with $\mathbb{E}[S_\tau] = N\,p_a$, so

$$\mathbb{E}[\tau] = \frac{N\,p_a - a}{p - q} = \frac{1}{p - q}\left(N\,\frac{(q/p)^a - 1}{(q/p)^N - 1} - a\right) \tag{10.13}$$

For a subfair game ($p < \frac{1}{2}$, so $r > 1$) the win probability (10.12) decays geometrically in the gap $N - a$: doubling the target roughly squares the already-small chance of reaching it. The house edge compounds, which is the mathematics behind why a small per-round disadvantage makes a large target nearly unreachable.

Between the symmetric example and its biased companion, optional stopping has now produced every quantity of the ruin problem in closed form, and the pattern is uniform: find a martingale whose value at the barriers is determined, certify one of the three hypotheses (bounded increments with $\mathbb{E}[\tau] < \infty$, or a bounded stopped process), and read off the answer from $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$. The same pattern drives the exercises, which push it to a branching-process barrier and to the maximal inequality that reappears, from the martingale side, in the next section.

Exercise 10.2.1

1. For the symmetric walk started at 0, let $\tau_{-b,c} = \inf\{n \big| S_n = -b \text{ or } S_n = c\}$ be the first

exit from the interval $(-b, c)$ with $b, c \geq 1$. Using the martingales S_n and $S_n^2 - n$ and optional stopping, compute $\mathbb{P}(S_{\tau_{-b,c}} = c)$ and $\mathbb{E}[\tau_{-b,c}]$. Check that $\mathbb{E}[\tau_{-b,c}] = bc$ and that this recovers (10.11) after recentring.

2. A gambler plays the symmetric one-dollar game and adopts the rule "quit the first time I am one dollar ahead", that is $\tau = \inf\{n | S_n = 1\}$ with $S_0 = 0$. Wald's first identity (10.5) with $\mu = 0$ would give $\mathbb{E}[S_\tau] = 0$, yet $S_\tau = 1$. Identify precisely which hypothesis of Wald's identity (corollary 10.2.3) fails, and connect your answer to example 10.2.2.

3. Let (S_n) be the symmetric walk from 0 and $\theta \in \mathbb{R}$. Show that $M_n = e^{\theta S_n}/(\cosh\theta)^n$ is a martingale (example 10.1.9). For the exit time $\tau = \tau_{-b,c}$ of Question 10.2.1 (1), apply optional stopping to M_n to derive a formula for $\mathbb{E}[(\cosh\theta)^{-\tau}\, \mathbb{1}_{\{S_\tau = c\}}]$ in terms of θ, b, c. (This is the generating function of the hitting time, the starting point for its full distribution.)

4. Let (Z_n) be a Galton-Watson branching process (definition 2.1.1 models each individual's offspring count): $Z_0 = 1$, and each of the Z_n individuals in generation n has, independently, a number of offspring with mean $\mu = 1$ and finite variance σ^2. It is a fact (proved in the applications section) that (Z_n) is itself a martingale when $\mu = 1$. Let $\tau = \inf\{n | Z_n = 0\}$ be the extinction time. Assuming $\mathbb{E}[\tau] < \infty$ fails here (extinction is certain but slow), explain why optional stopping does *not* give $\mathbb{E}[Z_\tau] = \mathbb{E}[Z_0] = 1$, and compute the true value of $\mathbb{E}[Z_\tau]$. Which of example 10.2.2 and example 10.2.4 is this closest to?

5. Let (M_n) be a nonnegative martingale with $M_0 = 1$, and fix $\lambda > 1$. Let $\tau = \inf\{n | M_n \geq \lambda\}$. Using optional stopping on the bounded time $\tau \wedge n$, show that $\mathbb{P}(\max_{k \leq n} M_k \geq \lambda) \leq 1/\lambda$. (This is the maximal inequality of the next section, obtained directly from optional stopping; it will reappear as a special case of theorem 10.3.1.)

6. Show by example that hypothesis (3) of theorem 10.2.1 cannot drop the condition $\mathbb{E}[\tau] < \infty$ even when increments are bounded: exhibit a martingale with $|M_{n+1} - M_n| \leq 1$ and a stopping time $\tau < \infty$ a.s. with $\mathbb{E}[M_\tau] \neq \mathbb{E}[M_0]$.

10.3 Doob's Inequalities

The optional stopping theorem controls a martingale at a single well-chosen time. The next question is coarser and, for most applications, more useful: how large can a martingale become over an entire stretch of time? A gambler who will be ruined if the fortune ever dips below a threshold cares not about the fortune at any one round but about its *running minimum*; a numerical analyst bounding an error process cares about the worst error seen so far. Both are statements about the maximum of $M_0, \ldots, M_n$, a random variable that mixes together the whole trajectory rather than reading off a single coordinate.

For a sum $M_n = X_1 + \cdots + X_n$ of independent mean-zero increments the tool is already in hand: Chebyshev's inequality (corollary 3.2.5) bounds $\mathbb{P}(|M_n| \geq \lambda)$ by $\mathrm{Var}(M_n)/\lambda^2$, and Kolmogorov's maximal inequality (theorem 4.1.1) extends this from the single endpoint M_n to the entire maximum $\max_{k \leq n} |M_k|$ at no extra cost. Doob's inequalities are the martingale generalization of exactly this phenomenon. They rest not on independence but on the submartingale property alone, and so apply far beyond sums of independent variables. The mechanism is the one already assembled in the two previous sections: a *first-passage stopping time*, the instant the process first reaches the level λ, together with optional stopping. This section proves three inequalities of increasing precision (a tail bound on the maximum, an L^p-norm bound, and a bound on the number of oscillations across a band) and reads Kolmogorov's inequality back out of the first as a special case. The upcrossing inequality is the analytic engine of the convergence theorem in section 10.4; the maximal and L^p inequalities are the standard tools for controlling a martingale uniformly in time.

The starting point is the tail bound on the running maximum. The heuristic is a conservation-of-mass argument. If the process reaches level λ somewhere in $[0, n]$, then at the first such instant it is at least λ; the submartingale property says that from that instant onward the process does not lose expected value, so the terminal expectation $\mathbb{E}[M_n]$ retains at least a λ-sized contribution from each trajectory that crossed. Comparing the total expected mass $\mathbb{E}[M_n]$ against λ times the probability of crossing yields the bound. The stopping time makes this precise.

Theorem 10.3.1: Doob's Maximal Inequality

Let $(M_n)_{n\geq 0}$ be a submartingale relative to a filtration $(\mathcal{F}_n)_{n\geq 0}$ (definition 10.1.3), and write

$$M_n^* = \max_{0\leq k\leq n} M_k$$

for its running maximum. Then for every $\lambda > 0$,

$$\lambda\,\mathbb{P}(M_n^* \geq \lambda) \leq \mathbb{E}\big[M_n\,\mathbb{1}_{\{M_n^*\geq\lambda\}}\big] \leq \mathbb{E}[M_n^+] \tag{10.14}$$

where $M_n^+ = \max(M_n, 0)$.

***Proof*:**
Fix $\lambda > 0$ and n. The strategy is to isolate the first time the process reaches λ, freeze it there, and compare expected values across that instant using the submartingale property.

Step 1: The first-passage stopping time. Define

$$\tau = \min\{k \geq 0 : M_k \geq \lambda\}$$

with the convention $\tau = \infty$ if no such $k \leq n$ exists. This is a stopping time (definition 10.1.5): the event $\{\tau = k\} = \{M_0 < \lambda, \ldots, M_{k-1} < \lambda,\ M_k \geq \lambda\}$ is determined by $M_0, \ldots, M_k$ and so lies in $\mathcal{F}_k$. The key observation is that the running maximum reaches λ by time n exactly when the first passage occurs by time n,

$$\{M_n^* \geq \lambda\} = \{\tau \leq n\} = \bigsqcup_{k=0}^{n}\{\tau = k\} \tag{10.15}$$

a disjoint union over the possible first-passage times.

Step 2: Comparing across the crossing. On the event $\{\tau = k\}$ with $k \leq n$, the process has just reached the level, so $M_k \geq \lambda$. The submartingale property propagates a lower bound on the expected terminal value: for each such k, since $\{\tau = k\} \in \mathcal{F}_k$ and M is a submartingale,

$$\mathbb{E}\big[M_n\,\mathbb{1}_{\{\tau=k\}}\big] = \mathbb{E}\big[\mathbb{E}[M_n|\mathcal{F}_k]\,\mathbb{1}_{\{\tau=k\}}\big] \geq \mathbb{E}\big[M_k\,\mathbb{1}_{\{\tau=k\}}\big] \geq \lambda\,\mathbb{P}(\tau = k)$$

The first equality is the tower property (proposition 9.2.2) together with the $\mathcal{F}_k$-measurability of $\mathbb{1}_{\{\tau=k\}}$, which lets it be pulled inside the conditional expectation (proposition 9.2.3); the first inequality is $\mathbb{E}[M_n | \mathcal{F}_k] \geq M_k$ a.s. for a submartingale, multiplied by the nonnegative indicator; the last is $M_k \geq \lambda$ on $\{\tau = k\}$.

Step 3: Summing over the first-passage times. Sum the display of Step 2 over $k = 0, \dots, n$ and use the disjoint decomposition (10.15),

$$\mathbb{E}\big[M_n \, \mathbb{1}_{\{M_n^* \geq \lambda\}}\big] = \sum_{k=0}^{n} \mathbb{E}\big[M_n \, \mathbb{1}_{\{\tau = k\}}\big] \geq \sum_{k=0}^{n} \lambda \, \mathbb{P}(\tau = k) = \lambda \, \mathbb{P}(\tau \leq n) = \lambda \, \mathbb{P}(M_n^* \geq \lambda)$$

This is the first inequality of (10.14). The second is immediate: on the event $\{M_n^* \geq \lambda\}$ one has $M_n \leq M_n^+$, and on its complement $M_n^+ \geq 0$, so $M_n \, \mathbb{1}_{\{M_n^* \geq \lambda\}} \leq M_n^+ \, \mathbb{1}_{\{M_n^* \geq \lambda\}} \leq M_n^+$; taking expectations gives $\mathbb{E}[M_n \, \mathbb{1}_{\{M_n^* \geq \lambda\}}] \leq \mathbb{E}[M_n^+]$, completing the proof.

The inequality is best read as a martingale upgrade of Markov's inequality (theorem 3.2.4). Markov bounds the tail of a single nonnegative variable by its mean; Doob bounds the tail of the entire running maximum by the mean of the terminal value, at no additional cost. The middle term of (10.14) is sharper than the right and is the form used in proofs: it says the contribution to $\mathbb{E}[M_n]$ coming from the crossing event already accounts for the whole λ-sized tail. The engine is the submartingale property used exactly once, in Step 2, to move expected value forward in time from the crossing instant to the terminal time. No independence is used anywhere, which is what makes the inequality so broadly applicable.

Two specializations are worth naming immediately, because they are the forms most often invoked. First, applied to a nonnegative submartingale, (10.14) gives $\lambda \, \mathbb{P}(M_n^* \geq \lambda) \leq \mathbb{E}[M_n]$ directly, since $M_n^+ = M_n$. Second, applied to $|M|$ when M is a martingale: for a martingale M the absolute value $|M_n|$ is a nonnegative submartingale (the convex function $x \mapsto |x|$ of a martingale, by proposition 10.1.4), so

$$\lambda \, \mathbb{P}\Big(\max_{k \leq n} |M_k| \geq \lambda\Big) \leq \mathbb{E}|M_n|$$

a one-sided-to-two-sided passage that is the martingale analogue of the elementary step from $\mathbb{P}(M_n \geq \lambda)$ to $\mathbb{P}(|M_n| \geq \lambda)$. The next example instantiates the inequality on the coin-tossing walk, where every quantity is computable by hand.

Example 10.3.2: The Maximal Inequality for the Symmetric Random Walk

Let $S_n = X_1 + \cdots + X_n$ with X_i independent ± 1 fair coin flips (example 1.3.6), so (S_n) is a martingale and $(S_n^2 - n)$ is a martingale as well (example 10.1.9). Two applications of Doob's inequality bound how far the walk strays.

Because $|S_n|$ is a nonnegative submartingale, the two-sided form gives, for integer $\lambda > 0$,

$$\mathbb{P}\Big(\max_{k \leq n} |S_k| \geq \lambda\Big) \leq \frac{\mathbb{E}|S_n|}{\lambda}$$

A cruder but more useful bound comes from squaring. The process S_n^2 is a nonnegative submartingale (a convex function of the martingale S_n), and $\mathbb{E}[S_n^2] = n$ since $S_n^2 - n$ has mean zero. Applying (10.14) to the submartingale S_n^2 at level λ^2,

$$\mathbb{P}\Big(\max_{k \leq n} |S_k| \geq \lambda\Big) = \mathbb{P}\Big(\max_{k \leq n} S_k^2 \geq \lambda^2\Big) \leq \frac{\mathbb{E}[S_n^2]}{\lambda^2} = \frac{n}{\lambda^2}$$

the running-maximum refinement of Chebyshev's inequality $\mathbb{P}(|S_n| \geq \lambda) \leq n/\lambda^2$. Taking $\lambda = c\sqrt{n}$ shows the walk stays within $c\sqrt{n}$ of the origin over the whole time interval $[0, n]$ with probability at least $1 - 1/c^2$: the entire trajectory, not just the endpoint, is confined to the diffusive scale $\sqrt{n}$. This is the maximal inequality the law of large numbers used in disguise (theorem 4.1.1), now seen as a one-line consequence of the martingale property.

The bound n/λ^2 just obtained is a moment bound on the maximum in disguise: integrating the tail against λ turns a tail estimate into a bound on $\mathbb{E}[(M_n^*)^2]$, hence on the L^2-norm of the maximum. The next result carries this out for every $p > 1$ and produces the sharpest possible constant. It is the single most-used consequence of the maximal inequality: it says that in L^p the running maximum of a martingale is controlled, up to a fixed constant, by the terminal value alone.

Theorem 10.3.3: Doob's L^p Inequality

Let $(M_n)_{n\geq 0}$ be a nonnegative submartingale and let $p > 1$. Writing $M_n^* = \max_{0\leq k\leq n} M_k$ and $\|\cdot\|_p$ for the L^p-norm (definition 3.3.1), one has

$$\|M_n^*\|_p \leq \frac{p}{p-1}\,\|M_n\|_p \tag{10.16}$$

In particular, if (M_n) is a martingale with $M_n \in L^p$, then $\big\|\max_{k\leq n}|M_k|\big\|_p \leq \frac{p}{p-1}\|M_n\|_p$, applying the bound to the nonnegative submartingale $|M_n|$.

***Proof*:**
The proof turns the tail bound (10.14) into a moment bound by integrating over the threshold λ (the layer-cake representation of the p-th moment), then closes the resulting inequality with Hölder.

Step 1: Reduce to a bounded maximum. Assume first that M_n^* is bounded, say $M_n^* \leq C$; the general case follows at the end by truncation. Both sides of (10.16) are then finite, so no term is ∞ during the manipulation.

Step 2: The layer-cake identity. For any nonnegative random variable Y and $p > 0$, the elementary identity

$$Y^p = \int_0^\infty p\,\lambda^{p-1}\,\mathbb{1}_{\{Y\geq\lambda\}}\,d\lambda \tag{10.17}$$

holds pointwise, since the right side is $\int_0^Y p\lambda^{p-1}\,d\lambda = Y^p$. Taking $Y = M_n^*$ and integrating against $\mathbb{P}$, Tonelli's theorem ([2]) exchanges the two integrations:

$$\mathbb{E}[(M_n^*)^p] = \int_0^\infty p\,\lambda^{p-1}\,\mathbb{P}(M_n^* \geq \lambda)\,d\lambda \tag{10.18}$$

Step 3: Insert the maximal inequality. The maximal inequality (10.14) bounds the tail: $\mathbb{P}(M_n^* \geq \lambda) \leq \lambda^{-1}\mathbb{E}[M_n\,\mathbb{1}_{\{M_n^*\geq\lambda\}}]$. Substituting into (10.18) and exchanging integrations once more by Tonelli,

$$\mathbb{E}[(M_n^*)^p] \leq \int_0^\infty p\,\lambda^{p-2}\,\mathbb{E}\big[M_n\,\mathbb{1}_{\{M_n^*\geq\lambda\}}\big]\,d\lambda = \mathbb{E}\Big[M_n\int_0^{M_n^*} p\,\lambda^{p-2}\,d\lambda\Big] = \frac{p}{p-1}\,\mathbb{E}\big[M_n\,(M_n^*)^{p-1}\big]$$

where the inner integral evaluates to $\frac{p}{p-1}(M_n^*)^{p-1}$ (this is where $p > 1$ is used, so that λ^{p-2} is integrable at 0; the indicator turns the upper limit into M_n^*).

Step 4: Close with Hölder. Apply Hölder's inequality (theorem 3.2.10) with conjugate exponents p and $q = p/(p-1)$ to the product $M_n \cdot (M_n^*)^{p-1}$:

$$\mathbb{E}\big[M_n\,(M_n^*)^{p-1}\big] \le \|M_n\|_p \,\big\|(M_n^*)^{p-1}\big\|_q = \|M_n\|_p \left(\mathbb{E}[(M_n^*)^{(p-1)q}]\right)^{1/q} = \|M_n\|_p \left(\mathbb{E}[(M_n^*)^p]\right)^{(p-1)/p}$$

since $(p-1)q = p$ and $1/q = (p-1)/p$. Combining Steps 3 and 4,

$$\mathbb{E}[(M_n^*)^p] \le \frac{p}{p-1}\,\|M_n\|_p \left(\mathbb{E}[(M_n^*)^p]\right)^{(p-1)/p}$$

Because $\mathbb{E}[(M_n^*)^p] \le C^p < \infty$ is finite by Step 1, it may be divided out. If $\mathbb{E}[(M_n^*)^p] > 0$, dividing both sides by $\left(\mathbb{E}[(M_n^*)^p]\right)^{(p-1)/p}$ leaves $\left(\mathbb{E}[(M_n^*)^p]\right)^{1/p} \le \frac{p}{p-1}\|M_n\|_p$, which is (10.16); if $\mathbb{E}[(M_n^*)^p] = 0$ the inequality is trivial.

Step 5: Remove the boundedness assumption. For general M_n^*, apply the established bound to the truncated maximum $M_n^* \wedge C$, which is bounded by C. The tail of $M_n^* \wedge C$ at a level $\lambda \le C$ agrees with that of M_n^* and vanishes above C, so the maximal inequality used in Step 3 still bounds it, $\mathbb{P}(M_n^* \wedge C \ge \lambda) \le \lambda^{-1}\mathbb{E}[M_n\,\mathbb{1}_{\{M_n^* \ge \lambda\}}]$ for $\lambda \le C$; carrying this through Steps 2 through 4 with M_n^* replaced by $M_n^* \wedge C$ on the left and $\|M_n\|_p$ kept on the right gives $\|M_n^* \wedge C\|_p \le \frac{p}{p-1}\|M_n\|_p$ for every C. Letting $C \to \infty$ and applying the monotone convergence theorem to $M_n^* \wedge C \uparrow M_n^*$ yields (10.16) in general, as we sought to show.

The constant $p/(p-1)$ is the conjugate exponent q, and it is sharp: no smaller constant works for all submartingales. As $p \downarrow 1$ the constant blows up, and this is not an artifact of the proof. At $p = 1$ the inequality fails outright: there exist martingales whose maximum is not integrable although the terminal value is, and the correct L^1 statement is weaker, involving an $L \log L$ correction rather than a constant multiple. The most important case is $p = 2$, where the constant is 2: the L^2-norm of the running maximum is at most twice the L^2-norm of the endpoint. This is the form that recovers Kolmogorov's inequality below and that drives the L^2-martingale convergence theory. The structural content of the inequality is that for $p > 1$ the map $M_n \mapsto M_n^*$ is bounded on L^p: passing from a martingale to its running maximum costs only a bounded factor, so any L^p-bound on the terminal values transfers, uniformly in n, to the whole trajectory.

The maximal and L^p inequalities bound the *size* of the trajectory. The upcrossing inequality bounds its *oscillation*: how many times the process can swing from below a level a up past a level b. This is a subtler quantity, and it is the one that governs convergence. A sequence of real numbers converges if and only if, for every rational band $[a, b]$, it crosses that band from bottom to top only finitely often; a sequence that oscillates forever fails to converge precisely because it upcrosses some band infinitely many times. The upcrossing inequality bounds the expected number of upcrossings by the terminal value, and feeding it into the Borel-Cantelli machinery will force almost-sure convergence in the next section. The mechanism is a betting strategy: give the process when it drops below a, hold until it rises above b, then sell, and repeat. Each completed give-low-sell-high cycle books a profit of at least $b - a$; since one cannot profit from a fair game, the number of such cycles is controlled.

Fix a submartingale (M_n) and reals $a < b$. Define the *upcrossing count* $U_n[a, b]$ to be the number of times the trajectory $M_0, \ldots, M_n$ completes a passage from at-or-below a to at-or-above b: formally, with $\sigma_0 = -1$ and, for $j \ge 1$,

$$s_j = \min\{k > \sigma_{j-1} : M_k \le a\}, \qquad t_j = \min\{k > s_j : M_k \ge b\}, \qquad \sigma_j = t_j$$

(each min of an empty set taken as ∞), set $U_n[a, b]$ equal to the number of completed upcrossings

$t_j \le n$. The betting strategy that harvests these upcrossings is the predictable process H that holds one unit exactly during each descent-to-ascent window,

$$H_k = \begin{cases} 1 & \text{if } s_j < k \le t_j \text{ for some } j \\ 0 & \text{otherwise} \end{cases} \tag{10.19}$$

that is, $H_k = 1$ once the process has dipped to a and until it next reaches b, and $H_k = 0$ the rest of the time. The strategy is predictable: whether to hold at step k depends only on the trajectory up to time $k-1$ (the last dip and the intervening history), so H_k is $\mathcal{F}_{k-1}$-measurable, and H takes values in $\{0, 1\}$.

Theorem 10.3.4: Doob's Upcrossing Inequality

Let $(M_n)_{n\ge 0}$ be a submartingale and $a < b$ reals. The number $U_n[a, b]$ of upcrossings of the band $[a, b]$ completed by time n satisfies

$$(b-a)\,\mathbb{E}\big[U_n[a,b]\big] \le \mathbb{E}\big[(M_n - a)^+\big] - \mathbb{E}\big[(M_0 - a)^+\big] \tag{10.20}$$

***Proof*:**
The argument compares the gambler's fortune under the betting strategy H against the raw process. It is cleanest for a nonnegative submartingale starting at 0, so the first step reduces to that case.

Step 1: Reduce to $Y_k = (M_k - a)^+$. The function $\varphi(x) = (x-a)^+$ is convex and nondecreasing, so $Y_k = (M_k - a)^+ = \varphi(M_k)$ is again a submartingale (proposition 10.1.4), and it is nonnegative. Crucially, Y upcrosses the band $[0, b-a]$ exactly when M upcrosses $[a, b]$: $M_k \le a \iff Y_k = 0 \le 0$ and $M_k \ge b \iff Y_k \ge b - a$, so the upcrossing counts coincide, $U_n^Y[0, b-a] = U_n[a, b]$. It therefore suffices to prove, for the nonnegative submartingale Y,

$$(b-a)\,\mathbb{E}\big[U_n^Y[0, b-a]\big] \le \mathbb{E}[Y_n] - \mathbb{E}[Y_0] \tag{10.21}$$

which is (10.20) rewritten through $Y_n = (M_n - a)^+$ and $Y_0 = (M_0 - a)^+$.

Step 2: The gain of the betting strategy. Let H be the predictable $\{0, 1\}$-strategy of (10.19), defined now with Y in place of M and the band $[0, b-a]$, and consider the martingale transform (definition 10.1.7)

$$(H \cdot Y)_n = \sum_{k=1}^{n} H_k\,(Y_k - Y_{k-1})$$

the accumulated winnings of holding one unit of Y exactly through each descent-to- ascent window. Each completed upcrossing contributes a gain of at least $b - a$: over a window from s_j (where $Y_{s_j} = 0$) to t_j (where $Y_{t_j} \ge b-a$) the increments telescope to $Y_{t_j} - Y_{s_j} \ge (b-a) - 0 = b-a$. There are $U_n^Y[0, b-a]$ completed upcrossings by time n. A possible final incomplete window, if the process has begun a descent but not yet completed the ascent by time n, contributes a gain of $Y_n - Y_{s_j} = Y_n \ge 0$ (again using $Y_{s_j} = 0$ at the start of a window and $Y_n \ge 0$). Adding the completed and incomplete contributions,

$$(H \cdot Y)_n \ge (b-a)\,U_n^Y[0, b-a] \tag{10.22}$$

the winnings are at least $b - a$ per completed upcrossing.

Step 3: The complementary strategy and its cost. Let $G_k = 1 - H_k$, the predictable $\{0,1\}$-strategy that holds exactly when H does not. Since $H + G \equiv 1$, the two transforms reconstruct the full increment,

$$(H \cdot Y)_n + (G \cdot Y)_n = \sum_{k=1}^{n} (H_k + G_k)(Y_k - Y_{k-1}) = \sum_{k=1}^{n} (Y_k - Y_{k-1}) = Y_n - Y_0$$

Now G is a bounded nonnegative predictable process and Y is a submartingale, so the transform $(G \cdot Y)$ is again a submartingale with $(G \cdot Y)_0 = 0$ (proposition 10.1.8). A submartingale starting at 0 has nonnegative expectation, $\mathbb{E}[(G \cdot Y)_n] \geq (G \cdot Y)_0 = 0$. Therefore

$$\mathbb{E}[(H \cdot Y)_n] = \mathbb{E}[Y_n - Y_0] - \mathbb{E}[(G \cdot Y)_n] \leq \mathbb{E}[Y_n] - \mathbb{E}[Y_0]$$

the winnings from the harvesting strategy cannot exceed the total drift of the submartingale.

Step 4: Combine. Taking expectations in (10.22) and inserting the bound of Step 3,

$$(b - a)\, \mathbb{E}\big[U_n^Y[0, b-a]\big] \leq \mathbb{E}[(H \cdot Y)_n] \leq \mathbb{E}[Y_n] - \mathbb{E}[Y_0]$$

which is (10.21). Undoing the reduction of Step 1 through $U_n^Y[0, b-a] = U_n[a,b]$, $Y_n = (M_n - a)^+$, and $Y_0 = (M_0 - a)^+$ gives (10.20), completing the proof.

The inequality quantifies the intuition that a fair (or favorable) game cannot be harvested for oscillation profit. The betting strategy H is the mathematical form of "give low, sell high"; the complementary strategy G absorbs the rest of the drift. Because $G \cdot Y$ is a submartingale, its expected gain is nonnegative, and the harvesting strategy H is left with at most the total drift $\mathbb{E}[Y_n] - \mathbb{E}[Y_0]$. Each completed upcrossing costs the process $b - a$ of that budget, so the expected number of upcrossings is bounded by drift over band-width. The reason this is the key to convergence is visible in the bound: for a martingale, or more generally an L^1-bounded submartingale, the right side stays bounded as $n \to \infty$ (it is at most $\mathbb{E}[(M_n - a)^+] \leq \mathbb{E}|M_n| + |a|$), so the expected total number of upcrossings of every fixed band is finite. A trajectory with finitely many upcrossings of every rational band cannot oscillate, and so must converge. That deduction is carried out in section 10.4; here the point is that the number of oscillations is controlled by the terminal value alone, with no independence hypothesis.

Example 10.3.5: Upcrossings of the Symmetric Random Walk

For the ± 1 walk S_n (example 1.3.6), take the band $[a, b] = [-1, 1]$. Since S_n is a martingale, $\mathbb{E}[(S_n - a)^+] = \mathbb{E}[(S_n + 1)^+]$, and the upcrossing inequality gives

$$2\, \mathbb{E}\big[U_n[-1,1]\big] = (b - a)\, \mathbb{E}[U_n[-1,1]] \leq \mathbb{E}[(S_n + 1)^+] - \mathbb{E}[(S_0 + 1)^+]$$

With $S_0 = 0$, one has $(S_0 + 1)^+ = 1$, and $\mathbb{E}[(S_n + 1)^+] \leq \mathbb{E}|S_n| + 1 \leq \sqrt{\mathbb{E}[S_n^2]} + 1 = \sqrt{n} + 1$ by the Cauchy-Schwarz bound $\mathbb{E}|S_n| \leq \|S_n\|_2$. Hence

$$\mathbb{E}\big[U_n[-1,1]\big] \leq \frac{\sqrt{n}}{2}$$

The expected number of upcrossings of a fixed band grows like $\sqrt{n}$: the walk oscillates, and the count of oscillations diverges, which is the analytic signature of the fact that the symmetric random walk does *not* converge. The upcrossing inequality does not force convergence here because the right side is unbounded in n; a martingale converges precisely when this fails to happen, that is, when the terminal expectations stay bounded, as the next section makes precise.

The three inequalities of this section are the analytic core of martingale theory, and their most direct dividend is that results proved earlier by hand for sums of independent variables now appear as special cases, valid because those sums are martingales rather than because of any independence. Kolmogorov's maximal inequality is the leading example.

Example 10.3.6: Recovering Kolmogorov's Maximal Inequality
Let $X_1, X_2, \ldots$ be independent, mean-zero, square-integrable random variables, and let $S_n = X_1 + \cdots + X_n$ with $S_0 = 0$. Independence and $\mathbb{E}[X_i] = 0$ make (S_n) a martingale relative to its natural filtration (proposition 10.1.4), and hence, by convexity of $x \mapsto x^2$, the squared process (S_n^2) is a nonnegative submartingale. Two of the inequalities above specialize to it.

1. *Kolmogorov's maximal inequality as the L^2 maximal bound.* Apply Doob's maximal inequality (10.14) to the nonnegative submartingale S_n^2 at level λ^2. Since $\{\max_{k \le n} |S_k| \ge \lambda\} = \{\max_{k \le n} S_k^2 \ge \lambda^2\}$ and $\mathbb{E}[S_n^2] = \sum_{k \le n} \operatorname{Var}(X_k)$ (cross-terms vanish by independence and mean-zero), the bound reads
$$\mathbb{P}\Big(\max_{k \le n} |S_k| \ge \lambda\Big) \le \frac{\mathbb{E}[S_n^2]}{\lambda^2} = \frac{1}{\lambda^2} \sum_{k=1}^{n} \operatorname{Var}(X_k)$$
which is exactly Kolmogorov's maximal inequality (theorem 4.1.1). That inequality, proved directly for sums of independent variables by an ad hoc first-entrance decomposition, is thus the L^2 case of Doob's maximal inequality: the ad hoc stopping time there is the first-passage time of the present proof, and the independence used there is only needed to identify $\mathbb{E}[S_n^2]$ with the variance sum. The inequality holds verbatim for any L^2-martingale, independent increments or not.

2. *The L^p bound on the maximal partial sum.* When the X_i lie in L^p for some $p > 1$, so does each S_n, and Doob's L^p inequality (10.16) applied to the martingale S_n gives
$$\Big\| \max_{k \le n} |S_k| \Big\|_p \le \frac{p}{p-1} \, \|S_n\|_p$$
The maximal partial sum has the same L^p-order as the final partial sum, up to the constant $p/(p-1)$. For $p = 2$ this is $\big\|\max_{k \le n} |S_k|\big\|_2 \le 2\|S_n\|_2 = 2\big(\sum_{k \le n} \operatorname{Var}(X_k)\big)^{1/2}$, the L^2 companion to part (1), controlling the maximum in norm rather than only its tail. These bounds are what the strong law of large numbers proved via maximal inequalities uses.

Seeing Kolmogorov's inequality drop out as the L^2 case exposes what the martingale formulation gives. The classical proof isolates a first-entrance event and exploits the independence of the increments after that event to discard a nonnegative remainder. The martingale proof replaces "independence of future increments" with the strictly weaker "the future does not decrease conditional expectation", which is all the first-passage argument ever used. Independence is not the source of the maximal inequality; the submartingale property is, and independence enters only to compute the particular bound $\sum \operatorname{Var}(X_k)$. This is the recurring lesson of the chapter: theorems that look like facts about independent sums are, at their core, facts about martingales, and hold in the far larger world where only the conditional mean is constrained.

Exercise 10.3.1

1. Let (M_n) be a supermartingale with $M_0 = x$, and let $\lambda > 0$. By applying Doob's maximal inequality (10.14) to the submartingale $(-M_n)$, or by adapting the first-passage argument directly, prove the minimum-tail bound

$$\lambda\, \mathbb{P}\Big(\min_{k\le n} M_k \le -\lambda\Big) \le \mathbb{E}[M_n^-]$$

where $M_n^- = \max(-M_n, 0)$. State the specialization to a nonnegative supermartingale with $M_0 = x$, giving the maximal-tail bound $\mathbb{P}(\max_{k\le n} M_k \ge \lambda) \le x/\lambda$ from the first-passage argument.

2. For the ± 1 walk S_n, use Doob's L^2 inequality to show $\mathbb{E}\big[\max_{k\le n} S_k^2\big] \le 4n$. Compare this with the exact endpoint value $\mathbb{E}[S_n^2] = n$ and interpret the factor 4: what does it say about the ratio of the expected squared maximum to the expected squared endpoint?

3. Let $\theta > 0$ and consider the exponential martingale $M_n = e^{\theta S_n}/(\cosh\theta)^n$ of example 10.1.9, a nonnegative martingale with $\mathbb{E}[M_n] = 1$. Apply Doob's maximal inequality to obtain, for $c > 0$,

$$\mathbb{P}\Big(\max_{k\le n}\big(S_k - k\tfrac{\log\cosh\theta}{\theta}\big) \ge c\Big) \le e^{-\theta c}$$

(Hint: $\{\max_k M_k \ge \lambda\}$ with $\lambda = e^{\theta c}$, then take logarithms inside the maximum.) This is a maximal version of the Chernoff bound of proposition 3.4.7.

4. Show that the layer-cake step degenerates at $p = 1$: the inner integral $\int_0^{M_n^*} \lambda^{-1}\, d\lambda$ in Step 3 of the proof of theorem 10.3.3 diverges. Explain in one or two sentences why this is the correct analytic reason that Doob's L^p inequality has no $p = 1$ analogue with a constant multiple, and name what replaces $\|M_n\|_1$ on the right-hand side (an $L\log L$ quantity).

5. Let (M_n) be a martingale bounded by $|M_n| \le B$ for all n. Show that for every band $[a, b] \subseteq [-B, B]$,

$$\mathbb{E}\big[U_n[a, b]\big] \le \frac{2B + |a|}{b - a}$$

uniformly in n, and deduce that $\sup_n \mathbb{E}[U_n[a, b]] < \infty$. (This uniform bound is the hypothesis under which the convergence theorem of the next section applies.)

6. Define the downcrossing count $D_n[a, b]$ as the number of completed passages from at-or-above b down to at-or-below a by time n. Argue that $|U_n[a, b] - D_n[a, b]| \le 1$ for every trajectory (each upcrossing is followed by at most one downcrossing before the next upcrossing, and conversely), and conclude that the same terminal-value bound controls $\mathbb{E}[D_n[a, b]]$ up to an additive 1. Why is the upcrossing count, rather than the downcrossing count, the natural one for the submartingale inequality?

10.4 Martingale Convergence

A martingale is a fair game, and the intuition that a fair game has no long-run drift suggests that if it also stays bounded in size then it ought to settle down. This is the content of the martingale convergence theorem, and it is the reason martingales are the natural tool for proving that a randomly evolving quantity has a limit. The defining equation $\mathbb{E}[M_{n+1} | \mathcal{F}_n] = M_n$ is a statement about one-step averages; it says nothing on its face about the pathwise behaviour of $n \mapsto M_n(\omega)$ for a fixed outcome

ω. What the convergence theorem extracts from it is a pathwise conclusion: for almost every ω the numerical sequence $M_n(\omega)$ converges to a finite limit. No integrability beyond a single uniform bound, no smoothness, no monotonicity is asked of the paths; the fairness constraint does all the work.

The mechanism that converts the one-step average constraint into a statement about paths is Doob's upcrossing inequality (theorem 10.3.4). A real sequence fails to converge in $[-\infty, \infty]$ exactly when it oscillates: there is some pair of rationals $a < b$ that the sequence crosses upward from below a to above b infinitely often. The upcrossing inequality bounds the expected number of such crossings for a submartingale, and an L^1 bound makes that expected number finite, hence the number of crossings is finite almost surely. Ruling this out for every rational interval at once rules out oscillation, and a bounded non-oscillating sequence converges. The remaining subtlety, that the limit be finite rather than $\pm\infty$, is settled by Fatou's lemma. This section proves the theorem in this form, isolates uniform integrability as the exact hypothesis that upgrades almost-sure convergence to convergence in L^1, and records the Doob decomposition, which splits any adapted integrable process into a martingale and a predictable drift.

Theorem 10.4.1: Martingale Convergence Theorem

Let $(M_n)_{n\geq 0}$ be a submartingale relative to $(\mathcal{F}_n)_{n\geq 0}$ that is bounded in L^1, meaning

$$\sup_{n\geq 0} \mathbb{E}|M_n| < \infty \tag{10.23}$$

Then there is an integrable random variable M_∞ such that $M_n \to M_\infty$ almost surely, and $\mathbb{E}|M_\infty| \leq \sup_n \mathbb{E}|M_n|$.

Two remarks fix the scope before the proof. First, the hypothesis is stated for submartingales, which is the natural generality: the upcrossing inequality is a submartingale statement, and a martingale is a submartingale (indeed both a sub- and a supermartingale), so the theorem applies to martingales as the special case. Second, for a martingale the L^1 bound (10.23) is often automatic in disguise. A nonnegative martingale satisfies $\mathbb{E}|M_n| = \mathbb{E}[M_n] = \mathbb{E}[M_0]$ for every n, a constant, so $\sup_n \mathbb{E}|M_n| = \mathbb{E}[M_0] < \infty$ with no further hypothesis; the same holds for any martingale bounded below by a constant. This is why the theorem so often applies without any verification beyond nonnegativity, as in the Galton-Watson process of the next section.

***Proof*:**
The argument has three steps: the upcrossing inequality forces finitely many upcrossings of each rational interval, this rules out oscillation and produces an almost-sure limit in $[-\infty, \infty]$, and Fatou's lemma shows the limit is finite and integrable.

Step 1: Finitely many upcrossings of each rational interval. Fix rationals $a < b$ and let $U_n[a, b]$ be the number of upcrossings of $[a, b]$ completed by the submartingale (M_k) by time n, as in theorem 10.3.4. That inequality states

$$(b - a)\,\mathbb{E}[U_n[a, b]] \leq \mathbb{E}[(M_n - a)^+] - \mathbb{E}[(M_0 - a)^+] \leq \mathbb{E}[(M_n - a)^+]$$

Since $(M_n - a)^+ \leq |M_n| + |a|$, the right side is at most $\sup_n \mathbb{E}|M_n| + |a|$, a bound independent of n. Let $U_\infty[a, b] = \lim_{n\to\infty} U_n[a, b]$ be the total number of upcrossings over the whole time axis;

the limit exists because $U_n[a,b]$ is nondecreasing in n. By the monotone convergence theorem,

$$(b-a)\,\mathbb{E}[U_\infty[a,b]] = \lim_{n\to\infty}(b-a)\,\mathbb{E}[U_n[a,b]] \le \sup_n \mathbb{E}|M_n| + |a| < \infty \tag{10.24}$$

A nonnegative random variable with finite expectation is finite almost surely, so $U_\infty[a,b] < \infty$ a.s. Thus, for each fixed pair of rationals $a<b$, the path $k \mapsto M_k(\omega)$ upcrosses $[a,b]$ only finitely often, outside a null set $N_{a,b}$.

Step 2: Almost-sure convergence in the extended reals. Let $N = \bigcup_{a<b \text{ rational}} N_{a,b}$ be the union over all pairs of rationals; as a countable union of null sets, $\mathbb{P}(N) = 0$. Fix $\omega \notin N$ and suppose, for contradiction, that the numerical sequence $M_n(\omega)$ does not converge in $[-\infty,\infty]$. Then its lower and upper limits differ,

$$\liminf_{n\to\infty} M_n(\omega) < \limsup_{n\to\infty} M_n(\omega)$$

and by density there are rationals $a<b$ strictly between them, $\liminf_n M_n(\omega) < a < b < \limsup_n M_n(\omega)$. Then $M_n(\omega) < a$ infinitely often and $M_n(\omega) > b$ infinitely often, so the path crosses from below a to above b infinitely many times, giving $U_\infty[a,b](\omega) = \infty$. This contradicts $\omega \notin N_{a,b}$. Hence for every $\omega \notin N$ the limit

$$M_\infty(\omega) = \lim_{n\to\infty} M_n(\omega)$$

exists in $[-\infty,\infty]$; setting $M_\infty = 0$ on the null set N gives a measurable function to which $M_n \to M_\infty$ almost surely.

Step 3: The limit is integrable. It remains to rule out the values $\pm\infty$ and to bound $\mathbb{E}|M_\infty|$. Since $|M_n| \to |M_\infty|$ almost surely and each $|M_n| \ge 0$, Fatou's lemma gives

$$\mathbb{E}|M_\infty| = \mathbb{E}\big[\liminf_{n\to\infty}|M_n|\big] \le \liminf_{n\to\infty}\mathbb{E}|M_n| \le \sup_n \mathbb{E}|M_n| < \infty$$

Thus $\mathbb{E}|M_\infty| < \infty$, so M_∞ is integrable and in particular finite almost surely. The displayed inequality is the asserted bound $\mathbb{E}|M_\infty| \le \sup_n \mathbb{E}|M_n|$, completing the proof.

The proof isolates precisely what the L^1 bound gives and what it does not. What it gives is the finiteness of the expected upcrossing count (10.24), and through it the almost-sure existence of a finite limit. What it does not give is any control over how M_n approaches M_∞ in the mean: the limit is pathwise, and $\mathbb{E}|M_n - M_\infty|$ need not go to zero. The gap between almost-sure convergence and L^1 convergence is the subject of the next result, and it has a clean cause. Fatou's lemma in Step 3 gives only an inequality $\mathbb{E}|M_\infty| \le \liminf_n \mathbb{E}|M_n|$, and this inequality can be strict: mass can escape to infinity along the sequence and be lost in the limit. The example to keep in mind is a martingale whose values grow in magnitude on a shrinking set, so that $\mathbb{E}|M_n|$ stays bounded while $M_n \to 0$ pointwise; then $\mathbb{E}|M_n| \not\to \mathbb{E}|M_\infty| = 0$, and M_n cannot converge in L^1. This is realized concretely in example 10.4.72 below.

A first application interprets the theorem for the running example. The symmetric ± 1 random walk S_n (example 10.1.9) is a martingale, but it is not L^1-bounded: $\mathbb{E}|S_n|$ grows like $\sqrt{n}$, so the theorem does not apply, and indeed S_n oscillates without limit, visiting every integer infinitely often. The theorem is not violated; its hypothesis simply fails, and the failure is exactly the unbounded growth of $\mathbb{E}|S_n|$. The lesson is that the L^1 bound is not a technical convenience but the precise dividing line between martingales that converge and martingales that wander.

Definition 10.4.2: Uniform Integrability

A family $\{X_i : i \in I\}$ of integrable random variables is *uniformly integrable* if

$$\lim_{K\to\infty} \sup_{i\in I} \mathbb{E}\big[|X_i|\,\mathbb{1}_{\{|X_i|>K\}}\big] = 0$$

that is, the contribution to $\mathbb{E}|X_i|$ from the region where $|X_i|$ exceeds K can be made small uniformly in i by taking K large.

Uniform integrability is the exact condition that closes the gap left by the convergence theorem: it forbids the escape of mass to infinity. The tail contribution $\mathbb{E}[|X_i|\mathbb{1}_{\{|X_i|>K\}}]$ is the part of the mean carried by large values, and demanding it be uniformly small prevents any single X_i from hiding a fixed amount of mass out at large magnitudes. Three facts about the definition are used repeatedly and are worth recording. A uniformly integrable family is bounded in L^1: taking K with the supremum below 1 gives $\mathbb{E}|X_i| \le K+1$ for all i. A single integrable random variable, or any finite family, is uniformly integrable, by dominated convergence applied to $|X|\mathbb{1}_{\{|X|>K\}} \to 0$. And a family dominated by a fixed integrable Y, meaning $|X_i| \le Y$ for all i, is uniformly integrable, since $\mathbb{E}[|X_i|\mathbb{1}_{\{|X_i|>K\}}] \le \mathbb{E}[Y\mathbb{1}_{\{Y>K\}}] \to 0$. The next example gives the criterion most often used in practice and a family that is L^1-bounded yet not uniformly integrable, marking the boundary.

Example 10.4.3: An L^p Bound Forces Uniform Integrability; an L^1 Bound Does Not

Two computations delimit the notion.

1. *An L^p bound suffices, $p > 1$.* Suppose $\sup_i \mathbb{E}[|X_i|^p] = C < \infty$ for some $p > 1$. On the event $\{|X_i| > K\}$ one has $|X_i| = |X_i|^p\,|X_i|^{1-p} \le |X_i|^p K^{1-p}$, so

$$\mathbb{E}\big[|X_i|\,\mathbb{1}_{\{|X_i|>K\}}\big] \le K^{1-p}\,\mathbb{E}[|X_i|^p] \le C\,K^{1-p}$$

   Since $p > 1$, the exponent $1-p$ is negative and $C\,K^{1-p} \to 0$ as $K \to \infty$, uniformly in i. Thus any family bounded in L^p for some $p > 1$ is uniformly integrable. A bound in L^1 alone is not enough, as the next item shows: the gain comes entirely from the extra power.

2. *An L^1-bounded family that is not uniformly integrable.* On the unit interval $([0,1], \text{Borel}, \text{Lebesgue})$ let $X_n = n\,\mathbb{1}_{(0,1/n)}$. Then $\mathbb{E}[X_n] = n \cdot \frac{1}{n} = 1$ for every n, so the family is bounded in L^1 with $\sup_n \mathbb{E}|X_n| = 1$. But for any fixed K and any $n > K$ the whole mass sits above K, since $X_n = n > K$ on $(0, 1/n)$, giving

$$\mathbb{E}\big[|X_n|\,\mathbb{1}_{\{|X_n|>K\}}\big] = \mathbb{E}[X_n] = 1$$

   so the supremum over n never drops below 1 and the family is not uniformly integrable. Here $X_n \to 0$ pointwise (indeed everywhere except at 0) while $\mathbb{E}[X_n] = 1 \not\to 0$: the mass escapes upward on the shrinking interval $(0, 1/n)$, the exact failure the definition forbids.

The counterexample in the second item is the template for every L^1-bounded martingale that fails to converge in L^1, and it is worth seeing why uniform integrability is tailored to close the gap. The bridge between almost-sure and L^1 convergence is the following standard fact from the prerequisite volume, which will be used in the proof of the next theorem: if $X_n \to X$ almost surely (or even in probability) and the family $\{X_n\}$ is uniformly integrable, then X is integrable and $X_n \to X$ in L^1, meaning $\mathbb{E}|X_n - X| \to 0$; conversely L^1 convergence implies uniform integrability of the sequence. This is the Vitali convergence theorem [2]. For a martingale, uniform integrability turns out to be equivalent to a structural property, that the martingale is the sequence of conditional expectations of a single terminal variable, and the next result makes this equivalence precise.

Theorem 10.4.4: Convergence of Uniformly Integrable Martingales

Let $(M_n)_{n\geq 0}$ be a martingale relative to $(\mathcal{F}_n)_{n\geq 0}$. The following are equivalent.

1. (M_n) is uniformly integrable.
2. (M_n) converges almost surely and in L^1 to a limit M_∞.
3. There is an integrable random variable Z with $M_n = \mathbb{E}[Z \big| \mathcal{F}_n]$ a.s. for every n; that is, (M_n) is *closed*.

When these hold, one may take $Z = M_\infty$, so $M_n = \mathbb{E}[M_\infty \big| \mathcal{F}_n]$ a.s. Conversely, for any integrable Z the sequence $M_n = \mathbb{E}[Z \big| \mathcal{F}_n]$ is a uniformly integrable martingale.

The word *closed* names the structural content: a uniformly integrable martingale is not an open-ended sequence but the record of a fixed random variable M_∞ seen through a growing sequence of σ-algebras, with M_n the best estimate of M_∞ available from the information $\mathcal{F}_n$. This is the martingale most often met in applications, because it arises whenever a fixed unknown Z is estimated as information accrues. The final sentence of the theorem, that every conditional-expectation sequence is a uniformly integrable martingale, is the converse that makes the class easy to recognize.

***Proof*:**
The last sentence is proved first, then the cycle (3)⇒(2)⇒(1)⇒(3).

The converse: $\mathbb{E}[Z \big| \mathcal{F}_n]$ is a uniformly integrable martingale. Let Z be integrable and set $M_n = \mathbb{E}[Z \big| \mathcal{F}_n]$. Each M_n is $\mathcal{F}_n$-measurable and integrable, and the tower property (proposition 9.2.2) with $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ gives

$$\mathbb{E}[M_{n+1} \big| \mathcal{F}_n] = \mathbb{E}\big[\mathbb{E}[Z \big| \mathcal{F}_{n+1}] \big| \mathcal{F}_n\big] = \mathbb{E}[Z \big| \mathcal{F}_n] = M_n$$

almost surely, so (M_n) is a martingale. For uniform integrability, fix $\varepsilon > 0$. Since Z is integrable it is uniformly integrable as a one-element family, so there is $\delta > 0$ with $\mathbb{E}[|Z| \mathbb{1}_A] < \varepsilon$ whenever $\mathbb{P}(A) < \delta$; this is absolute continuity of the integral of $|Z|$. Conditional Jensen (proposition 9.2.4) applied to the convex function $|\cdot|$ gives $|M_n| = |\mathbb{E}[Z \big| \mathcal{F}_n]| \leq \mathbb{E}[|Z| \big| \mathcal{F}_n]$ a.s. The event $A_n = \{|M_n| > K\}$ lies in $\mathcal{F}_n$, so by the defining property of conditional expectation,

$$\mathbb{E}\big[|M_n| \, \mathbb{1}_{A_n}\big] \leq \mathbb{E}\big[\mathbb{E}[|Z| \big| \mathcal{F}_n] \, \mathbb{1}_{A_n}\big] = \mathbb{E}[|Z| \, \mathbb{1}_{A_n}]$$

Markov's inequality (theorem 3.2.4) and $\mathbb{E}|M_n| \leq \mathbb{E}|Z|$ bound $\mathbb{P}(A_n) \leq \mathbb{E}|M_n|/K \leq \mathbb{E}|Z|/K$, which is below δ once $K > \mathbb{E}|Z|/\delta$, uniformly in n. For such K, $\mathbb{E}[|M_n| \mathbb{1}_{A_n}] \leq \mathbb{E}[|Z| \mathbb{1}_{A_n}] < \varepsilon$ for all n, so (M_n) is uniformly integrable.

(3)⇒(2). A closed martingale is uniformly integrable by the converse just proved, hence bounded in L^1. The martingale convergence theorem (theorem 10.4.1) gives an integrable M_∞ with $M_n \to M_\infty$ almost surely. Uniform integrability upgrades this to L^1 convergence by the Vitali convergence theorem [2]: $\mathbb{E}|M_n - M_\infty| \to 0$.

(2)⇒(1). If $M_n \to M_\infty$ in L^1, then $\{M_n\}$ is uniformly integrable: L^1 convergence implies uniform integrability of the sequence, again by the Vitali convergence theorem [2]. (Directly: $\mathbb{E}|M_n| \to \mathbb{E}|M_\infty|$ so the sequence is L^1-bounded, and the tails are controlled by combining the single-variable uniform integrability of M_∞ with the smallness of $\mathbb{E}|M_n - M_\infty|$.)

(1)⇒(3). Suppose (M_n) is uniformly integrable. Uniform integrability gives L^1 boundedness, hence an almost-sure and (by Vitali) L^1 limit M_∞. It remains to show $M_n = \mathbb{E}[M_\infty \big| \mathcal{F}_n]$ a.s., which realizes (3) with $Z = M_\infty$. Fix n and an event $A \in \mathcal{F}_n$. For every $m \ge n$ the martingale property and the tower property give $\mathbb{E}[M_m \mathbb{1}_A] = \mathbb{E}[M_n \mathbb{1}_A]$, because $M_n = \mathbb{E}[M_m \big| \mathcal{F}_n]$ and $A \in \mathcal{F}_n$. Letting $m \to \infty$, L^1 convergence lets one pass to the limit under the expectation,

$$\mathbb{E}[M_\infty \mathbb{1}_A] = \lim_{m\to\infty} \mathbb{E}[M_m \mathbb{1}_A] = \mathbb{E}[M_n \mathbb{1}_A]$$

since $|\mathbb{E}[M_m \mathbb{1}_A] - \mathbb{E}[M_\infty \mathbb{1}_A]| \le \mathbb{E}|M_m - M_\infty| \to 0$. Thus M_n and M_∞ have the same integral over every $A \in \mathcal{F}_n$, and M_n is $\mathcal{F}_n$-measurable, so M_n satisfies the defining identity for $\mathbb{E}[M_\infty \big| \mathcal{F}_n]$ (definition 9.1.1). Therefore $M_n = \mathbb{E}[M_\infty \big| \mathcal{F}_n]$ a.s. for every n, establishing (3) and completing the cycle.

The equivalence is the organizing structure of martingale convergence, and it deserves to be read as a dictionary. Uniform integrability is an analytic condition on tails; closedness is a structural condition, that the martingale springs from a single terminal variable; L^1 convergence is the mode-of-convergence conclusion. The theorem says these are three faces of one phenomenon. The practical value is that closedness is often visible by inspection, most of all in the sequence $\mathbb{E}[Z \big| \mathcal{F}_n]$ of successive estimates of a fixed quantity, and closedness then delivers both convergence and the identification of the limit as $M_\infty = Z$ whenever Z is $\mathcal{F}_\infty$-measurable. This last identification, that the limit of $\mathbb{E}[Z \big| \mathcal{F}_n]$ is $\mathbb{E}[Z \big| \mathcal{F}_\infty]$ rather than Z itself, is the content of Lévy's upward theorem, taken up in the next section.

The convergence theorem describes what a submartingale does asymptotically; the Doob decomposition describes its structure at every finite time. It expresses any adapted integrable process as a martingale plus a predictable drift, separating the fair-game part from the systematic tendency to rise or fall. For a submartingale the drift is increasing, and this increasing predictable part is the discrete precursor of the quadratic variation of continuous-time theory.

Definition 10.4.5: Doob Decomposition

Let $(X_n)_{n\ge0}$ be an adapted integrable process relative to $(\mathcal{F}_n)_{n\ge0}$. A *Doob decomposition* of (X_n) is a representation

$$X_n = X_0 + D_n + A_n, \qquad n \ge 0$$

in which (D_n) is a martingale with $D_0 = 0$ and (A_n) is a *predictable* process with $A_0 = 0$, meaning A_n is $\mathcal{F}_{n-1}$-measurable for each $n \ge 1$.

The word *predictable* carries the content: A_n is known one step in advance, at time $n-1$, so it represents the part of the increment $X_n - X_{n-1}$ that is forecastable from the past rather than random. The martingale D_n collects the unforecastable, fair-game part. The next proposition shows this split exists and is forced, and reads off when the drift is monotone.

Proposition 10.4.6: Existence and Uniqueness of the Doob Decomposition

Every adapted integrable process $(X_n)_{n\geq 0}$ has a Doob decomposition $X_n = X_0 + D_n + A_n$, and it is unique. The predictable part is given by

$$A_n = \sum_{k=1}^{n} \Big(\mathbb{E}[X_k \big| \mathcal{F}_{k-1}] - X_{k-1} \Big) \tag{10.25}$$

Moreover (A_n) is almost surely increasing if and only if (X_n) is a submartingale, and almost surely decreasing if and only if (X_n) is a supermartingale.

***Proof*:**
Existence and uniqueness are proved by extracting the only possible formula for A_n from the predictability and martingale requirements, then verifying it works.

Step 1: Uniqueness forces the formula. Suppose $X_n = X_0 + D_n + A_n$ is a Doob decomposition. Taking one-step conditional expectations of the increment $X_n - X_{n-1} = (D_n - D_{n-1}) + (A_n - A_{n-1})$ given $\mathcal{F}_{n-1}$, and using that D is a martingale (so $\mathbb{E}[D_n - D_{n-1} \big| \mathcal{F}_{n-1}] = 0$ a.s.) while A_n and A_{n-1} are both $\mathcal{F}_{n-1}$-measurable (so the increment $A_n - A_{n-1}$ is $\mathcal{F}_{n-1}$-measurable and comes out of the conditional expectation),

$$\mathbb{E}[X_n \big| \mathcal{F}_{n-1}] - X_{n-1} = \mathbb{E}[X_n - X_{n-1} \big| \mathcal{F}_{n-1}] = A_n - A_{n-1}$$

almost surely, where X_{n-1} is pulled out by proposition 9.2.3. Summing this identity from $k = 1$ to n and using $A_0 = 0$ telescopes to (10.25). Thus A_n is determined by (X_n), and then $D_n = X_n - X_0 - A_n$ is determined as well: the decomposition, if it exists, is unique.

Step 2: The formula gives a decomposition. Define A_n by (10.25) and $D_n = X_n - X_0 - A_n$. Then $A_0 = 0$ and $D_0 = 0$. Each summand $\mathbb{E}[X_k \big| \mathcal{F}_{k-1}] - X_{k-1}$ is $\mathcal{F}_{k-1}$-measurable, so the partial sum A_n is $\mathcal{F}_{n-1}$-measurable, that is (A_n) is predictable. For the martingale property of D, compute the conditional increment: using $A_n - A_{n-1} = \mathbb{E}[X_n \big| \mathcal{F}_{n-1}] - X_{n-1}$ from (10.25) and the $\mathcal{F}_{n-1}$-measurability of $A_n - A_{n-1}$,

$$\mathbb{E}[D_n - D_{n-1} \big| \mathcal{F}_{n-1}] = \mathbb{E}[X_n - X_{n-1} \big| \mathcal{F}_{n-1}] - (A_n - A_{n-1}) = 0$$

almost surely, so $\mathbb{E}[D_n \big| \mathcal{F}_{n-1}] = D_{n-1}$ and (D_n) is a martingale. Integrability of D_n and A_n follows from that of the X_k and their conditional expectations. Hence $X_n = X_0 + D_n + A_n$ is a Doob decomposition.

Step 3: Monotonicity of A matches the sub/supermartingale property. Each increment satisfies $A_n - A_{n-1} = \mathbb{E}[X_n \big| \mathcal{F}_{n-1}] - X_{n-1}$ almost surely. This is ≥ 0 a.s. for every n exactly when $\mathbb{E}[X_n \big| \mathcal{F}_{n-1}] \geq X_{n-1}$ a.s. for every n, which is the defining inequality of a submartingale (definition 10.1.3); so (A_n) is increasing if and only if (X_n) is a submartingale. The supermartingale case reverses the inequality identically, giving (A_n) decreasing, as we sought to show.

The decomposition is the discrete case of a construction that dominates continuous-time probability. There, a submartingale splits as a martingale plus an increasing predictable process by the Doob-Meyer theorem, and the increasing part of M_n^2 for a martingale M is its predictable quadratic variation, the discrete analogue of the object $\langle M \rangle_t$ that makes the Itô integral an isometry. Read against the running example, the decomposition recovers a computation from the first section: for the random walk martingale S_n, the process S_n^2 is a submartingale, and its Doob decomposition is $S_n^2 = n + D_n$, where the predictable increasing part is $A_n = n$ (each increment $\mathbb{E}[S_k^2 \big| \mathcal{F}_{k-1}] - S_{k-1}^2 = \mathbb{E}[(S_k - S_{k-1})^2 \big| \mathcal{F}_{k-1}] = 1$) and $D_n = S_n^2 - n$ is the martingale of example 10.1.9. The predictable part n is the discrete quadratic variation of the walk.

The three examples below instantiate the three regimes the section has mapped: convergence in both senses when the martingale is bounded, almost-sure convergence without L^1 convergence when it is L^1-bounded but not uniformly integrable, and the closed case where the limit is a prescribed terminal variable.

Example 10.4.7: Three Regimes of Martingale Convergence

Each example is a martingale relative to a natural filtration; the point in each case is which mode of convergence holds and why.

1. *A bounded martingale.* Let (M_n) be a martingale with $|M_n| \leq C$ almost surely for a single constant C. Then $\{M_n\}$ is dominated by the integrable constant C, hence uniformly integrable, so theorem 10.4.4 applies: $M_n \to M_\infty$ almost surely and in L^1, and $M_n = \mathbb{E}[M_\infty \big| \mathcal{F}_n]$ a.s., with $|M_\infty| \leq C$. A concrete instance is Pólya's urn. An urn starts with one black and one white ball; at each step a ball is drawn uniformly and returned together with a fresh ball of the same colour. If M_n is the fraction of black balls after n draws, a direct computation shows $\mathbb{E}[M_{n+1} \big| \mathcal{F}_n] = M_n$, so (M_n) is a martingale, and $0 \leq M_n \leq 1$ bounds it. Therefore $M_n \to M_\infty$ almost surely and in L^1; the limiting fraction M_∞ is a random variable, in fact uniform on $[0, 1]$, and each urn history locks in its own limiting colour balance.

2. *An L^1-bounded martingale that is not uniformly integrable.* On the coin-tossing space (example 1.3.6) let $\varepsilon_1, \varepsilon_2, \ldots$ be independent signs, $\mathbb{P}(\varepsilon_k = \pm 1) = \frac{1}{2}$, and define the *doubling martingale*

$$M_0 = 1, \qquad M_n = \prod_{k=1}^{n} (1 + \varepsilon_k)$$

   Each factor $1+\varepsilon_k$ is 0 or 2 with equal probability, so $\mathbb{E}[1+\varepsilon_k \big| \mathcal{F}_{k-1}] = 1$, giving $\mathbb{E}[M_n \big| \mathcal{F}_{n-1}] = M_{n-1}$ a.s.: (M_n) is a martingale, and $\mathbb{E}[M_n] = 1$ for all n, so it is bounded in L^1. But a factor $1 + \varepsilon_k = 0$ ever occurring makes $M_n = 0$ from then on, and the first zero occurs almost surely (the probability that all of the first n signs are $+1$ is $2^{-n} \to 0$). Hence $M_n \to M_\infty = 0$ almost surely, while $\mathbb{E}[M_n] = 1 \not\to 0 = \mathbb{E}[M_\infty]$. The convergence is not in L^1, and by theorem 10.4.4 the martingale is not uniformly integrable. On the surviving path where every sign is $+$, the value is $M_n = 2^n \to \infty$; this vanishing-probability, exploding-value branch is exactly the escape of mass forbidden by uniform integrability (example 10.4.3), and it is why L^1 convergence fails.

3. *A closed martingale.* Fix an integrable Z and a filtration $\mathcal{F}_n \uparrow \mathcal{F}_\infty = \sigma\big(\bigcup_n \mathcal{F}_n\big)$, and set $M_n = \mathbb{E}[Z \big| \mathcal{F}_n]$. By the converse in theorem 10.4.4, (M_n) is a uniformly integrable martingale, so $M_n \to M_\infty$ almost surely and in L^1. Concretely, take $Z = \mathbb{1}_B$ for an event $B \in \mathcal{F}_\infty$; then $M_n = \mathbb{P}(B \big| \mathcal{F}_n)$ is the running conditional probability of B given the information at time n, and it converges almost surely to a limit that (as the next section shows) equals $\mathbb{1}_B$ when $B \in \mathcal{F}_\infty$. The estimates of a fixed event sharpen, in the limit, to the indicator of the event itself.

The three examples exhaust the logical possibilities in a way worth stating plainly. A bounded martingale sits inside the uniformly integrable class and converges in every sense; a closed martingale is the same phenomenon presented through its terminal variable; and between the martingales that converge in L^1 and those that converge only almost surely lies exactly the boundary drawn by uniform integrability, with the doubling martingale showing that the boundary is real. What no example can show is a martingale that fails to converge almost surely while remaining L^1-bounded, because theorem 10.4.1 rules that out; almost-sure convergence is the cheap conclusion, and L^1 convergence is the one that costs a hypothesis.

Exercise 10.4.1

1. Show that every nonnegative supermartingale (M_n) converges almost surely to an integrable limit M_∞, with $\mathbb{E}[M_\infty] \le \mathbb{E}[M_0]$. (Deduce the L^1 bound from the supermartingale property and nonnegativity, then apply theorem 10.4.1 to $-M_n$, or directly.) Give an example of a nonnegative supermartingale for which the inequality $\mathbb{E}[M_\infty] \le \mathbb{E}[M_0]$ is strict.

2. Let (M_n) be a martingale bounded in L^2, meaning $\sup_n \mathbb{E}[M_n^2] < \infty$. Prove that (M_n) is uniformly integrable, hence converges almost surely and in L^1; then show it in fact converges in L^2. (Use example 10.4.3 for uniform integrability, and for the L^2 conclusion recall that a martingale has orthogonal increments, so $\mathbb{E}[M_n^2] = \mathbb{E}[M_0^2] + \sum_{k=1}^n \mathbb{E}[(M_k - M_{k-1})^2]$.)

3. Let (X_n) be a martingale and φ a convex function with $\varphi(X_n)$ integrable for all n, so that $(\varphi(X_n))$ is a submartingale (proposition 10.1.4). Write down its Doob decomposition, identifying the predictable increasing part A_n as a sum of conditional expectations. Specialize to $\varphi(x) = x^2$ and the random walk $X_n = S_n$ to recover $A_n = n$.

4. In the doubling martingale of example 10.4.72, compute $\mathbb{E}|M_n - M_\infty|$ exactly as a function of n and verify it does not tend to 0. Then exhibit the constant K-tail $\sup_n \mathbb{E}[|M_n|\mathbb{1}_{\{|M_n|>K\}}]$ for each fixed $K \ge 1$ and confirm it does not tend to 0 as $K \to \infty$, so the family is not uniformly integrable directly from the definition.

5. Let (M_n) be a martingale and suppose $M_n \to M_\infty$ in L^1. Prove directly, without invoking theorem 10.4.4, that $M_n = \mathbb{E}[M_\infty | \mathcal{F}_n]$ a.s. for every n. (This is the identification step of the theorem's proof, isolated: fix $A \in \mathcal{F}_n$ and pass to the limit in $\mathbb{E}[M_m \mathbb{1}_A] = \mathbb{E}[M_n \mathbb{1}_A]$.)

6. Let (Z_n) be a Galton-Watson branching process with offspring mean $\mu \in (0, \infty)$: $Z_0 = 1$, and $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n+1,i}$ with the ξ i.i.d. with mean μ, independent of $\mathcal{F}_n$. Show that $M_n = Z_n/\mu^n$ is a nonnegative martingale, and conclude from theorem 10.4.1 that M_n converges almost surely to an integrable limit M_∞. Explain why, in the subcritical case $\mu < 1$, the almost-sure limit must be $M_\infty = 0$, and interpret this as extinction.

10.5 Applications

The machinery assembled in the previous four sections is now brought to bear. The convergence theorem of section 10.4 guarantees that certain processes settle down; the point of this section is that the limits they settle to are objects of independent interest, and that the martingale is the right lens for producing them. Four applications organize the section. The first turns the uniformly integrable convergence of a closed martingale into a zero-one law that subsumes Kolmogorov's tail law (theorem 2.4.3). The second recovers the Radon-Nikodym theorem, the very tool that built conditional expectation, as a martingale limit, closing a circle. The third analyzes the Galton-Watson branching process, the canonical model of a population that reproduces at random, through the nonnegative martingale Z_n/μ^n. The fourth proves a concentration inequality, Azuma-Hoeffding, that bounds how far a martingale with controlled increments can stray from its start, and that is now standard in the probabilistic method. Each is a payoff the earlier theory was built to make.

The natural starting point is the process $\mathbb{E}[Z|\mathcal{F}_n]$ obtained by conditioning a fixed integrable Z on a growing filtration. This is the closed martingale of theorem 10.4.4: it is uniformly integrable, so it converges almost surely and in L^1. The content of Lévy's theorem is the identification of its limit. As the information $\mathcal{F}_n$ increases to the full information $\mathcal{F}_\infty$ generated by the whole filtration, the running estimate $\mathbb{E}[Z|\mathcal{F}_n]$ converges to the estimate $\mathbb{E}[Z|\mathcal{F}_\infty]$ made with all the information at once. The estimate does not just stabilize; it stabilizes at the answer.

Theorem 10.5.1: Lévy's Upward Theorem

Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration (definition 10.1.1) and let $\mathcal{F}_\infty = \sigma\big(\bigcup_{n\geq 0}\mathcal{F}_n\big)$ be the σ-algebra it generates. For every integrable random variable Z,

$$\mathbb{E}[Z|\mathcal{F}_n] \longrightarrow \mathbb{E}[Z|\mathcal{F}_\infty] \qquad \text{almost surely and in } L^1$$

In particular, for every event $A \in \mathcal{F}_\infty$,

$$\mathbb{P}(A|\mathcal{F}_n) \longrightarrow \mathbb{1}_A \qquad \text{almost surely and in } L^1$$

The second statement is *Lévy's zero-one law*: as the accumulated information grows to encompass $\mathcal{F}_\infty$, the conditional probability of any $\mathcal{F}_\infty$-event converges to the event's indicator, which takes only the values 0 and 1. Knowledge of an event of the limiting σ-algebra becomes, in the limit, certain. Its name and its power come from this dichotomy: the limit is not some intermediate probability but a $\{0,1\}$-valued random variable.

***Proof*:**
Set $M_n = \mathbb{E}[Z|\mathcal{F}_n]$. By theorem 10.4.4, M_n is a uniformly integrable martingale, so it converges almost surely and in L^1 to an integrable limit M_∞. It remains to identify M_∞ with $\mathbb{E}[Z|\mathcal{F}_\infty]$, and the tool is the uniqueness half of the conditional-expectation characterization (theorem 9.1.2): it suffices to check that M_∞ is $\mathcal{F}_\infty$-measurable and that $\int_A M_\infty\, d\mathbb{P} = \int_A Z\, d\mathbb{P}$ for every $A \in \mathcal{F}_\infty$.

Step 1: Measurability. Each M_n is $\mathcal{F}_n$-measurable, hence $\mathcal{F}_\infty$-measurable since $\mathcal{F}_n \subseteq \mathcal{F}_\infty$. An almost-sure limit of $\mathcal{F}_\infty$-measurable functions is $\mathcal{F}_\infty$-measurable (after modification on the null set where convergence fails, which lies in $\mathcal{F}_\infty$ once the space is complete; the modification does not affect the almost-sure equivalence class). Thus M_∞ is $\mathcal{F}_\infty$-measurable.

Step 2: The averaging identity on a generating algebra. The union $\mathcal{A} = \bigcup_{n\geq 0}\mathcal{F}_n$ is an algebra: it is closed under complements and finite unions because the $\mathcal{F}_n$ increase, so any two sets drawn from it lie in a common $\mathcal{F}_N$. Fix $A \in \mathcal{A}$, say $A \in \mathcal{F}_m$. For every $n \geq m$ the defining identity for $M_n = \mathbb{E}[Z|\mathcal{F}_n]$ applies to $A \in \mathcal{F}_m \subseteq \mathcal{F}_n$, giving

$$\int_A M_n\, d\mathbb{P} = \int_A Z\, d\mathbb{P} \tag{10.26}$$

The left side converges to $\int_A M_\infty\, d\mathbb{P}$ as $n \to \infty$: since $M_n \to M_\infty$ in L^1 and $\mathbb{1}_A$ is bounded, $\int_A M_n\, d\mathbb{P} = \mathbb{E}[M_n \mathbb{1}_A] \to \mathbb{E}[M_\infty \mathbb{1}_A] = \int_A M_\infty\, d\mathbb{P}$. Passing to the limit in (10.26) yields

$\int_A M_\infty \, d\mathbb{P} = \int_A Z \, d\mathbb{P}$ for every $A \in \mathcal{A}$.

Step 3: Extension to $\mathcal{F}_\infty$. It is enough to treat $Z \geq 0$; the general case follows by splitting $Z = Z^+ - Z^-$ and subtracting, since conditional expectation is linear. For $Z \geq 0$ both $A \mapsto \int_A M_\infty \, d\mathbb{P}$ and $A \mapsto \int_A Z \, d\mathbb{P}$ are finite measures on $\mathcal{F}_\infty$ of equal total mass $\mathbb{E}[Z]$ that agree on the algebra $\mathcal{A}$. Since $\mathcal{A}$ is a π-system generating $\mathcal{F}_\infty$, the two measures agree on all of $\mathcal{F}_\infty$ by the uniqueness theorem for finite measures determined on a generating π-system (the π-λ uniqueness established in the foundations of the book). Hence $\int_A M_\infty \, d\mathbb{P} = \int_A Z \, d\mathbb{P}$ for every $A \in \mathcal{F}_\infty$. Together with Step 1, this is exactly the pair of conditions defining $\mathbb{E}[Z \big| \mathcal{F}_\infty]$, so $M_\infty = \mathbb{E}[Z \big| \mathcal{F}_\infty]$ almost surely by theorem 9.1.2.

Step 4: The zero-one law. Apply the result to $Z = \mathbb{1}_A$ for an event $A \in \mathcal{F}_\infty$. Then $\mathbb{E}[\mathbb{1}_A \big| \mathcal{F}_n] = \mathbb{P}(A \big| \mathcal{F}_n)$ by definition 9.1.3, and $\mathbb{E}[\mathbb{1}_A \big| \mathcal{F}_\infty] = \mathbb{1}_A$ because $\mathbb{1}_A$ is $\mathcal{F}_\infty$-measurable and integrable, so it is its own conditional expectation given $\mathcal{F}_\infty$. Therefore $\mathbb{P}(A \big| \mathcal{F}_n) \to \mathbb{1}_A$ almost surely and in L^1, completing the proof.

The theorem has an immediate structural consequence that recovers a result proved much earlier by an entirely different route. Kolmogorov's zero-one law (theorem 2.4.3) asserts that the tail σ-algebra of a sequence of independent random variables is trivial: every tail event has probability 0 or 1. Lévy's law delivers this as a corollary, and the derivation is worth carrying out because it exhibits the mechanism.

Corollary 10.5.2: Kolmogorov's Zero-One Law via Martingales

Let $X_1, X_2, \ldots$ be independent random variables and let $\mathcal{T}$ be their tail σ-algebra (definition 2.4.1). Then every $A \in \mathcal{T}$ has $\mathbb{P}(A) \in \{0, 1\}$.

***Proof*:**
Let $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ be the natural filtration, so $\mathcal{F}_\infty = \sigma(X_1, X_2, \ldots)$. Fix a tail event $A \in \mathcal{T}$; since $\mathcal{T} \subseteq \mathcal{F}_\infty$, Lévy's law gives $\mathbb{P}(A \big| \mathcal{F}_n) \to \mathbb{1}_A$ almost surely. Now the point of the tail σ-algebra is its independence of every initial segment: A is measurable with respect to $\sigma(X_{n+1}, X_{n+2}, \ldots)$, which is independent of $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ (definition 2.2.2). Independence of the event A from the σ-algebra $\mathcal{F}_n$ makes the conditional probability constant,

$$\mathbb{P}(A \big| \mathcal{F}_n) = \mathbb{P}(A) \qquad \text{almost surely}$$

because the constant $\mathbb{P}(A)$ is $\mathcal{F}_n$-measurable and satisfies the defining identity $\int_B \mathbb{P}(A) \, d\mathbb{P} = \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A \cap B) = \int_B \mathbb{1}_A \, d\mathbb{P}$ for every $B \in \mathcal{F}_n$, the middle equality being independence. Letting $n \to \infty$, the left side is the constant $\mathbb{P}(A)$ while the right side tends to $\mathbb{1}_A$; hence $\mathbb{1}_A = \mathbb{P}(A)$ almost surely. A random variable equal almost surely to a constant yet taking only the values 0 and 1 forces that constant into $\{0, 1\}$, so $\mathbb{P}(A) \in \{0, 1\}$, as we sought to show.

The two proofs of Kolmogorov's law illuminate different facets of the same fact. The original argument (theorem 2.4.3) showed a tail event to be independent of itself, hence to have $\mathbb{P}(A) = \mathbb{P}(A)^2$. The martingale argument shows instead that the running estimate $\mathbb{P}(A \big| \mathcal{F}_n)$, which independence pins to the constant $\mathbb{P}(A)$ at every finite stage, must in the limit resolve to the $\{0, 1\}$-valued indicator $\mathbb{1}_A$. The tension between a constant sequence and a $\{0, 1\}$-valued limit is what forces triviality. This is the characteristic move of the martingale method: a quantity is computed two ways, once by a finite-stage identity and once as a convergence limit, and the two computations are forced to agree.

The next application returns to the foundation of the whole of Part III. Conditional expectation was constructed in chapter 9 out of the Radon-Nikodym theorem, imported from the prerequisite volume. Martingale convergence now runs the construction backward: it produces the Radon-Nikodym density itself as the limit of a martingale of finite-stage densities. The setting is two probability measures on a common space, filtered so that on each $\mathcal{F}_n$ the second measure has a density against the first; these densities form a nonnegative martingale, and its limit is the density on the full σ-algebra.

Before stating the result, the finite-stage density deserves a word of motivation. On a finite σ-algebra generated by a partition, computing a density is elementary: it is the ratio of the two measures block by block. The filtration refines these partitions as n grows, and the martingale property is exactly the consistency of the block-ratios under refinement, the statement that averaging a finer density over a coarser block returns the coarser density. That consistency is what makes the sequence of densities a martingale rather than an arbitrary sequence.

Theorem 10.5.3: Radon-Nikodym via Martingales

Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$, and let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration with $\mathcal{F}_\infty = \sigma\big(\bigcup_n \mathcal{F}_n\big) = \mathcal{F}$. Suppose that for each n the restriction $\mathbb{Q}|_{\mathcal{F}_n}$ is absolutely continuous with respect to $\mathbb{P}|_{\mathcal{F}_n}$, with density

$$M_n = \frac{d\mathbb{Q}|_{\mathcal{F}_n}}{d\mathbb{P}|_{\mathcal{F}_n}}$$

Then (M_n) is a nonnegative martingale under $\mathbb{P}$ relative to $(\mathcal{F}_n)$. If moreover (M_n) is uniformly integrable under $\mathbb{P}$, then $M_n \to M_\infty$ almost surely and in $L^1(\mathbb{P})$, and $M_\infty = d\mathbb{Q}/d\mathbb{P}$ is the Radon-Nikodym density of $\mathbb{Q}$ with respect to $\mathbb{P}$ on all of $\mathcal{F}$.

***Proof*:**
Step 1: The densities form a martingale. Each $M_n \geq 0$ is $\mathcal{F}_n$-measurable by construction, and $\mathbb{E}_\mathbb{P}[M_n] = \mathbb{Q}(\Omega) = 1 < \infty$, so M_n is a nonnegative integrable adapted process. For the martingale identity, fix $A \in \mathcal{F}_n$. Since $A \in \mathcal{F}_n \subseteq \mathcal{F}_{n+1}$, the defining property of the density M_{n+1} on $\mathcal{F}_{n+1}$ gives

$$\int_A M_{n+1}\, d\mathbb{P} = \mathbb{Q}(A) = \int_A M_n\, d\mathbb{P}$$

the outer equalities being the density relations for M_{n+1} and M_n on the respective σ-algebras, both evaluated at the set $A \in \mathcal{F}_n$. Thus M_n is a $\mathcal{F}_n$-measurable random variable with $\int_A M_{n+1}\, d\mathbb{P} = \int_A M_n\, d\mathbb{P}$ for all $A \in \mathcal{F}_n$, which is precisely the statement that $\mathbb{E}_\mathbb{P}[M_{n+1} \big| \mathcal{F}_n] = M_n$ almost surely (definition 9.1.1). Hence (M_n) is a nonnegative martingale under $\mathbb{P}$ (definition 10.1.3).

Step 2: The limit is a global density. Assume (M_n) is uniformly integrable under $\mathbb{P}$. By theorem 10.4.4, $M_n \to M_\infty$ almost surely and in $L^1(\mathbb{P})$ for some integrable $M_\infty \geq 0$. To see that M_∞ is the density of $\mathbb{Q}$ against $\mathbb{P}$ on $\mathcal{F}$, argue as in Lévy's theorem (theorem 10.5.1). For $A \in \bigcup_n \mathcal{F}_n$, say $A \in \mathcal{F}_m$, the density relation gives $\int_A M_n\, d\mathbb{P} = \mathbb{Q}(A)$ for every $n \geq m$; letting $n \to \infty$ and using $L^1(\mathbb{P})$ convergence to move the limit inside the integral yields

$$\int_A M_\infty\, d\mathbb{P} = \mathbb{Q}(A) \qquad \text{for all } A \in \textstyle\bigcup_n \mathcal{F}_n \tag{10.27}$$

Both $A \mapsto \int_A M_\infty\, d\mathbb{P}$ and $A \mapsto \mathbb{Q}(A)$ are probability measures on $\mathcal{F}$ agreeing on the generating π-system $\bigcup_n \mathcal{F}_n$, so they agree on all of $\mathcal{F}$ by the same π-λ uniqueness theorem. Thus $\mathbb{Q}(A) =$

$\int_A M_\infty \, d\mathbb{P}$ for every $A \in \mathcal{F}$, which says exactly that $M_\infty = d\mathbb{Q}/d\mathbb{P}$, completing the proof.

The uniform-integrability hypothesis is not decorative: it is precisely the condition separating the absolutely continuous case from the singular one. When $\mathbb{Q} \ll \mathbb{P}$ on $\mathcal{F}$, the densities M_n are uniformly integrable and the theorem produces the global density. When $\mathbb{Q}$ has a part singular to $\mathbb{P}$, the martingale M_n still converges almost surely (it is nonnegative, hence $L^1(\mathbb{P})$-bounded, so theorem 10.4.1 applies), but it fails to be uniformly integrable, the $L^1(\mathbb{P})$ convergence breaks, and the limit M_∞ is the density only of the absolutely continuous part of $\mathbb{Q}$. The general measure-theoretic Radon-Nikodym theorem, together with its Lebesgue-decomposition companion, is proved in the prerequisite volume [2]; what the martingale viewpoint adds is a dynamic construction of the density as the limit of its finite-resolution approximations, and an exact reading of uniform integrability as the dividing line between the absolutely continuous and singular regimes.

This closes a conceptual loop that has run through Part III. The Radon-Nikodym theorem built conditional expectation in chapter 9; conditional expectation built martingales; and martingale convergence now rebuilds the Radon-Nikodym density. Each layer of the theory turns out to encode the one below it.

The third application leaves the abstract theory for a concrete stochastic model in which the nonnegative-martingale limit is the object of interest rather than a technical device. The Galton-Watson process models a population reproducing at random: it was introduced to study the survival of family surnames, and it is the simplest nontrivial branching process. Start with one individual. Each individual, independently, produces a random number of offspring drawn from a fixed distribution; the offspring form the next generation, each reproduces in turn, and so on. Two questions dominate the analysis: does the population die out, and if it survives, how fast does it grow. The martingale Z_n/μ^n, where μ is the mean number of offspring, answers the second and organizes the first.

Example 10.5.4: The Galton-Watson Branching Process

Fix a probability distribution $(p_k)_{k\geq 0}$ on the nonnegative integers, the *offspring distribution*, with mean $\mu = \sum_k k\, p_k \in (0,\infty)$ and generating function

$$\varphi(s) = \sum_{k\geq 0} p_k s^k, \qquad s \in [0,1]$$

Let $\{\xi_{n,i} \,|\, n \geq 0,\ i \geq 1\}$ be independent random variables, each with law (p_k), where $\xi_{n,i}$ is the number of offspring of the i-th individual in generation n. Define the population sizes by $Z_0 = 1$ and

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i}$$

(an empty sum when $Z_n = 0$, so extinction is absorbing). Let $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$ be the natural filtration.

The mean and the martingale. Conditioning on generation n, the next generation is a sum of Z_n independent offspring counts, each of mean μ, so

$$\mathbb{E}[Z_{n+1} \,|\, \mathcal{F}_n] = \mu\, Z_n \qquad \text{almost surely}$$

Consequently $\mathbb{E}[Z_n] = \mu^n$, and the normalized process

$$M_n = \frac{Z_n}{\mu^n}$$

satisfies $\mathbb{E}[M_{n+1} \big| \mathcal{F}_n] = \mu^{-(n+1)} \mathbb{E}[Z_{n+1} \big| \mathcal{F}_n] = \mu^{-(n+1)} \mu Z_n = M_n$ almost surely. Since $M_n \geq 0$ and $\mathbb{E}[M_n] = 1$, it is a nonnegative martingale (proposition 10.1.4). Being nonnegative it is L^1-bounded, so by theorem 10.4.1 the limit

$$M_\infty = \lim_{n\to\infty} \frac{Z_n}{\mu^n}$$

exists almost surely and is integrable: the population, rescaled by its expected growth, settles to a random limit M_∞.

Extinction. Let $q = \mathbb{P}(\text{extinction}) = \mathbb{P}(Z_n = 0 \text{ for some } n)$. The extinction events $\{Z_n = 0\}$ increase in n (extinction is absorbing), so q is their limiting probability. Conditioning on the first generation, extinction of the whole tree occurs iff each of the Z_1 independent subtrees goes extinct, and each subtree is an independent copy of the whole process; hence q is a fixed point of φ,

$$q = \varphi(q) \tag{10.28}$$

The generating function φ is convex and increasing on $[0, 1]$ with $\varphi(1) = 1$ and $\varphi'(1) = \mu$; it is strictly convex whenever $p_k > 0$ for some $k \geq 2$, as is forced when $\mu > 1$. Analysis of (10.28) splits on μ. If $\mu \leq 1$ (the *subcritical* and *critical* cases), the only fixed point in $[0, 1]$ is $q = 1$: extinction is certain. If $\mu > 1$ (the *supercritical* case), strict convexity forces exactly one fixed point $q < 1$ in $[0, 1)$, and q is the smallest nonnegative solution of $s = \varphi(s)$: the population survives with positive probability $1 - q$. Iterating φ from 0 produces $\mathbb{P}(Z_n = 0) = \varphi^{\circ n}(0) \uparrow q$, which both proves the fixed-point claim and computes q numerically.

The supercritical growth rate. When $\mu > 1$ the limit M_∞ can carry real information, but only if it does not vanish. Here is the danger: M_∞ is an L^1-limit of a nonnegative martingale with $\mathbb{E}[M_n] = 1$, yet nothing so far prevents $M_\infty = 0$ almost surely, which would happen exactly when the convergence is not in L^1 and the unit mass escapes to infinity. A clean sufficient condition rules this out. Suppose the offspring distribution has finite variance $\sigma^2 = \mathrm{Var}(\xi) < \infty$ and $\mu > 1$. A direct second-moment computation, conditioning generation by generation, gives

$$\mathrm{Var}(Z_{n+1}) = \mu^2 \mathrm{Var}(Z_n) + \sigma^2 \mu^n$$

which solving with $\mathrm{Var}(Z_0) = 0$ yields $\mathrm{Var}(Z_n) = \sigma^2 \mu^{n-1}(\mu^n - 1)/(\mu - 1)$. Dividing by μ^{2n},

$$\mathrm{Var}(M_n) = \frac{\sigma^2}{\mu} \cdot \frac{\mu^n - 1}{\mu^n(\mu - 1)} \leq \frac{\sigma^2}{\mu(\mu - 1)}$$

uniformly in n. Hence $\sup_n \mathbb{E}[M_n^2] = \sup_n (1 + \mathrm{Var}(M_n)) < \infty$: the martingale is bounded in L^2. An L^2-bounded martingale is uniformly integrable, so $M_n \to M_\infty$ in L^1, and therefore $\mathbb{E}[M_\infty] = \lim_n \mathbb{E}[M_n] = 1 > 0$. A nonnegative random variable with mean 1 cannot vanish almost surely, so

$$\mathbb{P}(M_\infty > 0) > 0$$

Moreover extinction forces $M_\infty = 0$ (once $Z_n = 0$ the process stays 0), so $\{\text{extinction}\} \subseteq \{M_\infty = 0\}$ up to a null set. The reverse holds almost surely under the finite-variance hypothesis. Writing $p = \mathbb{P}(M_\infty = 0)$ and conditioning on the first generation, $M_\infty = 0$ for the whole tree exactly when the limit vanishes for each of the Z_1 independent subtrees, each an independent copy of the process, so $p = \varphi(p)$; the fixed points of φ in $[0, 1]$ are q and 1, and the L^2 bound $\mathbb{P}(M_\infty > 0) > 0$ rules out $p = 1$, leaving $p = q = \mathbb{P}(\text{extinction})$. Equal probabilities on nested events give $\{M_\infty = 0\} = \{\text{extinction}\}$ almost surely, that is $\{M_\infty > 0\} = \{Z_n \not\to 0\}$ (the Kesten-Stigum theorem sharpens this beyond finite variance: $M_\infty > 0$ almost surely on survival exactly when $\mathbb{E}[\xi \log^+ \xi] < \infty$). On survival, then, the population grows like $M_\infty \mu^n$ with a strictly positive random constant: the growth is exponential at rate μ, with the random prefactor M_∞ recording the accidents of the early generations.

The Galton-Watson analysis is a template for the martingale method applied to a concrete model. A quantity of interest, the population size, grows on average geometrically; dividing by its mean produces a nonnegative martingale whose limit is automatic; and the qualitative question, extinction versus survival, is settled by a fixed-point equation for the generating function, while the quantitative question, the growth rate on survival, is settled by an L^2 estimate that upgrades almost-sure convergence to L^1 and thereby keeps the limit from collapsing to zero. The dichotomy $\mu \leq 1$ versus $\mu > 1$ that governs extinction is the branching-process analogue of the recurrence-transience dichotomy for random walks: a fair or losing game dies out; a favorable one survives with positive probability.

The final application is a concentration inequality, and it changes the register from convergence to quantitative tail bounds. The laws of large numbers and the central limit theorem describe sums of *independent* variables. Many quantities of interest are not sums of independent variables but do have controlled dependence: they can be written as martingales whose increments are bounded. The Azuma-Hoeffding inequality gives such a martingale the same subgaussian tail bound that Hoeffding's inequality gives a sum of bounded independent variables, replacing the independence hypothesis by the martingale property together with a uniform bound on the increments. It is proved by the exponential-supermartingale method, the same Chernoff-bound mechanism (proposition 3.4.7) used for independent sums earlier in the book and for the maximal bound of the iterated-logarithm chapter (proposition 8.1.3).

The engine is a one-line estimate on a single bounded increment, Hoeffding's lemma, which controls the conditional moment generating function of a mean-zero variable by its range.

Lemma 10.5.5: Hoeffding's Lemma, Conditional Form

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra and let D be a random variable with $\mathbb{E}[D \big| \mathcal{G}] = 0$ almost surely and $|D| \leq c$ almost surely for a constant $c \geq 0$. Then for every $\theta \in \mathbb{R}$,

$$\mathbb{E}\big[e^{\theta D} \big| \mathcal{G}\big] \leq e^{\theta^2 c^2/2} \qquad \text{almost surely}$$

***Proof*:**
Fix θ. Since $x \mapsto e^{\theta x}$ is convex and $D \in [-c, c]$ almost surely, write D as the convex combination of the endpoints,

$$D = \frac{c - D}{2c}\,(-c) + \frac{c + D}{2c}\,(c)$$

(taking $c > 0$; the case $c = 0$ forces $D = 0$ and both sides equal 1). Convexity gives $e^{\theta D} \leq \frac{c-D}{2c} e^{-\theta c} + \frac{c+D}{2c} e^{\theta c}$ pointwise. Taking $\mathbb{E}[\cdot \big| \mathcal{G}]$ and using $\mathbb{E}[D \big| \mathcal{G}] = 0$, the D-linear terms drop out and

$$\mathbb{E}[e^{\theta D} \big| \mathcal{G}] \leq \frac{e^{-\theta c} + e^{\theta c}}{2} = \cosh(\theta c)$$

almost surely. The elementary inequality $\cosh(t) \leq e^{t^2/2}$ (compare the Taylor coefficients: $\frac{1}{(2k)!} \leq \frac{1}{2^k k!}$ for every k) with $t = \theta c$ gives $\cosh(\theta c) \leq e^{\theta^2 c^2/2}$, as we sought to show.

With the per-increment estimate in hand, the tail bound follows by building an exponential supermartingale and applying Markov's inequality to its terminal value, the standard Chernoff optimization. The motivation for reading the result is that it converts a purely qualitative fact, that a martingale with small steps does not move much, into a sharp exponential estimate matching the independent case.

Proposition 10.5.6: Azuma-Hoeffding Inequality

Let $(M_n)_{n\geq 0}$ be a martingale relative to $(\mathcal{F}_n)$ whose increments $D_k = M_k - M_{k-1}$ satisfy $|D_k| \leq c_k$ almost surely for constants $c_k \geq 0$. Then for every $\lambda > 0$,

$$\mathbb{P}\big(M_n - M_0 \geq \lambda\big) \leq \exp\left(-\frac{\lambda^2}{2\sum_{k=1}^n c_k^2}\right)$$

and, applying the same bound to $(-M_n)$,

$$\mathbb{P}\big(|M_n - M_0| \geq \lambda\big) \leq 2\exp\left(-\frac{\lambda^2}{2\sum_{k=1}^n c_k^2}\right)$$

***Proof*:**
Fix $\theta > 0$, to be optimized. The increments $D_k = M_k - M_{k-1}$ satisfy $\mathbb{E}[D_k \,|\, \mathcal{F}_{k-1}] = \mathbb{E}[M_k \,|\, \mathcal{F}_{k-1}] - M_{k-1} = 0$ almost surely by the martingale property (definition 10.1.3), and $|D_k| \leq c_k$ almost surely. So the conditional Hoeffding lemma (lemma 10.5.5) applies to each D_k with $\mathcal{G} = \mathcal{F}_{k-1}$,

$$\mathbb{E}\big[e^{\theta D_k} \,\big|\, \mathcal{F}_{k-1}\big] \leq e^{\theta^2 c_k^2/2} \qquad \text{almost surely} \tag{10.29}$$

Step 1: The exponential moment telescopes. Write $S_n = M_n - M_0 = \sum_{k=1}^n D_k$. Conditioning on $\mathcal{F}_{n-1}$ and using that $e^{\theta S_{n-1}}$ is $\mathcal{F}_{n-1}$-measurable (taking out what is known, proposition 9.2.3),

$$\mathbb{E}\big[e^{\theta S_n}\big] = \mathbb{E}\big[e^{\theta S_{n-1}}\,\mathbb{E}[e^{\theta D_n} \,\big|\, \mathcal{F}_{n-1}]\big] \leq e^{\theta^2 c_n^2/2}\,\mathbb{E}\big[e^{\theta S_{n-1}}\big]$$

the inequality by (10.29) and the nonnegativity of $e^{\theta S_{n-1}}$. Iterating this bound down to $S_0 = 0$, where $\mathbb{E}[e^{\theta S_0}] = 1$, gives

$$\mathbb{E}\big[e^{\theta(M_n - M_0)}\big] \leq \exp\left(\frac{\theta^2}{2}\sum_{k=1}^n c_k^2\right) \tag{10.30}$$

the subgaussian bound on the moment generating function of the increment sum.

Step 2: Chernoff optimization. By Markov's inequality applied to the nonnegative variable $e^{\theta(M_n - M_0)}$ (theorem 3.2.4, the Chernoff bound proposition 3.4.7), for every $\theta > 0$,

$$\mathbb{P}\big(M_n - M_0 \geq \lambda\big) = \mathbb{P}\big(e^{\theta(M_n - M_0)} \geq e^{\theta\lambda}\big) \leq e^{-\theta\lambda}\,\mathbb{E}\big[e^{\theta(M_n - M_0)}\big] \leq \exp\left(-\theta\lambda + \frac{\theta^2}{2}\sum_{k=1}^n c_k^2\right)$$

using (10.30). Let $C = \sum_{k=1}^n c_k^2$. The exponent $-\theta\lambda + \frac{1}{2}\theta^2 C$ is minimized over $\theta > 0$ at $\theta = \lambda/C$, giving exponent $-\lambda^2/(2C)$; substituting yields

$$\mathbb{P}\big(M_n - M_0 \geq \lambda\big) \leq \exp\left(-\frac{\lambda^2}{2\sum_{k=1}^n c_k^2}\right)$$

which is the one-sided bound. Applying it to the martingale $(-M_n)$, whose increments $-D_k$ satisfy the same bound $|-D_k| \le c_k$, controls the lower tail $\mathbb{P}(M_n - M_0 \le -\lambda)$ identically; the union of the two events gives the two-sided bound with the factor 2, completing the proof.

The inequality is the martingale generalization of Hoeffding's bound for independent sums, and it reduces to it exactly. If $M_n = X_1 + \cdots + X_n$ with the X_k independent, mean-zero, and bounded by $|X_k| \le c_k$, then (M_n) is a martingale for its natural filtration (proposition 10.1.4) with increments $D_k = X_k$, and Azuma-Hoeffding returns the classical two-sided subgaussian bound. The gain is that independence is never used in the proof: only the conditional mean-zero property $\mathbb{E}[D_k \big| \mathcal{F}_{k-1}] = 0$ and the increment bound enter. This is what makes the inequality so widely applicable in the probabilistic method. A function $f(Y_1, \ldots, Y_n)$ of independent inputs that changes by at most c_k when the k-th input is altered (a *bounded-differences* function) generates a martingale, the Doob martingale $M_k = \mathbb{E}[f \big| \mathcal{F}_k]$ with $\mathcal{F}_k = \sigma(Y_1, \ldots, Y_k)$, whose increments obey $|D_k| \le c_k$; Azuma-Hoeffding then concentrates f around its mean without any linearity or independence in f itself. The chromatic number of a random graph, the length of a longest common subsequence, and the empirical width of a random configuration are all concentrated by this single device.

The four applications of this section share a common shape and, together, close the book's development of the discrete theory. A martingale is built from the problem at hand; the convergence theorem or the exponential-supermartingale bound is applied to it; and the limit or the tail estimate is read back as the answer to the original question, be it a zero-one dichotomy, a Radon-Nikodym density, an extinction probability, or a concentration bound. The unifying reading is that a martingale is the language of information accumulating over time: $\mathcal{F}_n$ records what is known by stage n, the martingale property says the best current estimate is unbiased for the next, and the theorems describe how such estimates converge and concentrate. What lies beyond is the passage to continuous time. When the discrete index n is replaced by a continuous parameter t, the random walk becomes Brownian motion, the martingale transform becomes the Ito integral, and the increment bounds of this section become the quadratic variation that governs stochastic differential equations. That theory, the foundation of stochastic calculus and its applications to mathematical finance, is where the probabilistic story told in this book is continued.

Exercise 10.5.1

1. Lévy's zero-one law gives a slick proof of a conditional Borel-Cantelli lemma. Let $(\mathcal{F}_n)$ be a filtration and let $A_n \in \mathcal{F}_n$ be events. Show that, almost surely,

$$\{A_n \text{ occurs infinitely often}\} = \Big\{\sum_{n\ge1} \mathbb{P}(A_n \big| \mathcal{F}_{n-1}) = \infty\Big\}$$

(*Hint.* This is Lévy's extension of the Borel-Cantelli lemmas; both inclusions follow by comparing $\sum \mathbb{1}_{A_n}$ with $\sum \mathbb{P}(A_n \big| \mathcal{F}_{n-1})$ via the martingale $S_n = \sum_{k\le n}(\mathbb{1}_{A_k} - \mathbb{P}(A_k \big| \mathcal{F}_{k-1}))$ and its convergence set. State clearly which martingale result you use.)

2. Consider the Galton-Watson process with geometric offspring distribution $p_k = (1-a)a^k$ for $k \ge 0$, where $a \in (0, 1)$. Compute the mean μ and the generating function $\varphi(s)$. Find the extinction probability q as the smallest fixed point of φ, and determine for which a the process is supercritical.

3. Let (Z_n) be a Galton-Watson process with $\mu < 1$. Using the martingale $M_n = Z_n/\mu^n$ or otherwise, prove that $\mathbb{P}(Z_n \ge 1) \le \mu^n$, and deduce that extinction is certain $(q = 1)$ directly, without solving the fixed-point equation.

4. Let $Y_1, \dots, Y_n$ be independent and let $f\colon \mathbb{R}^n \to \mathbb{R}$ satisfy the bounded-differences condition: changing the k-th coordinate changes the value of f by at most c_k. Set $\mathcal{F}_k = \sigma(Y_1, \dots, Y_k)$ and $M_k = \mathbb{E}[f(Y_1, \dots, Y_n) \big| \mathcal{F}_k]$. Show that (M_k) is a martingale with $M_0 = \mathbb{E}[f]$ and $M_n = f(Y_1, \dots, Y_n)$, and that $|M_k - M_{k-1}| \le c_k$ almost surely. Conclude McDiarmid's inequality: $\mathbb{P}(|f - \mathbb{E}[f]| \ge \lambda) \le 2\exp(-\lambda^2/(2\sum_k c_k^2))$.

5. Apply Azuma-Hoeffding to the simple symmetric random walk $S_n = X_1 + \cdots + X_n$ with $X_k = \pm 1$ fair (example 1.3.6). Write the resulting tail bound on $\mathbb{P}(|S_n| \ge \lambda)$, and compare its exponent with the exponent predicted by the central limit theorem (theorem 7.1.3) and by the law of the iterated logarithm (theorem 8.2.3) for $\lambda = t\sqrt{n}$.

6. Prove the converse direction used implicitly in this section: for integrable Z, the process $M_n = \mathbb{E}[Z \big| \mathcal{F}_n]$ is uniformly integrable. (*Hint.* Bound $\mathbb{E}[|M_n| \mathbb{1}_{\{|M_n| > K\}}]$ using conditional Jensen $|M_n| \le \mathbb{E}[|Z| \big| \mathcal{F}_n]$ and the uniform integrability of the single variable $|Z|$; the event $\{|M_n| > K\}$ lies in $\mathcal{F}_n$.)

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What Next?

This book has carried the reader from the probability space and its events through random variables and independence, expectation and the inequalities it supports, the weak and strong laws of large numbers, weak convergence and the characteristic function, the central limit theorem and the law of the iterated logarithm, and finally conditional expectation and the discrete-time martingales built on it. Each of these opens into a subject larger than any chapter can hold. This closing chapter is a map of where to go next, pairing every direction with the companion EYNTKA volume where one exists and with the standard texts a working probabilist keeps at hand.

Continuous-time processes and stochastic calculus. The book stops, by design, at the threshold of continuous time. The martingales of chapter 10 have a continuous-parameter counterpart whose central object is Brownian motion, and the theory of integration against it, the Ito integral and the stochastic differential equations it makes sense of, is the direct sequel to this volume and the planned companion *Stochastic Calculus.* The standard texts are Karatzas and Shreve's *Brownian Motion and Stochastic Calculus*, Revuz and Yor's *Continuous Martingales and Brownian Motion*, Le Gall's *Brownian Motion, Martingales, and Stochastic Calculus*, and, for the applied side, Oksendal's *Stochastic Differential Equations.*

Markov processes and ergodic theory. The limit theorems of chapter 5 were proved for independent summands; the next step is dependence with structure. The ergodic theorem extends the strong law to stationary sequences, and Markov chains, evolving with memory only of the present, carry their own theory of stationary distributions, recurrence, and mixing. Durrett's *Probability: Theory and Examples* develops both alongside the material of this book; Norris's *Markov Chains* and Levin and Peres's *Markov Chains and Mixing Times* are the standard chain references, and Walters's *An Introduction to Ergodic Theory* the standard entry to the ergodic side.

Concentration and high-dimensional probability. The Azuma-Hoeffding inequality of section 10.5 was a first instance of a large theory: how a function of many independent or weakly dependent inputs concentrates about its mean. The systematic account of sub-Gaussian and sub-exponential tails, bounded-differences inequalities, and the phenomena peculiar to high dimension is in Vershynin's

High-Dimensional Probability and in Boucheron, Lugosi, and Massart's *Concentration Inequalities*. The refined asymptotics of rare events, sharpening the law of large numbers in the opposite direction from the central limit theorem, is the theory of large deviations, for which Dembo and Zeitouni's *Large Deviations Techniques and Applications* is standard.

Mathematical statistics. This book studies the laws that govern a sample; statistics runs the inference backward, from the sample to the law that produced it. Estimation, hypothesis testing, and their large-sample theory, consistency and asymptotic normality, rest directly on the strong law (chapter 5) and the central limit theorem (chapter 7). Rice's *Mathematical Statistics and Data Analysis* is a readable first course, Casella and Berger's *Statistical Inference* the standard graduate text, and van der Vaart's *Asymptotic Statistics* the measure-theoretic account of the limit theory.

The Fourier and number-theoretic threads. Two strands of this book are entrances to subjects of their own. The characteristic function of chapter 6 is Fourier analysis on the line, developed for its own sake in the companion *Harmonic Analysis* (codename `NateHarmonicAnalysis`). The probabilistic number theory that surfaced in the laws of large numbers and the central limit theorem, the Hardy-Ramanujan and Erdos-Kac theorems on the prime factors of a random integer, is a field in its own right: Tenenbaum's *Introduction to Analytic and Probabilistic Number Theory* is the standard account, and the analytic input rests on *Analytic Number Theory* [1].

A closing suggestion. The surest way to make this material one's own is to compute with it. Taking the fair coin and its random walk, the example that runs through every chapter, and pushing it through each major tool in turn, a Borel-Cantelli argument, a characteristic-function computation, an optional-stopping identity, a martingale-convergence limit, fixes the machinery as no second reading can. From there every direction above is open.

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Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Analytic Number Theory*. URL: https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-analytic-number-theory.pdf&name=Analytic%20Number%20Theory&source=books.

[2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Measure Theory*. 2023. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_measure_theory.pdf.